A Finite-Set Escape Program for the L^2 Benjamini-Schramm Conjecture

Working draft

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Abstract

This is a working manuscript for a long-term attempt to prove the Benjamini-Schramm non-uniqueness conjecture for Bernoulli bond percolation on nonamenable quasi-transitive graphs. The stronger target is the L^2 operator inequality $p_c(G) < p_{2\to 2}(G)$, where $p_{2\to 2}$ is the boundedness threshold of the two-point function $T_p(x,y) = \mathbf{P}_p(x \leftrightarrow y)$ on $\ell^2(V)$. A theorem of Hutchcroft reduces this stronger target to a mean-field susceptibility estimate and a finite-set escape condition for the subcritical two-point kernel. The present draft records the selected open problem, the verified status, the operator formulation, and several elementary reductions that isolate the remaining obstruction. It does not yet claim a solution of the global conjecture.

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1 Status of the project

The objective of this project is to solve a major open problem in probability theory, write the solution as a full LaTeX paper, and compile a PDF. The target selected here is the Benjamini-Schramm non-uniqueness conjecture for Bernoulli bond percolation on nonamenable quasi-transitive graphs [BS96]. The problem is both classical and active: recent work continues to solve important subclasses and threshold phenomena rather than the full generality [HP24b, HP24a, TT25, Li26].

The present manuscript is not a final solution. It is a structured research draft whose purpose is to make the target precise enough that subsequent proof attempts have a clear place to attach. The paper deliberately separates proved reductions from conjectural bridges.

1.1 Open-status audit

The current-status audit used for this draft has three layers. First, the local solved-problem register in the surrounding workspace records solved work on Gaussian phase retrieval, Balan-Wang type stability, Plucker reductions, and related random-matrix questions. It does not record a percolation theorem of the kind studied here. Second, the main modern reference for the operator route [Hut19] states Conjecture 2.1 and then solves it only under additional large-scale geometric assumptions, most prominently Gromov hyperbolicity. Third, the recent papers located during the audit continue to prove special classes or adjacent threshold phenomena [HP24b, HP24a, Li26]. No source found in the 2026-06-08 search announced a general proof of $p_c < p_u$, or of the stronger $p_c < p_{2\to 2}$, for every nonamenable quasi-transitive graph.

This matters methodologically. The project should not spend effort proving a statement already known under a different name, and it should not confuse a planar-site conjecture with the bond-percolation quasi-transitive conjecture. The target here is deliberately the general bond-percolation statement, with the L^2 threshold as the preferred route.

1.2 Success standard

A final solution would need at least the following components.

S1. A precise theorem statement covering all connected, locally finite, quasi-transitive, nonamenable graphs, or a clearly stronger theorem that implies that statement.

S2. A proof that does not assume hidden extra hypotheses such as hyperbolicity, planarity, nonunimodularity, product structure, property T, or existence of harmonic Dirichlet functions.

S3. A check against amenable quasi-transitive graphs, where $p_c = p_u$, to confirm that the argument uses nonamenability at the right point.

S4. A literature comparison explaining why the theorem is not subsumed by existing special-case results.

S5. A complete LaTeX manuscript and a compiled PDF.

The current draft satisfies only the setup and reduction parts of this list. The main proof is still missing.

2 Percolation background

Let $G = (V, E)$ be a connected, locally finite, infinite graph. Bernoulli bond percolation at parameter $p \in [0, 1]$ keeps each edge independently with probability p and deletes it otherwise. We write $G[p]$ for the resulting random subgraph and write $x \leftrightarrow y$ for the event that x and y lie in the same open cluster.

The critical probability and uniqueness threshold are

$$p_c(G) = \inf\{p : \mathbf{P}_p(G[p] \text{ has an infinite cluster}) = 1\}$$

and

$$p_u(G) = \inf\{p : \mathbf{P}_p(G[p] \text{ has a unique infinite cluster}) = 1\}.$$

For quasi-transitive graphs, the number of infinite clusters is almost surely 0, 1, or ∞, and the interval (p_c, p_u), when non-empty, is the non-uniqueness phase.

A graph is quasi-transitive if its automorphism group has finitely many orbits on V. It is nonamenable if its Cheeger constant is positive:

$$h(G) = \inf_{K \subset V \text{ finite}} \frac{|\partial_E K|}{\sum_{x \in K} \deg(x)} > 0.$$

Here $\partial_E K$ denotes the set of edges with one endpoint in K and one endpoint outside K.

Conjecture 2.1 (Benjamini-Schramm)**.** *Let G be a connected, locally finite, quasi-transitive graph. Then*

$$p_c(G) < p_u(G)$$

if and only if G is nonamenable.

The amenable direction is known: amenable quasi-transitive graphs satisfy $p_c = p_u$. The hard direction is to prove $p_c < p_u$ for every nonamenable quasi-transitive graph. Major positive cases include nonamenable planar quasi-transitive graphs, nonunimodular transitive graphs, hyperbolic quasi-transitive graphs, and several group-theoretic and product classes [Hut19, HH21, HP24b, HP24a].

3 The L^2 strengthening

For $p < p_c$, define the two-point function

$$T_p(x,y) = \tau_p(x,y) = \mathbf{P}_p(x \leftrightarrow y).$$

We view T_p as a positive symmetric kernel on V. Its susceptibility row sum is

$$\chi_p(x) = \sum_{y \in V} \tau_p(x,y), \qquad \overline{\chi}_p = \sup_{x \in V} \chi_p(x).$$

For quasi-transitive graphs, $\overline{\chi}_p < \infty$ for $p < p_c$ and diverges as $p \uparrow p_c$.

Following Hutchcroft [Hut19], define

$$p_{2\to 2}(G) = \sup\{p \in [0,1] : T_p \text{ is bounded on } \ell^2(V)\}.$$

For quasi-transitive graphs,

$$p_c(G) \le p_{2\to 2}(G) \le p_u(G).$$

Thus the following conjecture strengthens Conjecture 2.1.

Conjecture 3.1 (L^2 Benjamini-Schramm)**.** *Let G be connected, locally finite, quasi-transitive, and nonamenable. Then*

$$p_c(G) < p_{2\to 2}(G).$$

This formulation is attractive because $p_{2\to 2}$ can be attacked by functional analysis of the two-point operator. It also has a useful finite-set criterion, recalled next.

4 Hutchcroft's finite-set criterion

The next theorem is a restatement of the criterion developed in [Hut19]. It is included here to fix notation and to identify the precise obstruction that this project attempts to remove.

Theorem 4.1 (Finite-set criterion)**.** *Let G be connected, locally finite, and quasi-transitive. Suppose that*

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty \tag{MF}$$

and

$$\lim_{p \uparrow p_c} \sup_{K \subset V \text{ finite}} \frac{\sum_{x,y \in K} \tau_p(x,y)}{\overline{\chi}_p |K|} = 0. \tag{FE}$$

Then $p_c(G) < p_{2\to 2}(G)$. In particular $p_c(G) < p_u(G)$.

Remark 4.2. Condition (MF) is a mean-field susceptibility upper bound. Condition (FE) is the finite-set escape condition: after normalizing by the maximal row sum, no finite set can retain a positive fraction of the two-point mass as $p \uparrow p_c$.

The known hyperbolic proof verifies these two conditions using large-scale geometry. The open problem is whether nonamenability alone is enough.

5 Normalized kernels

Define the normalized two-point kernel

$$Q_p(x,y) = \frac{\tau_p(x,y)}{\overline{\chi}_p}.$$

Then Q_p is a positive symmetric kernel satisfying

$$\sup_x \sum_y Q_p(x,y) \le 1.$$

The finite-set escape condition becomes

$$\lim_{p \uparrow p_c} \eta_p = 0, \qquad \eta_p = \sup_{K \subset V \text{ finite}} \frac{1}{|K|} \sum_{x,y \in K} Q_p(x,y).$$

Lemma 5.1 (Elementary bounds for Q_p)**.** *For every $p < p_c$, the kernel Q_p is positive, symmetric, and satisfies*

$$\|Q_p\|_{1\to 1} \le 1, \qquad \|Q_p\|_{\infty\to\infty} \le 1, \qquad \|Q_p\|_{2\to 2} \le 1.$$

Proof. The first two bounds follow from symmetry and from the definition of $\overline{\chi}_p$:

$$\sup_x \sum_y Q_p(x,y) = \overline{\chi}_p^{-1} \sup_x \sum_y \tau_p(x,y) \le 1.$$

Since Q_p is symmetric, the same bound holds for column sums. The ℓ^2 bound follows by interpolation, or by the Schur test. □

Lemma 5.2 (Failure extraction)**.** *If finite-set escape fails, then there exist $\epsilon > 0$, parameters $p_n \uparrow p_c$, finite non-empty sets $K_n \subset V$, and functions*

$$f_n = |K_n|^{-1/2} \mathbf{1}_{K_n}$$

such that $\|f_n\|_2 = 1$ and

$$\langle f_n, Q_{p_n} f_n \rangle \ge \epsilon$$

for every n.

Proof. If finite-set escape fails, then for some $\epsilon > 0$ there are $p_n \uparrow p_c$ and finite non-empty K_n with

$$|K_n|^{-1} \sum_{x,y \in K_n} Q_{p_n}(x,y) \ge \epsilon.$$

For $f_n = |K_n|^{-1/2} \mathbf{1}_{K_n}$, symmetry and positivity give

$$\langle f_n, Q_{p_n} f_n \rangle = |K_n|^{-1} \sum_{x,y \in K_n} Q_{p_n}(x,y),$$

which is at least ϵ. □

Remark 5.3. Lemma 5.2 is elementary, but it is the correct starting point for a contradiction proof. The hard task is to convert large $\langle f_n, Q_{p_n} f_n \rangle$ into almost-invariance for the graph random walk, or into another object forbidden by nonamenability.

Lemma 5.4 (Failure has spectral content)**.** *If finite-set escape fails, then there are $p_n \uparrow p_c$ such that*

$$\liminf_{n\to\infty} \|Q_{p_n}\|_{2\to 2} > 0.$$

Equivalently,

$$\liminf_{n\to\infty} \frac{\|T_{p_n}\|_{2\to 2}}{\overline{\chi}_{p_n}} > 0.$$

Proof. By Lemma 5.2, failure of escape gives unit vectors f_n with $\langle f_n, Q_{p_n} f_n\rangle \geq \epsilon$. Hence

$$\|Q_{p_n}\|_{2\to 2} \geq \langle f_n, Q_{p_n} f_n\rangle \geq \epsilon.$$

The second display is just $Q_p = T_p/\overline{\chi}_p$. □

Remark 5.5. The converse is false in general: a positive lower bound on $\|Q_p\|_{2\to 2}$ need not be witnessed by normalized indicators of finite sets. The finite-set formulation is stronger and is tuned to the isoperimetric contradiction one wants from nonamenability.

6 A finite-volume boundary decomposition

The previous lemmas say what failure of finite-set escape gives at the level of the normalized kernel. The next proposition opens that event up using the geometry of a finite set. It is the first non-tautological reduction in this draft.

For a finite set $K \subset V$, let $\tau_p^K(x,y)$ be the probability that x and y are connected by an open path using only edges of the induced subgraph on K. Define

$$\chi_p^K(x) = \sum_{y\in K} \tau_p^K(x,y), \qquad S_p(K) = \sum_{x,y\in K} \tau_p^K(x,y).$$

Let

$$\partial^+ K = \{(a,b) : a \in K,\ b \notin K,\ a \sim b\}$$

be the oriented edge boundary, always oriented from inside K to outside K.

Proposition 6.1 (Last-exit boundary decomposition)**.** *For every finite non-empty $K \subset V$ and every $p \in [0,1]$,*

$$\sum_{x,y\in K} \tau_p(x,y) \leq S_p(K) + p^2 \sum_{\substack{(a,b)\in\partial^+ K\\(d,c)\in\partial^+ K}} \chi_p^K(a)\, \tau_p(b,c)\, \chi_p^K(d).$$

Proof. Fix $x, y \in K$. If x is connected to y by a path contained in K, the event contributes to $\tau_p^K(x,y)$. Otherwise choose a self-avoiding open path γ from x to y that is not contained in K. Let $(a,b) \in \partial^+ K$ be the first oriented boundary edge crossed by γ, and let $(d,c) \in \partial^+ K$ be the last oriented boundary edge crossed by γ, where the actual last crossing is from $c \notin K$ to $d \in K$.

The path γ decomposes into five edge-disjoint pieces: an internal path in K from x to a, the boundary edge ab, the middle segment from b to c, the boundary edge cd, and an internal path in K from d to y. The middle segment need not stay outside K; it may make further excursions through K. This is why the middle two-point function below is the ambient $\tau_p(b,c)$, not a complement-restricted two-point function.

Thus γ witnesses the disjoint occurrence of the following five increasing events: x is connected to a inside K, the edge ab is open, b is connected to c in G, the edge cd is open, and d is connected

to y inside K. By the BK inequality, or equivalently the standard disjoint-occurrence bound for increasing connection events [Gri99], the probability of this contribution is at most

$$\tau_p^K(x,a)\, p\, \tau_p(b,c)\, p\, \tau_p^K(d,y).$$

Taking the union bound over the possible first and last boundary crossings gives

$$\tau_p(x,y) \le \tau_p^K(x,y) + p^2 \sum_{\substack{(a,b)\in\partial^+K\\(d,c)\in\partial^+K}} \tau_p^K(x,a)\tau_p(b,c)\tau_p^K(d,y).$$

Summing over $x,y\in K$ and using symmetry of τ_p^K gives the stated inequality. □

Corollary 6.2 (Interior-boundary dichotomy)**.** *Let $Q_p = \tau_p/\overline{\chi}_p$. If, for some finite non-empty $K\subset V$,*

$$\frac{1}{|K|}\sum_{x,y\in K} Q_p(x,y) \ge \epsilon,$$

then at least one of the following two alternatives holds:

$$S_p(K) \ge \frac{\epsilon}{2}\overline{\chi}_p|K| \tag{I}$$

or

$$p^2 \sum_{\substack{(a,b)\in\partial^+K\\(d,c)\in\partial^+K}} \chi_p^K(a)\, Q_p(b,c)\, \chi_p^K(d) \ge \frac{\epsilon}{2}|K|. \tag{B}$$

Proof. Divide the inequality of Proposition 6.1 by $\overline{\chi}_p|K|$. If the normalized left-hand side is at least ϵ, then the normalized interior term or the normalized boundary-transfer term is at least $\epsilon/2$. □

Corollary 6.3 (Boundary L^2-mass obstruction)**.** *For $z\in V\setminus K$, define the boundary susceptibility weight*

$$\mu_{p,K}(z) = \sum_{\substack{a\in K\\ a\sim z}} \chi_p^K(a),$$

and set $\mu_{p,K}(z)=0$ for $z\in K$. If the boundary alternative (B) holds and $p>0$, then

$$\|\mu_{p,K}\|_2^2 \ge \frac{\epsilon}{2p^2}|K| \ge \frac{\epsilon}{2}|K|.$$

Proof. The boundary term in (B) can be written as

$$p^2 \sum_{b,c\in V} \mu_{p,K}(b) Q_p(b,c) \mu_{p,K}(c) = p^2 \left\langle \mu_{p,K}, Q_p\mu_{p,K}\right\rangle.$$

By Lemma 5.1, Q_p is a contraction on $\ell^2(V)$. Hence

$$\left\langle \mu_{p,K}, Q_p\mu_{p,K}\right\rangle \le \|\mu_{p,K}\|_2^2.$$

Combining this with (B) gives the first inequality. The second follows from $p\le 1$. □

Remark 6.4. Corollary 6.2 is useful because it splits the hard finite-set escape problem into two more concrete obstructions. The interior alternative says that the finite induced subgraph on K already has susceptibility comparable to the ambient near-critical susceptibility. The boundary alternative says that the finite-volume susceptibility weights on the edge boundary have large mutual interaction through the ambient graph. Corollary 6.3 shows that this can happen only if those boundary susceptibility weights have large ℓ^2-mass. A complete proof of finite-set escape must rule out the interior alternative and this boundary ℓ^2-mass obstruction uniformly near criticality on nonamenable quasi-transitive graphs.

6.1 Interior term via killed self-avoiding paths

For a finite $K \subset V$, let $c_n^K(x)$ be the number of self-avoiding paths of length n starting at x and using only vertices of K, and let

$$A_n(K) = \frac{1}{|K|} \sum_{x \in K} c_n^K(x).$$

Lemma 6.5 (Killed path-count bound)**.** *For every finite non-empty $K \subset V$ and every $p \in [0,1]$,*

$$\frac{S_p(K)}{|K|} \le \sum_{n \ge 0} p^n A_n(K).$$

Proof. If x is connected to y inside K, then there is an open self-avoiding path inside K from x to y. Taking the union bound over all such paths gives

$$\tau_p^K(x,y) \le \sum_{\gamma : x \to y \text{ s.a.w. in } K} p^{|\gamma|}.$$

Summing over $y \in K$ and then over $x \in K$ gives the claim. □

Proposition 6.6 (Interior escape from killed-growth control)**.** *Assume that there are constants $C < \infty$ and $\theta < 1/p_c(G)$ such that*

$$A_n(K) \le C\theta^n$$

for every finite non-empty $K \subset V$ and every $n \ge 0$. If $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, then

$$\limsup_{p \uparrow p_c} \sup_K \frac{S_p(K)}{\overline{\chi}_p |K|} = 0.$$

In particular, the interior alternative (I) *is ruled out uniformly near criticality.*

Proof. Choose $p_0 < p_c$ so close to p_c that $p_0\theta < 1$. For $p \in [p_0, p_c)$, Lemma 6.5 gives

$$\frac{S_p(K)}{|K|} \le \sum_{n \ge 0} p^n A_n(K) \le C \sum_{n \ge 0} (p\theta)^n \le \frac{C}{1 - p_c\theta}.$$

The right-hand side is independent of K and p. Dividing by $\overline{\chi}_p$ and using $\overline{\chi}_p \to \infty$ proves the claim. □

Remark 6.7. Proposition 6.6 gives a concrete intermediate target: prove a uniform killed-self-avoiding-walk growth rate strictly below $1/p_c$. This is much stronger than nonamenability as stated, but it is a clear route to a publishable special-case theorem if it can be verified in a new class of nonamenable quasi-transitive graphs.

6.2 Deterministic boundary-weight bounds

Let

$$\partial_{\text{in}} K = \{a \in K : \exists z \notin K,\ a \sim z\}, \qquad B_p(K) = \sum_{a \in \partial_{\text{in}} K} \chi_p^K(a),$$

and

$$M_p(K) = \max_{a \in \partial_{\text{in}} K} \chi_p^K(a),$$

with $M_p(K) = 0$ if $\partial_{\text{in}} K = \varnothing$.

Lemma 6.8 (Boundary-weight norm bounds)**.** *If G has maximum degree at most Δ, then for every finite non-empty $K \subset V$,*

$$\|\mu_{p,K}\|_1 \le \Delta B_p(K) \le \Delta S_p(K),$$
$$\|\mu_{p,K}\|_\infty \le \Delta M_p(K),$$

and

$$\|\mu_{p,K}\|_2^2 \le \Delta^2 \sum_{a \in \partial_{\mathrm{in}} K} \chi_p^K(a)^2 \le \Delta^2 M_p(K) B_p(K) \le \Delta^2 M_p(K) S_p(K).$$

Proof. For the L^1 bound,

$$\|\mu_{p,K}\|_1 = \sum_{z \notin K} \sum_{\substack{a \in K \\ a \sim z}} \chi_p^K(a) = \sum_{a \in \partial_{\mathrm{in}} K} \deg_{V \setminus K}(a) \chi_p^K(a) \le \Delta B_p(K).$$

The L^∞ bound follows from

$$\mu_{p,K}(z) = \sum_{\substack{a \in K \\ a \sim z}} \chi_p^K(a) \le \Delta M_p(K).$$

Finally, by Cauchy's inequality at each exterior vertex z,

$$\mu_{p,K}(z)^2 \le \deg_K(z) \sum_{\substack{a \in K \\ a \sim z}} \chi_p^K(a)^2 \le \Delta \sum_{\substack{a \in K \\ a \sim z}} \chi_p^K(a)^2.$$

Summing over $z \notin K$ gives

$$\|\mu_{p,K}\|_2^2 \le \Delta \sum_{a \in \partial_{\mathrm{in}} K} \deg_{V \setminus K}(a) \chi_p^K(a)^2 \le \Delta^2 \sum_{a \in \partial_{\mathrm{in}} K} \chi_p^K(a)^2.$$

The remaining inequalities use $\chi_p^K(a) \le M_p(K)$ on $\partial_{\mathrm{in}} K$ and $B_p(K) \le S_p(K)$. □

Corollary 6.9 (What the boundary alternative forces)**.** *Assume G has maximum degree at most Δ. If the boundary alternative* (B) *holds, then*

$$M_p(K) B_p(K) \ge \frac{\epsilon}{2p^2\Delta^2} |K|.$$

In particular,

$$M_p(K) S_p(K) \ge \frac{\epsilon}{2p^2\Delta^2} |K|.$$

Proof. Combine Corollary 6.3 with Lemma 6.8. □

Remark 6.10. Corollary 6.9 also explains a limitation of this route. Plain nonamenability gives a lower bound on the number of boundary edges, not an upper bound on the susceptibility-weighted boundary quantities $B_p(K)$ or $M_p(K)B_p(K)$. Thus a proof cannot close the boundary alternative by quoting the Cheeger inequality alone. It must use how Q_p distributes mass across the boundary, or it must prove a new estimate controlling the finite-volume susceptibility profile near the boundary.

6.3 A combined finite-set escape criterion

The preceding estimates can be summarized as a criterion that is weaker than the full Benjamini-Schramm conjecture but strong enough to define a concrete intermediate theorem.

Define the killed connective constant

$$\kappa_{\mathrm{kill}}(G) = \limsup_{n\to\infty} \left(\sup_{K\subset V \text{ finite}} A_n(K) \right)^{1/n}.$$

For a finite K, define the normalized boundary energy

$$\mathcal{E}_p(K) = \frac{p^2}{|K|} \langle \mu_{p,K}, Q_p \mu_{p,K} \rangle .$$

Theorem 6.11 (Finite-set escape criterion)**.** *Let G be connected, locally finite, quasi-transitive, and of bounded degree. Suppose that*

$$\kappa_{\mathrm{kill}}(G) < 1/p_{\mathrm{c}}(G), \tag{KCG}$$

$$\lim_{p\uparrow p_{\mathrm{c}}} \sup_{K\subset V \text{ finite}} \mathcal{E}_p(K) = 0, \tag{BE}$$

and $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Then finite-set escape holds:

$$\lim_{p\uparrow p_{\mathrm{c}}} \sup_{K\subset V \text{ finite}} \frac{1}{|K|} \sum_{x,y\in K} Q_p(x,y) = 0.$$

Proof. Fix $\epsilon > 0$. By (KCG), choose $\theta < 1/p_{\mathrm{c}}$ and $C < \infty$ such that $A_n(K) \le C\theta^n$ for all finite K and all $n \ge 0$. Proposition 6.6 then gives a $p_1 < p_{\mathrm{c}}$ such that, for $p \in [p_1, p_{\mathrm{c}})$ and every finite non-empty K,

$$\frac{S_p(K)}{\overline{\chi}_p |K|} < \epsilon/2.$$

By (BE), increasing p_1 if necessary, we also have

$$\sup_K \mathcal{E}_p(K) < \epsilon/2 \qquad p \in [p_1, p_{\mathrm{c}}).$$

For such p and K, Proposition 6.1, divided by $\overline{\chi}_p|K|$, gives

$$\frac{1}{|K|} \sum_{x,y\in K} Q_p(x,y) \le \frac{S_p(K)}{\overline{\chi}_p|K|} + \mathcal{E}_p(K) < \epsilon.$$

Taking the supremum over K proves the claim. □

Corollary 6.12 (Conditional L^2 non-uniqueness theorem)**.** *Let G be connected, locally finite, quasi-transitive, nonamenable, and of bounded degree. If (KCG), (BE), and the mean-field susceptibility bound (MF) hold, then*

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G) \le p_{\mathrm{u}}(G).$$

Proof. Theorem 6.11 gives finite-set escape. Combining finite-set escape with (MF), Theorem 4.1 gives $p_{\mathrm{c}}(G) < p_{2\to 2}(G)$. The inequality $p_{2\to 2} \le p_{\mathrm{u}}$ was recalled above in the L^2 formulation. □

Remark 6.13. The content of Theorem 6.11 is not that (KCG) and (BE) are easy. Rather, it separates the desired finite-set escape statement into a killed path-growth problem and a boundary-energy problem. This is a usable framework for searching for a publishable special case beyond the currently known hyperbolic, nonunimodular, planar, and product settings.

7 A self-avoiding operator criterion

The killed-growth condition controls averages over finite sets. A stronger but cleaner condition controls the full ℓ^2 operator norm of self-avoiding-path spheres.

For $n \geq 0$, let C_n be the kernel

$$C_n(x,y) = \#\{\text{length-}n \text{ self-avoiding paths from } x \text{ to } y\}.$$

Thus C_0 is the identity kernel. Define

$$\gamma_{\mathrm{SAW},2}(G) = \limsup_{n\to\infty} \|C_n\|_{2\to 2}^{1/n}.$$

Proposition 7.1 (SAW operator gap criterion)**.** *Let G be connected, locally finite, and of bounded degree. If*

$$\gamma_{\mathrm{SAW},2}(G) < 1/p_{\mathrm{c}}(G),$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

If, in addition, $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, then finite-set escape holds.

Proof. Choose θ with $\gamma_{\mathrm{SAW},2} < \theta < 1/p_{\mathrm{c}}$. By the definition of the limsup, there is $C < \infty$ such that

$$\|C_n\|_{2\to 2} \leq C\theta^n \qquad n \geq 0.$$

If x is connected to y, then there is an open self-avoiding path from x to y. Hence the union bound gives the entrywise estimate

$$T_p(x,y) = \tau_p(x,y) \leq \sum_{n\geq 0} p^n C_n(x,y).$$

The kernels involved are nonnegative and symmetric, so this entrywise domination implies domination of ℓ^2 operator norms. Indeed, for every finitely supported f,

$$|T_p f| \leq \left(\sum_{n\geq 0} p^n C_n\right) |f|$$

pointwise. Consequently, for every $p < 1/\theta$,

$$\|T_p\|_{2\to 2} \leq \sum_{n\geq 0} p^n \|C_n\|_{2\to 2} \leq \frac{C}{1-p\theta}.$$

Since $1/\theta > p_{\mathrm{c}}$, this proves $p_{2\to 2} > p_{\mathrm{c}}$.

For the finite-set escape conclusion, fix $p_* \in (p_{\mathrm{c}}, 1/\theta)$. The preceding bound is uniform for $p \leq p_{\mathrm{c}} < p_*$. Therefore, for every finite non-empty K,

$$\frac{1}{|K|}\sum_{x,y\in K} Q_p(x,y) = \frac{\langle \mathbf{1}_K, T_p \mathbf{1}_K\rangle}{\overline{\chi}_p |K|} \leq \frac{\|T_p\|_{2\to 2}}{\overline{\chi}_p} \leq \frac{C}{(1-p_*\theta)\overline{\chi}_p}.$$

Letting $p \uparrow p_{\mathrm{c}}$ proves finite-set escape. □

Remark 7.2. This criterion is deliberately strong. It replaces the finite-set problem by a global self-avoiding-walk operator norm gap. Its value is that it identifies one clean mechanism that is sufficient for the L^2 conjecture: an ℓ^2 self-avoiding growth exponent below $1/p_c$. The regular-tree calculation in Proposition 12.1 is the model example.

Proposition 7.3 (Amenable stress test for the SAW criterion)**.** *Let G be a connected transitive amenable graph of bounded degree. Let c_n be the number of length-n self-avoiding paths starting from a fixed vertex, and let*

$$\mu(G) = \lim_{n\to\infty} c_n^{1/n}$$

be the connective constant. Then

$$\gamma_{\mathrm{SAW},2}(G) = \mu(G).$$

Consequently the strict gap

$$\gamma_{\mathrm{SAW},2}(G) < 1/p_c(G)$$

cannot hold on such a graph.

Proof. For each n, the kernel C_n is nonnegative, symmetric, invariant under the transitive automorphism group, and has row and column sums equal to c_n. Schur's test gives

$$\|C_n\|_{2\to 2} \le c_n.$$

For the reverse inequality, fix n. Amenability gives finite sets F_k with $|\partial F_k|/|F_k| \to 0$. Since the graph has bounded degree, the union of all vertices within distance n of ∂F_k has size $o(|F_k|)$. Thus the n-interiors

$$F_k^{(-n)} = \{x \in F_k : B(x,n) \subseteq F_k\}$$

satisfy $|F_k^{(-n)}|/|F_k| \to 1$. If $x \in F_k^{(-n)}$, then every length-n self-avoiding path starting from x ends in F_k. Hence

$$\langle \mathbf{1}_{F_k}, C_n \mathbf{1}_{F_k} \rangle \ge c_n |F_k^{(-n)}|.$$

Dividing by $\|\mathbf{1}_{F_k}\|_2^2 = |F_k|$ and letting $k \to \infty$ gives $\|C_n\|_{2\to 2} \ge c_n$. Thus $\|C_n\|_{2\to 2} = c_n$ for every n, and taking nth roots gives $\gamma_{\mathrm{SAW},2} = \mu$.

Finally, the usual self-avoiding-path union bound implies

$$p_c(G) \ge 1/\mu(G).$$

Equivalently,

$$1/p_c(G) \le \mu(G),$$

so the strict operator gap cannot hold. □

Remark 7.4. This proposition is a useful guardrail. The SAW operator criterion uses nonamenability in exactly the place where amenable Folner sets would otherwise turn operator norm into row-sum growth. Thus the criterion does not risk proving $p_c < p_{2\to 2}$ on amenable transitive graphs, where $p_c = p_u$ and the desired non-uniqueness conclusion is false.

Proposition 7.5 (Cartesian product lower bound)**.** *Let G and H be connected infinite bounded-degree graphs. Then*

$$\gamma_{\mathrm{SAW},2}(G \square H) \ge \gamma_{\mathrm{SAW},2}(G) + \gamma_{\mathrm{SAW},2}(H),$$

where $G \square H$ denotes the Cartesian product graph.

Proof. Write C_n^X for the length-n SAW kernel on a graph X. First note that the sequence $\|C_n^X\|_{2\to 2}$ is submultiplicative up to the obvious inequality

$$C_{m+n}^X \le C_m^X C_n^X$$

entrywise: every length-$m+n$ self-avoiding path splits into a length-m self-avoiding path followed by a length-n self-avoiding path. Positivity then gives

$$\|C_{m+n}^X\|_{2\to 2} \le \|C_m^X\|_{2\to 2} \|C_n^X\|_{2\to 2}.$$

Hence $\|C_n^X\|_{2\to 2}^{1/n}$ has a limit equal to $\gamma_{\mathrm{SAW},2}(X)$.

Fix n and $0 \le k \le n$. Given a length-k self-avoiding path in G, a length-$n-k$ self-avoiding path in H, and a choice of the k times at which to advance in the G-coordinate, we obtain a self-avoiding path in $G\square H$. Indeed, if the product path visits the same vertex twice, then both coordinate paths have the same coordinate index at those two times; since the elapsed time is the sum of the two coordinate indices, the two times are equal. Therefore

$$C_n^{G\square H} \ge \binom{n}{k} C_k^G \otimes C_{n-k}^H$$

entrywise. Since the kernels are nonnegative, this implies

$$\|C_n^{G\square H}\|_{2\to 2} \ge \binom{n}{k} \|C_k^G\|_{2\to 2} \|C_{n-k}^H\|_{2\to 2}.$$

Let $k/n \to \alpha$. Taking nth roots gives

$$\gamma_{\mathrm{SAW},2}(G\square H) \ge \exp\{h(\alpha)\} \gamma_{\mathrm{SAW},2}(G)^{\alpha} \gamma_{\mathrm{SAW},2}(H)^{1-\alpha},$$

where $h(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha)$. Maximizing over $\alpha \in [0,1]$ yields the claimed sum. □

Corollary 7.6 (Product stress test for Route A7)**.** *Let T_d be the d-regular tree with $d \ge 3$, and let H be a connected transitive amenable graph of bounded degree. Then*

$$\gamma_{\mathrm{SAW},2}(T_d\square H) \ge \sqrt{d-1} + \mu(H).$$

Consequently, the SAW operator gap criterion can prove $p_c(T_d\square H) < p_{2\to 2}(T_d\square H)$ only if

$$p_c(T_d\square H) < \frac{1}{\sqrt{d-1} + \mu(H)}.$$

Proof. The lower bound combines Proposition 7.5, the tree calculation $\gamma_{\mathrm{SAW},2}(T_d) = \sqrt{d-1}$, and Proposition 7.3. On the tree this calculation is the same spherical-operator estimate used in Proposition 12.1, since $C_n = R_n$. The final displayed inequality is just the necessary condition $\gamma_{\mathrm{SAW},2}(T_d\square H) < 1/p_c(T_d\square H)$. □

Remark 7.7. This does not rule out Route A7 on products. It shows what the route would have to beat: mixed self-avoiding paths contribute at least additively to the ℓ^2 SAW growth exponent. If the displayed necessary inequality fails for a product class, the boundary-energy or scattering routes must be used instead of the global SAW operator gap.

Proposition 7.8 (Coarse high-branching product theorem)**.** *Let T_d be the d-regular tree, put $q = d-1$, and let H be a connected bounded-degree graph with adjacency operator A_H. If*

$$2\sqrt{q} + \|A_H\|_{2\to 2} < q,$$

then

$$p_{\mathrm{c}}(T_d \square H) < p_{2\to 2}(T_d \square H).$$

In particular, it is enough that

$$2\sqrt{d-1} + \Delta_H < d-1,$$

where Δ_H is the maximum degree of H.

Proof. Let A_T be the adjacency operator of T_d, and let A be the adjacency operator of $T_d \square H$. Then

$$A = A_T \otimes I + I \otimes A_H$$

and $\|A_T\|_{2\to 2} = 2\sqrt{q}$. Every self-avoiding path is a walk, so the SAW kernel is entrywise dominated by the walk kernel:

$$C_n^{T_d \square H} \le A^n.$$

Positivity gives

$$\left\|C_n^{T_d \square H}\right\|_{2\to 2} \le \|A^n\|_{2\to 2} \le \left(2\sqrt{q} + \|A_H\|_{2\to 2}\right)^n.$$

Thus

$$\gamma_{\mathrm{SAW},2}(T_d \square H) \le 2\sqrt{q} + \|A_H\|_{2\to 2}.$$

The graph $T_d \square H$ contains a copy of T_d, so

$$p_{\mathrm{c}}(T_d \square H) \le p_{\mathrm{c}}(T_d) = 1/q.$$

Under the displayed hypothesis,

$$\gamma_{\mathrm{SAW},2}(T_d \square H) < q \le 1/p_{\mathrm{c}}(T_d \square H).$$

Proposition 7.1 gives the claim. The final sufficient condition follows from $\|A_H\|_{2\to 2} \le \Delta_H$. □

Remark 7.9. This theorem is intentionally coarse and is not meant to compete with the known product results. Its role in this program is diagnostic: Route A7 is not empty beyond regular trees, but its easiest proof only reaches product graphs with sufficiently large tree branching. Low-branching products such as $T_3 \square \mathbb{Z}$ show why the SAW route is too crude; the line-factor case is handled later by a direct $p_{2\to 2} = p_u$ argument.

Proposition 7.10 (Finite-fiber high-branching tree bundles)**.** *Let T_d be the d-regular tree, put $b = d-1$, and let F be a finite set. Let A_0 be the adjacency matrix of a finite graph on F, and let P be a nonzero symmetric 0-1 matrix on F. Define a graph $G = G(T_d, F, A_0, P)$ on $V(T_d) \times F$ by declaring (u,i) adjacent to (u,j) when $A_0(i,j) = 1$, and (u,i) adjacent to (v,j) when $u \sim v$ in T_d and $P(i,j) = 1$. Suppose that G is connected and that $P(i_0,i_0) = 1$ for some $i_0 \in F$. If*

$$2\sqrt{b}\,\|P\|_{2\to 2} + \|A_0\|_{2\to 2} < b,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. The adjacency operator of G, under the identification $\ell^2(V(T_d) \times F) = \ell^2(V(T_d)) \otimes \mathbb{R}^F$, is

$$A_G = A_{T_d} \otimes P + I \otimes A_0.$$

Since A_0 and P are self-adjoint and $\|A_{T_d}\|_{2\to 2} = 2\sqrt{b}$,

$$\|A_G\|_{2\to 2} \le 2\sqrt{b}\, \|P\|_{2\to 2} + \|A_0\|_{2\to 2} < b.$$

Every self-avoiding path is a walk, so if C_n is the length-n self-avoiding path kernel on G, then $C_n \le A_G^n$ entrywise. Hence

$$\gamma_{\mathrm{SAW},2}(G) \le \limsup_{n\to\infty} \|A_G^n\|_{2\to 2}^{1/n} = \|A_G\|_{2\to 2} < b.$$

The assumption $P(i_0, i_0) = 1$ gives a copy of T_d inside G, namely the vertices $V(T_d) \times \{i_0\}$. Therefore

$$p_{\mathrm{c}}(G) \le p_{\mathrm{c}}(T_d) = 1/b,$$

so $b \le 1/p_{\mathrm{c}}(G)$. The strict SAW operator gap $\gamma_{\mathrm{SAW},2}(G) < 1/p_{\mathrm{c}}(G)$ follows, and Proposition 7.1 applies. □

Corollary 7.11 (High-branching finite cyclic fibers)**.** *The graph $T_d \square K_2$ satisfies*

$$p_{\mathrm{c}}(T_d \square K_2) < p_{2\to 2}(T_d \square K_2)$$

for every $d \ge 7$. For every $m \ge 3$, the graph $T_d \square C_m$ satisfies

$$p_{\mathrm{c}}(T_d \square C_m) < p_{2\to 2}(T_d \square C_m)$$

for every $d \ge 9$.

Proof. For a Cartesian product with a finite factor H, Proposition 7.10 applies with $P = I$ and $A_0 = A_H$. For K_2, $\|A_H\|_{2\to 2} = 1$, so the required inequality is

$$2\sqrt{b} + 1 < b,$$

which holds for $b \ge 6$, i.e. $d \ge 7$. For C_m, $\|A_H\|_{2\to 2} = 2$, and

$$2\sqrt{b} + 2 < b$$

holds for $b \ge 8$, i.e. $d \ge 9$. □

Remark 7.12. These examples have genuine short cycles: $T_d \square K_2$ contains a square over every tree edge, and $T_d \square C_m$ contains a cycle in every fibre. The estimate is deliberately rough and high-branching, but it is a useful cyclic calibration of the operator route. It also allows finite-fibre couplings that are not Cartesian products through the matrix P.

Lemma 7.13 (Nonbacktracking operator bound)**.** *Let X be an infinite connected D-regular graph with adjacency operator A_X, and let N_n be the kernel counting length-n nonbacktracking walks. Put $\rho = \|A_X\|_{2\to 2}$ and*

$$\beta(D, \rho) = \begin{cases} \sqrt{D-1}, & \rho \le 2\sqrt{D-1}, \\ \dfrac{\rho + \sqrt{\rho^2 - 4(D-1)}}{2}, & \rho > 2\sqrt{D-1}. \end{cases}$$

Then

$$\limsup_{n\to\infty} \|N_n\|_{2\to 2}^{1/n} \le \beta(D, \rho).$$

Proof. The kernels N_n are polynomials in A_X:

$$N_0 = I, \qquad N_1 = A_X, \qquad N_2 = A_X^2 - DI,$$

and, for $n \geq 3$,

$$N_n = A_X N_{n-1} - (D-1)N_{n-2}.$$

Indeed, $A_X N_{n-1}$ counts all paths obtained by prepending one edge to a nonbacktracking path of length $n-1$. The only bad paths immediately backtrack across the prepended edge. After deleting this two-edge backtrack, one obtains a nonbacktracking path of length $n-2$, and there are $D-1$ choices for the prepended neighbour, since it cannot be the first neighbour of the remaining path. The case $n = 2$ has D such choices, giving $N_2 = A_X^2 - DI$. Thus $N_n = P_n(A_X)$, where the scalar polynomials P_n satisfy the same recurrence. By the spectral theorem,

$$\|N_n\|_{2\to 2} \leq \sup_{|\lambda| \leq \rho} |P_n(\lambda)|.$$

For fixed λ, the recurrence has characteristic equation

$$r^2 - \lambda r + (D-1) = 0.$$

If $|\lambda| \leq 2\sqrt{D-1}$, both roots have modulus $\sqrt{D-1}$, and the possible double-root case gives only a polynomial factor in n. If $|\lambda| > 2\sqrt{D-1}$, the larger root has modulus at most the second branch in the definition of $\beta(D, \rho)$. Compactness of $[-\rho, \rho]$ gives, for every $\epsilon > 0$, a constant C_ϵ such that

$$|P_n(\lambda)| \leq C_\epsilon(\beta(D,\rho) + \epsilon)^n \qquad |\lambda| \leq \rho,\ n \geq 0.$$

Taking nth roots and then $\epsilon \downarrow 0$ proves the claim. □

Proposition 7.14 (Nonbacktracking product theorem)**.** *Let T_d be the d-regular tree, put $q = d-1$, and let H be an finite or infinite connected Δ-regular graph with adjacency operator A_H. Set*

$$D = d + \Delta, \qquad \rho = 2\sqrt{q} + \|A_H\|_{2\to 2}\,.$$

If

$$\beta(D, \rho) < q,$$

then

$$p_c(T_d \square H) < p_{2\to 2}(T_d \square H).$$

Proof. The graph $T_d \square H$ is D-regular, and its adjacency operator is

$$A_{T_d} \otimes I + I \otimes A_H.$$

Hence its adjacency norm is at most ρ. Every self-avoiding path is nonbacktracking, so

$$\gamma_{\mathrm{SAW},2}(T_d \square H) \leq \limsup_{n\to\infty} \|N_n\|_{2\to 2}^{1/n} \leq \beta(D, \rho)$$

by Lemma 7.13 and monotonicity of $\beta(D, \cdot)$. Since $T_d \square H$ contains a copy of T_d,

$$p_c(T_d \square H) \leq 1/q.$$

Thus $\beta(D, \rho) < q \leq 1/p_c(T_d \square H)$, and Proposition 7.1 applies. □

Corollary 7.15 (Nonbacktracking finite regular factors)**.** *Let H be a finite connected Δ-regular graph, let T_d be the d-regular tree, and put $b = d - 1$. If*

$$\beta(d + \Delta, 2\sqrt{b} + \Delta) < b,$$

then

$$p_c(T_d \square H) < p_{2\to 2}(T_d \square H).$$

Proof. Since H is finite, connected, and Δ-regular, $\|A_H\|_{2\to 2} = \Delta$. Proposition 7.14 applies. □

Corollary 7.16 (Sharper finite cyclic factors)**.** *The graph $T_d \square K_2$ satisfies*

$$p_c(T_d \square K_2) < p_{2\to 2}(T_d \square K_2)$$

for every $d \geq 5$. For every $m \geq 3$, the graph $T_d \square C_m$ satisfies

$$p_c(T_d \square C_m) < p_{2\to 2}(T_d \square C_m)$$

for every $d \geq 7$.

Proof. Put $b = d - 1$. For K_2, $\Delta = 1$, so

$$D - 1 = b + 1, \qquad \rho = 2\sqrt{b} + 1.$$

If $\rho \leq 2\sqrt{b+1}$, then

$$\beta(D, \rho) = \sqrt{b+1} < b$$

for $b \geq 4$. If $\rho > 2\sqrt{b+1}$, the larger root defining $\beta(D, \rho)$ is below b whenever

$$b > \rho/2 \quad \text{and} \quad b^2 - \rho b + (b+1) > 0.$$

For $b \geq 4$, the first inequality is immediate and the second is

$$b^2 - (2\sqrt{b} + 1)b + (b+1) = b^2 - 2b\sqrt{b} + 1 = b\sqrt{b}(\sqrt{b} - 2) + 1 > 0.$$

Thus $T_d \square K_2$ is covered for $d \geq 5$.

For C_m, $\Delta = 2$, so

$$D - 1 = b + 2, \qquad \rho = 2\sqrt{b} + 2.$$

If $\rho \leq 2\sqrt{b+2}$, then $\beta(D, \rho) = \sqrt{b+2} < b$ for $b \geq 6$. In the remaining case, it is enough to check

$$b > \rho/2 \quad \text{and} \quad b^2 - \rho b + (b+2) > 0.$$

For $b \geq 6$, the first inequality is immediate, and the second equals

$$b^2 - (2\sqrt{b} + 2)b + (b+2) = b^2 - 2b\sqrt{b} - b + 2 = b\big(b - 2\sqrt{b} - 1\big) + 2 > 0.$$

Thus $T_d \square C_m$ is covered for every $d \geq 7$. □

Corollary 7.17 (A spectral-factor family)**.** *Let T_d be the d-regular tree, put $s = \sqrt{d-1}$, and let H be an infinite connected Δ-regular bounded-degree graph. Set*

$$\alpha = \|A_H\|_{2\to 2}.$$

If

$$s \geq 2 + \max\{\sqrt{\alpha}, \Delta^{1/4}\},$$

then

$$p_c(T_d \square H) < p_{2\to 2}(T_d \square H).$$

Proof. It is enough to verify the hypothesis of Proposition 7.14. Put $q = s^2$, $D = d + \Delta$, and $\rho = 2s + \alpha$. If $\rho \le 2\sqrt{D-1} = 2\sqrt{s^2+\Delta}$, then

$$\beta(D,\rho) = \sqrt{s^2+\Delta} < s^2$$

because $s \ge 2 + \Delta^{1/4}$ implies $s^4 > s^2 + \Delta$.

It remains to consider $\rho > 2\sqrt{D-1}$. The larger root in the definition of $\beta(D,\rho)$ is less than s^2 if

$$s^2 > \rho/2$$

and

$$s^4 - \rho s^2 + (D-1) > 0.$$

The first inequality follows from $s \ge 2 + \sqrt{\alpha}$. For the second,

$$s^4 - \rho s^2 + (D-1) = s^2\big((s-1)^2 - \alpha\big) + \Delta > 0,$$

again by $s \ge 2 + \sqrt{\alpha}$. Hence $\beta(D,\rho) < s^2 = q$, and Proposition 7.14 applies. □

Corollary 7.18 (Regular amenable factors)**.** *Let T_d be the d-regular tree, put $s = \sqrt{d-1}$, and let H be an infinite connected amenable Δ-regular graph. If*

$$s \ge 1 + \sqrt{\Delta},$$

then

$$p_c(T_d \square H) < p_{2\to 2}(T_d \square H).$$

Proof. Since H is Δ-regular, $\|A_H\|_{2\to 2} \le \Delta$. Conversely, if F_j is a Folner sequence and $\partial_E F_j$ denotes the edge boundary, then

$$\frac{\langle \mathbf{1}_{F_j}, A_H \mathbf{1}_{F_j}\rangle}{\|\mathbf{1}_{F_j}\|_2^2} = \Delta - \frac{|\partial_E F_j|}{|F_j|} \longrightarrow \Delta.$$

Thus $\|A_H\|_{2\to 2} = \Delta$. It remains to verify Proposition 7.14. Put $a = \sqrt{\Delta}$. Then

$$D = d + \Delta, \qquad D - 1 = s^2 + \Delta, \qquad \rho = 2s + \Delta.$$

If $\rho \le 2\sqrt{D-1}$, then

$$\beta(D,\rho) = \sqrt{s^2+\Delta} < s^2,$$

since $s \ge 1 + a$ gives

$$s^4 - s^2 - \Delta \ge (1+a)^4 - (1+a)^2 - a^2 = a(a^3 + 4a^2 + 4a + 2) > 0.$$

If $\rho > 2\sqrt{D-1}$, the larger root defining $\beta(D,\rho)$ is below s^2 once

$$s^2 > \rho/2 \quad \text{and} \quad s^4 - \rho s^2 + (D-1) > 0.$$

These follow from

$$s^2 - \rho/2 = s^2 - s - \Delta/2 \ge \frac{a(a+2)}{2} > 0$$

and

$$s^4 - \rho s^2 + (D-1) = s^2((s-1)^2 - \Delta) + \Delta \ge \Delta > 0.$$

Thus $\beta(D,\rho) < s^2 = d - 1$, and Proposition 7.14 applies. □

Proposition 7.19 (Cluster-branching upper bound for tree products)**.** *Let T_d be the d-regular tree, put $q = d - 1$, and let H be an finite or infinite connected transitive graph. For Bernoulli bond percolation on H with parameter p, let $C_H(o)$ be the open cluster of a fixed vertex $o \in V(H)$, and define*

$$\varphi_H(p) = \mathbb{E}_p\left[1 - (1-p)^{|C_H(o)|}\right].$$

If

$$q\varphi_H(p) > 1,$$

then $p \geq p_c(T_d \square H)$. Consequently, if the inequality holds at some p_0, then $p_c(T_d \square H) < p_0$.

Proof. Root T_d, orient it away from the root, and keep any q root children and the q forward neighbours at each non-root vertex. Starting from a fiber $\{v\} \times H$, expose the open H-cluster of the current vertex inside that fiber. If this cluster has size M, then a fixed forward tree-neighbour fiber is reached whenever at least one of the M horizontal edges leading to that neighbouring fiber is open. Conditional on M, this has probability $1 - (1-p)^M$, independently for the q forward neighbours.

Let the offspring of a visited fiber be the number of forward neighbouring fibers reached in this way. Its mean is $q\varphi_H(p)$. When a neighbouring fiber is reached, choose one open horizontal edge by a fixed rule and then expose the H-cluster of its endpoint in the new fiber. Future clusters descended from distinct reached fibers lie in disjoint fibers, so the successive generation sizes have the law of a Galton-Watson process with this offspring distribution. If the mean is larger than one, the process survives with positive probability, producing an infinite open cluster in $T_d \square H$. The strict statement follows from left-continuity of φ_H at p_0 under the standard monotone uniform coupling. □

Remark 7.20. The quantity $\varphi_H(p)$ cannot be replaced by a function of the susceptibility $\chi_H(p) = \mathbb{E}_p|C_H(o)|$ alone without additional tail control. Indeed, for a general positive integer-valued random variable M with fixed mean $\mu > 1$, the expectation $\mathbb{E}[1 - (1-p)^M]$ can be made arbitrarily close to p by putting most of the mass at $M = 1$ and a vanishing amount of mass at a very large value. Thus the cluster-branching criterion records a genuinely distributional input from the factor H, not just its susceptibility.

Corollary 7.21 (Cluster-sharpened finite cyclic factors)**.** *The graph $T_d \square K_2$ satisfies*

$$p_c(T_d \square K_2) < p_{2\to 2}(T_d \square K_2)$$

for every $d \geq 4$. For every $m \geq 3$, the graph $T_d \square C_m$ satisfies

$$p_c(T_d \square C_m) < p_{2\to 2}(T_d \square C_m)$$

for every $d \geq 5$.

Proof. Corollary 7.16 already proves the K_2 claim for $d \geq 5$, so it remains to treat $T_4 \square K_2$. Put $p_0 = 3/10$. For K_2, the fibre cluster size is $M = 1 + X$, where X is Bernoulli with parameter p_0. Hence

$$\varphi_{K_2}(p_0) = p_0 + p_0^2 - p_0^3 = \frac{363}{1000},$$

and

$$3\varphi_{K_2}(p_0) = \frac{1089}{1000} > 1.$$

Proposition 7.19 gives $p_c(T_4 \square K_2) < 3/10$.

The graph $T_4 \square K_2$ is 5-regular and has adjacency norm at most

$$\rho = 2\sqrt{3} + 1.$$

For $D = 5$, the nonbacktracking exponent from Lemma 7.13 is at most $\beta(D, \rho)$. We claim that $\beta(5, 2\sqrt{3}+1) < 10/3$. Since $10/3 > \rho/2$, it is enough to check

$$\left(\frac{10}{3}\right)^2 - \rho\frac{10}{3} + 4 > 0.$$

This inequality is

$$\frac{106}{9} - \frac{20}{3}\sqrt{3} > 0,$$

which follows by squaring from $106 > 60\sqrt{3}$. Therefore $\gamma_{\mathrm{SAW},2}(T_4 \square K_2) < 10/3 < 1/p_c(T_4 \square K_2)$, and Proposition 7.1 applies.

For cycles, Corollary 7.16 already proves the claim for $d \geq 7$. It remains to treat $d = 5, 6$. For C_m, let X and Y be the indicators of the two fibre edges incident to the root. Then $M \geq 1 + X + Y$, and hence, for every p,

$$\varphi_{C_m}(p) \geq 1 - (1-p)(1-p^2)^2.$$

At $p_1 = 1/5$, this lower bound is $821/3125$, so

$$4\varphi_{C_m}(p_1) \geq \frac{3284}{3125} > 1.$$

Thus $p_c(T_5 \square C_m) < 1/5$. On the operator side, $T_5 \square C_m$ is 7-regular with adjacency norm at most 6, and

$$\beta(7, 6) = 3 + \sqrt{3} < 5.$$

The SAW operator criterion gives the result for $d = 5$.

For $d = 6$, take $p_2 = 19/100$. The same lower bound gives

$$5\varphi_{C_m}(p_2) \geq 5\left[1 - \frac{81}{100}\left(1 - \frac{361}{10000}\right)^2\right] = \frac{12371319995}{10000000000} > 1,$$

so $p_c(T_6 \square C_m) < 19/100$. The graph is 8-regular and has adjacency norm at most $2\sqrt{5} + 2$. Since $100/19 > (2\sqrt{5}+2)/2$, it is enough to check

$$\left(\frac{100}{19}\right)^2 - (2\sqrt{5}+2)\frac{100}{19} + 7 > 0.$$

The left hand side equals

$$\frac{8727}{361} - \frac{200}{19}\sqrt{5} > 0,$$

which follows by squaring from $8727 > 3800\sqrt{5}$. Hence $\beta(8, 2\sqrt{5}+2) < 100/19$, and the SAW operator criterion gives the result for $d = 6$. □

Corollary 7.22 (The two-layer finite factor)**.** *For every $d \geq 3$,*

$$p_c(T_d \square K_2) < p_{2\to 2}(T_d \square K_2).$$

Proof. Corollary 7.21 proves the claim for $d \geq 4$. It remains to treat $T_3 \square K_2$.

We first prove

$$p_c(T_3 \square K_2) < \frac{39}{100}.$$

Use the oriented tree-fibre exploration from Proposition 7.19, but keep the reached subset of the two-point fibre. There are two symmetry classes of non-empty subsets: singletons and doubletons. At $p = 39/100$, the mean matrix for the two forward child fibres is

$$M = \begin{pmatrix} 2379/5000 & 1521/5000 \\ 145119/250000 & 168831/250000 \end{pmatrix}.$$

Indeed, from a singleton, a fixed child is reached as a singleton with probability $p(1-p)$ and as a doubleton with probability p^2. From a doubleton, a fixed child is reached as a singleton with probability $2p(1-p)^2$ and as a doubleton with probability $3p^2 - 2p^3$. The factor two accounts for the two forward tree children.

With $v = (1, 7/4)^{\mathsf{T}}$, exact arithmetic gives

$$Mv = \begin{pmatrix} 20163/20000 \\ 1762293/1000000 \end{pmatrix} > v$$

coordinatewise. Thus the finite-type Galton–Watson lower process is supercritical, and

$$p_c(T_3 \square K_2) < 39/100.$$

It remains to prove an ℓ^2 SAW operator bound below $100/39$. Let C_n be the length-n self-avoiding endpoint kernel on $T_3 \square K_2$. For $n = 15$, quotienting tree labels in the exact enumeration gives the following coefficients for endpoints at tree distance $r = 0, \ldots, 15$:

$$192,\ 448,\ 592,\ 984,\ 1411,\ 1750,\ 1981,\ 2008,$$
$$1837,\ 1498,\ 1073,\ 649,\ 301,\ 91,\ 15,\ 1.$$

The endpoint layer is determined by parity. Combining these coefficients with the T_3 sphere-kernel norms, computed from

$$R_0 = I, \qquad \|R_1\|_{2\to 2} = 2\sqrt{2}, \qquad R_2 = A_{T_3}^2 - 3I, \qquad R_{r+1} = A_{T_3} R_r - 2R_{r-1} \quad (r \geq 2),$$

gives the exact bound

$$\|C_{15}\|_{2\to 2} \leq 639862 + 452048\sqrt{2}.$$

Using $\sqrt{2} < 99/70$,

$$\|C_{15}\|_{2\to 2} < \frac{44771546}{35} < \left(\frac{100}{39}\right)^{15}.$$

The kernels C_n are submultiplicative, so

$$\gamma_{\mathrm{SAW},2}(T_3 \square K_2) < \frac{100}{39}.$$

Since $p_c(T_3 \square K_2) < 39/100$, Proposition 7.1 applies. The exact enumeration and arithmetic checks are reproduced by `scripts/k2_saw_certificate.py`. □

Corollary 7.23 (Four-regular tree times finite cycles)**.** *For every $m \geq 3$,*

$$p_c(T_4 \square C_m) < p_{2\to 2}(T_4 \square C_m).$$

Consequently, for every $d \geq 4$ and $m \geq 3$,

$$p_c(T_d \square C_m) < p_{2\to 2}(T_d \square C_m).$$

Proof. The second statement follows from the first and Corollary 7.21, so it remains to prove the $d = 4$ statement. We split the cycle lengths into three cases.

First take $m = 3$. At $p_3 = 6/25$, the exact cycle-cluster distribution in Proposition 7.19 gives

$$3\varphi_{C_3}(p_3) = \frac{246948714}{244140625} > 1,$$

so $p_c(T_4 \square C_3) < 6/25$. A length-11 exact self-avoiding enumeration gives

$$\|C_{11}\|_{2\to 2} \leq 2649944 + 1532806\sqrt{3}.$$

Using $\sqrt{3} < 97/56$, this is less than $(25/6)^{11}$. Hence $\gamma_{\mathrm{SAW},2}(T_4 \square C_3) < 25/6 < 1/p_c(T_4 \square C_3)$.

For $m = 4$, take $p_4 = 19/80$. Exact cycle-cluster arithmetic gives

$$3\varphi_{C_4}(p_4) = \frac{1699239015678489}{1677721600000000} > 1,$$

and therefore $p_c(T_4 \square C_4) < 19/80$. The corresponding length-11 SAW certificate is

$$\|C_{11}\|_{2\to 2} \leq 3554176 + 2043694\sqrt{3} < \left(\frac{80}{19}\right)^{11},$$

again using $\sqrt{3} < 97/56$.

Finally suppose $m \geq 5$, and put $p_5 = 47/200$. For $m = 5, 6$, exact cycle-cluster arithmetic gives

$$3\varphi_{C_m}(p_5) > 1.$$

For $m \geq 7$, expose only the first three fibre edges in each direction from the root. If L and R are the corresponding truncated open runs, then

$$\varphi_{C_m}(p_5) \geq \mathbb{E}\big[1 - (1 - p_5)^{1+L+R}\big],$$

and exact arithmetic gives

$$3\mathbb{E}\big[1 - (1 - p_5)^{1+L+R}\big] = \frac{8208070810285966860790736889 81}{819200000000000000000000000000} > 1.$$

Thus $p_c(T_4 \square C_m) < 47/200$ for every $m \geq 5$.

The length-11 SAW enumeration is uniformly bounded for all $m \geq 5$ by the radius-coefficient vector

$$141050,\ 120742,\ 92376,\ 61444,\ 36592,\ 19364,$$
$$8918,\ 3442,\ 1018,\ 200,\ 22,\ 1.$$

For $m = 5, \ldots, 12$ this is checked directly, and for $m \geq 13$ a length-11 path cannot identify two distinct vertices of the line cover modulo m. Combining these coefficients with the T_4 sphere-kernel norms gives

$$\|C_{11}\|_{2\to 2} \leq 4074050 + 2347722\sqrt{3} < \left(\frac{200}{47}\right)^{11}.$$

The SAW operator criterion applies in all three cases. The exact quotient enumeration and arithmetic checks are reproduced by `scripts/cycle_saw_certificate.py`. □

Lemma 7.24 (A vertical-cluster bound for $T_d\square\mathbb{Z}$)**.** *Let $d \geq 3$, put $q = d-1$, and define*

$$\psi(p) = 1 - \frac{(1-p)^3}{(1-p+p^2)^2} = \frac{p(p^3-p^2+1)}{(p^2-p+1)^2}.$$

If

$$q\psi(p) > 1,$$

then $p \geq p_c(T_d\square\mathbb{Z})$. Consequently, if the inequality holds at some p_0, then it holds on an interval below p_0 and $p_c(T_d\square\mathbb{Z}) < p_0$.

Proof. Let M be the size of the open cluster of the origin in one-dimensional bond percolation on $\mathbb{Z}$ with parameter p. It remains only to compute $\varphi_{\mathbb{Z}}(p)$ in Proposition 7.19. The one-dimensional cluster size satisfies

$$\mathbb{P}(M = m) = m(1-p)^2 p^{m-1} \qquad m \geq 1,$$

and therefore

$$\mathbb{E}[t^M] = \frac{t(1-p)^2}{(1-pt)^2}.$$

Taking $t = 1-p$ gives

$$\mathbb{E}[1-(1-p)^M] = 1 - \frac{(1-p)^3}{(1-p+p^2)^2} = \psi(p).$$

The claim follows from Proposition 7.19. □

Lemma 7.25 (A two-step vertical-cluster bound)**.** *Let $d \geq 3$, put $q = d-1$, $b = 1-p$, and*

$$c = \frac{1-p}{1-p+p^2}.$$

For $k = 0, 1, 2$, define

$$A_k(z) = \frac{(1+c^k)bz}{1-pbz} - \frac{c^k b^2 z}{1-pb^2 z},$$

and

$$H(z) = A_2(z) + \frac{bA_1(z)^2}{1-bA_0(z)}.$$

Finally put

$$\Phi(p) = 1 - b^2 H'(p).$$

If

$$q^2\Phi(p) > 1,$$

then $p \geq p_c(T_d\square\mathbb{Z})$. Consequently, if the inequality holds at some p_0, then $p_c(T_d\square\mathbb{Z}) < p_0$.

Proof. We use the same oriented tree exploration as in Lemma 7.24, but look two tree generations ahead. Fix a parent vertex x, a child y, and a grandchild z. Suppose the available vertical cluster in the x-fiber has length m. A connection from x to z through y exists if some open vertical component in the y-fiber has at least one open source edge from the x-fiber and at least one open target edge to the z-fiber.

Consider a component whose intersection with the m-interval has length ℓ. If it touches $k \in \{0,1,2\}$ endpoints of the interval, it has k independent outside vertical tails. For one such tail G,

$$\mathbb{E}[b^G] = \sum_{j\geq 0}(1-p)p^j b^j = \frac{b}{1-p+p^2} = c.$$

Thus the probability that this component fails to have both a source and a target is

$$f_k(\ell) = b^\ell + c^k b^\ell - c^k b^{2\ell}.$$

The generating function for a component with k boundary tails, including its internal open vertical edges, is therefore

$$A_k(z) = \sum_{\ell \geq 1} p^{\ell-1} f_k(\ell) z^\ell.$$

Between consecutive components inside the interval there is one closed vertical edge, giving a factor b. Hence the generating function for the failure probability of the whole m-interval is

$$H(z) = A_2(z) + bA_1(z)\frac{A_1(z)}{1 - bA_0(z)}.$$

Writing $H(z) = \sum_{m \geq 1} F_m z^m$, the averaged failure probability over the one-dimensional parent cluster size distribution

$$\mathbb{P}(M = m) = mb^2 p^{m-1}$$

is $b^2 H'(p)$. Thus $\Phi(p)$ is the probability that a fixed grandchild is reached through its fixed parent child.

For each current individual there are q^2 forward grandchildren at tree distance two. Let the offspring count be the number of these grandchildren reached by the two-step procedure above. This offspring distribution has mean $q^2\Phi(p)$. When a grandchild is reached, choose one open target edge and only then expose the vertical cluster at its endpoint in the grandchild fiber; future vertical clusters descended from distinct reached grandchildren lie in disjoint fibers and have the same one-dimensional law as the initial cluster. Thus the even-generation exploration is a Galton-Watson process. If its mean is larger than one, the process survives with positive probability. The strict statement again follows from continuity. □

Proposition 7.26 (The line factor for $d \geq 4$)**.** *For every $d \geq 4$,*

$$p_c(T_d \square \mathbb{Z}) < p_{2\to 2}(T_d \square \mathbb{Z}).$$

Proof. First suppose $d \geq 5$. Put $q = d - 1 = s^2$. For $H = \mathbb{Z}$, the nonbacktracking exponent from Proposition 7.14 previously gave the sufficient condition

$$\beta(d + 2, 2\sqrt{d-1} + 2) < d - 1.$$

Here we use the same nonbacktracking bound together with the better critical-probability estimate of Lemma 7.24. The exponent is

$$\beta_d = \beta(d + 2, 2s + 2) = s + 1 + \sqrt{2s - 1}.$$

Let $r = \sqrt{2s-1}$ and $t = s+1+r = \beta_d$. Since $\gamma_{\mathrm{SAW},2}(T_d \square \mathbb{Z}) \leq t$, it is enough to prove $p_c(T_d \square \mathbb{Z}) < 1/t$.

Using the formula for ψ in Lemma 7.24 and the identity $r^2 = 2s - 1$, a direct calculation gives

$$s^2\psi(1/t) - 1 = \frac{s(3s^3 + 4s^2 - 13s - 6)r + (s+1)(s^4 + 7s^3 - 16s^2 + s + 1)}{(r^2 + 2rs + r + s^2 + s + 1)^2}.$$

For $d \geq 5$, we have $s \geq 2$. Writing $u = s - 2$, the two numerator coefficients become

$$s(3s^3 + 4s^2 - 13s - 6) = (u + 2)(3u^3 + 22u^2 + 39u + 8) > 0$$

and

$$(s+1)(s^4+7s^3-16s^2+s+1) = (u+3)(u^4+15u^3+50u^2+53u+11) > 0.$$

Thus $s^2\psi(1/t) > 1$. By Lemma 7.24,

$$p_c(T_d \square \mathbb{Z}) < 1/t.$$

The SAW operator gap criterion, Proposition 7.1, now gives the claim for $d \geq 5$.

It remains to handle $d = 4$. In this case $q = 3$, and the nonbacktracking bound gives

$$\gamma_{\mathrm{SAW},2}(T_4 \square \mathbb{Z}) \leq \beta_4 = \sqrt{3} + 1 + \sqrt{2\sqrt{3}-1}.$$

Let $p_0 = 23/100$. From Lemma 7.25, direct rational arithmetic gives

$$9\Phi(p_0) - 1 > \frac{1}{100}.$$

Therefore $p_c(T_4 \square \mathbb{Z}) < p_0$. On the other hand,

$$\sqrt{3} < \frac{26}{15}, \qquad \sqrt{2\sqrt{3}-1} < \frac{11}{7},$$

and hence

$$\beta_4 < \frac{26}{15} + 1 + \frac{11}{7} = \frac{452}{105} < \frac{100}{23} = \frac{1}{p_0}.$$

Thus

$$\gamma_{\mathrm{SAW},2}(T_4 \square \mathbb{Z}) < 1/p_c(T_4 \square \mathbb{Z}),$$

and Proposition 7.1 applies. □

Proposition 7.27 (A fiber-summed tree criterion)**.** *Let H be a transitive graph and let $d \geq 3$. Put $b = d - 1$, fix a vertex x_n of T_d at distance n from a root o_T, and define*

$$I_n(p) = \sum_{y \in V(H)} \mathbb{P}_p\big((o_T, o_H) \leftrightarrow (x_n, y)\big).$$

If $I_0(p) < \infty$ and

$$\limsup_{n\to\infty} I_n(p)^{1/n} < b^{-1/2},$$

then T_p is bounded on $\ell^2(T_d \square H)$. In particular $p \leq p_{2\to 2}(T_d \square H)$.

Proof. Choose $\theta < b^{-1/2}$ and $C < \infty$ such that $I_n(p) \leq C\theta^n$ for every $n \geq 1$. For $n \geq 0$, let R_n be the radius-n sphere kernel on T_d. Let K_n be the part of T_p that connects pairs whose tree coordinates are at distance n. We claim that

$$\|K_n\|_{2\to 2} \leq I_n(p)\, \|R_n\|_{2\to 2}\,.$$

Indeed, for fixed tree vertices u, v with $\mathrm{dist}_{T_d}(u, v) = n$, the kernel on the H-fibres

$$B_{u,v}(a, c) = \mathbb{P}_p\big((u, a) \leftrightarrow (v, c)\big)$$

has row sums equal to $I_n(p)$. By symmetry its column sums have the same bound, and Schur's test gives $\|B_{u,v}\|_{2\to 2} \leq I_n(p)$. For $f \in \ell^2(T_d \square H)$, put $g(v) = \|f(v, \cdot)\|_{\ell^2(H)}$. Then

$$\|(K_n f)(u, \cdot)\|_{\ell^2(H)} \leq I_n(p) \sum_{v : \mathrm{dist}(u,v)=n} g(v) = I_n(p)(R_n g)(u).$$

Taking the $\ell^2(T_d)$-norm proves the claim.

The tree sphere kernels satisfy

$$\|R_n\|_{2\to 2} \le C_d(n+1)b^{n/2}.$$

Thus

$$\|T_p\|_{2\to 2} \le I_0(p) + C' \sum_{n\ge 1}(n+1)(\theta\sqrt{b})^n < \infty,$$

as required. □

Proposition 7.28 (A base-sphere fiber criterion)**.** *Let B be a connected graph and let G be a graph whose vertex set is partitioned as*

$$V(G) = \bigsqcup_{u\in V(B)} F_u.$$

For $n \ge 0$, let S_n^B be the distance-n sphere kernel on B,

$$S_n^B(u,v) = \mathbf{1}\{\mathrm{dist}_B(u,v) = n\},$$

and define

$$J_n(p) = \sup_{\mathrm{dist}_B(u,v)=n} \max\left\{ \sup_{a\in F_u} \sum_{c\in F_v} \tau_p(a,c), \sup_{c\in F_v} \sum_{a\in F_u} \tau_p(a,c) \right\}.$$

If

$$\sum_{n\ge 0} J_n(p) \left\|S_n^B\right\|_{2\to 2} < \infty,$$

then T_p is bounded on $\ell^2(V(G))$. In particular $p \le p_{2\to 2}(G)$.

Proof. Let L_n be the part of T_p connecting pairs of vertices whose fiber indices are at distance n in B. For fixed $u, v \in V(B)$ with $\mathrm{dist}_B(u,v) = n$, the block

$$B_{u,v}(a,c) = \tau_p(a,c), \qquad a \in F_u,\ c \in F_v,$$

has row and column sums at most $J_n(p)$, so Schur's test gives

$$\|B_{u,v}\|_{2\to 2} \le J_n(p).$$

For $f \in \ell^2(V(G))$, put

$$g(u) = \|f\mathbf{1}_{F_u}\|_{\ell^2(F_u)}.$$

Then

$$\|(L_n f)\mathbf{1}_{F_u}\|_{\ell^2(F_u)} \le J_n(p) \sum_{v:\mathrm{dist}_B(u,v)=n} g(v) = J_n(p)(S_n^B g)(u).$$

Taking $\ell^2(V(B))$-norms gives

$$\|L_n\|_{2\to 2} \le J_n(p) \left\|S_n^B\right\|_{2\to 2}.$$

The assumed summability lets us sum the L_n's in operator norm, yielding a bounded operator dominating T_p, and hence T_p itself is bounded. □

Proposition 7.29 (Restricted-shell control of separator square mass)**.** *Use the notation of Proposition 7.28. For $\Sigma \subseteq V(B)$, let M_Σ be multiplication by $\mathbf{1}_\Sigma$ on $\ell^2(V(B))$, and let*

$$S_\Sigma = \bigcup_{u \in \Sigma} F_u.$$

If

$$A_\Sigma(p) := \sum_{n \geq 0} J_n(p) \left\| M_\Sigma S_n^B \right\|_{2\to 2} < \infty,$$

then every finite $K \subset V(G)$ satisfies

$$\sum_{z \in S_\Sigma} \left(T_p \mathbf{1}_K(z)\right)^2 \leq A_\Sigma(p)^2 |K|.$$

Consequently, for a family $\{\Sigma_a : a \in \mathcal{A}_L\}$,

$$\min_{a \in \mathcal{A}_L} \frac{1}{\overline{\chi}_p |K|} \sum_{z \in S_{\Sigma_a}} \left(T_p \mathbf{1}_K(z)\right)^2 \leq \frac{1}{\overline{\chi}_p} \min_{a \in \mathcal{A}_L} A_{\Sigma_a}(p)^2$$

uniformly over finite K.

Proof. Let L_n be the part of T_p connecting fibre indices at base distance n, as in the proof of Proposition 7.28. For a finite $K \subset V(G)$, put

$$g(u) = \left\| \mathbf{1}_{K \cap F_u} \right\|_{\ell^2(F_u)}.$$

Then $\|g\|_2^2 = |K|$. The block-Schur estimate from Proposition 7.28, followed by restriction to the target fibres above Σ, gives

$$\left\| (L_n \mathbf{1}_K) \mathbf{1}_{S_\Sigma} \right\|_2 \leq J_n(p) \left\| M_\Sigma S_n^B g \right\|_2 \leq J_n(p) \left\| M_\Sigma S_n^B \right\|_{2\to 2} \|g\|_2.$$

Summing over n and using the triangle inequality in $\ell^2(S_\Sigma)$ yields

$$\left\| (T_p \mathbf{1}_K) \mathbf{1}_{S_\Sigma} \right\|_2 \leq A_\Sigma(p) |K|^{1/2},$$

which is the first claim. Dividing by $\overline{\chi}_p |K|$ and minimizing over a gives the displayed consequence. □

Remark 7.30. This proposition is weaker than a global L^2 bound when $\sum_n J_n(p) \left\| S_n^B \right\|_{2\to 2} < \infty$, but it is aimed at the separator problem. The base set Σ may be a sparse grid, a horospherical slab, or another separator. Thus one can try to prove the layer-square estimate by showing that the restricted shell operators $M_\Sigma S_n^B$ are smaller than the full shell operators on the scales where the fibre masses $J_n(p)$ are large.

Proposition 7.31 (A limitation of periodic tree-level restrictions)**.** *Let T_d be the d-regular tree, $d \geq 3$, rooted at o, and put $q = d - 1$. Let R_n be the distance-n sphere kernel and, for $L \geq 1$ and $r \in \{0, \dots, L-1\}$, let*

$$S_r^{(L)} = \{x \in V(T_d) : \operatorname{dist}(o, x) \equiv r \pmod{L}\}$$

and let $M_r^{(L)}$ be multiplication by $\mathbf{1}_{S_r^{(L)}}$. Then for every $n \geq 1$,

$$\left\| M_r^{(L)} R_n \right\|_{2\to 2} \geq q^{n/2}.$$

Proof. Choose a non-root vertex y whose level satisfies

$$\operatorname{dist}(o, y) + n \equiv r \pmod{L}.$$

The q^n descendants of y at distance n from y all belong to $S_r^{(L)}$. Hence

$$M_r^{(L)} R_n \mathbf{1}_{\{y\}}$$

is equal to 1 on these q^n descendants. Since $\left\|\mathbf{1}_{\{y\}}\right\|_2 = 1$, the claimed lower bound follows. □

Remark 7.32. Together with the tree sphere bound $\|R_n\|_{2\to 2} \le C_d(n+1)q^{n/2}$, this shows that periodic tree-level restrictions can remove at most the Haagerup polynomial factor in an operator-norm estimate. They do not give a density factor tending to zero with L. Thus the restricted-shell route cannot rely on sparsity of the separator alone; it must also use the finite test set, averaging over shifts, or a genuinely percolative backscattering estimate for the fibre masses.

Corollary 7.33 (Exponential base-sphere test)**.** *With the notation of Proposition 7.28, suppose that there is $\rho > 0$ such that*

$$\limsup_{n\to\infty} \left\|S_n^B\right\|_{2\to 2}^{1/n} \le \rho$$

and

$$\limsup_{n\to\infty} J_n(p)^{1/n} < \rho^{-1}.$$

Then T_p is bounded on $\ell^2(V(G))$.

Proof. Choose θ with

$$\limsup_{n} J_n(p)^{1/n} < \theta < \rho^{-1}.$$

After increasing constants, $J_n(p) \le C\theta^n$, while for every $\rho' > \rho$ with $\theta\rho' < 1$, we have $\left\|S_n^B\right\|_{2\to 2} \le C'(\rho')^n$. Thus the summability hypothesis of Proposition 7.28 holds. □

Corollary 7.34 (A pointwise shell-decay criterion)**.** *Let G be a connected graph, and let S_n^G be its distance-n sphere kernel. Define*

$$\tau_p^{\max}(n) = \sup_{\operatorname{dist}_G(x,y)=n} \tau_p(x, y).$$

If

$$\sum_{n\ge 0} \tau_p^{\max}(n) \left\|S_n^G\right\|_{2\to 2} < \infty,$$

then T_p is bounded on $\ell^2(V(G))$. In particular $p \le p_{2\to 2}(G)$.

Proof. Apply Proposition 7.28 with $B = G$ and singleton fibres $F_x = \{x\}$. Then $J_n(p) = \tau_p^{\max}(n)$. □

Corollary 7.35 (Bounded-fiber pointwise decay)**.** *With the notation of Proposition 7.28, suppose that*

$$\sup_{u\in V(B)} |F_u| \le M < \infty$$

and define

$$\tau_{p,\mathrm{fib}}^{\max}(n) = \sup_{\operatorname{dist}_B(u,v)=n} \sup_{\substack{a\in F_u \\ c\in F_v}} \tau_p(a, c).$$

If

$$\sum_{n\geq 0} \tau_{p,\mathrm{fib}}^{\max}(n)\,\left\|S_n^B\right\|_{2\to 2} < \infty,$$

then T_p is bounded on $\ell^2(V(G))$.

Proof. For each pair of fibres at base distance n, all row and column sums between the two fibres are at most $M\tau_{p,\mathrm{fib}}^{\max}(n)$. Hence

$$J_n(p) \leq M\tau_{p,\mathrm{fib}}^{\max}(n),$$

and Proposition 7.28 applies. □

Proposition 7.36 (L^2-boundedness is open from the right)**.** *Let G have maximum degree at most $\Delta < \infty$. If T_{p_0} is bounded on $\ell^2(V)$ for some $p_0 < 1$, then T_q is bounded on $\ell^2(V)$ for every $q \in [p_0, 1]$ such that*

$$q < p_0 + \frac{1}{\Delta\,\|T_{p_0}\|_{2\to 2}}.$$

Consequently, if $p_c(G) < 1$ and T_{p_c} is bounded, then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. First work in a finite induced subgraph $W \subset V$. Let T_p^W be the two-point matrix for percolation restricted to W, and let A_W be the adjacency matrix of W. Russo's formula and the BK inequality give, entrywise,

$$\frac{d}{dp}T_p^W(x,y) \leq (T_p^W A_W T_p^W)(x,y).$$

Indeed, if an edge uv is pivotal for $x \leftrightarrow y$, then either $\{x \leftrightarrow u\}$ and $\{v \leftrightarrow y\}$, or $\{x \leftrightarrow v\}$ and $\{u \leftrightarrow y\}$, occur disjointly. Summing over oriented edges gives the displayed inequality.

Put $\lambda_W(p) = \left\|T_p^W\right\|_{2\to 2}$. Integrating the differential inequality and using positivity gives

$$\lambda_W(q) \leq \lambda_W(p_0) + \int_{p_0}^{q} \left\|T_s^W A_W T_s^W\right\|_{2\to 2}\,ds \leq \lambda_W(p_0) + \Delta\int_{p_0}^{q} \lambda_W(s)^2\,ds.$$

Since $T_{p_0}^W \leq T_{p_0}$ as positive kernels, $\lambda_W(p_0) \leq M := \|T_{p_0}\|_{2\to 2}$. The scalar comparison inequality yields

$$\lambda_W(q) \leq \frac{M}{1 - \Delta M(q - p_0)}$$

whenever the denominator is positive.

Let W increase to V. For finitely supported f, the quadratic forms $\langle f, T_q^W f\rangle$ increase to $\langle f, T_q f\rangle$, while the right-hand bound is uniform in W. Hence T_q is bounded on $\ell^2(V)$ with the same bound. Taking $p_0 = p_{\mathrm{c}}$ gives the final assertion. □

Corollary 7.37 (Critical shell decay gives a strict gap)**.** *Let G have bounded degree and $p_c(G) < 1$. If*

$$\sum_{n\geq 0} \tau_{p_{\mathrm{c}}}^{\max}(n)\,\left\|S_n^G\right\|_{2\to 2} < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 7.34 gives boundedness of T_{p_c}. Proposition 7.36 then gives boundedness of T_q for some $q > p_c$. □

Corollary 7.38 (Critical shell exponent criterion)**.** *Let G have bounded degree and $p_c(G) < 1$. Define*

$$\rho_{\mathrm{sph},2}(G) = \limsup_{n\to\infty} \left\|S_n^G\right\|_{2\to 2}^{1/n}$$

and

$$\alpha_c^{\max}(G) = \limsup_{n\to\infty} \tau_{p_c}^{\max}(n)^{1/n}.$$

If

$$\alpha_c^{\max}(G)\rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Choose α and ρ such that

$$\alpha_c^{\max}(G) < \alpha, \qquad \rho_{\mathrm{sph},2}(G) < \rho, \qquad \alpha\rho < 1.$$

By the definitions of the two limsups, after increasing constants if necessary,

$$\tau_{p_c}^{\max}(n) \le C\alpha^n \quad \text{and} \quad \left\|S_n^G\right\|_{2\to 2} \le C'\rho^n$$

for every $n \ge 0$. Hence

$$\sum_{n\ge 0} \tau_{p_c}^{\max}(n) \left\|S_n^G\right\|_{2\to 2} \le CC' \sum_{n\ge 0} (\alpha\rho)^n < \infty.$$

Corollary 7.37 applies. □

Remark 7.39. For the d-regular tree, $d \ge 3$, one has $\rho_{\mathrm{sph},2}(T_d) = \sqrt{d-1}$ and $\alpha_c^{\max}(T_d) = p_c(T_d) = 1/(d-1)$. Thus

$$\alpha_c^{\max}(T_d)\rho_{\mathrm{sph},2}(T_d) = \frac{1}{\sqrt{d-1}} < 1,$$

recovering the strict L^2 gap in this model from the critical shell criterion.

Corollary 7.40 (Endpoint SAW shell criterion)**.** *Let G be connected, locally finite, and of bounded degree, with $p_c(G) < 1$. For $n \ge 0$, define*

$$U_n^{\mathrm{SAW}}(p) = \sup_{\mathrm{dist}_G(x,y)=n} \sum_{m\ge n} p^m C_m(x,y),$$

where C_m is the length-m self-avoiding-walk endpoint kernel. Put

$$\beta_c^{\mathrm{SAW,end}}(G) = \limsup_{n\to\infty} \left(U_n^{\mathrm{SAW}}(p_c)\right)^{1/n}.$$

If

$$\beta_c^{\mathrm{SAW,end}}(G)\rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. If $\mathrm{dist}_G(x,y) = n$ and x is connected to y, then the open cluster connecting them contains an open self-avoiding path from x to y of some length $m \geq n$. The union bound therefore gives

$$\tau_{p_c}(x,y) \leq \sum_{m \geq n} p_c^m C_m(x,y) \leq U_n^{\mathrm{SAW}}(p_c).$$

Hence

$$\alpha_c^{\max}(G) \leq \beta_c^{\mathrm{SAW,end}}(G).$$

The claim follows from Corollary 7.38. □

Remark 7.41. This is a deliberately shell-resolved SAW condition. It only counts self-avoiding witnesses between two fixed vertices at distance n, rather than the full ℓ^2 operator norm of all length-m SAW endpoints. On the d-regular tree, the only self-avoiding path between two vertices is the geodesic, so $U_n^{\mathrm{SAW}}(p_c) = p_c^n$ and $\beta_c^{\mathrm{SAW,end}}(T_d) = 1/(d-1)$.

Corollary 7.42 (Excess-length endpoint SAW criterion)**.** *Let G be connected, locally finite, and of bounded degree, with $p_c(G) < 1$. For $n, k \geq 0$, define*

$$E_{n,k}^{\mathrm{SAW}}(G) = \sup_{\mathrm{dist}_G(x,y)=n} C_{n+k}(x,y).$$

Suppose that there are constants $A < \infty$, $a < \infty$, and $r < \infty$ such that

$$E_{n,k}^{\mathrm{SAW}}(G) \leq A a^n r^k \qquad \text{for every } n, k \geq 0,$$

and

$$p_c(G) r < 1 \qquad \text{and} \qquad p_c(G) a \, \rho_{\mathrm{sph},2}(G) < 1.$$

Then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Writing $m = n + k$, the endpoint shell sum satisfies

$$U_n^{\mathrm{SAW}}(p_c) \leq \sum_{k \geq 0} p_c^{n+k} E_{n,k}^{\mathrm{SAW}}(G) \leq A (p_c a)^n \sum_{k \geq 0} (p_c r)^k.$$

Since $p_c r < 1$, this gives

$$\beta_c^{\mathrm{SAW,end}}(G) \leq p_c(G) a.$$

The second displayed assumption and Corollary 7.40 complete the proof. □

Corollary 7.43 (Excess-length nonbacktracking criterion)**.** *Let G be connected, locally finite, and of bounded degree, with $p_c(G) < 1$. Let $N_m(x,y)$ be the number of length-m nonbacktracking walks from x to y, and define*

$$E_{n,k}^{\mathrm{NB}}(G) = \sup_{\mathrm{dist}_G(x,y)=n} N_{n+k}(x,y).$$

Suppose that there are constants $A < \infty$, $a < \infty$, and $r < \infty$ such that

$$E_{n,k}^{\mathrm{NB}}(G) \leq A a^n r^k \qquad \text{for every } n, k \geq 0,$$

and

$$p_c(G) r < 1 \qquad \text{and} \qquad p_c(G) a \, \rho_{\mathrm{sph},2}(G) < 1.$$

Then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Every self-avoiding walk is nonbacktracking, so

$$E_{n,k}^{\mathrm{SAW}}(G) \le E_{n,k}^{\mathrm{NB}}(G) \qquad n, k \ge 0.$$

The claim follows from Corollary 7.42. □

Corollary 7.44 (Nonbacktracking detour generating function criterion)**.** *Let G be connected, locally finite, and of bounded degree, with $p_c(G) < 1$. Let $N_m(x, y)$ be the number of length-m nonbacktracking walks from x to y. For $n \ge 0$, define*

$$V_n^{\mathrm{NB}}(p) = \sup_{\mathrm{dist}_G(x,y)=n} \sum_{k\ge 0} p^k N_{n+k}(x, y),$$

and put

$$\delta_c^{\mathrm{NB,end}}(G) = \limsup_{n\to\infty} \big(V_n^{\mathrm{NB}}(p_{\mathrm{c}})\big)^{1/n}.$$

If

$$p_{\mathrm{c}}(G)\, \delta_c^{\mathrm{NB,end}}(G) \rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. If $\mathrm{dist}_G(x, y) = n$, then the self-avoiding witness bound and the domination of self-avoiding walks by nonbacktracking walks give

$$\tau_{p_{\mathrm{c}}}(x, y) \le \sum_{k\ge 0} p_{\mathrm{c}}^{n+k} C_{n+k}(x, y) \le p_{\mathrm{c}}^n V_n^{\mathrm{NB}}(p_{\mathrm{c}}).$$

Consequently,

$$\alpha_c^{\max}(G) \le p_{\mathrm{c}}(G)\, \delta_c^{\mathrm{NB,end}}(G).$$

The claim follows from Corollary 7.38. □

Corollary 7.45 (Bounded-degree finite-state detour certificate)**.** *Let G be connected, locally finite, and of bounded degree, with $p_c(G) < 1$. Suppose that $q \in (0, 1)$ satisfies $p_{\mathrm{c}}(G) \le q$. Let $N_m(x, y)$ be the number of length-m nonbacktracking walks from x to y, and suppose that there are a finite set $\mathcal{S}$, a nonnegative $\mathcal{S} \times \mathcal{S}$ matrix M, and nonnegative vectors $a, b \in [0, \infty)^{\mathcal{S}}$ such that, for every $n \ge 0$,*

$$\sup_{\mathrm{dist}_G(x,y)=n} \sum_{k\ge 0} q^k N_{n+k}(x, y) \le a^{\mathsf{T}} M^n b.$$

If

$$q\, \rho(M) \rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Since $q \ge p_{\mathrm{c}}(G)$, the displayed certificate also bounds $V_n^{\mathrm{NB}}(p_{\mathrm{c}})$. For every $\theta > \rho(M)$, finite-dimensional linear algebra gives a constant $C_\theta < \infty$ such that

$$a^{\mathsf{T}} M^n b \le C_\theta \theta^n \qquad n \ge 0.$$

Thus

$$\delta_c^{\mathrm{NB,end}}(G) \le \rho(M).$$

Because $p_{\mathrm{c}}(G) \le q$, the displayed strict inequality implies

$$p_{\mathrm{c}}(G)\delta_c^{\mathrm{NB,end}}(G)\rho_{\mathrm{sph},2}(G) \le q\rho(M)\rho_{\mathrm{sph},2}(G) < 1.$$

Corollary 7.44 applies. □

Remark 7.46. This is the bounded-degree version of the later regular-graph bridge certificate. It uses raw nonbacktracking counts rather than nonbacktracking transition probabilities, so no regularity assumption is needed. The price is that the finite-state computation must estimate the full q-weighted detour series directly.

Corollary 7.47 (Collatz bounded-degree detour certificate)**.** *Let G be connected, locally finite, and of bounded degree, with $p_c(G) < 1$. Suppose that $q \in (0,1)$ satisfies $p_{\mathrm{c}}(G) \le q$, and let $R \ge \rho_{\mathrm{sph},2}(G)^2$. Let $N_m(x,y)$ be the number of length-m nonbacktracking walks from x to y, and suppose that there are a finite set $\mathcal{S}$, a nonnegative $\mathcal{S}\times\mathcal{S}$ matrix M, and nonnegative vectors $a, b \in [0,\infty)^{\mathcal{S}}$ such that, for every $n \ge 0$,*

$$\sup_{\mathrm{dist}_G(x,y)=n} \sum_{k\ge 0} q^k N_{n+k}(x,y) \le a^{\mathsf{T}} M^n b.$$

If there are $\lambda > 0$ and a vector $v \in (0,\infty)^{\mathcal{S}}$ such that

$$Mv \le \lambda v$$

coordinatewise and

$$(q\lambda)^2 R < 1,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. The Collatz–Wielandt inequality gives $\rho(M) \le \lambda$, and $\rho_{\mathrm{sph},2}(G) \le \sqrt{R}$. Therefore

$$q\rho(M)\rho_{\mathrm{sph},2}(G) \le q\lambda\sqrt{R} < 1.$$

Corollary 7.45 applies. □

Remark 7.48. This is the exact-arithmetic form of the bounded-degree count-kernel certificate. It differs from the regular bridge certificate below only in the normalizing factor: raw count kernels require $(q\lambda)^2 R < 1$, while the regular bridge-probability version requires $\big(q(D-1)\lambda\big)^2 R < 1$. The repository checker supports both criteria as separate exact certificate modes.

Proposition 7.49 (Bounded-degree path-sum detour certificate)**.** *Let G be connected, locally finite, and of bounded degree, with $p_c(G) < 1$, and suppose that $q \in (0,1)$ satisfies $p_{\mathrm{c}}(G) \le q$. Let $\mathcal{S}$ be finite, let M be a nonnegative $\mathcal{S}\times\mathcal{S}$ matrix, and let $a, b \in [0,\infty)^{\mathcal{S}}$. Suppose that for every $n \ge 0$ and every pair x, y with $\mathrm{dist}_G(x,y) = n$, there are nonnegative vectors $\mu_{x,y} \le a$ and $h_{x,y} \le b$, and, for each $1 \le r \le n$ and $\sigma, \tau \in \mathcal{S}$, a countable set $\mathcal{F}_{r,x,y}(\sigma,\tau)$ with weights $w_{r,x,y} : \mathcal{F}_{r,x,y}(\sigma,\tau) \to [0,\infty)$. Define*

$$K_{r,x,y}(\sigma,\tau) = \sum_{f\in\mathcal{F}_{r,x,y}(\sigma,\tau)} w_{r,x,y}(f),$$

and assume that these sums are finite and $K_{r,x,y} \le M$ coordinatewise.

Assume moreover that every nonbacktracking walk γ from x to y of length $n+k$, $k \geq 0$, is assigned injectively to data

$$\sigma_0, f_1, \sigma_1, \ldots, f_n, \sigma_n$$

with $f_r \in \mathcal{F}_{r,x,y}(\sigma_{r-1}, \sigma_r)$, where for $n = 0$ the data is just σ_0, and that

$$q^k \leq \mu_{x,y}(\sigma_0) \prod_{r=1}^{n} w_{r,x,y}(f_r) h_{x,y}(\sigma_n).$$

If

$$q\, \rho(M) \rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. For fixed x, y at distance n, sum the displayed bound over all nonbacktracking walks from x to y of all lengths $n+k$, $k \geq 0$. Injectivity gives

$$\begin{aligned} \sum_{k\geq 0} q^k N_{n+k}(x,y) &\leq \sum_{\sigma_0,\ldots,\sigma_n} \mu_{x,y}(\sigma_0) \prod_{r=1}^{n} K_{r,x,y}(\sigma_{r-1}, \sigma_r) h_{x,y}(\sigma_n) \\ &= \mu_{x,y}^{\mathsf{T}} K_{1,x,y} \cdots K_{n,x,y} h_{x,y}. \end{aligned}$$

For $n = 0$, the product is interpreted as the identity. Since all matrices and vectors are nonnegative and $K_{r,x,y} \leq M$, $\mu_{x,y} \leq a$, and $h_{x,y} \leq b$, the last expression is at most $a^{\mathsf{T}} M^n b$. Corollary 7.45 applies. □

Remark 7.50. This is the raw-count analogue of the regular path-sum transfer certificate below. The graph-specific task is to encode each weighted detour by phases whose transfer sums are uniformly dominated by M. No regularity or transition-probability normalization is used.

Example 7.51 (Semiregular-tree calibration)**.** Let $T_{a,b}$ be the infinite (a,b)-semiregular tree, where vertices in the two bipartition classes have degrees a and b, and put

$$r = (a-1)(b-1) > 1.$$

Then

$$p_{\mathrm{c}}(T_{a,b}) = \frac{1}{\sqrt{r}}, \qquad \rho_{\mathrm{sph},2}(T_{a,b}) \leq r^{1/4}.$$

Consequently

$$p_{\mathrm{c}}(T_{a,b}) \rho_{\mathrm{sph},2}(T_{a,b}) \leq r^{-1/4} < 1,$$

and hence

$$p_{\mathrm{c}}(T_{a,b}) < p_{2\to 2}(T_{a,b}).$$

Proof. The critical probability is the standard two-type branching-process threshold: after two generations the expected number of descendants is $p^2 r$, so $p_{\mathrm{c}}(T_{a,b}) = r^{-1/2}$.

For the spherical bound, use the standard two-type Schur estimate. Fix a root o. There are positive constants c_A, c_B, depending only on a, b, such that

$$\phi(x) = c_{\mathrm{type}(x)} r^{-d(o,x)/4}$$

satisfies the following bound uniformly in x and n. If $d(x,y)=n$ and z is the confluent point of x,y with respect to o, write $j=d(x,z)$. For each fixed j, the number of possible vertices y is bounded, up to a constant depending only on a,b, by $r^{(n-j)/2}$, while

$$\frac{\phi(y)}{\phi(x)} \le C_{a,b} r^{-n/4+j/2}.$$

Thus each value of j contributes at most $C_{a,b} r^{n/4}$, and summing over the at most $n+1$ possible values of j gives

$$\sup_x \phi(x)^{-1} \sum_{d(x,y)=n} \phi(y) \le C_{a,b}(n+1) r^{n/4}.$$

Since the sphere kernel S_n is symmetric, Schur's test gives $\|S_n\|_{2\to 2} \le C_{a,b}(n+1) r^{n/4}$, and hence $\rho_{\mathrm{sph},2}(T_{a,b}) \le r^{1/4}$. The strict gap follows from Corollary 7.45, since for a tree

$$\sup_{\mathrm{dist}(x,y)=n} \sum_{k\ge 0} p_{\mathrm{c}}^k N_{n+k}(x,y) = 1$$

for every $n \ge 0$. □

Remark 7.52. For the $(2,3)$-semiregular tree, $r=2$. The repository contains the rational exact certificate

$$q=\frac{3}{4}, \qquad R=\frac{3}{2}, \qquad M=(1),$$

for which

$$(q\lambda)^2 R = \frac{27}{32} < 1.$$

The checker verifies the Collatz inequality and the one-state tree detour majorization; the bounds $p_c \le 3/4$ and $\rho_{\mathrm{sph},2}^2 \le 3/2$ are supplied by the calculation above.

Proposition 7.53 (Finite-quotient tree calibration)**.** *Let H be a finite connected graph, let $T=\widetilde{H}$ be its universal cover, and let $\vec{E}(H)$ be the set of directed edges of H. Let B be the directed nonbacktracking matrix*

$$B_{e,f} = \mathbf{1}\{\mathrm{head}(e)=\mathrm{tail}(f),\ f\ne \bar{e}\}, \qquad e,f\in\vec{E}(H),$$

where $\bar{e}$ is e with the opposite orientation. Suppose that B is irreducible and has Perron–Frobenius radius $\beta>1$. Then

$$p_{\mathrm{c}}(T)=\frac{1}{\beta}, \qquad \rho_{\mathrm{sph},2}(T)\le\sqrt{\beta},$$

and consequently

$$p_{\mathrm{c}}(T) < p_{2\to 2}(T).$$

Proof. Let $u\in(0,\infty)^{\vec{E}(H)}$ satisfy $Bu=\beta u$. The equality $p_{\mathrm{c}}(T)=\beta^{-1}$ is the standard multitype branching-process calculation: starting from a directed lift of an edge of type e, the open nonbacktracking descendants have mean matrix pB, so survival starts exactly when $p\beta>1$.

It remains to bound the spherical kernels. Fix a root $o\in V(T)$. If $x\ne o$, let ϵ_x be the last directed edge on the geodesic from o to x, oriented away from o, and let $\mathrm{type}(\epsilon_x)\in\vec{E}(H)$ be its projection to H. Set

$$\phi(o)=1, \qquad \phi(x)=u_{\mathrm{type}(\epsilon_x)}\beta^{-d(o,x)/2} \quad (x\ne o).$$

Since there are finitely many directed edge types, all constants below may depend on H but not on x, y, n.

For any vertex z, any forbidden first step at z, and any $\ell \geq 0$, the sum of $u_{\mathrm{type}(\epsilon_y)}$ over vertices y reached by a length-ℓ nonbacktracking path from z avoiding that first step is at most $C_H \beta^\ell$. Indeed, for $\ell \geq 1$ this sum is bounded by a finite sum of coordinates of $B^{\ell-1}u = \beta^{\ell-1}u$, and $\ell = 0$ is absorbed into the constant.

Fix x and n. For y with $d(x,y) = n$, let z be the last common vertex of the geodesics $[o,x]$ and $[o,y]$, put $j = d(x,z)$, and $\ell = n - j = d(z,y)$. For fixed j, the vertex z is determined by x. The preceding weighted count gives

$$\sum_{\substack{d(x,y)=n \\ d(x,z)=j}} \frac{\phi(y)}{\phi(x)} \leq C_H \beta^{(j-\ell)/2} \beta^\ell = C_H \beta^{n/2}.$$

There are at most $n+1$ possible values of j, whence

$$\sup_x \phi(x)^{-1} \sum_{d(x,y)=n} \phi(y) \leq C_H (n+1) \beta^{n/2}.$$

The distance sphere kernel S_n is symmetric, so Schur's test yields

$$\|S_n\|_{2\to 2} \leq C_H (n+1) \beta^{n/2}.$$

Thus $\rho_{\mathrm{sph},2}(T) \leq \sqrt{\beta}$.

Finally, a nonbacktracking walk in a tree cannot revisit a vertex. Hence, if $d(x,y) = n$, then $N_n(x,y) = 1$ and $N_{n+k}(x,y) = 0$ for every $k > 0$. The bounded-degree detour certificate applies with $M = (1)$, giving

$$p_{\mathrm{c}}(T) \rho_{\mathrm{sph},2}(T) \leq \beta^{-1/2} < 1.$$

□

Remark 7.54. This proposition packages all finite-type tree calibrations in one place. It contains the regular and semiregular tree examples after choosing a finite regular or biregular base graph. The remaining obstruction in the main problem is not finite type on trees, but controlling genuine detours created by cycles in non-tree quasi-transitive graphs.

Corollary 7.55 (Nonbacktracking bridge-probability criterion)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$, with $p_c(G) < 1$. Let $(X_m)_{m\geq 0}$ be nonbacktracking random walk started at x: the first step is uniform among the D neighbours of x, and each later step is uniform among the $D-1$ choices that do not immediately return. For $s \geq 0$, define*

$$B_n^{\mathrm{NB}}(s) = \sup_{\mathrm{dist}_G(x,y)=n} \sum_{k\geq 0} s^k \mathbb{P}_x^{\mathrm{NB}}(X_{n+k} = y), \qquad n \geq 1,$$

and put

$$\lambda_c^{\mathrm{NB,br}}(G) = \limsup_{n\to\infty} \big(B_n^{\mathrm{NB}}(p_{\mathrm{c}}(G)(D-1))\big)^{1/n}.$$

If

$$p_{\mathrm{c}}(G)(D-1)\lambda_c^{\mathrm{NB,br}}(G)\rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. For $m \geq 1$,

$$N_m(x,y) = D(D-1)^{m-1}\mathbb{P}_x^{\mathrm{NB}}(X_m = y).$$

Thus, for $n \geq 1$,

$$V_n^{\mathrm{NB}}(p_c) = \sup_{\mathrm{dist}_G(x,y)=n} \sum_{k\geq 0} p_c^k N_{n+k}(x,y) = \frac{D}{D-1}(D-1)^n B_n^{\mathrm{NB}}(p_c(D-1)).$$

Consequently,

$$\delta_c^{\mathrm{NB,end}}(G) \leq (D-1)\lambda_c^{\mathrm{NB,br}}(G).$$

The claim follows from Corollary 7.44. □

Corollary 7.56 (Directed-edge spectral bridge bound)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$, with $p_c(G) < 1$. Let $\vec{E}$ be the set of directed edges, and let $\mathcal{P}_{\mathrm{NB}}$ be the nonbacktracking transition operator on $\ell^2(\vec{E})$,*

$$(\mathcal{P}_{\mathrm{NB}}F)(u,v) = \frac{1}{D-1}\sum_{\substack{w\sim v\\ w\neq u}} F(v,w).$$

Put

$$\sigma_{\mathrm{NB},2}(G) = \limsup_{m\to\infty} \|\mathcal{P}_{\mathrm{NB}}^m\|_{2\to 2}^{1/m}.$$

If

$$p_c(G)(D-1)\sigma_{\mathrm{NB},2}(G) < 1$$

and

$$p_c(G)(D-1)\sigma_{\mathrm{NB},2}(G)\rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Let $s_c = p_c(G)(D-1)$, and choose $\theta > \sigma_{\mathrm{NB},2}(G)$ so that $s_c\theta < 1$ and $s_c\theta\rho_{\mathrm{sph},2}(G) < 1$. By the definition of the limsup, there is $C_\theta < \infty$ such that

$$\|\mathcal{P}_{\mathrm{NB}}^m\|_{2\to 2} \leq C_\theta\theta^m \qquad m \geq 0.$$

For $m \geq 1$, write the first nonbacktracking edge as a uniform directed edge leaving x. If ν_x is this initial distribution and $\mathbf{1}_y^{\mathrm{in}}$ is the indicator of directed edges ending at y, then

$$\mathbb{P}_x^{\mathrm{NB}}(X_m = y) = \left\langle \nu_x, \mathcal{P}_{\mathrm{NB}}^{m-1}\mathbf{1}_y^{\mathrm{in}} \right\rangle.$$

Since $\|\nu_x\|_2 = D^{-1/2}$ and $\left\|\mathbf{1}_y^{\mathrm{in}}\right\|_2 = D^{1/2}$, this gives

$$\mathbb{P}_x^{\mathrm{NB}}(X_m = y) \leq C_\theta\theta^{m-1}.$$

Consequently, for $n \geq 1$,

$$B_n^{\mathrm{NB}}(s_c) \leq C_\theta\theta^{n-1}\sum_{k\geq 0}(s_c\theta)^k = \frac{C_\theta\theta^{n-1}}{1-s_c\theta}.$$

Thus

$$\lambda_c^{\mathrm{NB,br}}(G) \leq \theta.$$

Letting $\theta \downarrow \sigma_{\mathrm{NB},2}(G)$, the second displayed assumption and Corollary 7.55 imply the claim. □

Proposition 7.57 (Spherical growth from nonbacktracking growth)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$. With $\sigma_{\mathrm{NB},2}(G)$ as in Corollary 7.56,*

$$\rho_{\mathrm{sph},2}(G) \leq (D-1)\sigma_{\mathrm{NB},2}(G).$$

Proof. Let $\mathcal{A}_{\mathrm{NB}} = (D-1)\mathcal{P}_{\mathrm{NB}}$ be the unnormalized nonbacktracking transition operator on directed edges, and let $N_n(x,y)$ be the number of nonbacktracking walks of length n from x to y. If x and y have distance n, then every geodesic from x to y is nonbacktracking, so

$$S_n^G(x,y) \leq N_n(x,y) \qquad n \geq 1.$$

Since both kernels are nonnegative, this pointwise domination implies $\left\|S_n^G\right\|_{2\to 2} \leq \|N_n\|_{2\to 2}$.

Write e^- and e^+ for the tail and head of a directed edge e. Let $C : \ell^2(V) \to \ell^2(\vec{E})$ be $Cf(e) = f(e^+)$, and let $B : \ell^2(\vec{E}) \to \ell^2(V)$ be

$$Bh(x) = \sum_{e^-=x} h(e).$$

Then $N_n = B\mathcal{A}_{\mathrm{NB}}^{n-1}C$. Moreover $\|C\|_{2\to 2} = \sqrt{D}$ and $\|B\|_{2\to 2} \leq \sqrt{D}$. Hence

$$\|N_n\|_{2\to 2} \leq D(D-1)^{n-1}\left\|\mathcal{P}_{\mathrm{NB}}^{n-1}\right\|_{2\to 2}.$$

Taking n-th roots and then the limsup gives the claim. □

Corollary 7.58 (Adjacency-spectrum square criterion)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$, with $p_c(G) < 1$. Let A be its adjacency operator on $\ell^2(V)$, put $\rho_A = \|A\|_{2\to 2}$, and let $\beta(D,\rho_A)$ be the nonbacktracking root from Lemma 7.13. If*

$$p_{\mathrm{c}}(G)\beta(D,\rho_A)^2 < 1,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Choose $\theta > \beta(D,\rho_A)$ such that

$$p_{\mathrm{c}}(G)\theta < 1, \qquad p_{\mathrm{c}}(G)\theta^2 < 1.$$

By Lemma 7.13, after increasing a constant if needed,

$$\|N_m\|_{2\to 2} \leq C_\theta \theta^m \qquad m \geq 0.$$

Since every self-avoiding walk is nonbacktracking, for $\mathrm{dist}_G(x,y) = n$,

$$\tau_{p_{\mathrm{c}}}(x,y) \leq \sum_{m\geq n} p_{\mathrm{c}}(G)^m N_m(x,y) \leq C_\theta \sum_{m\geq n} \big(p_{\mathrm{c}}(G)\theta\big)^m \leq C'_\theta \big(p_{\mathrm{c}}(G)\theta\big)^n.$$

Hence $\alpha_c^{\max}(G) \leq p_{\mathrm{c}}(G)\theta$. Also, as in Proposition 7.57, $S_n^G(x,y) \leq N_n(x,y)$ for vertices at distance n, and therefore

$$\rho_{\mathrm{sph},2}(G) \leq \theta.$$

It follows that

$$\alpha_c^{\max}(G)\rho_{\mathrm{sph},2}(G) \leq p_{\mathrm{c}}(G)\theta^2 < 1.$$

Corollary 7.38 applies. □

Remark 7.59. This criterion is intentionally rough. For the D-regular tree, $\rho_A = 2\sqrt{D-1}$, so $\beta(D, \rho_A) = \sqrt{D-1}$ and $p_c(T_D)\beta(D, \rho_A)^2 = 1$. Thus pure adjacency-spectrum control reaches only the critical boundary in the model case, whereas the bridge-resolved criterion has a strict factor. The corollary is nevertheless a useful test: any proof relying only on the adjacency spectral radius cannot be the final general mechanism unless it improves this loss.

Corollary 7.60 (A global nonbacktracking square criterion)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$, with $p_c(G) < 1$. If*

$$p_c(G)\big((D-1)\sigma_{\mathrm{NB},2}(G)\big)^2 < 1,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Put $a = (D-1)\sigma_{\mathrm{NB},2}(G)$. Since $p_c(G) < 1$, the displayed assumption implies $p_c(G)a < 1$: this is immediate if $a \leq 1$, and follows from $p_c(G)a < 1/a$ if $a > 1$. Proposition 7.57 also gives

$$p_c(G)(D-1)\sigma_{\mathrm{NB},2}(G)\rho_{\mathrm{sph},2}(G) \leq p_c(G)a^2 < 1.$$

Corollary 7.56 applies. □

Remark 7.61. The square criterion is a useful diagnostic but is deliberately coarse. On regular trees it sits at the critical boundary rather than recovering the strict L^2 gap, whereas the bridge-resolved calibration in Example 7.72 does recover the gap. This is the reason the finite-state certificates above are aimed at bridge probabilities, not only at the global directed-edge operator.

Proposition 7.62 (Finite-state bridge certificate)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$, with $p_c(G) < 1$. Put $s_c = p_c(G)(D-1)$. Suppose that there are a finite set $\mathcal{S}$, a nonnegative $\mathcal{S} \times \mathcal{S}$ matrix M, and nonnegative vectors $a, b \in [0,\infty)^{\mathcal{S}}$ such that, for every $n \geq 1$,*

$$B_n^{\mathrm{NB}}(s_c) \leq a^{\mathsf{T}} M^n b.$$

If

$$p_c(G)(D-1)\rho(M)\rho_{\mathrm{sph},2}(G) < 1,$$

where $\rho(M)$ is the spectral radius of M, then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For every $\theta > \rho(M)$, finite-dimensional Perron–Frobenius theory, or equivalently the Jordan normal form, gives $C_\theta < \infty$ such that

$$a^{\mathsf{T}} M^n b \leq C_\theta \theta^n \qquad n \geq 1.$$

Thus

$$\lambda_c^{\mathrm{NB,br}}(G) \leq \rho(M).$$

The claim follows from Corollary 7.55. □

Remark 7.63. The proposition is intended as a certificate format. A radial decomposition, automaton, or transfer matrix may overcount nonbacktracking bridges, but if the overcount has finite-state growth rate $\rho(M)$ satisfying the displayed inequality, then it is already strong enough to imply the strict L^2 gap.

Corollary 7.64 (Bridge certificate with an upper bound on p_c)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$. Suppose that $q \in (0,1)$ satisfies $p_c(G) \leq q$. Put $s_q = q(D-1)$, and suppose that there are a finite set $\mathcal{S}$, a nonnegative $\mathcal{S} \times \mathcal{S}$ matrix M, and nonnegative vectors $a, b \in [0,\infty)^{\mathcal{S}}$ such that, for every $n \geq 1$,*

$$B_n^{\mathrm{NB}}(s_q) \leq a^{\mathsf{T}} M^n b.$$

If

$$q(D-1)\rho(M)\rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Since $B_n^{\mathrm{NB}}(s)$ is nondecreasing in s and $p_c(G) \leq q$, the same matrix bound holds at $s_c = p_c(G)(D-1)$. Moreover,

$$p_c(G)(D-1)\rho(M)\rho_{\mathrm{sph},2}(G) \leq q(D-1)\rho(M)\rho_{\mathrm{sph},2}(G) < 1.$$

Proposition 7.62 applies. □

Remark 7.65. This form is the one most likely to be useful computationally. It does not require knowing $p_c(G)$ exactly: an independently proved upper bound $p_c(G) \leq q$ and a finite-state bridge certificate at the larger parameter $q(D-1)$ suffice.

Proposition 7.66 (Nonhomogeneous block certificate)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$, and suppose that $q \in (0,1)$ satisfies $p_c(G) \leq q$. Put $s_q = q(D-1)$. Let $\mathcal{S}$ be finite, let M be a nonnegative $\mathcal{S} \times \mathcal{S}$ matrix, and let $a, b \in [0,\infty)^{\mathcal{S}}$. Suppose that for every $n \geq 1$ and every pair x, y with $\mathrm{dist}_G(x,y) = n$, there are a nonnegative row vector $\mu_{x,y}$, a nonnegative column vector $h_{x,y}$, and nonnegative $\mathcal{S} \times \mathcal{S}$ matrices*

$$K_{1,x,y}, \ldots, K_{n,x,y}$$

such that

$$\mu_{x,y} \leq a, \qquad h_{x,y} \leq b, \qquad K_{r,x,y} \leq M \quad (1 \leq r \leq n)$$

coordinatewise, and

$$\sum_{k \geq 0} s_q^k \mathbb{P}_x^{\mathrm{NB}}(X_{n+k} = y) \leq \mu_{x,y}^{\mathsf{T}} K_{1,x,y} K_{2,x,y} \cdots K_{n,x,y} h_{x,y}.$$

If

$$q(D-1)\rho(M)\rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For nonnegative matrices, coordinatewise domination is preserved under matrix multiplication. Hence

$$K_{1,x,y} K_{2,x,y} \cdots K_{n,x,y} \leq M^n$$

coordinatewise. Multiplying on the left and right by the dominated nonnegative vectors gives

$$\sum_{k \geq 0} s_q^k \mathbb{P}_x^{\mathrm{NB}}(X_{n+k} = y) \leq a^{\mathsf{T}} M^n b.$$

Taking the supremum over all pairs at distance n yields the hypothesis of Corollary 7.64. □

Remark 7.67. The proposition is the form expected from radial or first-descent computations. A graph-specific argument may produce level-dependent block matrices $K_{r,x,y}$ rather than one homogeneous transfer matrix. It is enough to dominate all of those blocks coordinatewise by a single finite matrix M, because the growth test only sees M^n.

Proposition 7.68 (Path-sum transfer certificate)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$, and suppose that $q \in (0,1)$ satisfies $p_c(G) \leq q$. Put $s_q = q(D-1)$. Let $\mathcal{S}$ be finite, let M be a nonnegative $\mathcal{S} \times \mathcal{S}$ matrix, and let $a, b \in [0,\infty)^{\mathcal{S}}$. Suppose that for every $n \geq 1$ and every pair x, y with $\mathrm{dist}_G(x,y) = n$, there are nonnegative vectors $\mu_{x,y} \leq a$ and $h_{x,y} \leq b$, and, for each $1 \leq r \leq n$ and $\sigma, \tau \in \mathcal{S}$, a countable set $\mathcal{F}_{r,x,y}(\sigma,\tau)$ with weights $w_{r,x,y} : \mathcal{F}_{r,x,y}(\sigma,\tau) \to [0,\infty)$. Define*

$$K_{r,x,y}(\sigma,\tau) = \sum_{f \in \mathcal{F}_{r,x,y}(\sigma,\tau)} w_{r,x,y}(f),$$

and assume that these sums are finite and $K_{r,x,y} \leq M$ coordinatewise.

Assume moreover that every nonbacktracking trajectory γ from x to y of length $n+k$, $k \geq 0$, is assigned injectively to data

$$\sigma_0, f_1, \sigma_1, \ldots, f_n, \sigma_n$$

with $f_r \in \mathcal{F}_{r,x,y}(\sigma_{r-1}, \sigma_r)$, and that

$$s_q^k \mathbb{P}_x^{\mathrm{NB}}(\gamma) \leq \mu_{x,y}(\sigma_0) \prod_{r=1}^{n} w_{r,x,y}(f_r) h_{x,y}(\sigma_n).$$

If

$$q(D-1)\rho(M)\rho_{\mathrm{sph},2}(G) < 1,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Sum the displayed bound over all nonbacktracking trajectories from x to y of all lengths $n+k$, $k \geq 0$. Since the assignment is injective, the resulting sum is bounded by the sum over all state sequences and all phase fragments:

$$\begin{aligned} \sum_{k \geq 0} s_q^k \mathbb{P}_x^{\mathrm{NB}}(X_{n+k} = y) &\leq \sum_{\sigma_0, \ldots, \sigma_n} \mu_{x,y}(\sigma_0) \prod_{r=1}^{n} K_{r,x,y}(\sigma_{r-1}, \sigma_r) h_{x,y}(\sigma_n) \\ &= \mu_{x,y}^{\mathsf{T}} K_{1,x,y} \cdots K_{n,x,y} h_{x,y}. \end{aligned}$$

Proposition 7.66 applies. □

Remark 7.69. This proposition spells out the computational obligation behind a radial or automaton certificate. The graph-specific part is the injective phase-decomposition and the local phase-weight bounds. Once those are proved, the remaining growth comparison is finite-dimensional.

Corollary 7.70 (Collatz–Wielandt bridge certificate)**.** *Let G be an infinite connected D-regular graph, $D \geq 3$. Suppose that $q \in (0,1)$ satisfies $p_c(G) \leq q$, and let $R \geq \rho_{\mathrm{sph},2}(G)^2$. Put $s_q = q(D-1)$. Suppose that there are a finite set $\mathcal{S}$, a nonnegative $\mathcal{S} \times \mathcal{S}$ matrix M, and nonnegative vectors $a, b \in [0,\infty)^{\mathcal{S}}$ such that, for every $n \geq 1$,*

$$B_n^{\mathrm{NB}}(s_q) \leq a^{\mathsf{T}} M^n b.$$

If there are $\lambda > 0$ and a vector $v \in (0,\infty)^{\mathcal{S}}$ such that

$$Mv \le \lambda v$$

coordinatewise and

$$\big(q(D-1)\lambda\big)^2 R < 1,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. The coordinatewise inequality $Mv \le \lambda v$ with $v > 0$ implies $\rho(M) \le \lambda$ by the Collatz–Wielandt formula, or by applying the inequality iteratively and comparing norms. Since $\rho_{\mathrm{sph},2}(G) \le \sqrt{R}$, the final displayed inequality gives

$$q(D-1)\rho(M)\rho_{\mathrm{sph},2}(G) < 1.$$

Corollary 7.64 applies. □

Remark 7.71. This is the exact-arithmetic form of the certificate. The inequalities $Mv \le \lambda v$ and $\big(q(D-1)\lambda\big)^2 R < 1$ can be checked using rational arithmetic once q, λ, R, M, a, b, and v are rational. The repository accompanying this working draft includes a small checker for this algebraic part and for the dimensions and signs of the boundary vectors, either certificate-by-certificate or for all JSON certificates in a directory; the graph-specific bridge majorization remains a separate mathematical certificate.

Example 7.72 (Regular-tree calibration)**.** For the D-regular tree T_D, $D \ge 3$,

$$\lambda_c^{\mathrm{NB,br}}(T_D) = \frac{1}{D-1}.$$

Indeed, if $\mathrm{dist}_{T_D}(x,y) = n$, a nonbacktracking walk from x can hit y only by following the unique geodesic from x to y, and it then hits y at time exactly n. Hence

$$B_n^{\mathrm{NB}}(s) = \frac{1}{D(D-1)^{n-1}} \qquad n \ge 1$$

for every $s \ge 0$. Since $p_{\mathrm{c}}(T_D) = 1/(D-1)$ and $\rho_{\mathrm{sph},2}(T_D) = \sqrt{D-1}$,

$$p_{\mathrm{c}}(T_D)(D-1)\lambda_c^{\mathrm{NB,br}}(T_D)\rho_{\mathrm{sph},2}(T_D) = \frac{1}{\sqrt{D-1}} < 1.$$

Thus Corollary 7.55 recovers the strict L^2 gap on regular trees.

Remark 7.73. The regular-tree certificate is also the one graph-specific majorization that the repository checker verifies directly. In the one-state certificate with $M = (1/(D-1))$, the bound

$$B_n^{\mathrm{NB}}(s) = \frac{1}{D(D-1)^{n-1}} \le \left(\frac{1}{D-1}\right)^n$$

is checked exactly for all $n \ge 1$. General radial or automaton majorizations still have to supply their own mathematical proof.

Remark 7.74. This criterion is weaker than the preceding self-avoiding version but is often easier to estimate, since nonbacktracking walk counts can be attacked through recurrences, transfer matrices, or spherical decompositions. Its usefulness depends on retaining fixed-endpoint and excess-length information; a global operator bound for all nonbacktracking endpoints is usually too coarse in low-branching product examples. The generating-function form in Corollary 7.44 is slightly more flexible than a separable estimate $Aa^n r^k$: it only asks for the p_c-weighted detour series after endpoints at distance n have been fixed. On regular graphs, Corollary 7.55 rewrites the same target as an exponentially small nonbacktracking bridge-probability estimate. Corollary 7.56 is a convenient sufficient condition, but Example 7.72 shows why the real target should remain bridge-resolved rather than just globally spectral.

Proposition 7.75 (A quotient-tree fiber criterion)**.** *Let G be a graph whose vertex set is partitioned as*

$$V(G) = \bigsqcup_{u \in V(T_d)} F_u$$

with $d \geq 3$. Put $b = d - 1$, and for $n \geq 0$ define

$$J_n(p) = \sup_{\mathrm{dist}_{T_d}(u,v)=n} \max\left\{ \sup_{a \in F_u} \sum_{c \in F_v} \tau_p(a,c), \sup_{c \in F_v} \sum_{a \in F_u} \tau_p(a,c) \right\}.$$

If $J_0(p) < \infty$ and

$$\limsup_{n \to \infty} J_n(p)^{1/n} < b^{-1/2},$$

then T_p is bounded on $\ell^2(V(G))$. In particular $p \leq p_{2\to 2}(G)$.

Proof. Choose $\theta < b^{-1/2}$ and $C < \infty$ such that $J_n(p) \leq C\theta^n$ for every $n \geq 1$. Let L_n be the part of T_p connecting pairs of vertices whose fiber indices in T_d are at tree distance n. For fixed $u, v \in V(T_d)$ with $\mathrm{dist}_{T_d}(u,v) = n$, the block

$$B_{u,v}(a,c) = \tau_p(a,c), \qquad a \in F_u,\ c \in F_v,$$

has row and column sums at most $J_n(p)$. Schur's test gives

$$\|B_{u,v}\|_{2\to 2} \leq J_n(p).$$

For $f \in \ell^2(V(G))$, put

$$g(u) = \|f \mathbf{1}_{F_u}\|_{\ell^2(F_u)}.$$

Then

$$\|(L_n f)\mathbf{1}_{F_u}\|_{\ell^2(F_u)} \leq J_n(p) \sum_{v : \mathrm{dist}_{T_d}(u,v)=n} g(v).$$

Taking the $\ell^2(T_d)$-norm and using the tree sphere kernel R_n gives

$$\|L_n\|_{2\to 2} \leq J_n(p)\, \|R_n\|_{2\to 2}.$$

Since $\|R_n\|_{2\to 2} \leq C_d(n+1)b^{n/2}$, we obtain

$$\|T_p\|_{2\to 2} \leq J_0(p) + C' \sum_{n \geq 1} (n+1)(\theta\sqrt{b})^n < \infty.$$

□

Proposition 7.76 (The full line factor)**.** *For every $d \geq 3$,*

$$p_c(T_d \square \mathbb{Z}) < p_{2\to 2}(T_d \square \mathbb{Z}).$$

Proof. We first prove the sharper identity

$$p_{2\to 2}(T_d \square \mathbb{Z}) = p_u(T_d \square \mathbb{Z}).$$

Put $b = d - 1$. Let x_n be a fixed vertex of T_d at tree distance n from the root and write

$$I_n(p) = \sum_{k \in \mathbb{Z}} \mathbb{P}_p\big((o, 0) \leftrightarrow (x_n, k)\big).$$

Yamamoto proves that for every $p < p_u$, the quantities $I_n(p)$ are finite, submultiplicative in n, and satisfy

$$\lim_{n\to\infty} I_n(p)^{1/n} = \alpha(p),$$

where $\alpha(p)$ is the fixed-direction connection exponent [Yam18]. The same work, using Schonmann's theorem and the strict monotonicity of α on $[0, p_u]$, gives

$$\alpha(p) < b^{-1/2} \qquad (p < p_u).$$

Hence for every $p < p_u$ there are $\theta < b^{-1/2}$ and $C < \infty$ such that $I_n(p) \leq C\theta^n$ for all $n \geq 1$. Yamamoto's estimate $\beta(p) < 1$ for vertical connections gives $I_0(p) < \infty$. Therefore Proposition 7.27 gives $p \leq p_{2\to 2}$ for every $p < p_u$, so $p_u \leq p_{2\to 2}$. The reverse inequality $p_{2\to 2} \leq p_u$ holds for every transitive graph, hence equality follows.

Hutchcroft proved $p_c(T_d \square \mathbb{Z}) < p_u(T_d \square \mathbb{Z})$ for all $d \geq 3$ [Hut20]. Combining this strict inequality with $p_{2\to 2} = p_u$ proves the claim. □

Remark 7.77. Hutchcroft and Pan prove a stronger product theorem: if T is a regular tree and H is an infinite amenable Cayley graph, then $p_{2\to 2}(T \square H) = p_u(T \square H)$; they also prove the analogous identity for the lamplighter graph over T, together with equality with the natural nonunimodular tilted thresholds [HP24a]. Proposition 7.76 is included here as a transparent line-factor stress test: Yamamoto's fixed-direction exponent supplies the fiber-summed decay, and Proposition 7.27 isolates the elementary operator step that converts that decay into L^2-boundedness.

Remark 7.78 (Finite-length SAW diagnostics for $T_3 \square \mathbb{Z}$)**.** Proposition 7.76 bypasses the following finite-length SAW bounds. They are retained as diagnostics for what a purely SAW-based proof would have to achieve. Let C_m be the length-m SAW endpoint kernel on $T_3 \square \mathbb{Z}$. Since the kernels C_m are submultiplicative,

$$\gamma_{\mathrm{SAW},2}(T_3 \square \mathbb{Z}) \leq \|C_m\|_{2\to 2}^{1/m}.$$

For fixed m, write $a_{r,h}^{(m)}$ for the number of length-m self-avoiding paths from the origin to a fixed vertex at tree distance r and height h. Then

$$C_m = \sum_{r,h} a_{r,h}^{(m)} R_r \otimes S^h,$$

where R_r is the radius-r sphere kernel on T_3 and S is the shift on $\mathbb{Z}$. At the edge of the tree spectrum,

$$\|R_0\| = 1, \qquad \|R_1\| = 2\sqrt{2}, \qquad \|R_2\| = 5, \qquad \|R_{r+1}\| = 2\sqrt{2}\,\|R_r\| - 2\,\|R_{r-1}\|.$$

The parity identity $r + h \equiv m \pmod 2$ makes the triangle bound sharp at a vertical frequency 0 or π. Exact enumeration gives

$$\|C_{12}\|_{2\to 2} = 3110764 + 2209048\sqrt{2},$$

and hence

$$\|C_{12}\|_{2\to 2}^{1/12} \le 3.683286.$$

Similarly, the same finite computation gives

$$\|C_{13}\|_{2\to 2}^{1/13} \le 3.669978, \qquad \|C_{14}\|_{2\to 2}^{1/14} \le 3.657847.$$

Quotienting out the tree labels in the exact enumeration reaches one further step and gives

$$\|C_{15}\|_{2\to 2} = 134618428 + 95108784\sqrt{2}, \qquad \|C_{15}\|_{2\to 2}^{1/15} \le 3.647512.$$

These improve the nonbacktracking exponent $\beta(5, 2\sqrt{2}+2) \approx 3.766408$, but are still much too large to combine with the vertical-branching p_c-upper bounds above.

Remark 7.79 (A finite-type interval bound for $T_3\square\mathbb{Z}$)**.** The one- and two-step estimates above keep only one vertical cluster after each tree move. A slightly stronger oriented lower process keeps one contiguous interval in each reached fiber. Fix $M \ge 1$. Type $m \in \{1, \dots, M\}$ means that the current fiber contains a reachable interval of length m, with $m = M$ meaning length at least M. From a parent interval of length m, expose the horizontal edges from its m sites to a fixed child fiber and then expose the vertical edges in that child fiber. Retain the capped length of the longest contiguous reached interval as the child type, or retain no child if there is no open horizontal seed. If $P_{m\ell}^{(M)}(p)$ is the single-child transition probability, then the mean matrix of this finite-type Galton-Watson lower process on $T_3\square\mathbb{Z}$ is

$$A_{m\ell}^{(M)}(p) = 2P_{m\ell}^{(M)}(p).$$

Thus $\rho(A^{(M)}(p)) > 1$ implies $p \ge p_{\mathrm{c}}(T_3\square\mathbb{Z})$, and by continuity a strict inequality at p_0 implies $p_{\mathrm{c}}(T_3\square\mathbb{Z}) < p_0$.

For $M = 6$ and $p_0 = 61/200$, exact enumeration gives a rational Perron-Frobenius certificate. With

$$v = (6271, 11456, 15785, 19374, 22333, 24781),$$

one has

$$\min_{1\le m\le 6} \frac{(A^{(6)}(p_0)v)_m}{v_m} > 1.00714.$$

Consequently

$$p_{\mathrm{c}}(T_3\square\mathbb{Z}) < \frac{61}{200}.$$

The finite calculation only inspects horizontal seed subsets and vertical edges in the window $[-(M-1), m-1+(M-1)]$, since reaching the boundary already certifies capped type M. It is reproducible with `scripts/interval_branching_bound.py`. This improves the available p_c-upper bound in the numerical route, but it would still need an operator estimate

$$\gamma_{\mathrm{SAW},2}(T_3\square\mathbb{Z}) < \frac{200}{61} \approx 3.27869,$$

whereas the current finite-length bound is 3.647512.

Remark 7.80 (A decorated-strip bound for $T_3\square\mathbb{Z}$)**.** Yamamoto's characterization of the critical point of $T_d\square\mathbb{Z}$ by the fixed-direction connection exponent gives another way to improve the p_c-upper side [Yam18]. Let x_n be a vertex of T_3 at distance n from a root o, and define

$$\alpha(p) = \lim_{n\to\infty} \mathbb{P}_p\big((o,0) \leftrightarrow (x_n,0)\big)^{1/n}.$$

For $T_3\square\mathbb{Z}$, Yamamoto's criterion says that $\alpha(p) > 1/2$ implies $p > p_c(T_3\square\mathbb{Z})$.

Consider the following finite decorated strip inside $T_3\square\mathbb{Z}$. Take the geodesic from o to x_n, keep the twelve heights $\{-5,\dots,6\}$, and attach at each internal geodesic fiber its one off-geodesic side fiber, again restricted to these twelve heights. After integrating out the thirty-four edges in one decorated column, the transfer matrix on the $2^{12}-1 = 4095$ non-empty subsets of the geodesic fiber has entries in $\mathbb{Q}$ when $p \in \mathbb{Q}$. At

$$p_0 = \frac{5863}{20000},$$

exact integer arithmetic gives a Perron-Frobenius certificate

$$\min_A \frac{2(M(p_0)v)_A}{v_A} > 1.000320$$

for a positive integer vector $v \in \mathbb{Z}^{4095}$. Hence $\rho(M(p_0)) > 1/2$, so the restricted-strip connection probability already has exponential rate larger than $1/2$ for connection to at least one of the twelve terminal heights. Since this is a finite height set, FKG and the event that the bounded vertical segment from that terminal height to height 0 is open show that the same lower exponential rate applies to the fixed-height connection defining α. Therefore

$$p_c(T_3\square\mathbb{Z}) < \frac{5863}{20000}.$$

The certificate is reproducible with `scripts/decorated_strip_fixedp_certificate.py`. This is still not enough for the current SAW criterion: it would require

$$\gamma_{\mathrm{SAW},2}(T_3\square\mathbb{Z}) < \frac{20000}{5863} \approx 3.41122,$$

while the best finite-length certificate above is 3.647512.

Remark 7.81 (Further diagnostics for $T_3\square\mathbb{Z}$)**.** Several stronger finite computations were tested after the decorated-strip bound. They are useful mostly because they identify which natural numerical routes would not by themselves reprove Proposition 7.76.

First, one may improve the interval branching process by retaining an arbitrary finite reached subset of a vertical fiber rather than only one interval. With type width W, the type is a non-empty subset of $\{0,\dots,W\}$ containing 0, normalized by vertical translation. A child type is chosen as a reached subset of span at most W. This one-step finite-subset lower process gives substantially better non-oriented p_c-upper diagnostics than the one-interval process:

W	7	8	9	10
threshold diagnostic	0.2963779	$(0.2961, 0.2962)$	≈ 0.2960	$> 0.29545.$

In particular, already at the older height-ten decorated-strip point $p = 5909/20000$, the width 10 ratio is still only 0.9975811. Thus this retention rule is close to, but does not beat, the current height-twelve decorated-strip certificate.

Second, depth-two tree-block explorations were tested. A naive Monte Carlo version looked promising, but it exposed vertical edges in both the incoming and outgoing boundary fibers of a block; iterating such blocks would reuse boundary-fiber randomness. Valid regenerative variants, where the state records either already-expanded sets or raw boundary seed sets, are much weaker. For type width 7 and margin 2, their ratios at $p = 0.296$ are about 0.956, and they cross one only near $p = 0.301$.

Finally, on the operator side, local finite-memory self-avoidance was tested. A memory-m walk is forbidden to step onto any of its previous m vertices and hence contains all self-avoiding walks. Row-sum growth is far too large: even memory 10 has growth 3.828133. Applying the same spherical ℓ^2 endpoint evaluation used in Remark 7.78 is stronger, but still not enough. For instance,

(m,n)	$(4,16)$	$(6,16)$	$(8,14)$
$\|D_n^{(m)}\|_{2\to2}^{1/n}$	3.699474	3.664311	3.666224.

These values remain above the exact finite-length SAW certificate 3.647512, and far above the current target

$$\frac{20000}{5863} \approx 3.41122.$$

Thus these numerical routes would still require either a genuinely asymptotic L^2 self-avoidance estimate or the direct $p_{2\to2}$ argument supplied by Proposition 7.76.

Corollary 7.82 (A lattice-factor family)**.** *Let $r \geq 1$. If*

$$\sqrt{d-1} \geq 2 + \sqrt{2r},$$

then

$$p_c(T_d \square \mathbb{Z}^r) < p_{2\to2}(T_d \square \mathbb{Z}^r).$$

Proof. Put $q = d - 1$ and $s = \sqrt{q}$. For $H = \mathbb{Z}^r$, we have $\Delta = 2r$ and $\|A_H\|_{2\to2} = 2r$. The nonbacktracking criterion therefore becomes

$$\beta(d + 2r, 2s + 2r) < s^2.$$

Since

$$2s + 2r > 2\sqrt{s^2 + 2r} = 2\sqrt{d + 2r - 1},$$

the second branch of β applies, and the condition is

$$s + r + \sqrt{r(2s + r - 2)} < s^2.$$

Let

$$F_r(s) = s^2 - s - r - \sqrt{r(2s + r - 2)}.$$

For $s \geq 2 + \sqrt{2r}$,

$$F_r'(s) = 2s - 1 - \frac{\sqrt{r}}{\sqrt{2s + r - 2}} > 0,$$

and

$$F_r(2 + \sqrt{2r}) = 2 + 2\sqrt{2r} > 0.$$

Thus $F_r(s) > 0$, and Proposition 7.14 applies. □

Remark 7.83. Corollary 7.82 is a deliberately crude family statement. It is useful here because it shows that the nonbacktracking SAW route handles tree-times-lattice products in a uniform high-branching regime. For $r = 1$, Proposition 7.26 is sharper because it also uses a line-specific upper bound on p_c. The real test for this route remains the low-branching regime, where the threshold above is far from sharp.

8 A finite-set escape abstraction

The following abstraction records a possible way to prove (FE). It is intentionally formulated independently of hyperbolicity or planarity.

Definition 8.1 (Kernel leakage profile)**.** Let Q be a positive symmetric kernel on V satisfying

$$\sup_x \sum_y Q(x,y) \le 1.$$

For a finite non-empty set $K \subset V$, define

$$\mathcal{M}_Q(K) = \frac{1}{|K|} \sum_{x,y \in K} Q(x,y).$$

For $\delta > 0$, say that K is δ-leaky for Q if $\mathcal{M}_Q(K) \le \delta$.

Proposition 8.2 (Uniform leakage implies finite-set escape)**.** *Let $p_n \uparrow p_c$. Suppose that for every $\delta > 0$ there is n_0 such that every finite non-empty $K \subset V$ is δ-leaky for Q_{p_n} whenever $n \ge n_0$. Then finite-set escape holds along p_n:*

$$\lim_{n\to\infty} \sup_K \mathcal{M}_{Q_{p_n}}(K) = 0.$$

Proof. This is just the definition of the supremum. Given $\delta > 0$, the assumption says $\mathcal{M}_{Q_{p_n}}(K) \le \delta$ for every finite non-empty K and every sufficiently large n. Taking the supremum over K gives $\sup_K \mathcal{M}_{Q_{p_n}}(K) \le \delta$, and δ was arbitrary. □

The proposition is tautological. Its value is conceptual: to solve Conjecture 3.1 via Theorem 4.1, one must prove uniform leakage for Q_p near p_c. Known proofs obtain such leakage from special geometry. The desired proof would derive it from nonamenability.

8.1 A separator version

The next criterion is also elementary, but it is closer to known geometric proofs. It says that if every finite set can be decomposed into many pieces whose Q_p-interaction is mostly external, then finite-set escape follows.

Definition 8.3 (Scattering partition)**.** Let Q be as above. A partition $K = K_1 \cup \cdots \cup K_m$ of a finite set is (a,b)-scattering for Q if

$$\frac{1}{|K|} \sum_{i=1}^{m} \sum_{x,y \in K_i} Q(x,y) \le a$$

and

$$\frac{1}{|K|} \sum_{i \ne j} \sum_{x \in K_i, y \in K_j} Q(x,y) \le b.$$

Proposition 8.4 (Scattering criterion)**.** *If, for every $\delta > 0$, all sufficiently near-critical Q_p admit a (δ,δ)-scattering partition of every finite $K \subset V$, then finite-set escape holds.*

Proof. For such a partition,

$$\mathcal{M}_{Q_p}(K) = \frac{1}{|K|}\sum_i \sum_{x,y\in K_i} Q_p(x,y) + \frac{1}{|K|}\sum_{i\neq j}\sum_{x\in K_i, y\in K_j} Q_p(x,y) \le 2\delta.$$

Since δ is arbitrary, the supremum over finite K tends to zero. □

In the hyperbolic proof, the role of the partition is played by large-scale half-space decompositions: most vertices see most other vertices far out in one or two separated directions, and subcritical connectivity to such distant regions is small after normalization. A possible path to a new theorem is to identify a weaker separator hypothesis implied by nonamenability alone, or by a new group-theoretic condition not covered by existing results.

8.2 A separator-convolution version

The preceding scattering criterion is intentionally terse. The next version records the percolation input that a geometric separator can exploit. It uses BK to turn cross-piece connections into a two-point convolution through separator sets.

Let

$$K = K_0 \sqcup K_1 \sqcup \cdots \sqcup K_m$$

be a buffered partition of a finite set, where K_0 is the part we do not try to separate. For $i, j \ge 1$, $i \neq j$, call a set $W_{ij} \subseteq V$ a proper path separator from K_i to K_j if

$$W_{ij} \cap (K_i \cup K_j) = \varnothing$$

and every graph path from a vertex of K_i to a vertex of K_j visits W_{ij}. Define, for $i \ge 1$,

$$U_i^p(z) = \sum_{x\in K_i} \tau_p(x,z)$$

and

$$\mathcal{C}_p(K_1,\dots,K_m;W) = \frac{1}{\overline{\chi}_p|K|}\sum_{i\neq j}\sum_{z\in W_{ij}} U_i^p(z)U_j^p(z).$$

Also define the normalized buffer mass

$$\mathcal{B}_p(K_0;K) = \frac{2}{\overline{\chi}_p|K|}\sum_{x\in K_0, y\in K} \tau_p(x,y).$$

Lemma 8.5 (Separator convolution controls cross mass)**.** *With the notation above, the cross mass between the non-buffer pieces satisfies*

$$\frac{1}{\overline{\chi}_p|K|}\sum_{\substack{i\neq j\\ i,j\ge 1}}\sum_{x\in K_i, y\in K_j} \tau_p(x,y) \le \mathcal{C}_p(K_1,\dots,K_m;W).$$

Proof. Fix $i, j \ge 1$, $i \neq j$, $x \in K_i$, and $y \in K_j$. If $x \leftrightarrow y$, then some open self-avoiding path from x to y visits W_{ij}. Let z be the first vertex of W_{ij} on this path. Since W_{ij} is disjoint from $K_i \cup K_j$, z is not an endpoint. The two path segments from x to z and from z to y are edge-disjoint. Thus the event $\{x \leftrightarrow y\}$ is contained in

$$\bigcup_{z\in W_{ij}} \big(\{x\leftrightarrow z\}\circ\{z\leftrightarrow y\}\big),$$

where $\circ$ denotes disjoint occurrence. The BK inequality gives

$$\tau_p(x,y) \le \sum_{z \in W_{ij}} \tau_p(x,z)\tau_p(z,y).$$

Summing over x, y, i, j, and using symmetry of τ_p, gives the displayed bound. □

Proposition 8.6 (Separator-convolution criterion)**.** *Suppose that for every $\delta > 0$ and all p sufficiently close to p_c, every finite non-empty $K \subset V$ admits a buffered partition*

$$K = K_0 \sqcup K_1 \sqcup \cdots \sqcup K_m$$

and proper path separators W_{ij}, $i, j \ge 1$, $i \ne j$, such that

$$\mathcal{B}_p(K_0; K) \le \delta,$$

$$\frac{1}{\overline{\chi}_p |K|} \sum_{i=1}^{m} \sum_{x,y \in K_i} \tau_p(x,y) \le \delta$$

and

$$\mathcal{C}_p(K_1, \dots, K_m; W) \le \delta.$$

Then finite-set escape holds.

Proof. The mass of pairs with at least one endpoint in K_0 is bounded above by $\mathcal{B}_p(K_0; K) \le \delta$. By Lemma 8.5, the normalized cross-piece mass among the non-buffer pieces is at most δ. The normalized same-piece mass is at most δ by assumption, so

$$\frac{1}{|K|} \sum_{x,y \in K} Q_p(x,y) \le 3\delta$$

for every finite non-empty K, once p is sufficiently close to p_c. Since δ is arbitrary, the supremum over K tends to zero. □

Remark 8.7. This criterion is still conditional, but it separates the two tasks in a way that mirrors geometric proofs. The partition controls local clusters of trapped mass, the buffer accounts for vertices lying in the separating interface, and the separator-convolution bound is where hyperbolicity, tilted mass transport, or a future nonamenability mechanism would enter. Requiring proper separators avoids the degenerate singleton partition in which each endpoint separates itself from every other endpoint. A plain small-cardinality separator is not enough: the relevant quantity is the two-point weight $U_i^p(z) U_j^p(z)$, normalized by $\overline{\chi}_p |K|$.

Definition 8.8 (Local separator-convolution property)**.** A bounded-degree graph G has the local separator-convolution property near p_c if for every $\delta > 0$ there are $R < \infty$ and $p_0 < p_c$ such that, for every $p \in [p_0, p_c)$ and every finite non-empty $K \subset V$, there is a buffered partition

$$K = K_0 \sqcup K_1 \sqcup \cdots \sqcup K_m$$

and proper path separators W_{ij}, $i, j \ge 1$, $i \ne j$, such that

$$\operatorname{diam}(K_i) \le R \qquad (1 \le i \le m),$$

$$\mathcal{B}_p(K_0; K) \le \delta, \qquad \mathcal{C}_p(K_1, \dots, K_m; W) \le \delta.$$

Proposition 8.9 (Local separator-convolution criterion)**.** *Let G be connected, locally finite, and of bounded degree. Suppose that*

$$\overline{\chi}_p \to \infty \qquad \text{as } p \uparrow p_c$$

and that G has the local separator-convolution property near p_c. Then finite-set escape holds.

Proof. Let Δ be a degree bound and fix $\delta > 0$. Let R and p_0 be supplied by Definition 8.8. Put

$$V_R = 1 + \Delta \sum_{\ell=0}^{R-1} (\Delta - 1)^\ell,$$

with the evident interpretation when $\Delta = 1$. Every non-buffer piece has at most V_R vertices. Since $\tau_p \le 1$,

$$\sum_{i=1}^{m} \sum_{x,y \in K_i} \tau_p(x,y) \le \sum_{i=1}^{m} |K_i|^2 \le V_R |K|.$$

Increasing p_0 if necessary, the divergence of $\overline{\chi}_p$ gives

$$\frac{V_R}{\overline{\chi}_p} \le \delta \qquad (p \in [p_0, p_c)).$$

For such p, the buffer mass and separator-convolution mass are at most δ, and the normalized same-piece mass is at most δ. The proof of Proposition 8.6 gives

$$\frac{1}{|K|} \sum_{x,y \in K} Q_p(x,y) \le 3\delta$$

uniformly over finite K. Since δ is arbitrary, finite-set escape follows. □

Theorem 8.10 (Local separator intermediate theorem)**.** *Let G be connected, locally finite, quasi-transitive, nonamenable, and of bounded degree. Assume:*

(i) G has the local separator-convolution property near p_c;

(ii) $\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty$.

Then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For quasi-transitive graphs, $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Proposition 8.9 gives finite-set escape. The mean-field assumption is (MF), so Theorem 4.1 applies. □

Remark 8.11. This local version removes the same-piece tautology from the abstract separator criterion. Once the pieces have uniformly bounded diameter, their internal contribution is killed by the divergence of $\overline{\chi}_p$. The remaining nontrivial work is to find a small buffer and to control the proper separator-convolution term.

Definition 8.12 (Two-replica finite-set profile)**.** For $p < p_c$ and finite non-empty $K \subset V$, define

$$\mathcal{R}_p(K) = \frac{1}{\overline{\chi}_p |K|} \sum_{z \in V} \left(\sum_{x \in K} \tau_p(x,z) \right)^2 = \frac{\|T_p \mathbf{1}_K\|_2^2}{\overline{\chi}_p |K|}.$$

Let

$$\mathcal{R}_p^* = \sup_{K \subset V \text{ finite non-empty}} \mathcal{R}_p(K).$$

Proposition 8.13 (Finite-set escape is sublinear two-replica escape)**.** *Let Q be a positive symmetric kernel on V satisfying $\sup_x \sum_y Q(x,y) \le 1$. Define*

$$\eta(Q) = \sup_{K \subset V \text{ finite non-empty}} \frac{\langle \mathbf{1}_K, Q\mathbf{1}_K \rangle}{|K|}$$

and

$$\rho(Q) = \sup_{K \subset V \text{ finite non-empty}} \frac{\|Q\mathbf{1}_K\|_2^2}{|K|}.$$

Then

$$\eta(Q)^2 \le \rho(Q) \le \sqrt{6\eta(Q)}.$$

Consequently, for the percolation kernels $Q_p = T_p/\overline{\chi}_p$,

$$\lim_{p \uparrow p_c} \eta(Q_p) = 0 \quad \Longleftrightarrow \quad \lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

Proof. For every finite non-empty K,

$$\frac{\langle \mathbf{1}_K, Q\mathbf{1}_K \rangle}{|K|} \le \frac{\|\mathbf{1}_K\|_2 \, \|Q\mathbf{1}_K\|_2}{|K|} = \left(\frac{\|Q\mathbf{1}_K\|_2^2}{|K|} \right)^{1/2}.$$

Taking suprema gives $\eta(Q)^2 \le \rho(Q)$.

For the converse bound, fix K and put $g = Q\mathbf{1}_K$ and

$$a = \frac{\|g\|_2^2}{|K|}.$$

Since Q has row and column sums at most one, $0 \le g \le 1$ and $\|g\|_1 \le |K|$, so $0 \le a \le 1$. If $a = 0$ there is nothing to prove. Let

$$H = \{y \in V : g(y) \ge a/2\}.$$

This set is finite, because $g \in \ell^1(V)$. Since $g^2 \le g$ on H and $g^2 \le (a/2)g$ on $V \setminus H$,

$$a|K| = \sum_y g(y)^2 \le \sum_{y \in H} g(y) + \frac{a}{2} \sum_{y \notin H} g(y) \le \sum_{y \in H} g(y) + \frac{a}{2}|K|.$$

Thus

$$\sum_{y \in H} g(y) \ge \frac{a}{2}|K|.$$

Also $|H| \le 2|K|/a$, because $g \ge a/2$ on H and $\|g\|_1 \le |K|$. Put $U = K \cup H$. Then

$$\begin{aligned} \eta(Q) &\ge \frac{\langle \mathbf{1}_U, Q\mathbf{1}_U \rangle}{|U|} \ge \frac{\sum_{y \in H} \sum_{x \in K} Q(y,x)}{|K| + |H|} \\ &= \frac{\sum_{y \in H} g(y)}{|K| + |H|} \ge \frac{(a/2)|K|}{|K|(1 + 2/a)} \ge \frac{a^2}{6}. \end{aligned}$$

Since this holds for every K, $\rho(Q) \le \sqrt{6\eta(Q)}$.

For $Q = Q_p$,

$$\rho(Q_p) = \sup_K \frac{\|Q_p\mathbf{1}_K\|_2^2}{|K|} = \frac{\mathcal{R}_p^*}{\overline{\chi}_p}.$$

The displayed equivalence follows from the two inequalities. □

Corollary 8.14 (Finite-set escape is stable under squaring)**.** *Let Q be as in Proposition 8.13, and let*

$$Q^2(x,y) = \sum_{z \in V} Q(x,z)Q(z,y).$$

Then Q^2 is again positive, symmetric, and has row sums at most one, and

$$\eta(Q)^2 \le \eta(Q^2) \le \sqrt{6\eta(Q)}.$$

Consequently, for any family Q_α of such kernels,

$$\eta(Q_\alpha) \to 0 \quad \Longleftrightarrow \quad \eta(Q_\alpha^2) \to 0.$$

For $Q_p = T_p/\overline{\chi}_p$,

$$\eta(Q_p^2) = \frac{\mathcal{R}_p^*}{\overline{\chi}_p}.$$

Proof. The row-sum bound follows from Tonelli's theorem:

$$\sum_y Q^2(x,y) = \sum_z Q(x,z) \sum_y Q(z,y) \le \sum_z Q(x,z) \le 1.$$

Symmetry and positivity are immediate. Moreover, for every finite K,

$$\left\langle \mathbf{1}_K, Q^2 \mathbf{1}_K \right\rangle = \langle Q\mathbf{1}_K, Q\mathbf{1}_K \rangle = \|Q\mathbf{1}_K\|_2^2 .$$

Thus $\eta(Q^2) = \rho(Q)$, and the two-sided estimate is exactly Proposition 8.13. The equivalence for a family follows from this estimate. The final identity is the computation already used in the proof of Proposition 8.13. □

Remark 8.15. The second half of the proof is constructive. If $\|Q\mathbf{1}_K\|_2^2/|K| = a > 0$, then the level set

$$H = \{Q\mathbf{1}_K \ge a/2\}$$

has $|H| \le 2|K|/a$, and $U = K \cup H$ satisfies

$$\frac{\langle \mathbf{1}_U, Q\mathbf{1}_U \rangle}{|U|} \ge \frac{a^2}{6}.$$

Thus large two-replica mass cannot hide only as an external L^2 effect: one thresholding step converts it into an ordinary finite-set trapping witness. Equivalently, the two-replica obstruction is precisely finite-set trapping for the squared kernel Q^2.

Proposition 8.16 (Square-kernel trapping is long range)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. For $R < \infty$, let*

$$V_R = \sup_{x \in V} |B(x,R)| < \infty.$$

Then, for $i = 1, 2$,

$$\sup_{K \subset V \text{ finite non-empty}} \frac{1}{|K|} \sum_{\substack{x,y \in K \\ d(x,y) \le R}} Q_p^i(x,y) \le \frac{V_R}{\overline{\chi}_p},$$

where $Q_p^1 = Q_p$ and Q_p^2 denotes the squared kernel. Consequently, if $\eta(Q_{p_n}^2) \ge \epsilon > 0$ along $p_n \uparrow p_c$, then there are finite non-empty $K_n \subset V$ such that, for every fixed $R < \infty$,

$$\liminf_{n \to \infty} \frac{1}{|K_n|} \sum_{\substack{x,y \in K_n \\ d(x,y) > R}} Q_{p_n}^2(x,y) \ge \frac{\epsilon}{2}.$$

Proof. For $i = 1$, $Q_p(x, y) = \tau_p(x, y)/\overline{\chi}_p \le 1/\overline{\chi}_p$. For the squared kernel,

$$Q_p^2(x, y) = \frac{1}{\overline{\chi}_p^2} \sum_{z \in V} \tau_p(x, z)\tau_p(z, y) \le \frac{1}{\overline{\chi}_p^2} \sum_{z \in V} \tau_p(x, z) \le \frac{1}{\overline{\chi}_p}.$$

Thus, for each fixed x, the contribution from $y \in K \cap B(x, R)$ is at most $V_R/\overline{\chi}_p$. Summing over $x \in K$ and dividing by $|K|$ proves the displayed local bound.

If $\eta(Q_{p_n}^2) \ge \epsilon$, choose K_n so that

$$\frac{1}{|K_n|} \sum_{x,y \in K_n} Q_{p_n}^2(x, y) \ge \frac{\epsilon}{2}.$$

Subtracting the local bound with fixed R, and using $\overline{\chi}_{p_n} \to \infty$, gives the claimed long-range lower bound. □

Proposition 8.17 (Long-range square trapping has a far root leg)**.** *In the setting of Proposition 8.16, suppose that $\eta(Q_{p_n}^2) \ge \epsilon > 0$ along $p_n \uparrow p_c$. Then there are finite non-empty $K_n \subset V$ such that, for every fixed $R < \infty$,*

$$\liminf_{n \to \infty} \frac{1}{|K_n|} \sum_{z \in V} \left(\sum_{\substack{x \in K_n \\ d(x,z) > R}} Q_{p_n}(x, z) \right) \left(\sum_{y \in K_n} Q_{p_n}(z, y) \right) \ge \frac{\epsilon}{4}.$$

Proof. Choose K_n as in the proof of Proposition 8.16, so that

$$\frac{1}{|K_n|} \sum_{x,y \in K_n} Q_{p_n}^2(x, y) \ge \frac{\epsilon}{2}.$$

The conclusion of Proposition 8.16 gives, for every fixed R,

$$\liminf_{n \to \infty} \frac{1}{|K_n|} \sum_{\substack{x,y \in K_n \\ d(x,y) > 2R}} Q_{p_n}^2(x, y) \ge \frac{\epsilon}{2}.$$

Now fix n and write $Q = Q_{p_n}$, $K = K_n$. Expanding the squared kernel and using the triangle inequality,

$$\begin{aligned} &\sum_{\substack{x,y \in K \\ d(x,y) > 2R}} Q^2(x, y) \\ &= \sum_{z \in V} \sum_{x,y \in K} \mathbf{1}\{d(x, y) > 2R\} Q(x, z) Q(z, y) \\ &\le 2 \sum_{z \in V} \left(\sum_{\substack{x \in K \\ d(x,z) > R}} Q(x, z) \right) \left(\sum_{y \in K} Q(z, y) \right). \end{aligned}$$

Indeed, if $d(x, y) > 2R$, then at least one of $d(x, z) > R$ and $d(y, z) > R$ holds. Divide by $|K_n|$, take lim inf, and use the previous display. □

Corollary 8.18 (Far-root square mass remains after deleting bounded neighborhoods)**.** *Under the assumptions of Proposition 8.17, the sets K_n may be chosen so that, for every fixed $R < \infty$,*

$$\liminf_{n\to\infty} \frac{1}{|K_n|} \sum_{z\in V} \left(\sum_{\substack{x\in K_n \\ d(x,z)>R}} Q_{p_n}(x,z) \right)^2 \geq \frac{\epsilon^2}{16}.$$

Equivalently, if

$$\mathcal{R}_p^{>R}(K) = \frac{1}{\overline{\chi}_p |K|} \sum_{z\in V} \left(\sum_{\substack{x\in K \\ d(x,z)>R}} \tau_p(x,z) \right)^2,$$

then

$$\liminf_{n\to\infty} \frac{\mathcal{R}_{p_n}^{>R}(K_n)}{\overline{\chi}_{p_n}} \geq \frac{\epsilon^2}{16}.$$

Proof. Use the sets K_n supplied by Proposition 8.17. Fix R, and write

$$u_n(z) = \sum_{y\in K_n} Q_{p_n}(z,y), \qquad u_{n,R}(z) = \sum_{\substack{x\in K_n \\ d(x,z)>R}} Q_{p_n}(x,z).$$

The proposition gives

$$\liminf_{n\to\infty} \frac{\langle u_{n,R}, u_n \rangle}{|K_n|} \geq \frac{\epsilon}{4}.$$

On the other hand,

$$\|u_n\|_2^2 = \langle \mathbf{1}_{K_n}, Q_{p_n}^2 \mathbf{1}_{K_n} \rangle \leq |K_n|,$$

because $Q_{p_n}^2$ has row sums at most one. Cauchy–Schwarz therefore gives

$$\frac{\langle u_{n,R}, u_n \rangle}{|K_n|} \leq \left(\frac{\|u_{n,R}\|_2^2}{|K_n|} \right)^{1/2}.$$

Taking lim inf proves the first display. The equivalence with $\mathcal{R}_p^{>R}$ follows by substituting $Q_p = \overline{\chi}_p^{-1} T_p$. □

Proposition 8.19 (Far-leg square criterion)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. For $R < \infty$, define*

$$\Theta_p(R) = \sup_{K\subset V \text{ finite non-empty}} \frac{\mathcal{R}_p^{>R}(K)}{\overline{\chi}_p}.$$

Then

$$\lim_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0$$

if and only if for every $\delta > 0$ there are $R < \infty$ and $p_0 < p_c$ such that

$$\Theta_p(R) \leq \delta \qquad (p \in [p_0, p_c)).$$

Proof. The forward implication is immediate from $\mathcal{R}_p^{>R}(K) \le \mathcal{R}_p(K)$.

For the converse, fix R, $p < p_c$, and a finite non-empty K. Write

$$Q_p \mathbf{1}_K = u_{\le R} + u_{>R},$$

where

$$u_{\le R}(z) = \sum_{\substack{x \in K \\ d(x,z) \le R}} Q_p(x,z), \qquad u_{>R}(z) = \sum_{\substack{x \in K \\ d(x,z) > R}} Q_p(x,z).$$

Let $V_R = \sup_x |B(x,R)|$. Since $Q_p(x,z) \le \overline{\chi}_p^{-1}$, the vector $u_{\le R}$ is supported on $\{z : d(z,K) \le R\}$, which has size at most $V_R|K|$, and satisfies $u_{\le R}(z) \le V_R/\overline{\chi}_p$. Hence

$$\frac{\|u_{\le R}\|_2^2}{|K|} \le \frac{V_R^3}{\overline{\chi}_p^2}.$$

Also

$$\frac{\|u_{>R}\|_2^2}{|K|} = \frac{\mathcal{R}_p^{>R}(K)}{\overline{\chi}_p} \le \Theta_p(R).$$

Therefore

$$\frac{\mathcal{R}_p(K)}{\overline{\chi}_p} = \frac{\|Q_p \mathbf{1}_K\|_2^2}{|K|} \le 2\frac{V_R^3}{\overline{\chi}_p^2} + 2\Theta_p(R).$$

If the displayed far-leg condition holds, first choose R, p_0 so that $\Theta_p(R) \le \delta/4$ for $p \in [p_0, p_c)$, and then increase p_0 so that $2V_R^3/\overline{\chi}_p^2 \le \delta/2$ on the same interval. Taking the supremum over K gives $\mathcal{R}_p^*/\overline{\chi}_p \le \delta$ near p_c. □

Corollary 8.20 (Tail-limit form of far-leg tightness)**.** *Under the hypotheses and notation of Proposition 8.19,*

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0$$

if and only if

$$\lim_{R \to \infty} \limsup_{p \uparrow p_c} \Theta_p(R) = 0.$$

Proof. For each fixed p, the function $R \mapsto \Theta_p(R)$ is non-increasing, because the summands defining $\mathcal{R}_p^{>R}(K)$ are nonnegative and the set $\{x \in K : d(x,z) > R\}$ decreases with R. Thus

$$L(R) = \limsup_{p \uparrow p_c} \Theta_p(R)$$

is non-increasing in R.

If $\mathcal{R}_p^*/\overline{\chi}_p \to 0$, the far-leg condition in Proposition 8.19 implies that for every $\delta > 0$ there is $R < \infty$ with $L(R) \le \delta$. Hence $\lim_{R\to\infty} L(R) = 0$.

Conversely, if $\lim_{R\to\infty} L(R) = 0$, then for every $\delta > 0$ there is $R < \infty$ with $L(R) < \delta$. By the definition of the limsup in p, there is $p_0 < p_c$ such that $\Theta_p(R) \le \delta$ for all $p \in [p_0, p_c)$. Proposition 8.19 gives the claim. □

Definition 8.21 (Annular square profiles)**.** For $I \subseteq [0,\infty)$, define

$$\mathcal{R}_p^I(K) = \frac{1}{\overline{\chi}_p |K|} \sum_{z \in V} \left(\sum_{\substack{x \in K \\ d(x,z) \in I}} \tau_p(x,z) \right)^2$$

and

$$\Theta_p(I) = \sup_{K \subset V \text{ finite non-empty}} \frac{\mathcal{R}_p^I(K)}{\overline{\chi}_p}.$$

Thus $\Theta_p((R,\infty)) = \Theta_p(R)$.

Lemma 8.22 (Annular profile size bounds)**.** *For every $I \subseteq [0,\infty)$, every $p < p_c$, and every finite non-empty $K \subset V$,*

$$\frac{\mathcal{R}_p^I(K)}{\overline{\chi}_p} \le \min\left\{1, \frac{|K|}{\overline{\chi}_p}\right\}.$$

Proof. Put

$$s_z = \sum_{\substack{x \in K \\ d(x,z) \in I}} \tau_p(x,z).$$

Since $\sum_x \tau_p(x,z) \le \overline{\chi}_p$ and $\sum_z s_z \le \overline{\chi}_p |K|$,

$$\sum_z s_z^2 \le \overline{\chi}_p \sum_z s_z \le \overline{\chi}_p^2 |K|.$$

This gives $\mathcal{R}_p^I(K)/\overline{\chi}_p \le 1$. Also, by Cauchy–Schwarz and $\tau_p(x,z)^2 \le \tau_p(x,z)$,

$$s_z^2 \le |K| \sum_{\substack{x \in K \\ d(x,z) \in I}} \tau_p(x,z)^2 \le |K| \sum_{\substack{x \in K \\ d(x,z) \in I}} \tau_p(x,z).$$

After summing over z, this gives

$$\sum_z s_z^2 \le \overline{\chi}_p |K|^2,$$

and hence $\mathcal{R}_p^I(K)/\overline{\chi}_p \le |K|/\overline{\chi}_p$. □

Lemma 8.23 (Weighted annular profiles reduce to level sets)**.** *Let $I \subseteq [0,\infty)$, let $p < p_c$, and let $w : V \to [0,\infty)$ be finitely supported and non-zero. Then*

$$\frac{1}{\overline{\chi}_p^2 \|w\|_1 \|w\|_\infty} \sum_{z \in V} \left(\sum_{\substack{x \in V \\ d(x,z) \in I}} w(x) \tau_p(x,z) \right)^2 \le \Theta_p(I).$$

In particular, if $0 \le w \le 1$, then the factor $\|w\|_\infty$ may be omitted.

Proof. Put $M = \|w\|_\infty$, and for $0 \le t < M$ let

$$K_t = \{x \in V : w(x) > t\}$$

and

$$g_t(z) = \sum_{\substack{x \in K_t \\ d(x,z) \in I}} \tau_p(x,z).$$

Then $w(x) = \int_0^M \mathbf{1}_{\{x \in K_t\}}\, dt$, and hence

$$s_z := \sum_{\substack{x \in V \\ d(x,z) \in I}} w(x) \tau_p(x,z) = \int_0^M g_t(z)\, dt.$$

Since

$$\int_0^M |K_t|\,dt = \|w\|_1\,,$$

Cauchy–Schwarz gives, for every z,

$$s_z^2 \le \|w\|_1 \int_{\{0\le t<M:K_t\neq\varnothing\}} \frac{g_t(z)^2}{|K_t|}\,dt.$$

After summing over z, Definition 8.21 gives

$$\sum_z s_z^2 \le \|w\|_1 \int_0^M \overline{\chi}_p^2 \Theta_p(I)\,dt = \overline{\chi}_p^2 \,\|w\|_1\,\|w\|_\infty\,\Theta_p(I).$$

Dividing by $\overline{\chi}_p^2\,\|w\|_1\,\|w\|_\infty$ proves the claim. □

Lemma 8.24 (Weighted annular mass lower bound)**.** *Let $I \subseteq [0,\infty)$, let $p < p_c$, let $A, Y \subset V$ be finite non-empty sets, and let $w : A \to [0,1]$ satisfy*

$$S := \sum_{a\in A} w(a) > 0.$$

If $m > 0$ and

$$\sum_{\substack{y\in Y\\ d(y,a)\in I}} \tau_p(y,a) \ge m \qquad (a \in A \text{ with } w(a) > 0),$$

then

$$\Theta_p(I) \ge \frac{m^2 S}{\overline{\chi}_p^2 |Y|}.$$

Proof. Extend w by zero off A, and put

$$s_y = \sum_{\substack{a\in A\\ d(y,a)\in I}} w(a)\tau_p(y,a) \qquad (y \in Y).$$

The mass assumption gives

$$\sum_{y\in Y} s_y \ge mS.$$

Thus Cauchy–Schwarz yields

$$\sum_{y\in Y} s_y^2 \ge \frac{m^2 S^2}{|Y|}.$$

Since $0 \le w \le 1$, Lemma 8.23 implies

$$\Theta_p(I) \ge \frac{1}{\overline{\chi}_p^2 S} \sum_{z\in V} \left(\sum_{\substack{a\in A\\ d(a,z)\in I}} w(a)\tau_p(a,z) \right)^2 \ge \frac{m^2 S}{\overline{\chi}_p^2 |Y|}.$$

□

Definition 8.25 (Annular root layers)**.** For $I \subseteq [0,\infty)$, $0 \le \lambda \le 1$, and a finite non-empty $K \subset V$, define

$$Z_p^I(K,\lambda) = \left\{ z \in V : \sum_{\substack{x \in K \\ d(x,z) \in I}} \tau_p(x,z) \ge \lambda \overline{\chi}_p \right\}.$$

Lemma 8.26 (Annular square mass creates root layers)**.** *For every $I \subseteq [0,\infty)$, every $0 \le \lambda \le 1$, every $p < p_c$, and every finite non-empty $K \subset V$,*

$$\frac{\mathcal{R}_p^I(K)}{\overline{\chi}_p} \le \lambda + \frac{|Z_p^I(K,\lambda)|}{|K|}.$$

Consequently, if $\mathcal{R}_p^I(K)/\overline{\chi}_p \ge \sigma > \lambda$, then

$$|Z_p^I(K,\lambda)| \ge (\sigma - \lambda)|K|.$$

Proof. Put

$$s_z = \sum_{\substack{x \in K \\ d(x,z) \in I}} \tau_p(x,z), \qquad Z = Z_p^I(K,\lambda).$$

As above, $s_z \le \overline{\chi}_p$ for every z, and $\sum_z s_z \le \overline{\chi}_p |K|$. On $V \setminus Z$, $s_z^2 \le \lambda \overline{\chi}_p s_z$, while on Z, $s_z^2 \le \overline{\chi}_p^2$. Hence

$$\sum_z s_z^2 \le \lambda \overline{\chi}_p^2 |K| + \overline{\chi}_p^2 |Z|.$$

Dividing by $\overline{\chi}_p^2 |K|$ gives the first claim, and the second follows immediately. □

Proposition 8.27 (Weighted annular far-leg criterion)**.** *Fix $R < \infty$, and let $\{I_j : j \in J\}$ be a countable partition of (R,∞). If $a_j > 0$ and*

$$A = \sum_{j \in J} a_j^{-1} < \infty,$$

then for every $p < p_c$,

$$\Theta_p(R) \le A \sum_{j \in J} a_j \Theta_p(I_j),$$

with the convention that the right-hand side may be infinite.

Proof. Fix a finite non-empty K. Write

$$u_j(z) = \sum_{\substack{x \in K \\ d(x,z) \in I_j}} Q_p(x,z).$$

Then

$$\sum_{\substack{x \in K \\ d(x,z) > R}} Q_p(x,z) = \sum_{j \in J} u_j(z)$$

pointwise. Weighted Cauchy–Schwarz gives

$$\left(\sum_{j \in J} u_j(z) \right)^2 \le \left(\sum_{j \in J} a_j^{-1} \right) \left(\sum_{j \in J} a_j u_j(z)^2 \right) = A \sum_{j \in J} a_j u_j(z)^2.$$

For countable J, apply this first to finite partial sums and then use monotone convergence. Summing over z, dividing by $|K|$, and taking the supremum over K gives the claim. □

Corollary 8.28 (Dyadic annular sufficient condition)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. For $j \geq J$, let $I_j = (2^j, 2^{j+1}]$. Suppose that for some $\alpha > 0$,*

$$\lim_{J\to\infty} \limsup_{p\uparrow p_c} \sum_{j\geq J} 2^{\alpha(j-J)} \Theta_p(I_j) = 0.$$

Then

$$\lim_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

Proof. Apply Proposition 8.27 with $R = 2^J$, $a_j = 2^{\alpha(j-J)}$, and

$$A = \sum_{j\geq J} 2^{-\alpha(j-J)} = \frac{1}{1-2^{-\alpha}}.$$

The displayed assumption implies $\lim_{J\to\infty} \limsup_{p\uparrow p_c} \Theta_p(2^J) = 0$. The tail-limit criterion, Corollary 8.20, gives the claim. □

Proposition 8.29 (Dyadic annular obstruction)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $\epsilon > 0$ such that for every $\alpha > 0$,

$$\liminf_{J\to\infty} \limsup_{p\uparrow p_c} \sum_{j\geq J} 2^{\alpha(j-J)} \Theta_p((2^j, 2^{j+1}]) \geq (1-2^{-\alpha})\epsilon.$$

More concretely, for every $\alpha > 0$, every integer $J \geq 0$, and every $p_0 < p_c$, there are $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$ such that, writing $I_j = (2^j, 2^{j+1}]$,

$$\sum_{j\geq J} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \geq (1-2^{-\alpha})\epsilon.$$

The same p and K may be chosen so that

$$\frac{\mathcal{R}_p^{>2^J}(K)}{\overline{\chi}_p} > 2\epsilon, \qquad |K| > 2\epsilon\overline{\chi}_p.$$

Proof. Let

$$L = \lim_{R\to\infty} \limsup_{p\uparrow p_c} \Theta_p(R).$$

If $L = 0$, Corollary 8.20 gives $\mathcal{R}_p^*/\overline{\chi}_p \to 0$, contradicting the hypothesis. Hence $L > 0$. Choose $\epsilon > 0$ with $2\epsilon < L$.

Fix $\alpha > 0$, an integer $J \geq 0$, and $p_0 < p_c$. Since

$$\limsup_{p\uparrow p_c} \Theta_p(2^J) \geq L > 2\epsilon,$$

there is $p \in [p_0, p_c)$ such that $\Theta_p(2^J) > 2\epsilon$. By the definition of the supremum, there is a finite non-empty $K \subset V$ with

$$\frac{\mathcal{R}_p^{>2^J}(K)}{\overline{\chi}_p} > 2\epsilon.$$

Lemma 8.22, applied to $I = (2^J, \infty)$, gives $|K| > 2\epsilon\overline{\chi}_p$. For $I_j = (2^j, 2^{j+1}]$, the annuli I_j, $j \geq J$, partition $(2^J, \infty)$. Applying the weighted Cauchy–Schwarz estimate from the proof of Proposition 8.27 to this same K, with $a_j = 2^{\alpha(j-J)}$, gives

$$\frac{\mathcal{R}_p^{\geq 2^J}(K)}{\overline{\chi}_p} \leq \frac{1}{1-2^{-\alpha}} \sum_{j \geq J} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p}.$$

The concrete lower bound follows. Since $\mathcal{R}_p^{I_j}(K)/\overline{\chi}_p \leq \Theta_p(I_j)$, the same construction also gives the stated $\liminf_J \limsup_p$ obstruction. □

Proposition 8.30 (Weighted annular root-layer obstruction)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $\epsilon > 0$ such that the following holds. For every $0 < \alpha < \beta$, every integer $J \geq 0$, and every $p_0 < p_c$, there are $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$ with

$$|K| > 2\epsilon\overline{\chi}_p$$

such that, writing $I_j = (2^j, 2^{j+1}]$ and

$$\lambda_j = \frac{\epsilon}{2}(1-2^{-\alpha})(1-2^{\alpha-\beta})2^{-\beta(j-J)},$$

we have

$$\sum_{j \geq J} 2^{\alpha(j-J)} \frac{|Z_p^{I_j}(K, \lambda_j)|}{|K|} \geq \frac{\epsilon}{2}(1-2^{-\alpha}).$$

Proof. Let ϵ be given by Proposition 8.29. Fix $0 < \alpha < \beta$, $J \geq 0$, and $p_0 < p_c$, and apply the concrete part of that proposition with this α. We get $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$ such that $|K| > 2\epsilon\overline{\chi}_p$ and

$$\sum_{j \geq J} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \geq (1-2^{-\alpha})\epsilon.$$

Set

$$c = (1-2^{-\alpha})\epsilon, \qquad \lambda_j = \frac{c}{2}(1-2^{\alpha-\beta})2^{-\beta(j-J)}.$$

Lemma 8.26 gives, for every $j \geq J$,

$$\frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \leq \lambda_j + \frac{|Z_p^{I_j}(K, \lambda_j)|}{|K|}.$$

Multiplying by $2^{\alpha(j-J)}$ and summing over $j \geq J$, we obtain

$$\begin{aligned} c &\leq \sum_{j \geq J} 2^{\alpha(j-J)} \lambda_j + \sum_{j \geq J} 2^{\alpha(j-J)} \frac{|Z_p^{I_j}(K, \lambda_j)|}{|K|} \\ &= \frac{c}{2} + \sum_{j \geq J} 2^{\alpha(j-J)} \frac{|Z_p^{I_j}(K, \lambda_j)|}{|K|}. \end{aligned}$$

This is the claim. □

Definition 8.31 (Annular incidence layers)**.** For $I \subseteq [0,\infty)$, $m \geq 0$, and finite non-empty $K \subset V$, define

$$\mathcal{N}^I(K,m) = \{z \in V : |\{x \in K : d(x,z) \in I\}| \geq m\}.$$

Lemma 8.32 (Root layers have large deterministic incidence)**.** *For every $I \subseteq [0,\infty)$, every $0 \leq \lambda \leq 1$, every $p < p_c$, and every finite non-empty $K \subset V$,*

$$Z_p^I(K,\lambda) \subseteq \mathcal{N}^I(K, \lambda\overline{\chi}_p).$$

Proof. If $z \in Z_p^I(K,\lambda)$, then

$$\lambda\overline{\chi}_p \leq \sum_{\substack{x \in K \\ d(x,z) \in I}} \tau_p(x,z) \leq |\{x \in K : d(x,z) \in I\}|,$$

since $\tau_p(x,z) \leq 1$. □

Proposition 8.33 (Weighted annular incidence obstruction)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $\epsilon > 0$ such that the following holds. For every $0 < \alpha < \beta$, every integer $J \geq 0$, and every $p_0 < p_c$, there are $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$ with

$$|K| > 2\epsilon\overline{\chi}_p$$

such that, writing $I_j = (2^j, 2^{j+1}]$ and

$$m_j = \frac{\epsilon}{2}(1-2^{-\alpha})(1-2^{\alpha-\beta})2^{-\beta(j-J)}\overline{\chi}_p,$$

we have

$$\sum_{j \geq J} 2^{\alpha(j-J)} \frac{|\mathcal{N}^{I_j}(K,m_j)|}{|K|} \geq \frac{\epsilon}{2}(1-2^{-\alpha}).$$

Proof. Apply Proposition 8.30. Its thresholds λ_j satisfy $m_j = \lambda_j\overline{\chi}_p$, and Lemma 8.32 gives

$$Z_p^{I_j}(K,\lambda_j) \subseteq \mathcal{N}^{I_j}(K,m_j) \qquad (j \geq J).$$

The stated lower bound follows from the lower bound in Proposition 8.30. □

Proposition 8.34 (Finite-window high-incidence alternative)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $0 < \epsilon \leq 1$ such that the following holds. Fix $\alpha > 0$, an integer $H \geq 0$, an integer $J \geq 0$, and $p_0 < p_c$. Put $I_j = (2^j, 2^{j+1}]$ and

$$c = (1-2^{-\alpha})\epsilon.$$

There are $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$, with

$$|K| > 2\epsilon\overline{\chi}_p,$$

such that at least one of the following alternatives holds:

$$\sum_{j>J+H} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \geq \frac{c}{2}, \tag{T}$$

or, writing

$$\lambda_j = \frac{c}{4(H+1)} 2^{-\alpha(j-J)} \qquad (J \leq j \leq J+H),$$

we have

$$\sum_{j=J}^{J+H} 2^{\alpha(j-J)} \frac{|\mathcal{N}^{I_j}(K, \lambda_j \overline{\chi}_p)|}{|K|} \geq \frac{c}{4}. \tag{W}$$

Moreover, for every $M < \infty$ there exists $p_1 < p_c$, depending on α, H, M, ϵ, such that if $p_0 \geq p_1$ and (W) *holds, then $\lambda_j \overline{\chi}_p \geq M$ for every $J \leq j \leq J+H$.*

Proof. Let ϵ be the constant from Proposition 8.29, shrunk if necessary so that $\epsilon \leq 1$. Applying the concrete part of that proposition gives $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$, with $|K| > 2\epsilon \overline{\chi}_p$, such that

$$\sum_{j \geq J} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \geq c.$$

If (T) holds, there is nothing to prove. If it does not hold, then

$$\sum_{j=J}^{J+H} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \geq \frac{c}{2}.$$

For $J \leq j \leq J+H$, Lemma 8.26 and Lemma 8.32 give

$$\frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \leq \lambda_j + \frac{|\mathcal{N}^{I_j}(K, \lambda_j \overline{\chi}_p)|}{|K|}.$$

Since

$$\sum_{j=J}^{J+H} 2^{\alpha(j-J)} \lambda_j = \frac{c}{4},$$

the finite-window incidence bound follows.

Finally, for $J \leq j \leq J+H$,

$$\lambda_j \geq \frac{c}{4(H+1)} 2^{-\alpha H}.$$

Since $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, choose $p_1 < p_c$ so that

$$\frac{c}{4(H+1)} 2^{-\alpha H} \overline{\chi}_p \geq M \qquad (p \in [p_1, p_c)).$$

This proves the last assertion. □

Lemma 8.35 (Incidence layer volume bound)**.** *Let G be locally finite. For $I \subseteq [0, \infty)$, put*

$$V(I) = \sup_{x \in V} |\{z \in V : d(x,z) \in I\}|.$$

Then for every finite non-empty $K \subset V$ and every $m > 0$,

$$\frac{|\mathcal{N}^I(K,m)|}{|K|} \le \frac{V(I)}{m},$$

with the convention that the bound is trivial if $V(I) = \infty$.

Proof. Each $z \in \mathcal{N}^I(K,m)$ contributes at least m incidences (x,z) with $x \in K$ and $d(x,z) \in I$. Therefore

$$m|\mathcal{N}^I(K,m)| \le |\{(x,z) \in K \times V : d(x,z) \in I\}| \le |K|V(I).$$

Divide by $m|K|$. □

Proposition 8.36 (Fixed windows cannot carry the obstruction near criticality)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $0 < \epsilon \le 1$ such that the following holds. Fix $\alpha > 0$, integers $J, H \ge 0$, and put

$$c = (1 - 2^{-\alpha})\epsilon, \qquad I_j = (2^j, 2^{j+1}].$$

There is $p_1 < p_c$ such that for every $p_0 \in [p_1, p_c)$ there are $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$, with

$$|K| > 2\epsilon\overline{\chi}_p,$$

such that

$$\sum_{j > J+H} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \ge \frac{c}{2}.$$

Proof. Let ϵ be the constant from Proposition 8.34. Since G has bounded degree, $V(I_j) < \infty$ for $J \le j \le J + H$. Choose $p_1 < p_c$ so that, for every $p \in [p_1, p_c)$,

$$\overline{\chi}_p > \frac{16(H+1)}{c^2} \sum_{j=J}^{J+H} 2^{2\alpha(j-J)} V(I_j).$$

This is possible because $\overline{\chi}_p \to \infty$.

Apply Proposition 8.34 with any $p_0 \in [p_1, p_c)$, and take the resulting p and K. If the tail alternative (T) holds, we are done. Suppose instead that the incidence alternative (W) holds. For its thresholds

$$\lambda_j = \frac{c}{4(H+1)} 2^{-\alpha(j-J)},$$

Lemma 8.35 gives

$$\begin{aligned} \sum_{j=J}^{J+H} 2^{\alpha(j-J)} \frac{|\mathcal{N}^{I_j}(K, \lambda_j \overline{\chi}_p)|}{|K|} &\le \sum_{j=J}^{J+H} 2^{\alpha(j-J)} \frac{V(I_j)}{\lambda_j \overline{\chi}_p} \\ &= \frac{4(H+1)}{c\overline{\chi}_p} \sum_{j=J}^{J+H} 2^{2\alpha(j-J)} V(I_j) < \frac{c}{4}, \end{aligned}$$

contradicting (W). Hence the tail alternative must hold. □

Proposition 8.37 (Single remote annulus obstruction)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $0 < \epsilon \le 1$ such that the following holds. For every $\alpha, \gamma > 0$, every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, and an integer $j > J + H$ such that

$$|K| > 2\epsilon \overline{\chi}_p$$

and, with $I_j = (2^j, 2^{j+1}]$,

$$\frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \ge \frac{\epsilon}{2}(1 - 2^{-\alpha})(1 - 2^{-\gamma}) 2^{-(\alpha+\gamma)(j-J)}.$$

Proof. Let ϵ be the constant from Proposition 8.36. Put

$$c = (1 - 2^{-\alpha})\epsilon.$$

For p_0 sufficiently close to p_c, Proposition 8.36 gives $p \in [p_0, p_c)$ and a finite non-empty $K \subset V$, with $|K| > 2\epsilon\overline{\chi}_p$, such that

$$\sum_{j > J+H} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} \ge \frac{c}{2}.$$

If the desired lower bound failed for every $j > J + H$, then

$$\begin{aligned} \sum_{j > J+H} 2^{\alpha(j-J)} \frac{\mathcal{R}_p^{I_j}(K)}{\overline{\chi}_p} &< \frac{c}{2}(1 - 2^{-\gamma}) \sum_{j > J+H} 2^{-\gamma(j-J)} \\ &\le \frac{c}{2}(1 - 2^{-\gamma}) \sum_{k \ge 1} 2^{-\gamma k} < \frac{c}{2}, \end{aligned}$$

where the inequality $j > J + H$ implies $j - J \ge 1$. This contradiction proves the claim. □

Corollary 8.38 (Single remote annulus has root and incidence layers)**.** *Under the assumptions of Proposition 8.37, there is $0 < \epsilon \le 1$ such that for every $\alpha, \gamma > 0$, every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, and an integer $j > J + H$ such that $|K| > 2\epsilon\overline{\chi}_p$ and, with*

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\alpha})(1 - 2^{-\gamma}) 2^{-(\alpha+\gamma)(j-J)}, \qquad I_j = (2^j, 2^{j+1}],$$

we have

$$|Z_p^{I_j}(K, \sigma_j/2)| \ge \frac{\sigma_j}{2}|K|$$

and

$$|\mathcal{N}^{I_j}(K, \sigma_j \overline{\chi}_p/2)| \ge \frac{\sigma_j}{2}|K|.$$

Proof. Choose p, K, j by Proposition 8.37. Lemma 8.26, applied with $\lambda = \sigma_j/2$, gives the root-layer lower bound. The incidence lower bound follows from Lemma 8.32. □

Corollary 8.39 (Escaping annuli have susceptibility-scale volume)**.** *Under the assumptions of Proposition 8.37, there is $0 < \epsilon \le 1$ such that for every $\alpha, \gamma > 0$, every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$ and an integer $j > J + H$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad V(I_j) = \sup_{x \in V} |\{z \in V : d(x,z) \in I_j\}|,$$

we have

$$V(I_j) \ge \frac{\epsilon^2}{16}(1 - 2^{-\alpha})^2(1 - 2^{-\gamma})^2 2^{-2(\alpha+\gamma)(j-J)} \overline{\chi}_p.$$

Proof. Choose p, K, j by Corollary 8.38, and let

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\alpha})(1 - 2^{-\gamma}) 2^{-(\alpha+\gamma)(j-J)}.$$

That corollary gives

$$\frac{|\mathcal{N}^{I_j}(K, \sigma_j \overline{\chi}_p/2)|}{|K|} \ge \frac{\sigma_j}{2}.$$

Lemma 8.35 gives the opposite bound

$$\frac{|\mathcal{N}^{I_j}(K, \sigma_j \overline{\chi}_p/2)|}{|K|} \le \frac{2V(I_j)}{\sigma_j \overline{\chi}_p}.$$

Thus $V(I_j) \ge \sigma_j^2 \overline{\chi}_p/4$, which is the displayed inequality. □

Corollary 8.40 (Penalized volume obstruction)**.** *Under the assumptions of Proposition 8.37, there is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$ and an integer $j > J + H$ such that*

$$V((2^j, 2^{j+1}]) \ge \frac{\epsilon^2}{16}(1 - 2^{-\eta/2})^4 2^{-2\eta(j-J)} \overline{\chi}_p.$$

Proof. Apply Corollary 8.39 with $\alpha = \gamma = \eta/2$. □

Corollary 8.41 (Logarithmic radius obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$. There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$ and an integer $j > J + H$ such that*

$$V((2^j, 2^{j+1}]) \ge \frac{\epsilon^2}{16}(1 - 2^{-\eta/2})^4 2^{-2\eta(j-J)} \overline{\chi}_p$$

and

$$2^j \ge \frac{1}{4} \log_\Delta \overline{\chi}_p.$$

Proof. Let

$$c = \frac{\epsilon^2}{16}(1 - 2^{-\eta/2})^4,$$

where ϵ is supplied by Corollary 8.40. Since $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, we may choose $p_* < p_c$ sufficiently close to p_c that Corollary 8.40 applies to every $p_0 \in [p_*, p_c)$, and every $p \in [p_*, p_c)$ satisfies

$$c\overline{\chi}_p > \Delta \overline{\chi}_p^{1/2} \left(\frac{1}{4} \log_\Delta \overline{\chi}_p \right)^{2\eta}$$

and $\log_\Delta \overline{\chi}_p > 4$. Fix $p_0 \in [p_*, p_c)$, and choose p, j by Corollary 8.40. Put $L = \log_\Delta \overline{\chi}_p$. If $2^j < L/4$, then

$$V((2^j, 2^{j+1}]) \le \sup_{x \in V} |B(x, 2^{j+1})| \le \Delta^{2^{j+1}+1} < \Delta \overline{\chi}_p^{1/2}$$

uniformly in the center x, while

$$2^{2\eta(j-J)} \le 2^{2\eta j} < \left(\frac{L}{4}\right)^{2\eta}.$$

The volume lower bound would then imply

$$c\overline{\chi}_p \le V((2^j, 2^{j+1}])\, 2^{2\eta(j-J)} < \Delta \overline{\chi}_p^{1/2} \left(\frac{1}{4} \log_\Delta \overline{\chi}_p\right)^{2\eta},$$

contradicting the choice of p_*. Hence $2^j \ge L/4$. □

Corollary 8.42 (Logarithmic backscatter fan)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$. There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z \subset V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)},$$

we have

$$|K| > 2\epsilon \overline{\chi}_p, \qquad 2^j \ge \frac{1}{4} \log_\Delta \overline{\chi}_p,$$

$$|Z| \ge \frac{\sigma_j^2}{4} \overline{\chi}_p, \qquad d(x, z) \in I_j \quad (z \in Z),$$

and every $z \in Z$ satisfies

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y, z) \ge \frac{\sigma_j}{2} \overline{\chi}_p.$$

In particular,

$$|\{y \in K : d(y, z) \in I_j\}| \ge \frac{\sigma_j}{2} \overline{\chi}_p \qquad (z \in Z).$$

Proof. Let ϵ be supplied by Corollary 8.38, and put

$$c = \frac{\epsilon^2}{16}(1 - 2^{-\eta/2})^4.$$

Choose $p_* < p_c$ sufficiently close to p_c that Corollary 8.38, with $\alpha = \gamma = \eta/2$, applies to every $p_0 \in [p_*, p_c)$, and every $p \in [p_*, p_c)$ satisfies

$$c\overline{\chi}_p > \Delta \overline{\chi}_p^{1/2} \left(\frac{1}{4} \log_\Delta \overline{\chi}_p\right)^{2\eta}$$

and $\log_\Delta \overline{\chi}_p > 4$.

Fix $p_0 \in [p_*, p_c)$, and choose p, K, j by Corollary 8.38. Let

$$Z_0 = Z_p^{I_j}(K, \sigma_j/2).$$

That corollary gives

$$|Z_0| \geq \frac{\sigma_j}{2}|K|$$

and, by Lemma 8.32,

$$|\{y \in K : d(y,z) \in I_j\}| \geq \frac{\sigma_j}{2}\overline{\chi}_p \qquad (z \in Z_0).$$

Hence

$$\begin{aligned}\sum_{x\in K} |\{z \in Z_0 : d(x,z) \in I_j\}| &= \sum_{z\in Z_0} |\{x \in K : d(x,z) \in I_j\}| \\ &\geq \frac{\sigma_j}{2}\overline{\chi}_p|Z_0| \geq \frac{\sigma_j^2}{4}\overline{\chi}_p|K|.\end{aligned}$$

Choose $x \in K$ for which the corresponding summand is at least the average, and define

$$Z = \{z \in Z_0 : d(x,z) \in I_j\}.$$

This proves the lower bound on $|Z|$, the annular relation $d(x,z) \in I_j$, and the two displayed lower bounds for each $z \in Z$.

It remains only to force the logarithmic lower bound on 2^j. Since $|Z| \leq V(I_j)$, the preceding lower bound gives

$$V(I_j) \geq c\,2^{-2\eta(j-J)}\overline{\chi}_p.$$

Put $L = \log_\Delta \overline{\chi}_p$. If $2^j < L/4$, then

$$V(I_j) \leq \sup_{v\in V} |B(v, 2^{j+1})| \leq \Delta^{2^{j+1}+1} < \Delta\overline{\chi}_p^{1/2},$$

while

$$2^{2\eta(j-J)} \leq 2^{2\eta j} < \left(\frac{L}{4}\right)^{2\eta}.$$

Thus

$$c\overline{\chi}_p \leq V(I_j)2^{2\eta(j-J)} < \Delta\overline{\chi}_p^{1/2}\left(\frac{1}{4}\log_\Delta \overline{\chi}_p\right)^{2\eta},$$

contradicting the choice of p_*. Therefore $2^j \geq L/4$. □

Corollary 8.43 (Volume-sensitive logarithmic backscatter fan)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B\log_\Delta \omega < 1,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z \subset V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

and

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)},$$

we have

$$|K| > 2\epsilon\overline{\chi}_p, \qquad 2^j > BL,$$

$$|Z| \geq \frac{\sigma_j^2}{4}\overline{\chi}_p, \qquad d(x,z) \in I_j \quad (z \in Z),$$

and every $z \in Z$ *satisfies*

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y,z) \geq \frac{\sigma_j}{2}\overline{\chi}_p.$$

In particular,

$$|\{y \in K : d(y,z) \in I_j\}| \geq \frac{\sigma_j}{2}\overline{\chi}_p \qquad (z \in Z).$$

If $\Delta \geq 3$*, one may take* $\omega = \Delta - 1$*, so the admissible range includes every*

$$B < \frac{1}{2\log_\Delta(\Delta - 1)}.$$

Proof. Let ϵ be supplied by Corollary 8.38, and put

$$c = \frac{\epsilon^2}{16}(1 - 2^{-\eta/2})^4.$$

Choose $p_* < p_c$ sufficiently close to p_c that Corollary 8.38, with $\alpha = \gamma = \eta/2$, applies to every $p_0 \in [p_*, p_c)$, and every $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$, satisfies $BL > 1$ and

$$c\overline{\chi}_p > C_{\mathrm{vol}}\overline{\chi}_p^{2B\log_\Delta \omega}(BL)^{2\eta}.$$

This is possible because $2B\log_\Delta \omega < 1$ and $\overline{\chi}_p \to \infty$.

Fix $p_0 \in [p_*, p_c)$, and choose p, K, j by Corollary 8.38. Let

$$Z_0 = Z_p^{I_j}(K, \sigma_j/2).$$

Exactly as in the proof of Corollary 8.42, choose $x \in K$ and

$$Z = \{z \in Z_0 : d(x,z) \in I_j\}$$

so that

$$|Z| \geq \frac{\sigma_j^2}{4}\overline{\chi}_p, \qquad d(x,z) \in I_j \quad (z \in Z),$$

and every $z \in Z$ has the displayed backscatter and incidence lower bounds.

It remains to prove $2^j > BL$. Suppose instead that $2^j \leq BL$. Since $|Z| \leq V(I_j)$, the volume-growth assumption gives

$$|Z| \leq \sup_{u \in V} |B(u, 2^{j+1})| \leq C_{\mathrm{vol}}\omega^{2^{j+1}} \leq C_{\mathrm{vol}}\overline{\chi}_p^{2B\log_\Delta \omega}.$$

Also $2^{2\eta(j-J)} \leq 2^{2\eta j} \leq (BL)^{2\eta}$. Therefore

$$c\overline{\chi}_p \leq |Z|2^{2\eta(j-J)} \leq C_{\mathrm{vol}}\overline{\chi}_p^{2B\log_\Delta \omega}(BL)^{2\eta},$$

contradicting the choice of p_*. Thus $2^j > BL$.

The final assertion follows from the same tree-volume estimate as in Corollary 8.761. □

Lemma 8.44 (Separated subfan extraction)**.** *Let G be locally finite, let $Z \subset V$ be finite, and let $r \geq 0$ be an integer. Put*

$$V_{2r} = \sup_{v \in V} |B(v, 2r)|.$$

Then there is a set $Z_r \subseteq Z$ such that

$$d(z, z') > 2r \qquad (z, z' \in Z_r,\ z \neq z')$$

and

$$|Z_r| \geq \frac{|Z|}{V_{2r}},$$

with the convention that the assertion is trivial if $V_{2r} = \infty$.

Proof. Let Z_r be maximal subject to the separation condition. Maximality implies that the balls $B(z, 2r)$, $z \in Z_r$, cover Z. Hence

$$|Z| \leq \sum_{z \in Z_r} |B(z, 2r)| \leq |Z_r| V_{2r}.$$

□

Lemma 8.45 (Weighted separated subfan extraction)**.** *Let G be locally finite, let $Z \subset V$ be finite, let $a_z \geq 0$ for $z \in Z$, and let $r \geq 0$ be an integer. Put*

$$V_{2r} = \sup_{v \in V} |B(v, 2r)|.$$

Then there is a set $Z_r \subseteq Z$ such that

$$d(z, z') > 2r \qquad (z, z' \in Z_r,\ z \neq z')$$

and

$$\sum_{z \in Z_r} a_z \geq \frac{1}{V_{2r}} \sum_{z \in Z} a_z,$$

with the convention that the assertion is trivial if $V_{2r} = \infty$.

Proof. Greedily choose vertices as follows. While unchosen vertices remain, choose a remaining vertex z of maximal weight a_z, add it to Z_r, and delete all remaining vertices in $B(z, 2r)$. The chosen vertices are $2r$-separated. If C_z is the set deleted when z is chosen, then $|C_z| \leq V_{2r}$ and $a_y \leq a_z$ for every $y \in C_z$. Thus

$$\sum_{y \in C_z} a_y \leq V_{2r} a_z.$$

Summing over the disjoint deleted clusters gives

$$\sum_{z \in Z} a_z \leq V_{2r} \sum_{z \in Z_r} a_z.$$

□

Corollary 8.46 (Volume-sensitive separated backscatter fan)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$1 - 2\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z_\theta \subset V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

we have

$$|K| > 2\epsilon\overline{\chi}_p, \qquad 2^j > BL,$$

$$|Z_\theta| \geq \frac{\sigma_j^2}{4C_{\mathrm{vol}}}\overline{\chi}_p^{1-2\theta\log_\Delta\omega}, \qquad d(x,z) \in I_j \quad (z \in Z_\theta),$$

$$d(z,z') > 2r_p \qquad (z, z' \in Z_\theta,\ z \neq z'),$$

and every $z \in Z_\theta$ satisfies

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y,z) \geq \frac{\sigma_j}{2}\overline{\chi}_p.$$

Consequently, the balls $B(z, r_p)$, $z \in Z_\theta$, are pairwise disjoint. If $\Delta \geq 3$, one may take $\omega = \Delta - 1$.

Proof. Choose p, K, j, x, Z by Corollary 8.43, and set

$$r_p = \lfloor \theta \log_\Delta \overline{\chi}_p \rfloor.$$

Apply Lemma 8.44 to Z and r_p. The resulting set Z_θ inherits $d(x,z) \in I_j$, the lower bound $2^j > BL$, and the pointwise backscatter lower bound from Z. Moreover,

$$|Z_\theta| \geq \frac{|Z|}{V_{2r_p}}.$$

The volume-growth assumption gives

$$V_{2r_p} \leq C_{\mathrm{vol}}\omega^{2r_p} \leq C_{\mathrm{vol}}\overline{\chi}_p^{2\theta\log_\Delta\omega}.$$

Combining this with

$$|Z| \geq \frac{\sigma_j^2}{4}\overline{\chi}_p$$

from Corollary 8.43 gives the displayed lower bound on $|Z_\theta|$. The separation $d(z,z') > 2r_p$ implies that the balls $B(z, r_p)$ are pairwise disjoint. The final assertion follows from the tree-volume estimate in Corollary 8.761. □

Corollary 8.47 (Separated backscatter fan)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/2$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z_\theta \subset V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

we have

$$|K| > 2\epsilon \overline{\chi}_p, \qquad 2^j \geq \frac{1}{4} \log_\Delta \overline{\chi}_p,$$

$$|Z_\theta| \geq \frac{\sigma_j^2}{4\Delta} \overline{\chi}_p^{1-2\theta}, \qquad d(x, z) \in I_j \quad (z \in Z_\theta),$$

$$d(z, z') > 2r_p \qquad (z, z' \in Z_\theta,\ z \neq z'),$$

and every $z \in Z_\theta$ satisfies

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y, z) \geq \frac{\sigma_j}{2} \overline{\chi}_p.$$

Consequently, the balls $B(z, r_p)$, $z \in Z_\theta$, are pairwise disjoint.

Proof. Choose p, K, j, x, Z by Corollary 8.42, and set

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor.$$

Apply Lemma 8.44 to Z and r_p. If Z_θ is the resulting separated subset, it inherits the annular relation and the pointwise mass lower bound from Z. Since G has maximum degree at most $\Delta \geq 2$,

$$V_{2r_p} \leq \Delta^{2r_p+1} \leq \Delta \overline{\chi}_p^{2\theta}.$$

Combining this estimate with

$$|Z| \geq \frac{\sigma_j^2}{4} \overline{\chi}_p$$

from Corollary 8.42 gives the displayed lower bound on $|Z_\theta|$. The separation $d(z, z') > 2r_p$ implies that the balls $B(z, r_p)$ are pairwise disjoint. □

Definition 8.48 (Rooted fan energy)**.** For $p < p_c$, $I \subseteq [0, \infty)$, $x \in V$, finite non-empty $K \subset V$, and $Z \subset V$, define

$$\mathcal{W}_p^I(x, K, Z) = \sum_{z \in Z} \tau_p(x, z) \sum_{\substack{y \in K \\ d(y,z) \in I}} \tau_p(y, z).$$

Definition 8.49 (Gated fan energy)**.** For $p < p_c$, $I \subseteq [0, \infty)$, $x \in V$, finite non-empty $K \subset V$, $Z \subset V$, and an integer $r \geq 0$, define

$$\mathcal{G}_{p,r}^I(x, K, Z) = \sum_{z \in Z} \left[\sum_{a : d(a,z) = r} \tau_p(x, a) \tau_p(a, z) \right] \cdot \left[\sum_{b : d(b,z) = r} \tau_p(b, z) \sum_{\substack{y \in K \\ d(y,z) \in I}} \tau_p(y, b) \right].$$

Definition 8.50 (Point-gated fan energy)**.** Let $p < p_c$, $I \subseteq [0,\infty)$, $x \in V$, let $K \subset V$ and $Z \subset V$ be finite non-empty sets, and let $r \geq 0$. If $a, b : Z \to V$ satisfy

$$d(a_z, z) = d(b_z, z) = r \qquad (z \in Z),$$

define

$$\mathcal{H}^I_{p,r}(x, K, Z, a, b) = \sum_{z \in Z} \tau_p(x, a_z)\tau_p(a_z, z)\tau_p(b_z, z) \sum_{\substack{y \in K \\ d(y,z) \in I}} \tau_p(y, b_z).$$

Definition 8.51 (Coalesced gate energy)**.** Let $p < p_c$, $I \subseteq [0,\infty)$, $x \in V$, let $K \subset V$ and $Z \subset V$ be finite non-empty sets, and let $r \geq 0$. If $a : Z \to V$ satisfies

$$d(a_z, z) = r \qquad (z \in Z),$$

define

$$\mathcal{J}^I_{p,r}(x, K, Z, a) = \sum_{z \in Z} \tau_p(x, a_z) \sum_{\substack{y \in K \\ d(y,z) \in I}} \tau_p(y, a_z).$$

Lemma 8.52 (Gate decomposition)**.** *Let G be locally finite, let $p < p_c$, let $r \geq 1$, and let $I \subseteq (r,\infty)$. If $x \in V$, $K \subset V$ is finite non-empty, $Z \subset V$, and $d(x, z) \in I$ for every $z \in Z$, then*

$$\mathcal{W}^I_p(x, K, Z) \leq \mathcal{G}^I_{p,r}(x, K, Z).$$

Proof. Fix $z \in Z$. Since $d(x, z) > r$, every open path from x to z has a first vertex a on the sphere $\{v : d(v, z) = r\}$. The initial segment from x to a and the terminal segment from a to z are edge-disjoint, so the BK inequality gives

$$\tau_p(x, z) \leq \sum_{a : d(a,z) = r} \tau_p(x, a)\tau_p(a, z).$$

Similarly, if $y \in K$ and $d(y, z) \in I$, then $d(y, z) > r$, and

$$\tau_p(y, z) \leq \sum_{b : d(b,z) = r} \tau_p(y, b)\tau_p(b, z).$$

Summing the second inequality over such y, multiplying by the first inequality, and then summing over $z \in Z$ gives the claim. □

Lemma 8.53 (Gate-pair selection)**.** *Let G be locally finite, let $p < p_c$, let $r \geq 0$, and put*

$$M_r = \sup_{z \in V} |\{v : d(v, z) = r\}|.$$

Let $I \subseteq [0,\infty)$, $x \in V$, let $K \subset V$ and $Z \subset V$ be finite non-empty sets, and suppose that the sphere $\{v : d(v, z) = r\}$ is non-empty for every $z \in Z$. Then there are maps $a, b : Z \to V$, with $d(a_z, z) = d(b_z, z) = r$, such that

$$\mathcal{H}^I_{p,r}(x, K, Z, a, b) \geq M_r^{-2} \mathcal{G}^I_{p,r}(x, K, Z).$$

Proof. For each $z \in Z$, write the z-summand of $\mathcal{G}^I_{p,r}(x, K, Z)$ as

$$\sum_{\substack{a:d(a,z)=r \\ b:d(b,z)=r}} \tau_p(x,a)\tau_p(a,z)\tau_p(b,z) \sum_{\substack{y\in K \\ d(y,z)\in I}} \tau_p(y,b).$$

There are at most M_r^2 non-negative summands. Choose (a_z, b_z) to maximize the displayed quantity. The chosen term is at least M_r^{-2} times the whole z-summand. Summing over z proves the claim. □

Lemma 8.54 (Gate coalescence)**.** *Let G be locally finite, let $p < p_c$, let $r \geq 0$, let $I \subseteq [0, \infty)$, $x \in V$, and let $K, Z \subset V$ be finite non-empty sets. If $a, b : Z \to V$ satisfy*

$$d(a_z, z) = d(b_z, z) = r \qquad (z \in Z),$$

then

$$\mathcal{J}^I_{p,r}(x, K, Z, a) \geq p^{2r}\mathcal{H}^I_{p,r}(x, K, Z, a, b).$$

Proof. Fix $z \in Z$, and let π_z be a geodesic from b_z to a_z. Since $d(a_z, b_z) \leq 2r$, the probability that every edge of π_z is open is at least p^{2r}. For every $y \in K$, the events $\{y \leftrightarrow b_z\}$ and $\{\pi_z \text{ is open}\}$ are increasing, and their intersection implies $\{y \leftrightarrow a_z\}$. Hence FKG gives

$$\tau_p(y, a_z) \geq p^{2r}\tau_p(y, b_z).$$

After summing over y with $d(y, z) \in I$ and using $\tau_p(a_z, z)\tau_p(b_z, z) \leq 1$, we obtain

$$\tau_p(x, a_z) \sum_{\substack{y\in K \\ d(y,z)\in I}} \tau_p(y, a_z) \geq p^{2r}\tau_p(x, a_z)\tau_p(a_z, z)\tau_p(b_z, z) \sum_{\substack{y\in K \\ d(y,z)\in I}} \tau_p(y, b_z).$$

Summing over z proves the lemma. □

Lemma 8.55 (High one-gate layer extraction)**.** *Let $p < p_c$, let $r \geq 0$, let $I \subseteq [0, \infty)$, let $x \in V$, let $K, Z \subset V$ be finite non-empty sets, and let $a : Z \to V$. Suppose that*

$$\sum_{z\in Z} \tau_p(x, a_z) \leq \overline{\chi}_p$$

and that

$$\mathcal{J}^I_{p,r}(x, K, Z, a) \geq L.$$

Put

$$Z^\sharp = \left\{ z \in Z : \sum_{\substack{y\in K \\ d(y,z)\in I}} \tau_p(y, a_z) \geq \frac{L}{2\overline{\chi}_p} \right\}.$$

Then

$$\sum_{z\in Z^\sharp} \tau_p(x, a_z) \geq \frac{L}{2\overline{\chi}_p}.$$

Proof. For $z \notin Z^\sharp$, the K-to-gate mass is less than $L/(2\overline{\chi}_p)$, while for every z it is at most $\overline{\chi}_p$. Therefore

$$\begin{aligned} L &\le \mathcal{J}^I_{p,r}(x, K, Z, a) \\ &\le \frac{L}{2\overline{\chi}_p} \sum_{z \in Z} \tau_p(x, a_z) + \overline{\chi}_p \sum_{z \in Z^\sharp} \tau_p(x, a_z) \\ &\le \frac{L}{2} + \overline{\chi}_p \sum_{z \in Z^\sharp} \tau_p(x, a_z). \end{aligned}$$

Rearranging gives the claim. □

Corollary 8.56 (Weighted separated fan)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$. There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every $0 < \theta < 1/2$, every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z_\theta \subset V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

we have

$$|K| > 2\epsilon\overline{\chi}_p, \qquad 2^j \ge \frac{1}{4}\log_\Delta \overline{\chi}_p,$$

$$d(x, z) \in I_j \quad (z \in Z_\theta), \qquad d(z, z') > 2r_p \quad (z, z' \in Z_\theta,\ z \ne z'),$$

$$\sum_{z \in Z_\theta} \tau_p(x, z) \ge \frac{\sigma_j^2}{4\Delta} \overline{\chi}_p^{1-2\theta},$$

and every $z \in Z_\theta$ satisfies

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y, z) \ge \frac{\sigma_j}{2}\overline{\chi}_p.$$

Moreover,

$$\mathcal{W}_p^{I_j}(x, K, Z_\theta) \ge \frac{\sigma_j^3}{8\Delta} \overline{\chi}_p^{2-2\theta}.$$

Consequently,

$$|Z_\theta| \ge \frac{\sigma_j^2}{4\Delta} \overline{\chi}_p^{1-2\theta},$$

and the balls $B(z, r_p)$, $z \in Z_\theta$, are pairwise disjoint.

Proof. Let ϵ be supplied by Corollary 8.38, and put

$$c = \frac{\epsilon^2}{16}(1 - 2^{-\eta/2})^4.$$

Choose $p_* < p_c$ sufficiently close to p_c that Corollary 8.38, with $\alpha = \gamma = \eta/2$, applies to every $p_0 \in [p_*, p_c)$, and every $p \in [p_*, p_c)$ satisfies

$$c\overline{\chi}_p > \Delta \overline{\chi}_p^{1/2} \left(\frac{1}{4} \log_\Delta \overline{\chi}_p \right)^{2\eta}$$

and $\log_\Delta \overline{\chi}_p > 4$.

Fix $p_0 \in [p_*, p_c)$, and choose p, K, j by Corollary 8.38. Let

$$Z_0 = Z_p^{I_j}(K, \sigma_j/2).$$

Then $|Z_0| \geq (\sigma_j/2)|K|$, and every $z \in Z_0$ satisfies

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y, z) \geq \frac{\sigma_j}{2} \overline{\chi}_p.$$

Consequently,

$$\begin{aligned} \sum_{x \in K} \sum_{\substack{z \in Z_0 \\ d(x,z) \in I_j}} \tau_p(x, z) &= \sum_{z \in Z_0} \sum_{\substack{x \in K \\ d(x,z) \in I_j}} \tau_p(x, z) \\ &\geq \frac{\sigma_j}{2} \overline{\chi}_p |Z_0| \geq \frac{\sigma_j^2}{4} \overline{\chi}_p |K|. \end{aligned}$$

Choose $x \in K$ for which the corresponding inner sum is at least the average, and set

$$Z = \{z \in Z_0 : d(x, z) \in I_j\}.$$

Then

$$\sum_{z \in Z} \tau_p(x, z) \geq \frac{\sigma_j^2}{4} \overline{\chi}_p.$$

Since $\tau_p(x, z) \leq 1$, this gives

$$V(I_j) \geq |Z| \geq \frac{\sigma_j^2}{4} \overline{\chi}_p = c\, 2^{-2\eta(j-J)} \overline{\chi}_p.$$

The same bounded-degree comparison as in Corollary 8.42 gives $2^j \geq (1/4) \log_\Delta \overline{\chi}_p$.

Now apply Lemma 8.45 to Z, with weights $a_z = \tau_p(x, z)$, and with

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor.$$

The resulting Z_θ is $2r_p$-separated and inherits the annular relation and the pointwise backscatter lower bound from Z. Since

$$V_{2r_p} \leq \Delta^{2r_p+1} \leq \Delta \overline{\chi}_p^{2\theta},$$

the weighted extraction gives

$$\sum_{z \in Z_\theta} \tau_p(x, z) \geq \frac{\sigma_j^2}{4\Delta} \overline{\chi}_p^{1-2\theta}.$$

Multiplying this by the pointwise lower bound $(\sigma_j/2)\overline{\chi}_p$ for the backscatter from K gives the displayed lower bound on $\mathcal{W}_p^{I_j}(x, K, Z_\theta)$. The cardinality bound follows from $\tau_p(x, z) \leq 1$, and the ball disjointness follows from the separation. □

Corollary 8.57 (Volume-sensitive weighted separated fan)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$1 - 2\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z_\theta \subset V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

we have

$$|K| > 2\epsilon \overline{\chi}_p, \qquad 2^j > BL,$$

$$d(x, z) \in I_j \quad (z \in Z_\theta), \qquad d(z, z') > 2r_p \quad (z, z' \in Z_\theta,\ z \ne z'),$$

$$\sum_{z \in Z_\theta} \tau_p(x, z) \ge \frac{\sigma_j^2}{4C_{\mathrm{vol}}} \overline{\chi}_p^{1-2\theta \log_\Delta \omega},$$

and every $z \in Z_\theta$ satisfies

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y, z) \ge \frac{\sigma_j}{2} \overline{\chi}_p.$$

Moreover,

$$\mathcal{W}_p^{I_j}(x, K, Z_\theta) \ge \frac{\sigma_j^3}{8C_{\mathrm{vol}}} \overline{\chi}_p^{2-2\theta \log_\Delta \omega}.$$

Consequently,

$$|Z_\theta| \ge \frac{\sigma_j^2}{4C_{\mathrm{vol}}} \overline{\chi}_p^{1-2\theta \log_\Delta \omega},$$

and the balls $B(z, r_p)$, $z \in Z_\theta$, are pairwise disjoint.

Proof. Let ϵ be supplied by Corollary 8.38, and put

$$c = \frac{\epsilon^2}{16}(1 - 2^{-\eta/2})^4.$$

Choose $p_* < p_c$ sufficiently close to p_c that Corollary 8.38, with $\alpha = \gamma = \eta/2$, applies to every $p_0 \in [p_*, p_c)$, and every $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$, satisfies $BL > 1$ and

$$c\overline{\chi}_p > C_{\mathrm{vol}} \overline{\chi}_p^{2B \log_\Delta \omega} (BL)^{2\eta}.$$

Fix $p_0 \in [p_*, p_c)$, and choose p, K, j by Corollary 8.38. Let

$$Z_0 = Z_p^{I_j}(K, \sigma_j/2).$$

As in the proof of Corollary 8.56, there are $x \in K$ and

$$Z = \{z \in Z_0 : d(x, z) \in I_j\}$$

such that every $z \in Z$ has the displayed backscatter lower bound and

$$\sum_{z \in Z} \tau_p(x,z) \geq \frac{\sigma_j^2}{4} \overline{\chi}_p.$$

Since $\tau_p(x,z) \leq 1$, this also gives

$$|Z| \geq c\, 2^{-2\eta(j-J)} \overline{\chi}_p.$$

If $2^j \leq BL$, then the volume-growth bound gives

$$|Z| \leq C_{\mathrm{vol}} \omega^{2^{j+1}} \leq C_{\mathrm{vol}} \overline{\chi}_p^{2B \log_\Delta \omega},$$

and $2^{2\eta(j-J)} \leq (BL)^{2\eta}$. This contradicts the choice of p_*. Hence $2^j > BL$.

Apply Lemma 8.45 to Z, with weights $a_z = \tau_p(x,z)$, and with

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor.$$

The resulting set Z_θ is $2r_p$-separated and inherits the annular relation, the scale bound $2^j > BL$, and the pointwise backscatter lower bound from Z. Since

$$V_{2r_p} \leq C_{\mathrm{vol}} \omega^{2r_p} \leq C_{\mathrm{vol}} \overline{\chi}_p^{2\theta \log_\Delta \omega},$$

the weighted extraction gives

$$\sum_{z \in Z_\theta} \tau_p(x,z) \geq \frac{\sigma_j^2}{4C_{\mathrm{vol}}} \overline{\chi}_p^{1-2\theta \log_\Delta \omega}.$$

Multiplying by the pointwise lower bound $(\sigma_j/2)\overline{\chi}_p$ for the backscatter from K gives the displayed lower bound on $\mathcal{W}_p^{I_j}(x, K, Z_\theta)$. The cardinality bound follows from $\tau_p(x,z) \leq 1$, and ball disjointness follows from separation. □

Definition 8.58 (Separated fans)**.** Assume G has maximum degree at most $\Delta \geq 2$. Fix $0 < \epsilon \leq 1$, $\eta > 0$, $0 < \theta < 1/2$, and integers $J, H \geq 0$. For $p < p_{\mathrm{c}}$, an $(\epsilon, \eta, \theta, J, H)$-separated fan at p is a quadruple (K, x, j, Z) such that $K \subset V$ is finite non-empty, $x \in K$, $j > J + H$, $Z \subset V$ is finite non-empty, and, writing

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

the following conditions hold:

$$|K| > 2\epsilon \overline{\chi}_p, \qquad 2^j \geq \frac{1}{4} \log_\Delta \overline{\chi}_p,$$

$$|Z| \geq \frac{\sigma_j^2}{4\Delta} \overline{\chi}_p^{1-2\theta}, \qquad d(x,z) \in I_j \quad (z \in Z),$$

$$d(z,z') > 2r_p \qquad (z, z' \in Z,\ z \neq z'),$$

and

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y,z) \geq \frac{\sigma_j}{2} \overline{\chi}_p \qquad (z \in Z).$$

Definition 8.59 (Root-weighted separated fans)**.** An $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p is root-weighted if, with σ_j as in Definition 8.58,

$$\sum_{z \in Z} \tau_p(x, z) \geq \frac{\sigma_j^2}{4\Delta} \overline{\chi}_p^{1-2\theta}.$$

Proposition 8.60 (Fan exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/2$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan at p exists for any $p \in [p_1, p_c)$. Then*

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. We prove the first claim by contraposition. Suppose that

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.47. Fix arbitrary $\eta > 0$, $0 < \theta < 1/2$, and integers $J, H \geq 0$. That corollary says that for every p_0 sufficiently close to p_c, there is $p \in [p_0, p_c)$ and an $(\epsilon, \eta, \theta, J, H)$-separated fan at p. Hence this ϵ violates the asserted fan-exclusion property. The contrapositive proves $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

Under the additional quasi-transitivity and mean-field susceptibility assumptions, the L^2-gap conclusion follows from Corollary 8.794. □

Proposition 8.61 (Weighted fan exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/2$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no root-weighted $(\epsilon, \eta, \theta, J, H)$-separated fan at p exists for any $p \in [p_1, p_c)$. Then*

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. The proof is the same contrapositive argument as for Proposition 8.60, using Corollary 8.56 in place of Corollary 8.47. □

Proposition 8.62 (Fan energy exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/2$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p satisfies*

$$\mathcal{W}_p^{(2^j, 2^{j+1}]}(x, K, Z) \geq \frac{\sigma_j^3}{8\Delta} \overline{\chi}_p^{2-2\theta}$$

for any $p \in [p_1, p_c)$, where σ_j is the parameter in Definition 8.58. Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Again argue by contraposition. If $\limsup_{p \uparrow p_c} \mathcal{R}_p^* / \overline{\chi}_p > 0$, then Corollary 8.56 supplies an $\epsilon > 0$ such that for every $\eta > 0$, every $0 < \theta < 1/2$, and every J, H, near-critical parameters contain a separated fan with the displayed fan-energy lower bound. This contradicts the asserted exclusion property for that ϵ. The L^2-gap conclusion follows from Corollary 8.794. □

Corollary 8.63 (Gated fan obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/4$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z_\theta \subset V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

(K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p, and

$$\mathcal{G}_{p, r_p}^{I_j}(x, K, Z_\theta) \geq \frac{\sigma_j^3}{8\Delta} \overline{\chi}_p^{2-2\theta}.$$

Proof. Choose $p_* < p_c$ sufficiently close to p_c that Corollary 8.56 applies to every $p_0 \in [p_*, p_c)$, and

$$\left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor \geq 1 \qquad (p \in [p_*, p_c)).$$

Fix $p_0 \in [p_*, p_c)$. Let p, K, j, x, Z_θ be supplied by Corollary 8.56. Since $2^j \geq (1/4) \log_\Delta \overline{\chi}_p$ and $\theta < 1/4$, we have

$$r_p < 2^j.$$

Thus $I_j \subseteq (r_p, \infty)$, and Lemma 8.52 applies. Combining

$$\mathcal{W}_p^{I_j}(x, K, Z_\theta) \geq \frac{\sigma_j^3}{8\Delta} \overline{\chi}_p^{2-2\theta}$$

with

$$\mathcal{W}_p^{I_j}(x, K, Z_\theta) \leq \mathcal{G}_{p, r_p}^{I_j}(x, K, Z_\theta)$$

gives the claim. □

Corollary 8.64 (Volume-sensitive gated fan obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 2\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, and a set $Z_\theta \subset V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

the data satisfy the conclusions of Corollary 8.57, and

$$\mathcal{G}_{p,r_p}^{I_j}(x, K, Z_\theta) \geq \frac{\sigma_j^3}{8C_{\mathrm{vol}}} \overline{\chi}_p^{2-2\theta \log_\Delta \omega}.$$

Proof. Choose $p_* < p_c$ sufficiently close to p_c that Corollary 8.57 applies to every $p_0 \in [p_*, p_c)$, and

$$\lfloor \theta \log_\Delta \overline{\chi}_p \rfloor \geq 1 \qquad (p \in [p_*, p_c)).$$

Fix $p_0 \in [p_*, p_c)$, and choose p, K, j, x, Z_θ by Corollary 8.57. Since

$$2^j > B \log_\Delta \overline{\chi}_p \qquad \text{and} \qquad r_p \leq \theta \log_\Delta \overline{\chi}_p < B \log_\Delta \overline{\chi}_p,$$

we have $r_p < 2^j$. Thus $I_j \subseteq (r_p, \infty)$, and Lemma 8.52 applies. Combining

$$\mathcal{W}_p^{I_j}(x, K, Z_\theta) \geq \frac{\sigma_j^3}{8C_{\mathrm{vol}}} \overline{\chi}_p^{2-2\theta \log_\Delta \omega}$$

with

$$\mathcal{W}_p^{I_j}(x, K, Z_\theta) \leq \mathcal{G}_{p,r_p}^{I_j}(x, K, Z_\theta)$$

gives the claim. □

Corollary 8.65 (Volume-sensitive point-gated fan obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 2\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, and maps $a, b : Z_\theta \to V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

the data satisfy the conclusions of Corollary 8.57,

$$d(a_z, z) = d(b_z, z) = r_p \qquad (z \in Z_\theta),$$

and

$$\mathcal{H}^{I_j}_{p,r_p}(x, K, Z_\theta, a, b) \geq \frac{\sigma_j^3}{8C_{\mathrm{vol}}^3} \overline{\chi}_p^{2-4\theta \log_\Delta \omega}.$$

Proof. Choose p, K, j, x, Z_θ by Corollary 8.64. Since $d(x, z) \in I_j$ for every $z \in Z_\theta$ and $r_p < 2^j$, each sphere $\{v : d(v, z) = r_p\}$ is non-empty. Let

$$M_{r_p} = \sup_{z \in V} |\{v : d(v, z) = r_p\}|.$$

Lemma 8.53 gives maps $a, b : Z_\theta \to V$ with

$$\mathcal{H}^{I_j}_{p,r_p}(x, K, Z_\theta, a, b) \geq M_{r_p}^{-2} \mathcal{G}^{I_j}_{p,r_p}(x, K, Z_\theta).$$

The volume-growth bound gives

$$M_{r_p} \leq C_{\mathrm{vol}} \omega^{r_p} \leq C_{\mathrm{vol}} \overline{\chi}_p^{\theta \log_\Delta \omega}.$$

Combining this with Corollary 8.64 gives the displayed point-gated lower bound. □

Corollary 8.66 (Volume-sensitive one-gate fan obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 2\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, and a map $a : Z_\theta \to V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1-2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

the data satisfy the conclusions of Corollary 8.57,

$$d(a_z, z) = r_p \qquad (z \in Z_\theta),$$

and

$$\mathcal{J}^{I_j}_{p,r_p}(x, K, Z_\theta, a) \geq p^{2r_p} \frac{\sigma_j^3}{8C_{\mathrm{vol}}^3} \overline{\chi}_p^{2-4\theta \log_\Delta \omega}.$$

Proof. Choose $p, K, j, x, Z_\theta, a, b$ by Corollary 8.65. The gate-coalescence lemma gives

$$\mathcal{J}^{I_j}_{p,r_p}(x, K, Z_\theta, a) \geq p^{2r_p} \mathcal{H}^{I_j}_{p,r_p}(x, K, Z_\theta, a, b),$$

and the point-gated lower bound gives the displayed one-gate lower bound. □

Corollary 8.67 (Volume-sensitive high one-gate obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 4\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are $p \in [p_0, p_{\mathrm{c}})$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1-2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

the tuple (K, x, j, Z_θ) satisfies the conclusions of Corollary 8.57,

$$d(a_z, z) = r_p \qquad (z \in Z^\sharp),$$

and, putting

$$\kappa_j^{\mathrm{vol}}(p) = \frac{p^{2r_p} \sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-4\theta \log_\Delta \omega},$$

we have

$$\sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j^{\mathrm{vol}}(p)$$

and

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y, a_z) \geq \kappa_j^{\mathrm{vol}}(p) \qquad (z \in Z^\sharp).$$

Proof. The condition $1 - 4\theta \log_\Delta \omega > 0$ implies the hypotheses of Corollary 8.66. Thus we may choose p, K, j, x, Z_θ and a map $a_0 : Z_\theta \to V$ by Corollary 8.66. Set

$$\Lambda = p^{2r_p} \frac{\sigma_j^3}{8C_{\mathrm{vol}}^3} \overline{\chi}_p^{2-4\theta \log_\Delta \omega}.$$

Since (K, x, j, Z_θ) satisfies the conclusions of Corollary 8.57, the balls $B(z, r_p)$, $z \in Z_\theta$, are pairwise disjoint. Since $d(a_{0,z}, z) = r_p$, the vertices $a_{0,z}$ are distinct, and hence

$$\sum_{z \in Z_\theta} \tau_p(x, a_{0,z}) \le \overline{\chi}_p.$$

Apply Lemma 8.55 to a_0 with $L = \Lambda$, and let $Z^\sharp$ be the resulting set. Restrict a_0 to $Z^\sharp$, and call the restriction a. Since

$$\frac{\Lambda}{2\overline{\chi}_p} = \kappa_j^{\mathrm{vol}}(p),$$

the two displayed lower bounds follow. The root lower bound makes $Z^\sharp$ non-empty. □

Corollary 8.68 (Volume-sensitive expanded gate-annulus obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \ge 1$ such that*

$$|B(u, R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 4\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

$$I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

and

$$\kappa_j^{\mathrm{vol}}(p) = \frac{p^{2r_p} \sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-4\theta \log_\Delta \omega},$$

the tuple (K, x, j, Z_θ) satisfies the conclusions of Corollary 8.57,

$$d(a_z, z) = r_p \qquad (z \in Z^\sharp),$$

$$d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z^\sharp),$$

$$\sum_{z \in Z^\sharp} \tau_p(x, a_z) \ge \kappa_j^{\mathrm{vol}}(p),$$

and

$$\sum_{\substack{y \in K \\ d(y, a_z) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a_z) \ge \kappa_j^{\mathrm{vol}}(p) \qquad (z \in Z^\sharp).$$

Proof. Choose the data by Corollary 8.67. If $z \in Z^\sharp$, then $d(a_z, z) = r_p$, while the weighted separated-fan conclusions give $d(x, z) \in I_j$. The triangle inequality gives

$$d(x, a_z) \in (2^j - r_p, 2^{j+1} + r_p] = I_{j,p}^{\mathrm{gate}}.$$

Similarly, every $y \in K$ with $d(y, z) \in I_j$ satisfies

$$d(y, a_z) \in I_{j,p}^{\mathrm{gate}}.$$

Thus the pointwise K-mass lower bound from Corollary 8.67 remains valid when the summation is enlarged to $d(y, a_z) \in I_{j,p}^{\mathrm{gate}}$. The remaining displayed properties are inherited. □

Corollary 8.69 (Capped anchor-count feasibility)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$ and

$$\rho(\theta, U, \omega) = 1 - 2\theta - 5\theta\alpha_\omega - 2U\alpha_\omega.$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$, every $U > B$, every $\theta > 0$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are data satisfying all conclusions of Corollary 8.68 such that, with $L = \log_\Delta \overline{\chi}_p$, either

$$2^j > UL,$$

or

$$A_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}\, L^{-3\eta} \overline{\chi}_p^{\rho(\theta,U,\omega)} \leq 1,$$

where

$$A_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{16 C_{\mathrm{vol}}^4} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J} U^{-3\eta}.$$

Consequently, for fixed admissible parameters, a capped branch occurring arbitrarily close to criticality forces $\rho(\theta, U, \omega) \leq 0$.

Proof. Use the ϵ from Corollary 8.68. Take p_0 close enough to p_{c} that the expanded gate-annulus obstruction applies, every selected $p \in [p_0, p_{\mathrm{c}})$ satisfies $p \geq 1/\Delta$, and $L = \log_\Delta \overline{\chi}_p \geq 1$. Choose the expanded gate-annulus data. If $2^j > UL$, the first alternative holds, so assume $2^j \leq UL$.

Let

$$A = \{a_z : z \in Z^\sharp\}.$$

The separated-fan conclusion gives disjointness of the balls $B(z, r_p)$, and since $d(a_z, z) = r_p$, the map $z \mapsto a_z$ is injective on $Z^\sharp$. The root-mass lower bound and $\tau_p \leq 1$ therefore give

$$|A| \geq \sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j^{\mathrm{vol}}(p).$$

The metric conclusions also give $A \subseteq B(x, 2^{j+1} + r_p)$. Under the cap $2^j \leq UL$,

$$|A| \leq C_{\mathrm{vol}} \omega^{2^{j+1} + r_p} \leq C_{\mathrm{vol}} \overline{\chi}_p^{(2U+\theta)\alpha_\omega}.$$

Since $p \geq 1/\Delta$, $r_p \leq \theta L$, and $j \leq \log_2(UL)$, we have

$$\kappa_j^{\mathrm{vol}}(p) \geq \frac{1}{16C_{\mathrm{vol}}^3}\left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J}U^{-3\eta}L^{-3\eta}\overline{\chi}_p^{1-2\theta-4\theta\alpha_\omega}.$$

Combining the last three displays gives the stated feasibility inequality.

If the capped branch occurs for a sequence $p \uparrow p_{\mathrm{c}}$, then the displayed inequality holds along that sequence. Since $\overline{\chi}_p \to \infty$ and $L = \log_\Delta \overline{\chi}_p$, this is impossible when $\rho(\theta, U, \omega) > 0$. Hence $\rho(\theta, U, \omega) \leq 0$. □

Corollary 8.70 (Optimized anchor-count gate escape)**.** *Under the graph and volume assumptions of Corollary 8.69, put*

$$\alpha_\omega = \log_\Delta \omega.$$

There is $0 < \epsilon \leq 1$ such that the following holds. If $\alpha_\omega > 0$, then for every

$$0 < U < \frac{1}{2\alpha_\omega},$$

every $\eta > 0$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are data satisfying all conclusions of Corollary 8.68 and

$$2^j > U \log_\Delta \overline{\chi}_p.$$

If $\alpha_\omega = 0$, the same conclusion holds for every finite $U > 0$.

Proof. Use the ϵ from Corollary 8.69. First suppose that $\alpha_\omega > 0$ and $0 < U < 1/(2\alpha_\omega)$. Choose $B = U/2$, and then choose $\theta > 0$ so small that

$$\theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

and

$$1 - 2\theta - 5\theta\alpha_\omega - 2U\alpha_\omega > 0.$$

Also $2B\alpha_\omega = U\alpha_\omega < 1/2 < 1$. By Corollary 8.69, for all p_0 sufficiently close to p_{c} there are expanded gate-annulus data satisfying either the desired inequality $2^j > U \log_\Delta \overline{\chi}_p$, or the capped anchor-count feasibility inequality with a positive exponent. Taking p_0 still closer to p_{c} makes the latter inequality impossible, so the desired escape holds.

If $\alpha_\omega = 0$, fix finite $U > 0$, choose $B = U/2$, and then choose $0 < \theta < \min\{B, 1/2\}$. The same argument applies because $\rho(\theta, U, \omega) = 1 - 2\theta > 0$. □

Corollary 8.71 (Half-logarithmic expanded gate obstruction)**.** *Under the graph and volume assumptions of Corollary 8.69, put*

$$\alpha_\omega = \log_\Delta \omega.$$

There is $0 < \epsilon \leq 1$ such that the following holds. If $\alpha_\omega > 0$, then for every $\delta > 0$, every $\eta > 0$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} - \delta\right) \log_\Delta \overline{\chi}_p.$$

If $\alpha_\omega = 0$, the same conclusion holds with the right-hand side replaced by $U \log_\Delta \overline{\chi}_p$ for every finite $U > 0$.

Proof. If $\alpha_\omega > 0$, choose

$$0 < U < \frac{1}{2\alpha_\omega} \quad \text{with} \quad U > \frac{1}{2\alpha_\omega} - \delta.$$

The first part follows from Corollary 8.70 with this U. The case $\alpha_\omega = 0$ is exactly the second part of that corollary. □

Remark 8.72. Corollary 8.70 is the strongest pure scale amplification available from the present volume-counting inputs. It uses only the number of selected anchors $A = \{a_z : z \in Z^\sharp\}$. The support-profile arguments below are therefore not needed to force logarithmic escape up to $1/(2\alpha_\omega)$; their role is to expose the additional two-replica support geometry of the remaining live branch.

The scale hierarchy is as follows. The normalized square-profile argument forces capped escape only up to $U < 1/(4\alpha_\omega)$, the Schur and support-profile arguments reach $U < 1/(3\alpha_\omega)$, and the anchor count reaches $U < 1/(2\alpha_\omega)$. These thresholds refer to different objects, so the weaker support-profile range below should be read as a diagnostic for Y and Θ_p, not as the best available pure scale bound.

Proposition 8.73 (Anchor-count large-scale criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $B > 0$, $U > B$, $\theta > 0$, integers $J, H \geq 0$, and $p_1 < p_c$ such that

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

$$\rho(\theta, U, \omega) = 1 - 2\theta - 5\theta\alpha_\omega - 2U\alpha_\omega > 0,$$

and no $p \in [p_1, p_c)$ admits data satisfying all conclusions of Corollary 8.68, with these parameters, and

$$2^j > U \log_\Delta \overline{\chi}_p.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. We prove the first claim by contraposition. Suppose that

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.69, and choose $\eta, B, U, \theta, J, H, p_1$ from the assumed criterion for this ϵ. Put $\rho = \rho(\theta, U, \omega) > 0$, and

$$A_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{16 C_{\mathrm{vol}}^4} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J} U^{-3\eta}.$$

Choose $p_* < p_c$ sufficiently close to p_c that $p_* \geq p_1$, the capped anchor-count feasibility corollary applies to every $p_0 \in [p_*, p_c)$, and whenever $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$,

$$A_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} L^{-3\eta} \overline{\chi}_p^{\rho} > 1.$$

This is possible because $\rho > 0$ and $\overline{\chi}_p \to \infty$.

Fix $p_0 \in [p_*, p_c)$. Corollary 8.69 gives expanded gate-annulus data. Its large-scale alternative contradicts the assumed exclusion. Its capped feasibility alternative contradicts the choice of p_*. This contradiction proves $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Proposition 8.74 (Half-logarithmic expanded-gate exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. If $\alpha_\omega > 0$, suppose that for every $0 < \epsilon \leq 1$ there exist

$$0 < \delta < \frac{1}{2\alpha_\omega}, \qquad \eta > 0, \qquad J, H \geq 0, \qquad p_1 < p_c$$

such that no $p \in [p_1, p_c)$ admits data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} - \delta \right) \log_\Delta \overline{\chi}_p.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If $\alpha_\omega = 0$, the same conclusion holds if for every $0 < \epsilon \leq 1$ there exist a finite $U > 0$, $\eta > 0$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no $p \in [p_1, p_c)$ admits data satisfying all conclusions of Corollary 8.68 and

$$2^j > U \log_\Delta \overline{\chi}_p.$$

In either case, if G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. We argue by contraposition. Suppose that

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.71.

First suppose $\alpha_\omega > 0$, and choose δ, η, J, H, p_1 from the assumed exclusion for this ϵ. Take $p_0 \geq p_1$ sufficiently close to p_c that Corollary 8.71 applies. It gives expanded gate-annulus data satisfying the displayed half-logarithmic lower bound, contradicting the exclusion.

If $\alpha_\omega = 0$, choose U, η, J, H, p_1 from the assumed exclusion for this ϵ, take $p_0 \geq p_1$ close enough to p_c, and apply the second part of Corollary 8.71. This again contradicts the exclusion. Hence $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

The final assertion follows from Corollary 8.794. □

Corollary 8.75 (Near-half-logarithmic anchor-density obstruction)**.** *Under the graph and volume assumptions of Corollary 8.69, suppose that* $\alpha_\omega = \log_\Delta \omega > 0$. *There is* $0 < \epsilon \le 1$ *such that for every* $\lambda > 0$, *every* $\eta > 0$, *every* $B > 0$, *every* $\theta > 0$ *satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \ge 0$, *and all* p_0 *sufficiently close to* p_c, *there are data satisfying all conclusions of Corollary 8.68 such that, with*

$$L = \log_\Delta \overline{\chi}_p, \qquad U_\lambda = \frac{1}{2\alpha_\omega} + \lambda, \qquad r_p = \lfloor \theta L \rfloor,$$

and with

$$A = \{a_z : z \in Z^\sharp\}, \qquad R^{\mathrm{gate}}_{j,p} = 2^{j+1} + r_p,$$

at least one of the following alternatives holds:

$$2^j > U_\lambda L,$$

or

$$\frac{|A|}{|B(x, R^{\mathrm{gate}}_{j,p})|} \ge D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where

$$D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega} = \frac{1}{16 C_{\mathrm{vol}}^4} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J} \left(\frac{1}{2\alpha_\omega} + \lambda\right)^{-3\eta}.$$

Proof. Use the ϵ from Corollary 8.68. Take p_0 close enough to p_c that the expanded gate-annulus obstruction applies, every selected $p \in [p_0, p_c)$ satisfies $p \ge 1/\Delta$, and $L \ge 1$. Choose the expanded gate-annulus data. If $2^j > U_\lambda L$, the first alternative holds, so assume $2^j \le U_\lambda L$.

As in the proof of Corollary 8.69, the map $z \mapsto a_z$ is injective on $Z^\sharp$, and

$$|A| \ge \sum_{z \in Z^\sharp} \tau_p(x, a_z) \ge \kappa_j^{\mathrm{vol}}(p).$$

Since $p \ge 1/\Delta$, $r_p \le \theta L$, and $2^j \le U_\lambda L$, we have

$$|A| \ge \frac{1}{16 C_{\mathrm{vol}}^3} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J} U_\lambda^{-3\eta} L^{-3\eta} \overline{\chi}_p^{1 - 2\theta - 4\theta\alpha_\omega}.$$

The metric conclusions give $A \subseteq B(x, R^{\mathrm{gate}}_{j,p})$, while the volume bound and the cap give

$$|B(x, R^{\mathrm{gate}}_{j,p})| \le C_{\mathrm{vol}} \omega^{2^{j+1} + r_p} \le C_{\mathrm{vol}} \overline{\chi}_p^{(2U_\lambda + \theta)\alpha_\omega}.$$

Dividing the last two displays and using

$$1 - 2\theta - 4\theta\alpha_\omega - (2U_\lambda + \theta)\alpha_\omega = -2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega$$

gives the stated density lower bound. □

Proposition 8.76 (Near-half-logarithmic density-exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 < \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that no $p \in [p_1, p_{\mathrm{c}})$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and either of the following two conditions:

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p,$$

or, with $L = \log_\Delta \overline{\chi}_p$,

$$2^j \leq \left(\frac{1}{2\alpha_\omega} + \lambda\right) L$$

and

$$\frac{|A|}{|B(x, R_{j,p}^{\mathrm{gate}})|} \geq D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where

$$A = \{a_z : z \in Z^\sharp\}, \qquad R_{j,p}^{\mathrm{gate}} = 2^{j+1} + \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

and $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ is the constant from Corollary 8.75. Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. We argue by contraposition. Suppose that

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.75, and choose the parameters from the assumed exclusion for this ϵ. Take $p_0 \geq p_1$ sufficiently close to p_{c} that Corollary 8.75 applies. The corollary gives expanded gate-annulus data satisfying either the displayed super-half-logarithmic scale condition or the capped density condition. Both are excluded by assumption, a contradiction. Hence $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Corollary 8.77 (Near-half-logarithmic root-ball mass obstruction)**.** *Under the graph and volume assumptions of Corollary 8.69, suppose that* $\alpha_\omega = \log_\Delta \omega > 0$. *There is* $0 < \epsilon \le 1$ *such that for every* $\lambda > 0$, *every* $\eta > 0$, *every* $B > 0$, *every* $\theta > 0$ *satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \ge 0$, *and all* p_0 *sufficiently close to* p_c, *there are data satisfying all conclusions of Corollary 8.68 such that, with*

$$L = \log_\Delta \overline{\chi}_p, \qquad U_\lambda = \frac{1}{2\alpha_\omega} + \lambda, \qquad r_p = \lfloor \theta L \rfloor, \qquad R^{\mathrm{gate}}_{j,p} = 2^{j+1} + r_p,$$

at least one of the following alternatives holds:

$$2^j > U_\lambda L,$$

or

$$\frac{1}{|B(x, R^{\mathrm{gate}}_{j,p})|} \sum_{a \in B(x, R^{\mathrm{gate}}_{j,p})} \tau_p(x, a) \ge D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ *is the constant from Corollary 8.75.*

Proof. Use the same data as in Corollary 8.75. If $2^j > U_\lambda L$, there is nothing to prove, so assume $2^j \le U_\lambda L$. Put $A = \{a_z : z \in Z^\sharp\}$. The expanded gate-annulus obstruction gives $A \subseteq B(x, R^{\mathrm{gate}}_{j,p})$. As in the proof of Corollary 8.69, the map $z \mapsto a_z$ is injective on $Z^\sharp$, and hence

$$\sum_{a \in A} \tau_p(x, a) = \sum_{z \in Z^\sharp} \tau_p(x, a_z) \ge \kappa^{\mathrm{vol}}_j(p).$$

The lower bound on $\kappa^{\mathrm{vol}}_j(p)$ and the upper bound on $|B(x, R^{\mathrm{gate}}_{j,p})|$ from the proof of Corollary 8.75 therefore give the displayed average-mass lower bound. □

Proposition 8.78 (Near-half-logarithmic root-ball density criterion)**.** *Let* G *have maximum degree at most* $\Delta \ge 2$, *suppose that* $\overline{\chi}_p \to \infty$ *as* $p \uparrow p_c$, *and suppose that there are constants* $C_{\mathrm{vol}} < \infty$ *and* $1 < \omega < \Delta$ *such that*

$$|B(u, R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. *Suppose that for every* $0 < \epsilon \le 1$ *there exist* $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, *integers* $J, H \ge 0$, *and* $p_1 < p_c$ *satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that the following two exclusions hold. First, no $p \in [p_1, p_c)$ *admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and*

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p.$$

Second, for every $p \in [p_1, p_c)$, *every vertex* x, *and every integer* $j > J + H$ *with*

$$B \log_\Delta \overline{\chi}_p < 2^j \le \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p,$$

if

$$L = \log_\Delta \overline{\chi}_p, \qquad R_{j,p}^{\mathrm{gate}} = 2^{j+1} + \lfloor \theta L \rfloor,$$

then

$$\frac{1}{|B(x, R_{j,p}^{\mathrm{gate}})|} \sum_{a \in B(x, R_{j,p}^{\mathrm{gate}})} \tau_p(x,a) < D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ *is the constant from Corollary 8.75. Then*

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G *is connected, locally finite, quasi-transitive, and*

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. We argue by contraposition. Suppose that

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.77, and choose the parameters from the assumed criterion for this ϵ. Take $p_0 \geq p_1$ sufficiently close to p_{c} that Corollary 8.77 applies. The corollary gives expanded gate-annulus data satisfying either the super-half-logarithmic scale condition, contradicting the first exclusion, or the displayed root-ball mass lower bound. In the latter case the inherited weighted separated-fan conclusion gives $2^j > B \log_\Delta \overline{\chi}_p$, so the second exclusion applies and gives the contradiction. Hence $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Corollary 8.79 (Near-half-logarithmic root-annulus mass obstruction)**.** *Under the graph and volume assumptions of Corollary 8.69, suppose that* $\alpha_\omega = \log_\Delta \omega > 0$*. There is* $0 < \epsilon \leq 1$ *such that for every* $\lambda > 0$*, every* $\eta > 0$*, every* $B > 0$*, every* $\theta > 0$ *satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$*, and all* p_0 *sufficiently close to* p_{c}*, there are data satisfying all conclusions of Corollary 8.68 such that, with*

$$L = \log_\Delta \overline{\chi}_p, \qquad U_\lambda = \frac{1}{2\alpha_\omega} + \lambda, \qquad r_p = \lfloor \theta L \rfloor, \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

at least one of the following alternatives holds:

$$2^j > U_\lambda L,$$

or

$$\frac{\sum_{\substack{a \in V \\ d(x,a) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(x,a)}{|\{a \in V : d(x,a) \in I_{j,p}^{\mathrm{gate}}\}|} \geq D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ *is the constant from Corollary 8.75.*

Proof. Use the same data as in Corollary 8.77. If $2^j > U_\lambda L$, there is nothing to prove, so assume $2^j \le U_\lambda L$. Put

$$S_{j,p}(x) = \{a \in V : d(x,a) \in I_{j,p}^{\mathrm{gate}}\}.$$

The expanded gate-annulus obstruction gives $A = \{a_z : z \in Z^\sharp\} \subseteq S_{j,p}(x)$, and the proof of Corollary 8.77 gives

$$\sum_{a \in A} \tau_p(x,a) \ge \kappa_j^{\mathrm{vol}}(p).$$

Hence $S_{j,p}(x)$ is non-empty and

$$\sum_{a \in S_{j,p}(x)} \tau_p(x,a) \ge \kappa_j^{\mathrm{vol}}(p).$$

Since $S_{j,p}(x) \subseteq B(x, 2^{j+1} + r_p)$, the same volume bound used in Corollary 8.77 gives

$$|S_{j,p}(x)| \le C_{\mathrm{vol}} \overline{\chi}_p^{(2U_\lambda+\theta)\alpha_\omega}.$$

Dividing by this upper bound and using the lower bound on $\kappa_j^{\mathrm{vol}}(p)$ from the capped case gives the displayed annular average lower bound. □

Proposition 8.80 (Near-half-logarithmic root-annulus density criterion)**.** *Let G have maximum degree at most $\Delta \ge 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 < \omega < \Delta$ such that*

$$|B(u,R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. Suppose that for every $0 < \epsilon \le 1$ there exist $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, integers $J, H \ge 0$, and $p_1 < p_c$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that the following two exclusions hold. First, no $p \in [p_1, p_c)$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p.$$

Second, for every $p \in [p_1, p_c)$, every vertex x, and every integer $j > J + H$ with

$$B \log_\Delta \overline{\chi}_p < 2^j \le \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p,$$

if

$$L = \log_\Delta \overline{\chi}_p, \qquad I_{j,p}^{\mathrm{gate}} = \left(2^j - \lfloor \theta L \rfloor, 2^{j+1} + \lfloor \theta L \rfloor\right],$$

then either $\{a \in V : d(x,a) \in I_{j,p}^{\mathrm{gate}}\}$ is empty or

$$\frac{\sum_{\substack{a \in V \\ d(x,a) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(x,a)}{|\{a \in V : d(x,a) \in I_{j,p}^{\mathrm{gate}}\}|} < D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ is the constant from Corollary 8.75. Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p\uparrow p_c}(p_c-p)\overline{\chi}_p<\infty,$$

then

$$p_c(G)<p_{2\to 2}(G).$$

Proof. We argue by contraposition. Suppose that

$$\limsup_{p\uparrow p_c}\frac{\mathcal{R}_p^*}{\overline{\chi}_p}>0.$$

Let ϵ be the constant supplied by Corollary 8.79, and choose the parameters from the assumed criterion for this ϵ. Take $p_0\geq p_1$ sufficiently close to p_c that Corollary 8.79 applies. The corollary gives expanded gate-annulus data satisfying either the super-half-logarithmic scale condition, contradicting the first exclusion, or the displayed root-annulus mass lower bound. In the latter case the inherited weighted separated-fan conclusion gives $2^j>B\log_\Delta\overline{\chi}_p$, so the second exclusion applies and gives the contradiction. Hence $\mathcal{R}_p^*/\overline{\chi}_p\to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Corollary 8.81 (Near-half-logarithmic root-sphere mass obstruction)**.** *Under the graph and volume assumptions of Corollary 8.69, suppose that $\alpha_\omega=\log_\Delta\omega>0$. There is $0<\epsilon\leq 1$ such that for every $\lambda>0$, every $\eta>0$, every $B>0$, every $\theta>0$ satisfying*

$$2B\alpha_\omega<1,\qquad \theta<B,\qquad 1-4\theta\alpha_\omega>0,$$

every pair of integers $J,H\geq 0$, and all p_0 sufficiently close to p_c, there are data satisfying all conclusions of Corollary 8.68 such that, with

$$L=\log_\Delta\overline{\chi}_p,\qquad U_\lambda=\frac{1}{2\alpha_\omega}+\lambda,\qquad r_p=\lfloor\theta L\rfloor,\qquad I_{j,p}^{\mathrm{gate}}=(2^j-r_p,2^{j+1}+r_p],$$

at least one of the following alternatives holds:

$$2^j>U_\lambda L,$$

or there is an integer $n\in I_{j,p}^{\mathrm{gate}}\cap\mathbb{N}$ such that

$$\{a\in V:d(x,a)=n\}\neq\varnothing$$

and

$$\frac{\sum_{\substack{a\in V\\ d(x,a)=n}}\tau_p(x,a)}{|\{a\in V:d(x,a)=n\}|}\geq D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol},\omega}}\,L^{-3\eta}\,\overline{\chi}_p^{-2\lambda\alpha_\omega-2\theta-5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol},\omega}}$ is the constant from Corollary 8.75.

Proof. Apply Corollary 8.79. If the super-half-logarithmic scale alternative fails, put

$$S_n(x)=\{a\in V:d(x,a)=n\},\qquad \mathcal{N}=I_{j,p}^{\mathrm{gate}}\cap\mathbb{N}.$$

The annulus $\{a:d(x,a)\in I_{j,p}^{\mathrm{gate}}\}$ is a disjoint union of the non-empty sets $S_n(x)$, $n\in\mathcal{N}$. Its average root mass is the corresponding size-weighted average of the exact-sphere averages. Since the annular average is at least the displayed threshold, one exact sphere must have average at least the same threshold. □

Proposition 8.82 (Near-half-logarithmic root-sphere density criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 < \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that the following two exclusions hold. First, no $p \in [p_1, p_{\mathrm{c}})$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda \right) \log_\Delta \overline{\chi}_p.$$

Second, for every $p \in [p_1, p_{\mathrm{c}})$, every vertex x, and every integer $j > J + H$ with

$$B \log_\Delta \overline{\chi}_p < 2^j \leq \left(\frac{1}{2\alpha_\omega} + \lambda \right) \log_\Delta \overline{\chi}_p,$$

put

$$L = \log_\Delta \overline{\chi}_p, \qquad I_{j,p}^{\mathrm{gate}} = \left(2^j - \lfloor \theta L \rfloor , 2^{j+1} + \lfloor \theta L \rfloor \right].$$

Then, for every $n \in I_{j,p}^{\mathrm{gate}} \cap \mathbb{N}$, either $\{a \in V : d(x,a) = n\}$ is empty or

$$\frac{\sum_{\substack{a \in V \\ d(x,a)=n}} \tau_p(x,a)}{|\{a \in V : d(x,a) = n\}|} < D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ is the constant from Corollary 8.75. Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. We argue by contraposition. Suppose that

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.81, and choose the parameters from the assumed criterion for this ϵ. Take $p_0 \geq p_1$ sufficiently close to p_{c} that Corollary 8.81 applies. The corollary gives expanded gate-annulus data satisfying either the super-half-logarithmic scale condition, contradicting the first exclusion, or an exact-sphere average lower bound. In the latter case the inherited weighted separated-fan conclusion gives $2^j > B \log_\Delta \overline{\chi}_p$, so the second exclusion applies and gives the contradiction. Hence $\mathcal{R}_p^* / \overline{\chi}_p \to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Definition 8.83 (Spherical-average two-point profile)**.** For $n \geq 0$, define

$$\bar{\tau}_p^{\mathrm{sph}}(n) = \sup_{\substack{x \in V \\ S_n(x) \neq \varnothing}} \frac{1}{|S_n(x)|} \sum_{a \in S_n(x)} \tau_p(x,a), \qquad S_n(x) = \{a \in V : d(x,a) = n\}.$$

If $S_n(x) = \varnothing$ for every $x \in V$, we set $\bar{\tau}_p^{\mathrm{sph}}(n) = 0$.

Corollary 8.84 (Near-half-logarithmic spherical-profile obstruction)**.** *Under the graph and volume assumptions of Corollary 8.69, suppose that $\alpha_\omega = \log_\Delta \omega > 0$. There is $0 < \epsilon \leq 1$ such that for every $\lambda > 0$, every $\eta > 0$, every $B > 0$, every $\theta > 0$ satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are data satisfying all conclusions of Corollary 8.68 such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad U_\lambda = \frac{1}{2\alpha_\omega} + \lambda, \qquad r_p = \lfloor \theta L \rfloor, \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

at least one of the following alternatives holds:

$$2^j > U_\lambda L,$$

or there is an integer $n \in I_{j,p}^{\mathrm{gate}} \cap \mathbb{N}$ such that

$$\bar{\tau}_p^{\mathrm{sph}}(n) \geq D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ is the constant from Corollary 8.75.

Proof. This is an immediate consequence of Corollary 8.81 and Definition 8.83. □

Proposition 8.85 (Near-half-logarithmic spherical-profile criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 < \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, integers $J, H \geq 0$, and $p_1 < p_c$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that the following two exclusions hold. First, no $p \in [p_1, p_c)$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p.$$

Second, for every $p \in [p_1, p_c)$ and every integer $j > J + H$ satisfying

$$B \log_\Delta \overline{\chi}_p < 2^j \leq \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p,$$

put

$$L = \log_\Delta \overline{\chi}_p, \qquad I_{j,p}^{\mathrm{gate}} = \left(2^j - \lfloor \theta L \rfloor, 2^{j+1} + \lfloor \theta L \rfloor\right].$$

Then, for every $n \in I_{j,p}^{\mathrm{gate}} \cap \mathbb{N}$,

$$\bar{\tau}_p^{\mathrm{sph}}(n) < D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ *is the constant from Corollary 8.75. Then*

$$\lim_{p\uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G *is connected, locally finite, quasi-transitive, and*

$$\limsup_{p\uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. We argue by contraposition. Suppose that

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.84, and choose the parameters from the assumed criterion for this ϵ. Take $p_0 \geq p_1$ sufficiently close to p_{c} that Corollary 8.84 applies. The corollary gives expanded gate-annulus data satisfying either the super-half-logarithmic scale condition, contradicting the first exclusion, or a spherical-profile lower bound. In the latter case the inherited weighted separated-fan conclusion gives $2^j > B \log_\Delta \overline{\chi}_p$, so the second exclusion applies and gives the contradiction. Hence $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Corollary 8.86 (Dyadic-free near-half-logarithmic spherical-profile criterion)**.** *Let* G, Δ, ω, C_{vol}, *and* α_ω *satisfy the assumptions of Proposition 8.85. Suppose that for every* $0 < \epsilon \leq 1$ *there exist* $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, *integers* $J, H \geq 0$, *and* $p_1 < p_{\mathrm{c}}$ *satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that no $p \in [p_1, p_{\mathrm{c}})$ *admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and*

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p.$$

Assume also that, for every $p \in [p_1, p_{\mathrm{c}})$, *with*

$$L = \log_\Delta \overline{\chi}_p, \qquad A_{\lambda,\theta} = 2\lambda\alpha_\omega + 2\theta + 5\theta\alpha_\omega,$$

every integer n *satisfying*

$$(B - \theta)L < n \leq \left(\frac{1}{\alpha_\omega} + 2\lambda + \theta\right) L$$

obeys

$$\bar{\tau}_p^{\mathrm{sph}}(n) < D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-A_{\lambda,\theta}},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ is the constant from Corollary 8.75. Then

$$\lim_{p\uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p\uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. It suffices to verify the second exclusion in Proposition 8.85. Let

$$BL < 2^j \le \left(\frac{1}{2\alpha_\omega} + \lambda\right) L$$

and let $n \in I_{j,p}^{\mathrm{gate}} \cap \mathbb{N}$. Since $r_p = \lfloor \theta L \rfloor \le \theta L$, the lower endpoint gives

$$n > 2^j - r_p > (B - \theta)L.$$

The upper endpoint gives

$$n \le 2^{j+1} + r_p \le \left(\frac{1}{\alpha_\omega} + 2\lambda + \theta\right) L.$$

Thus the dyadic-free profile assumption implies the second exclusion in Proposition 8.85. The conclusion follows from that proposition. □

Corollary 8.87 (Remote spherical-profile exponent criterion)**.** *Let G, Δ, ω, C_{vol}, and α_ω satisfy the assumptions of Proposition 8.85. Suppose that for every $0 < \epsilon \le 1$ there exist $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, integers $J, H \ge 0$, and $p_1 < p_{\mathrm{c}}$ satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that no $p \in [p_1, p_{\mathrm{c}})$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda\right) \log_\Delta \overline{\chi}_p.$$

For $p < p_{\mathrm{c}}$, put $L = \log_\Delta \overline{\chi}_p$,

$$A_{\lambda,\theta} = 2\lambda\alpha_\omega + 2\theta + 5\theta\alpha_\omega,$$

and

$$\Phi_p(B, \lambda, \theta) = \sup\left\{ \bar{\tau}_p^{\mathrm{sph}}(n) : n \in \mathbb{N},\ (B - \theta)L < n \le \left(\frac{1}{\alpha_\omega} + 2\lambda + \theta\right) L \right\},$$

with the convention that the supremum over an empty set is 0. Assume that

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta \Phi_p(B, \lambda, \theta)}{\log_\Delta \overline{\chi}_p} < -A_{\lambda,\theta},$$

where $\log_\Delta 0 = -\infty$. Then

$$\lim_{p\uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p\uparrow p_c}(p_c-p)\overline{\chi}_p<\infty,$$

then

$$p_c(G)<p_{2\to 2}(G).$$

Proof. Let $D=D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ be the constant from Corollary 8.75. The strict limsup hypothesis gives $\delta>0$ and, after increasing p_1, the bound

$$\Phi_p(B,\lambda,\theta)\le \overline{\chi}_p^{-A_{\lambda,\theta}-\delta}\qquad (p\in[p_1,p_c)).$$

Since $\overline{\chi}_p\to\infty$, we may increase p_1 again so that

$$\overline{\chi}_p^{-\delta}<DL^{-3\eta}\qquad (p\in[p_1,p_c)).$$

Thus the dyadic-free profile bound required by Corollary 8.86 holds. Applying that corollary gives the result. □

Corollary 8.88 (Half-to-full logarithmic spherical-profile criterion)**.** *Let G, Δ, ω, C_{vol}, and α_ω satisfy the assumptions of Proposition 8.85. Suppose that for every $0<\epsilon\le 1$ there exist*

$$0<s<\frac{1}{6\alpha_\omega},\qquad \eta>0,\qquad J,H\ge 0,\qquad 0<p_1<p_c,$$

such that no $p\in[p_1,p_c)$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j>\left(\frac{1}{2\alpha_\omega}+s\right)\log_\Delta\overline{\chi}_p.$$

Assume also that, with $L=\log_\Delta\overline{\chi}_p$,

$$\Psi_p(s)=\sup\left\{\bar{\tau}_p^{\mathrm{sph}}(n):n\in\mathbb{N},\ \left(\frac{1}{2\alpha_\omega}-3s\right)L<n\le\left(\frac{1}{\alpha_\omega}+3s\right)L\right\},$$

where the empty supremum is 0, satisfies

$$\limsup_{p\uparrow p_c}\frac{\log_\Delta\Psi_p(s)}{\log_\Delta\overline{\chi}_p}<-(2+7\alpha_\omega)s,$$

with $\log_\Delta 0=-\infty$. Then

$$\lim_{p\uparrow p_c}\frac{\mathcal{R}_p^*}{\overline{\chi}_p}=0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p\uparrow p_c}(p_c-p)\overline{\chi}_p<\infty,$$

then

$$p_c(G)<p_{2\to 2}(G).$$

Proof. Apply Corollary 8.87 with

$$\lambda = s, \qquad \theta = s, \qquad B = \frac{1}{2\alpha_\omega} - 2s.$$

The restriction $s < 1/(6\alpha_\omega)$ gives $B > 0$, $\theta < B$, $2B\alpha_\omega < 1$, and $1 - 4\theta\alpha_\omega > 0$. The remote interval in Corollary 8.87 becomes

$$\left(\frac{1}{2\alpha_\omega} - 3s\right) L < n \le \left(\frac{1}{\alpha_\omega} + 3s\right) L,$$

and its exponent is

$$2s\alpha_\omega + 2s + 5s\alpha_\omega = (2 + 7\alpha_\omega)s.$$

Thus the hypotheses are exactly those of the remote exponent criterion with this parameter choice. □

Remark 8.89 (Geodesic baseline for the half-to-full profile target)**.** Assume that G is connected and infinite. For every $p < p_c$, every root x, and every $a \in S_n(x)$, the event that one fixed geodesic from x to a is open gives

$$\tau_p(x, a) \ge p^n.$$

Consequently

$$\bar{\tau}_p^{\mathrm{sph}}(n) \ge p^n \qquad (n \ge 0),$$

and the half-to-full window in Corollary 8.88 necessarily satisfies

$$\liminf_{p \uparrow p_c} \frac{\log_\Delta \Psi_p(s)}{\log_\Delta \overline{\chi}_p} \ge \left(\frac{1}{2\alpha_\omega} - 3s\right) \log_\Delta p_c.$$

Thus the exponent target in that corollary is not ruled out by the geodesic baseline whenever

$$(2 + 7\alpha_\omega)s < -\left(\frac{1}{2\alpha_\omega} - 3s\right) \log_\Delta p_c.$$

Since $p_c < 1$ on nonamenable quasi-transitive graphs, this compatibility condition holds for all sufficiently small positive s, for instance for

$$s < \frac{-\log_\Delta p_c}{2\alpha_\omega\,(2 + 7\alpha_\omega - 3\log_\Delta p_c)}.$$

This does not prove the required upper bound; it only records that the spherical-average formulation avoids the immediate geodesic obstruction that afflicts strict pointwise-supremum criteria.

Corollary 8.90 (Geodesic-normalized half-to-full profile criterion)**.** *Let G, Δ, ω, C_{vol}, and α_ω satisfy the assumptions of Proposition 8.85, and suppose that G is connected and infinite. Suppose that for every $0 < \epsilon \le 1$ there exist*

$$0 < s < \frac{1}{6\alpha_\omega}, \qquad \eta > 0, \qquad J, H \ge 0, \qquad 0 < p_1 < p_c,$$

such that no $p \in [p_1, p_c)$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + s\right) \log_\Delta \overline{\chi}_p.$$

With $L = \log_\Delta \overline{\chi}_p$, *put*

$$c_s = \frac{1}{2\alpha_\omega} - 3s$$

and

$$\mathfrak{E}_p^{\mathrm{sph}}(s) = \sup\left\{ \frac{\bar{\tau}_p^{\mathrm{sph}}(n)}{p^n} : n \in \mathbb{N},\ c_s L < n \le \left(\frac{1}{\alpha_\omega} + 3s\right) L \right\},$$

with the convention that the empty supremum is 0. *Assume that*

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{E}_p^{\mathrm{sph}}(s)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2 + 7\alpha_\omega)s,$$

where $\log_\Delta 0 = -\infty$. *Then*

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G *is locally finite and quasi-transitive and*

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. For every n in the defining window of $\Psi_p(s)$, we have $n > c_s L$. Since $p < 1$,

$$p^n \le p^{c_s L},$$

and hence

$$\Psi_p(s) \le p^{c_s L} \mathfrak{E}_p^{\mathrm{sph}}(s).$$

Taking logarithms, using $p \uparrow p_{\mathrm{c}}$, and applying the assumed strict excess bound gives

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\log_\Delta \Psi_p(s)}{\log_\Delta \overline{\chi}_p} < -(2 + 7\alpha_\omega)s.$$

The conclusion follows from Corollary 8.88. □

Definition 8.91 (Spherical averaged self-avoiding envelope)**.** For $0 < p \le 1$ and $n \ge 0$, define

$$\mathfrak{S}_p^{\mathrm{saw,sph}}(n) = \sup_{\substack{x \in V \\ S_n(x) \neq \varnothing}} \frac{1}{|S_n(x)|} \sum_{a \in S_n(x)} \sum_{\gamma : x \to a \text{ SAW}} p^{|\gamma| - n},$$

where the inner sum is over finite self-avoiding paths from x to a, including the trivial path when $x = a$, and may take the value $+\infty$. If $S_n(x) = \varnothing$ for every x, we set $\mathfrak{S}_p^{\mathrm{saw,sph}}(n) = 0$.

Lemma 8.92 (Self-avoiding envelope controls the normalized spherical profile)**.** *Let* G *be connected, and let* $0 < p \le 1$. *For every* $n \ge 0$,

$$\frac{\bar{\tau}_p^{\mathrm{sph}}(n)}{p^n} \le \mathfrak{S}_p^{\mathrm{saw,sph}}(n).$$

Proof. Fix x with $S_n(x) \neq \varnothing$ and $a \in S_n(x)$. On the event $\{x \leftrightarrow a\}$, the open cluster contains an open self-avoiding path from x to a. The union bound gives

$$\tau_p(x,a) \leq \sum_{\gamma : x \to a \text{ SAW}} p^{|\gamma|}.$$

Divide by p^n, average over $a \in S_n(x)$, and take the supremum over x. □

Corollary 8.93 (Self-avoiding half-to-full envelope criterion)**.** *Let G, Δ, ω, C_{vol}, and α_ω satisfy the assumptions of Proposition 8.85, and suppose that G is connected and infinite. Suppose that for every $0 < \epsilon \leq 1$ there exist*

$$0 < s < \frac{1}{6\alpha_\omega}, \qquad \eta > 0, \qquad J, H \geq 0, \qquad 0 < p_1 < p_{\mathrm{c}},$$

such that no $p \in [p_1, p_{\mathrm{c}})$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + s \right) \log_\Delta \overline{\chi}_p.$$

With $L = \log_\Delta \overline{\chi}_p$, put

$$c_s = \frac{1}{2\alpha_\omega} - 3s$$

and

$$\mathfrak{S}_p^{\mathrm{saw}}(s) = \sup \left\{ \mathfrak{S}_p^{\mathrm{saw,sph}}(n) : n \in \mathbb{N},\ c_s L < n \leq \left(\frac{1}{\alpha_\omega} + 3s \right) L \right\},$$

where the empty supremum is 0. Assume that

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{S}_p^{\mathrm{saw}}(s)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2 + 7\alpha_\omega)s,$$

where $\log_\Delta 0 = -\infty$. Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is locally finite and quasi-transitive and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. By Lemma 8.92, the quantity $\mathfrak{E}_p^{\mathrm{sph}}(s)$ from Corollary 8.90 is at most $\mathfrak{S}_p^{\mathrm{saw}}(s)$. The stated exponent hypothesis therefore implies the exponent hypothesis of that corollary. □

Definition 8.94 (Spherical excess-length self-avoiding counts)**.** For $n, k \geq 0$, define

$$\mathfrak{N}_{n,k}^{\mathrm{saw,sph}} = \sup_{\substack{x \in V \\ S_n(x) \neq \varnothing}} \frac{1}{|S_n(x)|} \sum_{a \in S_n(x)} \#\{\gamma : x \to a \text{ SAW} : |\gamma| = n + k\},$$

with value 0 if $S_n(x) = \varnothing$ for every x.

Lemma 8.95 (Excess-length upper bound for the spherical SAW envelope)**.** *For every $0 < p \le 1$ and $n \ge 0$,*

$$\mathfrak{S}_p^{\mathrm{saw,sph}}(n) \le \sum_{k\ge 0} p^k\, \mathfrak{N}_{n,k}^{\mathrm{saw,sph}},$$

where both sides may be $+\infty$.

Proof. If $d(x,a) = n$, every self-avoiding path from x to a has length $n+k$ for some $k \ge 0$. For fixed x, grouping the inner sum in Definition 8.91 by this excess length gives

$$\frac{1}{|S_n(x)|}\sum_{a\in S_n(x)}\sum_{\gamma:x\to a\ \mathrm{SAW}} p^{|\gamma|-n} = \sum_{k\ge 0} p^k \frac{1}{|S_n(x)|}\sum_{a\in S_n(x)} \#\{\gamma : x\to a \text{ SAW} : |\gamma| = n+k\}.$$

Taking the supremum over x on the left and then bounding each summand by the corresponding supremum gives the stated upper bound. □

Corollary 8.96 (q-weighted excess-length SAW criterion)**.** *Let G, Δ, ω, C_{vol}, and α_ω satisfy the assumptions of Proposition 8.85, and suppose that G is connected and infinite. Suppose that for every $0 < \epsilon \le 1$ there exist*

$$0 < s < \frac{1}{6\alpha_\omega}, \qquad \eta > 0, \qquad J, H \ge 0, \qquad 0 < p_1 < p_{\mathrm{c}}, \qquad q \in (p_{\mathrm{c}}, 1),$$

such that no $p \in [p_1, p_{\mathrm{c}})$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + s\right)\log_\Delta \overline{\chi}_p.$$

With $L = \log_\Delta \overline{\chi}_p$, put

$$c_s = \frac{1}{2\alpha_\omega} - 3s$$

and

$$\mathfrak{Q}_q^{\mathrm{saw}}(s,p) = \sup\left\{\sum_{k\ge 0} q^k\, \mathfrak{N}_{n,k}^{\mathrm{saw,sph}} : n\in\mathbb{N},\ c_s L < n \le \left(\frac{1}{\alpha_\omega} + 3s\right)L\right\},$$

where the empty supremum is 0. Assume that

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{Q}_q^{\mathrm{saw}}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2 + 7\alpha_\omega)s,$$

where $\log_\Delta 0 = -\infty$. Then

$$\lim_{p\uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is locally finite and quasi-transitive and

$$\limsup_{p\uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Since $p < p_{\mathrm{c}} < q$, Lemma 8.95 gives

$$\mathfrak{S}_p^{\mathrm{saw,sph}}(n) \le \sum_{k\ge 0} q^k\, \mathfrak{N}_{n,k}^{\mathrm{saw,sph}}.$$

Thus $\mathfrak{S}_p^{\mathrm{saw}}(s)$ from Corollary 8.93 is at most $\mathfrak{Q}_q^{\mathrm{saw}}(s,p)$, and the result follows from that corollary. □

Corollary 8.97 (Finite-excess/tail SAW criterion)**.** *Let G, Δ, ω, C_{vol}, and α_ω satisfy the assumptions of Proposition 8.85, and suppose that G is connected and infinite. Suppose that for every $0 < \epsilon \le 1$ there exist*

$$0 < s < \frac{1}{6\alpha_\omega}, \qquad \eta > 0, \qquad J, H \ge 0, \qquad 0 < p_1 < p_{\mathrm{c}}, \qquad q \in (p_{\mathrm{c}}, 1), \qquad \rho > 0,$$

such that no $p \in [p_1, p_{\mathrm{c}})$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + s\right) \log_\Delta \overline{\chi}_p.$$

With $L = \log_\Delta \overline{\chi}_p$, put

$$c_s = \frac{1}{2\alpha_\omega} - 3s, \qquad K_{\rho,p} = \lfloor \rho L \rfloor.$$

Define

$$\mathfrak{A}_{q,\rho}^{\mathrm{saw}}(s,p) = \sup\left\{ q^k\, \mathfrak{N}_{n,k}^{\mathrm{saw,sph}} : n \in \mathbb{N},\ c_s L < n \le \left(\frac{1}{\alpha_\omega} + 3s\right) L,\ 0 \le k \le K_{\rho,p} \right\}$$

and

$$\mathfrak{T}_{q,\rho}^{\mathrm{saw}}(s,p) = \sup\left\{ \sum_{k > K_{\rho,p}} q^k\, \mathfrak{N}_{n,k}^{\mathrm{saw,sph}} : n \in \mathbb{N},\ c_s L < n \le \left(\frac{1}{\alpha_\omega} + 3s\right) L \right\},$$

where empty suprema and empty sums are 0. Assume that both

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{A}_{q,\rho}^{\mathrm{saw}}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2 + 7\alpha_\omega)s$$

and

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{T}_{q,\rho}^{\mathrm{saw}}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2 + 7\alpha_\omega)s$$

hold, with $\log_\Delta 0 = -\infty$. Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is locally finite and quasi-transitive and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. For every n in the half-to-full logarithmic window,

$$\sum_{k\geq 0} q^k\, \mathfrak{N}^{\mathrm{saw,sph}}_{n,k} \leq (K_{\rho,p}+1)\mathfrak{A}^{\mathrm{saw}}_{q,\rho}(s,p) + \mathfrak{T}^{\mathrm{saw}}_{q,\rho}(s,p).$$

The assumptions of Proposition 8.85 include $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, so $L \to \infty$ and

$$\lim_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta(K_{\rho,p}+1)}{L} = 0.$$

The two displayed exponent assumptions therefore imply the exponent hypothesis of Corollary 8.96. Applying that corollary gives the conclusion. □

Corollary 8.98 (Binned finite-excess SAW criterion)**.** *In the setting of Corollary 8.97, the exponent hypothesis on $\mathfrak{A}^{\mathrm{saw}}_{q,\rho}(s,p)$ may be replaced by the following finite family of sufficient hypotheses. Let $\delta > 0$, set*

$$m_{\rho,\delta} = \left\lfloor \frac{\rho}{\delta} \right\rfloor + 1,$$

and, for $0 \leq i < m_{\rho,\delta}$, define

$$\mathfrak{B}^{\mathrm{saw}}_{i,\delta}(s,p) = \sup\left\{ \mathfrak{N}^{\mathrm{saw,sph}}_{n,k} : n\in\mathbb{N},\ c_s L < n \leq \left(\frac{1}{\alpha_\omega}+3s\right)L,\ i\delta L \leq k < (i+1)\delta L \right\},$$

with empty supremum equal to 0. If the long-excess hypothesis on $\mathfrak{T}^{\mathrm{saw}}_{q,\rho}(s,p)$ from Corollary 8.97 holds and, for every $0 \leq i < m_{\rho,\delta}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{B}^{\mathrm{saw}}_{i,\delta}(s,p)}{\log_\Delta \overline{\chi}_p} + i\delta \log_\Delta q < -c_s \log_\Delta p_{\mathrm{c}} - (2+7\alpha_\omega)s,$$

then the conclusions of Corollary 8.97 hold.

Proof. If $i\delta L \leq k < (i+1)\delta L$, then $q^k \leq q^{i\delta L}$, since $q < 1$. Hence

$$\mathfrak{A}^{\mathrm{saw}}_{q,\rho}(s,p) \leq \max_{0\leq i<m_{\rho,\delta}} q^{i\delta L}\mathfrak{B}^{\mathrm{saw}}_{i,\delta}(s,p).$$

The number $m_{\rho,\delta}$ is independent of p. The displayed binned exponent inequalities therefore imply the exponent hypothesis on $\mathfrak{A}^{\mathrm{saw}}_{q,\rho}(s,p)$ in Corollary 8.97; the long-excess hypothesis is assumed. □

Remark 8.99 (Degree counting cannot control the long-excess tail)**.** The long-excess term in Corollary 8.97 cannot be verified by the raw bounded-degree count of self-avoiding paths. Indeed, for $n+k \geq 1$,

$$\mathfrak{N}^{\mathrm{saw,sph}}_{n,k} \leq \Delta(\Delta-1)^{n+k-1}.$$

Using this estimate in the q-weighted tail would require

$$q(\Delta-1) < 1$$

to obtain a geometric summation in k. But the standard tree-counting susceptibility bound gives $p_{\mathrm{c}}(G) \geq 1/(\Delta-1)$, and the criterion uses $q > p_{\mathrm{c}}$. Hence $q(\Delta-1) > 1$ in the relevant regime. Any proof of the long-excess hypothesis must therefore use the fixed exact sphere, endpoint constraints, averaging over $S_n(x)$, or additional geometry; ambient degree growth alone is too coarse.

Definition 8.100 (Radial-defect spherical SAW counts)**.** Let $\gamma = (v_0, \dots, v_m)$ be a path with $v_0 = x$. Define

$$I_x(\gamma) = \#\{1 \le i \le m : d(x, v_i) = d(x, v_{i-1}) - 1\}$$

and

$$H_x(\gamma) = \#\{1 \le i \le m : d(x, v_i) = d(x, v_{i-1})\}.$$

For $n, b, h \ge 0$, set

$$\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h} = \sup_{\substack{x \in V \\ S_n(x) \neq \varnothing}} \frac{1}{|S_n(x)|} \sum_{a \in S_n(x)} \#\{\gamma : x \to a \text{ SAW} : I_x(\gamma) = b,\ H_x(\gamma) = h\},$$

with value 0 if $S_n(x) = \varnothing$ for every x.

Lemma 8.101 (Radial-defect decomposition of excess length)**.** *For every $n, k \ge 0$,*

$$\mathfrak{N}^{\mathrm{saw,sph}}_{n,k} \le \sum_{\substack{b,h \ge 0 \\ 2b+h=k}} \mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h}.$$

Proof. Fix x with $S_n(x) \neq \varnothing$, and let $\gamma = (v_0, \dots, v_m)$ be a self-avoiding path from x to $a \in S_n(x)$. Let $O_x(\gamma)$ be the number of steps increasing the distance from x by one. Since adjacent vertices have distances from x that differ by at most one,

$$m = O_x(\gamma) + I_x(\gamma) + H_x(\gamma)$$

and, because $d(x, a) = n$,

$$n = O_x(\gamma) - I_x(\gamma).$$

Thus

$$m - n = 2I_x(\gamma) + H_x(\gamma).$$

Consequently every length-$n+k$ self-avoiding path from x to $S_n(x)$ contributes to exactly one pair (b, h) with $2b + h = k$. After averaging over $S_n(x)$, taking the supremum over x, and bounding the supremum of the sum by the sum of suprema, the displayed inequality follows. □

Corollary 8.102 (Radial-defect weighted SAW criterion)**.** *In the setting of Corollary 8.96, the exponent hypothesis on $\mathfrak{Q}^{\mathrm{saw}}_q(s,p)$ may be replaced by the following radial-defect hypothesis. With L and c_s as in that corollary, define*

$$\mathfrak{U}^{\mathrm{rad,saw}}_q(s,p) = \sup\left\{ \sum_{b,h \ge 0} q^{2b+h} \mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h} : n \in \mathbb{N},\ c_s L < n \le \left(\frac{1}{\alpha_\omega} + 3s\right) L \right\},$$

where the empty supremum is 0. If

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{U}^{\mathrm{rad,saw}}_q(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2 + 7\alpha_\omega)s,$$

with $\log_\Delta 0 = -\infty$, then the conclusions of Corollary 8.96 hold.

Proof. Lemma 8.101 gives, for each admissible n,

$$\sum_{k\ge 0} q^k\, \mathfrak{N}^{\mathrm{saw,sph}}_{n,k} \le \sum_{b,h\ge 0} q^{2b+h}\, \mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h}.$$

Thus $\mathfrak{Q}^{\mathrm{saw}}_q(s,p) \le \mathfrak{U}^{\mathrm{rad,saw}}_q(s,p)$, and the stated hypothesis implies the hypothesis of Corollary 8.96. □

Corollary 8.103 (Inward-restricted horizontal radial tail split)**.** *In the setting of Corollary 8.97, the exponent hypothesis on $\mathfrak{T}^{\mathrm{saw}}_{q,\rho}(s,p)$ may be replaced by an inward tail and a low-inward horizontal tail. With L, c_s, and $K_{\rho,p}$ as in that corollary, define*

$$\mathfrak{I}^{\mathrm{rad,saw}}_{q,\rho}(s,p) = \sup\left\{ \sum_{\substack{b>K_{\rho,p}/4\\ h\ge 0}} q^{2b+h}\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h} : n\in\mathbb{N},\ c_sL < n \le \left(\frac{1}{\alpha_\omega}+3s\right)L \right\}$$

and

$$\mathfrak{H}^{\mathrm{rad,saw}}_{q,\rho}(s,p) = \sup\left\{ \sum_{\substack{0\le b\le K_{\rho,p}/4\\ h>K_{\rho,p}/2}} q^{2b+h}\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h} : n\in\mathbb{N},\ c_sL < n \le \left(\frac{1}{\alpha_\omega}+3s\right)L \right\},$$

where empty suprema and empty sums are 0*. If the exponent hypothesis on $\mathfrak{A}^{\mathrm{saw}}_{q,\rho}(s,p)$ from Corollary 8.97 holds and both*

$$\limsup_{p\uparrow p_c} \frac{\log_\Delta \mathfrak{I}^{\mathrm{rad,saw}}_{q,\rho}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_c - (2+7\alpha_\omega)s$$

and

$$\limsup_{p\uparrow p_c} \frac{\log_\Delta \mathfrak{H}^{\mathrm{rad,saw}}_{q,\rho}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_c - (2+7\alpha_\omega)s$$

hold, with $\log_\Delta 0 = -\infty$, then the conclusions of Corollary 8.97 hold.

Proof. Let $K = K_{\rho,p}$. For every admissible n, Lemma 8.101 gives

$$\sum_{k>K} q^k \mathfrak{N}^{\mathrm{saw,sph}}_{n,k} \le \sum_{\substack{b,h\ge 0\\ 2b+h>K}} q^{2b+h}\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h}.$$

If $2b+h>K$, then either $b>K/4$, or else $b\le K/4$ and $h>K/2$. Therefore

$$\mathfrak{T}^{\mathrm{saw}}_{q,\rho}(s,p) \le \mathfrak{I}^{\mathrm{rad,saw}}_{q,\rho}(s,p) + \mathfrak{H}^{\mathrm{rad,saw}}_{q,\rho}(s,p).$$

The two displayed exponent hypotheses imply the exponent hypothesis on $\mathfrak{T}^{\mathrm{saw}}_{q,\rho}(s,p)$ in Corollary 8.97. The finite-excess hypothesis is assumed, so that corollary applies. □

Lemma 8.104 (No horizontal radial steps in bipartite graphs)**.** *If G is bipartite, then $H_x(\gamma)=0$ for every path γ and every starting vertex x. Consequently*

$$\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,h} = 0 \qquad (h\ge 1).$$

Proof. Let $V = V_0 \cup V_1$ be a bipartition, and fix x. After interchanging the colour classes if necessary, assume $x \in V_0$. Every vertex at even distance from x lies in V_0, and every vertex at odd distance from x lies in V_1. Adjacent vertices have opposite colours, so their distances from x have opposite parity. Since adjacent distances from x differ by at most one, they cannot be equal. Hence no step of any path keeps the distance from x fixed. □

Corollary 8.105 (Bipartite inward-tail criterion)**.** *In the setting of Corollary 8.97, assume in addition that G is bipartite. If the exponent hypothesis on $\mathfrak{A}^{\mathrm{saw}}_{q,\rho}(s,p)$ from that corollary holds and*

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{I}^{\mathrm{rad,saw}}_{q,\rho}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2+7\alpha_\omega)s,$$

with $\log_\Delta 0 = -\infty$, then the conclusions of Corollary 8.97 hold.

Proof. By Lemma 8.104, $\mathfrak{H}^{\mathrm{rad,saw}}_{q,\rho}(s,p) = 0$. Thus the hypotheses of Corollary 8.103 hold. □

Corollary 8.106 (Exact bipartite inward-tail criterion)**.** *In the setting of Corollary 8.97, assume in addition that G is bipartite. With L, c_s, and $K_{\rho,p}$ as in that corollary, define*

$$\mathfrak{L}^{\mathrm{in,saw}}_{q,\rho}(s,p) = \sup\left\{ \sum_{b>K_{\rho,p}/2} q^{2b}\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,0} : n\in\mathbb{N},\ c_sL < n \le \left(\frac{1}{\alpha_\omega}+3s\right)L \right\},$$

where empty suprema and empty sums are 0. If the exponent hypothesis on $\mathfrak{A}^{\mathrm{saw}}_{q,\rho}(s,p)$ from Corollary 8.97 holds and

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{L}^{\mathrm{in,saw}}_{q,\rho}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2+7\alpha_\omega)s,$$

with $\log_\Delta 0 = -\infty$, then the conclusions of Corollary 8.97 hold.

Proof. By Lemma 8.104, only the terms with $h = 0$ can contribute. Lemma 8.101 therefore gives, for every admissible n,

$$\sum_{k>K_{\rho,p}} q^k \mathfrak{N}^{\mathrm{saw,sph}}_{n,k} \le \sum_{b>K_{\rho,p}/2} q^{2b}\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,0}.$$

Thus the displayed exponent hypothesis implies the exponent hypothesis on $\mathfrak{T}^{\mathrm{saw}}_{q,\rho}(s,p)$ in Corollary 8.97. The finite-excess hypothesis is assumed, so that corollary applies. □

Corollary 8.107 (One-parameter bipartite inward criterion)**.** *In the setting of Corollary 8.97, assume in addition that G is bipartite. With L, c_s, and $K_{\rho,p}$ as in that corollary, define*

$$\mathfrak{F}^{\mathrm{in,saw}}_{q,\rho}(s,p) = \sup\left\{ q^{2b}\mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,0} : n\in\mathbb{N},\ c_sL < n \le \left(\frac{1}{\alpha_\omega}+3s\right)L,\ 0\le b\le K_{\rho,p}/2 \right\}$$

where empty suprema are 0, and let $\mathfrak{L}^{\mathrm{in,saw}}_{q,\rho}(s,p)$ be the tail profile from Corollary 8.106. If

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{F}^{\mathrm{in,saw}}_{q,\rho}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2+7\alpha_\omega)s$$

and

$$\limsup_{p\uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{L}^{\mathrm{in,saw}}_{q,\rho}(s,p)}{\log_\Delta \overline{\chi}_p} < -c_s \log_\Delta p_{\mathrm{c}} - (2+7\alpha_\omega)s$$

hold, with $\log_\Delta 0 = -\infty$, then the conclusions of Corollary 8.97 hold.

Proof. By Lemma 8.104, only $h = 0$ can occur. Lemma 8.101 gives, for each admissible n and each $0 \le k \le K_{\rho,p}$,

$$q^k \mathfrak{N}^{\mathrm{saw,sph}}_{n,k} \le \sum_{\substack{b \ge 0 \\ 2b = k}} q^{2b} \mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,0} \le \mathfrak{F}^{\mathrm{in,saw}}_{q,\rho}(s,p).$$

Thus the displayed exponent hypothesis on $\mathfrak{F}^{\mathrm{in,saw}}_{q,\rho}$ implies the finite-excess hypothesis on $\mathfrak{A}^{\mathrm{saw}}_{q,\rho}$. The displayed exponent hypothesis on $\mathfrak{L}^{\mathrm{in,saw}}_{q,\rho}$ implies the tail hypothesis by Corollary 8.106's proof. Applying Corollary 8.97 gives the conclusion. □

Corollary 8.108 (Binned bipartite finite inward criterion)**.** *In the setting of Corollary 8.107, the exponent hypothesis on $\mathfrak{F}^{\mathrm{in,saw}}_{q,\rho}(s,p)$ may be replaced by the following finite family of sufficient hypotheses. Let $\delta > 0$, set*

$$m^{\mathrm{in}}_{\rho,\delta} = \left\lfloor \frac{\rho}{2\delta} \right\rfloor + 1,$$

and, for $0 \le i < m^{\mathrm{in}}_{\rho,\delta}$, define

$$\mathfrak{C}^{\mathrm{in,saw}}_{i,\delta}(s,p) = \sup \left\{ \mathfrak{M}^{\mathrm{rad,saw,sph}}_{n,b,0} : n \in \mathbb{N},\ c_s L < n \le \left(\frac{1}{\alpha_\omega} + 3s \right) L,\ i\delta L \le b < (i+1)\delta L \right\},$$

with empty suprema equal to 0. If the exponent hypothesis on $\mathfrak{L}^{\mathrm{in,saw}}_{q,\rho}(s,p)$ from Corollary 8.107 holds and, for every $0 \le i < m^{\mathrm{in}}_{\rho,\delta}$,

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\log_\Delta \mathfrak{C}^{\mathrm{in,saw}}_{i,\delta}(s,p)}{\log_\Delta \overline{\chi}_p} + 2i\delta \log_\Delta q < -c_s \log_\Delta p_{\mathrm{c}} - (2 + 7\alpha_\omega)s,$$

then the conclusions of Corollary 8.97 hold.

Proof. If $0 \le b \le K_{\rho,p}/2$, then $b \le \rho L/2$, so b belongs to one of the bins $0 \le i < m^{\mathrm{in}}_{\rho,\delta}$. In the ith bin, $q^{2b} \le q^{2i\delta L}$, since $q < 1$. Hence

$$\mathfrak{F}^{\mathrm{in,saw}}_{q,\rho}(s,p) \le \max_{0 \le i < m^{\mathrm{in}}_{\rho,\delta}} q^{2i\delta L} \mathfrak{C}^{\mathrm{in,saw}}_{i,\delta}(s,p).$$

The number $m^{\mathrm{in}}_{\rho,\delta}$ is independent of p. The displayed binned exponent hypotheses therefore imply the exponent hypothesis on $\mathfrak{F}^{\mathrm{in,saw}}_{q,\rho}(s,p)$. The tail hypothesis is assumed, so Corollary 8.107 applies. □

Proposition 8.109 (Subdivision reduction to bipartite graphs)**.** *Let G be a connected locally finite graph of maximum degree at most $\Delta < \infty$, and suppose that G is infinite. Let $\mathrm{Sub}(G)$ be the graph obtained by subdividing every edge of G once. Then*

$$p_{\mathrm{c}}(\mathrm{Sub}(G)) = \sqrt{p_{\mathrm{c}}(G)}$$

and

$$p_{2\to 2}(\mathrm{Sub}(G)) = \sqrt{p_{2\to 2}(G)}.$$

Consequently,

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G) \quad \Longleftrightarrow \quad p_{\mathrm{c}}(\mathrm{Sub}(G)) < p_{2\to 2}(\mathrm{Sub}(G)).$$

Moreover, if G is quasi-transitive and nonamenable, then $\mathrm{Sub}(G)$ is bipartite, quasi-transitive, and nonamenable.

Proof. Write $H = \mathrm{Sub}(G)$. Its vertex set is $V(G) \cup E(G)$, where an edge-vertex $e \in E(G)$ is adjacent in H to the two endpoints of e. In Bernoulli-p percolation on H, an original edge $e = \{u, v\}$ is open as a two-step connection between u and v exactly when both subdivided edges incident to e are open. These events are independent over $e \in E(G)$ and have probability p^2. Thus the open clusters of old vertices in $H[p]$ have the same law as the open clusters of $G[p^2]$. Since every infinite cluster of $H[p]$ contains infinitely many old vertices, this proves

$$p_{\mathrm{c}}(H) = \sqrt{p_{\mathrm{c}}(G)}.$$

It remains to compare the L^2 thresholds. Let $q = p^2$, and let $T_G(q)$ and $T_H(p)$ be the two-point operators on G and H. For old vertices $x, y \in V(G)$,

$$\tau_p^H(x, y) = \tau_q^G(x, y).$$

Let $R : \ell^2(E(G)) \to \ell^2(V(G))$ be the incidence operator

$$(Rf)(v) = \sum_{e \ni v} f(e).$$

Since G has maximum degree at most Δ and each edge has two endpoints, R is bounded. If $e \in E(G)$ is viewed as a midpoint, then

$$\tau_p^H(x, e) \le p \sum_{z \in e} \tau_q^G(x, z)$$

and, for midpoints e, f,

$$\tau_p^H(e, f) \le \mathbf{1}_{\{e=f\}} + p^2 \sum_{\substack{z \in e \\ w \in f}} \tau_q^G(z, w).$$

These inequalities give the block domination

$$T_H(p) \preceq \begin{pmatrix} T_G(q) & pT_G(q)R \\ pR^* T_G(q) & I + p^2 R^* T_G(q) R \end{pmatrix}$$

in the sense of positive kernels. Hence $T_G(q)$ bounded on $\ell^2(V(G))$ implies $T_H(p)$ bounded on $\ell^2(V(H))$. Conversely, if $T_H(p)$ is bounded, then restricting it to functions supported on old vertices and then projecting back to old vertices gives $T_G(q)$, so $T_G(q)$ is bounded. Therefore

$$p < p_{2\to 2}(H) \quad \Longleftrightarrow \quad p^2 < p_{2\to 2}(G),$$

which proves the claimed identity for $p_{2\to 2}$.

The graph H is bipartite with parts $V(G)$ and $E(G)$. If G is quasi-transitive, then H is quasi-transitive because a locally finite quasi-transitive graph has finitely many edge orbits. Finally, G and H are bounded-degree quasi-isometric graphs, so nonamenability is preserved. □

Lemma 8.110 (Subdivision susceptibility comparison)**.** *Let G be a connected infinite graph of maximum degree at most $\Delta < \infty$, let $H = \mathrm{Sub}(G)$, and put $q = p^2$. There is a constant $C_\Delta < \infty$ such that, for every $0 \le p < 1$,*

$$\overline{\chi}_q^G \le \overline{\chi}_p^H \le C_\Delta \big(1 + \overline{\chi}_q^G\big).$$

Consequently, if

$$\limsup_{q \uparrow p_{\mathrm{c}}(G)} (p_{\mathrm{c}}(G) - q) \overline{\chi}_q^G < \infty,$$

then

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} (p_{\mathrm{c}}(H) - p) \overline{\chi}_p^H < \infty.$$

Proof. The lower bound follows by restricting the susceptibility in H to old vertices and using $\tau_p^H(x,y) = \tau_q^G(x,y)$ for old vertices x, y.

For the upper bound, first let x be an old vertex. The old-vertex contribution to $\chi_p^H(x)$ is $\chi_q^G(x)$. For a midpoint $e \in E(G)$,

$$\tau_p^H(x,e) \le p \sum_{z \in e} \tau_q^G(x,z),$$

so the midpoint contribution is at most

$$p \sum_{e \in E(G)} \sum_{z \in e} \tau_q^G(x,z) \le p\Delta\, \chi_q^G(x).$$

Thus $\chi_p^H(x) \le (1+\Delta)\overline{\chi}_q^G$.

Now let $e \in E(G)$ be a midpoint. For old vertices y,

$$\tau_p^H(e,y) \le p \sum_{z \in e} \tau_q^G(z,y),$$

so the old-vertex contribution is at most $2p\overline{\chi}_q^G$. For midpoint vertices f,

$$\tau_p^H(e,f) \le \mathbf{1}_{\{e=f\}} + p^2 \sum_{\substack{z \in e \\ w \in f}} \tau_q^G(z,w),$$

and summing over $f \in E(G)$ gives at most

$$1 + p^2\Delta \sum_{z \in e} \chi_q^G(z) \le 1 + 2\Delta\overline{\chi}_q^G.$$

Combining the two cases gives the comparison.

Since $p_c(H) = \sqrt{p_c(G)}$ by Proposition 8.109, the final implication follows from

$$p_c(G) - p^2 = (p_c(H) - p)(p_c(H) + p)$$

and the fact that $p_c(H) > 0$ for bounded-degree graphs. □

Lemma 8.111 (Even-sphere subdivision radial identity)**.** *Let G be connected and locally finite, let $H = \mathrm{Sub}(G)$, and let $x \in V(G)$. For every self-avoiding path $\pi = (x_0, \dots, x_\ell)$ in G with $x_0 = x$, let $\widehat{\pi}$ be its canonical lift to H. Then*

$$I_x^H(\widehat{\pi}) = 2I_x^G(\pi) + H_x^G(\pi).$$

Consequently, for every $n, B \ge 0$ such that $S_n^G(x) \ne \varnothing$,

$$\begin{aligned} &\frac{1}{|S_{2n}^H(x)|} \sum_{y \in S_{2n}^H(x)} \#\{\gamma : x \to y \text{ SAW in } H : I_x^H(\gamma) = B\} \\ &\quad = \frac{1}{|S_n^G(x)|} \sum_{y \in S_n^G(x)} \sum_{\substack{i,h \ge 0 \\ 2i+h=B}} \#\{\pi : x \to y \text{ SAW in } G : I_x^G(\pi) = i,\ H_x^G(\pi) = h\}. \end{aligned}$$

In particular, if the supremum defining $\mathfrak{M}_{2n,B,0}^{H,\mathrm{rad,saw,sph}}$ is restricted to old roots, then the resulting even-sphere quantity is at most

$$\sum_{\substack{i,h \ge 0 \\ 2i+h=B}} \mathfrak{M}_{n,i,h}^{G,\mathrm{rad,saw,sph}}.$$

Proof. Every path in H from an old vertex to an old vertex has even length and alternates old vertices and midpoints. Projecting such a path to its old vertices gives a path in G, and self-avoidance in H implies self-avoidance of the projected old-vertex path. Conversely, every self-avoiding path in G has a unique canonical lift to a self-avoiding path in H. Thus canonical lift is a bijection between self-avoiding paths in G from x to y and self-avoiding paths in H from the old vertex x to the old vertex y.

For an old vertex $z \in V(G)$,

$$d_H(x, z) = 2d_G(x, z).$$

If $e = \{u, v\}$ is a midpoint of H, then

$$d_H(x, e) = 2\min\{d_G(x, u), d_G(x, v)\} + 1.$$

Consider one step $u \to v$ of π, and write $r = d_G(x, u)$. If $d_G(x, v) = r+1$, its lifted two-step segment has no inward step in H. If $d_G(x, v) = r$, the lifted segment first moves outward to the midpoint and then inward to v, contributing one inward step. If $d_G(x, v) = r - 1$, both lifted half-steps are inward, contributing two inward steps. Summing over the steps of π gives

$$I_x^H(\widehat{\pi}) = 2I_x^G(\pi) + H_x^G(\pi).$$

Finally, $S_{2n}^H(x) = S_n^G(x)$ as sets of old vertices, while Lemma 8.104 gives no horizontal radial steps in H. The displayed average identity follows from the bijection and the inward-count identity. Taking the supremum over old roots and then bounding the supremum of the sum by the sum of suprema gives the final inequality. □

Lemma 8.112 (Odd-sphere old-root subdivision bound)**.** *Let G have maximum degree at most Δ, let $H = \mathrm{Sub}(G)$, fix $x \in V(G)$, and define*

$$A_{m,i,h}^G(x) = \frac{1}{|S_m^G(x)|} \sum_{y \in S_m^G(x)} \#\{\pi : x \to y \text{ SAW in } G : I_x^G(\pi) = i,\ H_x^G(\pi) = h\}.$$

Suppose that every vertex in $S_n^G(x)$ has a neighbour in $S_{n+1}^G(x)$. Then, for every $B \geq 0$,

$$\begin{aligned}\frac{1}{|S_{2n+1}^H(x)|} \sum_{e \in S_{2n+1}^H(x)} &\#\{\gamma : x \to e \text{ SAW in } H : I_x^H(\gamma) = B\} \\ &\leq \Delta \sum_{\substack{i,h \geq 0 \\ 2i+h=B}} A_{n,i,h}^G(x) + \Delta \sum_{\substack{i,h \geq 0 \\ 2i+h=B-1}} A_{n+1,i,h}^G(x),\end{aligned}$$

where the second sum is 0 when $B = 0$. In particular, under the same local no-dead-end assumption, the old-root odd-sphere quantity is at most

$$\Delta \sum_{\substack{i,h \geq 0 \\ 2i+h=B}} \mathfrak{M}_{n,i,h}^{G,\mathrm{rad,saw,sph}} + \Delta \sum_{\substack{i,h \geq 0 \\ 2i+h=B-1}} \mathfrak{M}_{n+1,i,h}^{G,\mathrm{rad,saw,sph}}.$$

Proof. Let

$$E_n(x) = \{e = \{u, v\} \in E(G) : \min(d_G(x, u), d_G(x, v)) = n\}.$$

The midpoint vertices $E_n(x)$ are exactly $S_{2n+1}^H(x)$. Since every vertex of $S_n^G(x)$ has an edge to $S_{n+1}^G(x)$, choosing one such edge for each vertex gives $|E_n(x)| \geq |S_n^G(x)|$. Also every vertex of $S_{n+1}^G(x)$ has an edge to $S_n^G(x)$ along a geodesic to x, so $|E_n(x)| \geq |S_{n+1}^G(x)|$.

Let γ be a self-avoiding path in H from x to a midpoint $e \in S^H_{2n+1}(x)$, and let z be the old vertex immediately preceding e. Removing the final half-edge and projecting the remaining old-to-old path gives a self-avoiding path π in G from x to z. Since $e \in S^H_{2n+1}(x)$, the vertex z lies either in $S^G_n(x)$ or in $S^G_{n+1}(x)$. In the first case the final half-edge is outward and

$$I^H_x(\gamma) = 2I^G_x(\pi) + H^G_x(\pi).$$

In the second case the final half-edge is inward and

$$I^H_x(\gamma) = 2I^G_x(\pi) + H^G_x(\pi) + 1.$$

For each fixed projected path π, there are at most Δ choices for the final midpoint e. Therefore the odd-sphere numerator is bounded by

$$\Delta \sum_{z \in S^G_n(x)} \sum_{\substack{i,h \geq 0 \\ 2i+h=B}} \#\{\pi : x \to z : I^G_x(\pi) = i,\ H^G_x(\pi) = h\}$$

$$+\Delta \sum_{z \in S^G_{n+1}(x)} \sum_{\substack{i,h \geq 0 \\ 2i+h=B-1}} \#\{\pi : x \to z : I^G_x(\pi) = i,\ H^G_x(\pi) = h\}.$$

Dividing the first term by $|E_n(x)| \geq |S^G_n(x)|$ and the second by $|E_n(x)| \geq |S^G_{n+1}(x)|$ gives the displayed bound. The final inequality follows by taking the appropriate suprema in the definition of $\mathfrak{M}^{G,\mathrm{rad,saw,sph}}$. □

Corollary 8.113 (Nonamenable old-root odd-sphere subdivision bound)**.** *In Lemma 8.112, the local no-dead-end assumption may be replaced by $h(G) > 0$. More precisely, if G has maximum degree at most Δ and $h(G) > 0$, then, for every old root $x \in V(G)$ and every $n, B \geq 0$,*

$$\frac{1}{|S^H_{2n+1}(x)|} \sum_{e \in S^H_{2n+1}(x)} \#\{\gamma : x \to e \text{ SAW in } H : I^H_x(\gamma) = B\}$$

$$\leq \frac{\Delta}{h(G)} \sum_{\substack{i,h \geq 0 \\ 2i+h=B}} \mathfrak{M}^{G,\mathrm{rad,saw,sph}}_{n,i,h} + \Delta \sum_{\substack{i,h \geq 0 \\ 2i+h=B-1}} \mathfrak{M}^{G,\mathrm{rad,saw,sph}}_{n+1,i,h},$$

where the second sum is 0 when $B = 0$.

Proof. Use the notation $E_n(x)$ from the proof of Lemma 8.112. The numerator bound from that proof does not use the local no-dead-end assumption. Since $E_n(x)$ contains the edge boundary of the ball $B^G_n(x)$, nonamenability gives

$$|E_n(x)| \geq |\partial_E B^G_n(x)| \geq h(G) \sum_{z \in B^G_n(x)} \deg(z) \geq h(G)|S^G_n(x)|.$$

Also $|E_n(x)| \geq |S^G_{n+1}(x)|$, because each vertex in $S^G_{n+1}(x)$ has an edge to $S^G_n(x)$ along a geodesic to x. Dividing the two terms in the numerator bound by these two lower bounds gives the claim. □

Lemma 8.114 (Midpoint-root subdivision bound under shifted sphere comparison)**.** *Let $H = \mathrm{Sub}(G)$, let $r \in E(G)$ be a midpoint of H, and let $u_0, u_1 \in V(G)$ be its two neighbouring old vertices. Fix $m \geq 1$ and $C \geq 1$, and suppose that*

$$|S^H_{m-1}(u_j)| \leq C|S^H_m(r)| \quad \text{and} \quad |S^H_{m+1}(u_j)| \leq C|S^H_m(r)| \qquad (j = 0, 1).$$

Then, for every $B \geq 0$,

$$\frac{1}{|S_m^H(r)|} \sum_{a \in S_m^H(r)} \#\{\gamma : r \to a \text{ SAW in } H : I_r^H(\gamma) = B\}$$
$$\leq C \sum_{j=0}^{1} \left(\mathfrak{M}_{m-1,B,0}^{H,\mathrm{rad,saw,sph}} + \mathfrak{M}_{m+1,B-1,0}^{H,\mathrm{rad,saw,sph}} \right),$$

where the term with $B-1$ is interpreted as 0 when $B = 0$.

Proof. Let γ be a self-avoiding path in H from r to $a \in S_m^H(r)$, and suppose its first step is to u_j. Delete this first half-edge to obtain a self-avoiding path γ' from u_j to a. Since $d_H(r, u_j) = 1$, the distance $d_H(u_j, a)$ is either $m-1$ or $m+1$. Writing $|\gamma|$ for the length of γ, and using that H is bipartite, we have

$$I_r^H(\gamma) = \frac{|\gamma| - m}{2}$$

and

$$I_{u_j}^H(\gamma') = \frac{|\gamma| - 1 - d_H(u_j, a)}{2}.$$

Thus $I_{u_j}^H(\gamma') = I_r^H(\gamma)$ when $d_H(u_j, a) = m-1$, and $I_{u_j}^H(\gamma') = I_r^H(\gamma) - 1$ when $d_H(u_j, a) = m+1$.

The map $\gamma \mapsto (j, \gamma')$ is injective after recording the first old vertex u_j. Therefore the numerator in the displayed average is at most the sum, over $j = 0, 1$, of the old-root counts from u_j to $S_{m-1}^H(u_j)$ with inward count B, plus the old-root counts from u_j to $S_{m+1}^H(u_j)$ with inward count $B-1$. Dividing by $|S_m^H(r)|$, using the shifted sphere comparisons, and bounding the old-root averages by the corresponding global $\mathfrak{M}^H$ profiles gives the claim. □

Lemma 8.115 (Cheeger shifted-sphere comparison)**.** *Let X be a connected locally finite graph of maximum degree at most $D < \infty$, and suppose that $h(X) > 0$. If $a \sim b$, then, for every $m \geq 1$,*

$$|S_{m-1}^X(a)| \leq \frac{D^4}{h(X)} |S_m^X(b)|$$

and

$$|S_{m+1}^X(a)| \leq \frac{D^4}{h(X)} |S_m^X(b)|.$$

Consequently, Lemma 8.114 applies in every bounded-degree nonamenable subdivision graph H, with

$$C = \frac{D_H^4}{h(H)}$$

where D_H is the maximum degree of H.

Proof. For every $r \geq 0$ and $v \in V(X)$, nonamenability applied to the finite ball $B_r^X(v)$ gives

$$h(X)|B_r^X(v)| \leq |\partial_E B_r^X(v)| \leq D|S_{r+1}^X(v)|.$$

Thus

$$|B_r^X(v)| \leq \frac{D}{h(X)} |S_{r+1}^X(v)|.$$

Since $a \sim b$,

$$S_{m-1}^X(a) \subseteq B_m^X(b) \qquad \text{and} \qquad S_{m+1}^X(a) \subseteq B_{m+2}^X(b).$$

Also, for $k \geq 0$, each vertex of $S^X_{k+1}(b)$ has a geodesic predecessor in $S^X_k(b)$, and each vertex has at most D neighbours, so

$$|S^X_{k+1}(b)| \leq D|S^X_k(b)|.$$

Combining these estimates gives

$$|S^X_{m-1}(a)| \leq |B^X_m(b)| \leq \frac{D}{h(X)}|S^X_{m+1}(b)| \leq \frac{D^2}{h(X)}|S^X_m(b)|$$

and

$$|S^X_{m+1}(a)| \leq |B^X_{m+2}(b)| \leq \frac{D}{h(X)}|S^X_{m+3}(b)| \leq \frac{D^4}{h(X)}|S^X_m(b)|.$$

The common constant $D^4/h(X)$ proves the two displayed inequalities. □

Corollary 8.116 (Restricted subdivision radial-profile comparison)**.** *Let G be connected and locally finite with maximum degree at most $\Delta < \infty$, let $H = \mathrm{Sub}(G)$, and suppose that $h(G) > 0$ and $h(H) > 0$. Put*

$$C_H = \frac{D_H^4}{h(H)},$$

where D_H is the maximum degree of H. For $\ell, B \geq 0$, let $\mathfrak{M}^{H,\mathrm{old}}_{\ell,B}$ and $\mathfrak{M}^{H,\mathrm{mid}}_{\ell,B}$ denote the quantities obtained from $\mathfrak{M}^{H,\mathrm{rad,saw,sph}}_{\ell,B,0}$ by restricting the root supremum to old vertices and midpoint vertices, respectively. Set these quantities equal to 0 when $B < 0$. Then

$$\mathfrak{M}^{H,\mathrm{rad,saw,sph}}_{\ell,B,0} = \max\{\mathfrak{M}^{H,\mathrm{old}}_{\ell,B}, \mathfrak{M}^{H,\mathrm{mid}}_{\ell,B}\}.$$

Moreover, for every $n, B \geq 0$,

$$\mathfrak{M}^{H,\mathrm{old}}_{2n,B} \leq \sum_{\substack{i,h\geq 0\\ 2i+h=B}} \mathfrak{M}^{G,\mathrm{rad,saw,sph}}_{n,i,h},$$

and

$$\mathfrak{M}^{H,\mathrm{old}}_{2n+1,B} \leq \frac{\Delta}{h(G)} \sum_{\substack{i,h\geq 0\\ 2i+h=B}} \mathfrak{M}^{G,\mathrm{rad,saw,sph}}_{n,i,h} + \Delta \sum_{\substack{i,h\geq 0\\ 2i+h=B-1}} \mathfrak{M}^{G,\mathrm{rad,saw,sph}}_{n+1,i,h}.$$

Finally, for every $m \geq 1$ and $B \geq 0$,

$$\mathfrak{M}^{H,\mathrm{mid}}_{m,B} \leq 2C_H \left(\mathfrak{M}^{H,\mathrm{old}}_{m-1,B} + \mathfrak{M}^{H,\mathrm{old}}_{m+1,B-1}\right).$$

Proof. Since H is bipartite with parts $V(G)$ and $E(G)$, every root is either old or a midpoint, and Lemma 8.104 gives $h = 0$ for the radial profile on H. This proves the displayed maximum identity.

The even old-root bound is the final inequality of Lemma 8.111. The odd old-root bound is Corollary 8.113. For midpoint roots, Lemma 8.115 verifies the shifted sphere hypothesis in Lemma 8.114 with $C = C_H$. Restricting the old-root averages appearing in its proof by the old-root supremum gives

$$\mathfrak{M}^{H,\mathrm{mid}}_{m,B} \leq C_H \sum_{j=0}^{1} \left(\mathfrak{M}^{H,\mathrm{old}}_{m-1,B} + \mathfrak{M}^{H,\mathrm{old}}_{m+1,B-1}\right),$$

which is the claimed bound. □

Definition 8.117 (Collapsed radial-defect profiles)**.** For $n \geq 0$ and $d \in \mathbb{Z}$, define

$$\mathfrak{R}_{n,d}^{G,\mathrm{rad,saw,sph}} = 0 \qquad (d < 0),$$

and, for $d \geq 0$, define

$$\mathfrak{R}_{n,d}^{G,\mathrm{rad,saw,sph}} = \sum_{\substack{i,j \geq 0 \\ 2i+j=d}} \mathfrak{M}_{n,i,j}^{G,\mathrm{rad,saw,sph}},$$

Thus d is the original excess defect $2I + H$, with the two radial coordinates summed out.

Definition 8.118 (Property-restricted collapsed radial profiles)**.** Let $\mathcal{P}$ be a property of rooted paths. For $n \geq 0$ and $d \in \mathbb{Z}$, define

$$\mathfrak{R}_{n,d}^{G,\mathcal{P}} = 0 \qquad (d < 0),$$

and, for $d \geq 0$, define

$$\mathfrak{R}_{n,d}^{G,\mathcal{P}} = \sup_{\substack{x \in V \\ S_n(x) \neq \varnothing}} \frac{1}{|S_n(x)|} \sum_{a \in S_n(x)} \#\{\gamma : x \to a \text{ SAW} : 2I_x(\gamma) + H_x(\gamma) = d,\ \mathcal{P}(x,\gamma)\}.$$

The unrestricted profile $\mathfrak{R}_{n,d}^{G,\mathrm{rad,saw,sph}}$ corresponds to the always-true property.

Corollary 8.119 (Weighted subdivision-to-original profile comparison)**.** *In the setting of Corollary 8.116, let $0 < q < 1$, put $Q = q^2$, and set*

$$C_\star = \max\left\{1, \frac{\Delta}{h(G)}, \Delta\right\}(1 + 2C_H)\,.$$

Then, for every $m, B \geq 0$,

$$q^{2B}\mathfrak{M}_{m,B,0}^{H,\mathrm{rad,saw,sph}} \leq C_\star \sum_{\substack{r \geq 0 \\ |2r-m| \leq 2}} \sum_{a=0}^{2} Q^{B-a}\mathfrak{R}_{r,B-a}^{G,\mathrm{rad,saw,sph}}.$$

Here and below, every term with a negative defect $B - a < 0$ is interpreted as 0.

Proof. Let

$$C_0 = \max\left\{1, \frac{\Delta}{h(G)}, \Delta\right\}.$$

The even and odd old-root bounds in Corollary 8.116, together with $q^{2B} \leq Q^{B-a}$ when $a \in \{0,1\}$ and the defect is $B - a$, give

$$q^{2B}\mathfrak{M}_{m,B}^{H,\mathrm{old}} \leq C_0 \sum_{\substack{r \geq 0 \\ |2r-m| \leq 1}} \sum_{a=0}^{1} Q^{B-a}\mathfrak{R}_{r,B-a}^{G,\mathrm{rad,saw,sph}}.$$

For midpoint roots, Corollary 8.116 gives

$$\mathfrak{M}_{m,B}^{H,\mathrm{mid}} \leq 2C_H\left(\mathfrak{M}_{m-1,B}^{H,\mathrm{old}} + \mathfrak{M}_{m+1,B-1}^{H,\mathrm{old}}\right).$$

Multiplying by q^{2B}, using

$$q^{2B} \leq q^{2(B-1)}$$

for the second term when $B \geq 1$, and applying the old-root estimate to the two shifted radii yields the same right-hand side with $|2r - m| \leq 2$ and $a \in \{0,1,2\}$, multiplied by $2C_HC_0$. The $B = 0$ second term is zero by convention. Since the full H-profile is the maximum of the old and midpoint restricted profiles, the displayed bound follows. □

Corollary 8.120 (Subdivision inward profiles from original radial profiles)**.** *Let G, H, q, Q, and $C_\star$ be as in Corollary 8.119. Let D_H be the maximum degree of H, let α_ω be the volume-growth parameter used when applying the bipartite inward criterion to H, and for $p < p_c(H)$ put*

$$L_H = \log_{D_H} \overline{\chi}_p^H, \qquad K_{\rho,p}^H = \lfloor \rho L_H \rfloor, \qquad c_s = \frac{1}{2\alpha_\omega} - 3s.$$

Define the shifted original finite profile

$$\mathfrak{F}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p) = \sup\left\{ Q^d \mathfrak{R}_{r,d}^{G,\mathrm{rad,saw,sph}} : \begin{array}{l} m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right) L_H, \\ r\ge 0,\ |2r-m|\le 2,\ 0\le d\le K_{\rho,p}^H/2 \end{array} \right\}$$

and the shifted original tail profile

$$\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p) = \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2 \\ d > K_{\rho,p}^H/2-2}} Q^d \mathfrak{R}_{r,d}^{G,\mathrm{rad,saw,sph}} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right) L_H \right\},$$

with empty suprema and sums equal to 0. Then the bipartite inward profiles of Corollary 8.107, computed in H, satisfy

$$\mathfrak{F}_{q,\rho}^{H,\mathrm{in,saw}}(s,p) \le 9C_\star\, \mathfrak{F}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$$

and

$$\mathfrak{L}_{q,\rho}^{H,\mathrm{in,saw}}(s,p) \le 3C_\star\, \mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p).$$

Consequently, the exponent hypotheses of Corollary 8.107 for H follow from the same exponent hypotheses on $\mathfrak{F}_{Q,\rho}^{G\to H,\mathrm{rad}}$ and $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}$.

Proof. For the finite profile, apply Corollary 8.119 with $0 \le B \le K_{\rho,p}^H/2$. Then every defect $d = B - a$ appearing on the right-hand side satisfies $d \le K_{\rho,p}^H/2$. There are at most three choices of a and at most three integers r with $|2r-m|\le 2$, giving the factor $9C_\star$.

For the tail profile, sum the same comparison over $B > K_{\rho,p}^H/2$. For each $a\in\{0,1,2\}$, the defect $d = B - a$ satisfies

$$d > K_{\rho,p}^H/2 - 2,$$

and each fixed d is produced by at most one $B = d + a$ for that a. Summing over a gives the factor $3C_\star$. Multiplicative constants do not affect the displayed logarithmic exponent hypotheses because $\overline{\chi}_p^H \to \infty$ as $p \uparrow p_c(H)$. □

Remark 8.121 (Subdivision inward weights match original radial weights)**.** Let $H = \mathrm{Sub}(G)$, let $0 < q < 1$, and put $Q = q^2$. The weights in the bipartite inward profiles on H are compatible with the two-coordinate radial weights on G. Indeed, in the even-sphere identity of Lemma 8.111, the relation

$$B = 2i + h$$

gives

$$q^{2B} = Q^{2i+h}.$$

In the old-root odd-sphere comparison of Lemma 8.112, the two alternatives are $B = 2i + h$ and $B = 2i + h + 1$, and hence

$$q^{2B} \le Q^{2i+h}.$$

Similarly, in Lemma 8.114, deleting the first half-edge either preserves the inward count or lowers it by one, and the factor q^{2B} is no larger than the corresponding old-root weight. Thus the subdivision radius shifts introduce constants and neighbouring radii, but no adverse power of q.

Corollary 8.122 (Subdivision one-parameter inward criterion)**.** *Let G be connected, locally finite, quasi-transitive, and nonamenable, with bounded degree, and let $H = \mathrm{Sub}(G)$. Suppose that*

$$\limsup_{p\uparrow p_c(G)} (p_c(G) - p)\overline{\chi}_p^G < \infty.$$

If the hypotheses of Corollary 8.107 are satisfied for H, then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. By Proposition 8.109, H is bipartite, quasi-transitive, and nonamenable, and $p_c(H) = \sqrt{p_c(G)}$. Lemma 8.110 transfers the mean-field susceptibility input from G to H. Applying Corollary 8.107 to H gives

$$p_c(H) < p_{2\to 2}(H).$$

The equivalence in Proposition 8.109 then yields $p_c(G) < p_{2\to 2}(G)$. □

Corollary 8.123 (Original radial-profile subdivision criterion)**.** *Let G and $H = \mathrm{Sub}(G)$ be as in Corollary 8.122. Suppose that*

$$\limsup_{p\uparrow p_c(G)} (p_c(G) - p)\overline{\chi}_p^G < \infty.$$

Assume that the non-profile hypotheses of Corollary 8.107 are satisfied for H with some $q \in (p_c(H), 1)$, and put $Q = q^2$. If the two logarithmic exponent hypotheses in that corollary, for the inward profiles on H, hold instead for the shifted original profiles

$$\mathfrak{F}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p) \quad \text{and} \quad \mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$$

from Corollary 8.120, then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.120 shows that the shifted original radial-profile exponent hypotheses imply the corresponding exponent hypotheses on the one-parameter bipartite inward profiles of H. Thus the hypotheses of Corollary 8.107 hold for H. Applying Corollary 8.122 gives the claim. □

Corollary 8.124 (Binned shifted original finite radial criterion)**.** *In the setting of Corollary 8.123, use the notation $D_H, L_H, c_s, \alpha_\omega, Q$ from Corollary 8.120. The exponent hypothesis on $\mathfrak{F}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ may be replaced by the following finite family of sufficient hypotheses. Let $\delta > 0$, set*

$$m_{\rho,\delta}^{G\to H} = \left\lfloor \frac{\rho}{2\delta} \right\rfloor + 1,$$

and define, for $0 \le i < m_{\rho,\delta}^{G\to H}$,

$$\mathfrak{D}_{i,\delta}^{G\to H,\mathrm{rad}}(s,p) = \sup\left\{ \mathfrak{R}_{r,d}^{G,\mathrm{rad,saw,sph}} : \begin{array}{l} m \in \mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H, \\ r \ge 0,\ |2r - m| \le 2,\ i\delta L_H \le d < (i+1)\delta L_H \end{array} \right\},$$

where empty suprema are 0. If the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ from Corollary 8.123 holds and, for every $0 \le i < m_{\rho,\delta}^{G\to H}$,

$$\limsup_{p\uparrow p_c(H)} \frac{\log_{D_H} \mathfrak{D}_{i,\delta}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + i\delta \log_{D_H} Q < -c_s \log_{D_H} p_c(H) - (2 + 7\alpha_\omega)s,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. If $0 \le d \le K^H_{\rho,p}/2$, then $d \le \rho L_H/2$, so d lies in one of the bins $0 \le i < m^{G\to H}_{\rho,\delta}$. In the ith bin,

$$Q^d \le Q^{i\delta L_H},$$

since $Q < 1$. Hence

$$\mathfrak{F}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p) \le \max_{0\le i<m^{G\to H}_{\rho,\delta}} Q^{i\delta L_H}\mathfrak{D}^{G\to H,\mathrm{rad}}_{i,\delta}(s,p).$$

The number of bins is independent of p. The displayed binned exponent hypotheses therefore imply the exponent hypothesis on $\mathfrak{F}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The tail exponent hypothesis is assumed, so Corollary 8.123 applies. □

Corollary 8.125 (Binned shifted original radial tail criterion)**.** *In the setting of Corollary 8.123, use the notation $D_H, L_H, c_s, \alpha_\omega, Q$ from Corollary 8.120. The exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ may be replaced by the following finite-bin plus far-tail hypotheses. Let $\lambda > \rho/2$ and $\delta > 0$, and set*

$$m^{G\to H}_{\lambda,\delta} = \left\lfloor \frac{\lambda}{\delta} \right\rfloor + 1.$$

For $0 \le i < m^{G\to H}_{\lambda,\delta}$, define

$$\mathfrak{E}^{G\to H,\mathrm{rad}}_{i,\delta}(s,p) = \sup\left\{ \mathfrak{R}^{G,\mathrm{rad,saw,sph}}_{r,d} : \begin{array}{l} m\in\mathbb{N},\ c_sL_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H, \\ r\ge 0,\ |2r-m|\le 2,\ i\delta L_H \le d < (i+1)\delta L_H \end{array} \right\},$$

and define the far-tail profile

$$\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda}(s,p) = \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathrm{rad,saw,sph}}_{r,d} : m\in\mathbb{N},\ c_sL_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H \right\}.$$

Assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le i < m^{G\to H}_{\lambda,\delta}$,

$$\limsup_{p\uparrow p_c(H)} \frac{\log_{D_H}\mathfrak{E}^{G\to H,\mathrm{rad}}_{i,\delta}(s,p)}{\log_{D_H}\overline{\chi}^H_p} + i\delta\log_{D_H}Q < -c_s\log_{D_H}p_c(H) - (2+7\alpha_\omega)s,$$

and that

$$\limsup_{p\uparrow p_c(H)} \frac{\log_{D_H}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda}(s,p)}{\log_{D_H}\overline{\chi}^H_p} < -c_s\log_{D_H}p_c(H) - (2+7\alpha_\omega)s.$$

Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Fix an admissible m. Split the sum defining $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}$ into terms with

$$d \le \lambda L_H$$

and terms with $d > \lambda L_H$. The second part is bounded by $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda}(s,p)$. The first part is covered by the finitely many bins

$$0 \le i < m^{G\to H}_{\lambda,\delta}.$$

For each bin, $Q^d \le Q^{i\delta L_H}$, and there are at most $\lfloor \delta L_H \rfloor + 2$ integer values of d in the bin and at most three integers r with $|2r-m| \le 2$. Hence the finite part is bounded by

$$3(\lfloor \delta L_H \rfloor + 2) \sum_{0 \le i < m_{\lambda,\delta}^{G\to H}} Q^{i\delta L_H}\, \mathfrak{E}_{i,\delta}^{G\to H,\mathrm{rad}}(s,p).$$

The number of bins is independent of p, and

$$\frac{\log_{D_H}(\lfloor \delta L_H \rfloor + 2)}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)).$$

The displayed assumptions therefore imply the desired logarithmic exponent bound for the shifted tail profile. □

Corollary 8.126 (Overshoot and slab localization of the shifted far tail)**.** *In the setting of Corollary 8.125, fix $\sigma, \beta > 0$. Put*

$$W_{\beta,p} = \max\{1, \lfloor \beta L_H \rfloor\}, \qquad J_{\sigma,\beta} = \left\lfloor \frac{\frac{1}{2\alpha_\omega} + \frac{3s}{2} + \sigma + 2}{\beta} \right\rfloor + 2,$$

and

$$\theta_{\lambda,\sigma,\beta} = \frac{\lambda}{J_{\sigma,\beta}}.$$

For a rooted path $\gamma = (v_0, \dots, v_\ell)$, write

$$M_x(\gamma) = \max_{0 \le t \le \ell} d(x, v_t)$$

and, for an interval $A \subseteq \mathbb{N}$,

$$\begin{aligned} D_x(\gamma; A) = &\sum_{u \in A} 2\#\{1 \le t \le \ell : d(x,v_{t-1}) = u,\ d(x,v_t) = u-1\} \\ &+ \sum_{u \in A} \#\{1 \le t \le \ell : d(x,v_{t-1}) = u,\ d(x,v_t) = u\}. \end{aligned}$$

Define the overshoot profile

$$\mathfrak{O}_{Q,\lambda,\sigma}^{G\to H,\mathrm{rad}}(s,p) = \sup\left\{ \sum_{\substack{r \ge 0,\ |2r-m| \le 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}_{r,d}^{G,\mathcal{O}_{r,\sigma,p}} : m \in \mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \right\},$$

where $\mathcal{O}_{r,\sigma,p}(x,\gamma)$ is the property

$$M_x(\gamma) > r + \sigma L_H.$$

For $0 \le j < J_{\sigma,\beta}$, define the heavy-slab profile

$$\mathfrak{V}_{Q,\lambda,\sigma,\beta,j}^{G\to H,\mathrm{rad}}(s,p) = \sup\left\{ \sum_{\substack{r \ge 0,\ |2r-m| \le 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}_{r,d}^{G,\mathcal{V}_{j,p}} : m \in \mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \right\},$$

where $\mathcal{V}_{j,p}(x,\gamma)$ is the property

$$D_x\big(\gamma;[jW_{\beta,p},(j+1)W_{\beta,p})\big)>\theta_{\lambda,\sigma,\beta}L_H.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda}(s,p)\le\mathfrak{O}^{G\to H,\mathrm{rad}}_{Q,\lambda,\sigma}(s,p)+\sum_{j=0}^{J_{\sigma,\beta}-1}\mathfrak{V}^{G\to H,\mathrm{rad}}_{Q,\lambda,\sigma,\beta,j}(s,p).$$

Consequently, the exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda}$ follows from the same exponent hypothesis on the overshoot profile and on each of the finitely many heavy-slab profiles.

Proof. Let m,r,d be in the range defining $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda}$, and let γ be a path counted by $\mathfrak{R}^{G,\mathrm{rad,saw,sph}}_{r,d}$. If $M_x(\gamma)>r+\sigma L_H$, then γ is counted by the overshoot profile. Otherwise,

$$M_x(\gamma)\le r+\sigma L_H.$$

Since $|2r-m|\le 2$ and

$$m\le\left(\frac{1}{\alpha_\omega}+3s\right)L_H,$$

all radial levels visited by γ are contained, for p close enough to $p_{\mathrm{c}}(H)$, in the union of the slabs

$$[jW_{\beta,p},(j+1)W_{\beta,p}),\qquad 0\le j<J_{\sigma,\beta}.$$

Moreover,

$$\sum_{j=0}^{J_{\sigma,\beta}-1}D_x\big(\gamma;[jW_{\beta,p},(j+1)W_{\beta,p})\big)=2I_x(\gamma)+H_x(\gamma)=d.$$

Since $d>\lambda L_H$, one of the $J_{\sigma,\beta}$ slabs has

$$D_x\big(\gamma;[jW_{\beta,p},(j+1)W_{\beta,p})\big)>\theta_{\lambda,\sigma,\beta}L_H.$$

Thus every non-overshooting path counted by the far tail is counted by at least one heavy-slab profile. Taking the union bound after averaging over spheres and then taking the displayed suprema gives the claimed inequality. The final assertion follows because $J_{\sigma,\beta}$ is independent of p. □

Lemma 8.127 (Overshoot forces inward defect)**.** *Let $\gamma=(v_0,\dots,v_\ell)$ be a path started at x, ending at a vertex $a\in S_r(x)$, and let*

$$M_x(\gamma)=\max_{0\le t\le\ell}d(x,v_t).$$

Then

$$I_x(\gamma)\ge M_x(\gamma)-r.$$

Consequently,

$$2I_x(\gamma)+H_x(\gamma)\ge 2\big(M_x(\gamma)-r\big).$$

Proof. Let t_0 be a time at which $d(x,v_{t_0})=M_x(\gamma)$. Along the suffix $(v_{t_0},\dots,v_\ell)$, the distance from x drops from $M_x(\gamma)$ to r. The number of inward steps in this suffix minus the number of outward steps in this suffix is therefore $M_x(\gamma)-r$. In particular, the suffix has at least $M_x(\gamma)-r$ inward steps, and so does the whole path. The second inequality follows from $H_x(\gamma)\ge 0$. □

Corollary 8.128 (Overshoot tail amplification)**.** *In the setting of Corollary 8.126, put*

$$\lambda' = \max\{\lambda, 2\sigma\}.$$

Let $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda'}(s,p)$ *denote the far-tail profile from Corollary 8.125 with* λ *replaced by* λ'*. Then*

$$\mathfrak{O}^{G\to H,\mathrm{rad}}_{Q,\lambda,\sigma}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda'}(s,p).$$

In particular, if $\sigma > \lambda/2$*, then the overshoot term is controlled by a strictly deeper shifted far tail.*

Proof. Every path counted by $\mathfrak{O}^{G\to H,\mathrm{rad}}_{Q,\lambda,\sigma}(s,p)$ ends at radius r, has defect $d > \lambda L_H$, and satisfies

$$M_x(\gamma) > r + \sigma L_H.$$

By Lemma 8.127,

$$d = 2I_x(\gamma) + H_x(\gamma) > 2\sigma L_H.$$

Thus $d > \lambda' L_H$. Removing the overshoot restriction can only increase the corresponding weighted sum, which gives the displayed inequality. □

Corollary 8.129 (Finite far-tail ladder)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*, and let* $\beta_0, \dots, \beta_{N-1} > 0$*. For each* $0 \le i < N$*, put*

$$\sigma_i = \frac{\lambda_{i+1}}{2}.$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p) + \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{V}^{G\to H,\mathrm{rad}}_{Q,\lambda_i,\sigma_i,\beta_i,j}(s,p).$$

Consequently, the exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ *follows from the same exponent hypothesis on the terminal far tail* $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}$ *and on each of the finitely many heavy-slab profiles in the displayed sum.*

Proof. Apply Corollary 8.126 with $(\lambda,\sigma,\beta) = (\lambda_i,\sigma_i,\beta_i)$. Since $\lambda_{i+1} > \lambda_i$ and $\sigma_i = \lambda_{i+1}/2$, Corollary 8.128 gives

$$\mathfrak{O}^{G\to H,\mathrm{rad}}_{Q,\lambda_i,\sigma_i}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_{i+1}}(s,p).$$

Thus

$$\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_i}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_{i+1}}(s,p) + \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{V}^{G\to H,\mathrm{rad}}_{Q,\lambda_i,\sigma_i,\beta_i,j}(s,p).$$

Iterating this inequality for $0 \le i < N$ gives the displayed ladder bound. The exponent conclusion follows because all sums are finite and independent of p. □

Corollary 8.130 (Heavy-slab token dichotomy)**.** *In the setting of Corollary 8.126, fix $\sigma, \beta > 0$, $0 \le j < J_{\sigma,\beta}$, and put*

$$A_{j,p} = [jW_{\beta,p}, (j+1)W_{\beta,p}).$$

For a rooted path $\gamma = (v_0, \dots, v_\ell)$, define

$$I_x(\gamma; A) = \sum_{u \in A} \#\{1 \le t \le \ell : d(x, v_{t-1}) = u,\ d(x, v_t) = u - 1\}$$

and

$$H_x(\gamma; A) = \sum_{u \in A} \#\{1 \le t \le \ell : d(x, v_{t-1}) = u,\ d(x, v_t) = u\}.$$

Thus $D_x(\gamma; A) = 2I_x(\gamma; A) + H_x(\gamma; A)$. Let $\mathcal{V}^{\mathrm{in}}_{j,p}(x, \gamma)$ be the property

$$I_x(\gamma; A_{j,p}) > \frac{\theta_{\lambda,\sigma,\beta}}{4} L_H$$

and let $\mathcal{V}^{\mathrm{hor}}_{j,p}(x, \gamma)$ be the property

$$H_x(\gamma; A_{j,p}) > \frac{\theta_{\lambda,\sigma,\beta}}{2} L_H.$$

Define the corresponding inward-token and horizontal-token profiles by

$$\mathfrak{V}^{G\to H,\mathrm{rad,in}}_{Q,\lambda,\sigma,\beta,j}(s,p) = \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{V}^{\mathrm{in}}_{j,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \right\},$$

and

$$\mathfrak{V}^{G\to H,\mathrm{rad,hor}}_{Q,\lambda,\sigma,\beta,j}(s,p) = \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{V}^{\mathrm{hor}}_{j,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \right\}.$$

Then

$$\mathfrak{V}^{G\to H,\mathrm{rad}}_{Q,\lambda,\sigma,\beta,j}(s,p) \le \mathfrak{V}^{G\to H,\mathrm{rad,in}}_{Q,\lambda,\sigma,\beta,j}(s,p) + \mathfrak{V}^{G\to H,\mathrm{rad,hor}}_{Q,\lambda,\sigma,\beta,j}(s,p).$$

Consequently, the exponent hypothesis on the heavy-slab profile follows from the same exponent hypothesis on its inward-token and horizontal-token subprofiles.

Proof. If a path satisfies $\mathcal{V}_{j,p}$, then

$$2I_x(\gamma; A_{j,p}) + H_x(\gamma; A_{j,p}) > \theta_{\lambda,\sigma,\beta} L_H.$$

If neither $\mathcal{V}^{\mathrm{in}}_{j,p}$ nor $\mathcal{V}^{\mathrm{hor}}_{j,p}$ held, then

$$2I_x(\gamma; A_{j,p}) + H_x(\gamma; A_{j,p}) \le 2 \cdot \frac{\theta_{\lambda,\sigma,\beta}}{4} L_H + \frac{\theta_{\lambda,\sigma,\beta}}{2} L_H = \theta_{\lambda,\sigma,\beta} L_H,$$

a contradiction. Thus each path counted by the heavy-slab profile is counted by at least one of the two token profiles. Taking the union bound inside each weighted sum and then taking the supremum gives the displayed inequality. Finite sums and fixed multiplicative constants do not affect the logarithmic exponent condition. □

Lemma 8.131 (Inward level tokens give exterior returns)**.** *Let $\gamma = (v_0, \dots, v_\ell)$ be a path with $v_0 = x$, and fix $u \geq 1$. For every time t such that*

$$d(x, v_{t-1}) = u, \qquad d(x, v_t) = u - 1,$$

there is a time $s < t$ such that

$$d(x, v_s) = u - 1, \qquad d(x, v_{s+1}) = u,$$

and

$$d(x, v_a) \geq u \qquad (s < a < t).$$

Moreover, for fixed u, the intervals $[s, t]$ obtained by choosing s maximal have disjoint interiors as t ranges over all inward level-u tokens.

Proof. Since $v_0 = x$ has distance $0 < u$, the path must visit level $u - 1$ before the time $t - 1$ at which it is at level u. Let $s < t$ be the last time before t at which $d(x, v_s) = u - 1$. Then v_{s+1} cannot have distance $u - 1$, by maximality of s, and it cannot have distance less than $u - 1$, since the path later reaches level u before time t and would then have to visit level $u - 1$ again. Hence $d(x, v_{s+1}) = u$. The same maximality argument gives $d(x, v_a) \geq u$ for every $s < a < t$.

If $t_1 < t_2$ are two inward level-u tokens and s_i is chosen maximal for t_i, then after time t_1 the path is back at level $u-1$. To have another inward level-u token at time t_2, it must cross from level $u-1$ to level u at some time after or at t_1. Thus $s_2 \geq t_1$, proving the disjoint-interior assertion. □

Corollary 8.132 (Inward-heavy slabs: spread or congestion)**.** *In the setting and notation of Corollary 8.130, fix an integer $K \geq 1$. Let $\mathcal{V}^{\mathrm{spr}}_{j,K,p}(x, \gamma)$ be the property*

$$\#\{u \in A_{j,p} : I_x(\gamma; \{u\}) > 0\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

and let $\mathcal{V}^{\mathrm{con}}_{j,K,p}(x, \gamma)$ be the property

$$\max_{u \in A_{j,p}} I_x(\gamma; \{u\}) > K.$$

Define $\mathfrak{V}^{G\to H,\mathrm{rad,in,spr}}_{Q,\lambda,\sigma,\beta,j,K}(s, p)$ and $\mathfrak{V}^{G\to H,\mathrm{rad,in,con}}_{Q,\lambda,\sigma,\beta,j,K}(s, p)$ from the same formula as $\mathfrak{V}^{G\to H,\mathrm{rad,in}}_{Q,\lambda,\sigma,\beta,j}(s, p)$, with $\mathcal{V}^{\mathrm{in}}_{j,p}$ replaced by $\mathcal{V}^{\mathrm{spr}}_{j,K,p}$ and $\mathcal{V}^{\mathrm{con}}_{j,K,p}$, respectively. Then

$$\mathfrak{V}^{G\to H,\mathrm{rad,in}}_{Q,\lambda,\sigma,\beta,j}(s, p) \leq \mathfrak{V}^{G\to H,\mathrm{rad,in,spr}}_{Q,\lambda,\sigma,\beta,j,K}(s, p) + \mathfrak{V}^{G\to H,\mathrm{rad,in,con}}_{Q,\lambda,\sigma,\beta,j,K}(s, p).$$

In the congested alternative, Lemma 8.131 produces more than K exterior return segments through a single level of the slab.

Proof. Write

$$N_x(\gamma; A_{j,p}) = \#\{u \in A_{j,p} : I_x(\gamma; \{u\}) > 0\}.$$

If a path satisfies $\mathcal{V}^{\mathrm{in}}_{j,p}$, then

$$\sum_{u \in A_{j,p}} I_x(\gamma; \{u\}) > \frac{\theta_{\lambda,\sigma,\beta}}{4} L_H.$$

If neither the spread nor the congestion property held, then

$$\sum_{u \in A_{j,p}} I_x(\gamma; \{u\}) \leq K\, N_x(\gamma; A_{j,p}) \leq \frac{\theta_{\lambda,\sigma,\beta}}{4} L_H,$$

contradicting the previous display. Hence every inward-token path is counted by at least one of the two subprofiles. The displayed profile inequality follows by the same union-bound argument as in Corollary 8.130. The final sentence is exactly Lemma 8.131. □

Corollary 8.133 (Tokenized finite far-tail ladder)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*. Let* $\beta_0, \dots, \beta_{N-1} > 0$*, let* $K_0, \dots, K_{N-1} \geq 1$ *be integers, and put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\begin{aligned}
\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \leq\ & \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{Y}^{G\to H,\mathrm{rad,hor}}_{Q,\lambda_i,\sigma_i,\beta_i,j}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{Y}^{G\to H,\mathrm{rad,in,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{Y}^{G\to H,\mathrm{rad,in,con}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ *follows from the same exponent hypothesis on the terminal far tail and on each tokenized slab profile appearing on the right-hand side.*

Proof. Apply Corollary 8.129. For each heavy-slab term in its right-hand side, apply Corollary 8.130 and then Corollary 8.132 with $K = K_i$. The displayed inequality follows after summing the resulting bounds over i and j. All sums are finite and independent of p, so the logarithmic exponent conclusion follows. □

Definition 8.134 (Exact-level exterior-return profiles)**.** Let $u \geq 1$ and $K \geq 0$. For a rooted path $\gamma = (v_0, \dots, v_\ell)$, let $\mathcal{X}_{u,K}(x,\gamma)$ be the property that γ contains more than K subpaths

$$(v_{s_a}, v_{s_a+1}, \dots, v_{t_a}), \qquad a = 1, \dots, K+1,$$

with pairwise disjoint open time intervals (s_a, t_a), such that

$$d(x, v_{s_a}) = d(x, v_{t_a}) = u - 1, \qquad d(x, v_{s_a+1}) = d(x, v_{t_a-1}) = u,$$

and

$$d(x, v_b) \geq u \qquad (s_a < b < t_a).$$

In the setting of Corollary 8.130, for every integer $j \geq 0$ write

$$A_{j,p} = [jW_{\beta,p}, (j+1)W_{\beta,p})$$

and define

$$\mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda,\sigma,\beta,j,K}(s,p)$$
$$= \sup_{\substack{u\in A_{j,p}\cap\mathbb{N}\\ u\ge 1}} \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{X}_{u,K}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right) L_H \right\},$$

with the value 0 if $A_{j,p}\cap\mathbb{N}\cap[1,\infty)$ is empty.

Definition 8.135 (Spread exterior-return profiles)**.** In the setting and notation of Corollary 8.132, for every integer $j\ge 0$ let $\mathcal{X}^{\mathrm{spr}}_{j,K,p}(x,\gamma)$ be the property

$$\#\{u\in A_{j,p}\cap\mathbb{N} : u\ge 1,\ \mathcal{X}_{u,0}(x,\gamma)\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{X}^{\mathrm{spr}}_{j,K,p}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right) L_H \right\}.$$

Corollary 8.136 (Spread-inward slabs localize to spread-return levels)**.** *In the setting and notation of Corollary 8.132,*

$$\mathfrak{V}^{G\to H,\mathrm{rad},\mathrm{in},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \le \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p).$$

Consequently, the exponent hypothesis on the spread-inward slab profile follows from the same exponent hypothesis on the spread exterior-return profile.

Proof. If $I_x(\gamma;\{u\})>0$, then $u\ge 1$. Applying Lemma 8.131 to one inward level-u token gives an exterior return through level u, i.e. $\mathcal{X}_{u,0}(x,\gamma)$. Hence

$$\mathcal{V}^{\mathrm{spr}}_{j,K,p}(x,\gamma) \quad\Longrightarrow\quad \mathcal{X}^{\mathrm{spr}}_{j,K,p}(x,\gamma).$$

The displayed profile inequality follows immediately from monotonicity of the property-restricted profiles. □

Corollary 8.137 (Level-congested slabs localize to exterior-return levels)**.** *In the setting and notation of Corollary 8.132,*

$$\mathfrak{V}^{G\to H,\mathrm{rad},\mathrm{in},\mathrm{con}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \le W_{\beta,p}\, \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda,\sigma,\beta,j,K}(s,p).$$

Consequently, the exponent hypothesis on the level-congested inward profile follows from the same exponent hypothesis on the exact-level exterior-return profile.

Proof. If γ satisfies $\mathcal{V}^{\mathrm{con}}_{j,K,p}(x,\gamma)$, then for some

$$u\in A_{j,p}\cap\mathbb{N}, \qquad u\ge 1,$$

we have $I_x(\gamma;\{u\})>K$. Applying Lemma 8.131 to each inward level-u token and choosing the last preceding outward crossing gives more than K exterior return subpaths with disjoint open time intervals. Hence $\mathcal{X}_{u,K}(x,\gamma)$ holds. Therefore, for each r,d,

$$\mathfrak{R}^{G,\mathcal{V}^{\mathrm{con}}_{j,K,p}}_{r,d} \le \sum_{\substack{u\in A_{j,p}\cap\mathbb{N}\\ u\ge 1}} \mathfrak{R}^{G,\mathcal{X}_{u,K}}_{r,d}.$$

After multiplying by Q^d, summing over the same r, d range, and taking the supremum in m, the right-hand side is bounded by

$$|A_{j,p} \cap \mathbb{N}|\, \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \le W_{\beta,p}\, \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda,\sigma,\beta,j,K}(s,p).$$

Finally,

$$\frac{\log_{D_H} W_{\beta,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

so the extra factor $W_{\beta,p}$ does not affect logarithmic exponent criteria. □

Corollary 8.138 (Exterior-return finite far-tail ladder)**.** *In the setting of Corollary 8.133, write*

$$X^{\mathrm{ret}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p)$$

for $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$. Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le{}& \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)\\ &+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{V}^{G\to H,\mathrm{rad},\mathrm{hor}}_{Q,\lambda_i,\sigma_i,\beta_i,j}(s,p)\\ &+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{V}^{G\to H,\mathrm{rad},\mathrm{in},\mathrm{spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p)\\ &+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, X^{\mathrm{ret}}_{i,j}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, on the horizontal-heavy and spread-inward slab profiles, and on the exact-level exterior-return profiles $X^{\mathrm{ret}}_{i,j}$.

Proof. Apply Corollary 8.133. Each level-congested inward term in its right-hand side is bounded by Corollary 8.137, with $(\lambda,\sigma,\beta,K) = (\lambda_i,\sigma_i,\beta_i,K_i)$. This gives the displayed inequality. The number of indices i, j is independent of p, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Thus the slab-width factors do not affect the logarithmic exponent hypothesis. □

Corollary 8.139 (Return-spread exterior-return far-tail ladder)**.** *In the setting of Corollary 8.138, write*

$$S^{\mathrm{ret}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p)$$

and

$$X^{\mathrm{ret}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{V}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{V}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \mathfrak{V}^{G\to H,\mathrm{rad,hor}}_{Q,\lambda_i,\sigma_i,\beta_i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, X^{\mathrm{ret}}_{i,j}(s,p).&\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, on the horizontal-heavy slab profiles, on the spread exterior-return profiles $S^{\mathrm{ret}}_{i,j}$, and on the exact-level exterior-return profiles $X^{\mathrm{ret}}_{i,j}$.

Proof. Apply Corollary 8.138. Each spread-inward term is bounded by Corollary 8.136, with

$$(\lambda,\sigma,\beta,K) = (\lambda_i,\sigma_i,\beta_i,K_i).$$

The displayed inequality follows. The exponent conclusion is unchanged because the sums are finite and independent of p, and the only nonconstant multiplicative losses are the already exponent-negligible factors $W_{\beta_i,p}$. □

Definition 8.140 (Exact-level same-sphere profiles)**.** In the setting of Corollary 8.130, let $u, K \ge 0$. Let $\mathcal{Z}_{u,K}(x,\gamma)$ be the property

$$H_x(\gamma;\{u\}) > K.$$

For $K \ge 1$, let $\mathcal{Z}^{\mathrm{spr}}_{j,K,p}(x,\gamma)$ be the property

$$\#\{u \in A_{j,p}\cap\mathbb{N} : H_x(\gamma;\{u\}) > 0\} > \frac{\theta_{\lambda,\sigma,\beta}}{2K} L_H.$$

Define

$$\begin{aligned}&\mathfrak{Z}^{G\to H,\mathrm{rad,hor,lev}}_{Q,\lambda,\sigma,\beta,j,K}(s,p)\\ &\quad= \sup_{u\in A_{j,p}\cap\mathbb{N}} \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{Z}_{u,K}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right) L_H \right\},\end{aligned}$$

with value 0 if $A_{j,p}\cap\mathbb{N}$ is empty, and

$$\mathfrak{Z}^{G\to H,\mathrm{rad,hor,spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{Z}^{\mathrm{spr}}_{j,K,p}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right) L_H \right\}.$$

Corollary 8.141 (Horizontal-heavy slabs: spread or same-sphere congestion)**.** *In the setting of Corollary 8.130, fix $K \ge 1$. Then*

$$\mathfrak{V}^{G\to H,\mathrm{rad,hor}}_{Q,\lambda,\sigma,\beta,j}(s,p) \le \mathfrak{Z}^{G\to H,\mathrm{rad,hor,spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) + W_{\beta,p}\, \mathfrak{Z}^{G\to H,\mathrm{rad,hor,lev}}_{Q,\lambda,\sigma,\beta,j,K}(s,p).$$

Consequently, the exponent hypothesis on the horizontal-heavy slab profile follows from the same exponent hypothesis on the same-sphere spread and exact-level same-sphere congestion profiles.

Proof. If a path satisfies $\mathcal{V}^{\mathrm{hor}}_{j,p}$, then

$$\sum_{u\in A_{j,p}\cap\mathbb{N}} H_x(\gamma;\{u\}) = H_x(\gamma;A_{j,p}) > \frac{\theta_{\lambda,\sigma,\beta}}{2}L_H.$$

Let

$$N^{\mathrm{hor}}_x(\gamma;A_{j,p}) = \#\{u\in A_{j,p}\cap\mathbb{N} : H_x(\gamma;\{u\})>0\}.$$

If $\mathcal{Z}^{\mathrm{spr}}_{j,K,p}$ fails and no level in $A_{j,p}\cap\mathbb{N}$ satisfies $\mathcal{Z}_{u,K}$, then

$$H_x(\gamma;A_{j,p}) \le K\, N^{\mathrm{hor}}_x(\gamma;A_{j,p}) \le \frac{\theta_{\lambda,\sigma,\beta}}{2}L_H,$$

a contradiction. Hence every horizontal-heavy path is counted either by the spread same-sphere profile or by $\mathcal{Z}_{u,K}$ for at least one exact level $u\in A_{j,p}\cap\mathbb{N}$. Taking the union bound over such levels gives the displayed inequality, since

$$|A_{j,p}\cap\mathbb{N}| \le W_{\beta,p}.$$

The exponent conclusion follows because $\log_{D_H} W_{\beta,p} = o(L_H)$. □

Definition 8.142 (Tangential run profiles)**.** Let $u\ge 0$. A level-u tangential run of a rooted path $\gamma=(v_0,\dots,v_\ell)$ is a maximal interval of times $[a,b]\subseteq[0,\ell]$ such that $b>a$ and

$$d(x,v_t)=u \qquad (a\le t\le b).$$

Its length is $b-a$. Let $R^u_x(\gamma)$ be the number of level-u tangential runs, and let

$$T^u_x(\gamma) = \max\{b-a : [a,b] \text{ is a level-}u \text{ tangential run of } \gamma\},$$

with $T^u_x(\gamma)=0$ if there is no such run. For $L\ge 1$ and $R\ge 0$, let $\mathcal{T}^{\mathrm{long}}_{u,L}(x,\gamma)$ be the property

$$T^u_x(\gamma) > L$$

and let $\mathcal{T}^{\mathrm{many}}_{u,R}(x,\gamma)$ be the property

$$R^u_x(\gamma) > R.$$

In the setting of Corollary 8.130, define

$$\mathfrak{T}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p) = \sup_{u\in A_{j,p}\cap\mathbb{N}} \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{T}^{\mathrm{long}}_{u,L}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H \right\},$$

and

$$\mathfrak{T}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{many}}_{Q,\lambda,\sigma,\beta,j,R}(s,p) = \sup_{u\in A_{j,p}\cap\mathbb{N}} \sup\left\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{T}^{\mathrm{many}}_{u,R}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H \right\},$$

with value 0 if $A_{j,p}\cap\mathbb{N}$ is empty.

Definition 8.143 (Adjacent-level crossing tangential profiles)**.** Let $u \geq 0$, and let $\eta, \zeta \in \{-1, +1\}$. A level-$u$ tangential run $[a, b]$ of $\gamma = (v_0, \dots, v_\ell)$ is an (η, ζ)-crossing run if

$$a > 0, \qquad b < \ell, \qquad d(x, v_{a-1}) = u + \eta, \qquad d(x, v_{b+1}) = u + \zeta.$$

Let $R^u_{x,\eta,\zeta}(\gamma)$ be the number of such runs. If either $u+\eta < 0$ or $u+\zeta < 0$, this number is understood to be zero. For $R \geq 0$, let $\mathcal{C}^{\eta,\zeta}_{u,R}(x, \gamma)$ be the property

$$R^u_{x,\eta,\zeta}(\gamma) > R.$$

In the setting of Corollary 8.130, define

$$\mathfrak{C}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{cross}}_{Q,\lambda,\sigma,\beta,j,R,\eta,\zeta}(s,p)$$
$$= \sup_{u\in A_{j,p}\cap\mathbb{N}} \sup\left\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{C}^{\eta,\zeta}_{u,R}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \right\},$$

with value 0 if $A_{j,p} \cap \mathbb{N}$ is empty.

Definition 8.144 (Outward-through tangential spread profiles)**.** In the setting of Corollary 8.130, for $K \geq 1$ let $\mathcal{U}^{\mathrm{out}}_{j,K,p}(x, \gamma)$ be the property

$$\#\{u \in A_{j,p} \cap \mathbb{N} : R^u_{x,-1,+1}(\gamma) > 0\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{U}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{out},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{U}^{\mathrm{out}}_{j,K,p}}_{r,d}$$
$$: m\in\mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.145 (Same-sphere spread: returns or outward-through spread)**.** *In the setting of Corollary 8.141, fix $K \geq 1$. Then, for all p sufficiently close to $p_c(H)$,*

$$\begin{aligned}\mathfrak{Z}^{G\to H,\mathrm{rad},\mathrm{hor},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \leq\ & \mathfrak{U}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{out},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p)\\ &+ \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,8K}(s,p)\\ &+ \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j+1,8K}(s,p).\end{aligned}$$

Consequently, the same-sphere spread exponent hypothesis follows from the same exponent hypothesis on the outward-through tangential spread profile and on the same and next spread exterior-return profiles.

Proof. Let N be the number of levels $u \in A_{j,p} \cap \mathbb{N}$ with $H_x(\gamma; \{u\}) > 0$. If $\mathcal{Z}^{\mathrm{spr}}_{j,K,p}$ holds, then

$$N > \frac{\theta_{\lambda,\sigma,\beta}}{2K} L_H.$$

For every such level except possibly the initial level of the whole path and the terminal level of the whole path, there is a two-sided level-u tangential run. Thus at least $N - 2$ of these levels have a two-sided tangential run.

If more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H$$

levels have an outward-through $(-1,+1)$-run, then $\mathcal{U}^{\mathrm{out}}_{j,K,p}$ holds. Otherwise, for p close enough that

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K}L_H > 2,$$

more than $\theta_{\lambda,\sigma,\beta}L_H/(8K)$ levels have a two-sided tangential run of one of the other three types.

Each of these other three types gives an exterior return through level u or through level $u+1$: the two types with exit to $S_{u-1}(x)$ give an inward level-u token, while type $(+1,+1)$ gives an inward level-$u+1$ token. By Lemma 8.131, each such token gives the corresponding $\mathcal{X}_{\cdot,0}$ event. The map from a same-sphere level u to the resulting return level has at most two preimages, so more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K}L_H$$

distinct return levels occur in $A_{j,p}\cup A_{j+1,p}$. One of the two slabs therefore contains more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{32K}L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(8K)}L_H$$

return levels, which is exactly the spread-return property with parameter $8K$ in slab j or $j+1$. This proves the property inclusion behind the displayed profile inequality. Taking the union bound inside the restricted profiles gives the result. □

Definition 8.146 (Record outward-through tangential spread profiles)**.** In the setting of Corollary 8.130, say that a level $u\ge 1$ is record outward-through for (x,γ) if there is a $(-1,+1)$-crossing level-u tangential run $[a,b]$ such that

$$d(x,v_t)\neq u \qquad (0\le t<a).$$

For $K\ge 1$, let $\mathcal{U}^{\mathrm{rec}}_{j,K,p}(x,\gamma)$ be the property

$$\#\{u\in A_{j,p}\cap\mathbb{N} : u\ge 1,\ u \text{ is record outward-through for } (x,\gamma)\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{rec}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{U}^{\mathrm{rec}}_{j,K,p}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H \Bigg\}.$$

Corollary 8.147 (Outward-through spread: record or return)**.** *In the setting of Corollary 8.130, fix $K\ge 1$. Then*

$$\begin{aligned}\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) &\le \mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{rec}}_{Q,\lambda,\sigma,\beta,j,2K}(s,p)\\ &\quad + \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,2K}(s,p)\\ &\le \mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{rec}}_{Q,\lambda,\sigma,\beta,j,2K}(s,p) + \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,8K}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on outward-through spread follows from the same exponent hypothesis on record outward-through spread and on the spread exterior-return profile.

Proof. Suppose that $\mathcal{U}^{\mathrm{out}}_{j,K,p}(x,\gamma)$ holds, and let

$$N_{\mathrm{out}} = \#\{u \in A_{j,p} \cap \mathbb{N} : R^u_{x,-1,+1}(\gamma) > 0\}.$$

If more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K} L_H$$

of these levels are record outward-through, then $\mathcal{U}^{\mathrm{rec}}_{j,2K,p}$ holds. Otherwise, more than the same number of levels admit a $(-1,+1)$-crossing run $[a,b]$ that is not the first visit to level u.

For such a non-record level u, choose a time $t_0 < a$ with $d(x, v_{t_0}) = u$. Since the chosen run is entered from level $u-1$, we have $d(x, v_{a-1}) = u-1$. Let t be the first time after t_0 at which the path is at level $u-1$. Then $d(x, v_{t-1}) = u$ and $d(x, v_t) = u-1$, so t is an inward level-u token. By Lemma 8.131, $\mathcal{X}_{u,0}(x,\gamma)$ holds. Therefore the non-record alternative implies $\mathcal{X}^{\mathrm{spr}}_{j,2K,p}(x,\gamma)$.

This proves the first property inclusion, and hence the first displayed profile inequality. The second inequality follows from monotonicity in the spread threshold, since

$$\mathcal{X}^{\mathrm{spr}}_{j,2K,p} \subseteq \mathcal{X}^{\mathrm{spr}}_{j,8K,p}.$$

□

Definition 8.148 (Subterminal record outward-through profiles)**.** In the setting of Corollary 8.130, for a rooted path $\gamma = (v_0, \dots, v_\ell)$ write

$$e_x(\gamma) = d(x, v_\ell).$$

For $K \geq 1$, let $\mathcal{U}^{\mathrm{sub}}_{j,K,p}(x,\gamma)$ be the property

$$\#\{u \in A_{j,p} \cap \mathbb{N} : u \geq 1,\ u < e_x(\gamma),\ u \text{ is record outward-through for } (x,\gamma)\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{sub}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{U}^{\mathrm{sub}}_{j,K,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.149 (Record outward-through: subterminal or return)**.** *In the setting of Corollary 8.130, fix $K \geq 1$. Then*

$$\begin{aligned}\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{rec}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \leq\ & \mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{sub}}_{Q,\lambda,\sigma,\beta,j,2K}(s,p)\\ &+ \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,4K}(s,p)\\ &+ \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j+1,4K}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on record outward-through spread follows from the same exponent hypothesis on subterminal record outward-through spread and on the same and next spread exterior-return profiles.

Proof. Suppose that $\mathcal{U}^{\mathrm{rec}}_{j,K,p}(x,\gamma)$ holds. If more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K}L_H$$

of the record outward-through levels in $A_{j,p}$ are subterminal, then $\mathcal{U}^{\mathrm{sub}}_{j,2K,p}$ holds. Otherwise, more than the same number of record outward-through levels $u \in A_{j,p}$ satisfy

$$u \geq e_x(\gamma).$$

For such a level, choose a record $(-1,+1)$-crossing run $[a,b]$. Since $d(x,v_{b+1}) = u+1$ and the endpoint has distance at most u, there is a first time $t > b+1$ at which $d(x,v_t) = u$. By the choice of t, $d(x,v_{t-1}) = u+1$, so t is an inward level-$u+1$ token. Lemma 8.131 gives $\mathcal{X}_{u+1,0}(x,\gamma)$.

The map $u \mapsto u+1$ is injective, and if $u \in A_{j,p}$ then

$$u+1 \in A_{j,p} \cup A_{j+1,p}.$$

Thus more than $\theta_{\lambda,\sigma,\beta}L_H/(8K)$ terminal or post-terminal record levels give the same number of return levels in the union of the two slabs. One of the two slabs contains more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K}L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(4K)}L_H$$

such return levels, which is precisely the spread-return property with parameter $4K$ in slab j or $j+1$. This gives the property inclusion and hence the displayed profile inequality. □

Definition 8.150 (Persistent subterminal record profiles)**.** In the setting of Corollary 8.130, let

$$\tau^{\mathrm{end}}_x(\gamma) = \min\{t : d(x,v_t) = e_x(\gamma)\}.$$

A subterminal record outward-through level u is persistent for (x,γ) if its record $(-1,+1)$-crossing run $[a,b]$ satisfies

$$d(x,v_t) > u \qquad (b < t \leq \tau^{\mathrm{end}}_x(\gamma)).$$

For $K \geq 1$, let $\mathcal{U}^{\mathrm{per}}_{j,K,p}(x,\gamma)$ be the property

$$\#\{u \in A_{j,p} \cap \mathbb{N} : u \text{ is persistent subterminal record for } (x,\gamma)\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{per}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{U}^{\mathrm{per}}_{j,K,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.151 (Subterminal record: persistent or return)**.** *In the setting of Corollary 8.130, fix $K \geq 1$. Then*

$$\begin{aligned}\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{sub}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \leq\ & \mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{per}}_{Q,\lambda,\sigma,\beta,j,2K}(s,p)\\ &+ \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,4K}(s,p)\\ &+ \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j+1,4K}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on subterminal record spread follows from the same exponent hypothesis on persistent subterminal record spread and on the same and next spread exterior-return profiles.

Proof. Suppose that $\mathcal{U}^{\mathrm{sub}}_{j,K,p}(x,\gamma)$ holds. If more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K}L_H$$

of the subterminal record levels in $A_{j,p}$ are persistent, then $\mathcal{U}^{\mathrm{per}}_{j,2K,p}$ holds. Otherwise, more than the same number of subterminal record levels $u \in A_{j,p}$ have a record run $[a,b]$ after which the path returns to level u before the first hit of the endpoint sphere.

For each such nonpersistent level, let $t > b+1$ be the first time with $d(x,v_t) = u$. Since $d(x,v_{b+1}) = u+1$, the first-return choice gives $d(x,v_{t-1}) = u+1$, so t is an inward level-$u+1$ token. Lemma 8.131 gives $\mathcal{X}_{u+1,0}(x,\gamma)$.

As in Corollary 8.149, the map $u \mapsto u+1$ is injective and sends $A_{j,p}$ into $A_{j,p} \cup A_{j+1,p}$. Hence one of these two slabs contains more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K}L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(4K)}L_H$$

return levels, giving the spread-return property with parameter $4K$ in slab j or $j+1$. The displayed profile inequality follows. □

Definition 8.152 (Deep persistent subterminal profiles)**.** In the setting of Corollary 8.130, for $K \geq 1$ let $\mathcal{U}^{\mathrm{deep}}_{j,K,p}(x,\gamma)$ be the property

$$\#\left\{u \in A_{j,p} \cap \mathbb{N} : \begin{array}{l} u \text{ is persistent subterminal record for } (x,\gamma), \\ e_x(\gamma) - u > \dfrac{\theta_{\lambda,\sigma,\beta}}{4K}L_H \end{array}\right\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{deep}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2 \\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{U}^{\mathrm{deep}}_{j,K,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.153 (Persistent levels are deeply persistent)**.** *In the setting of Corollary 8.130, fix $K \geq 1$. Then*

$$\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{per}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \leq \mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{deep}}_{Q,\lambda,\sigma,\beta,j,2K}(s,p).$$

Consequently, the exponent hypothesis on persistent subterminal record spread follows from the same exponent hypothesis on deep persistent subterminal record spread.

Proof. Suppose that $\mathcal{U}^{\mathrm{per}}_{j,K,p}(x,\gamma)$ holds and put

$$B = \frac{\theta_{\lambda,\sigma,\beta}}{8K}L_H.$$

There are at most B integers u satisfying

$$0 < e_x(\gamma) - u \leq B.$$

Since the persistent event gives more than $2B$ persistent subterminal record levels in $A_{j,p}$, more than B of them satisfy

$$e_x(\gamma) - u > B.$$

This is exactly $\mathcal{U}^{\mathrm{deep}}_{j,2K,p}(x,\gamma)$. The displayed profile inequality follows by monotonicity of property-restricted profiles. □

Definition 8.154 (Endpoint-prefix spread profiles)**.** In the setting of Corollary 8.130, for a rooted path $\gamma = (v_0, \dots, v_\ell)$ let

$$\tau_x^{\mathrm{end}}(\gamma) = \min\{t : d(x, v_t) = e_x(\gamma)\}.$$

For $K \geq 1$, let $\mathcal{P}^{\mathrm{pre}}_{j,K,p}(x,\gamma)$ be the property

$$\#\left\{u \in A_{j,p} \cap \mathbb{N} : \begin{array}{l} \gamma \text{ has a level-}u \text{ tangential run } [a,b] \text{ with } b < \tau_x^{\mathrm{end}}(\gamma), \\ e_x(\gamma) - u > \dfrac{\theta_{\lambda,\sigma,\beta}}{4K} L_H \end{array}\right\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{deep}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre}}_{j,K,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.155 (Deep persistent levels are prefix-horizontal)**.** *In the setting of Corollary 8.130, fix $K \geq 1$. Then*

$$\mathfrak{U}^{G\to H,\mathrm{rad},\tan,\mathrm{out},\mathrm{deep}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \leq \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{deep}}_{Q,\lambda,\sigma,\beta,j,K}(s,p).$$

Consequently, the exponent hypothesis on deep persistent subterminal record spread follows from the same exponent hypothesis on endpoint-prefix deep horizontal spread.

Proof. Every level counted in $\mathcal{U}^{\mathrm{deep}}_{j,K,p}$ is persistent subterminal record, hence has a record $(-1,+1)$-crossing tangential run $[a,b]$. Since the level is subterminal, the endpoint sphere has radius strictly larger than u; after the run exits to level $u+1$, the path first reaches the endpoint sphere at a time strictly after b. Thus $[a,b]$ is a level-u tangential run with

$$b < \tau_x^{\mathrm{end}}(\gamma).$$

The depth condition in $\mathcal{U}^{\mathrm{deep}}_{j,K,p}$ is exactly the depth condition in $\mathcal{P}^{\mathrm{pre}}_{j,K,p}$. Hence

$$\mathcal{U}^{\mathrm{deep}}_{j,K,p}(x,\gamma) \Longrightarrow \mathcal{P}^{\mathrm{pre}}_{j,K,p}(x,\gamma),$$

and the property-restricted profile inequality follows. □

Definition 8.156 (Short endpoint-prefix spread profiles)**.** In the setting of Corollary 8.130, for $K, L \geq 1$, let $\mathcal{P}^{\mathrm{pre,short}}_{j,K,L,p}(x,\gamma)$ be the property

$$\#\left\{u \in A_{j,p} \cap \mathbb{N} : \begin{array}{l} \gamma \text{ has a level-}u \text{ tangential run } [a,b], \\ b < \tau_x^{\mathrm{end}}(\gamma) \text{ and } b - a \leq L, \\ e_x(\gamma) - u > \dfrac{\theta_{\lambda,\sigma,\beta}}{4K} L_H \end{array}\right\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{deep},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,short}}_{j,K,L,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.157 (Endpoint-prefix spread is long or short)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$. Then*

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{deep}}_{Q,\lambda,\sigma,\beta,j,K}(s,p)\\&\quad\leq W_{\beta,p}\,\mathfrak{T}^{G\to H,\mathrm{rad},\tan,\mathrm{long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p)+\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{deep},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on endpoint-prefix deep horizontal spread follows from the same exponent hypothesis on long tangential runs and short endpoint-prefix deep horizontal spread.

Proof. If $\mathcal{P}^{\mathrm{pre}}_{j,K,p}(x,\gamma)$ holds but $\mathcal{P}^{\mathrm{pre},\mathrm{short}}_{j,K,L,p}(x,\gamma)$ fails, then among the levels counted by $\mathcal{P}^{\mathrm{pre}}_{j,K,p}$ there is at least one level $u \in A_{j,p}\cap\mathbb{N}$ that is not counted by the short endpoint-prefix property. This level has a prefix tangential run witnessing its membership in $\mathcal{P}^{\mathrm{pre}}_{j,K,p}$, and no such witnessing run has length at most L. Hence $T^u_x(\gamma) > L$, so $\mathcal{T}^{\mathrm{long}}_{u,L}(x,\gamma)$ holds. Thus

$$\mathcal{P}^{\mathrm{pre}}_{j,K,p} \subseteq \mathcal{P}^{\mathrm{pre},\mathrm{short}}_{j,K,L,p} \cup \bigcup_{u\in A_{j,p}\cap\mathbb{N}} \mathcal{T}^{\mathrm{long}}_{u,L}.$$

Taking the weighted property-restricted union bound and then the supremum in the definition of the profiles gives the displayed inequality, since $|A_{j,p}\cap\mathbb{N}| \leq W_{\beta,p}$. The exponent conclusion follows because $\log_{D_H} W_{\beta,p} = o(L_H)$. □

Definition 8.158 (Short endpoint-prefix outward profiles)**.** In the setting of Corollary 8.130, for $K, L \geq 1$, let $\mathcal{P}^{\mathrm{pre},\mathrm{out},\mathrm{short}}_{j,K,L,p}(x,\gamma)$ be the property

$$\#\left\{u\in A_{j,p}\cap\mathbb{N}:\ \begin{array}{l}\gamma \text{ has a level-}u\ (-1,+1)\text{-crossing run } [a,b],\\ b<\tau^{\mathrm{end}}_x(\gamma) \text{ and } b-a\leq L,\\ e_x(\gamma)-u>\dfrac{\theta_{\lambda,\sigma,\beta}}{4K}L_H\end{array}\right\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p)\\&\quad=\sup\Bigg\{\sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d\mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{out},\mathrm{short}}_{j,K,L,p}}_{r,d}\\&\qquad\qquad:m\in\mathbb{N},\ c_sL_H<m\leq\left(\frac{1}{\alpha_\omega}+3s\right)L_H\Bigg\}.\end{aligned}$$

Corollary 8.159 (Short prefix spread is outward or returning)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$. Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,*

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{deep},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p)\\&\quad\leq\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,2K,L}(s,p)\\&\qquad+\mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,16K}(s,p)+\mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j+1,16K}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on short endpoint-prefix deep horizontal spread follows from the same exponent hypothesis on short endpoint-prefix outward-through spread and on the same and next spread exterior-return profiles.

Proof. Let N be the number of levels counted by $\mathcal{P}^{\mathrm{pre,short}}_{j,K,L,p}(x,\gamma)$. If this property holds, then

$$N > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

If $\mathcal{P}^{\mathrm{pre,out,short}}_{j,2K,L,p}(x,\gamma)$ also fails, fewer than or equal to $\theta_{\lambda,\sigma,\beta}L_H/(8K)$ of these counted levels have a witnessing $(-1,+1)$-crossing run. Thus more than $\theta_{\lambda,\sigma,\beta}L_H/(8K)$ counted levels have a short endpoint-prefix tangential run that is not of type $(-1,+1)$.

For p close enough, $\theta_{\lambda,\sigma,\beta}L_H/(16K) > 1$. After discarding the possible initial level $u = 0$, more than $\theta_{\lambda,\sigma,\beta}L_H/(16K)$ such levels remain. Each remaining run is two-sided. If it exits to level $u-1$, its last step gives an inward level-u token; if it enters from level $u+1$, its first step gives an inward level-$u+1$ token. By Lemma 8.131, each token yields an exterior return through level u or $u+1$.

The map from the original level u to the resulting return level has at most two preimages. Hence more than $\theta_{\lambda,\sigma,\beta}L_H/(32K)$ distinct return levels occur in $A_{j,p} \cup A_{j+1,p}$, provided p is close enough that $W_{\beta,p} > 1$. One of the two slabs therefore contains more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{64K} L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(16K)} L_H$$

return levels. This is exactly $\mathcal{X}^{\mathrm{spr}}_{j,16K,p}(x,\gamma)$ or $\mathcal{X}^{\mathrm{spr}}_{j+1,16K,p}(x,\gamma)$. Therefore, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\mathcal{P}^{\mathrm{pre,short}}_{j,K,L,p} \subseteq \mathcal{P}^{\mathrm{pre,out,short}}_{j,2K,L,p} \cup \mathcal{X}^{\mathrm{spr}}_{j,16K,p} \cup \mathcal{X}^{\mathrm{spr}}_{j+1,16K,p}.$$

Taking the corresponding property-restricted union bound gives the displayed profile inequality. □

Definition 8.160 (Short prefix-record outward profiles)**.** In the setting of Corollary 8.130, for $K, L \geq 1$, let $\mathcal{P}^{\mathrm{pre,rec,short}}_{j,K,L,p}(x,\gamma)$ be the property

$$\#\left\{ u \in A_{j,p} \cap \mathbb{N} : \begin{array}{l} \gamma \text{ has a level-}u\ (-1,+1)\text{-crossing run } [a,b], \\ d(x,v_t) \neq u\ (0 \leq t < a), \\ b < \tau^{\mathrm{end}}_x(\gamma) \text{ and } b - a \leq L, \\ e_x(\gamma) - u > \dfrac{\theta_{\lambda,\sigma,\beta}}{4K} L_H \end{array} \right\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) \\ &\quad = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,rec,short}}_{j,K,L,p}}_{r,d} \\ &\qquad\qquad : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}. \end{aligned}$$

Corollary 8.161 (Short outward-prefix spread is record or returning)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$. Then*

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) \\ &\quad \leq \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda,\sigma,\beta,j,2K,L}(s,p) + \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j,2K}(s,p). \end{aligned}$$

Consequently, the exponent hypothesis on short outward-prefix spread follows from the same exponent hypothesis on short prefix-record outward spread and on the spread exterior-return profile.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,out,short}}_{j,K,L,p}(x,\gamma)$ holds, and let N be the number of levels counted by it. If more than $\theta_{\lambda,\sigma,\beta}L_H/(8K)$ of these levels have a witnessing run that is record, then $\mathcal{P}^{\mathrm{pre,rec,short}}_{j,2K,L,p}(x,\gamma)$ holds. Otherwise, more than the same number of counted levels admit a short endpoint-prefix $(-1,+1)$-crossing run $[a,b]$ that is not the first visit to level u.

For each such non-record level u, choose a time $t_0 < a$ with $d(x,v_{t_0}) = u$. Since the chosen run is entered from level $u-1$, let t be the first time after t_0 at which $d(x,v_t) = u-1$. Then $t < a$, and

$$d(x,v_{t-1}) = u, \qquad d(x,v_t) = u-1.$$

Thus t is an inward level-u token. By Lemma 8.131, $\mathcal{X}_{u,0}(x,\gamma)$ holds. Therefore the non-record alternative implies $\mathcal{X}^{\mathrm{spr}}_{j,2K,p}(x,\gamma)$. Taking the property-restricted union bound gives the displayed profile inequality. □

Definition 8.162 (Short persistent prefix-record profiles)**.** In the setting of Corollary 8.130, for $K,L \geq 1$, let $\mathcal{P}^{\mathrm{pre,per,short}}_{j,K,L,p}(x,\gamma)$ be the property

$$\#\left\{ u \in A_{j,p}\cap\mathbb{N} : \begin{array}{l} \gamma \text{ has a level-}u\ (-1,+1)\text{-crossing run } [a,b],\\ d(x,v_t) \neq u\ (0 \leq t < a),\\ d(x,v_t) > u\ (b < t \leq \tau^{\mathrm{end}}_x(\gamma)),\\ b < \tau^{\mathrm{end}}_x(\gamma) \text{ and } b-a \leq L,\\ e_x(\gamma) - u > \dfrac{\theta_{\lambda,\sigma,\beta}}{4K}L_H \end{array} \right\} > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p)\\ &\quad = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,short}}_{j,K,L,p}}_{r,d}\\ &\qquad\qquad : m\in\mathbb{N},\ c_sL_H < m \leq \left(\frac{1}{\alpha_\omega}+3s\right)L_H \Bigg\}. \end{aligned}$$

Corollary 8.163 (Short prefix-record spread is persistent or returning)**.** *In the setting of Corollary 8.130, fix $K,L \geq 1$. Then, for all p sufficiently close to $p_c(H)$,*

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p)\\ &\quad \leq \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda,\sigma,\beta,j,2K,L}(s,p)\\ &\qquad + \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j,4K}(s,p) + \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j+1,4K}(s,p). \end{aligned}$$

Consequently, the exponent hypothesis on short prefix-record spread follows from the same exponent hypothesis on short persistent prefix-record spread and on the same and next spread exterior-return profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,rec,short}}_{j,K,L,p}(x,\gamma)$ holds, and let N be the number of levels counted by it. If more than $\theta_{\lambda,\sigma,\beta}L_H/(8K)$ of these levels have a witnessing record run that stays strictly above level u until $\tau^{\mathrm{end}}_x(\gamma)$, then $\mathcal{P}^{\mathrm{pre,per,short}}_{j,2K,L,p}(x,\gamma)$ holds. Otherwise, more than the same number of counted levels have a witnessing record run $[a,b]$ for which the path returns to level u before or at $\tau^{\mathrm{end}}_x(\gamma)$.

For each such nonpersistent level, let $t > b+1$ be the first time with $d(x,v_t) = u$. Since $d(x,v_{b+1}) = u+1$, the first-return choice gives

$$d(x,v_{t-1}) = u+1, \qquad d(x,v_t) = u.$$

Thus t is an inward level-$u+1$ token. By Lemma 8.131, $\mathcal{X}_{u+1,0}(x,\gamma)$ holds.

The map $u \mapsto u+1$ is injective and sends $A_{j,p}$ into $A_{j,p} \cup A_{j+1,p}$, for all p sufficiently close to $p_{\mathrm{c}}(H)$. Thus one of the two slabs contains more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(4K)} L_H$$

return levels. This gives $\mathcal{X}^{\mathrm{spr}}_{j,4K,p}(x,\gamma)$ or $\mathcal{X}^{\mathrm{spr}}_{j+1,4K,p}(x,\gamma)$. The displayed profile inequality follows by taking the property-restricted union bound. □

Definition 8.164 (Ordered prefix-ladder profiles)**.** In the setting of Corollary 8.130, for $K, L \geq 1$, let $\mathcal{P}^{\mathrm{pre,per,lad,short}}_{j,K,L,p}(x,\gamma)$ be the property that there are an integer $N_\star \geq 1$, levels

$$u_1 < \cdots < u_{N_\star} \qquad (u_q \in A_{j,p} \cap \mathbb{N}),$$

and level-u_q $(-1,+1)$-crossing runs $[a_q, b_q]$ such that

$$N_\star > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

and, for every $1 \leq q \leq N_\star$,

$$d(x, v_t) \neq u_q \quad (0 \leq t < a_q), \qquad d(x, v_t) > u_q \quad (b_q < t \leq \tau^{\mathrm{end}}_x(\gamma)),$$

$$b_q < \tau^{\mathrm{end}}_x(\gamma), \qquad b_q - a_q \leq L, \qquad e_x(\gamma) - u_q > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

and the witnessing runs are chronologically ordered:

$$b_q < a_{q+1} \qquad (1 \leq q < N_\star).$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,lad,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,lad,short}}_{j,K,L,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.165 (Persistent prefix levels form ordered ladders)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) \leq \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,lad,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).$$

Consequently, the exponent hypothesis on short persistent prefix-record spread follows from the same exponent hypothesis on ordered short persistent prefix ladders.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,short}}_{j,K,L,p}(x,\gamma)$ holds. Sort the counted levels increasingly and write them as $u_1 < \cdots < u_{N_\star}$. For each q, choose a witnessing record $(-1,+1)$-crossing run $[a_q, b_q]$. All displayed conditions in Definition 8.164 except the chronological ordering are exactly the defining conditions of $\mathcal{P}^{\mathrm{pre,per,short}}_{j,K,L,p}$.

It remains to check the order. Let $u_q < u_{q+1}$. Since $[a_q, b_q]$ is record, a_q is the first time that the path visits level u_q. Before the path can first visit level u_{q+1}, its radial distance sequence must visit level u_q. Hence $a_q < a_{q+1}$. The time a_{q+1} is not inside the level-u_q run, because $d(x, v_{a_{q+1}}) = u_{q+1} > u_q$. Therefore $b_q < a_{q+1}$. Thus the sorted witnesses form an ordered prefix ladder, proving the property inclusion and hence the profile inequality. □

Definition 8.166 (Prefix-pair profiles)**.** In the setting of Corollary 8.130, fix $K, L, B \geq 1$. Let $\mathcal{P}^{\mathrm{pre,per,pair,short}}_{j,K,L,B,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ for which there exist $v \in A_{j,p} \cap \mathbb{N}$ and level-u, level-v $(-1,+1)$-crossing runs $[a,b]$, $[a',b']$ satisfying

$$u < v \leq u + B, \qquad b < a',$$

$$d(x, v_t) \neq u \quad (0 \leq t < a), \qquad d(x, v_t) > u \quad (b < t \leq \tau_x^{\mathrm{end}}(\gamma)),$$
$$d(x, v_t) \neq v \quad (0 \leq t < a'), \qquad d(x, v_t) > v \quad (b' < t \leq \tau_x^{\mathrm{end}}(\gamma)),$$

and

$$b, b' < \tau_x^{\mathrm{end}}(\gamma), \qquad b - a \leq L, \quad b' - a' \leq L,$$
$$e_x(\gamma) - u > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H, \qquad e_x(\gamma) - v > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{Q,\lambda,\sigma,\beta,j,K,L,B}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,pair,short}}_{j,K,L,B,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.167 (Ordered ladders have many bounded successor gaps)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$, and put*

$$B_{\lambda,\sigma,\beta,K} = \left\lceil \frac{16K\beta}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,lad,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) \leq \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{Q,\lambda,\sigma,\beta,j,2K,L,B_{\lambda,\sigma,\beta,K}}(s,p).$$

Consequently, the exponent hypothesis on ordered short persistent prefix ladders follows from the same exponent hypothesis on bounded-gap short persistent prefix-pair profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,lad,short}}_{j,K,L,p}(x,\gamma)$ holds, and choose witnesses $u_1 < \cdots < u_{N_\star}$ and $[a_q, b_q]$ as in Definition 8.164. Let $B = B_{\lambda,\sigma,\beta,K}$. Since all levels lie in the same slab,

$$\sum_{q=1}^{N_\star - 1} (u_{q+1} - u_q) \leq W_{\beta,p}.$$

For p close enough to $p_{\mathrm{c}}(H)$, $W_{\beta,p} \leq \beta L_H$, and hence the number of indices q with $u_{q+1} - u_q > B$ is at most

$$\frac{\beta}{B} L_H < \frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H.$$

Since

$$N_\star > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

and p may be taken close enough that $\theta_{\lambda,\sigma,\beta}L_H/(16K) > 1$, more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K}L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(2K)}L_H$$

indices q satisfy $u_{q+1} - u_q \le B$. For every such q, the pair (u_q, u_{q+1}), together with the witnessing runs $[a_q, b_q]$, $[a_{q+1}, b_{q+1}]$, satisfies the bounded-gap prefix-pair conditions with parameter $2K$. The depth requirements are inherited because

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H > \frac{\theta_{\lambda,\sigma,\beta}}{4(2K)}L_H.$$

The lower levels u_q are distinct, so $\mathcal{P}^{\mathrm{pre,per,pair,short}}_{j,2K,L,B,p}(x,\gamma)$ holds. This property inclusion gives the displayed profile inequality. □

Definition 8.168 (Fixed-gap prefix-pair profiles)**.** In the setting of Corollary 8.130, fix $K, L, h \ge 1$. Let $\mathcal{P}^{\mathrm{pre,per,gap,short}}_{j,K,L,h,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ such that $u+h \in A_{j,p} \cap \mathbb{N}$ and there are level-u, level-$u+h$ $(-1,+1)$-crossing runs $[a,b]$, $[a',b']$ satisfying

$$b < a',$$

$$\begin{aligned} d(x,v_t) &\neq u \quad (0 \le t < a), & d(x,v_t) &> u \quad (b < t \le \tau_x^{\mathrm{end}}(\gamma)),\\ d(x,v_t) &\neq u+h \quad (0 \le t < a'), & d(x,v_t) &> u+h \quad (b' < t \le \tau_x^{\mathrm{end}}(\gamma)), \end{aligned}$$

and

$$\begin{aligned} b, b' &< \tau_x^{\mathrm{end}}(\gamma), \qquad b-a \le L, \quad b'-a' \le L,\\ e_x(\gamma) - u &> \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H, \qquad e_x(\gamma) - (u+h) > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H \end{aligned}$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p)\\ &\quad= \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,gap,short}}_{j,K,L,h,p}}_{r,d}\\ &\qquad\qquad : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H \Bigg\}. \end{aligned}$$

Corollary 8.169 (Bounded-gap pairs have a fixed gap)**.** *In the setting of Corollary 8.130, fix $K, L, B \ge 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{Q,\lambda,\sigma,\beta,j,K,L,B}(s,p) \le \sum_{h=1}^{B} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{Q,\lambda,\sigma,\beta,j,KB,L,h}(s,p).$$

Consequently, the exponent hypothesis on bounded-gap short persistent prefix-pair profiles follows from the same exponent hypothesis on the finite family of fixed-gap short persistent prefix-pair profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,pair,short}}_{j,K,L,B,p}(x,\gamma)$ holds. For $1 \le h \le B$, let N_h be the number of levels u for which the fixed gap h has witnesses satisfying the conditions in Definition 8.168, except with the stronger depth threshold $\theta_{\lambda,\sigma,\beta} L_H/(4K)$. Every level counted by the bounded-gap pair property contributes to at least one N_h, so

$$\sum_{h=1}^{B} N_h > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Thus some h has

$$N_h > \frac{\theta_{\lambda,\sigma,\beta}}{4KB} L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(KB)} L_H.$$

The depth assumptions with parameter K imply those with parameter KB. Hence, for this h,

$$\mathcal{P}^{\mathrm{pre,per,gap,short}}_{j,KB,L,h,p}(x,\gamma)$$

holds. Taking the property-restricted union bound over $1 \le h \le B$ gives the displayed profile inequality. □

Definition 8.170 (Confined fixed-gap bridge profiles)**.** In the setting of Corollary 8.130, fix $K, L, h \ge 1$. Let $\mathcal{P}^{\mathrm{pre,per,bridge,short}}_{j,K,L,h,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ satisfying the fixed-gap conditions from Definition 8.168, with witnesses $[a,b]$ at level u and $[a',b']$ at level $u+h$, and such that

$$u < d(x, v_t) < u + h \qquad (b < t < a')$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p) \\
&\quad = \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2 \\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,bridge,short}}_{j,K,L,h,p}}_{r,d} \\
&\qquad\qquad : m \in \mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.
\end{aligned}$$

Corollary 8.171 (Fixed-gap pairs have confined bridges)**.** *In the setting of Corollary 8.130, fix $K, L, h \ge 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p).$$

Consequently, the exponent hypothesis on fixed-gap short persistent prefix-pair profiles follows from the same exponent hypothesis on confined fixed-gap bridge profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,gap,short}}_{j,K,L,h,p}(x,\gamma)$ holds, and take a counted level u with witnesses $[a,b]$ at level u and $[a',b']$ at level $u+h$. For $b < t < a'$, lower persistence gives

$$d(x, v_t) > u.$$

The upper run is record at level $u+h$, so

$$d(x, v_t) \ne u + h \qquad (0 \le t < a').$$

Since $d(x, v_b) = u$ and $d(x, v_{b+1}) = u + 1$, the radial distance sequence cannot exceed $u + h$ before first visiting $u + h$. Hence

$$d(x, v_t) < u + h \qquad (b < t < a').$$

Thus every fixed-gap witness is a confined bridge witness. The profile inequality follows by monotonicity of property-restricted profiles. □

Lemma 8.172 (Large fixed gaps do not have enough slab starts)**.** *In the setting of Corollary 8.130, fix $K, L, h \geq 1$. If*

$$W_{\beta,p} - h \leq \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

then

$$\mathcal{P}^{\mathrm{pre,per,bridge,short}}_{j,K,L,h,p}(x, \gamma)$$

is false for every rooted path (x, γ). Consequently,

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s, p) = 0.$$

In particular, the profile is zero whenever $h \geq W_{\beta,p}$.

Proof. The only possible counted lower levels u are those for which both

$$u \in A_{j,p} \cap \mathbb{N} \qquad \text{and} \qquad u + h \in A_{j,p} \cap \mathbb{N}.$$

The slab $A_{j,p}$ contains at most $W_{\beta,p}$ integer levels, so there are at most $\max\{W_{\beta,p} - h, 0\}$ such lower levels. If the displayed assumption holds, this maximum is at most $\theta_{\lambda,\sigma,\beta} L_H/(4K)$. The bridge property requires strictly more than this many counted levels, which is impossible. The final assertion follows because $h \geq W_{\beta,p}$ makes the maximum equal to zero. □

Definition 8.173 (Inward-free confined fixed-gap bridge profiles)**.** In the setting of Corollary 8.130, fix $K, L, h \geq 1$. Let $\mathcal{P}^{\mathrm{pre,per,bridge,free,short}}_{j,K,L,h,p}(x, \gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ satisfying the confined fixed-gap bridge conditions from Definition 8.170, with witnesses $[a, b]$ at level u and $[a', b']$ at level $u + h$, and such that

$$d(x, v_{t+1}) \geq d(x, v_t) \qquad (b \leq t < a')$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s, p) \\
&\quad = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,bridge,free,short}}_{j,K,L,h,p}}_{r,d} \\
&\qquad\qquad : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.
\end{aligned}$$

Corollary 8.174 (Confined bridges are inward-free or returning)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$ and $h \geq 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p) \leq \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda,\sigma,\beta,j,2K,L,h}(s,p) + \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j,2K\max\{h-1,1\}}(s,p).$$

Consequently, the exponent hypothesis on confined fixed-gap bridge profiles follows from the same exponent hypothesis on inward-free confined bridge profiles and on the corresponding spread exterior-return profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,bridge,short}}_{j,K,L,h,p}(x,\gamma)$ holds, and let N be the number of counted lower levels u. If more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K}L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(2K)}L_H$$

of these starts admit inward-free bridge witnesses, then the inward-free bridge property with parameter $2K$ holds.

Otherwise, more than $\theta_{\lambda,\sigma,\beta}L_H/(8K)$ counted starts have no inward-free bridge witness. For such a start, choose a confined bridge witness $[a,b]$, $[a',b']$. Since it is not inward-free, there is some t with $b \leq t < a'$ such that

$$d(x,v_{t+1}) = d(x,v_t) - 1.$$

The confinement condition rules out a step down to level u before a', and the upper run is record at level $u+h$. Thus this is an inward token at some level

$$w \in \{u+1,\dots,u+h-1\}.$$

By Lemma 8.131, $\mathcal{X}_{w,0}(x,\gamma)$ holds. Since u and $u+h$ both lie in the slab, so does every such intermediate w.

For a fixed return level w, the possible lower starts u that can map to it satisfy $u < w < u+h$, so there are at most $\max\{h-1,1\}$ of them. Hence the number of distinct return levels in $A_{j,p}$ is larger than

$$\frac{1}{\max\{h-1,1\}}\frac{\theta_{\lambda,\sigma,\beta}}{8K}L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(2K\max\{h-1,1\})}L_H.$$

This is the spread-return property with parameter $2K\max\{h-1,1\}$. The displayed profile inequality follows from the property-restricted union bound. □

Remark 8.175 (Adjacent fixed gaps are immediate)**.** When $h=1$, the confinement condition

$$u < d(x,v_t) < u+1$$

has no integer solution. Hence every confined fixed-gap bridge with $h=1$ satisfies $a' = b+1$: the upper record run starts immediately after the lower run exits upward.

Definition 8.176 (Adjacent stacked prefix-pair profiles)**.** In the setting of Corollary 8.130, fix $K, L \geq 1$. Let $\mathcal{P}^{\mathrm{pre,per,adj,short}}_{j,K,L,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ satisfying the fixed-gap conditions from Definition 8.168 with $h=1$, with witnesses $[a,b]$ at level u and $[a',b']$ at level $u+1$, and such that

$$a' = b+1$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{adj},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{adj},\mathrm{short}}_{j,K,L,p}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega}+3s\right) L_H \Bigg\}.$$

Corollary 8.177 (Adjacent confined bridges are stacked)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,1}(s,p) \leq \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{adj},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).$$

Consequently, the exponent hypothesis on adjacent confined fixed-gap bridge profiles follows from the same exponent hypothesis on adjacent stacked prefix-pair profiles.

Proof. For $h=1$, Remark 8.175 shows that every confined fixed-gap bridge witness has $a' = b+1$. Thus every level counted by $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{j,K,L,1,p}(x,\gamma)$ is counted by $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{adj},\mathrm{short}}_{j,K,L,p}(x,\gamma)$. The profile inequality follows by monotonicity of property-restricted profiles. □

Definition 8.178 (Two-level outward-staircase profiles)**.** In the setting of Corollary 8.130, fix $K, L \geq 1$. Let $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{short}}_{j,K,L,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p}\cap\mathbb{N}$ such that $u+1 \in A_{j,p}\cap\mathbb{N}$ and there are a level-u $(-1,+1)$-crossing run $[a,b]$ and a level-$u+1$ $(-1,+1)$-crossing run $[b+1,b']$ satisfying

$$d(x,v_t)\neq u \quad (0\leq t<a), \qquad d(x,v_t)>u \quad (b<t\leq \tau^{\mathrm{end}}_x(\gamma)),$$

$$d(x,v_t)\neq u+1 \quad (0\leq t<b+1), \qquad d(x,v_t)>u+1 \quad (b'<t\leq \tau^{\mathrm{end}}_x(\gamma)),$$

$$b,b' < \tau^{\mathrm{end}}_x(\gamma), \qquad b'-a\leq L,$$

and

$$e_x(\gamma)-u > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H, \qquad e_x(\gamma)-(u+1) > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) = \sup\Bigg\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{short}}_{j,K,L,p}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega}+3s\right) L_H \Bigg\}.$$

Definition 8.179 (Locally short two-level staircase profiles)**.** In the setting of Corollary 8.130, fix $K, L \geq 1$. Let $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{short}}_{j,K,L,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p}\cap\mathbb{N}$ such that

$u+1 \in A_{j,p} \cap \mathbb{N}$ and there are a level-u $(-1,+1)$-crossing run $[a,b]$ and a level-$u+1$ $(-1,+1)$-crossing run $[b+1,b']$ satisfying

$$d(x,v_t) \neq u \quad (0 \le t < a), \qquad d(x,v_t) > u \quad (b < t \le \tau_x^{\mathrm{end}}(\gamma)),$$

$$d(x,v_t) \neq u+1 \quad (0 \le t < b+1), \qquad d(x,v_t) > u+1 \quad (b' < t \le \tau_x^{\mathrm{end}}(\gamma)),$$

$$b, b' < \tau_x^{\mathrm{end}}(\gamma), \qquad b-a \le L, \quad b'-(b+1) \le L,$$

and

$$e_x(\gamma) - u > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H, \qquad e_x(\gamma) - (u+1) > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p)$$
$$= \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{short}}_{j,K,L,p}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.180 (Adjacent stacks are outward staircases)**.** *In the setting of Corollary 8.130, fix $K, L \ge 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{adj},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,2L+1}(s,p).$$

Consequently, the exponent hypothesis on adjacent stacked prefix-pair profiles follows from the same exponent hypothesis on two-level outward staircase profiles with the length cutoff $2L+1$.

Proof. Every adjacent stacked witness has $a' = b+1$. The two witnessing runs have lengths at most L, so the combined staircase interval satisfies

$$b' - a = (b-a) + 1 + (b'-a') \le 2L+1.$$

All record, persistence, and endpoint-clearance conditions are the same as in the adjacent stacked definition, with $u+1$ in place of the upper fixed-gap level. Hence

$$\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{adj},\mathrm{short}}_{j,K,L,p}(x,\gamma) \Longrightarrow \mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{short}}_{j,K,2L+1,p}(x,\gamma).$$

The profile inequality follows by monotonicity. □

Corollary 8.181 (Staircases are locally short or long)**.** *In the setting of Corollary 8.130, fix $K, L \ge 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,2L+1}(s,p)$$
$$\le W_{\beta,p}\, \mathfrak{T}^{G\to H,\mathrm{rad},\tan,\mathrm{long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p) + \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).$$

Consequently, the exponent hypothesis on two-level outward-staircase profiles with cutoff $2L+1$ follows from the same exponent hypothesis on long tangential runs and locally short two-level staircase profiles.

Proof. If $\mathcal{P}^{\mathrm{pre,per,stair,short}}_{j,K,2L+1,p}(x,\gamma)$ holds but $\mathcal{P}^{\mathrm{pre,per,stair,loc,short}}_{j,K,L,p}(x,\gamma)$ fails, then some counted staircase level u is not locally short. For its witnessing runs, either

$$b-a>L \qquad \text{or} \qquad b'-(b+1)>L.$$

In the first case $\mathcal{T}^{\mathrm{long}}_{u,L}(x,\gamma)$ holds; in the second case $\mathcal{T}^{\mathrm{long}}_{u+1,L}(x,\gamma)$ holds. Since the staircase definition requires both u and $u+1$ to lie in $A_{j,p}\cap\mathbb{N}$, we have

$$\mathcal{P}^{\mathrm{pre,per,stair,short}}_{j,K,2L+1,p} \subseteq \mathcal{P}^{\mathrm{pre,per,stair,loc,short}}_{j,K,L,p} \cup \bigcup_{w\in A_{j,p}\cap\mathbb{N}} \mathcal{T}^{\mathrm{long}}_{w,L}.$$

Taking the weighted property-restricted union bound and then the supremum in the profile definitions gives the displayed inequality, because $|A_{j,p}\cap\mathbb{N}|\le W_{\beta,p}$. □

Definition 8.182 (Bounded-gap locally short staircase-pair profiles)**.** In the setting of Corollary 8.130, fix $K,L,B\ge 1$. Let $\mathcal{P}^{\mathrm{pre,per,stair,loc,pair,short}}_{j,K,L,B,p}(x,\gamma)$ be the property that the number of $u\in A_{j,p}\cap\mathbb{N}$ for which there is a level $v\in A_{j,p}\cap\mathbb{N}$ with

$$u<v\le u+B$$

such that the locally short two-level staircase conditions from Definition 8.179 hold at lower levels u and v, with the same parameters K,L, is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\begin{aligned}\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,pair,short}}_{Q,\lambda,\sigma,\beta,j,K,L,B}(s,p)&\\ =\sup\Bigg\{&\sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d\mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,stair,loc,pair,short}}_{j,K,L,B,p}}_{r,d}\\ &: m\in\mathbb{N},\ c_sL_H<m\le\left(\frac{1}{\alpha_\omega}+3s\right)L_H\Bigg\}.\end{aligned}$$

Corollary 8.183 (Locally short staircases have bounded successor gaps)**.** *In the setting of Corollary 8.130, fix $K,L\ge 1$, and put*

$$B_{\lambda,\sigma,\beta,K}=\left\lceil\frac{16K\beta}{\theta_{\lambda,\sigma,\beta}}\right\rceil+1.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p)\le\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,pair,short}}_{Q,\lambda,\sigma,\beta,j,2K,L,B_{\lambda,\sigma,\beta,K}}(s,p).$$

Consequently, the exponent hypothesis on locally short two-level staircase profiles follows from the same exponent hypothesis on bounded-gap locally short staircase-pair profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,stair,loc,short}}_{j,K,L,p}(x,\gamma)$ holds, and sort the counted lower levels as

$$u_1<\cdots<u_{N_\star}.$$

Let $B = B_{\lambda,\sigma,\beta,K}$. Since all lower levels lie in the same slab,

$$\sum_{q=1}^{N_\star-1} (u_{q+1} - u_q) \le W_{\beta,p}.$$

For p close enough to $p_{\mathrm{c}}(H)$, $W_{\beta,p} \le \beta L_H$, and hence the number of indices q with $u_{q+1} - u_q > B$ is at most

$$\frac{\beta}{B} L_H < \frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H.$$

As in Corollary 8.167, more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K} L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(2K)} L_H$$

successor pairs have gap at most B. For each such pair, the two locally short staircase witnesses satisfy the bounded-gap staircase-pair conditions with parameter $2K$; the endpoint-clearance assumptions are inherited because replacing K by $2K$ weakens the required depth. Thus the bounded-gap staircase-pair property holds, giving the displayed profile inequality. □

Definition 8.184 (Fixed-gap locally short staircase-pair profiles)**.** In the setting of Corollary 8.130, fix $K, L, h \ge 1$. Let $\mathcal{P}^{\mathrm{pre,per,stair,loc,gap,short}}_{j,K,L,h,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ such that $u+h \in A_{j,p} \cap \mathbb{N}$ and the locally short two-level staircase conditions from Definition 8.179 hold at lower levels u and $u+h$, with the same parameters K, L, is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\begin{aligned}\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,gap,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p) \\ = \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2 \\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,stair,loc,gap,short}}_{j,K,L,h,p}}_{r,d} \\ : m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.\end{aligned}$$

Corollary 8.185 (Bounded-gap staircase pairs have a fixed gap)**.** *In the setting of Corollary 8.130, fix $K, L, B \ge 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,pair,short}}_{Q,\lambda,\sigma,\beta,j,K,L,B}(s,p) \le \sum_{h=1}^{B} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,gap,short}}_{Q,\lambda,\sigma,\beta,j,KB,L,h}(s,p).$$

Consequently, the exponent hypothesis on bounded-gap locally short staircase-pair profiles follows from the same exponent hypothesis on the finite family of fixed-gap locally short staircase-pair profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,stair,loc,pair,short}}_{j,K,L,B,p}(x,\gamma)$ holds. For $1 \le h \le B$, let N_h be the number of levels u for which the fixed gap h has locally short staircase witnesses at lower levels u and $u+h$, except with the stronger depth threshold $\theta_{\lambda,\sigma,\beta} L_H/(4K)$. Every level counted by the bounded-gap staircase-pair property contributes to at least one N_h, so

$$\sum_{h=1}^{B} N_h > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Thus some h has

$$N_h > \frac{\theta_{\lambda,\sigma,\beta}}{4KB} L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(KB)} L_H.$$

The depth assumptions with parameter K imply those with parameter KB. For this h,

$$\mathcal{P}^{\mathrm{pre,per,stair,loc,gap,short}}_{j,KB,L,h,p}(x,\gamma)$$

holds. Taking the property-restricted union bound over $1 \le h \le B$ gives the displayed profile inequality. □

Definition 8.186 (Locally short outward staircase-chain profiles)**.** In the setting of Corollary 8.130, fix $K, L \ge 1$ and $\ell \ge 2$. Let $\mathcal{P}^{\mathrm{pre,per,stair,loc,chain,short}}_{j,K,L,\ell,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ such that $u + r \in A_{j,p} \cap \mathbb{N}$ for every $0 \le r < \ell$, and such that there are level-$u+r$ $(-1,+1)$-crossing runs $[a_r, b_r]$, $0 \le r < \ell$, satisfying

$$a_{r+1} = b_r + 1 \qquad (0 \le r < \ell - 1),$$

$$d(x, v_t) \ne u + r \quad (0 \le t < a_r), \qquad d(x, v_t) > u + r \quad (b_r < t \le \tau_x^{\mathrm{end}}(\gamma)) \qquad (0 \le r < \ell),$$

$$b_r < \tau_x^{\mathrm{end}}(\gamma), \qquad b_r - a_r \le L \qquad (0 \le r < \ell),$$

and

$$e_x(\gamma) - (u + r) > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H \qquad (0 \le r < \ell)$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) = \sup\Bigg\{ \sum_{\substack{n\ge 0,\ |2n-m|\le 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre,per,stair,loc,chain,short}}_{j,K,L,\ell,p}}_{n,d} : m \in \mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega} + 3s\right) L_H \Bigg\}.$$

Corollary 8.187 (Inward-free bridges are chains or long runs)**.** *In the setting of Corollary 8.130, fix $K, L, h \ge 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h+1}(s,p) + W_{\beta,p}\, \mathfrak{T}^{G\to H,\mathrm{rad,tan,long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p).$$

Consequently, the exponent hypothesis on inward-free confined bridge profiles follows from the same exponent hypothesis on locally short staircase-chain profiles of length $h+1$ and on the long-run profile.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre,per,bridge,free,short}}_{j,K,L,h,p}(x,\gamma)$ holds, and take a counted level u with inward-free bridge witnesses $[a,b]$ at level u and $[a',b']$ at level $u+h$. If $h = 1$, the confinement condition has no intermediate integer level and the bridge is adjacent, so the two witnesses form a locally short chain of length $2 = h + 1$.

Assume $h \geq 2$. Since the radial distance is nondecreasing on $[b, a']$, starts at level u, and first reaches level $u+h$ at a', for every $1 \leq r < h$ there is a unique maximal interval $[a_r, b_r] \subseteq [b+1, a'-1]$ on which

$$d(x, v_t) = u + r.$$

Moreover these intervals are consecutive and are $(-1, +1)$-crossing runs: if $[a_0, b_0] = [a, b]$ and $[a_h, b_h] = [a', b']$, then

$$a_{r+1} = b_r + 1 \qquad (0 \leq r < h).$$

The lower bridge run is record at level u. Hence no level $u + r$, $1 \leq r < h$, can have been visited before a_r, since any such visit would force an earlier visit to level u. Thus the intermediate runs are record runs. Persistence after each intermediate run follows from monotonicity until a' and from the upper-run persistence after b'. The endpoint-clearance inequalities for intermediate levels follow from the upper endpoint-clearance inequality.

Therefore, if all intermediate runs have length at most L, the counted start u also witnesses the locally short staircase-chain property of length $h + 1$. If not, then for some

$$w \in \{u + 1, \ldots, u + h - 1\} \subseteq A_{j,p} \cap \mathbb{N}$$

the long-run event $\mathcal{T}^{\mathrm{long}}_{w,L}(x, \gamma)$ holds.

Consequently,

$$\mathcal{P}^{\mathrm{pre,per,bridge,free,short}}_{j,K,L,h,p} \subseteq \mathcal{P}^{\mathrm{pre,per,stair,loc,chain,short}}_{j,K,L,h+1,p} \cup \bigcup_{w \in A_{j,p} \cap \mathbb{N}} \mathcal{T}^{\mathrm{long}}_{w,L}.$$

Taking the property-restricted union bound in the weighted sums and then the supremum in the profile definitions gives the displayed inequality, since $|A_{j,p} \cap \mathbb{N}| \leq W_{\beta,p}$. □

Definition 8.188 (Confined fixed-gap locally short staircase-bridge profiles)**.** In the setting of Corollary 8.130, fix $K, L \geq 1$ and $h \geq 3$. Let $\mathcal{P}^{\mathrm{pre,per,stair,loc,bridge,short}}_{j,K,L,h,p}(x, \gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ satisfying the fixed-gap locally short staircase-pair conditions from Definition 8.184, with lower staircase witnesses $[a, b]$, $[b+1, b']$ at levels $u, u+1$ and upper staircase witnesses $[c, d]$, $[d+1, d']$ at levels $u + h, u + h + 1$, and such that

$$u + 1 < d(x, v_t) < u + h \qquad (b' < t < c)$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G \to H,\mathrm{rad,tan,pre,out,per,stair,loc,bridge,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s, p) = \sup \Bigg\{ \sum_{\substack{r \geq 0,\ |2r-m| \leq 2 \\ d > \lambda L_H}} Q^d \mathfrak{R}^{G, \mathcal{P}^{\mathrm{pre,per,stair,loc,bridge,short}}_{j,K,L,h,p}}_{r,d} : m \in \mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega} + 3s \right) L_H \Bigg\}.$$

Corollary 8.189 (Fixed-gap staircase pairs split into chains and bridges)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$. Then*

$$\mathfrak{P}^{G \to H,\mathrm{rad,tan,pre,out,per,stair,loc,gap,short}}_{Q,\lambda,\sigma,\beta,j,K,L,1}(s, p) \leq \mathfrak{P}^{G \to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,3}(s, p),$$

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,gap,short}}_{Q,\lambda,\sigma,\beta,j,K,L,2}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,4}(s,p),$$

and, for every $h \ge 3$,

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,gap,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,bridge,short}}_{Q,\lambda,\sigma,\beta,j,K,L,h}(s,p).$$

Consequently, the exponent hypothesis on fixed-gap locally short staircase-pair profiles follows from the same exponent hypothesis on the three-level and four-level locally short staircase-chain profiles and the confined fixed-gap locally short staircase-bridge profiles with $h \ge 3$.

Proof. First suppose $h = 1$, and take a counted level u in $\mathcal{P}^{\mathrm{pre,per,stair,loc,gap,short}}_{j,K,L,1,p}(x,\gamma)$. Let the lower staircase witnesses be $[a_0,b_0]$, $[a_1,b_1]$ at levels $u, u+1$, where $a_1 = b_0+1$. Let the upper staircase witnesses be $[c,d]$, $[d+1,d']$ at levels $u+1, u+2$. Both $[a_1,b_1]$ and $[c,d]$ are record level-$u+1$ tangential runs, so their initial times are the first visit to level $u+1$. Hence $c = a_1$, and maximality of tangential runs gives $d = b_1$. Thus

$$[a_0,b_0], \quad [a_1,b_1], \quad [b_1+1,d']$$

are consecutive locally short outward staircase-chain witnesses at the levels $u, u+1, u+2$. All record, persistence, length, and endpoint-clearance conditions are exactly those supplied by the two fixed-gap staircase witnesses.

Now suppose $h = 2$. With the same notation for the lower staircase and with upper witnesses $[c,d]$, $[d+1,d']$ at levels $u+2, u+3$, the level-$u+1$ run $[a_1,b_1]$ exits outward, so

$$d(x, v_{b_1+1}) = u+2.$$

Before a_1 the path has not visited level $u+1$, and during $[a_1,b_1]$ it is at level $u+1$. Since the radial distance changes by at most one at each step, a previous visit to level $u+2$ would force a previous visit to level $u+1$. Hence b_1+1 is the first visit to level $u+2$. Since the upper lower run is record at level $u+2$, we have $c = b_1+1$. The four runs

$$[a_0,b_0], \quad [a_1,b_1], \quad [c,d], \quad [d+1,d']$$

therefore satisfy the locally short staircase-chain conditions at the levels $u, u+1, u+2, u+3$.

Finally suppose $h \ge 3$, and write the lower staircase witnesses as $[a,b]$, $[b+1,b']$ and the upper staircase witnesses as $[c,d]$, $[d+1,d']$. The upper lower run is record at level $u+h$, so c is the first time the path visits level $u+h$. A visit to level $u+h$ before $b+1$ would force a visit to level $u+1$ before $b+1$, while the times $b+1,\dots,b'$ themselves are at level $u+1$. Thus $b' < c$. For $b' < t < c$, persistence of the level-$u+1$ run gives

$$d(x, v_t) > u+1.$$

The path has not yet visited level $u+h$, and the radial distance sequence cannot jump over level $u+h$. Hence

$$d(x, v_t) < u+h \qquad (b' < t < c).$$

Thus every fixed-gap staircase-pair witness with $h \ge 3$ is a confined staircase-bridge witness. Each displayed profile inequality follows by monotonicity of property-restricted profiles. □

Definition 8.190 (Bounded-gap locally short staircase-chain-pair profiles)**.** In the setting of Corollary 8.130, fix $K, L, \ell, B \ge 1$ with $\ell \ge 2$. Let $\mathcal{P}^{\mathrm{pre,per,stair,loc,chain,pair,short}}_{j,K,L,\ell,B,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ for which there is a level $v \in A_{j,p} \cap \mathbb{N}$ with

$$u < v \le u + B$$

such that the locally short outward staircase-chain conditions from Definition 8.186 hold at lower levels u and v, with the same parameters K, L, ℓ, is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{pair},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,B}(s,p)$$
$$= \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{pair},\mathrm{short}}_{j,K,L,\ell,B,p}}_{r,d}$$
$$: m\in\mathbb{N},\ c_s L_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right) L_H \Bigg\}.$$

Corollary 8.191 (Staircase chains have bounded successor gaps)**.** *In the setting of Corollary 8.130, fix $K, L \ge 1$ and $\ell \ge 2$, and put*

$$B_{\lambda,\sigma,\beta,K} = \left\lceil \frac{16K\beta}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{pair},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,2K,L,\ell,B_{\lambda,\sigma,\beta,K}}(s,p).$$

Consequently, the exponent hypothesis on locally short staircase-chain profiles follows from the same exponent hypothesis on bounded-gap locally short staircase-chain-pair profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{j,K,L,\ell,p}(x,\gamma)$ holds, and sort the counted chain-start levels as

$$u_1 < \cdots < u_{N_\star}.$$

The same bounded-gap count as in Corollary 8.183 shows that, for p close enough to $p_{\mathrm{c}}(H)$, more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K} L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(2K)} L_H$$

successor pairs have gap at most $B_{\lambda,\sigma,\beta,K}$. For each such pair, the two chain witnesses satisfy the bounded-gap staircase-chain-pair conditions with parameter $2K$. The endpoint clearance assumptions are inherited because replacing K by $2K$ weakens the required depth. This proves the property inclusion and hence the profile inequality. □

Definition 8.192 (Fixed-gap locally short staircase-chain-pair profiles)**.** In the setting of Corollary 8.130, fix $K, L, h \ge 1$ and $\ell \ge 2$. Let $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{gap},\mathrm{short}}_{j,K,L,\ell,h,p}(x,\gamma)$ be the property that the number of $u \in A_{j,p} \cap \mathbb{N}$ such that $u+h \in A_{j,p} \cap \mathbb{N}$ and the locally short outward staircase-chain conditions from Definition 8.186 hold at lower levels u and $u+h$, with the same parameters K, L, ℓ, is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{gap},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,h}(s,p)$$
$$= \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{gap},\mathrm{short}}_{j,K,L,\ell,h,p}}_{r,d}$$
$$: m\in\mathbb{N},\ c_sL_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H\Bigg\}.$$

Corollary 8.193 (Bounded-gap staircase-chain pairs have a fixed gap)**.** *In the setting of Corollary 8.130, fix $K,L,\ell,B\ge 1$ with $\ell\ge 2$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{pair},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,B}(s,p)$$
$$\le \sum_{h=1}^{B} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{gap},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,KB,L,\ell,h}(s,p).$$

Consequently, the exponent hypothesis on bounded-gap locally short staircase-chain-pair profiles follows from the same exponent hypothesis on the finite family of fixed-gap locally short staircase-chain-pair profiles.

Proof. The proof is the same finite-gap pigeonhole used in Corollary 8.185. If the bounded-gap staircase-chain-pair property holds, write N_h for the number of counted starts whose successor chain starts at distance h, still using the stronger depth threshold $\theta_{\lambda,\sigma,\beta}L_H/(4K)$. Then

$$\sum_{h=1}^{B} N_h > \frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H,$$

so some h has

$$N_h > \frac{\theta_{\lambda,\sigma,\beta}}{4KB}L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(KB)}L_H.$$

The depth assumptions with parameter K imply those with parameter KB, so the fixed-gap staircase-chain-pair property with parameters (KB,L,ℓ,h) holds. Taking the property-restricted union bound over $1\le h\le B$ gives the displayed inequality. □

Definition 8.194 (Confined fixed-gap staircase-chain bridge profiles)**.** In the setting of Corollary 8.130, fix $K,L,h\ge 1$ and $\ell\ge 2$. Let $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{j,K,L,\ell,h,p}(x,\gamma)$ be the property that the number of $u\in A_{j,p}\cap\mathbb{N}$ satisfying the fixed-gap staircase-chain-pair conditions from Definition 8.192, with lower-chain witnesses $[a_r,b_r]$, $0\le r<\ell$, at levels $u+r$, and upper-chain witnesses $[c_r,d_r]$, $0\le r<\ell$, at levels $u+h+r$, and such that

$$u+\ell-1 < d(x,v_t) < u+h \qquad (b_{\ell-1}<t<c_0)$$

is larger than

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H.$$

Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,h}(s,p)$$
$$= \sup\Bigg\{ \sum_{\substack{r\ge 0,\ |2r-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{j,K,L,\ell,h,p}}_{r,d}$$
$$: m\in\mathbb{N},\ c_sL_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H \Bigg\}.$$

Corollary 8.195 (Fixed-gap staircase-chain pairs split)**.** *In the setting of Corollary 8.130, fix $K, L \ge 1$, $\ell \ge 2$, and $h \ge 1$. If $1 \le h \le \ell$, then*

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{gap},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,h}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell+h}(s,p).$$

If $h \ge \ell+1$, then

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{gap},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,h}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,h}(s,p).$$

Consequently, the exponent hypothesis on fixed-gap locally short staircase-chain-pair profiles follows from the same exponent hypothesis on longer locally short staircase chains and on confined fixed-gap staircase-chain bridge profiles.

Proof. Take a fixed-gap staircase-chain-pair witness with lower chain $[a_r, b_r]$, $0 \le r < \ell$, and upper chain $[c_r, d_r]$, $0 \le r < \ell$.

Suppose first that $1 \le h \le \ell$. For $0 \le r < \ell - h$, the lower run $[a_{h+r}, b_{h+r}]$ and the upper run $[c_r, d_r]$ are both record tangential runs at level $u + h + r$. Hence they have the same initial time, and maximality gives

$$[c_r, d_r] = [a_{h+r}, b_{h+r}] \qquad (0 \le r < \ell - h).$$

If $h < \ell$, the identity at $r = \ell - h - 1$, together with the consecutive condition in the upper chain, gives

$$c_{\ell-h} = d_{\ell-h-1} + 1 = b_{\ell-1} + 1.$$

If $h = \ell$, the lower last run exits outward at time $b_{\ell-1} + 1$; the radial distance cannot have visited level $u+\ell$ earlier without a previous visit to level $u+\ell-1$, while $[a_{\ell-1}, b_{\ell-1}]$ is the first level-$u+\ell-1$ run. Thus $c_0 = b_{\ell-1} + 1$. In both cases, concatenating the lower chain with the non-overlapping tail of the upper chain gives a locally short outward staircase chain of length $\ell + h$, with all record, persistence, length, and endpoint-clearance conditions inherited from the two witnesses.

Now suppose $h \ge \ell + 1$. The upper first run is record at level $u + h$, so c_0 is the first time the path visits level $u + h$. A visit to level $u + h$ before $a_{\ell-1}$ would force a visit to level $u + \ell - 1$ before $a_{\ell-1}$, and the interval $[a_{\ell-1}, b_{\ell-1}]$ itself is at level $u + \ell - 1$. Hence $b_{\ell-1} < c_0$. For $b_{\ell-1} < t < c_0$, persistence of the last lower-chain run gives

$$d(x, v_t) > u + \ell - 1,$$

while the record property of the upper first run and nearest-neighbour radial motion give

$$d(x, v_t) < u + h.$$

Thus the fixed-gap chain-pair witness is a confined staircase-chain bridge. The two displayed profile inequalities follow by monotonicity. □

Corollary 8.196 (Staircase-chain bridges project to ordinary bridges)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$, $\ell \geq 2$, and $h \geq \ell + 1$. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,h}(s,p) \leq \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,h-\ell+1}(s,p).$$

Consequently, the exponent hypothesis on confined fixed-gap staircase-chain bridge profiles follows from the corresponding exponent hypothesis on ordinary confined fixed-gap bridge profiles.

Proof. Suppose that $\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{j,K,L,\ell,h,p}(x,\gamma)$ holds, and take a counted lower-chain start u. Let $[a_r, b_r]$, $0 \leq r < \ell$, be the lower-chain witnesses at levels $u + r$, and let $[c_r, d_r]$, $0 \leq r < \ell$, be the upper-chain witnesses at levels $u + h + r$. Put

$$U = u + \ell - 1, \qquad g = h - \ell + 1.$$

Then $g \geq 2$, $U \in A_{j,p} \cap \mathbb{N}$, and

$$U + g = u + h \in A_{j,p} \cap \mathbb{N},$$

because the lower and upper chain conditions require the starts $u + \ell - 1$ and $u + h$ to lie in the slab.

The last lower-chain run $[a_{\ell-1}, b_{\ell-1}]$ is a record $(-1,+1)$-crossing run at level U, and the first upper-chain run $[c_0, d_0]$ is a record $(-1,+1)$-crossing run at level $U + g$. Their length, persistence, and endpoint-clearance assumptions are exactly the corresponding fixed-gap prefix-pair assumptions with gap g. The chain-bridge confinement gives

$$U < d(x, v_t) < U + g \qquad (b_{\ell-1} < t < c_0),$$

which is the ordinary confined bridge condition. Thus every counted chain-bridge start u yields the counted ordinary bridge level U. The map $u \mapsto U = u + \ell - 1$ is injective, so the counted-level lower bound is preserved with the same parameter K. This proves the property inclusion and hence the profile inequality. □

Corollary 8.197 (One-step staircase-chain amplification)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$ and $\ell \geq 2$. Put*

$$B_{\lambda,\sigma,\beta,K} = \left\lceil \frac{16K\beta}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1, \qquad K^+ = 2KB_{\lambda,\sigma,\beta,K}.$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad \leq \sum_{h=1}^{\min\{\ell, B_{\lambda,\sigma,\beta,K}\}} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K^+,L,\ell+h}(s,p) \\
&\qquad + \sum_{h=\ell+1}^{B_{\lambda,\sigma,\beta,K}} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K^+,L,\ell,h}(s,p),
\end{aligned}$$

where the second sum is absent if $B_{\lambda,\sigma,\beta,K} \leq \ell$. More generally, if $\widetilde{B} \geq B_{\lambda,\sigma,\beta,K}$ and $\widetilde{K} = 2K\widetilde{B}$, then

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad \leq \sum_{h=1}^{\ell} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,\widetilde{K},L,\ell+h}(s,p) \\
&\qquad + \sum_{h=\ell+1}^{\widetilde{B}} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,\widetilde{K},L,\ell,h}(s,p).
\end{aligned}$$

Consequently, the exponent hypothesis on locally short staircase-chain profiles follows from the same exponent hypothesis on the displayed longer-chain and confined chain-bridge profiles.

Proof. Apply Corollary 8.191 to obtain a bounded-gap chain-pair profile with parameter $2K$ and cutoff $B_{\lambda,\sigma,\beta,K}$. Then apply Corollary 8.193, which replaces this by fixed-gap chain-pair profiles with parameter

$$K^{+} = 2KB_{\lambda,\sigma,\beta,K}.$$

Finally split each fixed gap using Corollary 8.195: gaps $1 \le h \le \ell$ glue into chains of length $\ell + h$, and gaps $h \ge \ell + 1$ become confined chain bridges. Combining the three displayed profile inequalities gives the claim. The more general displayed inequality follows from the first one by monotonicity in the depth parameter and by adding finitely many nonnegative chain and bridge terms. □

Corollary 8.198 (Finite staircase-chain amplification)**.** *In the setting of Corollary 8.130, fix $K, L \ge 1$, $\ell \ge 2$, and $n \ge 0$. Put $K_0 = K$. For $1 \le r \le n$, choose integers B_r recursively so that*

$$B_r \ge B_{\lambda,\sigma,\beta,K_{r-1}} = \left\lceil \frac{16K_{r-1}\beta}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1, \qquad K_r = 2K_{r-1}B_r.$$

Let $\mathcal{A}_0(\ell) = \{(\ell)\}$, and for $r \ge 1$ let $\mathcal{A}_r(\ell)$ be the set of sequences $\mathbf{a} = (a_0, \dots, a_r)$ such that $(a_0, \dots, a_{r-1}) \in \mathcal{A}_{r-1}(\ell)$ and

$$a_{r-1} + 1 \le a_r \le 2a_{r-1}.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad \le \sum_{\mathbf{a}\in\mathcal{A}_n(\ell)} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a_n}(s,p) \\
&\qquad + \sum_{r=1}^{n} \sum_{\mathbf{a}\in\mathcal{A}_{r-1}(\ell)} \sum_{h=a_{r-1}+1}^{B_r} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,a_{r-1},h}(s,p).
\end{aligned}$$

Here an inner bridge sum is interpreted as zero when $B_r \le a_{r-1}$, and the sums over $\mathcal{A}_r(\ell)$ keep multiplicity through the sequence $\mathbf{a}$. Consequently, the exponent hypothesis on the initial chain profile follows from the same exponent hypothesis on the terminal chain profiles and the accumulated confined chain-bridge profiles displayed above.

Proof. The case $n = 0$ is the identity inequality. Suppose the claim has been proved for $n - 1$. There are finitely many terminal sequences $\mathbf{a} = (a_0, \dots, a_{n-1}) \in \mathcal{A}_{n-1}(\ell)$, so Corollary 8.197 may be applied to each terminal chain profile with

$$(\widetilde{B}, \widetilde{K}) = (B_n, K_n)$$

and with chain length a_{n-1}. This gives, for each such $\mathbf{a}$,

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{n-1},L,a_{n-1}}(s,p) \\
&\quad \le \sum_{a_n=a_{n-1}+1}^{2a_{n-1}} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a_n}(s,p) \\
&\qquad + \sum_{h=a_{n-1}+1}^{B_n} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a_{n-1},h}(s,p).
\end{aligned}$$

Substituting these finitely many inequalities into the induction hypothesis for $n-1$ gives exactly the displayed bound for n. Since all index sets are finite and independent of p, the exponent implication follows from the union-bound form of the displayed inequality. □

Corollary 8.199 (Multiplicity-free finite chain amplification)**.** *In the setting of Corollary 8.198, define*

$$I_r(\ell) = \{\ell + r, \ell + r + 1, \dots, 2^r \ell\} \qquad (r \geq 0),$$

where $I_0(\ell) = \{\ell\}$*. Then, for all* p *sufficiently close to* $p_c(H)$*,*

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad \leq \sum_{a\in I_n(\ell)} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a}(s,p) \\
&\qquad + \sum_{r=1}^{n} \sum_{a\in I_{r-1}(\ell)} \sum_{h=a+1}^{B_r} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,a,h}(s,p).
\end{aligned}$$

As usual, an inner bridge sum is zero when $B_r \leq a$*. Consequently, the exponent hypothesis on the initial chain profile follows from the same exponent hypothesis on these distinct terminal chain and accumulated chain-bridge profiles.*

Proof. We use the property inclusions underlying Corollary 8.197, rather than the already-summed inequality. For every $a \geq 2$, the one-step proof gives

$$\begin{aligned}
\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{j,K_{r-1},L,a,p} \subseteq &\bigcup_{b=a+1}^{2a} \mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{j,K_r,L,b,p} \\
&\cup \bigcup_{h=a+1}^{B_r} \mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{j,K_r,L,a,h,p}.
\end{aligned}$$

The possible chain lengths after r chain continuations are exactly the integer interval $I_r(\ell)$, since

$$\bigcup_{a\in I_{r-1}(\ell)} \{a+1, \dots, 2a\} = I_r(\ell).$$

Iterating the displayed property inclusion n times therefore gives a single union over the distinct terminal lengths in $I_n(\ell)$, together with the distinct bridge profiles created at each intermediate step. Taking the property-restricted union bound gives the displayed profile inequality. All displayed index sets are finite and independent of p, so the exponent implication follows. □

Corollary 8.200 (Multiplicity-free chain amplification with ordinary bridges)**.** *In the setting of Corollary 8.199, for all* p *sufficiently close to* $p_c(H)$*,*

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad \leq \sum_{a\in I_n(\ell)} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a}(s,p) \\
&\qquad + \sum_{r=1}^{n} \sum_{g=2}^{B_r-\ell-r+2} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,g}(s,p).
\end{aligned}$$

An inner ordinary-bridge sum is interpreted as zero when $B_r - \ell - r + 2 < 2$*. Consequently, the exponent hypothesis on the initial chain profile follows from the same exponent hypothesis on the distinct terminal chain profiles and the displayed ordinary confined fixed-gap bridge profiles.*

Proof. Starting from Corollary 8.199, each bridge residual at step r has the form

$$\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,a,h}(s,p),$$

where $a \in I_{r-1}(\ell)$ and $h \geq a+1$. By Corollary 8.196, this term is bounded by the ordinary bridge profile with gap

$$g = h - a + 1.$$

For $a \in I_{r-1}(\ell)$, we have $a \geq \ell + r - 1$, so

$$2 \leq g \leq B_r - a + 1 \leq B_r - \ell - r + 2.$$

Thus the union of all projected bridge residuals created at stage r is contained in the union of the ordinary bridge properties with

$$2 \leq g \leq B_r - \ell - r + 2.$$

Taking the property-restricted union bound, with duplicate ordinary bridge gaps counted only once, gives the displayed inequality. The exponent implication follows because the displayed index sets are finite for fixed n. □

Corollary 8.201 (Ordinary-bridge truncation in finite chain amplification)**.** *In the setting of Corollary 8.200, for all p sufficiently close to $p_{\mathrm{c}}(H)$,*

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad\leq \sum_{a\in I_n(\ell)} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a}(s,p) \\
&\qquad+\sum_{r=1}^{n} \sum_{\substack{2\leq g\leq B_r-\ell-r+2 \\ g<W_{\beta,p}-\frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,g}(s,p).
\end{aligned}$$

In particular, the bridge sum at stage r may always be truncated to gaps $2 \leq g < W_{\beta,p}$.

Proof. Start from Corollary 8.200. For each stage r, Lemma 8.172 shows that every ordinary bridge term with

$$g \geq W_{\beta,p} - \frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H$$

is zero. Removing these zero terms gives the displayed inequality. The last assertion is the weaker special case $g < W_{\beta,p}$. □

Corollary 8.202 (Finite chain amplification with inward-free bridges)**.** *In the setting of Corollary 8.201, for all p sufficiently close to $p_{\mathrm{c}}(H)$,*

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad\leq \sum_{a\in I_n(\ell)} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a}(s,p) \\
&\qquad+\sum_{r=1}^{n} \sum_{\substack{2\leq g\leq B_r-\ell-r+2 \\ g<W_{\beta,p}-\frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{free},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,2K_r,L,g}(s,p) \\
&\qquad+\sum_{r=1}^{n} \sum_{\substack{2\leq g\leq B_r-\ell-r+2 \\ g<W_{\beta,p}-\frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H}} \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda,\sigma,\beta,j,2K_r(g-1)}(s,p).
\end{aligned}$$

Consequently, for fixed n, the exponent hypothesis on the initial chain profile follows from uniform exponent hypotheses on the displayed terminal chain profiles, inward-free bridge profiles, and spread-return profiles over the displayed gap ranges.

Proof. Apply Corollary 8.174 to each ordinary bridge term in Corollary 8.201. Since every displayed gap satisfies $g \geq 2$, the return-profile parameter is $2K_r(g-1)$. This gives the displayed inequality.

For fixed n, the number of displayed gaps at each stage is at most $W_{\beta,p}$, by Corollary 8.201. Thus the total number of nonterminal summands is $O_n(W_{\beta,p})$, and

$$\frac{\log_{D_H} W_{\beta,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)).$$

Uniform logarithmic exponent bounds over the displayed ranges therefore imply the same bound for their finite union. □

Corollary 8.203 (Finite chain amplification with chain-only bridge residuals)**.** *In the setting of Corollary 8.202, for all p sufficiently close to $p_{\mathrm{c}}(H)$,*

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad\leq \sum_{a\in I_n(\ell)} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a}(s,p) \\
&\qquad + \sum_{r=1}^{n} \sum_{\substack{2\leq g\leq B_r-\ell-r+2 \\ g<W_{\beta,p}-\frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H}} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,2K_r,L,g+1}(s,p) \\
&\qquad + \sum_{r=1}^{n} \sum_{\substack{2\leq g\leq B_r-\ell-r+2 \\ g<W_{\beta,p}-\frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H}} \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j,2K_r(g-1)}(s,p) \\
&\qquad + n\,W_{\beta,p}^2\,\mathfrak{T}^{G\to H,\mathrm{rad,tan,long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p).
\end{aligned}$$

Consequently, for fixed n, the bridge residuals in finite chain amplification can be replaced by locally short chain profiles of lengths $g+1$, spread-return profiles, and the long-run profile, over the displayed slab-truncated gap ranges.

Proof. Apply Corollary 8.187 to each inward-free bridge term in Corollary 8.202, with $(K,h) = (2K_r, g)$. This replaces that bridge term by a chain profile of length $g+1$ and a term

$$W_{\beta,p}\,\mathfrak{T}^{G\to H,\mathrm{rad,tan,long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p).$$

At each stage r, the gap range contains at most $W_{\beta,p}$ integers by Corollary 8.201. Summing the long-run contributions over $1 \leq r \leq n$ and all displayed gaps gives the coefficient $nW_{\beta,p}^2$. The remaining terms are exactly those displayed. □

Lemma 8.204 (Long chains do not fit in one slab)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$ and $\ell \geq 2$. If*

$$\ell > W_{\beta,p},$$

then

$$\mathcal{P}^{\mathrm{pre,per,stair,loc,chain,short}}_{j,K,L,\ell,p}(x,\gamma)$$

is false for every rooted path (x,γ). Consequently,

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) = 0.$$

Proof. The slab

$$A_{j,p} = [jW_{\beta,p}, (j+1)W_{\beta,p})$$

contains at most $W_{\beta,p}$ integer levels. A locally short staircase-chain witness of length ℓ starting at u requires the ℓ distinct integer levels

$$u, u+1, \dots, u+\ell-1$$

all to belong to $A_{j,p} \cap \mathbb{N}$. This is impossible when $\ell > W_{\beta,p}$. Hence the defining property is empty, and the corresponding property-restricted profile is zero. □

Lemma 8.205 (Long chains do not have enough slab starts)**.** *In the setting of Corollary 8.130, fix $K, L \geq 1$ and $\ell \geq 2$. If*

$$W_{\beta,p} - \ell + 1 \leq \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

then

$$\mathcal{P}^{\mathrm{pre,per,stair,loc,chain,short}}_{j,K,L,\ell,p}(x,\gamma)$$

is false for every rooted path (x,γ). Consequently,

$$\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) = 0.$$

In particular, the profile is zero whenever $\ell > W_{\beta,p}$.

Proof. A chain witness of length ℓ starting at u requires all levels

$$u, u+1, \dots, u+\ell-1$$

to belong to $A_{j,p} \cap \mathbb{N}$. Since this slab contains at most $W_{\beta,p}$ integer levels, there are at most

$$\max\{W_{\beta,p} - \ell + 1, 0\}$$

possible starts. Under the displayed assumption this number is at most $\theta_{\lambda,\sigma,\beta} L_H/(4K)$, while the chain property requires strictly more than that many counted starts. Hence the defining property is empty. The final assertion is the special case in which the maximum is zero. □

Corollary 8.206 (Start-count truncation of bridge-free finite chains)**.** *In the setting of Corollary 8.203, for all p sufficiently close to $p_{\mathrm{c}}(H)$,*

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\
&\quad\leq \sum_{\substack{a\in I_n(\ell) \\ a<W_{\beta,p}+1-\frac{\theta_{\lambda,\sigma,\beta}}{4K_n}L_H}} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K_n,L,a}(s,p) \\
&\qquad+\sum_{r=1}^{n} \sum_{\substack{2\leq g\leq B_r-\ell-r+2 \\ g<W_{\beta,p}-\frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H}} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,2K_r,L,g+1}(s,p) \\
&\qquad+\sum_{r=1}^{n} \sum_{\substack{2\leq g\leq B_r-\ell-r+2 \\ g<W_{\beta,p}-\frac{\theta_{\lambda,\sigma,\beta}}{4K_r}L_H}} \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j,2K_r(g-1)}(s,p) \\
&\qquad+ n\,W_{\beta,p}^2\,\mathfrak{T}^{G\to H,\mathrm{rad,tan,long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p).
\end{aligned}$$

Consequently, terminal chain lengths in the top $\theta_{\lambda,\sigma,\beta}L_H/(4K_n)$ portion of the slab do not contribute to the bridge-free finite chain reduction.

Proof. Apply Corollary 8.203. For any terminal length $a \in I_n(\ell)$ satisfying

$$a \geq W_{\beta,p} + 1 - \frac{\theta_{\lambda,\sigma,\beta}}{4K_n} L_H,$$

we have

$$W_{\beta,p} - a + 1 \leq \frac{\theta_{\lambda,\sigma,\beta}}{4K_n} L_H.$$

Lemma 8.205 therefore makes the corresponding terminal chain profile zero. Removing these zero terminal terms gives the displayed inequality. □

Remark 8.207 (The remaining iteration obstruction)**.** Lemma 8.204 shows that terminal chain profiles disappear once the amplified chain length exceeds the slab width. Since $W_{\beta,p} \asymp L_H$, reaching this deterministic cutoff through the lower growth bound $a \geq \ell + n$ requires an iteration depth of order L_H, not a fixed depth. Thus the remaining obstruction is no longer a local gluing issue: it is to control the accumulated bridge families under a depth that grows with $p \uparrow p_{\mathrm{c}}(H)$, or to replace the linear-depth iteration by a sharper amplification.

Lemma 8.208 (Counting range of the successor-gap step)**.** *In the proof of Corollary 8.191, the successor-gap conclusion follows from the counted-start lower bound only in the range*

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H > 1.$$

More precisely, if

$$N_\star > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H$$

chain-start levels lie in one slab and

$$B_{\lambda,\sigma,\beta,K} = \left\lceil \frac{16K\beta}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1,$$

then the argument proves that more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K} L_H$$

successor pairs have gap at most $B_{\lambda,\sigma,\beta,K}$, provided $\theta_{\lambda,\sigma,\beta} L_H/(16K) > 1$. If instead

$$K \geq \frac{\theta_{\lambda,\sigma,\beta}}{4} L_H,$$

then the lower bound

$$N_\star > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H$$

is compatible with $N_\star = 1$, and no successor-pair conclusion can be deduced from this counting input alone.

Proof. Let $B = B_{\lambda,\sigma,\beta,K}$. Since all counted starts lie in one slab,

$$\sum_{q=1}^{N_\star - 1} (u_{q+1} - u_q) \leq W_{\beta,p}.$$

For p close enough to $p_{\mathrm{c}}(H)$, $W_{\beta,p} \le \beta L_H$, so the number of successor gaps larger than B is at most

$$\frac{\beta}{B} L_H < \frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H.$$

Thus the number of successor gaps at most B is greater than

$$N_\star - 1 - \frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H.$$

Set

$$a = \frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H.$$

The hypothesis $N_\star > 4a$, together with $a > 1$, gives

$$N_\star - 1 - a > 2a = \frac{\theta_{\lambda,\sigma,\beta}}{8K} L_H.$$

This is the claimed successor-pair lower bound. The final assertion is just the observation that, when $K \ge \theta_{\lambda,\sigma,\beta} L_H/4$, the defining lower bound on $N_\star$ is at most 1. A single counted start has no successor pair, so the count alone cannot force a bounded-gap successor. □

Lemma 8.209 (Residual-width successor-gap count for chains)**.** *In the setting of Corollary 8.130, fix $K, L \ge 1$ and $\ell \ge 2$. Put*

$$R_{\ell,p} = \max\{W_{\beta,p} - \ell + 1, 0\}$$

and

$$B^{\mathrm{res}}_{K,\ell,p} = \left\lceil \frac{16K R_{\ell,p}}{\theta_{\lambda,\sigma,\beta} L_H} \right\rceil + 1.$$

If

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H > 1,$$

then

$$\mathcal{P}^{\mathrm{pre,per,stair,loc,chain,short}}_{j,K,L,\ell,p} \subseteq \mathcal{P}^{\mathrm{pre,per,stair,loc,chain,pair,short}}_{j,2K,L,\ell,B^{\mathrm{res}}_{K,\ell,p},p}.$$

Consequently, for such p,

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p)\\ &\quad\le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,pair,short}}_{Q,\lambda,\sigma,\beta,j,2K,L,\ell,B^{\mathrm{res}}_{K,\ell,p}}(s,p).\end{aligned}$$

Proof. Suppose that the chain property holds, and list the counted chain starts as

$$u_1 < \cdots < u_{N_\star}.$$

For every counted start u_q, the whole interval

$$u_q, u_q + 1, \ldots, u_q + \ell - 1$$

lies in $A_{j,p} \cap \mathbb{N}$. Hence the starts lie in an interval of integer length at most $R_{\ell,p}$, and therefore

$$\sum_{q=1}^{N_\star - 1} (u_{q+1} - u_q) \le R_{\ell,p}.$$

The number of successor gaps larger than $B^{\mathrm{res}}_{K,\ell,p}$ is at most

$$\frac{R_{\ell,p}}{B^{\mathrm{res}}_{K,\ell,p}} < \frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H.$$

Since the chain property gives

$$N_\star > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

the same calculation as in Lemma 8.208 shows that more than

$$\frac{\theta_{\lambda,\sigma,\beta}}{8K} L_H = \frac{\theta_{\lambda,\sigma,\beta}}{4(2K)} L_H$$

successor pairs have gap at most $B^{\mathrm{res}}_{K,\ell,p}$. Each such successor pair gives the bounded-gap staircase-chain-pair property with parameter $2K$; endpoint-clearance conditions are inherited because the depth requirement is weakened. This proves the property inclusion, and the profile inequality follows by the property-restricted union bound. □

Corollary 8.210 (Near-top chains have bounded residual successor cutoffs)**.** *In the setting of Lemma 8.209, fix $C > 0$ and put*

$$B_C = \left\lceil \frac{16C}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1.$$

If

$$R_{\ell,p} \le \frac{C}{K} L_H \qquad \text{and} \qquad \frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H > 1,$$

then

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\ &\quad \le \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{pair},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,2K,L,\ell,B_C}(s,p). \end{aligned}$$

Proof. The residual bound gives

$$B^{\mathrm{res}}_{K,\ell,p} \le B_C.$$

Apply Lemma 8.209 and then use monotonicity of the bounded-gap chain-pair property in the gap cutoff. □

Corollary 8.211 (Residual-width one-step amplification)**.** *In the setting of Lemma 8.209, put*

$$K^{\mathrm{res}}_{K,\ell,p} = 2K B^{\mathrm{res}}_{K,\ell,p}.$$

If

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K} L_H > 1,$$

then

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \\ &\quad \le \sum_{h=1}^{\min\{\ell,B^{\mathrm{res}}_{K,\ell,p}\}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K^{\mathrm{res}}_{K,\ell,p},L,\ell+h}(s,p) \\ &\qquad + \sum_{h=\ell+1}^{B^{\mathrm{res}}_{K,\ell,p}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K^{\mathrm{res}}_{K,\ell,p},L,\ell,h}(s,p), \end{aligned}$$

where the second sum is absent if $B^{\mathrm{res}}_{K,\ell,p} \le \ell$.

Proof. Apply Lemma 8.209 to obtain a bounded-gap chain-pair profile with parameter $2K$ and cutoff $B^{\mathrm{res}}_{K,\ell,p}$. Corollary 8.193 then replaces this profile by fixed-gap chain-pair profiles with parameter

$$(2K)B^{\mathrm{res}}_{K,\ell,p} = K^{\mathrm{res}}_{K,\ell,p}.$$

Finally apply Corollary 8.195: gaps $1 \le h \le \ell$ glue into chains of length $\ell + h$, and gaps $h \ge \ell + 1$ become confined chain bridges. □

Corollary 8.212 (Near-top one-step amplification has bounded cost)**.** *In the setting of Corollary 8.210, put*

$$K_C = 2KB_C.$$

If

$$R_{\ell,p} \le \frac{C}{K}L_H \qquad \text{and} \qquad \frac{\theta_{\lambda,\sigma,\beta}}{16K}L_H > 1,$$

then

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p)\\
&\quad\le \sum_{h=1}^{\min\{\ell,B_C\}} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_C,L,\ell+h}(s,p)\\
&\qquad+\sum_{h=\ell+1}^{B_C} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_C,L,\ell,h}(s,p),
\end{aligned}$$

where the second sum is absent if $B_C \le \ell$.

Proof. Apply Corollary 8.210 to get a bounded-gap chain-pair profile with parameter $2K$ and cutoff B_C. The fixed-gap pigeonhole and split steps are exactly those used in the proof of Corollary 8.211, with B_C in place of $B^{\mathrm{res}}_{K,\ell,p}$. □

Corollary 8.213 (Bridge-free residual descent near the slab top)**.** *In the setting of Corollary 8.211, suppose that*

$$B^{\mathrm{res}}_{K,\ell,p} \le \ell.$$

Then

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p)\\
&\quad\le \sum_{h=1}^{B^{\mathrm{res}}_{K,\ell,p}} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K^{\mathrm{res}}_{K,\ell,p},L,\ell+h}(s,p).
\end{aligned}$$

Moreover, every displayed chain term has residual width

$$R_{\ell+h,p} = \max\{R_{\ell,p} - h, 0\} \le \max\{R_{\ell,p} - 1, 0\}.$$

The hypothesis $B^{\mathrm{res}}_{K,\ell,p} \le \ell$ is implied by

$$R_{\ell,p} \le \frac{\theta_{\lambda,\sigma,\beta}(\ell-1)}{16K}L_H.$$

In particular, if

$$R_{\ell,p} \le \frac{C}{K}L_H, \qquad C \le \frac{\theta_{\lambda,\sigma,\beta}(\ell-1)}{16},$$

then the near-top one-step amplification of Corollary 8.212 has no confined chain-bridge residuals.

Proof. Under $B^{\mathrm{res}}_{K,\ell,p} \le \ell$, the bridge sum in Corollary 8.211 is absent. This gives the displayed inequality. The residual identity follows directly from the definition:

$$R_{\ell+h,p} = \max\{W_{\beta,p} - (\ell+h) + 1, 0\} = \max\{R_{\ell,p} - h, 0\}.$$

If

$$R_{\ell,p} \le \frac{\theta_{\lambda,\sigma,\beta}(\ell-1)}{16K} L_H,$$

then

$$\frac{16KR_{\ell,p}}{\theta_{\lambda,\sigma,\beta}L_H} \le \ell - 1,$$

so $B^{\mathrm{res}}_{K,\ell,p} \le \ell$. The final assertion is the same criterion after using the bound $R_{\ell,p} \le CL_H/K$. □

Corollary 8.214 (Finite bridge-free residual descent iteration)**.** *In the setting of Corollary 8.213, fix $m \ge 0$. Let $\mathcal{H}_0 = \{\varnothing\}$, and set*

$$\ell(\varnothing) = \ell, \qquad K(\varnothing) = K.$$

For each word $\mathbf{h} \in \mathcal{H}_r$, put

$$B(\mathbf{h}) = B^{\mathrm{res}}_{K(\mathbf{h}),\ell(\mathbf{h}),p}.$$

Assume that, for every $0 \le r < m$ and every $\mathbf{h} \in \mathcal{H}_r$,

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K(\mathbf{h})} L_H > 1, \qquad B(\mathbf{h}) \le \ell(\mathbf{h}).$$

Define $\mathcal{H}_{r+1}$ to consist of all words $\mathbf{h} \frown h$, where $\mathbf{h} \in \mathcal{H}_r$ and

$$1 \le h \le B(\mathbf{h}),$$

and set

$$\ell(\mathbf{h} \frown h) = \ell(\mathbf{h}) + h, \qquad K(\mathbf{h} \frown h) = 2K(\mathbf{h})B(\mathbf{h}).$$

Then

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p)\\ &\quad\le \sum_{\mathbf{h}\in\mathcal{H}_m} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K(\mathbf{h}),L,\ell(\mathbf{h})}(s,p).\end{aligned}$$

Moreover, every terminal word $\mathbf{h} \in \mathcal{H}_m$ satisfies

$$R_{\ell(\mathbf{h}),p} \le \max\{R_{\ell,p} - m, 0\}.$$

Consequently, if the displayed hypotheses hold through some $m \ge R_{\ell,p}$, then

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) = 0.$$

Proof. The case $m = 0$ is the identity inequality. Suppose the inequality has been proved at depth $r < m$. For each terminal word $\mathbf{h} \in \mathcal{H}_r$, the hypotheses allow Corollary 8.213 to be applied with (K,ℓ) replaced by $(K(\mathbf{h}),\ell(\mathbf{h}))$. This replaces the terminal chain profile by the sum over children $\mathbf{h} \frown h$, $1 \le h \le B(\mathbf{h})$, with the displayed recursive parameters. Substituting these finitely many inequalities gives the depth $r+1$ statement, and induction proves the displayed bound.

At each child step, Corollary 8.213 gives

$$R_{\ell(\mathbf{h}\frown h),p} = \max\{R_{\ell(\mathbf{h}),p} - h, 0\} \le \max\{R_{\ell(\mathbf{h}),p} - 1, 0\}.$$

Iterating this estimate along a word of length m gives the residual bound. If $m \geq R_{\ell,p}$, then every terminal word has

$$R_{\ell(\mathbf{h}),p} = 0,$$

so

$$W_{\beta,p} - \ell(\mathbf{h}) + 1 \leq 0 \leq \frac{\theta_{\lambda,\sigma,\beta}}{4K(\mathbf{h})} L_H.$$

Lemma 8.205 makes every terminal profile in the displayed sum equal to zero. □

Lemma 8.215 (Bridge-free residual exhaustion has logarithmic depth obstruction)**.** *In the setting of Corollary 8.214, suppose that $R_{\ell,p} \geq 1$ and that the hypotheses of that corollary hold through some depth $m \geq R_{\ell,p}$. Then*

$$4^{R_{\ell,p}-1} K < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H.$$

Equivalently,

$$R_{\ell,p} < 1 + \log_4 \left(\frac{\theta_{\lambda,\sigma,\beta} L_H}{16K} \right).$$

Thus, under the current bridge-free descent recursion, residual exhaustion from a fixed K can only be certified in this way when the initial residual start width is $O(\log L_H)$.

Proof. Let $\mathbf{1}_r$ denote the word of length r all of whose letters are equal to 1. Since $1 \leq B(\mathbf{h})$ for every node, the words $\mathbf{1}_r$ belong to $\mathcal{H}_r$ for all $r \leq R_{\ell,p} - 1$. Along this branch,

$$\ell(\mathbf{1}_r) = \ell + r,$$

and therefore

$$R_{\ell(\mathbf{1}_r),p} = \max\{R_{\ell,p} - r, 0\} = R_{\ell,p} - r \qquad (0 \leq r \leq R_{\ell,p} - 1).$$

In particular, for $0 \leq r \leq R_{\ell,p} - 2$, the residual width is positive, so

$$B(\mathbf{1}_r) = B^{\mathrm{res}}_{K(\mathbf{1}_r),\ell(\mathbf{1}_r),p} \geq 2.$$

The recursive definition then gives

$$K(\mathbf{1}_{r+1}) = 2K(\mathbf{1}_r)B(\mathbf{1}_r) \geq 4K(\mathbf{1}_r) \qquad (0 \leq r \leq R_{\ell,p} - 2),$$

and hence

$$K(\mathbf{1}_{R_{\ell,p}-1}) \geq 4^{R_{\ell,p}-1} K.$$

Because $m \geq R_{\ell,p}$, the hypotheses of Corollary 8.214 include the counting-range condition at the node $\mathbf{1}_{R_{\ell,p}-1}$:

$$\frac{\theta_{\lambda,\sigma,\beta}}{16K(\mathbf{1}_{R_{\ell,p}-1})} L_H > 1.$$

Combining the last two displays gives the claimed inequality. The logarithmic form is just its base-4 logarithm. □

Corollary 8.216 (Wide residual entry forces a first descent obstruction)**.** *In the setting of Corollary 8.214, put $R = R_{\ell,p}$. Suppose that $R \geq 1$ and*

$$4^{R-1} K \geq \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H.$$

Define the unit-branch parameters by

$$\ell_r = \ell + r, \qquad K_0 = K, \qquad B_r = B^{\mathrm{res}}_{K_r,\ell_r,p} \qquad (0 \le r \le R-1),$$

and

$$K_{r+1} = 2K_r B_r \qquad (0 \le r \le R-1).$$

Then there is a least $r_* \in \{0,\dots,R-1\}$ for which either

$$K_{r_*} \ge \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H \qquad \text{or} \qquad B_{r_*} > \ell_{r_*}.$$

For every $r < r_*$, one has

$$K_r < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H, \qquad B_r \le \ell_r,$$

so the bridge-free residual one-step descent is valid at the corresponding unit-branch node. If

$$K_{r_*} < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H,$$

then $B_{r_*} > \ell_{r_*}$, and the residual-width one-step amplification at (K_{r_*}, ℓ_{r_*}) contains the non-empty confined chain-bridge sum

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*},L,\ell_{r_*}}(s,p) \\
&\quad \le \sum_{h=1}^{\ell_{r_*}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*+1},L,\ell_{r_*}+h}(s,p) \\
&\qquad + \sum_{h=\ell_{r_*}+1}^{B_{r_*}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*+1},L,\ell_{r_*},h}(s,p).
\end{aligned}$$

Thus a residual entry wider than the logarithmic threshold forces, along the unit descent branch, either saturation of the counted-start range or the appearance of a genuine chain-bridge packet before residual exhaustion.

Proof. For $0 \le r \le R-1$,

$$R_{\ell_r,p} = \max\{W_{\beta,p} - (\ell + r) + 1, 0\} = R - r.$$

In particular $R_{\ell_r,p} > 0$ for $0 \le r \le R-1$. Hence

$$B_r = \left\lceil \frac{16 K_r R_{\ell_r,p}}{\theta_{\lambda,\sigma,\beta} L_H} \right\rceil + 1 \ge 2 \qquad (0 \le r \le R-1).$$

It follows that

$$K_{r+1} = 2K_r B_r \ge 4K_r \qquad (0 \le r \le R-1),$$

and in particular

$$K_{R-1} \ge 4^{R-1} K \ge \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H.$$

Therefore the displayed set of failing indices is non-empty, so the least r_* exists. Its minimality gives the two inequalities for every $r < r_*$. Those inequalities are exactly the counting-range and bridge-free hypotheses needed to apply Corollary 8.213 at the corresponding unit-branch node.

If

$$K_{r_*} < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H,$$

then the first alternative in the definition of r_* fails, so $B_{r_*} > \ell_{r_*}$. The counting-range hypothesis still holds at (K_{r_*}, ℓ_{r_*}), and Corollary 8.211 applies. Since $B_{r_*} > \ell_{r_*}$, its bridge sum is non-empty, and

$$K^{\mathrm{res}}_{K_{r_*},\ell_{r_*},p} = 2K_{r_*}B_{r_*} = K_{r_*+1}.$$

Substituting these parameters gives the displayed inequality. □

Corollary 8.217 (First bridge-active obstruction projects to ordinary bridges)**.** *In the setting and notation of Corollary 8.216, suppose that the first obstruction is bridge-active rather than counting-saturating, that is,*

$$K_{r_*} < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H.$$

Then $B_{r_*} > \ell_{r_*}$ and

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*},L,\ell_{r_*}}(s,p) \\
&\quad \le \sum_{h=1}^{\ell_{r_*}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*+1},L,\ell_{r_*}+h}(s,p) \\
&\qquad + \sum_{g=2}^{B_{r_*}-\ell_{r_*}+1} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*+1},L,g}(s,p).
\end{aligned}$$

Moreover, the bridge-active alternative implies the quantitative lower bound

$$R_{\ell_{r_*},p} > \frac{\theta_{\lambda,\sigma,\beta}(\ell_{r_*}-1)}{16K_{r_*}} L_H.$$

Thus any first obstruction of this type is an ordinary confined-bridge packet occurring while the current residual width is still too large for the bridge-free range.

Proof. The hypothesis rules out the counting-saturation alternative in Corollary 8.216, so $B_{r_*} > \ell_{r_*}$. The same corollary gives the residual-width one-step bound at (K_{r_*}, ℓ_{r_*}) with a non-empty chain-bridge sum. For each $h \in \{\ell_{r_*}+1, \dots, B_{r_*}\}$, Corollary 8.196 gives

$$\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*+1},L,\ell_{r_*},h}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*+1},L,h-\ell_{r_*}+1}(s,p).$$

Reindexing the bridge gaps by $g = h - \ell_{r_*} + 1$ gives the displayed ordinary-bridge sum.

Finally,

$$B_{r_*} = \left\lceil \frac{16K_{r_*}R_{\ell_{r_*},p}}{\theta_{\lambda,\sigma,\beta}L_H} \right\rceil + 1 > \ell_{r_*}.$$

Hence the ceiling is at least ℓ_{r_*}, which forces

$$\frac{16K_{r_*}R_{\ell_{r_*},p}}{\theta_{\lambda,\sigma,\beta}L_H} > \ell_{r_*} - 1.$$

This is the claimed lower bound. □

Corollary 8.218 (First counting saturation has a last large bridge-free step)**.** *In the setting and notation of Corollary 8.216, put*

$$T_{\lambda,\sigma,\beta,p} = \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H.$$

Suppose that the first obstruction index r_* is counting-saturating, in the sense that

$$K_{r_*} \geq T_{\lambda,\sigma,\beta,p}.$$

If $r_* = 0$, then the initial chain profile is already outside the successor-counting range. If $r_* \geq 1$, put $r_0 = r_* - 1$. Then

$$K_{r_0} < T_{\lambda,\sigma,\beta,p}, \qquad B_{r_0} \leq \ell_{r_0},$$

and the predecessor step is bridge-free:

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_0},L,\ell_{r_0}}(s,p)\\ &\quad\leq \sum_{h=1}^{B_{r_0}} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r_*},L,\ell_{r_0}+h}(s,p).\end{aligned}$$

Moreover,

$$B_{r_0} \geq \frac{T_{\lambda,\sigma,\beta,p}}{2K_{r_0}} = \frac{\theta_{\lambda,\sigma,\beta}}{32K_{r_0}} L_H,$$

and every displayed child with increment h has residual width

$$R_{\ell_{r_0}+h,p} = \max\{R_{\ell_{r_0},p} - h, 0\}.$$

In particular the largest-increment child satisfies

$$R_{\ell_{r_0}+B_{r_0},p} \leq \max\left\{R_{\ell_{r_0},p} - \frac{\theta_{\lambda,\sigma,\beta}}{32K_{r_0}} L_H, 0\right\}.$$

Thus a non-initial counting-saturation obstruction is preceded by a bridge-free one-step packet with an L_H/K_{r_0}-scale successor window.

Proof. If $r_* = 0$, the displayed counting-saturation hypothesis says exactly that the initial parameter K is not in the successor-counting range.

Assume $r_* \geq 1$, and set $r_0 = r_* - 1$. By minimality of r_* in Corollary 8.216, the two obstruction alternatives both fail at r_0, so

$$K_{r_0} < T_{\lambda,\sigma,\beta,p}, \qquad B_{r_0} \leq \ell_{r_0}.$$

These are precisely the counting-range and bridge-free hypotheses for Corollary 8.213 at (K_{r_0}, ℓ_{r_0}). Applying that corollary gives the displayed bridge-free inequality, and the recursive definition gives

$$K_{r_*} = K_{r_0+1} = 2K_{r_0}B_{r_0}.$$

Combining this identity with $K_{r_*} \geq T_{\lambda,\sigma,\beta,p}$ proves

$$B_{r_0} \geq \frac{T_{\lambda,\sigma,\beta,p}}{2K_{r_0}} = \frac{\theta_{\lambda,\sigma,\beta}}{32K_{r_0}} L_H.$$

The residual identity for the children is the residual identity in Corollary 8.213. Taking $h = B_{r_0}$ and using the lower bound on B_{r_0} gives the final display. □

Corollary 8.219 (Wide residual first-obstruction normal form)**.** *In the setting and notation of Corollary 8.216, put*

$$T_{\lambda,\sigma,\beta,p} = \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H.$$

Assume $R = R_{\ell,p} \geq 1$ *and*

$$4^{R-1} K \geq T_{\lambda,\sigma,\beta,p}.$$

Then at least one of the following three alternatives holds.

(i) Initial saturation:

$$K \geq T_{\lambda,\sigma,\beta,p}.$$

(ii) Ordinary bridge packet: there is $r \in \{0, \dots, R-1\}$ *such that*

$$K_r < T_{\lambda,\sigma,\beta,p}, \qquad B_r > \ell_r,$$

and

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p) \\ &\quad \leq \sum_{h=1}^{\ell_r} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p) \\ &\qquad + \sum_{g=2}^{B_r-\ell_r+1} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{bridge},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,g}(s,p). \end{aligned}$$

Moreover

$$R_{\ell_r,p} > \frac{\theta_{\lambda,\sigma,\beta}(\ell_r - 1)}{16 K_r} L_H.$$

(iii) Last valid large-window step: there is an r *with* $0 \leq r \leq R-2$ *such that*

$$K_r < T_{\lambda,\sigma,\beta,p}, \qquad B_r \leq \ell_r, \qquad K_{r+1} \geq T_{\lambda,\sigma,\beta,p},$$

and

$$\begin{aligned} &\mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p) \\ &\quad \leq \sum_{h=1}^{B_r} \mathfrak{P}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p), \end{aligned}$$

with

$$B_r \geq \frac{\theta_{\lambda,\sigma,\beta}}{32 K_r} L_H.$$

Consequently, the largest-increment child in this packet has residual width at most

$$\max\left\{ R_{\ell_r,p} - \frac{\theta_{\lambda,\sigma,\beta}}{32 K_r} L_H, 0 \right\}.$$

Thus every wide residual entry is reduced to initial loss of the counting range, an ordinary confined-bridge packet, or the last bridge-free step before the counting range is lost.

Proof. Let r_* be the least obstruction index given by Corollary 8.216. If $K \geq T_{\lambda,\sigma,\beta,p}$, then alternative (i) holds. Assume therefore that $K < T_{\lambda,\sigma,\beta,p}$.

If $K_{r_*} < T_{\lambda,\sigma,\beta,p}$, then the first obstruction is bridge-active. Alternative (ii), with $r = r_*$, is exactly Corollary 8.217.

It remains to consider $K_{r_*} \geq T_{\lambda,\sigma,\beta,p}$. Since $K_0 = K < T_{\lambda,\sigma,\beta,p}$, this forces $r_* \geq 1$. Applying Corollary 8.218 and setting $r = r_* - 1$ gives alternative (iii). Since $r_* \leq R - 1$, this index satisfies $r \leq R - 2$. □

Corollary 8.220 (Bridge-packet branch has truncated return form)**.** *In the setting and notation of Corollary 8.219, suppose that the ordinary bridge packet alternative holds at some index r. Define*

$$\mathcal{G}_r = \left\{ g \in \mathbb{N} : 2 \leq g \leq B_r - \ell_r + 1, \quad g < W_{\beta,p} - \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}} L_H \right\}.$$

Then

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p) \\
&\quad \leq \sum_{h=1}^{\ell_r} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p) \\
&\qquad + \sum_{g\in\mathcal{G}_r} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda,\sigma,\beta,j,2K_{r+1},L,g}(s,p) \\
&\qquad + \sum_{g\in\mathcal{G}_r} \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j,2K_{r+1}(g-1)}(s,p).
\end{aligned}$$

Consequently, the bridge-packet branch contains only ordinary bridge gaps strictly below the slab-start cutoff, and those remaining gaps are reduced to inward-free bridge profiles and spread-return profiles.

Proof. The ordinary bridge packet alternative in Corollary 8.219 gives the same display with the ordinary bridge sum over

$$2 \leq g \leq B_r - \ell_r + 1.$$

If

$$g \geq W_{\beta,p} - \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}} L_H,$$

then Lemma 8.172, with $K = K_{r+1}$, makes the corresponding ordinary bridge profile equal to zero. Hence only $g \in \mathcal{G}_r$ remain.

For every $g \in \mathcal{G}_r$, apply Corollary 8.174 with $(K, h) = (K_{r+1}, g)$. Since $g \geq 2$, the spread-return parameter is $2K_{r+1}(g-1)$. Substituting these inequalities into the truncated ordinary bridge sum gives the displayed bound. □

Corollary 8.221 (Large-window branch has subexponential child cost)**.** *In the setting and notation of Corollary 8.219, suppose that the last valid large-window step alternative holds at some index r. Then*

$$B_r \leq \ell_r < W_{\beta,p},$$

and hence the bridge-free child sum in that alternative contains at most $W_{\beta,p}$ terms. Moreover,

$$\frac{\log_{D_H} W_{\beta,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)).$$

Consequently, the exponent hypothesis on

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)$$

follows from the same uniform exponent hypothesis on the child profiles

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p), \qquad 1\le h\le B_r.$$

The largest child in this subexponential family has residual width at most

$$\max\left\{R_{\ell_r,p}-\frac{\theta_{\lambda,\sigma,\beta}}{32K_r}L_H,0\right\}.$$

Proof. The large-window alternative gives $0\le r\le R-2$ and $B_r\le \ell_r$. Since $R=R_{\ell,p}\ge 1$, we have

$$R=W_{\beta,p}-\ell+1.$$

Therefore

$$\ell_r=\ell+r\le \ell+R-2=W_{\beta,p}-1,$$

and hence $B_r\le \ell_r<W_{\beta,p}$. The bridge-free sum in Corollary 8.219 therefore has at most $W_{\beta,p}$ summands.

Since $W_{\beta,p}=\max\{1,\lfloor \beta L_H\rfloor\}$, we have

$$\log_{D_H} W_{\beta,p}=o(L_H) \qquad (p\uparrow p_{\mathrm{c}}(H)).$$

Thus multiplying a uniform logarithmic exponent bound by the displayed number of child summands does not change the exponent. This proves the stated exponent implication. The residual-width bound for the largest child is the corresponding conclusion of Corollary 8.219. □

Definition 8.222 (Few-start chain profiles)**.** In the setting of Definition 8.186, fix $K,L\ge 1$, $\ell\ge 2$, and $q\ge 1$. Let

$$\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{j,K,L,\ell,q,p}(x,\gamma)$$

be the property obtained from Definition 8.186 by replacing the final lower bound on the number of admissible chain starts by the requirement that there are at least q such starts. Equivalently, there are at least q distinct $u\in A_{j,p}\cap\mathbb{N}$ for which the length-ℓ locally short outward staircase-chain witnesses and the endpoint-clearance inequalities with parameter K all hold. Define

$$\begin{aligned}\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,q}(s,p)\\ =\sup\Bigg\{\sum_{\substack{n\ge 0,\ |2n-m|\le 2\\ d>\lambda L_H}} Q^d\mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{j,K,L,\ell,q,p}}_{n,d}\\ : m\in\mathbb{N},\ c_sL_H<m\le\left(\frac{1}{\alpha_\omega}+3s\right)L_H\Bigg\}.\end{aligned}$$

Definition 8.223 (Exact-start chain profiles)**.** In the setting of Definition 8.186, fix $K,L\ge 1$, $\ell\ge 2$, and $u\in\mathbb{N}$. Let

$$\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{j,K,L,\ell,u,p}(x,\gamma)$$

be the property that u is an admissible chain start in Definition 8.186: namely, $u+r\in A_{j,p}\cap\mathbb{N}$ for every $0\le r<\ell$, and there are level-$u+r$ $(-1,+1)$-crossing runs $[a_r,b_r]$, $0\le r<\ell$, satisfying all

consecutive-run, record, persistence, shortness, and endpoint-clearance conditions in that definition. Define

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p)$$
$$= \sup_{u\in A_{j,p}\cap\mathbb{N}} \sup\Bigg\{ \sum_{\substack{n\ge 0,\ |2n-m|\le 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{j,K,L,\ell,u,p}}_{n,d}$$
$$: m\in\mathbb{N},\ c_sL_H < m \le \left(\frac{1}{\alpha_\omega}+3s\right)L_H \Bigg\},$$

with value 0 if $A_{j,p}\cap\mathbb{N}$ is empty.

Corollary 8.224 (Few-start chains localize to exact starts)**.** *In the setting of Definition 8.222, for every $q\ge 1$, put*

$$R_{\ell,p} = \max\{W_{\beta,p}-\ell+1,0\}.$$

Then

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,q}(s,p)$$
$$\le R_{\ell,p}\,\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p).$$

Consequently, the exponent hypothesis on the few-start chain profile follows from the same exponent hypothesis on the exact-start chain profile.

Proof. If the few-start property holds, at least one admissible chain start $u\in A_{j,p}\cap\mathbb{N}$ exists. Hence

$$\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{j,K,L,\ell,q,p} \subseteq \bigcup_{\substack{u\in A_{j,p}\cap\mathbb{N}\\ u,u+1,\dots,u+\ell-1\in A_{j,p}\cap\mathbb{N}}} \mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{j,K,L,\ell,u,p}.$$

The displayed union has at most $R_{\ell,p}$ possible starts, since a length-ℓ chain must fit entirely inside a slab of integer width $W_{\beta,p}$. Taking the property-restricted union bound in the weighted sums and then the supremum in m gives the displayed inequality. The exponent conclusion follows from

$$\frac{\log_{D_H} W_{\beta,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)).$$

□

Corollary 8.225 (Large-window children reduce to few-start chains)**.** *In the setting and notation of Corollary 8.219, suppose that the last valid large-window step alternative holds at some index r. Put*

$$q_{r,p} = \left\lfloor \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}} L_H \right\rfloor + 1.$$

Then $1\le q_{r,p}\le 5$, and for every $1\le h\le B_r$,

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p)$$
$$\le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h,q_{r,p}}(s,p)$$
$$\le \sum_{q=1}^{5} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h,q}(s,p).$$

Consequently,

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)$$
$$\le \sum_{h=1}^{B_r}\sum_{q=1}^{5}\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h,q}(s,p).$$

The number of displayed summands is at most $5W_{\beta,p}$. Hence the exponent hypothesis on the parent chain profile follows from a uniform exponent hypothesis on these few-start child profiles. The largest increment $h = B_r$ still has residual width at most

$$\max\left\{R_{\ell_r,p} - \frac{\theta_{\lambda,\sigma,\beta}}{32K_r}L_H, 0\right\}.$$

Proof. The large-window alternative gives

$$K_{r+1} \ge \frac{\theta_{\lambda,\sigma,\beta}}{16}L_H.$$

Therefore

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}}L_H \le 4,$$

and so $1 \le q_{r,p} \le 5$.

For a fixed h, the ordinary chain property with parameter K_{r+1} requires the number of admissible chain starts to be strictly larger than $\theta_{\lambda,\sigma,\beta}L_H/(4K_{r+1})$. Since this number is an integer, it is at least $q_{r,p}$. This is exactly the few-start property with $q = q_{r,p}$, giving the first displayed inequality. The second follows because $q_{r,p} \in \{1,\dots,5\}$.

Substituting these child bounds into the large-window inequality in Corollary 8.219 gives the parent inequality. By Corollary 8.221, $B_r < W_{\beta,p}$, so the total number of summands is at most $5W_{\beta,p}$, and

$$\frac{\log_{D_H}(5W_{\beta,p})}{L_H} \to 0 \qquad (p \uparrow p_c(H)).$$

Thus uniform few-start child exponent bounds pass to the parent. The residual-width bound for $h = B_r$ is again the corresponding conclusion of Corollary 8.219. □

Corollary 8.226 (Large-window branch localizes to exact starts)**.** *In the setting and notation of Corollary 8.219, suppose that the last valid large-window step alternative holds at some index r. Then*

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)$$
$$\le \sum_{h=1}^{B_r} 5R_{\ell_r+h,p}\,\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p).$$

Here

$$R_{\ell_r+h,p} = \max\{R_{\ell_r,p} - h, 0\}.$$

Since $B_r < W_{\beta,p}$, the total residual start cost is at most

$$5\sum_{h=1}^{B_r} R_{\ell_r+h,p} \le 5W_{\beta,p}^2,$$

and hence its logarithm is $o(L_H)$. Consequently, the exponent hypothesis on the parent chain profile follows from a uniform exponent hypothesis on the exact-start child profiles in the displayed sum. The child with $h = B_r$ retains the residual-width bound

$$R_{\ell_r+B_r,p} \le \max\left\{R_{\ell_r,p} - \frac{\theta_{\lambda,\sigma,\beta}}{32K_r}L_H, 0\right\}.$$

Proof. Apply Corollary 8.225, and then apply Corollary 8.224 to each few-start child profile. This gives the displayed inequality. The bound

$$R_{\ell_r+h,p} = \max\{R_{\ell_r,p} - h, 0\}$$

is the residual identity for child length $\ell_r + h$. The bound $B_r < W_{\beta,p}$ is Corollary 8.221, and also $R_{\ell_r+h,p} \le W_{\beta,p}$. Therefore the total residual start cost is bounded by $5W_{\beta,p}^2$. Because

$$\frac{\log_{D_H}(5W_{\beta,p}^2)}{L_H} \to 0,$$

uniform exponent bounds over those exact-start children pass to the parent. The residual-width statement is the largest-child conclusion of Corollary 8.219. □

Corollary 8.227 (Large-window exact-start branch has a low-residual tail)**.** *In the setting and notation of Corollary 8.226, put*

$$t_r = \left\lceil \frac{B_r}{2} \right\rceil.$$

Then

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)\\
&\quad\le 5\sum_{h=1}^{t_r-1} R_{\ell_r+h,p}\,\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p)\\
&\qquad + 5\max\{R_{\ell_r,p} - t_r, 0\}\sum_{h=t_r}^{B_r}\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p),
\end{aligned}$$

where the first sum is interpreted as zero when $t_r = 1$. Every exact-start child in the second sum has residual width at most

$$\max\left\{R_{\ell_r,p} - \frac{\theta_{\lambda,\sigma,\beta}}{64K_r}L_H, 0\right\}.$$

The total expanded union cost in the two sums is still at most $5W_{\beta,p}^2$, and hence is subexponential on the L_H scale.

Proof. Start from Corollary 8.226. Split its sum over $1 \le h \le B_r$ into $h < t_r$ and $h \ge t_r$. For the tail, the residual identity gives

$$R_{\ell_r+h,p} = \max\{R_{\ell_r,p} - h, 0\} \le \max\{R_{\ell_r,p} - t_r, 0\}.$$

Substituting this uniform tail bound gives the displayed inequality.

Since $t_r \ge B_r/2$, the large-window lower bound

$$B_r \ge \frac{\theta_{\lambda,\sigma,\beta}}{32K_r}L_H$$

from Corollary 8.219 implies

$$t_r \geq \frac{\theta_{\lambda,\sigma,\beta}}{64K_r} L_H.$$

This proves the residual-width bound for every tail child. Finally, Corollary 8.221 gives $B_r < W_{\beta,p}$, and every displayed residual factor is at most $W_{\beta,p}$. Thus the total expanded union cost is bounded by $5W_{\beta,p}^2$, whose logarithm is $o(L_H)$. □

Corollary 8.228 (Dyadic residual layers for the exact-start branch)**.** *In the setting and notation of Corollary 8.226, put*

$$M_r = \max\{1, \lceil \log_2 B_r \rceil\}, \qquad s_{r,0} = B_r + 1, \qquad s_{r,m} = \left\lceil \frac{B_r}{2^m} \right\rceil \quad (1 \leq m \leq M_r).$$

Then $s_{r,M_r} = 1$, and the intervals

$$I_{r,m} = \{s_{r,m}, s_{r,m}+1, \ldots, s_{r,m-1}-1\}, \qquad 1 \leq m \leq M_r,$$

partition $\{1, \ldots, B_r\}$. Moreover,

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p) \\
&\quad \leq 5 \sum_{m=1}^{M_r} \max\{R_{\ell_r,p} - s_{r,m}, 0\} \sum_{h \in I_{r,m}} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,exact,short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p).
\end{aligned}$$

Every exact-start child in the m-th layer has residual width at most

$$\max\left\{ R_{\ell_r,p} - \frac{\theta_{\lambda,\sigma,\beta}}{2^{m+5}K_r} L_H, 0 \right\}.$$

The total expanded union cost over all dyadic layers is at most $5W_{\beta,p}^2$, and hence is subexponential on the L_H scale.

Proof. Since $M_r \geq \lceil \log_2 B_r \rceil$, we have

$$0 < \frac{B_r}{2^{M_r}} \leq 1,$$

and hence $s_{r,M_r} = 1$. The displayed intervals are consecutive and therefore partition $\{1, \ldots, B_r\}$.

Start from Corollary 8.226. For $h \in I_{r,m}$, the residual identity gives

$$R_{\ell_r+h,p} = \max\{R_{\ell_r,p} - h, 0\} \leq \max\{R_{\ell_r,p} - s_{r,m}, 0\}.$$

Replacing each residual factor by this layerwise upper bound gives the displayed dyadic-layer inequality.

For the residual-width estimate, use $s_{r,m} \geq B_r/2^m$ and the large-window lower bound

$$B_r \geq \frac{\theta_{\lambda,\sigma,\beta}}{32K_r} L_H$$

from Corollary 8.219. Thus

$$s_{r,m} \geq \frac{\theta_{\lambda,\sigma,\beta}}{2^{m+5}K_r} L_H,$$

and every child in $I_{r,m}$ has the stated residual-width bound.

Finally, the dyadic decomposition only repartitions the weighted exact-start sum from Corollary 8.226. Since $B_r < W_{\beta,p}$ and every residual factor is at most $W_{\beta,p}$, the expanded cost is at most $5W_{\beta,p}^2$, whose logarithm is $o(L_H)$. □

Lemma 8.229 (Exact-start chains with enough levels are record spread)**.** *In the setting of Definition 8.223, if*

$$\ell > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

then

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).$$

Consequently, if

$$K \ge \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H,$$

then the displayed inequality holds for every $\ell \ge 5$.

Proof. Fix an admissible exact start u. If

$$\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{j,K,L,\ell,u,p}(x,\gamma)$$

holds, then for each $0 \le r < \ell$ there is a level-$u+r$ $(-1,+1)$-crossing run satisfying the record, shortness, and endpoint-clearance conditions from Definition 8.160. The levels $u, u+1, \dots, u+\ell-1$ are distinct and all lie in $A_{j,p} \cap \mathbb{N}$. Since their number is larger than $\theta_{\lambda,\sigma,\beta} L_H/(4K)$, the property $\mathcal{P}^{\mathrm{pre},\mathrm{rec},\mathrm{short}}_{j,K,L,p}(x,\gamma)$ holds. Thus, for every u,

$$\mathcal{P}^{\mathrm{pre},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{j,K,L,\ell,u,p} \subseteq \mathcal{P}^{\mathrm{pre},\mathrm{rec},\mathrm{short}}_{j,K,L,p}.$$

Taking the corresponding property-restricted sums and then the suprema gives the profile inequality.

If $K \ge \theta_{\lambda,\sigma,\beta} L_H/16$, then

$$\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H \le 4,$$

so every $\ell \ge 5$ satisfies the first displayed hypothesis. □

Corollary 8.230 (Few-start chains with enough levels are record spread)**.** *In the setting of Definition 8.222, if*

$$\ell > \frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H,$$

then, for every $q \ge 1$,

$$\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,q}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).$$

Proof. The few-start property with any $q \ge 1$ includes at least one admissible exact start u. For that start, the exact-start chain property with length ℓ holds. Since ℓ is larger than the displayed threshold, Lemma 8.229 makes the corresponding path a prefix-record outward-spread path. This gives the property inclusion and hence the profile inequality. □

Corollary 8.231 (Large-window exact starts are record spread except short children)**.** *In the setting and notation of Corollary 8.226, put*

$$\mathcal{S}_r = \{h \in \{1,\dots,B_r\} : \ell_r + h \le 4\}.$$

Then

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)\\&\quad\le 5\sum_{h\in\mathcal{S}_r} R_{\ell_r+h,p}\,\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p)\\&\qquad+5\left(\sum_{h=1}^{B_r} R_{\ell_r+h,p}\right)\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L}(s,p).\end{aligned}$$

Moreover, $\mathcal{S}_r\subseteq\{1,\dots,4-\ell_r\}$, so $|\mathcal{S}_r|\le 2$, and $\mathcal{S}_r=\varnothing$ whenever $\ell_r\ge 4$. The total coefficient in the second line is at most $10W_{\beta,p}$, while the coefficient of the record-spread profile is at most $5W_{\beta,p}^2$. Hence both costs are subexponential on the L_H scale.

Proof. Start from Corollary 8.226. Split the sum over $1\le h\le B_r$ according to whether $h\in\mathcal{S}_r$. The large-window alternative gives

$$K_{r+1}\ge\frac{\theta_{\lambda,\sigma,\beta}}{16}L_H,$$

as used in Corollary 8.225. For every $h\notin\mathcal{S}_r$, we have $\ell_r+h\ge 5$, and Lemma 8.229 therefore gives

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p)\\&\quad\le\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L}(s,p).\end{aligned}$$

Substituting these bounds gives the displayed inequality.

Since Definition 8.186 has $\ell_r\ge 2$, the set $\mathcal{S}_r$ is contained in $\{1,\dots,4-\ell_r\}$ and has at most two elements. Each residual factor is at most $W_{\beta,p}$, so the total short-child coefficient is at most $10W_{\beta,p}$. Also,

$$\sum_{h=1}^{B_r} R_{\ell_r+h,p}\le W_{\beta,p}^2$$

by Corollary 8.226. Since $\log_{D_H} W_{\beta,p}=o(L_H)$, both displayed costs are subexponential. □

Corollary 8.232 (Threshold-tight short exact-start exceptions)**.** *In the setting and notation of Corollary 8.226, put*

$$\mathcal{E}_r=\left\{h\in\{1,\dots,B_r\}:\ell_r+h\le\frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}}L_H\right\}.$$

Then

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)\\&\quad\le 5\sum_{h\in\mathcal{E}_r} R_{\ell_r+h,p}\,\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{exact},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h}(s,p)\\&\qquad+5\left(\sum_{h=1}^{B_r} R_{\ell_r+h,p}\right)\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L}(s,p).\end{aligned}$$

Moreover, $\mathcal{E}_r\subseteq\mathcal{S}_r$, where $\mathcal{S}_r$ is the set from Corollary 8.231. In particular $|\mathcal{E}_r|\le 2$, and $\mathcal{E}_r=\varnothing$ if

$$K_{r+1}>\frac{\theta_{\lambda,\sigma,\beta}}{12}L_H.$$

If $h\in\mathcal{E}_r$ and $\ell_r+h=4$, then

$$K_{r+1}=\frac{\theta_{\lambda,\sigma,\beta}}{16}L_H.$$

Thus length-four exceptions can occur only at exact saturation, and all exceptions occur while

$$\frac{\theta_{\lambda,\sigma,\beta}}{16}L_H \le K_{r+1} \le \frac{\theta_{\lambda,\sigma,\beta}}{12}L_H.$$

The exceptional exact-start coefficient is at most $10W_{\beta,p}$, while the record-spread coefficient is at most $5W_{\beta,p}^2$; both are subexponential on the L_H scale.

Proof. Start from Corollary 8.226. If $h \notin \mathcal{E}_r$, then

$$\ell_r + h > \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}}L_H,$$

so Lemma 8.229, with $K = K_{r+1}$ and $\ell = \ell_r + h$, bounds the corresponding exact-start child by the prefix-record outward-spread profile at parameter K_{r+1}. Substituting this estimate for all $h \notin \mathcal{E}_r$ gives the displayed inequality.

The large-window branch has

$$K_{r+1} \ge \frac{\theta_{\lambda,\sigma,\beta}}{16}L_H.$$

Hence every $h \in \mathcal{E}_r$ satisfies

$$\ell_r + h \le 4,$$

so $\mathcal{E}_r \subseteq \mathcal{S}_r$. The bound $|\mathcal{E}_r| \le 2$ follows from Corollary 8.231. Since every child length satisfies $\ell_r + h \ge 3$, the inequality defining $\mathcal{E}_r$ is impossible when

$$3 > \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}}L_H,$$

which is equivalent to the displayed condition $K_{r+1} > \theta_{\lambda,\sigma,\beta}L_H/12$.

If $h \in \mathcal{E}_r$ and $\ell_r + h = 4$, then the defining inequality gives

$$K_{r+1} \le \frac{\theta_{\lambda,\sigma,\beta}}{16}L_H.$$

Together with the large-window lower bound on K_{r+1}, this gives the claimed equality. The final interval for possible exceptions follows by combining the same lower bound with the emptiness criterion.

Finally, every residual coefficient is at most $W_{\beta,p}$. Since $|\mathcal{E}_r| \le 2$, the exceptional coefficient is at most $10W_{\beta,p}$. The record coefficient is bounded by

$$5\sum_{h=1}^{B_r} R_{\ell_r+h,p} \le 5W_{\beta,p}^2$$

as in Corollary 8.226. The subexponential statements follow from $\log_{D_H} W_{\beta,p} = o(L_H)$. □

Corollary 8.233 (Large-window exceptions retain surplus starts)**.** *In the setting and notation of Corollary 8.225, put*

$$A_r = \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}}L_H, \qquad q_{r,p} = \lfloor A_r \rfloor + 1,$$

and

$$\mathcal{E}_r = \{h \in \{1,\dots,B_r\} : \ell_r + h \le A_r\}.$$

Then

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)\\&\quad\le\sum_{h\in\mathcal{E}_r}\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h,q_{r,p}}(s,p)\\&\qquad+B_r\,\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L}(s,p).\end{aligned}$$

Moreover, if $h\in\mathcal{E}_r$, then

$$q_{r,p}>\ell_r+h,$$

and $q_{r,p}\in\{4,5\}$. Thus the only exceptional few-start types are

$$(\ell_r+h,q_{r,p})\in\{(3,4),(3,5),(4,5)\}.$$

The exceptional sum has at most two terms, while the record-spread coefficient is at most $W_{\beta,p}$; both costs are subexponential on the L_H scale.

Proof. Start from the sharper first inequality in Corollary 8.225, namely the child bound with the single value $q=q_{r,p}$, and substitute it into the large-window parent sum from Corollary 8.219. This gives

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)\\&\quad\le\sum_{h=1}^{B_r}\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h,q_{r,p}}(s,p).\end{aligned}$$

If $h\notin\mathcal{E}_r$, then $\ell_r+h>A_r$. Applying Corollary 8.230 with $K=K_{r+1}$ and $\ell=\ell_r+h$ bounds the corresponding summand by the prefix-record outward-spread profile at parameter K_{r+1}. There are at most B_r such summands, giving the displayed inequality.

For $h\in\mathcal{E}_r$, the integer ℓ_r+h satisfies

$$\ell_r+h\le A_r<\lfloor A_r\rfloor+1=q_{r,p},$$

so $q_{r,p}>\ell_r+h$. The large-window branch also gives

$$K_{r+1}\ge\frac{\theta_{\lambda,\sigma,\beta}}{16}L_H,$$

and hence $A_r\le 4$, so $q_{r,p}\le 5$. If $\mathcal{E}_r$ is nonempty, then some child length $\ell_r+h\ge 3$ is at most A_r, so $A_r\ge 3$ and $q_{r,p}\ge 4$. This proves $q_{r,p}\in\{4,5\}$, and the displayed list of possible pairs follows from $\ell_r+h\in\{3,4\}$ and $q_{r,p}>\ell_r+h$.

Finally, $|\mathcal{E}_r|\le 2$ follows from Corollary 8.232, while $B_r<W_{\beta,p}$ follows from Corollary 8.221. Since $\log_{D_H}W_{\beta,p}=o(L_H)$, both costs are subexponential. □

Corollary 8.234 (Surplus-start chains are record spread)**.** *In the setting of Definition 8.222, if*

$$q>\frac{\theta_{\lambda,\sigma,\beta}}{4K}L_H,$$

then

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell,q}(s,p)\\&\quad\le\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).\end{aligned}$$

Proof. If the q-start chain property holds, there are at least q distinct admissible start levels $u \in A_{j,p} \cap \mathbb{N}$. For each such u, the first run in the length-ℓ staircase-chain witness is a level-u $(-1,+1)$-crossing run satisfying the record, shortness, and endpoint-clearance conditions in Definition 8.160. Since the number of these distinct start levels is larger than $\theta_{\lambda,\sigma,\beta} L_H/(4K)$, the prefix-record outward-spread property holds. Taking the property-restricted union bound gives the profile inequality. □

Corollary 8.235 (Large-window few-start branch collapses to record spread)**.** *In the setting and notation of Corollary 8.225, one has*

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)\\ &\quad\le B_r\,\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L}(s,p).\end{aligned}$$

Since $B_r < W_{\beta,p}$, the coefficient is subexponential on the L_H scale. Consequently, the exponent hypothesis on the large-window parent chain profile follows from the same exponent hypothesis on the single prefix-record outward-spread profile at parameter K_{r+1}.

Proof. Put

$$A_r = \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}} L_H, \qquad q_{r,p} = \lfloor A_r \rfloor + 1.$$

Then $q_{r,p} > A_r$. Start from the sharper first inequality in Corollary 8.225, and substitute it into the large-window parent sum from Corollary 8.219. This gives

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p)\\ &\quad\le \sum_{h=1}^{B_r} \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{qstart},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L,\ell_r+h,q_{r,p}}(s,p).\end{aligned}$$

For every h, apply Corollary 8.234 with

$$K = K_{r+1}, \qquad \ell = \ell_r + h, \qquad q = q_{r,p}.$$

This bounds each child summand by the same prefix-record outward-spread profile. There are B_r summands, giving the displayed inequality. Finally, $B_r < W_{\beta,p}$ is Corollary 8.221, and $\log_{D_H} W_{\beta,p} = o(L_H)$. □

Corollary 8.236 (Locally short staircase chains are prefix-record spread)**.** *In the setting of Definition 8.186, for every $K, L \ge 1$ and $\ell \ge 2$,*

$$\begin{aligned}&\mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{per},\mathrm{stair},\mathrm{loc},\mathrm{chain},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L,\ell}(s,p)\\ &\quad\le \mathfrak{P}^{G\to H,\mathrm{rad},\tan,\mathrm{pre},\mathrm{out},\mathrm{rec},\mathrm{short}}_{Q,\lambda,\sigma,\beta,j,K,L}(s,p).\end{aligned}$$

Proof. If the locally short staircase-chain property holds, then the number of admissible chain starts is larger than $\theta_{\lambda,\sigma,\beta} L_H/(4K)$. For each such start u, the first run in the chain witness is a level-u $(-1,+1)$-crossing run satisfying the record, shortness, and endpoint-clearance conditions in Definition 8.160. The admissible starts are distinct levels in $A_{j,p} \cap \mathbb{N}$, so these first runs witness the prefix-record outward-spread property. This gives a property inclusion, and the displayed profile inequality follows by monotonicity of property-restricted profiles. □

Corollary 8.237 (Wide residual normal form with chain branches absorbed)**.** *In the setting and notation of Corollary 8.219, assume* $R = R_{\ell,p} \geq 1$ *and*

$$4^{R-1}K \geq T_{\lambda,\sigma,\beta,p}.$$

Then at least one of the following alternatives holds.

(i) Initial saturation:

$$K \geq T_{\lambda,\sigma,\beta,p}.$$

(ii) Reduced bridge packet: there is $r \in \{0,\dots,R-1\}$ *such that*

$$K_r < T_{\lambda,\sigma,\beta,p}, \qquad B_r > \ell_r,$$

and, with

$$\mathcal{G}_r = \left\{ g \in \mathbb{N} : 2 \leq g \leq B_r - \ell_r + 1, \quad g < W_{\beta,p} - \frac{\theta_{\lambda,\sigma,\beta}}{4K_{r+1}} L_H \right\},$$

one has

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p) \\
&\quad \leq \ell_r\, \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L}(s,p) \\
&\qquad + \sum_{g\in\mathcal{G}_r} \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda,\sigma,\beta,j,2K_{r+1},L,g}(s,p) \\
&\qquad + \sum_{g\in\mathcal{G}_r} \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda,\sigma,\beta,j,2K_{r+1}(g-1)}(s,p).
\end{aligned}$$

(iii) Reduced large-window packet: there is r *with* $0 \leq r \leq R-2$ *such that*

$$K_r < T_{\lambda,\sigma,\beta,p}, \qquad B_r \leq \ell_r, \qquad K_{r+1} \geq T_{\lambda,\sigma,\beta,p},$$

and

$$\begin{aligned}
&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda,\sigma,\beta,j,K_r,L,\ell_r}(s,p) \\
&\quad \leq B_r\, \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda,\sigma,\beta,j,K_{r+1},L}(s,p).
\end{aligned}$$

In alternatives (ii) and (iii), all displayed index sets and coefficients have logarithm $o(L_H)$*: indeed* $\ell_r \leq W_{\beta,p}$*,* $B_r < W_{\beta,p}$*, and* $|\mathcal{G}_r| \leq W_{\beta,p}$*.*

Proof. Apply Corollary 8.219. The initial-saturation alternative is unchanged.

If the ordinary bridge-packet alternative holds, apply Corollary 8.220. Each chain child in its first sum is bounded by the prefix-record outward-spread profile at parameter K_{r+1} by Corollary 8.236. There are ℓ_r such children, giving the reduced bridge-packet display.

If the last valid large-window alternative holds, apply Corollary 8.235, which gives the reduced large-window display.

It remains only to record the size bounds. In the bridge-packet alternative $r \leq R-1$, and hence

$$\ell_r = \ell + r \leq \ell + R - 1 = W_{\beta,p}.$$

The set $\mathcal{G}_r$ is contained in $\{2,\dots,W_{\beta,p}-1\}$. In the large-window alternative, Corollary 8.221 gives $B_r < W_{\beta,p}$. Since $\log_{D_H} W_{\beta,p} = o(L_H)$, all displayed finite costs are subexponential on the L_H scale. □

Remark 8.238 (Why slab-exhausting depth is not automatic)**.** The recursive depth parameters in the finite chain amplification satisfy

$$K_r = 2K_{r-1}B_r, \qquad B_r \geq \left\lceil \frac{16K_{r-1}\beta}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1.$$

Thus the current one-step mechanism increases K_r very rapidly. Lemma 8.208 says that the bounded-successor step is justified by the present counting argument only while

$$K_{r-1} < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H.$$

Lemma 8.209 improves the successor cutoff near the top of a slab by replacing the crude width $W_{\beta,p}$ with the residual start width $R_{\ell,p} = W_{\beta,p} - \ell + 1$. In particular, once a chain length is already within $O(L_H/K)$ of the top of the slab, the successor cutoff can be bounded independently of p, and Corollary 8.212 gives a bounded-cost one-step amplification in that regime. In the still narrower range of Corollary 8.213, this step is bridge-free and decreases the residual chain-start width. These refinements do not by themselves give a route from a fixed initial ℓ into the near-top regime. Corollary 8.214 shows that, if the bridge-free hypotheses persisted until the residual start-width was exhausted, the chain profile would vanish; the missing point is to justify those hypotheses along such a descent from the chain profiles produced by the earlier reductions. Lemma 8.215 shows that the current K-recursion can certify such an exhaustion only when the residual start-width at entry is already logarithmic in L_H. Corollary 8.216 packages the contrapositive as a first-obstruction alternative: outside that logarithmic range, the unit descent branch must either leave the counted-start range or create a confined chain-bridge packet. Corollary 8.217 then projects the latter alternative to the ordinary confined-bridge families already used elsewhere in the reduction. Corollary 8.218 handles the remaining alternative: a non-initial counting saturation is preceded by a bridge-free step with a successor window of order L_H/K. Corollary 8.219 combines these observations into a single normal form for wide residual entries. Corollary 8.220 then pushes the ordinary-bridge alternative through the existing bridge truncation and inward-free-or-return split. Corollary 8.221 records that the remaining large-window branch carries only a subexponential child union, while one child has a residual drop of order L_H/K. Corollary 8.225 further shows that, once this branch has saturated the counting range, each child can be replaced by a bounded finite-start chain profile with $q \leq 5$. Corollary 8.226 removes the remaining slab-level start choice at residual cost $\sum_{h=1}^{B_r} R_{\ell_r+h,p}$, reducing the branch to exact-start chain profiles. Corollary 8.227 then separates the upper half of this exact-start branch, where every child has an L_H/K_r-scale residual-width drop. Corollary 8.228 refines this into dyadic residual layers, exposing the residual drop available at each child-window scale. Corollary 8.231 gives a complementary escape from the exact-start branch: once the saturated child parameter is in force, every exact-start child of total length at least five is already one of the prefix-record outward-spread profiles treated by the earlier reduction tree, leaving only exact-start children of total length three or four as genuinely new finite exceptions. Corollary 8.232 sharpens this finite exception set further: the actual exceptional increments are only those for which the short child length is still below the saturated counting threshold $\theta_{\lambda,\sigma,\beta}L_H/(4K_{r+1})$. Thus the exact-start exception set is empty once $K_{r+1} > \theta_{\lambda,\sigma,\beta}L_H/12$, and any length-four exception can occur only at exact counting saturation. Corollary 8.233 records the same branch before the final exact-start localization is spent: all non-exceptional children go directly to the prefix-record outward-spread profile, while each exceptional child remains a few-start profile with more admissible starts than chain levels. The only remaining surplus-start types are $(3,4)$, $(3,5)$, and $(4,5)$. Corollary 8.235 then removes even this finite surplus-start family: the $q_{r,p}$

distinct start levels themselves already exceed the prefix-record counting threshold, so every large-window few-start child is a prefix-record outward-spread event. Thus the large-window branch no longer leaves a new chain-type residual; it returns to the existing prefix-record reduction tree with only a single $B_r < W_{\beta,p}$ coefficient. Corollary 8.236 then observes that the same start-level argument applies to every locally short staircase-chain profile, not only to the saturated few-start children. Consequently Corollary 8.237 absorbs the remaining chain sums in the wide-residual normal form: outside initial counting saturation, the two first-obstruction branches now reduce to prefix-record outward-spread profiles, inward-free bridge profiles, and spread-return profiles, all with subexponential coefficients. Still, one cannot simply choose n of order $W_{\beta,p} \asymp L_H$ in Corollary 8.199 and invoke Lemma 8.204. A valid growing-depth argument must either keep the effective depth parameter below the counting range, add a new source of many successor pairs after K_r becomes large, or amplify chain length without the quadratic K-loss.

Lemma 8.239 (Counting-range-valid iterations stay subslab)**.** *In the setting of Corollary 8.199, fix $K = K_0 \geq 1$ and $\ell \geq 2$, independent of p. Suppose that a finite chain-amplification iteration of depth $n = n(p)$ remains inside the successor-counting range at every step, in the sense that*

$$K_{r-1} < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H \qquad (1 \leq r \leq n).$$

Then every terminal chain length $a \in I_n(\ell)$ satisfies

$$a \leq 2^n \ell \leq \ell \sqrt{\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H}.$$

Consequently, for all p sufficiently close to $p_c(H)$, every such terminal length satisfies

$$a < W_{\beta,p}.$$

Thus the slab-width vanishing criterion of Lemma 8.204 cannot eliminate any terminal chain length produced by a successor-counting-valid iteration.

Proof. Since

$$B_r \geq \left\lceil \frac{16 K_{r-1} \beta}{\theta_{\lambda,\sigma,\beta}} \right\rceil + 1 \geq 2,$$

the recursive depth parameters satisfy

$$K_r = 2 K_{r-1} B_r \geq 4 K_{r-1}.$$

Hence, for $n \geq 1$,

$$K_{n-1} \geq 4^{n-1} K.$$

The counting-range assumption at step n gives

$$4^{n-1} K < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H,$$

and therefore

$$2^n < \sqrt{\frac{\theta_{\lambda,\sigma,\beta}}{4K} L_H}.$$

The case $n = 0$ is trivial after enlarging the final p-neighbourhood. Since $I_n(\ell) \subseteq \{\ell + n, \ldots, 2^n \ell\}$, the displayed bound on terminal lengths follows.

Finally,

$$W_{\beta,p} = \max\{1, \lfloor \beta L_H \rfloor\} \geq \frac{\beta}{2} L_H$$

for all p sufficiently close to $p_c(H)$, while the terminal-length bound is $O(\sqrt{L_H})$ with constants fixed. Thus

$$2^n \ell < W_{\beta,p}$$

for all sufficiently large L_H, proving the final assertion. □

Remark 8.240 (Consequence for the present chain program)**.** Rounds 267–269 show that the present chain-amplification program has a structural gap, not just a bookkeeping gap. The slab support cutoff is at linear scale $W_{\beta,p} \asymp L_H$, but the terminal lengths reachable before the successor-counting proof exits its range are only $O(\sqrt{L_H})$. Closing the argument therefore requires a genuinely new ingredient: either a bounded-successor extraction with much smaller depth-parameter loss, a way to restart the counting lower bound after renormalizing large K_r, or a direct estimate for the bridge families that accumulate before terminal chains can be exhausted.

Lemma 8.241 (Ideal doubled-count exhaustion has a constant obstruction)**.** *In the setting of Corollary 8.130, fix $K \geq 1$ and $\ell \geq 1$, independent of p. Suppose that, along a sequence $p \uparrow p_c(H)$, there are integers $n = n(p) \geq 0$ such that the idealized doubled-count parameters remain in the successor-counting range, namely*

$$2^{r-1} K < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H \qquad (1 \leq r \leq n),$$

and that the ideal terminal length reaches the slab cutoff,

$$2^n \ell > W_{\beta,p}.$$

Then

$$K\beta \leq \frac{\theta_{\lambda,\sigma,\beta}}{8} \ell.$$

Consequently, if

$$K\beta > \frac{\theta_{\lambda,\sigma,\beta}}{8} \ell,$$

then, for all p sufficiently close to $p_c(H)$, no such n exists.

Proof. For all sufficiently large L_H, the second displayed condition forces $n \geq 1$. Applying the counting-range assumption at step n gives

$$2^{n-1} K < \frac{\theta_{\lambda,\sigma,\beta}}{16} L_H,$$

and hence

$$2^n < \frac{\theta_{\lambda,\sigma,\beta}}{8K} L_H.$$

Since $W_{\beta,p} < 2^n \ell$, we obtain

$$\frac{K W_{\beta,p}}{L_H} < \frac{\theta_{\lambda,\sigma,\beta}}{8} \ell.$$

Taking $p \uparrow p_c(H)$ along the chosen sequence and using $W_{\beta,p}/L_H \to \beta$ gives the asserted necessary condition. The final assertion is the contrapositive. □

Remark 8.242 (What the ideal doubling test rules out)**.** Lemma 8.241 tests the most optimistic version of a lower-loss successor extraction that merely removes the fixed-gap pigeonhole loss $K \mapsto KB$, while retaining the unavoidable halving of the counted successor starts. For the raw chain terms in the finite far-tail ladder, where $q \in \{3, 4\}$ and

$$K_{i,0}^{\mathrm{ch}} = 512 M_i B_i C_i,$$

such an idealized route would still need

$$512 M_i B_i C_i \, \beta_i \le \frac{\theta_{\lambda_i,\sigma_i,\beta_i}}{8} q.$$

This constant inequality is not supplied by the present reductions. Thus a successful lower-loss route must do more than replace the step $K \mapsto 2KB$ by $K \mapsto 2K$: it must also improve the initial depth-to-length ratio, restart the counted-start supply with a fresh ratio, or estimate the bridge packets without waiting for terminal chains to exceed the slab width.

Corollary 8.243 (Many tangential runs: adjacent crossing split)**.** *In the setting of Corollary 8.141, let $R \ge 2$ and put*

$$P_R = \left\lfloor \frac{R-2}{4} \right\rfloor.$$

Then

$$\mathfrak{T}^{G\to H,\mathrm{rad},\tan,\mathrm{many}}_{Q,\lambda,\sigma,\beta,j,R}(s,p) \le \sum_{\eta,\zeta\in\{-1,+1\}} \mathfrak{C}^{G\to H,\mathrm{rad},\tan,\mathrm{cross}}_{Q,\lambda,\sigma,\beta,j,P_R,\eta,\zeta}(s,p).$$

Consequently, the exponent hypothesis on the many-run tangential profile follows from the same exponent hypothesis on the four adjacent-crossing tangential profile families.

Proof. Fix u and a rooted path $\gamma = (v_0, \dots, v_\ell)$. A level-$u$ tangential run that is not two-sided must contain time 0 or time ℓ; hence there are at most two such runs. Every two-sided level-u tangential run is of exactly one of the four types $(\eta, \zeta) \in \{-1, +1\}^2$, since maximality forces the adjacent vertices just before and just after the run to lie outside $S_u(x)$, while adjacency to $S_u(x)$ forces their distances from x to be $u-1$ or $u+1$. Therefore

$$R_x^u(\gamma) \le 2 + \sum_{\eta,\zeta\in\{-1,+1\}} R^u_{x,\eta,\zeta}(\gamma).$$

If $\mathcal{T}^{\mathrm{many}}_{u,R}$ holds and all four $\mathcal{C}^{\eta,\zeta}_{u,P_R}$ fail, then

$$R_x^u(\gamma) \le 2 + 4P_R \le R,$$

contradicting $R_x^u(\gamma) > R$. Thus

$$\mathcal{T}^{\mathrm{many}}_{u,R} \subseteq \bigcup_{\eta,\zeta\in\{-1,+1\}} \mathcal{C}^{\eta,\zeta}_{u,P_R}.$$

Taking the corresponding union bound inside the property-restricted profiles and then the supremum over u gives the displayed inequality. The exponent conclusion follows because there are only four terms. □

Definition 8.244 (Upper-shifted exterior-return profiles)**.** In the setting of Corollary 8.130, for $K \geq 0$ define

$$\mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{up}}_{Q,\lambda,\sigma,\beta,j,K}(s,p)$$
$$= \sup_{u\in A_{j,p}\cap\mathbb{N}} \sup\left\{ \sum_{\substack{r\geq 0,\ |2r-m|\leq 2\\ d>\lambda L_H}} Q^d \mathfrak{R}^{G,\mathcal{X}_{u+1,K}}_{r,d} : m\in\mathbb{N},\ c_s L_H < m \leq \left(\frac{1}{\alpha_\omega}+3s\right) L_H \right\},$$

with value 0 if $A_{j,p}\cap\mathbb{N}$ is empty.

Corollary 8.245 (Upper-shifted returns move at most one slab)**.** *In the setting of Corollary 8.130, for every $j \geq 0$ and $K \geq 0$,*

$$\mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{up}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \leq \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) + \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda,\sigma,\beta,j+1,K}(s,p).$$

Consequently, the upper-shifted exterior-return exponent hypothesis follows from the same exponent hypothesis on the same and next ordinary exterior-return slabs.

Proof. Let $W = W_{\beta,p}$. If $u \in A_{j,p}\cap\mathbb{N}$, then

$$u+1 \in A_{j,p}\cap\mathbb{N} \quad \text{or} \quad u+1 \in A_{j+1,p}\cap\mathbb{N}.$$

Indeed, the only way $u+1$ is not still in $[jW,(j+1)W)$ is that $u = (j+1)W-1$, in which case $u+1 = (j+1)W \in A_{j+1,p}$. Hence the supremum over $\mathcal{X}_{u+1,K}$ with $u \in A_{j,p}$ is bounded by the sum of the ordinary return-profile suprema over $A_{j,p}$ and $A_{j+1,p}$. The displayed inequality follows. The exponent conclusion follows because the right-hand side has two terms. □

Corollary 8.246 (Adjacent crossings localize to exterior returns)**.** *In the setting of Corollary 8.243, let $R \geq 1$ and set $K = R-1$. Then*

$$\mathfrak{C}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{cross}}_{Q,\lambda,\sigma,\beta,j,R,\eta,\zeta}(s,p) \leq \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda,\sigma,\beta,j,K}(s,p)$$

for

$$(\eta,\zeta) \in \{(-1,+1),(-1,-1),(+1,-1)\},$$

and

$$\mathfrak{C}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{cross}}_{Q,\lambda,\sigma,\beta,j,R,+1,+1}(s,p) \leq \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{up}}_{Q,\lambda,\sigma,\beta,j,K}(s,p).$$

Consequently, all adjacent-crossing exponent hypotheses follow from the corresponding same-level and upper-shifted exterior-return exponent hypotheses.

Proof. Fix u and write the relevant level-u crossing runs in increasing order of their starting times. If the exit type is $\zeta = -1$, then every such run ends with an inward level-u token. Thus more than R runs give more than R inward level-u tokens, and hence, by Lemma 8.131, the property $\mathcal{X}_{u,K}$.

It remains to consider type $(-1,+1)$. If there are $q > R$ such runs, then between the end of each run and the start of the next one the path must move from level $u+1$ to level $u-1$. In each of the $q-1$ intervening time intervals there is therefore an inward level-u token. Since $q-1 \geq R > K$, Lemma 8.131 again gives $\mathcal{X}_{u,K}$.

For type $(+1,+1)$, each run begins with an inward level-$u+1$ token. More than R such runs therefore imply $\mathcal{X}_{u+1,K}$, again by Lemma 8.131. Taking the corresponding property inclusions inside the profile definitions and then the suprema gives the displayed inequalities. □

Corollary 8.247 (Exact-level same-sphere congestion: long run or many runs)**.** *In the setting of Corollary 8.141, let $K, L \geq 1$ and put*

$$R_{K,L} = \left\lfloor \frac{K}{L} \right\rfloor.$$

Then

$$\begin{aligned}\mathfrak{Z}^{G\to H,\mathrm{rad},\mathrm{hor},\mathrm{lev}}_{Q,\lambda,\sigma,\beta,j,K}(s,p) \leq\ & \mathfrak{T}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{long}}_{Q,\lambda,\sigma,\beta,j,L}(s,p) \\ & + \mathfrak{T}^{G\to H,\mathrm{rad},\mathrm{tan},\mathrm{many}}_{Q,\lambda,\sigma,\beta,j,R_{K,L}}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on the exact-level same-sphere congestion profile follows from the same exponent hypothesis on the long-run and many-run tangential profiles.

Proof. Fix an exact level u. The total number of same-sphere steps at level u is the sum of the lengths of all level-u tangential runs. If $\mathcal{Z}_{u,K}$ holds but $\mathcal{T}^{\mathrm{long}}_{u,L}$ and $\mathcal{T}^{\mathrm{many}}_{u,R_{K,L}}$ both fail, then every level-u tangential run has length at most L, and there are at most $R_{K,L}$ such runs. Hence

$$H_x(\gamma;\{u\}) \leq LR_{K,L} \leq K,$$

contradicting $\mathcal{Z}_{u,K}$. Thus

$$\mathcal{Z}_{u,K} \subseteq \mathcal{T}^{\mathrm{long}}_{u,L} \cup \mathcal{T}^{\mathrm{many}}_{u,R_{K,L}}.$$

Taking the corresponding union bound inside the property-restricted profiles, then summing with the same weights, gives the displayed profile inequality after taking suprema. The exponent conclusion follows because the sum has two terms. □

Corollary 8.248 (Fully localized finite far-tail ladder)**.** *In the setting of Corollary 8.139, let $M_0, \dots, M_{N-1} \geq 1$ be integers and write*

$$\begin{aligned}Z^{\mathrm{spr}}_{i,j}(s,p) &= \mathfrak{Z}^{G\to H,\mathrm{rad},\mathrm{hor},\mathrm{spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,M_i}(s,p),\\ Z^{\mathrm{lev}}_{i,j}(s,p) &= \mathfrak{Z}^{G\to H,\mathrm{rad},\mathrm{hor},\mathrm{lev}}_{Q,\lambda_i,\sigma_i,\beta_i,j,M_i}(s,p),\\ S^{\mathrm{ret}}_{i,j}(s,p) &= \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p),\end{aligned}$$

and

$$X^{\mathrm{ret}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad},\mathrm{ret}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \leq\ & \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p) \\ & + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} Z^{\mathrm{spr}}_{i,j}(s,p) \\ & + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, Z^{\mathrm{lev}}_{i,j}(s,p) \\ & + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p) \\ & + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, X^{\mathrm{ret}}_{i,j}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail and on the four localized profile families appearing on the right-hand side.

Proof. Apply Corollary 8.139. Each horizontal-heavy slab term is bounded by Corollary 8.141, with

$$(\lambda, \sigma, \beta, K) = (\lambda_i, \sigma_i, \beta_i, M_i).$$

The displayed inequality follows after summing over i, j. All sums are finite and independent of p, and the slab-width factors are exponent-negligible. □

Corollary 8.249 (Tangentially localized finite far-tail ladder)**.** *In the setting of Corollary 8.248, let $L_0, \dots, L_{N-1} \geq 1$ be integers and put*

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor \qquad (0 \leq i < N).$$

Define $Z_{i,j}^{\mathrm{spr}}$, $S_{i,j}^{\mathrm{ret}}$, and $X_{i,j}^{\mathrm{ret}}$ as in Corollary 8.248, and write

$$T_{i,j}^{\mathrm{long}}(s,p) = \mathfrak{T}_{Q,\lambda_i,\sigma_i,\beta_i,j,L_i}^{G\to H,\mathrm{rad},\tan,\mathrm{long}}(s,p),$$

$$T_{i,j}^{\mathrm{many}}(s,p) = \mathfrak{T}_{Q,\lambda_i,\sigma_i,\beta_i,j,R_i}^{G\to H,\mathrm{rad},\tan,\mathrm{many}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \leq\ & \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} Z_{i,j}^{\mathrm{spr}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T_{i,j}^{\mathrm{many}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, X_{i,j}^{\mathrm{ret}}(s,p).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the same-sphere spread profiles, the two tangential run-profile families, and the two exterior-return profile families.

Proof. Apply Corollary 8.248. Each exact-level same-sphere congestion term

$$Z_{i,j}^{\mathrm{lev}}$$

is bounded by Corollary 8.247, with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, M_i, L_i).$$

Substituting these bounds gives the displayed inequality. The exponent conclusion is unchanged because all sums are finite and the only variable multiplicative losses are slab-width factors $W_{\beta_i,p}$, which have logarithmic size $o(L_H)$. □

Corollary 8.250 (Crossing finite far-tail ladder)**.** *In the setting of Corollary 8.249, assume that*

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor \geq 2 \qquad (0 \leq i < N),$$

and put

$$P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor \qquad (0 \leq i < N).$$

Define $Z^{\mathrm{spr}}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $X^{\mathrm{ret}}_{i,j}$ as in Corollary 8.249, and write

$$C^{\eta,\zeta}_{i,j}(s,p) = \mathfrak{C}^{G\to H,\mathrm{rad,tan,cross}}_{Q,\lambda_i,\sigma_i,\beta_i,j,P_i,\eta,\zeta}(s,p) \qquad (\eta,\zeta \in \{-1,+1\}).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \leq\ & \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} Z^{\mathrm{spr}}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p} \sum_{\eta,\zeta\in\{-1,+1\}} C^{\eta,\zeta}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, X^{\mathrm{ret}}_{i,j}(s,p).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the same-sphere spread profiles, the long-run profiles, the four adjacent-crossing tangential profile families, and the two exterior-return profile families.

Proof. Apply Corollary 8.249. For each i, j, bound the many-run term

$$T^{\mathrm{many}}_{i,j}$$

by Corollary 8.243, with

$$(\lambda, \sigma, \beta, R) = (\lambda_i, \sigma_i, \beta_i, R_i).$$

Substitution gives the displayed inequality. The exponent conclusion is unchanged because the added crossing-type sum has four terms and the only variable multiplicative losses remain the factors $W_{\beta_i,p}$, whose logarithmic size is $o(L_H)$. □

Corollary 8.251 (Return-absorbed crossing finite far-tail ladder)**.** *In the setting of Corollary 8.250, assume that*

$$P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor \geq 1 \qquad (0 \leq i < N),$$

and put

$$K_i^{\times} = P_i - 1 \qquad (0 \leq i < N).$$

Define $Z_{i,j}^{\mathrm{spr}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $X_{i,j}^{\mathrm{ret}}$ *as in Corollary 8.250, and write*

$$X_{i,j}^{\times}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\times}}^{G\to H,\mathrm{rad},\mathrm{ret}}(s,p),$$

$$U_{i,j}^{\times}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\times}}^{G\to H,\mathrm{rad},\mathrm{ret},\mathrm{up}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \leq{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} Z_{i,j}^{\mathrm{spr}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p} \left(3X_{i,j}^{\times}(s,p) + U_{i,j}^{\times}(s,p)\right) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, X_{i,j}^{\mathrm{ret}}(s,p).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the same-sphere spread profiles, the long-run profiles, the spread-return profiles, and the same-level, upper-shifted, and repeated exterior-return profiles.

Proof. Apply Corollary 8.250. For each i, j, use Corollary 8.246 with

$$R = P_i, \qquad K = K_i^{\times}.$$

The three crossing types other than $(+1,+1)$ are bounded by $X_{i,j}^{\times}$, while the $(+1,+1)$ type is bounded by $U_{i,j}^{\times}$. Substitution gives the displayed inequality. The exponent conclusion is unchanged because all sums and the coefficient 3 are fixed, and the only variable multiplicative losses remain the exponent-negligible factors $W_{\beta_i,p}$. □

Corollary 8.252 (Return-only crossing finite far-tail ladder)**.** *In the setting of Corollary 8.251, define* $Z_{i,j}^{\mathrm{spr}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, $X_{i,j}^{\mathrm{ret}}$, *and* $K_i^{\times}$ *as there, and for* $0 \leq j \leq J_{\sigma_i,\beta_i}$ *write*

$$B_{i,j}^{\times}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\times}}^{G\to H,\mathrm{rad},\mathrm{ret}}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & Z^{\mathrm{spr}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(4B^{\times}_{i,j}(s,p) + B^{\times}_{i,j+1}(s,p)\right)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, X^{\mathrm{ret}}_{i,j}(s,p).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the same-sphere spread profiles, the long-run profiles, the spread-return profiles, and the two ordinary exact-level exterior-return profile families $B^{\times}$ and X^{ret}.

Proof. Apply Corollary 8.251. For each i, j, use Corollary 8.245 to bound

$$U^{\times}_{i,j} \le B^{\times}_{i,j} + B^{\times}_{i,j+1}.$$

The term $3X^{\times}_{i,j} + U^{\times}_{i,j}$ is therefore bounded by

$$4B^{\times}_{i,j} + B^{\times}_{i,j+1}.$$

Substitution gives the displayed inequality. The exponent conclusion is unchanged because only one adjacent slab has been added for each fixed i, j, and all sums, coefficients, and numbers of slabs are independent of p, apart from the already exponent-negligible $W_{\beta_i,p}$ factors. □

Corollary 8.253 (Threshold-dominated return finite far-tail ladder)**.** *In the setting of Corollary 8.252, assume that*

$$K^{\times}_i \ge K_i \qquad (0 \le i < N).$$

Define $Z^{\mathrm{spr}}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, and $S^{\mathrm{ret}}_{i,j}$ as in Corollary 8.252, and for $0 \le j \le J_{\sigma_i,\beta_i}$ write

$$\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} Z^{\mathrm{spr}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).&\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the same-sphere spread profiles, the long-run profiles, the spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.252. Since $K_i^{\times} \ge K_i$, monotonicity of the event $\mathcal{X}_{u,K}$ in K gives

$$B^{\times}_{i,j}(s,p) \le \widehat{X}^{\mathrm{ret}}_{i,j}(s,p) \qquad (0 \le j \le J_{\sigma_i,\beta_i}).$$

For $j < J_{\sigma_i,\beta_i}$, the repeated-return term $X^{\mathrm{ret}}_{i,j}$ in Corollary 8.252 is exactly $\widehat{X}^{\mathrm{ret}}_{i,j}$. Substituting the displayed monotonicity bound gives the claimed inequality. The exponent conclusion is unchanged because all added coefficients and slab counts are fixed, while the factors $W_{\beta_i,p}$ are exponent-negligible. □

Corollary 8.254 (Outward-through finite ladder)**.** *In the setting of Corollary 8.253, define $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$U^{\mathrm{out}}_{i,j}(s,p) = \mathfrak{U}^{G\to H,\mathrm{rad,tan,out,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,M_i}(s,p),$$

and for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R^{\mathrm{spr}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,8M_i}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & U^{\mathrm{out}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \left(R^{\mathrm{spr}}_{i,j}(s,p) + R^{\mathrm{spr}}_{i,j+1}(s,p)\right)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, outward-through tangential spread profiles, long-run profiles, spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.253. For each i, j, bound the same-sphere spread term

$$Z^{\mathrm{spr}}_{i,j}$$

by Corollary 8.145, with

$$(\lambda, \sigma, \beta, K) = (\lambda_i, \sigma_i, \beta_i, M_i).$$

Substitution gives the displayed inequality. The exponent conclusion is unchanged because the added current/next spread-return sums have finitely many terms independent of p, and the only nonconstant multiplicative losses are still the exponent-negligible factors $W_{\beta_i,p}$. □

Corollary 8.255 (Record-outward finite ladder)**.** *In the setting of Corollary 8.254, define $R^{\mathrm{spr}}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{rec}}_{i,j}(s,p) = \mathfrak{U}^{G\to H,\mathrm{rad,tan,out,rec}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2M_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{rec}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(2R^{\mathrm{spr}}_{i,j}(s,p)+R^{\mathrm{spr}}_{i,j+1}(s,p)\big)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, record outward-through spread profiles, long-run profiles, spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.254. For each i,j, Corollary 8.147 with

$$(\lambda,\sigma,\beta,K)=(\lambda_i,\sigma_i,\beta_i,M_i)$$

gives

$$U^{\mathrm{out}}_{i,j}(s,p)\le P^{\mathrm{rec}}_{i,j}(s,p)+\mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2M_i}(s,p).$$

The last profile is bounded by $R^{\mathrm{spr}}_{i,j}$, by monotonicity in the spread threshold and the definition of $R^{\mathrm{spr}}_{i,j}$ with parameter $8M_i$. Substituting this bound in the outward-through ladder gives the displayed inequality. The exponent conclusion is unchanged because only a fixed coefficient on an already listed spread-return family has changed. □

Corollary 8.256 (Subterminal record finite ladder)**.** *In the setting of Corollary 8.255, define $R^{\mathrm{spr}}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0\le j<J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{sub}}_{i,j}(s,p)=\mathfrak{U}^{G\to H,\mathrm{rad,tan,out,sub}}_{Q,\lambda_i,\sigma_i,\beta_i,j,4M_i}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{sub}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \left(3R^{\mathrm{spr}}_{i,j}(s,p) + 2R^{\mathrm{spr}}_{i,j+1}(s,p)\right)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, subterminal record outward-through spread profiles, long-run profiles, spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.255. For each i, j, Corollary 8.149 with

$$(\lambda, \sigma, \beta, K) = (\lambda_i, \sigma_i, \beta_i, 2M_i)$$

gives

$$P^{\mathrm{rec}}_{i,j}(s,p) \le P^{\mathrm{sub}}_{i,j}(s,p) + R^{\mathrm{spr}}_{i,j}(s,p) + R^{\mathrm{spr}}_{i,j+1}(s,p).$$

Substituting this in the record-outward ladder adds one more current spread-return term and one more next-slab spread-return term, giving the displayed coefficients 3 and 2. The exponent conclusion is unchanged because only fixed coefficients on already listed spread-return families have changed. □

Corollary 8.257 (Persistent subterminal finite ladder)**.** *In the setting of Corollary 8.256, define $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{per}}_{i,j}(s,p) = \mathfrak{U}^{G\to H,\mathrm{rad,tan,out,per}}_{Q,\lambda_i,\sigma_i,\beta_i,j,8M_i}(s,p),$$

and for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R^{\mathrm{spr},+}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,16M_i}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{per}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \left(4R^{\mathrm{spr},+}_{i,j}(s,p) + 3R^{\mathrm{spr},+}_{i,j+1}(s,p)\right)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, persistent subterminal record profiles, long-run profiles, spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.256. For each i, j, Corollary 8.151 with

$$(\lambda, \sigma, \beta, K) = (\lambda_i, \sigma_i, \beta_i, 4M_i)$$

gives

$$P^{\mathrm{sub}}_{i,j}(s,p) \le P^{\mathrm{per}}_{i,j}(s,p) + R^{\mathrm{spr},+}_{i,j}(s,p) + R^{\mathrm{spr},+}_{i,j+1}(s,p).$$

Also, by monotonicity in the spread threshold,

$$R^{\mathrm{spr}}_{i,j}(s,p) \le R^{\mathrm{spr},+}_{i,j}(s,p), \qquad 0 \le j \le J_{\sigma_i,\beta_i}.$$

Substituting these inequalities into the subterminal record ladder gives the displayed coefficients 4 and 3. The exponent conclusion is unchanged because only fixed coefficients and a fixed threshold choice for the same spread-return family have changed. □

Corollary 8.258 (Deep persistent finite ladder)**.** *In the setting of Corollary 8.257, define $R^{\mathrm{spr},+}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{deep}}_{i,j}(s,p) = \mathfrak{U}^{G\to H,\mathrm{rad,tan,out,deep}}_{Q,\lambda_i,\sigma_i,\beta_i,j,16M_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{deep}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(4R^{\mathrm{spr},+}_{i,j}(s,p)+3R^{\mathrm{spr},+}_{i,j+1}(s,p)\big)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, deep persistent subterminal record profiles, long-run profiles, spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.257. For each i,j, Corollary 8.153 with

$$(\lambda,\sigma,\beta,K)=(\lambda_i,\sigma_i,\beta_i,8M_i)$$

gives

$$P^{\mathrm{per}}_{i,j}(s,p)\le P^{\mathrm{deep}}_{i,j}(s,p).$$

Substitution gives the displayed inequality, and the exponent conclusion is unchanged. □

Corollary 8.259 (Endpoint-prefix finite ladder)**.** *In the setting of Corollary 8.258, define $R^{\mathrm{spr},+}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0\le j<J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre}}_{i,j}(s,p)=\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,deep}}_{Q,\lambda_i,\sigma_i,\beta_i,j,16M_i}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{pre}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \left(4R^{\mathrm{spr},+}_{i,j}(s,p) + 3R^{\mathrm{spr},+}_{i,j+1}(s,p)\right)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, endpoint-prefix deep horizontal spread profiles, long-run profiles, spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.258. For each i, j, Corollary 8.155 with

$$(\lambda, \sigma, \beta, K) = (\lambda_i, \sigma_i, \beta_i, 16M_i)$$

gives

$$P^{\mathrm{deep}}_{i,j}(s,p) \le P^{\mathrm{pre}}_{i,j}(s,p).$$

Substitution gives the displayed inequality, and the exponent conclusion is unchanged. □

Corollary 8.260 (Short endpoint-prefix finite ladder)**.** *In the setting of Corollary 8.259, define $R^{\mathrm{spr},+}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre,sh}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,deep,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,16M_i,L_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{pre,sh}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \big(4R^{\mathrm{spr},+}_{i,j}(s,p) + 3R^{\mathrm{spr},+}_{i,j+1}(s,p)\big)\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, short endpoint-prefix deep horizontal spread profiles, long-run profiles, spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.259. For each i, j, Corollary 8.157 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,16M_i,L_i)$$

gives

$$P^{\mathrm{pre}}_{i,j}(s,p) \le W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p) + P^{\mathrm{pre,sh}}_{i,j}(s,p).$$

Substitution gives the displayed inequality, with the existing long-run term and the new long-run term combined into the factor 2. The exponent conclusion is unchanged because the factor 2 is fixed and $\log_{D_H} W_{\beta_i,p} = o(L_H)$. □

Corollary 8.261 (Short outward-prefix finite ladder)**.** *In the setting of Corollary 8.260, define $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre,out,sh}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,32M_i,L_i}(s,p),$$

and for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R^{\mathrm{spr},++}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_i}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{pre,out,sh}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \left(5R^{\mathrm{spr},++}_{i,j}(s,p) + 4R^{\mathrm{spr},++}_{i,j+1}(s,p)\right)\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, short outward-prefix spread profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.260. For each i, j, Corollary 8.159 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 16M_i, L_i)$$

gives

$$P^{\mathrm{pre,sh}}_{i,j}(s,p) \le P^{\mathrm{pre,out,sh}}_{i,j}(s,p) + R^{\mathrm{spr},++}_{i,j}(s,p) + R^{\mathrm{spr},++}_{i,j+1}(s,p).$$

Moreover,

$$R^{\mathrm{spr},+}_{i,j}(s,p) \le R^{\mathrm{spr},++}_{i,j}(s,p) \qquad (0 \le j \le J_{\sigma_i,\beta_i})$$

by monotonicity in the spread-return threshold. Substitution gives the displayed coefficients 5 and 4. The exponent conclusion is unchanged because only fixed coefficients and fixed threshold choices for the same spread-return family have changed. □

Corollary 8.262 (Short prefix-record finite ladder)**.** *In the setting of Corollary 8.261, define $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre,rec,sh}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,64M_i,L_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{pre,rec,sh}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \big(6R^{\mathrm{spr},++}_{i,j}(s,p) + 4R^{\mathrm{spr},++}_{i,j+1}(s,p)\big)\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, short prefix-record spread profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.261. For each i, j, Corollary 8.161 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,32M_i,L_i)$$

gives

$$P^{\mathrm{pre,out,sh}}_{i,j}(s,p) \le P^{\mathrm{pre,rec,sh}}_{i,j}(s,p) + \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,64M_i}(s,p).$$

The last term is bounded by $R^{\mathrm{spr},++}_{i,j}(s,p)$ by monotonicity in the spread-return threshold. Substitution adds one current-slab spread-return term, giving the displayed coefficients 6 and 4. The exponent conclusion is unchanged. □

Corollary 8.263 (Short persistent-prefix finite ladder)**.** *In the setting of Corollary 8.262, define $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre,per,sh}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,128M_i,L_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{pre,per,sh}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \big(7R^{\mathrm{spr},++}_{i,j}(s,p) + 5R^{\mathrm{spr},++}_{i,j+1}(s,p)\big)\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, short persistent-prefix spread profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.262. For each i, j, Corollary 8.163 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,64M_i,L_i)$$

gives

$$P^{\mathrm{pre,rec,sh}}_{i,j}(s,p) \le P^{\mathrm{pre,per,sh}}_{i,j}(s,p) + R^{\mathrm{spr},++}_{i,j}(s,p) + R^{\mathrm{spr},++}_{i,j+1}(s,p),$$

since $4K = 256M_i$. Substitution adds one current-slab and one next-slab spread-return term, giving the displayed coefficients 7 and 5. The exponent conclusion is unchanged. □

Corollary 8.264 (Ordered prefix-ladder finite ladder)**.** *In the setting of Corollary 8.263, define $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre,lad,sh}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,lad,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,128M_i,L_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{pre,lad,sh}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \left(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\right)\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, ordered short persistent prefix-ladder profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.263. For each i,j, Corollary 8.165 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,128M_i,L_i)$$

gives

$$P^{\mathrm{pre,per,sh}}_{i,j}(s,p)\le P^{\mathrm{pre,lad,sh}}_{i,j}(s,p).$$

Substitution gives the displayed inequality with the same return coefficients. The exponent conclusion is unchanged. □

Corollary 8.265 (Bounded-gap prefix-pair finite ladder)**.** *In the setting of Corollary 8.264, define $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. Put*

$$\Theta_i=\theta_{\lambda_i,\sigma_i,\beta_i},\qquad B_i=\left\lceil\frac{2048M_i\beta_i}{\Theta_i}\right\rceil+1.$$

For $0\le j<J_{\sigma_i,\beta_i}$, write

$$P^{\mathrm{pre,pair,sh}}_{i,j}(s,p)=\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_i,L_i,B_i}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & P^{\mathrm{pre,pair,sh}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & \left(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\right)\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & S^{\mathrm{ret}}_{i,j}(s,p)\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} & W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, bounded-gap short persistent prefix-pair profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.264. For each i,j, Corollary 8.167 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,128M_i,L_i)$$

gives

$$P^{\mathrm{pre,lad,sh}}_{i,j}(s,p)\le P^{\mathrm{pre,pair,sh}}_{i,j}(s,p),$$

because $2K=256M_i$ and

$$B_{\lambda_i,\sigma_i,\beta_i,128M_i}=\left\lceil\frac{2048M_i\beta_i}{\Theta_i}\right\rceil+1=B_i.$$

Substitution gives the displayed inequality. No coefficient changes occur in this reduction, and the exponent conclusion is unchanged. □

Corollary 8.266 (Fixed-gap prefix-pair finite ladder)**.** *In the setting of Corollary 8.265, define B_i, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0\le j<J_{\sigma_i,\beta_i}$ and $1\le h\le B_i$, write*

$$P^{\mathrm{pre,gap,sh}}_{i,j,h}(s,p)=\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,L_i,h}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i} P^{\mathrm{pre,gap,sh}}_{i,j,h}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\right)&\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).&\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, fixed-gap short persistent prefix-pair profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.265. For each i,j, Corollary 8.169 with

$$(\lambda,\sigma,\beta,K,L,B)=(\lambda_i,\sigma_i,\beta_i,256M_i,L_i,B_i)$$

gives

$$P^{\mathrm{pre,pair,sh}}_{i,j}(s,p)\le\sum_{h=1}^{B_i}P^{\mathrm{pre,gap,sh}}_{i,j,h}(s,p).$$

Substitution gives the displayed inequality. The sums over h are finite and independent of p, so the exponent conclusion is unchanged. □

Corollary 8.267 (Confined fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.266, define B_i, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0\le j<J_{\sigma_i,\beta_i}$ and $1\le h\le B_i$, write*

$$P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)=\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,L_i,h}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\right)&\\ + 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).&\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, confined fixed-gap bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.266. For each i,j,h, Corollary 8.171 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,256M_iB_i,L_i)$$

gives

$$P^{\mathrm{pre,gap,sh}}_{i,j,h}(s,p)\le P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p).$$

Substitution gives the displayed inequality. The exponent conclusion is unchanged. □

Corollary 8.268 (Adjacent-stacked bridge finite ladder)**.** *In the setting of Corollary 8.267, define B_i, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0\le j<J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre,adj,sh}}_{i,j}(s,p)=\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,adj,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,L_i}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{pre,adj,sh}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\right)\\
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, adjacent stacked prefix-pair profiles, confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.267. For each i, j, Corollary 8.177 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,256M_iB_i,L_i)$$

gives

$$P^{\mathrm{pre,br,sh}}_{i,j,1}(s,p) \le P^{\mathrm{pre,adj,sh}}_{i,j}(s,p).$$

Substitution gives the displayed inequality. The exponent conclusion is unchanged. □

Corollary 8.269 (Outward-staircase bridge finite ladder)**.** *In the setting of Corollary 8.268, define B_i, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{pre,stair,sh}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,2L_i+1}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,stair,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, two-level outward staircase profiles, confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.268. For each i, j, Corollary 8.180 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,256M_iB_i,L_i)$$

gives

$$P_{i,j}^{\mathrm{pre,adj,sh}}(s,p) \le P_{i,j}^{\mathrm{pre,stair,sh}}(s,p).$$

Substitution gives the displayed inequality. The exponent conclusion is unchanged. □

Corollary 8.270 (Locally short staircase finite ladder)**.** *In the setting of Corollary 8.269, define B_i, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr},++}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, and $\widehat{X}_{i,j}^{\mathrm{ret}}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P_{i,j}^{\mathrm{pre,stair,loc,sh}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,L_i}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le{}& \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)\\ &+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{pre,stair,loc,sh}}_{i,j}(s,p)\\ &+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)\\ &+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\right)\\ &+3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)\\ &+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)\\ &+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, locally short two-level staircase profiles, confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.269. For each i, j, Corollary 8.181 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,256M_iB_i,L_i)$$

gives

$$P^{\mathrm{pre,stair,sh}}_{i,j}(s,p) \le W_{\beta_i,p}T^{\mathrm{long}}_{i,j}(s,p) + P^{\mathrm{pre,stair,loc,sh}}_{i,j}(s,p).$$

Substitution gives the displayed inequality, with the old coefficient 2 on the long-run term increased to 3. The exponent conclusion is unchanged because $\log_{D_H} W_{\beta_i,p} = o(L_H)$. □

Corollary 8.271 (Fixed-gap staircase-pair finite ladder)**.** *In the setting of Corollary 8.270, define B_i, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there, and put*

$$C_i = \left\lceil \frac{4096M_iB_i\beta_i}{\theta_{\lambda_i,\sigma_i,\beta_i}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le C_i$, write

$$P^{\mathrm{pre,stair,loc,gap,sh}}_{i,j,h}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,gap,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,512M_iB_iC_i,L_i,h}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{C_i} P^{\mathrm{pre,stair,loc,gap,sh}}_{i,j,h}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\big(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\big)&\\ + 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)&\\ + \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).&\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, fixed-gap locally short staircase-pair profiles, confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.270. For each i, j, Corollary 8.183 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,256M_iB_i,L_i)$$

gives a bounded-gap staircase-pair profile with parameter $512M_iB_i$ and gap cutoff

$$C_i = B_{\lambda_i,\sigma_i,\beta_i,256M_iB_i}.$$

Applying Corollary 8.185 with

$$(\lambda,\sigma,\beta,K,L,B) = (\lambda_i,\sigma_i,\beta_i,512M_iB_i,L_i,C_i)$$

then gives

$$P^{\mathrm{pre,stair,loc,sh}}_{i,j}(s,p) \le \sum_{h=1}^{C_i} P^{\mathrm{pre,stair,loc,gap,sh}}_{i,j,h}(s,p).$$

Substitution gives the displayed inequality. The sums over h are finite and independent of p, so the exponent conclusion is unchanged. □

Corollary 8.272 (Staircase-chain/bridge finite ladder)**.** *In the setting of Corollary 8.271, define B_i, C_i, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. For $q \in \{3,4\}$, write*

$$P^{\mathrm{pre,stair,loc,chain,sh}}_{i,j,q}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,512M_iB_iC_i,L_i,q}(s,p),$$

and, for $3 \le h \le C_i$*, write*

$$P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,bridge,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,512M_iB_iC_i,L_i,h}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\begin{aligned}
\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le{}& \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(P^{\mathrm{pre,stair,loc,chain,sh}}_{i,j,3}(s,p) + P^{\mathrm{pre,stair,loc,chain,sh}}_{i,j,4}(s,p)\right) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(7R^{\mathrm{spr},++}_{i,j}(s,p) + 5R^{\mathrm{spr},++}_{i,j+1}(s,p)\right) \\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, locally short staircase-chain profiles with $q = 3, 4$*, confined fixed-gap locally short staircase-bridge profiles with* $3 \le h \le C_i$*, confined fixed-gap bridge profiles with* $h \ge 2$*, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.271. For each i, j, Corollary 8.189 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 512M_iB_iC_i, L_i)$$

gives

$$\begin{aligned}
&\sum_{h=1}^{C_i} P^{\mathrm{pre,stair,loc,gap,sh}}_{i,j,h}(s,p) \\
&\quad\le P^{\mathrm{pre,stair,loc,chain,sh}}_{i,j,3}(s,p) + P^{\mathrm{pre,stair,loc,chain,sh}}_{i,j,4}(s,p) + \sum_{h=3}^{C_i} P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}(s,p).
\end{aligned}$$

Substitution gives the displayed inequality. The number of new chain and bridge profiles is finite and independent of p, so the exponent conclusion is unchanged. □

Corollary 8.273 (Chain-absorbed staircase-chain/bridge finite ladder)**.** *In the setting of Corollary 8.272, define* B_i, C_i, $P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}$, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, *and* $\widehat{X}^{\mathrm{ret}}_{i,j}$ *as there. For* $0 \le j < J_{\sigma_i,\beta_i}$, *write*

$$P^{\mathrm{pre,rec,ch}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,512M_iB_iC_i,L_i}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{P}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{pre,rec,ch}}_{i,j}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R^{\mathrm{spr},++}_{i,j}(s,p) + 5R^{\mathrm{spr},++}_{i,j+1}(s,p)\big)\\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the chain-depth prefix-record outward-spread profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, *confined fixed-gap bridge profiles with* $h \ge 2$, *long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.272. For each i, j and $q \in \{3, 4\}$, the chain profile

$$P^{\mathrm{pre,stair,loc,chain,sh}}_{i,j,q}(s,p)$$

is bounded by $P^{\mathrm{pre,rec,ch}}_{i,j}(s,p)$ by Corollary 8.236, with

$$(\lambda, \sigma, \beta, K, L, \ell) = (\lambda_i, \sigma_i, \beta_i, 512M_iB_iC_i, L_i, q).$$

Substituting these two bounds for $q = 3$ and $q = 4$ gives the displayed inequality. The coefficient 2 is fixed, all other sums are those of Corollary 8.272, and the only nonconstant multiplicative losses remain the factors $W_{\beta_i,p}$, whose logarithms are $o(L_H)$. □

Corollary 8.274 (Persistent chain-absorbed finite ladder)**.** *In the setting of Corollary 8.273, define* B_i, C_i, $P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}$, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, *and* $\widehat{X}^{\mathrm{ret}}_{i,j}$ *as there. For* $0 \le j < J_{\sigma_i,\beta_i}$, *write*

$$P^{\mathrm{pre,per,ch}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,1024M_iB_iC_i,L_i}(s,p),$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, *write*

$$R^{\mathrm{spr,ch}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2048M_iB_iC_i}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le{}& \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)\\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{pre,per,ch}}_{i,j}(s,p)\\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R^{\mathrm{spr,ch}}_{i,j}(s,p) + R^{\mathrm{spr,ch}}_{i,j+1}(s,p)\big)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R^{\mathrm{spr},++}_{i,j}(s,p) + 5R^{\mathrm{spr},++}_{i,j+1}(s,p)\big)\\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the chain-depth persistent-prefix profiles, the chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, *confined fixed-gap bridge profiles with* $h \ge 2$, *long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.273. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 512M_iB_iC_i, L_i).$$

This gives

$$P^{\mathrm{pre,rec,ch}}_{i,j}(s,p) \le P^{\mathrm{pre,per,ch}}_{i,j}(s,p) + R^{\mathrm{spr,ch}}_{i,j}(s,p) + R^{\mathrm{spr,ch}}_{i,j+1}(s,p).$$

Substituting this inequality into the coefficient-2 prefix-record term of Corollary 8.273 gives the displayed bound. The newly added coefficients are fixed, and all remaining nonconstant multiplicative losses are the same slab-width factors as before, whose logarithms are $o(L_H)$. □

Corollary 8.275 (Bounded-gap persistent chain-absorbed finite ladder)**.** *In the setting of Corollary 8.274, define* B_i, C_i, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{ch}} = \left\lceil \frac{16384 M_i B_i C_i \beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, *write*

$$P_{i,j}^{\mathrm{pre,pair,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2048M_iB_iC_i,L_i,B_i^{\mathrm{ch}}}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,pair,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\right) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(7R_{i,j}^{\mathrm{spr,++}}(s,p) + 5R_{i,j+1}^{\mathrm{spr,++}}(s,p)\right) \\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap chain-depth persistent prefix-pair profiles, the chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, *confined fixed-gap bridge profiles with* $h \ge 2$, *long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.274. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 1024 M_i B_i C_i, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since

$$2K = 2048 M_i B_i C_i$$

and

$$B_{\lambda_i, \sigma_i, \beta_i, 1024 M_i B_i C_i} = \left\lceil \frac{16384 M_i B_i C_i \beta_i}{\Theta_i} \right\rceil + 1 = B_i^{\mathrm{ch}},$$

the two inequalities give

$$P_{i,j}^{\mathrm{pre,per,ch}}(s,p) \le P_{i,j}^{\mathrm{pre,pair,ch}}(s,p).$$

Substituting this into Corollary 8.274 gives the displayed bound. The reduction introduces no new variable coefficient. The cutoff B_i^{ch} is fixed once the finite ladder parameters are fixed, and the exponent conclusion follows as before. □

Corollary 8.276 (Fixed-gap persistent chain-absorbed finite ladder)**.** *In the setting of Corollary 8.275, define B_i, C_i, B_i^{ch}, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, and $\widehat{X}_{i,j}^{\mathrm{ret}}$ as there, and put*

$$K_i^{\mathrm{gap,ch}} = 2048 M_i B_i C_i B_i^{\mathrm{ch}}.$$

For $0 \le j < J_{\sigma_i, \beta_i}$ and $1 \le h \le B_i^{\mathrm{ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,gap,ch}}(s,p) = \mathfrak{P}_{Q, \lambda_i, \sigma_i, \beta_i, j, K_i^{\mathrm{gap,ch}}, L_i, h}^{G \to H, \mathrm{rad,tan,pre,out,per,gap,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i^{\mathrm{ch}}} P_{i,j,h}^{\mathrm{pre,gap,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\big(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\big(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of fixed-gap chain-depth persistent prefix-pair profiles, the chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.275. For each i, j, apply Corollary 8.169 with

$$(\lambda, \sigma, \beta, K, L, B) = (\lambda_i, \sigma_i, \beta_i, 2048M_iB_iC_i, L_i, B_i^{\mathrm{ch}}).$$

Since $KB = K_i^{\mathrm{gap,ch}}$, this gives

$$P_{i,j}^{\mathrm{pre,pair,ch}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{ch}}} P_{i,j,h}^{\mathrm{pre,gap,ch}}(s,p).$$

Substitute this bound into the coefficient-2 bounded-gap pair term in Corollary 8.275. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.277 (Confined-bridge persistent chain-absorbed finite ladder)**.** *In the setting of Corollary 8.276, define* B_i, C_i, B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{ch}}$, *write*

$$P_{i,j,h}^{\mathrm{pre,br,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i^{\mathrm{ch}}} P_{i,j,h}^{\mathrm{pre,br,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\right) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(7R_{i,j}^{\mathrm{spr,++}}(s,p) + 5R_{i,j+1}^{\mathrm{spr,++}}(s,p)\right) \\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of chain-depth confined fixed-gap bridge profiles, the chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, *the earlier confined fixed-gap bridge profiles with* $h \ge 2$, *long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.276. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,gap,ch}}(s,p) \le P_{i,j,h}^{\mathrm{pre,br,ch}}(s,p).$$

Substituting into the coefficient-2 fixed-gap chain-depth prefix-pair sum gives the displayed inequality. The exponent conclusion is unchanged because no new variable coefficient is introduced. □

Corollary 8.278 (Inward-free bridge persistent chain-absorbed finite ladder)**.** *In the setting of Corollary 8.277, define* B_i, C_i, B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{ch}}$, *write*

$$P_{i,j,h}^{\mathrm{pre,br,free,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}(s,p)$$

and

$$R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,ch}}\max\{h-1,1\}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i^{\mathrm{ch}}} P_{i,j,h}^{\mathrm{pre,br,free,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr,++}}(s,p) + 5R_{i,j+1}^{\mathrm{spr,++}}(s,p)\big) \\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of chain-depth inward-free confined bridge profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, *the earlier confined fixed-gap bridge profiles with* $h \ge 2$, *long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.277. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,br,ch}}(s,p) \le P_{i,j,h}^{\mathrm{pre,br,free,ch}}(s,p) + R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p).$$

Substituting into the coefficient-2 chain-depth confined-bridge sum gives the displayed inequality. The additional h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.279 (Record-absorbed inward-free bridge persistent chain-absorbed finite ladder)**.** *In the setting of Corollary 8.278, define* B_i, C_i, B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. For* $0 \le j < J_{\sigma_i,\beta_i}$, *write*

$$P_{i,j}^{\mathrm{pre,rec,br,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,rec,br,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr,++}}(s,p) + 5R_{i,j+1}^{\mathrm{spr,++}}(s,p)\big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p} \big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the chain-depth bridge-record prefix-record outward-spread profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.278. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,ch}}, L_i).$$

This gives

$$\begin{aligned} P_{i,j,h}^{\mathrm{pre,br,free,ch}}(s,p) \le & \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,ch}},L_i,h+1}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}(s,p) \\ & + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{pre,rec,br,ch}}$. Substituting these inequalities into the coefficient-2 inward-free bridge sum gives $2B_i^{\mathrm{ch}}$ copies of the prefix-record profile and $2B_i^{\mathrm{ch}}$ additional copies of $W_{\beta_i,p} T_{i,j}^{\mathrm{long}}$. Combining these with the existing coefficient 3 on the long-run term gives the displayed inequality. □

Corollary 8.280 (Persistent bridge-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.279, define B_i, C_i, B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, and $\widehat{X}_{i,j}^{\mathrm{ret}}$ as there. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P_{i,j}^{\mathrm{pre,per,br,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}(s,p),$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,ch}}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,per,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}}+3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record persistent prefix profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.279. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,ch}}, L_i).$$

This gives

$$P_{i,j}^{\mathrm{pre,rec,br,ch}}(s,p) \le P_{i,j}^{\mathrm{pre,per,br,ch}}(s,p) + R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p),$$

because the return threshold $4K$ in the general split is $8K_i^{\mathrm{gap,ch}}$. Substituting into the coefficient $2B_i^{\mathrm{ch}}$ bridge-record prefix-record term gives the displayed inequality. The new coefficients are fixed by the finite ladder parameters, and the only nonconstant losses remain the existing slab-width factors. □

Corollary 8.281 (Bounded-gap bridge-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.280, define* B_i, C_i, B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{br,ch}} = \left\lceil \frac{64K_i^{\mathrm{gap,ch}}\beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, *write*

$$P_{i,j}^{\mathrm{pre,pair,br,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,ch}},L_i,B_i^{\mathrm{br,ch}}}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,pair,br,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big) \\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr,++}}(s,p) + 5R_{i,j+1}^{\mathrm{spr,++}}(s,p)\big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}}+3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p} \big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap bridge-record prefix-pair profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.280. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,ch}}} = \left\lceil \frac{64K_i^{\mathrm{gap,ch}}\beta_i}{\Theta_i} \right\rceil + 1 = B_i^{\mathrm{br,ch}},$$

the two inequalities give

$$P_{i,j}^{\mathrm{pre,per,br,ch}}(s,p) \le P_{i,j}^{\mathrm{pre,pair,br,ch}}(s,p).$$

Substituting this into the coefficient-$2B_i^{\mathrm{ch}}$ persistent bridge-record term gives the displayed bound. The new cutoff $B_i^{\mathrm{br,ch}}$ is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.282 (Fixed-gap bridge-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.281, define B_i, C_i, B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $B_i^{\mathrm{br,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, $R_{i,j}^{\mathrm{spr,ch}}$, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, and $\widehat{X}_{i,j}^{\mathrm{ret}}$ as there, and put*

$$K_i^{\mathrm{gap,br,ch}} = 8K_i^{\mathrm{gap,ch}} B_i^{\mathrm{br,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,gap,br,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}}+3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of fixed-gap bridge-record prefix-pair profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.281. For each i, j, apply Corollary 8.169 with

$$(\lambda, \sigma, \beta, K, L, B) = (\lambda_i, \sigma_i, \beta_i, 8K_i^{\mathrm{gap,ch}}, L_i, B_i^{\mathrm{br,ch}}).$$

Since $KB = K_i^{\mathrm{gap,br,ch}}$, this gives

$$P_{i,j}^{\mathrm{pre,pair,br,ch}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,ch}}(s,p).$$

Substitute this bound into the coefficient-$2B_i^{\mathrm{ch}}$ bounded-gap bridge-record pair term. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.283 (Confined bridge-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.282, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}}+3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record confined fixed-gap bridge profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.282. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,gap,br,ch}}(s,p) \le P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}(s,p).$$

Substituting into the coefficient-$2B_i^{\mathrm{ch}}$ fixed-gap bridge-record pair sum gives the displayed inequality. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.284 (Inward-free bridge-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.283, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}(s,p)$$

and

$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,ch}}\max\{h-1,1\}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{V}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{V}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}}+3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record inward-free confined bridge profiles, the bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.283. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}(s,p) \le P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}}(s,p) + R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p).$$

Substituting into the coefficient-$2B_i^{\mathrm{ch}}$ bridge-record confined bridge sum gives the displayed inequality. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.285 (Record-absorbed inward-free bridge-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.284, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,rec,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, \big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record prefix-record outward-spread profiles, the bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.284. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,ch}}, L_i).$$

This gives

$$P^{\mathrm{pre,br,free,recbr,ch}}_{i,j,h}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,ch}},L_i,h+1}(s,p) + W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p).$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P^{\mathrm{pre,rec,br,recbr,ch}}_{i,j}$. Substituting these inequalities into the coefficient-$2B_i^{\mathrm{ch}}$ inward-free bridge-record sum gives $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ copies of the prefix-record profile and $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ additional copies of $W_{\beta_i,p}T^{\mathrm{long}}_{i,j}$. Combining these with the existing coefficient $2B_i^{\mathrm{ch}}+3$ on the long-run term gives the displayed inequality. □

Corollary 8.286 (Persistent bridge-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.285, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$R^{\mathrm{spr,br,recbr,ch}}_{i,j,h},\ R^{\mathrm{spr,recbr,ch}}_{i,j},\ R^{\mathrm{spr,br,ch}}_{i,j,h},\ R^{\mathrm{spr,ch}}_{i,j},$$
$$P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h},\ P^{\mathrm{pre,br,sh}}_{i,j,h},\ R^{\mathrm{spr,++}}_{i,j},$$
$$T^{\mathrm{long}}_{i,j},\ S^{\mathrm{ret}}_{i,j},\ \widehat{X}^{\mathrm{ret}}_{i,j}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P^{\mathrm{pre,per,br,recbr,ch}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,br,ch}},L_i}(s,p),$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R^{\mathrm{spr,recbr,recbr,ch}}_{i,j}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,ch}}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,per,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record persistent prefix profiles, the new bridge-record persistent-prefix return-spread profiles, the bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.285. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,ch}}, L_i).$$

This gives

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,ch}}(s,p) \le P_{i,j}^{\mathrm{pre,per,br,recbr,ch}}(s,p) + R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p),$$

because the return threshold $4K$ in the general split is $8K_i^{\mathrm{gap,br,ch}}$. Substituting into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ bridge-record prefix-record term gives the displayed inequality. The new coefficients are fixed by the finite ladder parameters, and the only nonconstant losses remain the existing slab-width factors. □

Corollary 8.287 (Bounded-gap bridge-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.286, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

Put

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{br,recbr,ch}} = \left\lceil \frac{64K_i^{\mathrm{gap,br,ch}}\beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,ch}},L_i,B_i^{\mathrm{br,recbr,ch}}}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap bridge-record-record prefix-pair profiles, the new bridge-record persistent-prefix return-spread profiles, the bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.286. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,br,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,ch}}} = \left\lceil \frac{64K_i^{\mathrm{gap,br,ch}}\beta_i}{\Theta_i} \right\rceil + 1 = B_i^{\mathrm{br,recbr,ch}},$$

the two inequalities give

$$P_{i,j}^{\mathrm{pre,per,br,recbr,ch}}(s,p) \le P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}(s,p).$$

Substituting this into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ persistent bridge-record-record term gives the displayed bound. The new cutoff $B_i^{\mathrm{br,recbr,ch}}$ is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.288 (Fixed-gap bridge-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.287, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\ B_i^{\mathrm{br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

Put

$$K_i^{\mathrm{gap,br,recbr,ch}} = 8K_i^{\mathrm{gap,br,ch}}B_i^{\mathrm{br,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of fixed-gap bridge-record-record prefix-pair profiles, the new bridge-record persistent-prefix return-spread profiles, the bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.287. For each i, j, apply Corollary 8.169 with

$$(\lambda, \sigma, \beta, K, L, B) = (\lambda_i, \sigma_i, \beta_i, 8K_i^{\mathrm{gap,br,ch}}, L_i, B_i^{\mathrm{br,recbr,ch}}).$$

Since $KB = K_i^{\mathrm{gap,br,recbr,ch}}$, this gives

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}(s,p).$$

Substitute this bound into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ bounded-gap bridge-record-record pair term. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.289 (Confined bridge-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.288, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\ B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record confined fixed-gap bridge profiles, the new bridge-record persistent-prefix return-spread profiles, the bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.288. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}(s,p) \le P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,ch}}(s,p).$$

Substituting into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ fixed-gap bridge-record-record pair sum gives the displayed inequality. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.290 (Inward-free bridge-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.289, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\ B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}(s,p)$$

and

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,ch}}\max\{h-1,1\}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record inward-free confined bridge profiles, the new bridge-record-record bridge-return spread profiles, the new bridge-record persistent-prefix return-spread profiles, the earlier bridge-record bridge-return spread profiles, the bridge-record

spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.289. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,ch}}(s,p) \le P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,ch}}(s,p) + R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p).$$

Substituting into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ bridge-record-record confined bridge sum gives the displayed inequality. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.291 (Record-absorbed inward-free bridge-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.290, use the following notation from that corollary:*

$$\begin{gathered}
B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},\\
P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},\\
T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.
\end{gathered}$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$
\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} + 2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record prefix-record outward-spread profiles, the bridge-record-record bridge-return spread profiles, the bridge-record persistent-prefix return-spread

profiles, the earlier bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.290. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,ch}}(s,p) \le & \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,ch}},L_i,h+1}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}(s,p) \\ & + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,ch}}$. Substituting these inequalities into the coefficient-$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}$ inward-free bridge-record-record sum gives $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}$ copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3$ on the long-run term gives the displayed inequality. □

Corollary 8.292 (Persistent bridge-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.291, use the following notation from that corollary:*

$$\begin{gathered} B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}}, \\ B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}}, \\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}, \\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}}, \\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}, \\ P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}}, \\ T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}. \end{gathered}$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,br,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}(s,p),$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,ch}}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned} \mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le & \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\ & + 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,ch}}(s,p) \end{aligned}$$

$$
\begin{aligned}
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7 R_{i,j}^{\mathrm{spr},++}(s,p) + 5 R_{i,j+1}^{\mathrm{spr},++}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2 B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} + 2 B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2 B_i^{\mathrm{ch}} + 3) W_{\beta_i,p} T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p} \big(5 \widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p) \big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record persistent prefix profiles, the new bridge-record-record persistent-prefix return-spread profiles, the bridge-record-record bridge-return spread profiles, the bridge-record persistent-prefix return-spread profiles, the earlier bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-

bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.291. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,ch}}(s,p) \le & P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,ch}}(s,p) \\ & + R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p), \end{aligned}$$

because the return threshold $4K$ in the general split is $8K_i^{\mathrm{gap,br,recbr,ch}}$. Substituting into the coefficient $2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}}$ bridge-record-record prefix-record term gives the displayed inequality. The new coefficients are fixed by the finite ladder parameters, and the only nonconstant losses remain the existing slab-width factors. □

Corollary 8.293 (Bounded-gap bridge-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.292, use the following notation from that corollary:*

$$\begin{gathered} B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}}, \\ B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}, \\ P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}}, \\ T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}. \end{gathered}$$

Put

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{br,recbr,recbr,ch}} = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,ch}},L_i,B_i^{\mathrm{br,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned} \mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le & \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\ & + 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,ch}}(s,p) \\ & + 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\ & + 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3) W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap bridge-record-record-record prefix-pair profiles, the new bridge-record-record persistent-prefix return-spread profiles, the bridge-record-record bridge-return spread profiles, the bridge-record persistent-prefix return-spread profiles, the earlier bridge-record bridge-return spread profiles, the bridge-record spread-return profiles, the chain-depth bridge-return spread profiles, the earlier chain-depth spread-return profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, the earlier confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.292. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,br,recbr,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,ch}}} = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1 = B_i^{\mathrm{br,recbr,recbr,ch}},$$

the two inequalities give

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,ch}}(s,p) \le P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,ch}}(s,p).$$

Substituting this into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}$ persistent bridge-record-record term gives the displayed bound. The new cutoff $B_i^{\mathrm{br,recbr,recbr,ch}}$ is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.294 (Fixed-gap bridge-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.293, use the following notation from that corollary:*

$$\begin{gathered}
B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},\\
P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},\\
T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.
\end{gathered}$$

Put

$$K_i^{\mathrm{gap,br,recbr,recbr,ch}} = 8K_i^{\mathrm{gap,br,recbr,ch}}B_i^{\mathrm{br,recbr,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7 R_{i,j}^{\mathrm{spr},++}(s,p) + 5 R_{i,j+1}^{\mathrm{spr},++}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} (2 B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} + 2 B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} + 2 B_i^{\mathrm{ch}} + 3) W_{\beta_i,p} T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p} \big(5 \widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p) \big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of fixed-gap bridge-record-record-record prefix-pair profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.293. For each i, j, apply Corollary 8.169 with

$$
(\lambda, \sigma, \beta, K, L, B) = (\lambda_i, \sigma_i, \beta_i, 8K_i^{\mathrm{gap,br,recbr,ch}}, L_i, B_i^{\mathrm{br,recbr,recbr,ch}}).
$$

Since $KB = K_i^{\mathrm{gap,br,recbr,recbr,ch}}$, this gives

$$
P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,ch}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,ch}}(s,p).
$$

Substitute this bound into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}$ bounded-gap bridge-record-record-record pair term. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.295 (Confined bridge-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.294, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i,h}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{P}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{P}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\ \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\ \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\ \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\ \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i}P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i}P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr,++}}(s,p)+5R_{i,j+1}^{\mathrm{spr,++}}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}(2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3)W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record confined fixed-gap bridge profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.294. For each i,j,h, apply Corollary 8.171 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,ch}}(s,p)\le P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,ch}}(s,p).$$

Substituting into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}$ fixed-gap bridge-record-record-record pair sum gives the displayed inequality. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.296 (Inward-free bridge-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.295, use the following notation from that corollary:*

$$
\begin{gathered}
B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},\\
P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},\\
T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.
\end{gathered}
$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}(s,p)$$

and

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,ch}}\max\{h-1,1\}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big) \\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)
\end{aligned}$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}(2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3)W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record inward-free confined bridge profiles, the new bridge-record-record-record bridge-return spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.295. For each i,j,h, apply Corollary 8.174 with

$$
(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i).
$$

This gives

$$
P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,ch}}(s,p)\le P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,ch}}(s,p)+R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p).
$$

Substituting into the coefficient $2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}$ bridge-record-record-record confined bridge sum gives the displayed inequality. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.297 (Record-absorbed inward-free bridge-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.296, use the following notation from that corollary:*

$$
\begin{gathered}
B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},\\
P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},\\
T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.
\end{gathered}
$$

For $0\le j<J_{\sigma_i,\beta_i}$, write

$$
P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,ch}}(s,p)=\mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}(s,p).
$$

Also put

$$
\begin{aligned}
A_i^{\mathrm{long,br,recbr,recbr,ch}}:=\;&2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}B_i^{\mathrm{br,recbr,recbr,ch}}\\
&+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3.
\end{aligned}
$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} B_i^{\mathrm{br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)
\end{aligned}$$

$$+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record prefix-record profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.296. For each i,j,h, apply Corollary 8.187 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,2K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i).$$

This gives

$$\begin{aligned}P^{\mathrm{pre,br,free,recbr,recbr,recbr,ch}}_{i,j,h}(s,p)\le&\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i,h+1}(s,p)\\&+W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p).\end{aligned}$$

Since $h+1\ge 2$, Corollary 8.236 with $\ell=h+1$ bounds the chain profile by $P^{\mathrm{pre,rec,br,recbr,recbr,recbr,ch}}_{i,j}$. Substituting these inequalities into the coefficient-$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}$ inward-free bridge-record-record-record sum gives

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}B_i^{\mathrm{br,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T^{\mathrm{long}}_{i,j}$. Combining these with the existing coefficient

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

on the long-run term gives the displayed inequality. □

Corollary 8.298 (Persistent bridge-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.297, use the following notation from that corollary:*

$$\begin{gathered}B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\ B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,recbr,ch}},\\ A_i^{\mathrm{long,br,recbr,recbr,ch}},\\ R^{\mathrm{spr,br,recbr,recbr,recbr,ch}}_{i,j,h},\ R^{\mathrm{spr,recbr,recbr,recbr,ch}}_{i,j},\\ R^{\mathrm{spr,br,recbr,recbr,ch}}_{i,j,h},\ R^{\mathrm{spr,recbr,recbr,ch}}_{i,j},\\ R^{\mathrm{spr,br,recbr,ch}}_{i,j,h},\ R^{\mathrm{spr,recbr,ch}}_{i,j},\\ R^{\mathrm{spr,br,ch}}_{i,j,h},\ R^{\mathrm{spr,ch}}_{i,j},\\ P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h},\ P^{\mathrm{pre,br,sh}}_{i,j,h},\ R^{\mathrm{spr,++}}_{i,j},\\ T^{\mathrm{long}}_{i,j},\ S^{\mathrm{ret}}_{i,j},\ \widehat{X}^{\mathrm{ret}}_{i,j}.\end{gathered}$$

For $0\le j<J_{\sigma_i,\beta_i}$, write

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,ch}}_{i,j}(s,p)=\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i}(s,p),$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$*, write*

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Also put

$$D_i^{\mathrm{rec,br,recbr,recbr,ch}} := 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}B_i^{\mathrm{br,recbr,recbr,ch}}.$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \bigl(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\bigr)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \bigl(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\bigr)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \bigl(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\bigr)\\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record persistent prefix profiles, the new bridge-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.297. For each i, j, apply Corollary 8.163 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,2K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i).$$

This gives

$$
\begin{aligned}
P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,ch}}(s,p)\le{}&P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,ch}}(s,p)\\
&+R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p),
\end{aligned}
$$

because the return threshold $4K$ in the general split is $8K_i^{\mathrm{gap,br,recbr,recbr,ch}}$. Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,ch}}=2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}B_i^{\mathrm{br,recbr,recbr,ch}}$$

bridge-record-record-record prefix-record term gives the displayed inequality. The new coefficients are fixed by the finite ladder parameters, and the only nonconstant losses remain the existing slab-width factors. □

Corollary 8.299 (Bounded-gap bridge-record-record-record-record chain-absorbed finite ladder)**.**

In the setting of Corollary 8.298, use the following notation from that corollary:

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

Put

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,ch}} = \left\lceil \frac{64 K_i^{\mathrm{gap,br,recbr,recbr,ch}} \beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i,B_i^{\mathrm{br,recbr,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{P}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{P}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p) \right)
\end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap bridge-record-record-record-record prefix-pair profiles, the new bridge-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.298. For each i,j, apply Corollary 8.165 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,ch}},L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,ch}}}=\left\lceil\frac{64K_i^{\mathrm{gap,br,recbr,recbr,ch}}\beta_i}{\Theta_i}\right\rceil+1=B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$

the two inequalities give

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,ch}}(s,p)\le P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,ch}}(s,p).$$

Substituting this into the coefficient $D_i^{\mathrm{rec,br,recbr,recbr,ch}}$ persistent bridge-record-record-record term gives the displayed bound. The new cutoff $B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$ is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.300 (Fixed-gap bridge-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.299, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

Put

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}} = 8K_i^{\mathrm{gap,br,recbr,recbr,ch}} B_i^{\mathrm{br,recbr,recbr,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le\ & \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of fixed-gap bridge-record-record-record-record prefix-pair profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.299. For each i, j, apply Corollary 8.169 with

$$(\lambda, \sigma, \beta, K, L, B) = (\lambda_i, \sigma_i, \beta_i, 8K_i^{\mathrm{gap,br,recbr,recbr,ch}}, L_i, B_i^{\mathrm{br,recbr,recbr,recbr,ch}}).$$

Since $KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}$, this gives

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,ch}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,ch}}(s,p).$$

Substitute this bound into the coefficient $D_i^{\mathrm{rec,br,recbr,recbr,ch}}$ bounded-gap bridge-record-record-record-record pair term. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.301 (Confined bridge-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.300, use the following notation from that corollary:*

$$\begin{gathered}
B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\\
A_i^{\mathrm{long,br,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},\\
P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},\\
T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.
\end{gathered}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i,h}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le{}& \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7 R_{i,j}^{\mathrm{spr},++}(s,p) + 5 R_{i,j+1}^{\mathrm{spr},++}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p} \big(5 \widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p) \big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record-record confined fixed-gap bridge profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.300. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,ch}}(s,p) \le P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,ch}}(s,p).$$

Substituting into the coefficient $D_i^{\mathrm{rec,br,recbr,recbr,ch}}$ fixed-gap bridge-record-record-record-record pair sum gives the displayed inequality. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.302 (Inward-free bridge-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.301, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}(s,p)$$

and

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}\,\max\{h-1,1\}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

To keep the displayed finite ladder short, put

$$P_{i,j,h}^{\mathrm{ifr4}} := P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,ch}}, \qquad R_{i,j,h}^{\mathrm{ret4}} := R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}}.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}\mathfrak{P}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{P}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)&\\ + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} & P_{i,j,h}^{\mathrm{ifr4}}(s,p)\\ + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} & R_{i,j,h}^{\mathrm{ret4}}(s,p)\end{aligned}$$

$$
\begin{aligned}
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,recbr,ch}}} R_{i,j,h}^{\text{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,ch}}} R_{i,j,h}^{\text{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\text{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\text{spr,recbr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,ch}}} R_{i,j,h}^{\text{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\text{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\text{spr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{ch}}} R_{i,j,h}^{\text{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\text{spr,ch}}(s,p) + R_{i,j+1}^{\text{spr,ch}}(s,p) \right) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\text{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\text{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(7 R_{i,j}^{\text{spr,++}}(s,p) + 5 R_{i,j+1}^{\text{spr,++}}(s,p) \right) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\text{long,br,recbr,recbr,ch}} W_{\beta_i,p} \, T_{i,j}^{\text{long}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\text{ret}}(s,p)
\end{aligned}
$$

$$+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\right).$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record-record inward-free confined bridge profiles, the new bridge-record-record-record-record bridge-return spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.301. For each i,j,h, apply Corollary 8.174 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i).$$

This gives

$$P^{\mathrm{ifr4}}_{i,j,h}=P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,ch}}_{i,j,h},\qquad R^{\mathrm{ret4}}_{i,j,h}=R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}}_{i,j,h},$$

and

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p)\le P^{\mathrm{ifr4}}_{i,j,h}(s,p)+R^{\mathrm{ret4}}_{i,j,h}(s,p).$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,ch}}$$

confined bridge sum gives the displayed inequality. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.303 (Record-absorbed inward-free bridge-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.302, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$R^{\mathrm{ret4}}_{i,j,h},\ R^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}_{i,j},$$
$$R^{\mathrm{spr,br,recbr,recbr,recbr,ch}}_{i,j,h},\ R^{\mathrm{spr,recbr,recbr,recbr,ch}}_{i,j},$$
$$R^{\mathrm{spr,br,recbr,recbr,ch}}_{i,j,h},\ R^{\mathrm{spr,recbr,recbr,ch}}_{i,j},$$
$$R^{\mathrm{spr,br,recbr,ch}}_{i,j,h},\ R^{\mathrm{spr,recbr,ch}}_{i,j},$$
$$R^{\mathrm{spr,br,ch}}_{i,j,h},\ R^{\mathrm{spr,ch}}_{i,j},$$
$$P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h},\ P^{\mathrm{pre,br,sh}}_{i,j,h},\ R^{\mathrm{spr,++}}_{i,j},$$
$$T^{\mathrm{long}}_{i,j},\ S^{\mathrm{ret}}_{i,j},\ \widehat{X}^{\mathrm{ret}}_{i,j}.$$

For $0\le j<J_{\sigma_i,\beta_i}$, write

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,ch}}_{i,j}(s,p)=\mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i}(s,p).$$

Put

$$P_{i,j}^{\mathrm{rec4}} := P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,ch}}$$

and define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} &:= D_i^{\mathrm{rec,br,recbr,recbr,ch}}\, B_i^{\mathrm{br,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}} &:= A_i^{\mathrm{long,br,recbr,recbr,ch}} + D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned} \mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec4}}(s,p) \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret4}}(s,p) \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) \\ &+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\ &+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\ &+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) \\ &+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big) \\ &+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\ &+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big) \\ &+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i}P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i}P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}}W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record-record prefix-record profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.302. For each i,j,h, apply Corollary 8.187 with

$$
(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i).
$$

This gives

$$
\begin{aligned}
P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,ch}}(s,p)\le&\mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i,h+1}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}(s,p)\\
&+W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p).
\end{aligned}
$$

Since $h+1\ge 2$, Corollary 8.236 with $\ell=h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec4}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$
D_i^{\mathrm{rec,br,recbr,recbr,ch}}
$$

gives $D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}$ copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$
A_i^{\mathrm{long,br,recbr,recbr,ch}}
$$

gives the displayed inequality. □

Corollary 8.304 (Persistent bridge-record-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.303, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}(s,p),$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Put

$$P_{i,j}^{\mathrm{per4}} := P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,ch}}, \qquad R_{i,j}^{\mathrm{spr5}} := R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per4}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret4}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr,++}}(s,p) + 5R_{i,j+1}^{\mathrm{spr,++}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record-record-record persistent prefix profiles, the new bridge-record-record-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.303. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec4}}(s,p) \le & P_{i,j}^{\mathrm{per4}}(s,p) \\ & + R_{i,j}^{\mathrm{spr5}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr5}}(s,p), \end{aligned}$$

because the return threshold $4K$ in the general split is $8K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}$. Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}$$

bridge-record-record-record-record prefix-record term gives the displayed inequality. The new coefficients are fixed by the finite ladder parameters, and the only nonconstant losses remain the existing slab-width factors. □

Corollary 8.305 (Bounded-gap bridge-record-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.304, use the following notation from that corollary:*

$$\begin{gathered} B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}}, \\ B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}}, \\ K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}}, \\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}, \\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},\ P_{i,j}^{\mathrm{per4}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}}, \\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}, \\ P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}}, \\ T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}. \end{gathered}$$

Put

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}} = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair5}} := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,ch}},$$

and define this common profile by

$$P_{i,j}^{\mathrm{pair5}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i,B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{P}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{P}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret4}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \bigl(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\bigr)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \bigl(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\bigr)
\end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap bridge-record-record-record-record-record prefix-pair profiles, the new bridge-record-record-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.304. For each i, j, apply Corollary 8.165 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}}=\left\lceil\frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i}\right\rceil+1=B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$

the two inequalities give

$$P_{i,j}^{\mathrm{per4}}(s,p)\le P_{i,j}^{\mathrm{pair5}}(s,p).$$

Substituting this into the coefficient $D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}$ persistent bridge-record-record-record-record term gives the displayed bound. The new cutoff $B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$ is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.306 (Fixed-gap bridge-record-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.305, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\ P_{i,j}^{\mathrm{pair5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

Put

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}} = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{gap5}} := P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,ch}},$$

and define this common profile by

$$P_{i,j,h}^{\mathrm{gap5}}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i,h}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\begin{aligned}
\mathfrak{P}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{P}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret4}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of fixed-gap bridge-record-record-record-record-record prefix-pair profiles, the new bridge-record-record-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.305. For each i, j, apply Corollary 8.169 with

$$(\lambda, \sigma, \beta, K, L, B) = (\lambda_i, \sigma_i, \beta_i, 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}, L_i, B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}).$$

Since $KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}$, this gives

$$P_{i,j}^{\mathrm{pair5}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap5}}(s,p).$$

Substitute this bound into the coefficient $D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}$ bounded-gap bridge-record-record-record-record-record pair term. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.307 (Confined bridge-record-record-record-record-record chain-absorbed finite ladder)**.** *In the setting of Corollary 8.306, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{gap5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$, write

$$P_{i,j,h}^{\mathrm{br5}} := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

and define this common profile by

$$P_{i,j,h}^{\mathrm{br5}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$
\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret4}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record-record-record confined fixed-gap bridge profiles, the new bridge-record-record-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.306. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{gap5}}(s,p) \le P_{i,j,h}^{\mathrm{br5}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient $D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}$ gives the displayed inequality. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.308 (Inward-free bridge-record-record-record-record-record chain-absorbed finite lad-

der). *In the setting of Corollary 8.307, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{br5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$, *write*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i,h}(s,p)$$

and

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}\,\max\{h-1,1\}}(s,p).$$

To keep the displayed finite ladder short, put

$$P_{i,j,h}^{\mathrm{ifr5}} := P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad R_{i,j,h}^{\mathrm{ret5}} := R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{ifr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\text{ret4}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,recbr,ch}}} R_{i,j,h}^{\text{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,ch}}} R_{i,j,h}^{\text{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\text{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\text{spr,recbr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,ch}}} R_{i,j,h}^{\text{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\text{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\text{spr,recbr,ch}}(s,p) \big) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{ch}}} R_{i,j,h}^{\text{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\text{spr,ch}}(s,p) + R_{i,j+1}^{\text{spr,ch}}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\text{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\text{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7 R_{i,j}^{\text{spr,++}}(s,p) + 5 R_{i,j+1}^{\text{spr,++}}(s,p) \big) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\text{long,br,recbr,recbr,recbr,ch}} W_{\beta_i,p} \, T_{i,j}^{\text{long}}(s,p)
\end{aligned}
$$

$$+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)$$
$$+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record-record-record inward-free confined bridge profiles, the new bridge-record-record-record-record-record bridge-return spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.307. For each i,j,h, apply Corollary 8.174 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{ifr5}}=P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,ch}},\qquad R_{i,j,h}^{\mathrm{ret5}}=R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

and

$$P_{i,j,h}^{\mathrm{br5}}(s,p)\le P_{i,j,h}^{\mathrm{ifr5}}(s,p)+R_{i,j,h}^{\mathrm{ret5}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}$$

gives the displayed inequality. The added h-sum is finite and independent of p, and no new non-constant multiplicative loss is introduced. □

Corollary 8.309 (Fifth record-absorbed inward-free bridge finite ladder)**.** *In the setting of Corollary 8.308, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{ret5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}(s,p).$$

Put

$$P_{i,j}^{\mathrm{rec5}} := P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

and define the fixed coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} := D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}\, B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}} := A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}} + D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret4}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record-record-record prefix-record profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.308. For each i, j, h, apply Corollary 8.187 with

$$
(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}, L_i).
$$

This gives

$$
\begin{aligned}
P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \le{}& \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i,h+1}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}(s,p)\\
&+ W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p).
\end{aligned}
$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec5}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}$$

gives $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}$ copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p} T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}}$$

gives the displayed inequality. □

Corollary 8.310 (Sixth persistent bridge finite ladder)**.** *In the setting of Corollary 8.309, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{ret5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}(s,p),$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,ret,spr}}(s,p).$$

Put

$$P_{i,j}^{\mathrm{per5}} := P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad R_{i,j}^{\mathrm{spr6}} := R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)$$

$$
\begin{aligned}
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\text{per5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\text{ret5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\text{ret4}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\text{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,recbr,ch}}} R_{i,j,h}^{\text{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\text{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} B_i^{\text{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\text{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,recbr,ch}}} R_{i,j,h}^{\text{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} B_i^{\text{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\text{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\text{spr,recbr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\text{br,ch}}} R_{i,j,h}^{\text{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\text{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\text{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\text{spr,recbr,ch}}(s,p) \right)
\end{aligned}
$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record-record-record-record persistent prefix profiles, the new bridge-record-record-record-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.309. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$
\begin{aligned}
P_{i,j}^{\mathrm{rec5}}(s,p) \le & P_{i,j}^{\mathrm{per5}}(s,p)\\
&+ R_{i,j}^{\mathrm{spr6}}(s,p)\\
&+ R_{i,j+1}^{\mathrm{spr6}}(s,p),
\end{aligned}
$$

because the return threshold $4K$ in the general split is $8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}$. Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}$$

bridge-record-record-record-record-record prefix-record term gives the displayed inequality. The new coefficients are fixed by the finite ladder parameters, and the only nonconstant losses remain the existing slab-width factors. □

Corollary 8.311 (Sixth bounded-gap bridge finite ladder)**.** *In the setting of Corollary 8.310, use the following notation from that corollary:*

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{per5}},\ R_{i,j}^{\mathrm{spr6}},\ R_{i,j,h}^{\mathrm{ret5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

Put

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}} = \left\lceil \frac{64 K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}} \beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair6}} := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

and define this common profile by

$$P_{i,j}^{\mathrm{pair6}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i,B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{P}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{P}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr6}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret4}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p) \right) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap bridge-record-record-record-record-record-record prefix-pair profiles, the new bridge-record-record-record-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.310. For each i, j, apply Corollary 8.165 with

$$(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$
\begin{aligned}
B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}} &= \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i}\right\rceil+1\\
&= B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}},
\end{aligned}
$$

the two inequalities give

$$P_{i,j}^{\mathrm{per5}}(s,p)\le P_{i,j}^{\mathrm{pair6}}(s,p).$$

Substituting this into the coefficient $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}$ persistent bridge-record-record-record-record-record term gives the displayed bound. The new cutoff $B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$ is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.312 (Sixth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.311, use the*

following notation from that corollary:

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pair6}},\ R_{i,j}^{\mathrm{spr6}},\ R_{i,j,h}^{\mathrm{ret5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

Put

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}} = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{gap6}} := P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

and define this common profile by

$$P_{i,j,h}^{\mathrm{gap6}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le\ & \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret5}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret4}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p) \right) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\left(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of fixed-gap bridge-record-record-record-record-record-record prefix-pair profiles, the new bridge-record-record-record-record-record-record persistent-prefix return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.311. For each i,j, apply Corollary 8.169 with

$$
(\lambda,\sigma,\beta,K,L,B)=(\lambda_i,\sigma_i,\beta_i,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},L_i,B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}).
$$

Since $KB=K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}$, this gives

$$
P_{i,j}^{\mathrm{pair6}}(s,p)\le\sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}}P_{i,j,h}^{\mathrm{gap6}}(s,p).
$$

Substitute this bound into the coefficient $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}$ bounded-gap bridge-record-record-record-record-record-record pair term. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.313 (Sixth confined bridge finite ladder)**.** *In the setting of Corollary 8.312, use the*

following notation from that corollary:

$$B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{gap6}},\ R_{i,j}^{\mathrm{spr6}},\ R_{i,j,h}^{\mathrm{ret5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},$$
$$T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{br6}} := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

and define this common profile by

$$P_{i,j,h}^{\mathrm{br6}}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\begin{aligned}
\mathfrak{P}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le\ & \mathfrak{P}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr6}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret4}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p) \right) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p) \\
&+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(R_{i,j}^{\mathrm{spr,ch}}(s,p) + R_{i,j+1}^{\mathrm{spr,ch}}(s,p) \right) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p) \\
&+ \sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(7 R_{i,j}^{\mathrm{spr,++}}(s,p) + 5 R_{i,j+1}^{\mathrm{spr,++}}(s,p) \right)
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record-record-record-record confined fixed-gap bridge profiles, the new bridge-record-record-record-record-record-record adjacent return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.312. For each i,j,h, apply Corollary 8.171 with

$$
(\lambda,\sigma,\beta,K,L)=(\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i).
$$

This gives

$$
P_{i,j,h}^{\mathrm{gap6}}(s,p)\le P_{i,j,h}^{\mathrm{br6}}(s,p).
$$

Substituting this bound in the fixed-gap sum with coefficient $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}$ gives the displayed inequality. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.314 (Sixth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.313, use the following notation from that corollary:*

$$
\begin{gathered}
B_i,\ C_i,\ B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},\\
A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},\\
D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\ D_i^{\mathrm{rec,br,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{br6}},\ R_{i,j}^{\mathrm{spr6}},\\
R_{i,j,h}^{\mathrm{ret5}},\ R_{i,j}^{\mathrm{spr5}},\ R_{i,j,h}^{\mathrm{ret4}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}},\\
P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\ P_{i,j,h}^{\mathrm{pre,br,sh}},\ R_{i,j}^{\mathrm{spr,++}},\\
T_{i,j}^{\mathrm{long}},\ S_{i,j}^{\mathrm{ret}},\ \widehat{X}_{i,j}^{\mathrm{ret}}.
\end{gathered}
$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$*, write*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h}(s,p)$$

and

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) = \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}\max\{h-1,1\}}}(s,p).$$

To keep the displayed finite ladder short, put

$$P_{i,j,h}^{\mathrm{ifr6}} := P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad R_{i,j,h}^{\mathrm{ret6}} := R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*,*

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{ifr6}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret6}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr6}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr6}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr5}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret4}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+ 2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}}(s,p)
\end{aligned}$$

$$
\begin{aligned}
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} B_i^{\mathrm{br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} B_i^{\mathrm{br,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,ch}}} R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} B_i^{\mathrm{ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,recbr,ch}}(s,p)\big)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{ch}}} R_{i,j,h}^{\mathrm{spr,br,ch}}(s,p)\\
&+2\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(R_{i,j}^{\mathrm{spr,ch}}(s,p)+R_{i,j+1}^{\mathrm{spr,ch}}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R_{i,j}^{\mathrm{spr},++}(s,p)+5R_{i,j+1}^{\mathrm{spr},++}(s,p)\big)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+\sum_{i=0}^{N-1} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p)+\widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\big).
\end{aligned}
$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite family of bridge-record-record-record-record-record-record inward-free confined bridge profiles, the new bridge-record-record-record-record-record-record bridge-return spread profiles, the adjacent return-spread profiles, the retained return-spread and bridge-return profiles named in the display, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.313. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{ifr6}} = P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad R_{i,j,h}^{\mathrm{ret6}} = R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

and

$$P_{i,j,h}^{\mathrm{br6}}(s,p) \le P_{i,j,h}^{\mathrm{ifr6}}(s,p) + R_{i,j,h}^{\mathrm{ret6}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed inequality. The added h-sum is finite and independent of p, and no new non-constant multiplicative loss is introduced. □

Corollary 8.315 (Sixth record-absorbed inward-free bridge finite ladder)**.** *In the setting of Corollary 8.314, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{rec6}}(s,p) &:= P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} &:= D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}\, B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}} &:= A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}} + D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.314 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr6}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec6}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the bridge-record-record-record-record-record-record bridge-return spread family $R_{i,j,h}^{\mathrm{ret6}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr6}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record-record-record-record prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.314. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} &P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &\quad \le \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}(s,p) + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec6}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}}$$

gives $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$ copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.316 (Seventh persistent bridge finite ladder)**.** *In the setting of Corollary 8.315, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per6}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr7}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}}^{G\to H,\mathrm{rad,ret,spr}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.315 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec6}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per6}} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr7}} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr7}}. \end{aligned}$$

All other terms from that finite ladder, including the bridge-record-record-record-record-record-record bridge-return spread family $R^{\mathrm{ret6}}_{i,j,h}$, the adjacent return-spread family $R^{\mathrm{spr6}}_{i,j}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bridge-record-record-record-record-record-record-record persistent prefix profiles, the new bridge-record-record-record-record-record-record-record return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.315. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} P^{\mathrm{rec6}}_{i,j}(s,p) \le & P^{\mathrm{per6}}_{i,j}(s,p) \\ & + R^{\mathrm{spr7}}_{i,j}(s,p) \\ & + R^{\mathrm{spr7}}_{i,j+1}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}} \quad \text{and} \quad 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.317 (Seventh bounded-gap bridge finite ladder)**.** *In the setting of Corollary 8.316, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} P^{\mathrm{pair7}}_{i,j}(s,p) &:= P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},L_i,B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.316 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{per6}}_{i,j}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{pair7}}_{i,j}.$$

All other terms from that finite ladder, including the new adjacent return-spread family $R_{i,j}^{\mathrm{spr7}}$, the bridge-record-record-record-record-record-record bridge-return spread family $R_{i,j,h}^{\mathrm{ret6}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr6}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the bounded-gap bridge-record-record-record-record-record-record-record prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.316. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}} &= \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1 \\ &= B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per6}}(s,p) \le P_{i,j}^{\mathrm{pair7}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.318 (Seventh fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.317, use the notation from that corollary. Put*

$$\begin{aligned} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &= 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$, write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap7}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.317 remains valid after replacing the bounded-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair7}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap7}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr7}}$, the bridge-record-record-record-record-record- record bridge-return spread family $R_{i,j,h}^{\mathrm{ret6}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr6}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite fixed-gap bridge-record-record-record-record-record-record-record prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.317. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair7}}(s,p) \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap7}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.319 (Seventh confined bridge finite ladder)**.** *In the setting of Corollary 8.318, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$, write*

$$P_{i,j,h}^{\mathrm{br7}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.318 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap7}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br7}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr7}}_{i,j}$, the bridge-record-record-record-record-record- record bridge-return spread family $R^{\mathrm{ret6}}_{i,j,h}$, the adjacent return-spread family $R^{\mathrm{spr6}}_{i,j}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite bridge-record-record-record-record-record-record confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.318. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P^{\mathrm{gap7}}_{i,j,h}(s,p) \le P^{\mathrm{br7}}_{i,j,h}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.320 (Seventh inward-free bridge finite ladder)**.** *In the setting of Corollary 8.319, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$, write*

$$\begin{aligned} P^{\mathrm{ifr7}}_{i,j,h}(s,p) &:= P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h}(s,p), \end{aligned}$$

and

$$\begin{aligned} R^{\mathrm{ret7}}_{i,j,h}(s,p) &:= R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\max\{h-1,1\}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.319 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P^{\mathrm{br7}}_{i,j,h}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P^{\mathrm{ifr7}}_{i,j,h} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} R^{\mathrm{ret7}}_{i,j,h}. \end{aligned}$$

All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr7}}_{i,j}$, the bridge-record-record-record-record-record- record bridge-return spread family $R^{\mathrm{ret6}}_{i,j,h}$, the adjacent return-spread family $R^{\mathrm{spr6}}_{i,j}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite bridge-record-record-record-record-record-record-record inward-free bridge profiles, the new bridge-record-record-record-record-record-record-record bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.319. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P^{\mathrm{br7}}_{i,j,h}(s,p) \le P^{\mathrm{ifr7}}_{i,j,h}(s,p) + R^{\mathrm{ret7}}_{i,j,h}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.321 (Seventh record-absorbed finite ladder)**.** *In the setting of Corollary 8.320, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P^{\mathrm{rec7}}_{i,j}(s,p) &:= P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad + D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.320 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{ifr7}}_{i,j,h}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{rec7}}_{i,j},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the seventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret7}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr7}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the seventh prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.320. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} &P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &\quad \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}(s,p) + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec7}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$ copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.322 (Eighth persistent finite ladder)**.** *In the setting of Corollary 8.321, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per7}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr8}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{Q,\lambda_i,\sigma_i,\beta_i,j,8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.321 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec7}}$$

by

$$\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per7}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr8}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr8}}.
\end{aligned}$$

All other terms from that finite ladder, including the seventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret7}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr7}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighth persistent prefix profiles, the new eighth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.321. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned}
P_{i,j}^{\mathrm{rec7}}(s,p) \le\ & P_{i,j}^{\mathrm{per7}}(s,p) \\
&+ R_{i,j}^{\mathrm{spr8}}(s,p) \\
&+ R_{i,j+1}^{\mathrm{spr8}}(s,p),
\end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \quad \text{and} \quad 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.323 (Eighth bounded-gap finite ladder)**.** *In the setting of Corollary 8.322, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} = \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} P_{i,j}^{\mathrm{pair8}}(s,p) &:= P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.322 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per7}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair8}}.$$

All other terms from that finite ladder, including the new adjacent return-spread family $R_{i,j}^{\mathrm{spr8}}$, the seventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret7}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.322. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &\quad = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1 \\ &\quad = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per7}}(s,p) \le P_{i,j}^{\mathrm{pair8}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.324 (Eighth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.323, use the notation from that corollary. Put*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$*, write*

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap8}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.323 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair8}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap8}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr8}}$*, the seventh bridge-return spread family* $R_{i,j,h}^{\mathrm{ret7}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite fixed-gap bridge-record-record-record-record-record-record-record-record prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.323. For each i, j, apply Corollary 8.169 with

$$\begin{aligned} &\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i, \\ &K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ &B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair8}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap8}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.325 (Eighth confined bridge finite ladder)**.** *In the setting of Corollary 8.324, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$, *write*

$$\begin{aligned} P_{i,j,h}^{\mathrm{br8}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$, *the finite ladder of Corollary 8.324 remains valid after replacing the fixed-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap8}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br8}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr8}}$, *the seventh bridge-return spread family* $R_{i,j,h}^{\mathrm{ret7}}$, *all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite bridge-record-record-record-record-record-record-record-record confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.324. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{gap8}}(s,p) \le P_{i,j,h}^{\mathrm{br8}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.326 (Eighth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.325, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$, *write*

$$\begin{aligned} P_{i,j,h}^{\mathrm{ifr8}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p), \end{aligned}$$

and

$$R^{\mathrm{ret8}}_{i,j,h}(s,p) := R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p)$$
$$:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\max\{h-1,1\}}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.325 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{br8}}_{i,j,h}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{ifr8}}_{i,j,h}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R^{\mathrm{ret8}}_{i,j,h}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr8}}_{i,j}$, the seventh bridge-return spread family $R^{\mathrm{ret7}}_{i,j,h}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite bridge-record-record-record-record-record-record-record-record inward-free bridge profiles, the new bridge-record-record-record-record-record-record- record-record bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.325. For each i,j,h, apply Corollary 8.174 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

This gives

$$P^{\mathrm{br8}}_{i,j,h}(s,p) \le P^{\mathrm{ifr8}}_{i,j,h}(s,p) + R^{\mathrm{ret8}}_{i,j,h}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.327 (Eighth record-absorbed finite ladder)**.** *In the setting of Corollary 8.326, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P^{\mathrm{rec8}}_{i,j}(s,p) := P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p).$$

Define the fixed coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} + D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then the finite ladder of Corollary 8.326 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr8}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec8}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the eighth bridge-return spread family $R_{i,j,h}^{\mathrm{ret8}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr8}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.326. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}(s,p) + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p).$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec8}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$ copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.328 (Ninth persistent finite ladder)**.** *In the setting of Corollary 8.327, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per8}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr9}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.327 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec8}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per8}} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr9}} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr9}}. \end{aligned}$$

All other terms from that finite ladder, including the eighth bridge-return spread family $R_{i,j,h}^{\mathrm{ret8}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr8}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the ninth persistent prefix profiles, the new ninth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.327. For each i, j, apply Corollary 8.163 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec8}}(s,p) \le\ & P_{i,j}^{\mathrm{per8}}(s,p) \\ & + R_{i,j}^{\mathrm{spr9}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr9}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \quad \text{and} \quad 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.329 (Ninth bounded-gap finite ladder)**.** *In the setting of Corollary 8.328, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} = \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} P_{i,j}^{\mathrm{pair9}}(s,p) &:= P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.328 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per8}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair9}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr9}}$, the eighth bridge-return spread family $R_{i,j,h}^{\mathrm{ret8}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the ninth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.328. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &\quad = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1 \\ &\quad = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per8}}(s,p) \le P_{i,j}^{\mathrm{pair9}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.330 (Ninth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.329, use the notation from that corollary. Put*

$$\begin{aligned} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &= 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap9}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.329 remains valid after replacing the bounded-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair9}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap9}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr9}}$, the eighth bridge-return spread family $R_{i,j,h}^{\mathrm{ret8}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and

exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite fixed-gap bridge-record-record-record-record-record-record-record-record-record prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.329. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair9}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap9}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.331 (Ninth confined bridge finite ladder)**.** *In the setting of Corollary 8.330, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{br9}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.330 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap9}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{br9}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr9}}_{i,j}$, the eighth bridge-return spread family $R^{\mathrm{ret8}}_{i,j,h}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite bridge-record-record-record-record-record-record-record-record-record confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.330. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P^{\mathrm{gap9}}_{i,j,h}(s,p) \le P^{\mathrm{br9}}_{i,j,h}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.332 (Ninth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.331, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P^{\mathrm{ifr9}}_{i,j,h}(s,p) &:= P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p), \end{aligned}$$

and

$$\begin{aligned} R^{\mathrm{ret9}}_{i,j,h}(s,p) &:= R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ \max\{h-1,1\}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.331 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h} P^{\mathrm{br9}}_{i,j,h}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h} P^{\mathrm{ifr9}}_{i,j,h} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h} R^{\mathrm{ret9}}_{i,j,h}. \end{aligned}$$

Here each $\sum_h$ is over the h-range displayed above. All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr9}}_{i,j}$, the eighth bridge-return spread family $R^{\mathrm{ret8}}_{i,j,h}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the finite bridge-record-record-record-record-record-record-record-record-record inward-free bridge profiles, the new bridge-record-record-record-record- record-record-record-record-record bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.331. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P^{\mathrm{br9}}_{i,j,h}(s,p) \le P^{\mathrm{ifr9}}_{i,j,h}(s,p) + R^{\mathrm{ret9}}_{i,j,h}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.333 (Ninth record-absorbed finite ladder)**.** *In the setting of Corollary 8.332, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P^{\mathrm{rec9}}_{i,j}(s,p) &:= P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad + D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.332 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_h P^{\mathrm{ifr9}}_{i,j,h}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{rec9}}_{i,j},$$

where $\sum_h$ is over the h-range in Corollary 8.332, and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the ninth bridge-return spread family $R_{i,j,h}^{\mathrm{ret9}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr9}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the ninth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.332. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} &P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &\quad \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}}(s,p) \\ &\qquad + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec9}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives $D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$ copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.334 (Tenth persistent finite ladder)**.** *In the setting of Corollary 8.333, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per9}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr10}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_c(H)$, the finite ladder of Corollary 8.333 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec9}}$$

by

$$\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per9}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr10}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr10}}.
\end{aligned}$$

All other terms from that finite ladder, including the ninth bridge-return spread family $R_{i,j,h}^{\mathrm{ret9}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr9}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the tenth persistent prefix profiles, the new tenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.333. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned}
P_{i,j}^{\mathrm{rec9}}(s,p) \le\ & P_{i,j}^{\mathrm{per9}}(s,p) \\
&+ R_{i,j}^{\mathrm{spr10}}(s,p) \\
&+ R_{i,j+1}^{\mathrm{spr10}}(s,p),
\end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$\begin{aligned}
&4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\
&8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.
\end{aligned}$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.335 (Tenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.334, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair10}}(s,p) := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) := \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.334 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per9}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair10}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr10}}$, the ninth bridge-return spread family $R_{i,j,h}^{\mathrm{ret9}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the tenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.334. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} &= \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1 \\ &= B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per9}}(s,p) \le P_{i,j}^{\mathrm{pair10}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.336 (Tenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.335, use the notation from that corollary. Put*

$$\begin{aligned} &K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &:= \\ &8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap10}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{\substack{G\to H,\mathrm{rad,tan,pre,out,per,gap,short} \\ Q,\lambda_i,\sigma_i,\beta_i,j, \\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.335 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair10}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap10}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr10}}$*, the ninth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret9}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the tenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.335. For each i, j, apply Corollary 8.169 with

$$\begin{aligned} \lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i, \\ K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair10}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap10}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.337 (Tenth confined bridge finite ladder)**.** *In the setting of Corollary 8.336, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{br10}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{\substack{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short} \\ Q,\lambda_i,\sigma_i,\beta_i,j, \\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.336 remains valid after replacing the fixed-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap10}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{br10}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr10}}$*, the ninth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret9}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the tenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.336. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{gap10}}(s,p) \le P_{i,j,h}^{\mathrm{br10}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.338 (Tenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.337, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{ifr10}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p), \end{aligned}$$

and

$$\begin{aligned} R_{i,j,h}^{\mathrm{ret10}}(s,p) &:= R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\\ \max\{h-1,1\}}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.337 remains valid after replacing the confined bridge contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{br10}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{ifr10}} \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret10}}. \end{aligned}$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr10}}_{i,j}$, the ninth bridge-return spread family $R^{\mathrm{ret9}}_{i,j,h}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the tenth inward-free bridge profiles, the new tenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.337. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P^{\mathrm{br10}}_{i,j,h}(s,p) \le P^{\mathrm{ifr10}}_{i,j,h}(s,p) + R^{\mathrm{ret10}}_{i,j,h}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.339 (Tenth record-absorbed finite ladder)**.** *In the setting of Corollary 8.338, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P^{\mathrm{rec10}}_{i,j}(s,p) &:= P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad + \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.338 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{ifr10}}_{i,j,h}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec10}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr10}}$, the ninth bridge-return spread family $R_{i,j,h}^{\mathrm{ret9}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the tenth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.338. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} &P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &\quad \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}(s,p) \\ &\qquad + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec10}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p} T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.340 (Eleventh persistent finite ladder)**.** *In the setting of Corollary 8.339, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per10}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,\ 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, *write*

$$\begin{aligned} R_{i,j}^{\mathrm{spr11}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\,Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$, *the finite ladder of Corollary 8.339 remains valid after replacing the prefix-record contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec10}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per10}} \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr11}} \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr11}}. \end{aligned}$$

All other terms from that finite ladder, including the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, *the adjacent return-spread family* $R_{i,j}^{\mathrm{spr10}}$, *the ninth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret9}}$, *all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eleventh persistent prefix profiles, the new eleventh return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.339. For each i, j, apply Corollary 8.163 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec10}}(s,p) \le\ & P_{i,j}^{\mathrm{per10}}(s,p) \\ & + R_{i,j}^{\mathrm{spr11}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr11}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$\begin{aligned} & 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ & 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.341 (Eleventh bounded-gap finite ladder)**.** *In the setting of Corollary 8.340, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$\begin{aligned} & B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ & \quad := \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} P_{i,j}^{\mathrm{pair11}}(s,p) &:= P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.340 remains valid after replacing the persistent-prefix contribution

$$\begin{aligned} & \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ & \quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per10}} \end{aligned}$$

by

$$\begin{aligned} & \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ & \quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair11}}. \end{aligned}$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr10}}$, all retained older

return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eleventh bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.340. For each i, j, apply Corollary 8.165 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &\quad = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1 \\ &\quad = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per10}}(s,p) \le P_{i,j}^{\mathrm{pair11}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.342 (Eleventh fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.341, use the notation from that corollary. Put*

$$\begin{aligned} &K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad := \\ &\quad 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap11}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.341 remains valid after replacing the bounded-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair11}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap11}}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent return-spread family $R_{i,j}^{\mathrm{spr10}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eleventh fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.341. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair11}}(s,p) \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap11}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.343 (Eleventh confined bridge finite ladder)**.** *In the setting of Corollary 8.342, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P^{\mathrm{br11}}_{i,j,h}(s,p) := P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.342 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{gap11}}_{i,j,h}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{br11}}_{i,j,h}.$$

All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr11}}_{i,j}$, the tenth bridge-return spread family $R^{\mathrm{ret10}}_{i,j,h}$, the adjacent tenth return-spread family $R^{\mathrm{spr10}}_{i,j}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eleventh confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.342. For each i, j, h, apply Corollary 8.171 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

This gives

$$P^{\mathrm{gap11}}_{i,j,h}(s,p) \le P^{\mathrm{br11}}_{i,j,h}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.344 (Eleventh inward-free bridge finite ladder)**.** *In the setting of Corollary 8.343, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P^{\mathrm{ifr11}}_{i,j,h}(s,p) := P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p),$$

and

$$R^{\mathrm{ret11}}_{i,j,h}(s,p) := R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}(s,p)$$
$$:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\\ \max\{h-1,1\}}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.343 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{br11}}_{i,j,h}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{ifr11}}_{i,j,h}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R^{\mathrm{ret11}}_{i,j,h}.$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent return-spread family $R^{\mathrm{spr11}}_{i,j}$, the tenth bridge-return spread family $R^{\mathrm{ret10}}_{i,j,h}$, the adjacent tenth return-spread family $R^{\mathrm{spr10}}_{i,j}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eleventh inward-free bridge profiles, the new eleventh bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.343. For each i,j,h, apply Corollary 8.174 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

This gives

$$P^{\mathrm{br11}}_{i,j,h}(s,p) \le P^{\mathrm{ifr11}}_{i,j,h}(s,p) + R^{\mathrm{ret11}}_{i,j,h}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.345 (Eleventh record-absorbed finite ladder)**.** *In the setting of Corollary 8.344, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P^{\mathrm{rec11}}_{i,j}(s,p) &:= P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad + \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.344 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P^{\mathrm{ifr11}}_{i,j,h}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P^{\mathrm{rec11}}_{i,j},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the eleventh bridge-return spread family $R^{\mathrm{ret11}}_{i,j,h}$, the adjacent return-spread family $R^{\mathrm{spr11}}_{i,j}$, the tenth bridge-return spread family $R^{\mathrm{ret10}}_{i,j,h}$, the adjacent tenth return-spread family $R^{\mathrm{spr10}}_{i,j}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles,

are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eleventh prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.344. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} &P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &\quad \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}}(s,p) \\ &\qquad + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec11}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.346 (Twelfth persistent finite ladder)**.** *In the setting of Corollary 8.345, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per11}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr12}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.345 remains valid after replacing the prefix-record contribution

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec11}} \end{aligned}$$

by

$$
\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\
&\quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per11}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\
&\quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr12}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\
&\quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr12}}.
\end{aligned}
$$

All other terms from that finite ladder, including the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent tenth return-spread family $R_{i,j}^{\mathrm{spr10}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the twelfth persistent prefix profiles, the new twelfth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.345. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$
\begin{aligned}
P_{i,j}^{\mathrm{rec11}}(s,p) \le\ & P_{i,j}^{\mathrm{per11}}(s,p) \\
& + R_{i,j}^{\mathrm{spr12}}(s,p) \\
& + R_{i,j+1}^{\mathrm{spr12}}(s,p),
\end{aligned}
$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$
\begin{aligned}
&4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\
&8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.
\end{aligned}
$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.347 (Twelfth bounded-gap finite ladder)**.** *In the setting of Corollary 8.346, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair12}}(s,p) := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) := \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.346 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per11}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair12}}.$$

All other terms from that finite ladder, including the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent tenth return-spread family $R_{i,j}^{\mathrm{spr10}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the twelfth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.346. For each i, j, apply Corollary 8.165 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction

doubles K, and

$$\begin{aligned}
&B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\
&\quad = \left\lceil \frac{64K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\beta_i}{\Theta_i} \right\rceil + 1 \\
&\quad = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},
\end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per11}}(s,p) \le P_{i,j}^{\mathrm{pair12}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.348 (Twelfth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.347, use the notation from that corollary. Put*

$$\begin{aligned}
&K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\
&\quad := \\
&\quad 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\
&\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.
\end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned}
P_{i,j,h}^{\mathrm{gap12}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\
&:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p).
\end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.347 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\
&\quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair12}}
\end{aligned}$$

by

$$\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\
&\quad \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap12}}.
\end{aligned}$$

All other terms from that finite ladder, including the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent tenth return-spread family $R_{i,j}^{\mathrm{spr10}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the twelfth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.347. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair12}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap12}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.349 (Twelfth confined bridge finite ladder)**.** *In the setting of Corollary 8.348, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{br12}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{\substack{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short} \\ Q,\lambda_i,\sigma_i,\beta_i,j, \\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ L_i,h}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.348 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap12}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br12}}.$$

All other terms from that finite ladder, including the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent tenth return-spread family $R_{i,j}^{\mathrm{spr10}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the twelfth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.348. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{gap12}}(s,p) \le P_{i,j,h}^{\mathrm{br12}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, so the exponent conclusion is unchanged. □

Corollary 8.350 (Twelfth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.349, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{ifr12}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p), \end{aligned}$$

and

$$\begin{aligned} R_{i,j,h}^{\mathrm{ret12}}(s,p) &:= R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ Q,\lambda_i,\sigma_i,\beta_i,j,\ \max\{h-1,1\}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.349 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br12}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr12}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret12}}.$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, the adjacent tenth return-spread family $R_{i,j}^{\mathrm{spr10}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the twelfth inward-free bridge profiles, the new twelfth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.349. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{br12}}(s,p) \le P_{i,j,h}^{\mathrm{ifr12}}(s,p) + R_{i,j,h}^{\mathrm{ret12}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.351 (Twelfth record-absorbed finite ladder)**.** *In the setting of Corollary 8.350, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{rec12}}(s,p) &:= P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &+ \\ &D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.350 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr12}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec12}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, the tenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret10}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the twelfth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.350. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned}
&P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&\quad\le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}}(s,p)\\
&\qquad + W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p).
\end{aligned}$$

Since $h+1\ge 2$, Corollary 8.236 with $\ell=h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec12}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.352 (Thirteenth persistent finite ladder)**.** *In the setting of Corollary 8.351, use the notation from that corollary. For $0\le j<J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned}
P_{i,j}^{\mathrm{per12}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p),
\end{aligned}$$

and, for $0\le j\le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned}
R_{i,j}^{\mathrm{spr13}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p).
\end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.351 remains valid after replacing the prefix-record contribution

$$\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\\
&\quad\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec12}}
\end{aligned}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per12}} + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr13}} + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr13}}.$$

All other terms from that finite ladder, including the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$*, the adjacent twelfth return-spread family* $R_{i,j}^{\mathrm{spr12}}$*, the eleventh bridge-return spread family* $R_{i,j,h}^{\mathrm{ret11}}$*, the adjacent eleventh return-spread family* $R_{i,j}^{\mathrm{spr11}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the thirteenth persistent prefix profiles, the new thirteenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.351. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec12}}(s,p) \le\ & P_{i,j}^{\mathrm{per12}}(s,p) \\ & + R_{i,j}^{\mathrm{spr13}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr13}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.353 (Thirteenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.352, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair13}}(s,p) := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) := \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.352 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per12}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair13}}.$$

All other terms from that finite ladder, including the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the thirteenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.352. For each i, j, apply Corollary 8.165 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &= \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1 \\ &= B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per12}}(s,p) \le P_{i,j}^{\mathrm{pair13}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.354 (Thirteenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.353, use the notation from that corollary. Put*

$$\begin{aligned} &K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &:= \\ &8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap13}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.353 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair13}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap13}}.$$

All other terms from that finite ladder, including the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$*, the twelfth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret12}}$*, the adjacent twelfth return-spread family* $R_{i,j}^{\mathrm{spr12}}$*, the eleventh bridge-return spread family* $R_{i,j,h}^{\mathrm{ret11}}$*, the adjacent eleventh return-spread family* $R_{i,j}^{\mathrm{spr11}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are*

unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the thirteenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.353. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair13}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap13}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.355 (Thirteenth confined bridge finite ladder)**.** *In the setting of Corollary 8.354, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{br13}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.354 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap13}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br13}}.$$

All other terms from that finite ladder, including the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$*, the twelfth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret12}}$*, the adjacent twelfth return-spread family* $R_{i,j}^{\mathrm{spr12}}$*, the eleventh bridge-return spread family* $R_{i,j,h}^{\mathrm{ret11}}$*, the adjacent eleventh return-spread family* $R_{i,j}^{\mathrm{spr11}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the thirteenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.354. For each i, j, h, apply Corollary 8.171 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{gap13}}(s,p) \le P_{i,j,h}^{\mathrm{br13}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, and the finite h-range is unchanged. □

Corollary 8.356 (Thirteenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.355, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{ifr13}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p), \end{aligned}$$

and

$$\begin{aligned} R_{i,j,h}^{\mathrm{ret13}}(s,p) &:= R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\\ \max\{h-1,1\}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.355 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br13}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr13}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret13}}.$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, the eleventh bridge-return spread family $R_{i,j,h}^{\mathrm{ret11}}$, the adjacent eleventh return-spread family $R_{i,j}^{\mathrm{spr11}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the thirteenth inward-free bridge profiles, the new thirteenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.355. For each i, j, h, apply Corollary 8.174 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$P_{i,j,h}^{\mathrm{br13}}(s,p) \le P_{i,j,h}^{\mathrm{ifr13}}(s,p) + R_{i,j,h}^{\mathrm{ret13}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.357 (Thirteenth record-absorbed finite ladder)**.** *In the setting of Corollary 8.356, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{rec13}}(s,p) &:= P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &\quad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad + \\ &\quad D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.356 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr13}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec13}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the thirteenth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.356. For each i, j, h, apply Corollary 8.187 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned}
&P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&\quad\le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}}(s,p)\\
&\qquad + W_{\beta_i,p}\,T_{i,j}^{\mathrm{long}}(s,p).
\end{aligned}$$

Since $h+1\ge 2$, Corollary 8.236 with $\ell=h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec13}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.358 (Fourteenth persistent finite ladder)**.** *In the setting of Corollary 8.357, use the notation from that corollary. For $0\le j<J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned}
P_{i,j}^{\mathrm{per13}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}}(s,p),
\end{aligned}$$

and, for $0\le j\le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned}
R_{i,j}^{\mathrm{spr14}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)\\
&:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p).
\end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.357 remains valid after replacing the prefix-record contribution

$$\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\\
&\quad\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec13}}
\end{aligned}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per13}} + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr14}} + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr14}}.$$

All other terms from that finite ladder, including the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the persistent prefix profiles P^{per13}, the new fourteenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.357. For each i, j, apply Corollary 8.163 with

$$(\lambda, \sigma, \beta, K, L) = (\lambda_i, \sigma_i, \beta_i, 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, L_i).$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec13}}(s,p) \le\ & P_{i,j}^{\mathrm{per13}}(s,p) \\ & + R_{i,j}^{\mathrm{spr14}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr14}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.359 (Fourteenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.358, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair14}}(s,p) := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) := \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.358 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per13}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair14}}.$$

All other terms from that finite ladder, including the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fourteenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.358. For each i, j, apply Corollary 8.165 with

$$(\lambda,\sigma,\beta,K,L) = (\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i).$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &\quad = \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1 \\ &\quad = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per13}}(s,p) \le P_{i,j}^{\mathrm{pair14}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.360 (Fourteenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.359, use the notation from that corollary. Put*

$$\begin{aligned} &K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad := \\ &\quad 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap14}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ Q,\lambda_i,\sigma_i,\beta_i,j,\ L_i,h}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.359 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair14}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap14}}.$$

All other terms from that finite ladder, including the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$*, the thirteenth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret13}}$*, the adjacent thirteenth return-spread family* $R_{i,j}^{\mathrm{spr13}}$*, the twelfth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret12}}$*, the adjacent twelfth return-spread family* $R_{i,j}^{\mathrm{spr12}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are*

unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fourteenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.359. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair14}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap14}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.361 (Fourteenth confined bridge finite ladder)**.** *In the setting of Corollary 8.360, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{br14}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{\substack{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.360 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap14}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br14}}.$$

All other terms from that finite ladder, including the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$*, the thirteenth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret13}}$*, the adjacent thirteenth return-spread family* $R_{i,j}^{\mathrm{spr13}}$*, the twelfth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret12}}$*, the adjacent twelfth return-spread family* $R_{i,j}^{\mathrm{spr12}}$*, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fourteenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.360. For each i, j, h, apply Corollary 8.171 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{gap14}}(s,p) \le P_{i,j,h}^{\mathrm{br14}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, and the finite h-range is unchanged. □

Corollary 8.362 (Fourteenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.361, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{ifr14}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ L_i,h}(s,p),$$

and

$$R_{i,j,h}^{\mathrm{ret14}}(s,p) := R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \max\{h-1,1\}}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$, the finite ladder of Corollary 8.361 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br14}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr14}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret14}}.$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fourteenth inward-free bridge profiles, the new fourteenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.361. For each i, j, h, apply Corollary 8.174 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{br14}}(s,p) \le P_{i,j,h}^{\mathrm{ifr14}}(s,p) + R_{i,j,h}^{\mathrm{ret14}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.363 (Fourteenth record-absorbed finite ladder)**.** *In the setting of Corollary 8.362, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{rec14}}(s,p) &:= P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &+ \\ &D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.362 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr14}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec14}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the fourteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret14}}$, the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, the twelfth bridge-return spread family $R_{i,j,h}^{\mathrm{ret12}}$, the adjacent twelfth return-spread family $R_{i,j}^{\mathrm{spr12}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fourteenth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.362. For each i, j, h, apply Corollary 8.187 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned} P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}(s,p) \\ + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec14}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.364 (Fifteenth persistent finite ladder)**.** *In the setting of Corollary 8.363, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per14}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr15}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.363 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec14}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$
$$\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per14}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$
$$\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr15}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$
$$\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr15}}.$$

All other terms from that finite ladder, including the fourteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret14}}$, the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the persistent prefix profiles P^{per14}, the new fifteenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.363. For each i, j, apply Corollary 8.163 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec14}}(s,p) \le\ & P_{i,j}^{\mathrm{per14}}(s,p) \\ & + R_{i,j}^{\mathrm{spr15}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr15}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.365 (Fifteenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.364, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair15}}(s,p) := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) := \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\,B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\,Q,\lambda_i,\sigma_i,\beta_i,j,L_i,}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.364 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per14}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair15}}.$$

All other terms from that finite ladder, including the adjacent fifteenth return-spread family, the fourteenth bridge-return and adjacent return-spread families, the thirteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fifteenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.364. For each i, j, apply Corollary 8.165 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &= \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1 \\ &= B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per14}}(s,p) \le P_{i,j}^{\mathrm{pair15}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.366 (Fifteenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.365, use the notation from that corollary. Put*

$$\begin{aligned} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ := \\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap15}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ L_i,h}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.365 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair15}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap15}}.$$

All other terms from that finite ladder, including the adjacent fifteenth return-spread family, the fourteenth bridge-return and adjacent return-spread families, the thirteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail

follows from the same exponent hypothesis on the terminal far tail, the fifteenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.365. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair15}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap15}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.367 (Fifteenth confined bridge finite ladder)**.** *In the setting of Corollary 8.366, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{br15}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ L_i,h}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.366 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap15}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br15}}.$$

All other terms from that finite ladder, including the adjacent fifteenth return-spread family, the fourteenth bridge-return and adjacent return-spread families, the thirteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fifteenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.366. For each i, j, h, apply Corollary 8.171 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{gap15}}(s,p) \le P_{i,j,h}^{\mathrm{br15}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, and the finite h-range is unchanged. □

Corollary 8.368 (Fifteenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.367, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{ifr15}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ L_i,h}(s,p),$$

and

$$R_{i,j,h}^{\mathrm{ret15}}(s,p) := R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\max\{h-1,1\}}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$, the finite ladder of Corollary 8.367 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{br15}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr15}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret15}}.$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent fifteenth return-spread family $R_{i,j}^{\mathrm{spr15}}$, the fourteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret14}}$, the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fifteenth inward-free bridge profiles, the new fifteenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.367. For each i, j, h, apply Corollary 8.174 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{br15}}(s,p) \le P_{i,j,h}^{\mathrm{ifr15}}(s,p) + R_{i,j,h}^{\mathrm{ret15}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.369 (Fifteenth record-absorbed finite ladder)**.** *In the setting of Corollary 8.368, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{rec15}}(s,p) &:= P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\ Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p). \end{aligned}$$

Define the fixed coefficients

$$\begin{aligned} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} &:= \\ &A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &+ \\ &D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

Then the finite ladder of Corollary 8.368 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr15}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec15}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the fifteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret15}}$, the adjacent fifteenth return-spread family $R_{i,j}^{\mathrm{spr15}}$, the fourteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret14}}$, the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, the thirteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret13}}$, the adjacent thirteenth return-spread family $R_{i,j}^{\mathrm{spr13}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the fifteenth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.368. For each i, j, h, apply Corollary 8.187 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned} P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},Q,\lambda_i,\sigma_i,\beta_i,j,L_i,h+1}(s,p) \\ + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec15}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.370 (Sixteenth persistent finite ladder)**.** *In the setting of Corollary 8.369, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per15}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},Q,\lambda_i,\sigma_i,\beta_i,j,L_i}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr16}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.369 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec15}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$
$$\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per15}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$
$$\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr16}}$$
$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$
$$\cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr16}}.$$

All other terms from that finite ladder, including the fifteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret15}}$, the adjacent fifteenth return-spread family $R_{i,j}^{\mathrm{spr15}}$, the fourteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret14}}$, the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the persistent prefix profiles P^{per15}, the new sixteenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.369. For each i, j, apply Corollary 8.163 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec15}}(s,p) \le\ & P_{i,j}^{\mathrm{per15}}(s,p) \\ & + R_{i,j}^{\mathrm{spr16}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr16}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.371 (Sixteenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.370, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair16}}(s,p) := P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) := \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ Q,\lambda_i,\sigma_i,\beta_i,j,L_i,}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.370 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per15}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair16}}.$$

All other terms from that finite ladder, including the adjacent sixteenth return-spread family, the fifteenth bridge-return and adjacent return-spread families, the fourteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the sixteenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.370. For each i, j, apply Corollary 8.165 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &= \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1 \\ &= B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per15}}(s,p) \le P_{i,j}^{\mathrm{pair16}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.372 (Sixteenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.371, use the notation from that corollary. Put*

$$\begin{aligned} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ := \\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch},}$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap16}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ L_i,h}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.371 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair16}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap16}}.$$

All other terms from that finite ladder, including the adjacent sixteenth return-spread family, the fifteenth bridge-return and adjacent return-spread families, the fourteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the sixteenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.371. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair16}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap16}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.373 (Sixteenth confined bridge finite ladder)**.** *In the setting of Corollary 8.372, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{br16}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\ L_i,h}}(s,p).$$

Then, for all p *sufficiently close to* $p_c(H)$*, the finite ladder of Corollary 8.372 remains valid after replacing the fixed-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap16}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{br16}}.$$

All other terms from that finite ladder, including the adjacent sixteenth return-spread family, the fifteenth bridge-return and adjacent return-spread families, the fourteenth bridge-return and adjacent

return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the sixteenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.372. For each i, j, h, apply Corollary 8.171 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{gap16}}(s,p) \le P_{i,j,h}^{\mathrm{br16}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, and the finite h-range is unchanged. □

Corollary 8.374 (Sixteenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.373, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{ifr16}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{\substack{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ L_i,h}}(s,p), \end{aligned}$$

and

$$\begin{aligned} R_{i,j,h}^{\mathrm{ret16}}(s,p) &:= R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{\substack{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \\ \max\{h-1,1\}}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_c(H)$, the finite ladder of Corollary 8.373 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{br16}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr16}}$$

$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} R_{i,j,h}^{\mathrm{ret16}}.$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent sixteenth return-spread family $R_{i,j}^{\mathrm{spr16}}$, the fifteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret15}}$, the adjacent fifteenth return-spread family $R_{i,j}^{\mathrm{spr15}}$, the fourteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret14}}$, the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the sixteenth inward-free bridge profiles, the new sixteenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.373. For each i, j, h, apply Corollary 8.174 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{br16}}(s,p) \le P_{i,j,h}^{\mathrm{ifr16}}(s,p) + R_{i,j,h}^{\mathrm{ret16}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.375 (Sixteenth record-absorbed finite ladder)**.** *In the setting of Corollary 8.374, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P_{i,j}^{\mathrm{rec16}}(s,p) := P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\,Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p).$$

Define the fixed coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} + D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then the finite ladder of Corollary 8.374 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{ifr16}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec16}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the sixteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret16}}$, the adjacent sixteenth return-spread family $R_{i,j}^{\mathrm{spr16}}$, the fifteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret15}}$, the adjacent fifteenth return-spread family $R_{i,j}^{\mathrm{spr15}}$, the fourteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret14}}$, the adjacent fourteenth return-spread family $R_{i,j}^{\mathrm{spr14}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the sixteenth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.374. For each i, j, h, apply Corollary 8.187 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned} P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,\ 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,h+1}(s,p) \\ + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec16}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.376 (Seventeenth persistent finite ladder)**.** *In the setting of Corollary 8.375, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per16}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i} \\ &\qquad {\scriptstyle Q,\lambda_i,\sigma_i,\beta_i,j,} \\ &\qquad (s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr17}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &\qquad {\scriptstyle Q,\lambda_i,\sigma_i,\beta_i,j,} \\ &\qquad (s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.375 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec16}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per16}} \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr17}} \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr17}}. \end{aligned}$$

All other terms from that finite ladder, including the sixteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret16}}$, the adjacent sixteenth return-spread family $R_{i,j}^{\mathrm{spr16}}$, the fifteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret15}}$, the adjacent fifteenth return-spread family $R_{i,j}^{\mathrm{spr15}}$, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return

profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the persistent prefix profiles P^{per16}*, the new seventeenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.375. For each i, j, apply Corollary 8.163 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec16}}(s,p) \le\ & P_{i,j}^{\mathrm{per16}}(s,p) \\ & + R_{i,j}^{\mathrm{spr17}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr17}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.377 (Seventeenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.376, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and

$$\begin{aligned} & B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ & \quad := \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$*, write*

$$\begin{aligned} P_{i,j}^{\mathrm{pair17}}(s,p) &:= P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_c(H)$*, the finite ladder of Corollary 8.376 remains valid after replacing the persistent-prefix contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per16}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair17}}.$$

All other terms from that finite ladder, including the adjacent seventeenth return-spread family, the sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the seventeenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.376. For each i, j, apply Corollary 8.165 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &\quad = \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \right\rceil + 1 \\ &\quad = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per16}}(s,p) \le P_{i,j}^{\mathrm{pair17}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.378 (Seventeenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.377, use the notation from that corollary. Put*

$$\begin{aligned} &K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad := \\ &\quad 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \\ &\quad \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}. \end{aligned}$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{gap17}}(s,p) := P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$, the finite ladder of Corollary 8.377 remains valid after replacing the bounded-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair17}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap17}}.$$

All other terms from that finite ladder, including the adjacent seventeenth return-spread family, the sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the seventeenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.377. For each i,j, apply Corollary 8.169 with

$$\lambda=\lambda_i,\qquad \sigma=\sigma_i,\qquad \beta=\beta_i,\qquad L=L_i,$$
$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B = B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair17}}(s,p) \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap17}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.379 (Seventeenth confined bridge finite ladder)**.** *In the setting of Corollary 8.378, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{br17}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.378 remains valid after replacing the fixed-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{gap17}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{br17}}.$$

All other terms from that finite ladder, including the adjacent seventeenth return-spread family, the sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the seventeenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.378. For each i, j, h, apply Corollary 8.171 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{gap17}}(s,p) \le P_{i,j,h}^{\mathrm{br17}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, and the finite h-range is unchanged. □

Corollary 8.380 (Seventeenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.379, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{ifr17}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p), \end{aligned}$$

and

$$\begin{aligned} R_{i,j,h}^{\mathrm{ret17}}(s,p) &:= R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ Q,\lambda_i,\sigma_i,\beta_i,j,\ \max\{h-1,1\}}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.379 remains valid after replacing the confined bridge contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{br17}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} P_{i,j,h}^{\mathrm{ifr17}} \\ &+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot \sum_{h=1}^{B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} R_{i,j,h}^{\mathrm{ret17}}. \end{aligned}$$

All h*-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent seventeenth return-spread family* $R_{i,j}^{\mathrm{spr17}}$*, the sixteenth bridge-return spread family* $R_{i,j,h}^{\mathrm{ret16}}$*, the adjacent sixteenth return-spread family* $R_{i,j}^{\mathrm{spr16}}$*, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the seventeenth inward-free bridge profiles, the new seventeenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.379. For each i, j, h, apply Corollary 8.174 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{br17}}(s,p) \le P_{i,j,h}^{\mathrm{ifr17}}(s,p) + R_{i,j,h}^{\mathrm{ret17}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.381 (Seventeenth record-absorbed finite ladder)**.** *In the setting of Corollary 8.380, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P_{i,j}^{\mathrm{rec17}}(s,p) := P_{i,j}^{\mathrm{pre,rec,br,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i}(s,p).$$

Define the fixed coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} :=$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} :=$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} + D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Then the finite ladder of Corollary 8.380 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{ifr17}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec17}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All other terms from that finite ladder, including the seventeenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret17}}$, the adjacent seventeenth return-spread family $R_{i,j}^{\mathrm{spr17}}$, the sixteenth bridge-return spread family

$R^{\mathrm{ret16}}_{i,j,h}$, *the adjacent sixteenth return-spread family* $R^{\mathrm{spr16}}_{i,j}$, *the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the seventeenth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.380. For each i, j, h, apply Corollary 8.187 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned} &P^{\mathrm{pre,br,free,recbr,ch}_{i,j,h}(s,p) \\ &\quad \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\,Q,\lambda_i,\sigma_i,\beta_i,j,\,L_i,h+1}(s,p) \\ &\qquad + W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p). \end{aligned}$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P^{\mathrm{rec17}}_{i,j}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p}T^{\mathrm{long}}_{i,j}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.382 (Eighteenth persistent finite ladder)**.** *In the setting of Corollary 8.381, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$, *write*

$$\begin{aligned} P^{\mathrm{per17}}_{i,j}(s,p) &:= P^{\mathrm{pre,per,br,recbr,ch}_{i,j}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\,Q,\lambda_i,\sigma_i,\beta_i,j,\,L_i}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, *write*

$$\begin{aligned} R^{\mathrm{spr18}}_{i,j}(s,p) &:= R^{\mathrm{spr,recbr,ch}_{i,j}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{8K_i^{\mathrm{gap,br,recbr,ch}},\,Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_c(H)$, the finite ladder of Corollary 8.381 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec17}}$$

by

$$\begin{aligned}
&\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per17}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr18}} \\
&\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr18}}.
\end{aligned}$$

All other terms from that finite ladder, including the seventeenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret17}}$, the adjacent seventeenth return-spread family $R_{i,j}^{\mathrm{spr17}}$, the sixteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret16}}$, the adjacent sixteenth return-spread family $R_{i,j}^{\mathrm{spr16}}$, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the persistent prefix profiles P^{per17}, the new eighteenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.381. For each i, j, apply Corollary 8.163 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$\begin{aligned}
P_{i,j}^{\mathrm{rec17}}(s,p) \le\ & P_{i,j}^{\mathrm{per17}}(s,p) \\
& + R_{i,j}^{\mathrm{spr18}}(s,p) \\
& + R_{i,j+1}^{\mathrm{spr18}}(s,p),
\end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.383 (Eighteenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.382, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and define

$$B_i^{\mathrm{br,recbr,ch}}$$
$$:= \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$, write

$$P_{i,j}^{\mathrm{pair18}}(s,p) := P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{8K_i^{\mathrm{gap,br,recbr,ch}},\\ B_i^{\mathrm{br,recbr,ch}}}}_{Q,\lambda_i,\sigma_i,\beta_i,j,L_i,}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.382 remains valid after replacing the persistent-prefix contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per17}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair18}}.$$

All other terms from that finite ladder, including the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighteenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.382. For each i, j, apply Corollary 8.165 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = 4K_i^{\mathrm{gap,br,recbr,ch}}.$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,ch}}}$$
$$= \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,ch}} \right\rceil + 1$$
$$= B_i^{\mathrm{br,recbr,ch}},$$

the two inequalities give

$$P_{i,j}^{\mathrm{per17}}(s,p) \le P_{i,j}^{\mathrm{pair18}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.384 (Eighteenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.383, use the notation from that corollary. Put*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} := 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} B_i^{\mathrm{br,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch},}$$

write

$$P_{i,j,h}^{\mathrm{gap18}}(s,p) := P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}(s,p) := \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.383 remains valid after replacing the bounded-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair18}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap18}}.$$

All other terms from that finite ladder, including the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighteenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.383. For each i,j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = 8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$B = B_i^{\mathrm{br,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair18}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{gap18}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.385 (Eighteenth confined bridge finite ladder)**.** *In the setting of Corollary 8.384, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{br18}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{K_i^{\mathrm{gap,br,recbr,ch}},\ L_i,h}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.384 remains valid after replacing the fixed-gap prefix-pair contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{gap18}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{br18}}.$$

All other terms from that finite ladder, including the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return

profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighteenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.384. For each i, j, h, apply Corollary 8.171 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$
$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{gap18}}(s,p) \le P_{i,j,h}^{\mathrm{br18}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, and the finite h-range is unchanged. □

Corollary 8.386 (Eighteenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.385, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{ifr18}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ L_i,h}(s,p),$$

and

$$R_{i,j,h}^{\mathrm{ret18}}(s,p) := R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p)$$
$$:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \max\{h-1,1\}}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.385 remains valid after replacing the confined bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{br18}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{ifr18}}$$

$$+ \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \cdot \sum_{h=1} R_{i,j,h}^{\mathrm{ret18}}.$$

All h-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighteenth inward-free bridge profiles, the new eighteenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.385. For each i, j, h, apply Corollary 8.174 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{br18}}(s,p) \le P_{i,j,h}^{\mathrm{ifr18}}(s,p) + R_{i,j,h}^{\mathrm{ret18}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.387 (Eighteenth record-absorbed finite ladder)**.** *In the setting of Corollary 8.386, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$P_{i,j}^{\mathrm{rec18}}(s,p) := P_{i,j}^{\mathrm{pre,rec,br,recbr,ch}}(s,p)$$

$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,rec,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i,\ Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p).$$

Define the fixed coefficients

$$D_i^{\mathrm{rec,br,recbr,ch}} := D_i^{\mathrm{rec,br,recbr,ch}} \cdot B_i^{\mathrm{br,recbr,ch}},$$

$$A_i^{\mathrm{long,br,recbr,ch}} := A_i^{\mathrm{long,br,recbr,ch}} + D_i^{\mathrm{rec,br,recbr,ch}}.$$

Then the finite ladder of Corollary 8.386 remains valid after replacing the inward-free bridge contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{ifr18}}$$

by

$$\textstyle\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec18}},$$

and after replacing the long-run coefficient

$$A_i^{\mathrm{long,br,recbr,ch}}$$

by

$$A_i^{\mathrm{long,br,recbr,ch}}.$$

All other terms from that finite ladder, including the eighteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret18}}$, the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the eighteenth prefix-record profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.386. For each i, j, h, apply Corollary 8.187 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = 2K_i^{\mathrm{gap,br,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}(s,p) \le \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{2K_i^{\mathrm{gap,br,recbr,ch}},L_i,h+1} Q,\lambda_i,\sigma_i,\beta_i,j,(s,p) + W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p).$$

Since $h+1 \ge 2$, Corollary 8.236 with $\ell = h+1$ bounds the chain profile by $P_{i,j}^{\mathrm{rec18}}$. Substituting these inequalities into the inward-free bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

copies of the prefix-record profile and the same number of additional copies of $W_{\beta_i,p} T_{i,j}^{\mathrm{long}}$. Combining these with the existing long-run coefficient

$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. □

Corollary 8.388 (Nineteenth persistent finite ladder)**.** *In the setting of Corollary 8.387, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$, write*

$$\begin{aligned} P_{i,j}^{\mathrm{per18}}(s,p) &:= P_{i,j}^{\mathrm{pre,per,br,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,short}}_{4K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},L_i} \\ &\qquad {}_{Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p), \end{aligned}$$

and, for $0 \le j \le J_{\sigma_i,\beta_i}$, write

$$\begin{aligned} R_{i,j}^{\mathrm{spr19}}(s,p) &:= R_{i,j}^{\mathrm{spr,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{8K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}} \\ &\qquad {}_{Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.387 remains valid after replacing the prefix-record contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{rec18}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per18}} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j}^{\mathrm{spr19}} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} R_{i,j+1}^{\mathrm{spr19}}. \end{aligned}$$

All other terms from that finite ladder, including the eighteenth bridge-return spread family $R_{i,j,h}^{\mathrm{ret18}}$, the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread

and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the persistent prefix profiles P^{per18}*, the new nineteenth return-spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.387. For each i, j, apply Corollary 8.163 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = 2K_i^{\mathrm{gap,br,recbr,ch}}.$$

This gives

$$\begin{aligned} P_{i,j}^{\mathrm{rec18}}(s,p) \le\ & P_{i,j}^{\mathrm{per18}}(s,p) \\ & + R_{i,j}^{\mathrm{spr19}}(s,p) \\ & + R_{i,j+1}^{\mathrm{spr19}}(s,p), \end{aligned}$$

because the persistent-prefix threshold $2K$ and return threshold $4K$ in the general split become

$$4K_i^{\mathrm{gap,br,recbr,ch}},$$
$$8K_i^{\mathrm{gap,br,recbr,ch}}.$$

Substituting into the coefficient

$$D_i^{\mathrm{rec,br,recbr,ch}}$$

prefix-record term gives the displayed replacement. No long-run coefficient changes in this split. □

Corollary 8.389 (Nineteenth bounded-gap finite ladder)**.** *In the setting of Corollary 8.388, use the notation from that corollary. Put*

$$\Theta_i = \theta_{\lambda_i,\sigma_i,\beta_i}$$

and define

$$B_i^{\mathrm{br,recbr,ch}}$$
$$:= \left\lceil \frac{64\beta_i}{\Theta_i}\, K_i^{\mathrm{gap,br,recbr,ch}} \right\rceil + 1.$$

For $0 \le j < J_{\sigma_i,\beta_i}$*, write*

$$\begin{aligned} P_{i,j}^{\mathrm{pair19}}(s,p) &:= P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,pair,short}}_{\substack{8K_i^{\mathrm{gap,br,recbr,ch}}, \\ B_i^{\mathrm{br,recbr,ch}}}}_{Q,\lambda_i,\sigma_i,\beta_i,j,L_i,}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.388 remains valid after replacing the persistent-prefix contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{per18}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair19}}.$$

All other terms from that finite ladder, including the adjacent nineteenth return-spread family, the eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the seventeenth bridge-return and adjacent return-spread families, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the nineteenth bounded-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.388. For each i, j, apply Corollary 8.165 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = 4K_i^{\mathrm{gap,br,recbr,ch}}.$$

Then apply Corollary 8.167 with the same parameters. Since the bounded-successor reduction doubles K, and

$$\begin{aligned} &B_{\lambda_i,\sigma_i,\beta_i,4K_i^{\mathrm{gap,br,recbr,ch}}} \\ &\quad = \left\lceil \frac{64\beta_i}{\Theta_i} K_i^{\mathrm{gap,br,recbr,ch}} \right\rceil + 1 \\ &\quad = B_i^{\mathrm{br,recbr,ch}}, \end{aligned}$$

the two inequalities give

$$P_{i,j}^{\mathrm{per18}}(s,p) \le P_{i,j}^{\mathrm{pair19}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,ch}}$$

persistent-prefix term gives the displayed replacement. The new cutoff is fixed by the finite ladder parameters, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.390 (Nineteenth fixed-gap bridge finite ladder)**.** *In the setting of Corollary 8.389, use the notation from that corollary. Put*

$$K_i^{\mathrm{gap,br,recbr,ch}} := 8K_i^{\mathrm{gap,br,recbr,ch}} B_i^{\mathrm{br,recbr,ch}}.$$

For $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{gap19}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,gap,short}}_{K_i^{\mathrm{gap,br,recbr,ch}}, \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p). \end{aligned}$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.389 remains valid after replacing the bounded-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} P_{i,j}^{\mathrm{pair19}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \sum_{h=1} P_{i,j,h}^{\mathrm{gap19}}.$$

All other terms from that finite ladder, including the adjacent nineteenth return-spread family, the eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the seventeenth bridge-return and adjacent return-spread families, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the nineteenth fixed-gap prefix-pair profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.389. For each i, j, apply Corollary 8.169 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = 8K_i^{\mathrm{gap,br,recbr,ch}},$$

$$B = B_i^{\mathrm{br,recbr,ch}}.$$

Since

$$KB = K_i^{\mathrm{gap,br,recbr,ch}},$$

this gives

$$P_{i,j}^{\mathrm{pair19}}(s,p) \le \sum_{h=1}^{B_i^{\mathrm{br,recbr,ch}}} P_{i,j,h}^{\mathrm{gap19}}(s,p).$$

Substituting this into the coefficient

$$D_i^{\mathrm{rec,br,recbr,ch}}$$

bounded-gap pair term gives the displayed replacement. The added sum over h is finite and independent of p, and all other coefficients and variable losses are unchanged. □

Corollary 8.391 (Nineteenth confined bridge finite ladder)**.** *In the setting of Corollary 8.390, use the notation from that corollary. For $0 \le j < J_{\sigma_i,\beta_i}$ and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch}},$$

write

$$P_{i,j,h}^{\mathrm{br19}}(s,p) := P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}(s,p)$$
$$:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,short}}_{K_i^{\mathrm{gap,br,recbr,ch}},\ L_i,h}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$, the finite ladder of Corollary 8.390 remains valid after replacing the fixed-gap prefix-pair contribution

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{gap19}}$$

by

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} B_i^{\mathrm{br,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{br19}}.$$

All other terms from that finite ladder, including the adjacent nineteenth return-spread family, the eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the seventeenth bridge-return and adjacent return-spread families, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the nineteenth confined bridge profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.390. For each i, j, h, apply Corollary 8.171 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = K_i^{\mathrm{gap,br,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{gap19}}(s,p) \le P_{i,j,h}^{\mathrm{br19}}(s,p).$$

Substituting this bound in the fixed-gap sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,ch}}$$

gives the displayed replacement. No new variable coefficient is introduced, and the finite h-range is unchanged. □

Corollary 8.392 (Nineteenth inward-free bridge finite ladder)**.** *In the setting of Corollary 8.391, use the notation from that corollary. For* $0 \le j < J_{\sigma_i,\beta_i}$ *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch}},$$

write

$$\begin{aligned} P_{i,j,h}^{\mathrm{ifr19}}(s,p) &:= P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}}(s,p) \\ &:= \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,bridge,free,short}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\ \substack{Q,\lambda_i,\sigma_i,\beta_i,j,\\ L_i,h}}(s,p), \end{aligned}$$

and

$$\begin{aligned} R_{i,j,h}^{\mathrm{ret19}}(s,p) &:= R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}(s,p) \\ &:= \mathfrak{X}^{G\to H,\mathrm{rad,ret,spr}}_{2K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}\ \max\{h-1,1\},\ Q,\lambda_i,\sigma_i,\beta_i,j,}(s,p). \end{aligned}$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$*, the finite ladder of Corollary 8.391 remains valid after replacing the confined bridge contribution*

$$\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \cdot B_i^{\mathrm{br,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{br19}}$$

by

$$\begin{aligned} &\sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \\ &\quad \cdot B_i^{\mathrm{br,recbr,ch}} \cdot \sum_{h=1} P_{i,j,h}^{\mathrm{ifr19}} \\ &\quad + \sum_{i=0}^{N-1} D_i^{\mathrm{rec,br,recbr,ch}} \sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \\ &\quad \cdot B_i^{\mathrm{br,recbr,ch}} \cdot \sum_{h=1} R_{i,j,h}^{\mathrm{ret19}}. \end{aligned}$$

All h*-sums are over the displayed finite range. All other terms from that finite ladder, including the adjacent nineteenth return-spread family, the eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the seventeenth bridge-return and adjacent return-spread families, the sixteenth bridge-return and adjacent return-spread families, the fifteenth bridge-return and adjacent return-spread families, all retained older return-spread and bridge-return families, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and exact-level return profiles, are unchanged. Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the nineteenth inward-free bridge profiles, the new nineteenth bridge-return spread profiles, the retained return-spread and bridge-return profiles named above, confined fixed-gap staircase and bridge profiles, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.391. For each i, j, h, apply Corollary 8.174 with

$$\lambda = \lambda_i, \qquad \sigma = \sigma_i, \qquad \beta = \beta_i, \qquad L = L_i,$$

$$K = K_i^{\mathrm{gap,br,recbr,ch}}.$$

This gives

$$P_{i,j,h}^{\mathrm{br19}}(s,p) \le P_{i,j,h}^{\mathrm{ifr19}}(s,p) + R_{i,j,h}^{\mathrm{ret19}}(s,p).$$

Substituting into the confined bridge sum with coefficient

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

gives the displayed replacement. The added h-sum is finite and independent of p, and no new nonconstant multiplicative loss is introduced. □

Corollary 8.393 (Staircase-chain-pair finite ladder)**.** *In the setting of Corollary 8.272, define* B_i, C_i, $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr,++}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. Put*

$$D_i = \left\lceil \frac{8192 M_i B_i C_i \beta_i}{\theta_{\lambda_i,\sigma_i,\beta_i}} \right\rceil + 5$$

and

$$K_i^{\sharp} = 1024 M_i B_i C_i D_i.$$

For $q \in \{3,4\}$ *and* $1 \le h \le q$, *write*

$$P_{i,j,q,h}^{\mathrm{pre,stair,loc,longchain,sh}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\sharp},L_i,q+h}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}(s,p),$$

and, for $q \in \{3,4\}$ *and* $q < h \le D_i$, *write*

$$P_{i,j,q,h}^{\mathrm{pre,stair,loc,chainbr,sh}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^{\sharp},L_i,q,h}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,bridge,short}}(s,p).$$

Then, for all p sufficiently close to $p_c(H)$,

$$\begin{aligned}
\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p) \le \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{q=3}^{4}\sum_{h=1}^{q} P_{i,j,q,h}^{\mathrm{pre,stair,loc,longchain,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{q=3}^{4}\sum_{h=q+1}^{D_i} P_{i,j,q,h}^{\mathrm{pre,stair,loc,chainbr,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P_{i,j,h}^{\mathrm{pre,br,sh}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \left(7R_{i,j}^{\mathrm{spr},++}(s,p) + 5R_{i,j+1}^{\mathrm{spr},++}(s,p)\right)\\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T_{i,j}^{\mathrm{long}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S_{i,j}^{\mathrm{ret}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\left(5\widehat{X}_{i,j}^{\mathrm{ret}}(s,p) + \widehat{X}_{i,j+1}^{\mathrm{ret}}(s,p)\right).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, longer locally short staircase-chain profiles with lengths $4,\dots,8$, confined fixed-gap staircase-chain bridge profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.272. For each i,j and $q \in \{3,4\}$, apply Corollary 8.191 with

$$(\lambda,\sigma,\beta,K,L,\ell) = (\lambda_i,\sigma_i,\beta_i,512M_iB_iC_i,L_i,q).$$

By the definition of D_i,

$$D_i \ge B_{\lambda_i,\sigma_i,\beta_i,512M_iB_iC_i}.$$

Thus monotonicity in the bounded-gap cutoff gives a bounded-gap staircase-chain-pair profile with parameter $1024M_iB_iC_i$ and cutoff D_i. Applying Corollary 8.193 with this cutoff gives

$$P_{i,j,q}^{\mathrm{pre,stair,loc,chain,sh}}(s,p) \le \sum_{h=1}^{D_i} \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^\sharp,L_i,q,h}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,gap,short}}(s,p).$$

Now split each fixed gap using Corollary 8.195. The terms with $1 \le h \le q$ become the longer-chain profiles $P_{i,j,q,h}^{\mathrm{pre,stair,loc,longchain,sh}}$, and the terms with $q < h \le D_i$ become the confined chain-bridge

profiles $P^{\mathrm{pre,stair,loc,chainbr,sh}}_{i,j,q,h}$. Substituting these two bounds for the $q=3$ and $q=4$ chain terms in Corollary 8.272 gives the displayed inequality. All new sums are finite and independent of p, so the exponent conclusion is unchanged. □

Corollary 8.394 (Second-stage staircase-chain finite ladder)**.** *In the setting of Corollary 8.393, define* B_i, C_i, D_i, $K_i^\sharp$, $P^{\mathrm{pre,stair,loc,chainbr,sh}}_{i,j,q,h}$, $P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}$, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, *and* $\widehat{X}^{\mathrm{ret}}_{i,j}$ *as there. Put*

$$E_i = \left\lceil \frac{16K_i^\sharp \beta_i}{\theta_{\lambda_i,\sigma_i,\beta_i}} \right\rceil + 9, \qquad K_i^\flat = 2K_i^\sharp E_i.$$

For $q \in \{3,4\}$, $1 \le h \le q$, *write* $\ell_{q,h} = q + h$. *For* $1 \le a \le \ell_{q,h}$, *define*

$$P^{\mathrm{pre,stair,loc,nextchain,sh}}_{i,j,q,h,a}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^\flat,L_i,\ell_{q,h}+a}(s,p),$$

and, for $\ell_{q,h} < a \le E_i$, *define*

$$P^{\mathrm{pre,stair,loc,nextchainbr,sh}}_{i,j,q,h,a}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,bridge,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K_i^\flat,L_i,\ell_{q,h},a}(s,p).$$

Then, for all p *sufficiently close to* $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{P}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le{}& \mathfrak{P}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{q=3}^{4}\sum_{h=1}^{q}\sum_{a=1}^{\ell_{q,h}} P^{\mathrm{pre,stair,loc,nextchain,sh}}_{i,j,q,h,a}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{q=3}^{4}\sum_{h=1}^{q}\sum_{a=\ell_{q,h}+1}^{E_i} P^{\mathrm{pre,stair,loc,nextchainbr,sh}}_{i,j,q,h,a}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{q=3}^{4}\sum_{h=q+1}^{D_i} P^{\mathrm{pre,stair,loc,chainbr,sh}}_{i,j,q,h}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} \big(7R^{\mathrm{spr},++}_{i,j}(s,p) + 5R^{\mathrm{spr},++}_{i,j+1}(s,p)\big) \\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\, T^{\mathrm{long}}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p) \\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p) + \widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, locally short staircase-chain profiles with lengths $5,\dots,16$*, confined fixed-gap staircase-chain bridge profiles, confined fixed-gap locally short staircase-bridge profiles with* $3 \le h \le C_i$*, confined fixed-gap bridge profiles with* $h \ge 2$*, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.*

Proof. Apply Corollary 8.393. For each i, j, q, h with $q \in \{3,4\}$ and $1 \le h \le q$, put $\ell = \ell_{q,h}$. By the definition of E_i,

$$E_i \ge B_{\lambda_i,\sigma_i,\beta_i,K_i^\sharp}.$$

Applying Corollary 8.197 with

$$(\lambda,\sigma,\beta,K,L,\ell,\widetilde{B}) = (\lambda_i,\sigma_i,\beta_i,K_i^\sharp,L_i,\ell,E_i)$$

gives

$$\begin{aligned} &P^{\mathrm{pre,stair,loc,longchain,sh}}_{i,j,q,h}(s,p) \\ &\quad\le \sum_{a=1}^{\ell_{q,h}} P^{\mathrm{pre,stair,loc,nextchain,sh}}_{i,j,q,h,a}(s,p) + \sum_{a=\ell_{q,h}+1}^{E_i} P^{\mathrm{pre,stair,loc,nextchainbr,sh}}_{i,j,q,h,a}(s,p). \end{aligned}$$

Substituting this bound for every longer-chain term in Corollary 8.393 gives the displayed inequality. All new sums are finite and independent of p, so the exponent conclusion is unchanged. □

Corollary 8.395 (Finite-stage staircase-chain finite ladder)**.** *In the setting of Corollary 8.272, define* B_i*,* C_i*,* $P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}$*,* $P^{\mathrm{pre,br,sh}}_{i,j,h}$*,* $R^{\mathrm{spr},++}_{i,j}$*,* $T^{\mathrm{long}}_{i,j}$*,* $S^{\mathrm{ret}}_{i,j}$*, and* $\widehat{X}^{\mathrm{ret}}_{i,j}$ *as there. Fix* $n \ge 0$*. For* $0 \le i < N$*, put*

$$K^{\mathrm{ch}}_{i,0} = 512 M_i B_i C_i.$$

For $1 \le r \le n$*, choose integers* $B^{\mathrm{ch}}_{i,r}$ *so that*

$$B^{\mathrm{ch}}_{i,r} \ge B_{\lambda_i,\sigma_i,\beta_i,K^{\mathrm{ch}}_{i,r-1}} = \left\lceil \frac{16 K^{\mathrm{ch}}_{i,r-1}\beta_i}{\theta_{\lambda_i,\sigma_i,\beta_i}} \right\rceil + 1, \qquad K^{\mathrm{ch}}_{i,r} = 2K^{\mathrm{ch}}_{i,r-1}B^{\mathrm{ch}}_{i,r}.$$

For $q \in \{3,4\}$ *and* $\mathbf{a} = (a_0,\dots,a_n) \in \mathcal{A}_n(q)$*, write*

$$P^{\mathrm{pre,stair,loc,nchain,sh}}_{i,j,q,\mathbf{a}}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K^{\mathrm{ch}}_{i,n},L_i,a_n}(s,p).$$

For $q \in \{3,4\}$*,* $1 \le r \le n$*,* $\mathbf{a} = (a_0,\dots,a_{r-1}) \in \mathcal{A}_{r-1}(q)$*, and* $a_{r-1} < h \le B^{\mathrm{ch}}_{i,r}$*, write*

$$P^{\mathrm{pre,stair,loc,nchainbr,sh}}_{i,j,q,r,\mathbf{a},h}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,chain,bridge,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,K^{\mathrm{ch}}_{i,r},L_i,a_{r-1},h}(s,p).$$

Then, for all p sufficiently close to $p_{\mathrm{c}}(H)$,

$$\begin{aligned}
\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p) \le \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{q=3}^{4}\sum_{\mathbf{a}\in\mathcal{A}_n(q)} P^{\mathrm{pre,stair,loc,nchain,sh}}_{i,j,q,\mathbf{a}}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{q=3}^{4}\sum_{r=1}^{n}\sum_{\mathbf{a}\in\mathcal{A}_{r-1}(q)}\sum_{h=a_{r-1}+1}^{B^{\mathrm{ch}}_{i,r}} P^{\mathrm{pre,stair,loc,nchainbr,sh}}_{i,j,q,r,\mathbf{a},h}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=3}^{C_i} P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\sum_{h=2}^{B_i} P^{\mathrm{pre,br,sh}}_{i,j,h}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1}\big(7R^{\mathrm{spr},++}_{i,j}(s,p)+5R^{\mathrm{spr},++}_{i,j+1}(s,p)\big)\\
&+ 3\sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,T^{\mathrm{long}}_{i,j}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} S^{\mathrm{ret}}_{i,j}(s,p)\\
&+ \sum_{i=0}^{N-1}\sum_{j=0}^{J_{\sigma_i,\beta_i}-1} W_{\beta_i,p}\,\big(5\widehat{X}^{\mathrm{ret}}_{i,j}(s,p)+\widehat{X}^{\mathrm{ret}}_{i,j+1}(s,p)\big).
\end{aligned}$$

Consequently, the exponent hypothesis on the initial far tail follows from the same exponent hypothesis on the terminal far tail, the terminal finite-stage staircase-chain profiles, the accumulated finite-stage confined staircase-chain bridge profiles, confined fixed-gap locally short staircase-bridge profiles with $3 \le h \le C_i$, confined fixed-gap bridge profiles with $h \ge 2$, long-run profiles, enlarged spread-return profiles, and ordinary exact-level exterior-return profiles.

Proof. Apply Corollary 8.272. For each i, j and $q \in \{3, 4\}$, apply Corollary 8.198 with

$$(\lambda, \sigma, \beta, K, L, \ell) = (\lambda_i, \sigma_i, \beta_i, K^{\mathrm{ch}}_{i,0}, L_i, q)$$

and with the chosen cutoffs $B_1, \dots, B_n$ equal to

$$B^{\mathrm{ch}}_{i,1}, \dots, B^{\mathrm{ch}}_{i,n}.$$

This bounds the raw chain profile $P^{\mathrm{pre,stair,loc,chain,sh}}_{i,j,q}$ by the terminal chain profiles indexed by $\mathcal{A}_n(q)$ and the accumulated chain-bridge profiles indexed by $1 \le r \le n$, $\mathcal{A}_{r-1}(q)$, and $a_{r-1} < h \le B^{\mathrm{ch}}_{i,r}$. Substituting these two bounds for the $q = 3$ and $q = 4$ chain terms gives the displayed inequality. All added families are finite for fixed n and independent of p, so the exponent conclusion is unchanged. □

Corollary 8.396 (Fully localized shifted original radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \geq 1, \qquad M_0, \dots, M_{N-1} \geq 1$$

be integers. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N)$$

and define the localized profiles $Z^{\mathrm{spr}}_{i,j}$, $Z^{\mathrm{lev}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $X^{\mathrm{ret}}_{i,j}$ as in Corollary 8.248. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \leq a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

and that, for every $0 \leq i < N$ and $0 \leq j < J_{\sigma_i,\beta_i}$, each of

$$Z^{\mathrm{spr}}_{i,j}, \qquad Z^{\mathrm{lev}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}, \qquad X^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ denotes the profile under consideration. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.248 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the four localized profile families, with the only nonconstant multiplicative losses equal to $W_{\beta_i,p}$. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for every fixed i, the displayed exponent hypotheses imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the assumed near-tail bin estimates for $\mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}$, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$. Hence the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ follows. □

Corollary 8.397 (Fully localized original radial subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.396 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.396 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are exactly the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. That corollary therefore applies and gives $p_{\mathrm{c}}(G) < p_{2\to 2}(G)$. □

Corollary 8.398 (Tangentially localized shifted original radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*, let* $\delta > 0$*, let* $\beta_0, \dots, \beta_{N-1} > 0$*, and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2}, \qquad R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor \qquad (0 \le i < N),$$

and define $Z^{\mathrm{spr}}_{i,j}$*,* $T^{\mathrm{long}}_{i,j}$*,* $T^{\mathrm{many}}_{i,j}$*,* $S^{\mathrm{ret}}_{i,j}$*, and* $X^{\mathrm{ret}}_{i,j}$ *as in Corollary 8.249. Assume, with* $\log_{D_H} 0 = -\infty$*, that for every*

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

and that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$Z^{\mathrm{spr}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad T^{\mathrm{many}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}, \qquad X^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.249 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the five tangentially localized profile families, with the only nonconstant multiplicative losses equal to $W_{\beta_i,p}$. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i, the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the assumed near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.399 (Tangentially localized original radial subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.398 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.398 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.400 (Adjacent-crossing localized shifted original radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor,$$

one has $R_i \ge 2$ for every $0 \le i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $Z^{\mathrm{spr}}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $C^{\eta,\zeta}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $X^{\mathrm{ret}}_{i,j}$ as in Corollary 8.250. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

and that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$Z^{\mathrm{spr}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}, \qquad X^{\mathrm{ret}}_{i,j}$$

and each of the four profiles

$$C^{\eta,\zeta}_{i,j} \qquad (\eta,\zeta \in \{-1,+1\})$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.250 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the same-sphere spread, long-run, adjacent-crossing, spread-return, and repeated-return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.401 (Adjacent-crossing localized original radial subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.400 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.400 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.402 (Return-absorbed shifted original radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \geq 1, \qquad M_0, \dots, M_{N-1} \geq 1, \qquad L_0, \dots, L_{N-1} \geq 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $P_i \geq 1$ for every $0 \leq i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $Z_{i,j}^{\mathrm{spr}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, $X_{i,j}^{\mathrm{ret}}$, $X_{i,j}^{\times}$, and $U_{i,j}^{\times}$ as in Corollary 8.251. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \leq a < m_{\lambda_0,\delta}^{G \to H}$$

one has

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G \to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G \to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

and that, for every $0 \leq i < N$ and $0 \leq j < J_{\sigma_i,\beta_i}$, each of

$$Z_{i,j}^{\mathrm{spr}}, \qquad T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}, \qquad X_{i,j}^{\mathrm{ret}}, \qquad X_{i,j}^{\times}, \qquad U_{i,j}^{\times}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

where $\mathfrak{A}$ denotes the profile under consideration. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G \to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.251 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G \to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the same-sphere spread, long-run, spread-return, same-level-return, upper-shifted-return, and repeated-return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G \to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.403 (Return-absorbed original radial subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.402 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.402 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.404 (Return-only shifted original radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $P_i \ge 1$ for every $0 \le i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $Z_{i,j}^{\mathrm{spr}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, $X_{i,j}^{\mathrm{ret}}$, and $B_{i,j}^{\times}$ as in Corollary 8.252. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \le a < m_{\lambda_0,\delta}^{G\to H}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, each of

$$Z_{i,j}^{\mathrm{spr}}, \qquad T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}, \qquad X_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$B^{\times}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.252 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the same-sphere spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.405 (Return-only original radial subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.404 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$, *set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, *that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.404 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.406 (Threshold-dominated return shifted original radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, *let* $\delta > 0$, *let* $\beta_0, \dots, \beta_{N-1} > 0$, *and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K^{\times}_i = P_i - 1,$$

one has

$$K_i^{\times} \geq K_i \qquad (0 \leq i < N).$$

Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $Z_{i,j}^{\mathrm{spr}}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as in Corollary 8.253. Assume, with* $\log_{D_H} 0 = -\infty$, *that for every*

$$0 \leq a < m_{\lambda_0,\delta}^{G\to H}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \leq i < N$ *and* $0 \leq j < J_{\sigma_i,\beta_i}$, *each of*

$$Z_{i,j}^{\mathrm{spr}}, \qquad T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ *and* $0 \leq j \leq J_{\sigma_i,\beta_i}$,

$$\widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.253 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the same-sphere spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.407 (Threshold-dominated return original radial subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.406 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$, *set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.406 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.408 (Outward-through reduced shifted original radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*, let* $\delta > 0$*, let* $\beta_0, \dots, \beta_{N-1} > 0$*, and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \ge K_i$ *for every* $0 \le i < N$*. Put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $U^{\mathrm{out}}_{i,j}$*,* $R^{\mathrm{spr}}_{i,j}$*,* $T^{\mathrm{long}}_{i,j}$*,* $S^{\mathrm{ret}}_{i,j}$*, and* $\widehat{X}^{\mathrm{ret}}_{i,j}$ *as in Corollary 8.254. Assume, with* $\log_{D_H} 0 = -\infty$*, that for every*

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$U^{\mathrm{out}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$*, each of*

$$R^{\mathrm{spr}}_{i,j}, \qquad \widehat{X}^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.254 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the outward-through spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.409 (Outward-through reduced original radial subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.408 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.408 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.410 (Record-outward shifted radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \ge K_i$ for every $0 \le i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $P^{\mathrm{rec}}_{i,j}$, $R^{\mathrm{spr}}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as in Corollary 8.255. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$P^{\mathrm{rec}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$*, each of*

$$R^{\mathrm{spr}}_{i,j}, \qquad \widehat{X}^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.255 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the record outward-through spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.411 (Record-outward subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.410 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.410 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.412 (Subterminal record radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*, let* $\delta > 0$*, let* $\beta_0, \dots, \beta_{N-1} > 0$*, and let*

$$K_0, \dots, K_{N-1} \geq 1, \qquad M_0, \dots, M_{N-1} \geq 1, \qquad L_0, \dots, L_{N-1} \geq 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \geq K_i$ *for every* $0 \leq i < N$*. Put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $P_{i,j}^{\mathrm{sub}}$*,* $R_{i,j}^{\mathrm{spr}}$*,* $T_{i,j}^{\mathrm{long}}$*,* $S_{i,j}^{\mathrm{ret}}$*, and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as in Corollary 8.256. Assume, with* $\log_{D_H} 0 = -\infty$*, that for every*

$$0 \leq a < m_{\lambda_0,\delta}^{G \to H}$$

one has

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G \to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G \to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that, for every $0 \leq i < N$ *and* $0 \leq j < J_{\sigma_i,\beta_i}$*, each of*

$$P_{i,j}^{\mathrm{sub}}, \qquad T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ *and* $0 \leq j \leq J_{\sigma_i,\beta_i}$*, each of*

$$R_{i,j}^{\mathrm{spr}}, \qquad \widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G \to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.256 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G \to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the subterminal record outward-through spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G \to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.413 (Subterminal record subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.412 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.412 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.414 (Persistent subterminal radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*, let* $\delta > 0$*, let* $\beta_0, \dots, \beta_{N-1} > 0$*, and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \ge K_i$ *for every* $0 \le i < N$*. Put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $P^{\mathrm{per}}_{i,j}$*,* $R^{\mathrm{spr},+}_{i,j}$*,* $T^{\mathrm{long}}_{i,j}$*,* $S^{\mathrm{ret}}_{i,j}$*, and* $\widehat{X}^{\mathrm{ret}}_{i,j}$ *as in Corollary 8.257. Assume, with* $\log_{D_H} 0 = -\infty$*, that for every*

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$P^{\mathrm{per}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$, *each of*

$$R_{i,j}^{\mathrm{spr},+}, \qquad \widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.257 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the persistent subterminal record spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.415 (Persistent subterminal subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.414 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$, *set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, *that for every* $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.414 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.416 (Deep persistent radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, *let* $\delta > 0$, *let* $\beta_0, \dots, \beta_{N-1} > 0$, *and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \geq K_i$ *for every* $0 \leq i < N$. *Put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $P_{i,j}^{\mathrm{deep}}$, $R_{i,j}^{\mathrm{spr},+}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as in Corollary 8.258. Assume, with* $\log_{D_H} 0 = -\infty$, *that for every*

$$0 \leq a < m_{\lambda_0,\delta}^{G\to H}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \leq i < N$ *and* $0 \leq j < J_{\sigma_i,\beta_i}$, *each of*

$$P_{i,j}^{\mathrm{deep}}, \qquad T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ *and* $0 \leq j \leq J_{\sigma_i,\beta_i}$, *each of*

$$R_{i,j}^{\mathrm{spr},+}, \qquad \widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.258 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the deep persistent subterminal record spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.417 (Deep persistent subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.416 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$, *set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, *that for every* $0 \leq a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.416 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.418 (Endpoint-prefix radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \geq 1, \qquad M_0, \dots, M_{N-1} \geq 1, \qquad L_0, \dots, L_{N-1} \geq 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \geq K_i$ for every $0 \leq i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $P^{\mathrm{pre}}_{i,j}$, $R^{\mathrm{spr},+}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as in Corollary 8.259. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \leq a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that, for every $0 \leq i < N$ and $0 \leq j < J_{\sigma_i,\beta_i}$, each of

$$P^{\mathrm{pre}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ and $0 \leq j \leq J_{\sigma_i,\beta_i}$, each of

$$R^{\mathrm{spr},+}_{i,j}, \qquad \widehat{X}^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

where $\mathfrak{A}$ denotes the profile under consideration. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.259 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the endpoint-prefix deep horizontal spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H))$$

for each fixed i. Hence the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.419 (Endpoint-prefix subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.418 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.418 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.420 (Short endpoint-prefix radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \ge K_i$ for every $0 \le i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $P^{\mathrm{pre,sh}}_{i,j}$, $R^{\mathrm{spr},+}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as in Corollary 8.260. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$P^{\mathrm{pre,sh}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$*, each of*

$$R^{\mathrm{spr},+}_{i,j}, \qquad \widehat{X}^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.260 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the short endpoint-prefix deep horizontal spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.421 (Short endpoint-prefix subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.420 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.420 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.422 (Short outward-prefix radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \geq 1, \qquad M_0, \dots, M_{N-1} \geq 1, \qquad L_0, \dots, L_{N-1} \geq 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \geq K_i$ for every $0 \leq i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $P^{\mathrm{pre,out,sh}}_{i,j}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as in Corollary 8.261. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \leq a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that, for every $0 \leq i < N$ and $0 \leq j < J_{\sigma_i,\beta_i}$, each of

$$P^{\mathrm{pre,out,sh}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ and $0 \leq j \leq J_{\sigma_i,\beta_i}$, each of

$$R^{\mathrm{spr},++}_{i,j}, \qquad \widehat{X}^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

where $\mathfrak{A}$ denotes the profile under consideration. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.261 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the short outward-prefix spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.423 (Short outward-prefix subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.422 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.422 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.424 (Short prefix-record radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \ge K_i$ for every $0 \le i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $P^{\mathrm{pre,rec,sh}}_{i,j}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as in Corollary 8.262. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$P^{\mathrm{pre,rec,sh}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$*, each of*

$$R^{\mathrm{spr},++}_{i,j}, \qquad \widehat{X}^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.262 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the short prefix-record spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.425 (Short prefix-record subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.424 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.424 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.426 (Short persistent-prefix radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \geq 1, \qquad M_0, \dots, M_{N-1} \geq 1, \qquad L_0, \dots, L_{N-1} \geq 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \geq K_i$ for every $0 \leq i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $P_{i,j}^{\mathrm{pre,per,sh}}$, $R_{i,j}^{\mathrm{spr},++}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, and $\widehat{X}_{i,j}^{\mathrm{ret}}$ as in Corollary 8.263. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \leq a < m_{\lambda_0,\delta}^{G \to H}$$

one has

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G \to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G \to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

that, for every $0 \leq i < N$ and $0 \leq j < J_{\sigma_i,\beta_i}$, each of

$$P_{i,j}^{\mathrm{pre,per,sh}}, \qquad T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ and $0 \leq j \leq J_{\sigma_i,\beta_i}$, each of

$$R_{i,j}^{\mathrm{spr},++}, \qquad \widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p \uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2 + 7\alpha_\omega)s,$$

where $\mathfrak{A}$ denotes the profile under consideration. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G \to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.263 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G \to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the short persistent-prefix spread, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G \to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.427 (Short persistent-prefix subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.426 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.426 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.428 (Ordered prefix-ladder radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, let $\delta > 0$, let $\beta_0, \dots, \beta_{N-1} > 0$, and let

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \ge K_i$ for every $0 \le i < N$. Put

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define $P^{\mathrm{pre,lad,sh}}_{i,j}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as in Corollary 8.264. Assume, with $\log_{D_H} 0 = -\infty$, that for every

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, each of

$$P^{\mathrm{pre,lad,sh}}_{i,j}, \qquad T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$, *each of*

$$R_{i,j}^{\mathrm{spr},++}, \qquad \widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.264 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the ordered short persistent prefix-ladder, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.429 (Ordered prefix-ladder subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.428 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$, *set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, *that for every* $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.428 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.430 (Bounded-gap prefix-pair radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$, *let* $\delta > 0$, *let* $\beta_0, \dots, \beta_{N-1} > 0$, *and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \geq K_i$ *for every* $0 \leq i < N$. *Put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define $P_{i,j}^{\mathrm{pre,pair,sh}}$, $R_{i,j}^{\mathrm{spr},++}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as in Corollary 8.265. Assume, with* $\log_{D_H} 0 = -\infty$, *that for every*

$$0 \leq a < m_{\lambda_0,\delta}^{G\to H}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \leq i < N$ *and* $0 \leq j < J_{\sigma_i,\beta_i}$, *each of*

$$P_{i,j}^{\mathrm{pre,pair,sh}}, \qquad T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ *and* $0 \leq j \leq J_{\sigma_i,\beta_i}$, *each of*

$$R_{i,j}^{\mathrm{spr},++}, \qquad \widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.265 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bounded-gap short persistent prefix-pair, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.431 (Bounded-gap prefix-pair subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.430 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$, *set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.430 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.432 (Fixed-gap prefix-pair radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*, let* $\delta > 0$*, let* $\beta_0, \dots, \beta_{N-1} > 0$*, and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \ge K_i$ *for every* $0 \le i < N$*. Put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \le i < N),$$

and define B_i*,* $P^{\mathrm{pre,gap,sh}}_{i,j,h}$*,* $R^{\mathrm{spr},++}_{i,j}$*,* $T^{\mathrm{long}}_{i,j}$*,* $S^{\mathrm{ret}}_{i,j}$*, and* $\widehat{X}^{\mathrm{ret}}_{i,j}$ *as in Corollary 8.266. Assume, with* $\log_{D_H} 0 = -\infty$*, that for every*

$$0 \le a < m^{G\to H}_{\lambda_0,\delta}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}^{G\to H,\mathrm{rad}}_{a,\delta}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_N}(s,p)}{\log_{D_H} \overline{\chi}^H_p} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \le i < N$*,* $0 \le j < J_{\sigma_i,\beta_i}$*, and* $1 \le h \le B_i$*,*

$$P^{\mathrm{pre,gap,sh}}_{i,j,h}$$

satisfies the same logarithmic exponent bound, that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, each of*

$$T^{\mathrm{long}}_{i,j}, \qquad S^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$*, each of*

$$R^{\mathrm{spr},++}_{i,j}, \qquad \widehat{X}^{\mathrm{ret}}_{i,j}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.266 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the fixed-gap short persistent prefix-pair, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. The additional sums over $1 \le h \le B_i$ are finite and independent of p. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.433 (Fixed-gap prefix-pair subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.432 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.432 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.434 (Confined fixed-gap bridge radial tail criterion)**.** *In the setting of Corollary 8.125, let*

$$\lambda_0 < \lambda_1 < \cdots < \lambda_N$$

with $\lambda_0 > \rho/2$*, let* $\delta > 0$*, let* $\beta_0, \dots, \beta_{N-1} > 0$*, and let*

$$K_0, \dots, K_{N-1} \ge 1, \qquad M_0, \dots, M_{N-1} \ge 1, \qquad L_0, \dots, L_{N-1} \ge 1$$

be integers such that, with

$$R_i = \left\lfloor \frac{M_i}{L_i} \right\rfloor, \qquad P_i = \left\lfloor \frac{R_i - 2}{4} \right\rfloor, \qquad K_i^{\times} = P_i - 1,$$

one has $K_i^{\times} \geq K_i$ *for every* $0 \leq i < N$. *Put*

$$\sigma_i = \frac{\lambda_{i+1}}{2} \qquad (0 \leq i < N),$$

and define B_i, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr},++}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as in Corollary 8.267. Assume, with* $\log_{D_H} 0 = -\infty$, *that for every*

$$0 \leq a < m_{\lambda_0,\delta}^{G\to H}$$

one has

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{E}_{a,\delta}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{Y}_{Q,\lambda_N}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

that, for every $0 \leq i < N$, $0 \leq j < J_{\sigma_i,\beta_i}$, *and* $1 \leq h \leq B_i$,

$$P_{i,j,h}^{\mathrm{pre,br,sh}}$$

satisfies the same logarithmic exponent bound, that, for every $0 \leq i < N$ *and* $0 \leq j < J_{\sigma_i,\beta_i}$, *each of*

$$T_{i,j}^{\mathrm{long}}, \qquad S_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \leq i < N$ *and* $0 \leq j \leq J_{\sigma_i,\beta_i}$, *each of*

$$R_{i,j}^{\mathrm{spr},++}, \qquad \widehat{X}_{i,j}^{\mathrm{ret}}$$

satisfies the same logarithmic exponent bound

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{A}(s,p)}{\log_{D_H} \overline{\chi}_p^H} < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s,$$

where $\mathfrak{A}$ *denotes the profile under consideration. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.267 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the confined fixed-gap bridge, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. The sums over $1 \leq h \leq B_i$ are finite and independent of p. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.435 (Confined bridge subdivision criterion). *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.434 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.434 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.436 (Adjacent-stacked bridge radial tail criterion). *In the setting of Corollary 8.434, use the same parameters and define B_i, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. Also define*

$$P^{\mathrm{pre,adj,sh}}_{i,j}(s,p) = \mathfrak{P}^{G\to H,\mathrm{rad,tan,pre,out,per,adj,short}}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,L_i}(s,p).$$

Assume all hypotheses of Corollary 8.434, except that the bound on $P^{\mathrm{pre,br,sh}}_{i,j,h}$ is required only for $2 \le h \le B_i$, and for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$ the adjacent stacked profile

$$P^{\mathrm{pre,adj,sh}}_{i,j}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.268 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the adjacent stacked, confined fixed-gap bridge $h \ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.437 (Adjacent-stacked bridge subdivision criterion). *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.436 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.436 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.438 (Outward-staircase bridge radial tail criterion)**.** *In the setting of Corollary 8.436, use the same parameters and define* B_i, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr},++}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. Also define*

$$P_{i,j}^{\mathrm{pre,stair,sh}}(s,p) = \mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,2L_i+1}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,short}}(s,p).$$

Assume all hypotheses of Corollary 8.436, except that the adjacent stacked profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,stair,sh}}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.269 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the two-level outward staircase, confined fixed-gap bridge $h \ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the extra factor 2 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.439 (Staircase subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.438 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$, *set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, *that for every* $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.438 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.440 (Locally short staircase radial tail criterion)**.** *In the setting of Corollary 8.438, use the same parameters and define* B_i, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr},++}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. Also define*

$$P_{i,j}^{\mathrm{pre,stair,loc,sh}}(s,p)=\mathfrak{P}_{Q,\lambda_i,\sigma_i,\beta_i,j,256M_iB_i,L_i}^{G\to H,\mathrm{rad,tan,pre,out,per,stair,loc,short}}(s,p).$$

Assume all hypotheses of Corollary 8.438, except that the two-level outward staircase profile bound is replaced by the requirement that, for every $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,stair,loc,sh}}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.270 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the locally short staircase, confined fixed-gap bridge $h\ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the coefficient 3 on the long-run family is fixed. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.441 (Locally short staircase subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.440 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}}>0$, *set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}=\left\lfloor\frac{\rho}{2\delta_{\mathrm{fin}}}\right\rfloor+1,$$

and assume, with $\log_{D_H}0=-\infty$, *that for every* $0\le a<m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)}\frac{\log_{D_H}\mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H}\overline{\chi}_p^H}+a\delta_{\mathrm{fin}}\log_{D_H}Q<-c_s\log_{D_H}p_{\mathrm{c}}(H)-(2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G)<p_{2\to 2}(G).$$

Proof. Corollary 8.440 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta=\delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.442 (Fixed-gap staircase-pair radial tail criterion)**.** *In the setting of Corollary 8.440, use the same parameters and define* B_i, $P_{i,j,h}^{\mathrm{pre,br,sh}}$, $R_{i,j}^{\mathrm{spr},++}$, $T_{i,j}^{\mathrm{long}}$, $S_{i,j}^{\mathrm{ret}}$, *and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. Put*

$$C_i=\left\lceil\frac{4096M_iB_i\beta_i}{\theta_{\lambda_i,\sigma_i,\beta_i}}\right\rceil+1$$

and define $P_{i,j,h}^{\mathrm{pre,stair,loc,gap,sh}}$ *as in Corollary 8.271. Assume all hypotheses of Corollary 8.440, except that the locally short staircase profile bound is replaced by the requirement that, for every* $0\le i<N$, $0\le j<J_{\sigma_i,\beta_i}$, *and* $1\le h\le C_i$,

$$P_{i,j,h}^{\mathrm{pre,stair,loc,gap,sh}}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.271 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the fixed-gap locally short staircase-pair, confined fixed-gap bridge $h\ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the coefficient 3 on the long-run family is fixed. The sums over $1\le h\le C_i$ are finite and independent of p. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.443 (Fixed-gap staircase-pair subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.442 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}}>0$*, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}=\left\lfloor\frac{\rho}{2\delta_{\mathrm{fin}}}\right\rfloor+1,$$

and assume, with $\log_{D_H}0=-\infty$*, that for every* $0\le a<m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)}\frac{\log_{D_H}\mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H}\overline{\chi}_p^H}+a\delta_{\mathrm{fin}}\log_{D_H}Q<-c_s\log_{D_H}p_{\mathrm{c}}(H)-(2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G)<p_{2\to 2}(G).$$

Proof. Corollary 8.442 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta=\delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.444 (Staircase-chain/bridge radial tail criterion)**.** *In the setting of Corollary 8.442, use the same parameters and define* B_i*,* C_i*,* $P_{i,j,h}^{\mathrm{pre,br,sh}}$*,* $R_{i,j}^{\mathrm{spr},++}$*,* $T_{i,j}^{\mathrm{long}}$*,* $S_{i,j}^{\mathrm{ret}}$*, and* $\widehat{X}_{i,j}^{\mathrm{ret}}$ *as there. Define* $P_{i,j,q}^{\mathrm{pre,stair,loc,chain,sh}}$ *and* $P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}}$ *as in Corollary 8.272. Assume all hypotheses of Corollary 8.442, except that the fixed-gap locally short staircase-pair profile bound is replaced by the requirement that, for every* $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$*,*

$$P_{i,j,q}^{\mathrm{pre,stair,loc,chain,sh}} \qquad (q=3,4)$$

and

$$P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}} \qquad (3\le h\le C_i)$$

satisfy the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.272 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the locally short staircase-chain, confined locally short staircase-bridge, confined fixed-gap bridge $h\ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative

losses are the factors $W_{\beta_i,p}$, and the coefficient 3 on the long-run family is fixed. The additional sums over $q \in \{3,4\}$ and $3 \le h \le C_i$ are finite and independent of p. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.445 (Chain-absorbed staircase-chain/bridge radial tail criterion)**.** *In the setting of Corollary 8.442, use the same parameters and define B_i, C_i, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. Define $P^{\mathrm{pre,rec,ch}}_{i,j}$ and $P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}$ as in Corollary 8.273. Assume all hypotheses of Corollary 8.442, except that the fixed-gap locally short staircase-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,*

$$P^{\mathrm{pre,rec,ch}}_{i,j}$$

and, for every $3 \le h \le C_i$,

$$P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}$$

satisfy the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.273 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the chain-depth prefix-record outward-spread, confined locally short staircase-bridge, confined fixed-gap bridge $h \ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The coefficient 2 on the prefix-record family is fixed, the additional sums over $3 \le h \le C_i$ are finite and independent of p, and the only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.446 (Persistent chain-absorbed staircase-chain/bridge radial tail criterion)**.** *In the setting of Corollary 8.445, use the same parameters and define $P^{\mathrm{pre,per,ch}}_{i,j}$ and $R^{\mathrm{spr,ch}}_{i,j}$ as in Corollary 8.274. Assume all hypotheses of Corollary 8.445, except that the chain-depth prefix-record profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,*

$$P^{\mathrm{pre,per,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R^{\mathrm{spr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.274 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the chain-depth persistent-prefix profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, confined fixed-gap bridge $h \geq 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. The new coefficients are fixed, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.447 (Bounded-gap persistent chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.446, use the same parameters and define B_i^{ch}, $P_{i,j}^{\mathrm{pre,pair,ch}}$, and $R_{i,j}^{\mathrm{spr,ch}}$ as in Corollary 8.275. Assume all hypotheses of Corollary 8.446, except that the chain-depth persistent-prefix profile bound is replaced by the requirement that, for every $0 \leq i < N$ and $0 \leq j < J_{\sigma_i,\beta_i}$,*

$$P_{i,j}^{\mathrm{pre,pair,ch}}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.275 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bounded-gap chain-depth prefix-pair profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, confined fixed-gap bridge $h \geq 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. The new cutoff B_i^{ch} is fixed for each i, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.448 (Fixed-gap persistent chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.447, use the same parameters and define B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $P_{i,j,h}^{\mathrm{pre,gap,ch}}$, and $R_{i,j}^{\mathrm{spr,ch}}$ as in Corollary 8.276. Assume all hypotheses of Corollary 8.447, except that the bounded-gap chain-depth prefix-pair profile bound is replaced by the requirement that, for every $0 \leq i < N$, $0 \leq j < J_{\sigma_i,\beta_i}$, and $1 \leq h \leq B_i^{\mathrm{ch}}$,*

$$P_{i,j,h}^{\mathrm{pre,gap,ch}}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.276 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the fixed-gap chain-depth prefix-pair profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, confined fixed-gap bridge $h \geq 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the sum over $1 \leq h \leq B_i^{\mathrm{ch}}$ is finite and independent of

p, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.449 (Confined-bridge persistent chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.448, use the same parameters and define* B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $P_{i,j,h}^{\mathrm{pre,br,ch}}$, *and* $R_{i,j}^{\mathrm{spr,ch}}$ *as in Corollary 8.277. Assume all hypotheses of Corollary 8.448, except that the fixed-gap chain-depth prefix-pair profile bound is replaced by the requirement that, for every* $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,ch}}$$

satisfies the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.277 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the chain-depth confined fixed-gap bridge profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the sum over $1 \le h \le B_i^{\mathrm{ch}}$ is finite and independent of p, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.450 (Inward-free bridge persistent chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.449, use the same parameters and define* B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $P_{i,j,h}^{\mathrm{pre,br,free,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, *and* $R_{i,j}^{\mathrm{spr,ch}}$ *as in Corollary 8.278. Assume all hypotheses of Corollary 8.449, except that the chain-depth confined fixed-gap bridge profile bound is replaced by the requirement that, for every* $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{ch}}$, *both*

$$P_{i,j,h}^{\mathrm{pre,br,free,ch}} \qquad \textit{and} \qquad R_{i,j,h}^{\mathrm{spr,br,ch}}$$

satisfy the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.278 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the chain-depth inward-free confined bridge profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. The two new chain-depth sums range over $1 \le h \le B_i^{\mathrm{ch}}$, which is finite and independent of p. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.451 (Record-absorbed inward-free bridge persistent chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.450, use the same parameters and define* B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $P_{i,j}^{\mathrm{pre,rec,br,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, *and* $R_{i,j}^{\mathrm{spr,ch}}$ *as in Corollary 8.279. Assume all hypotheses of Corollary 8.450, except that the chain-depth inward-free confined bridge profile bound is replaced by the requirement that, for every* $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.279 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the chain-depth bridge-record prefix-record profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h\ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficient $2B_i^{\mathrm{ch}}$ and the enlarged long-run coefficient $2B_i^{\mathrm{ch}}+3$ are fixed. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.452 (Persistent bridge-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.451, use the same parameters and define* B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $P_{i,j}^{\mathrm{pre,per,br,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, *and* $R_{i,j}^{\mathrm{spr,ch}}$ *as in Corollary 8.280. Assume all hypotheses of Corollary 8.451, except that the bridge-record prefix-record profile bound is replaced by the requirement that, for every* $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,ch}}$$

satisfies the same logarithmic exponent bound, and that, for every $0\le i<N$ *and* $0\le j\le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.280 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record persistent prefix profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h\ge 2$, long-run profiles, spread-return profiles, and ordinary

exact-level return profiles. For each i, the coefficients $2B_i^{\mathrm{ch}}$ and $2B_i^{\mathrm{ch}}+3$ are fixed. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.453 (Bounded-gap bridge-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.452, use the same parameters and define* B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $B_i^{\mathrm{br,ch}}$, $P_{i,j}^{\mathrm{pre,pair,br,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, *and* $R_{i,j}^{\mathrm{spr,ch}}$ *as in Corollary 8.281. Assume all hypotheses of Corollary 8.452, except that the bridge-record persistent prefix profile bound is replaced by the requirement that, for every* $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,ch}}$ *and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.281 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bounded-gap bridge-record prefix-pair profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h\ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients $2B_i^{\mathrm{ch}}$, $B_i^{\mathrm{br,ch}}$, and $2B_i^{\mathrm{ch}}+3$ are fixed. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.454 (Fixed-gap bridge-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.453, use the same parameters and define* B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $B_i^{\mathrm{br,ch}}$, $K_i^{\mathrm{gap,br,ch}}$, $P_{i,j,h}^{\mathrm{pre,gap,br,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, *and* $R_{i,j}^{\mathrm{spr,ch}}$ *as in Corollary 8.282. Assume all hypotheses of Corollary 8.453, except that the bounded-gap bridge-record prefix-pair profile bound is replaced by the requirement that, for every* $0\le i<N$, $0\le j<J_{\sigma_i,\beta_i}$, *and* $1\le h\le B_i^{\mathrm{br,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,ch}}$ *and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.282 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the fixed-gap bridge-record prefix-pair profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles,

earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients $2B_i^{\mathrm{ch}}$, $B_i^{\mathrm{br,ch}}$, and $2B_i^{\mathrm{ch}}+3$ are fixed, as is the sum over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.455 (Confined bridge-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.454, use the same parameters and define* B_i^{ch}, $K_i^{\mathrm{gap,ch}}$, $B_i^{\mathrm{br,ch}}$, $K_i^{\mathrm{gap,br,ch}}$, $P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,ch}}$, *and* $R_{i,j}^{\mathrm{spr,ch}}$ *as in Corollary 8.283. Assume all hypotheses of Corollary 8.454, except that the fixed-gap bridge-record prefix-pair profile bound is replaced by the requirement that, for every* $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,ch}}$ *and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.283 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record confined fixed-gap bridge profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients $2B_i^{\mathrm{ch}}$, $B_i^{\mathrm{br,ch}}$, and $2B_i^{\mathrm{ch}}+3$ are fixed, as is the sum over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.456 (Inward-free bridge-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.455, use the same parameters and define the families*

$$\begin{gathered} B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}}, \\ P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}} \end{gathered}$$

as in Corollary 8.284. Assume all hypotheses of Corollary 8.455, except that the bridge-record confined fixed-gap bridge profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,ch}}$, *both*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,ch}} \qquad \textit{and} \qquad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,ch}}$ *and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.284 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record inward-free confined bridge profiles, bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients $2B_i^{\mathrm{ch}}$, $B_i^{\mathrm{br,ch}}$, and $2B_i^{\mathrm{ch}}+3$ are fixed, as are the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.457 (Record-absorbed inward-free bridge-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.456, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$P_{i,j}^{\mathrm{pre,rec,br,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.285. Assume all hypotheses of Corollary 8.456, except that the bridge-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}$*,* $R_{i,j}^{\mathrm{spr,recbr,ch}}$*, and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.285 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record prefix-record profiles, bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}, \qquad 2B_i^{\mathrm{ch}}, \qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3$$

are fixed, as are the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.458 (Persistent bridge-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.457, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$P_{i,j}^{\mathrm{pre,per,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.286. Assume all hypotheses of Corollary 8.457, except that the bridge-record prefix-record profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, and $R_{i,j,h}^{\mathrm{spr,br,ch}}$ are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.286 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record persistent prefix profiles, the new bridge-record persistent-prefix return-spread profiles, bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}, \qquad 2B_i^{\mathrm{ch}}, \qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3$$

are fixed, as are the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.459 (Bounded-gap bridge-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.458, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.287. Assume all hypotheses of Corollary 8.458, except that the bridge-record persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, *and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.287 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bounded-gap bridge-record-record-record prefix-pair profiles, the new bridge-record persistent-prefix return-spread profiles, bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}, \qquad 2B_i^{\mathrm{ch}}, \qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3$$

are fixed, as are the cutoff $B_i^{\mathrm{br,recbr,ch}}$ and the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.460 (Fixed-gap bridge-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.459, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.288. Assume all hypotheses of Corollary 8.459, except that the bounded-gap bridge-record-record prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, *and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.288 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the fixed-gap bridge-record-record prefix-pair profiles, the new bridge-record persistent-prefix return-spread profiles, bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth

bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}, \qquad 2B_i^{\mathrm{ch}}, \qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, the sums over $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.461 (Confined bridge-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.460, use the same parameters and define the families*

$$\begin{gathered} B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}}, \\ B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}, \\ R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}} \end{gathered}$$

as in Corollary 8.289. Assume all hypotheses of Corollary 8.460, except that the fixed-gap bridge-record-record prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, and $R_{i,j,h}^{\mathrm{spr,br,ch}}$ are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.289 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record-record confined fixed-gap bridge profiles, the new bridge-record persistent-prefix return-spread profiles, bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}, \qquad 2B_i^{\mathrm{ch}}, \qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, the sums over $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.462 (Inward-free bridge-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.461, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.290. Assume all hypotheses of Corollary 8.461, except that the bridge-record-record confined fixed-gap bridge profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, *both*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,ch}} \qquad \text{and} \qquad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The hypotheses on $R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}$, $R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}$, $R_{i,j}^{\mathrm{spr,recbr,ch}}$, *and* $R_{i,j,h}^{\mathrm{spr,br,ch}}$ *are retained. Then the exponent hypothesis on* $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.290 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record-record inward-free confined bridge profiles, bridge-record-record bridge-return spread profiles, the new bridge-record persistent-prefix return-spread profiles, earlier bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}, \qquad 2B_i^{\mathrm{ch}}, \qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}} + 2B_i^{\mathrm{ch}} + 3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, the sums over $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.463 (Record-absorbed inward-free bridge-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.462, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.291. Assume all hypotheses of Corollary 8.462, except that the bridge-record-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.291 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record-record prefix-record profiles, bridge-record-record bridge-return spread profiles, the bridge-record persistent-prefix return-spread profiles, earlier bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}},\qquad 2B_i^{\mathrm{ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, the sums over $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.464 (Persistent bridge-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.463, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.292. Assume all hypotheses of Corollary 8.463, except that the bridge-record-record prefix-record profile bound is replaced by the requirements that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.292 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the bridge-record-record persistent prefix profiles and the new bridge-record-record persistent-prefix return-spread profiles. It also retains the bridge-record-record bridge-return spread profiles, the bridge-record persistent-prefix return-spread profiles, earlier bridge-record bridge-return spread profiles, bridge-record spread-return profiles, chain-depth bridge-return spread profiles, chain-depth spread-return profiles, confined locally short staircase-bridge profiles, earlier confined fixed-gap bridge profiles $h \ge 2$, long-run profiles, spread-return profiles, and ordinary exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}},\qquad 2B_i^{\mathrm{ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, the sums over $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.465 (Bounded-gap bridge-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.464, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.293. Assume all hypotheses of Corollary 8.464, except that the bridge-record-record persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.293 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the bounded-gap prefix-pair family $P^{\mathrm{pre,pair,br,recbr,recbr,ch}}$, the new return-spread family $R^{\mathrm{spr,recbr,recbr,recbr,ch}}$, all retained bridge-return and spread-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}},\qquad 2B_i^{\mathrm{ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, $B_i^{\mathrm{br,recbr,recbr,ch}}$, the sums over $1\le h\le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1\le h\le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.466 (Fixed-gap bridge-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.465, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.294. Assume all hypotheses of Corollary 8.465, except that the bounded-gap bridge-record-record-record prefix-pair profile bound is replaced by the requirement that, for every $0\le i<N$, $0\le j<J_{\sigma_i,\beta_i}$, *and* $1\le h\le B_i^{\mathrm{br,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.294 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the families named in its conclusion: the fixed-gap family $P^{\mathrm{pre,gap,br,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}},\qquad 2B_i^{\mathrm{ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, $B_i^{\mathrm{br,recbr,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,recbr,ch}}$, the sums over $1\le h\le B_i^{\mathrm{br,recbr,recbr,ch}}$, the sums over $1\le h\le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1\le h\le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.467 (Confined bridge-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.466, use the same parameters and define the families*

$$\begin{gathered}
B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}
\end{gathered}$$

as in Corollary 8.295. Assume all hypotheses of Corollary 8.466, except that the fixed-gap bridge-record-record-record prefix-pair profile bound is replaced by the requirement that, for every $0\le i<N$, $0\le j<J_{\sigma_i,\beta_i}$, and $1\le h\le B_i^{\mathrm{br,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$\begin{gathered}
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.
\end{gathered}$$

Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.295 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the families named in its conclusion: the confined bridge family $P^{\mathrm{pre,br,recbr,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}},\qquad 2B_i^{\mathrm{ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

are fixed, as are the cutoffs $B_i^{\mathrm{br,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,ch}}$, $B_i^{\mathrm{br,recbr,recbr,ch}}$, $K_i^{\mathrm{gap,br,recbr,recbr,ch}}$, the sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,ch}}$, the sums over $1 \le h \le B_i^{\mathrm{br,recbr,ch}}$, and the sums over $1 \le h \le B_i^{\mathrm{br,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.468 (Inward-free bridge-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.467, use the same parameters and define the families*

$$\begin{gathered}
B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}
\end{gathered}$$

as in Corollary 8.296. Assume all hypotheses of Corollary 8.467, except that the bridge-record-record-record confined fixed-gap bridge profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,ch}}$, both

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,ch}} \qquad \text{and} \qquad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$\begin{gathered}
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.
\end{gathered}$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.296 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the inward-free bridge family $P^{\mathrm{pre,br,free,recbr,recbr,recbr,ch}}$, the new bridge-return family $R^{\mathrm{spr,br,recbr,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}},\qquad 2B_i^{\mathrm{ch}},\qquad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

are fixed, as are all B_i- and K_i-cutoffs appearing in the finite ladder. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.469 (Record-absorbed inward-free bridge-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.468, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.297. Assume all hypotheses of Corollary 8.468, except that the bridge-record-record-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.297 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the families named in its conclusion: the prefix-record family $P^{\mathrm{pre,rec,br,recbr,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}B_i^{\mathrm{br,recbr,recbr,ch}},\quad 2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}},$$
$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}},\quad 2B_i^{\mathrm{ch}},$$
$$2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}B_i^{\mathrm{br,recbr,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}B_i^{\mathrm{br,recbr,ch}}+2B_i^{\mathrm{ch}}B_i^{\mathrm{br,ch}}+2B_i^{\mathrm{ch}}+3$$

are fixed, as are all B_i- and K_i-cutoffs appearing in the finite ladder. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.470 (Persistent bridge-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.469, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.298. Assume all hypotheses of Corollary 8.469, except that the bridge-record-record-record prefix-record profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.298 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the persistent prefix family $P^{\mathrm{pre,per,br,recbr,recbr,recbr,ch}}$, the new adjacent return-spread family $R^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients $D_i^{\mathrm{rec,br,recbr,recbr,ch}}$, $A_i^{\mathrm{long,br,recbr,recbr,ch}}$, and all retained finite-ladder coefficients are fixed, as are all B_i- and K_i-cutoffs. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.471 (Bounded-gap bridge-record-record-record-record chain-absorbed radial tail cri-

terion). *In the setting of Corollary 8.470, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.299. Assume all hypotheses of Corollary 8.470, except that the bridge-record-record-record persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.299 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the bounded-gap prefix-pair family $P^{\mathrm{pre,pair,br,recbr,recbr,recbr,ch}}$, the new adjacent return-spread family $R^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients $D_i^{\mathrm{rec,br,recbr,recbr,ch}}$, $A_i^{\mathrm{long,br,recbr,recbr,ch}}$, and all retained finite-ladder coefficients are fixed, as are all B_i- and K_i-cutoffs including $B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.472 (Fixed-gap bridge-record-record-record-record chain-absorbed radial tail crite-

rion). *In the setting of Corollary 8.471, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.300. Assume all hypotheses of Corollary 8.471, except that the bridge-record-record-record bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.300 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the fixed-gap prefix-pair family $P^{\mathrm{pre,gap,br,recbr,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,ch}},$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including $K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}$, and of the finite sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.473 (Confined bridge-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.472, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.301. Assume all hypotheses of Corollary 8.472, except that the fixed-gap bridge-record-record-record-record prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.301 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the confined bridge family $P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,ch}}$, all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,ch}},$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including $K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}$, and of the finite sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.474 (Inward-free bridge-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.473, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.302. Assume all hypotheses of Corollary 8.473, except that the bridge-record-record-record-record confined fixed-gap bridge profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$, both

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,ch}} \qquad \text{and} \qquad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.302 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the inward-free bridge family

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,ch}},$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including $K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}$, and of the finite sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.475 (Record-absorbed inward-free bridge-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.474, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\ R_{i,j}^{\mathrm{spr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.303. Assume all hypotheses of Corollary 8.474, except that the bridge-record-record-record-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.303 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the prefix-record family

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\qquad D_i^{\mathrm{rec,br,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including $K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.476 (Persistent bridge-record-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.475, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.304. Assume all hypotheses of Corollary 8.475, except that the bridge-record-record-record-record prefix-record profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.304 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the persistent prefix family

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\qquad D_i^{\mathrm{rec,br,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including $K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}$. The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.477 (Bounded-gap bridge-record-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.476, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.305. Assume all hypotheses of Corollary 8.476, except that the bridge-record-record-record-record persistent prefix profile bound is replaced by the requirement that, for every $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.305 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the families named in its conclusion: the bounded-gap prefix-pair family

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\qquad D_i^{\mathrm{rec,br,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}, \qquad K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}}.$$

The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.478 (Fixed-gap bridge-record-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.477, use the same parameters and define the families*

$$\begin{gathered}
B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}
\end{gathered}$$

as in Corollary 8.306. Assume all hypotheses of Corollary 8.477, except that the bridge-record-record-record-record-record bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$\begin{gathered}
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.
\end{gathered}$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.306 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the fixed-gap prefix-pair family

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}}, \qquad D_i^{\mathrm{rec,br,recbr,recbr,ch}}, \qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.479 (Confined bridge-record-record-record-record-record chain-absorbed radial tail criterion)**.** *In the setting of Corollary 8.478, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.307. Assume all hypotheses of Corollary 8.478, except that the bridge-record-record-record-record-record fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.307 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the confined bridge family

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\qquad D_i^{\mathrm{rec,br,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1\le h\le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.480 (Inward-free bridge-record-record-record-record-record chain-absorbed radial tail

criterion). *In the setting of Corollary 8.479, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.308. Assume all hypotheses of Corollary 8.479, except that the bridge-record-record-record-record-record confined fixed-gap bridge profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$, *both*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,ch}} \qquad \text{and} \qquad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.308 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the inward-free bridge family

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,ch}},$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.481 (Fifth record-absorbed inward-free bridge radial tail criterion)**.** *In the setting of Corollary 8.480, use the same parameters and define the families*

$$\begin{gathered}
B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}
\end{gathered}$$

as in Corollary 8.309. Assume all hypotheses of Corollary 8.480, except that the bridge-record-record-record-record-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$\begin{gathered}
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.
\end{gathered}$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.309 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the prefix-record family

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},\qquad D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},\qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}.$$

The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.482 (Sixth persistent bridge radial tail criterion)**.** *In the setting of Corollary 8.481, use the same parameters and define the families*

$$\begin{gathered}
B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},\\
B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},\\
K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\\
P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\\
R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}
\end{gathered}$$

as in Corollary 8.310. Assume all hypotheses of Corollary 8.481, except that the bridge-record-record-record-record-record prefix-record profile bound is replaced by the requirement that, for every $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and that, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}, \quad R_{i,j}^{\mathrm{spr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.310 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the persistent prefix family

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}}.$$

The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.483 (Sixth bounded-gap bridge radial tail criterion)**.** *In the setting of Corollary 8.482, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.311. Assume all hypotheses of Corollary 8.482, except that the bridge-record-record-record-record-record persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.311 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the bounded-gap prefix-pair family

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}}.$$

The only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.484 (Sixth fixed-gap bridge radial tail criterion)**.** *In the setting of Corollary 8.483, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.312. Assume all hypotheses of Corollary 8.483, except that the bridge-record-record-record-record-record-record bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}, \quad R_{i,j}^{\mathrm{spr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.312 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the fixed-gap prefix-pair family

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.485 (Sixth confined bridge radial tail criterion)**.** *In the setting of Corollary 8.484, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.313. Assume all hypotheses of Corollary 8.484, except that the bridge-record-record-record-record-record-record fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},\quad R_{i,j}^{\mathrm{spr,recbr,ch}},\quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.313 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the confined bridge family

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.486 (Sixth inward-free bridge radial tail criterion)**.** *In the setting of Corollary 8.485, use the same parameters and define the families*

$$B_i^{\mathrm{ch}},\ K_i^{\mathrm{gap,ch}},\ B_i^{\mathrm{br,ch}},\ K_i^{\mathrm{gap,br,ch}},$$
$$B_i^{\mathrm{br,recbr,ch}},\ K_i^{\mathrm{gap,br,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,ch}},\ B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,ch}},\ R_{i,j,h}^{\mathrm{spr,br,ch}},\ R_{i,j}^{\mathrm{spr,ch}}$$

as in Corollary 8.314. Assume all hypotheses of Corollary 8.485, except that the bridge-record-record-record-record-record-record confined fixed-gap bridge profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and* $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$, *both*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \quad \textit{and} \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The retained spread-profile hypotheses are those for

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,recbr,ch}}, \quad R_{i,j}^{\mathrm{spr,recbr,ch}}, \quad R_{i,j,h}^{\mathrm{spr,br,ch}}.$$

Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.314 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the families named in its conclusion: the inward-free bridge family

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.487 (Sixth record-absorbed inward-free bridge radial tail criterion)**.** *In the setting of Corollary 8.486, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.315. Assume all hypotheses of Corollary 8.486, except that the bridge-record-record-record-record-record-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The bridge-record-record-record-record-record-record bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.486 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.315 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.488 (Seventh persistent bridge radial tail criterion)**.** *In the setting of Corollary 8.487, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.316. Assume all hypotheses of Corollary 8.487, except that the bridge-record-record-record-record-record-record prefix-record profile bound is replaced by the following two requirements: for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$*,*

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The bridge-record-record-record-record-record-record bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.487 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.316 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.489 (Seventh bounded-gap bridge radial tail criterion)**.** *In the setting of Corollary 8.488, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.317. Assume all hypotheses of Corollary 8.488, except that the bridge-record-record-record-record-record-record persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The new adjacent return-spread profiles

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.488 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.317 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.490 (Seventh fixed-gap bridge radial tail criterion)**.** *In the setting of Corollary 8.489, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.318. Assume all hypotheses of Corollary 8.489, except that the bridge-record-record-record-record-record-record bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.489 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.318 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new fixed-gap h-range and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.491 (Seventh confined bridge radial tail criterion)**.** *In the setting of Corollary 8.490, use the same parameters and define*

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

as in Corollary 8.319. Assume all hypotheses of Corollary 8.490, except that the bridge-record-record-record-record-record-record fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$,

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

and all retained spread-profile hypotheses from Corollary 8.490 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.319 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new bridge h-range and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.492 (Seventh inward-free radial criterion)**.** *In the setting of Corollary 8.491, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.320. Assume all hypotheses of Corollary 8.491, except that the bridge-record-record-record-record-record-record-record confined fixed-gap bridge profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$, both

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}} \quad \text{and} \quad R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.491 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.320 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new bridge h-range and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.493 (Seventh record-absorbed radial criterion)**.** *In the setting of Corollary 8.492, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.321. Assume all hypotheses of Corollary 8.492, except that the bridge-record-record-record-record-record-record-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The bridge-record-record-record-record-record-record-record bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.492 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.321 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$ are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.494 (Eighth persistent radial criterion)**.** *In the setting of Corollary 8.493, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.322. Assume all hypotheses of Corollary 8.493, except that the seventh prefix-record profile bound is replaced by the following two requirements: for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The seventh bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.493 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.322 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.495 (Eighth bounded-gap radial criterion)**.** *In the setting of Corollary 8.494, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.323. Assume all hypotheses of Corollary 8.494, except that the eighth persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread profiles

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.494 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.323 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.496 (Eighth fixed-gap bridge radial tail criterion)**.** *In the setting of Corollary 8.495, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.324. Assume all hypotheses of Corollary 8.495, except that the eighth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.495 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.324 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new fixed-gap h-range and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.497 (Eighth confined bridge radial tail criterion)**.** *In the setting of Corollary 8.496, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.325. Assume all hypotheses of Corollary 8.496, except that the eighth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$,

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.496 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.325 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new bridge h-range and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.498 (Eighth inward-free radial criterion)**.** *In the setting of Corollary 8.497, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.326. Assume all hypotheses of Corollary 8.497, except that the bridge-record-record-record-record-record-record-record-record confined fixed-gap bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.497 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.326 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new bridge h-range and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.499 (Eighth record-absorbed radial criterion)**.** *In the setting of Corollary 8.498, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.327. Assume all hypotheses of Corollary 8.498, except that the bridge-record-record-record-record-record-record-record-record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The bridge-record-record-record-record-record-record-record-record bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.498 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.327 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.500 (Ninth persistent radial criterion)**.** *In the setting of Corollary 8.499, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.328. Assume all hypotheses of Corollary 8.499, except that the bridge-record-record-record-record-record-record-record-record prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The bridge-record-record-record-record-record-record-record-record bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.499 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.328 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.501 (Ninth bounded-gap radial criterion)**.** *In the setting of Corollary 8.500, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.329. Assume all hypotheses of Corollary 8.500, except that the ninth persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread profiles

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.500 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.329 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.502 (Ninth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.501, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.330. Assume all hypotheses of Corollary 8.501, except that the ninth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, and for every*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.501 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.330 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.503 (Ninth confined bridge radial criterion)**.** *In the setting of Corollary 8.502, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.331. Assume all hypotheses of Corollary 8.502, except that the ninth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0\le i<N$ and $0\le j<J_{\sigma_i,\beta_i}$, and for every

$$1\le h\le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.502 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.331 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},\qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.504 (Ninth inward-free bridge radial criterion)**.** *In the setting of Corollary 8.503, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.332. Assume all hypotheses of Corollary 8.503, except that the bridge-record-record-record-record-record-record-record-record- record confined fixed-gap bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.503 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.332 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new bridge h-range and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.505 (Ninth record-absorbed radial criterion)**.** *In the setting of Corollary 8.504, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.333. Assume all hypotheses of Corollary 8.504, except that the bridge-record-record-record-record-record-record-record-record- record inward-free confined bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The bridge-record-record-record-record-record-record-record-record-record bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.504 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.333 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.506 (Tenth persistent radial criterion)**.** *In the setting of Corollary 8.505, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.334. Assume all hypotheses of Corollary 8.505, except that the bridge-record-record-record-record-record-record-record-record- record prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The bridge-record-record-record-record-record-record-record-record-record bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.505 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.334 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}, \qquad B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.507 (Tenth bounded-gap radial criterion)**.** *In the setting of Corollary 8.506, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.335. Assume all hypotheses of Corollary 8.506, except that the tenth persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.506 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.335 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.508 (Tenth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.507, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.336. Assume all hypotheses of Corollary 8.507, except that the tenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, and for every*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.507 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.336 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.509 (Tenth confined bridge radial criterion)**.** *In the setting of Corollary 8.508, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.337. Assume all hypotheses of Corollary 8.508, except that the tenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, and for every*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.508 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.337 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.510 (Tenth inward-free bridge radial criterion)**.** *In the setting of Corollary 8.509, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.338. Assume all hypotheses of Corollary 8.509, except that the tenth confined fixed-gap bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.509 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.338 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase profiles, long-run pro-

files, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the new bridge and bridge-return h-ranges and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.511 (Tenth record-absorbed radial criterion)**.** *In the setting of Corollary 8.510, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.339. Assume all hypotheses of Corollary 8.510, except that the tenth inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The tenth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.510 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.339 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.512 (Eleventh persistent radial criterion)**.** *In the setting of Corollary 8.511, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.340. Assume all hypotheses of Corollary 8.511, except that the tenth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The tenth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.511 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.340 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.513 (Eleventh bounded-gap radial criterion)**.** *In the setting of Corollary 8.512, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.341. Assume all hypotheses of Corollary 8.512, except that the eleventh persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.512 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.341 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.514 (Eleventh fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.513, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.342. Assume all hypotheses of Corollary 8.513, except that the eleventh bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, and for every

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.513 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.342 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.515 (Eleventh confined bridge radial criterion)**.** *In the setting of Corollary 8.514, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.343. Assume all hypotheses of Corollary 8.514, except that the eleventh fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, and for every

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.514 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.343 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles,

and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.516 (Eleventh inward-free bridge radial criterion)**.** *In the setting of Corollary 8.515, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.344. Assume all hypotheses of Corollary 8.515, except that the eleventh confined fixed-gap bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.515 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.344 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. For each i, the coefficients

$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained finite-ladder coefficients are fixed. The same is true of all B_i- and K_i-cutoffs, including

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}.$$

All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.517 (Eleventh record-absorbed radial criterion)**.** *In the setting of Corollary 8.516, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.345. Assume all hypotheses of Corollary 8.516, except that the eleventh inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The eleventh bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.516 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.345 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.518 (Twelfth persistent radial criterion)**.** *In the setting of Corollary 8.517, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.346. Assume all hypotheses of Corollary 8.517, except that the eleventh prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The eleventh bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.517 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.346 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. The prefix-record coefficient, the long-run coefficient, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.519 (Twelfth bounded-gap radial criterion)**.** *In the setting of Corollary 8.518, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.347. Assume all hypotheses of Corollary 8.518, except that the twelfth persistent prefix profile bound is replaced by the requirement that, for every $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent twelfth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.518 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.347 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent twelfth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained bridge-return and older return-spread families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.520 (Twelfth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.519, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.348. Assume all hypotheses of Corollary 8.519, except that the twelfth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, and for every

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent twelfth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.519 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.348 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent twelfth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained bridge-return and older return-spread families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the new fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.521 (Twelfth confined bridge radial criterion)**.** *In the setting of Corollary 8.520, use the same parameters and define*

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

as in Corollary 8.349. Assume all hypotheses of Corollary 8.520, except that the twelfth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, and for every

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

satisfies the same logarithmic exponent bound. The adjacent twelfth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

and all retained spread-profile hypotheses from Corollary 8.520 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.349 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent twelfth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained bridge-return and older return-spread families, confined staircase profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.522 (Twelfth inward-free bridge radial criterion)**.** *In the setting of Corollary 8.521, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.350. Assume all hypotheses of Corollary 8.521, except that the twelfth confined fixed-gap bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent twelfth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.521 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.350 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent twelfth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, confined staircase profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.523 (Twelfth record-absorbed radial criterion)**.** *In the setting of Corollary 8.522, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.351. Assume all hypotheses of Corollary 8.522, except that the twelfth inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The twelfth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.522 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.351 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent twelfth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.524 (Thirteenth persistent radial criterion)**.** *In the setting of Corollary 8.523, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.352. Assume all hypotheses of Corollary 8.523, except that the twelfth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The twelfth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.523 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.352 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. The prefix-record coefficient, the long-run coefficient, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.525 (Thirteenth bounded-gap radial criterion)**.** *In the setting of Corollary 8.524, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.353. Assume all hypotheses of Corollary 8.524, except that the thirteenth persistent prefix profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent thirteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.524 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.353 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained bridge-return and older return-spread families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.526 (Thirteenth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.525, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.354. Assume all hypotheses of Corollary 8.525, except that the thirteenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0\le i<N$ *and* $0\le j<J_{\sigma_i,\beta_i}$*, and for every*

$$1\le h\le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent thirteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.525 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.354 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained bridge-return and older return-spread families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the new fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.527 (Thirteenth confined bridge radial criterion)**.** *In the setting of Corollary 8.526, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.355. Assume all hypotheses of Corollary 8.526, except that the thirteenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, and for every*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent thirteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.526 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.355 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained bridge-return and older return-spread families, confined staircase profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap pair term, the

long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.528 (Thirteenth inward-free bridge radial criterion)**.** *In the setting of Corollary 8.527, use the same parameters and define the families*

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

as in Corollary 8.356. Assume all hypotheses of Corollary 8.527, except that the thirteenth confined fixed-gap bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

satisfy the same logarithmic exponent bound. The adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

and all retained spread-profile hypotheses from Corollary 8.527 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.356 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all retained return-spread and bridge-return families, confined staircase profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run

coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.529 (Thirteenth record-absorbed radial criterion)**.** *In the setting of Corollary 8.528, use the same parameters and define*

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j},$$
$$D^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i},$$
$$A^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i}$$

as in Corollary 8.357. Assume all hypotheses of Corollary 8.528, except that the thirteenth inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. The thirteenth bridge-return spread profiles

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

over the same finite h-range, the adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

and all retained spread-profile hypotheses from Corollary 8.528 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.357 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained thirteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained

finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.530 (Fourteenth persistent radial criterion)**.** *In the setting of Corollary 8.529, use the same parameters and define*

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j},$$
$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

as in Corollary 8.358. Assume all hypotheses of Corollary 8.529, except that the thirteenth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. The thirteenth bridge-return spread profiles

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

and all retained spread-profile hypotheses from Corollary 8.529 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.358 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained thirteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, the confined staircase profiles, long-run profiles, and exact-level return profiles. The prefix-record coefficient, the long-run coefficient, all retained

finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.531 (Fourteenth bounded-gap radial criterion)**.** *In the setting of Corollary 8.530, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.359. Assume all hypotheses of Corollary 8.530, except that the persistent-prefix profile P^{per13} bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent fourteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.530 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.359 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained thirteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.532 (Fourteenth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.531, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.360. Assume all hypotheses of Corollary 8.531, except that the fourteenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, and for every*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent fourteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.531 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.360 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained thirteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the new fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.533 (Fourteenth confined bridge radial criterion)**.** *In the setting of Corollary 8.532, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.361. Assume all hypotheses of Corollary 8.532, except that the fourteenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*, and for every*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent fourteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.532 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.361 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained thirteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent thirteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap pair term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.534 (Fourteenth inward-free bridge radial criterion)**.** *In the setting of Corollary 8.533, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.362. Assume all hypotheses of Corollary 8.533, except that the fourteenth confined fixed-gap bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent fourteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.533 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.362 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new fourteenth bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained thirteenth bridge-return spread family, the adjacent thirteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.535 (Fourteenth record-absorbed radial criterion)**.** *In the setting of Corollary 8.534, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.363. Assume all hypotheses of Corollary 8.534, except that the fourteenth inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The fourteenth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

over the same finite h-range, the adjacent fourteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.534 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.363 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fourteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained thirteenth bridge-return spread family, the adjacent thirteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.536 (Fifteenth persistent radial criterion)**.** *In the setting of Corollary 8.535, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.364. Assume all hypotheses of Corollary 8.535, except that the fourteenth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$*,*

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The fourteenth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

over the same finite h*-range, the adjacent fourteenth return-spread family*

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.535 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.364 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fourteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase profiles, long-run profiles, and exact-level return profiles. The prefix-record coefficient, the long-run coefficient, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.537 (Fifteenth bounded-gap radial criterion)**.** *In the setting of Corollary 8.536, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.365. Assume all hypotheses of Corollary 8.536, except that the persistent-prefix profile P^{per14} bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent fifteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.536 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.365 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fourteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.538 (Fifteenth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.537, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.366. Assume all hypotheses of Corollary 8.537, except that the fifteenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, and for every

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent fifteenth return-spread family and all retained spread-profile hypotheses from Corollary 8.537 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.366 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fourteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the new fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_c(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.539 (Fifteenth confined bridge radial criterion)**.** *In the setting of Corollary 8.538, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.367. Assume all hypotheses of Corollary 8.538, except that the fifteenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent fifteenth return-spread family and all retained spread-profile hypotheses from Corollary 8.538 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.367 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fourteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fourteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap term, the long-run coefficient, all retained finite-ladder coefficients, and all K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.540 (Fifteenth inward-free bridge radial criterion)**.** *In the setting of Corollary 8.539, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.368. Assume all hypotheses of Corollary 8.539, except that the fifteenth confined bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent fifteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.539 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.368 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new fifteenth bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fourteenth bridge-return spread family, the adjacent fourteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.541 (Fifteenth record-absorbed radial criterion)**.** *In the setting of Corollary 8.540, use the same parameters and define*

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j},$$
$$D^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i},$$
$$A^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i}$$

as in Corollary 8.369. Assume all hypotheses of Corollary 8.540, except that the fifteenth inward-free bridge profile bound is replaced by the requirement that, for every $0\le i<N$ and $0\le j<J_{\sigma_i,\beta_i}$,

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. The fifteenth bridge-return spread profiles

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

over the same finite h-range, the adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

and all retained spread-profile hypotheses from Corollary 8.540 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.369 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fourteenth bridge-return spread family, the adjacent fourteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.542 (Sixteenth persistent radial criterion)**.** *In the setting of Corollary 8.541, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.370. Assume all hypotheses of Corollary 8.541, except that the fifteenth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The fifteenth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

over the same finite h-range, the adjacent fifteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.541 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.370 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase profiles, long-run profiles, and exact-level return profiles. The prefix-record coefficient, the long-run coefficient, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.543 (Sixteenth bounded-gap radial criterion)**.** *In the setting of Corollary 8.542, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.371. Assume all hypotheses of Corollary 8.542, except that the persistent-prefix profile P^{per15} bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent sixteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.542 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.371 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.544 (Sixteenth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.543, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.372. Assume all hypotheses of Corollary 8.543, except that the sixteenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, and for every

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent sixteenth return-spread family and all retained spread-profile hypotheses from Corollary 8.543 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.372 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the new fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.545 (Sixteenth confined bridge radial criterion)**.** *In the setting of Corollary 8.544, use the same parameters and define*

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

as in Corollary 8.373. Assume all hypotheses of Corollary 8.544, except that the sixteenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

satisfies the same logarithmic exponent bound. The adjacent sixteenth return-spread family and all retained spread-profile hypotheses from Corollary 8.544 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.373 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent fifteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap term, the long-run coefficient, all retained finite-ladder coefficients, and all K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.546 (Sixteenth inward-free bridge radial criterion)**.** *In the setting of Corollary 8.545, use the same parameters and define the families*

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.374. Assume all hypotheses of Corollary 8.545, except that the sixteenth confined bridge profile bound is replaced by the following requirement. For every $0\le i<N$, $0\le j<J_{\sigma_i,\beta_i}$, and

$$1\le h\le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P_{i,j,h}^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfy the same logarithmic exponent bound. The adjacent sixteenth return-spread family

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.545 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.374 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new sixteenth bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0\qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.547 (Sixteenth record-absorbed radial criterion)**.** *In the setting of Corollary 8.546, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$D_i^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$A_i^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.375. Assume all hypotheses of Corollary 8.546, except that the sixteenth inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$*,*

$$P_{i,j}^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The sixteenth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

over the same finite h*-range, the adjacent sixteenth return-spread family*

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.546 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.375 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained sixteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.548 (Seventeenth persistent radial criterion)**.** *In the setting of Corollary 8.547, use the same parameters and define*

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.376. Assume all hypotheses of Corollary 8.547, except that the sixteenth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The sixteenth bridge-return spread profiles

$$R_{i,j,h}^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

over the same finite h*-range, the adjacent sixteenth return-spread family*

$$R_{i,j}^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

and all retained spread-profile hypotheses from Corollary 8.547 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.376 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained sixteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The prefix-record coefficient, the long-run coefficient, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.549 (Seventeenth bounded-gap radial criterion)**.** *In the setting of Corollary 8.548, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.377. Assume all hypotheses of Corollary 8.548, except that the persistent-prefix profile P^{per16} bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent seventeenth return-spread family and all retained spread-profile hypotheses from Corollary 8.548 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.377 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent seventeenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained sixteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.550 (Seventeenth fixed-gap bridge radial criterion)**.** *In the setting of Corollary 8.549, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$
$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.378. Assume all hypotheses of Corollary 8.549, except that the seventeenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$, and for every

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent seventeenth return-spread family and all retained spread-profile hypotheses from Corollary 8.549 are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.378 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent seventeenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained sixteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the new fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.551 (Seventeenth confined bridge radial criterion)**.** *In the setting of Corollary 8.550, use the same parameters and define*

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

as in Corollary 8.379. Assume all hypotheses of Corollary 8.550, except that the seventeenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

satisfies the same logarithmic exponent bound. The adjacent seventeenth return-spread family and all retained spread-profile hypotheses from Corollary 8.550 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.379 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent seventeenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained sixteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap term, the long-run coefficient, all retained finite-ladder coefficients, and all K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.552 (Seventeenth inward-free radial)**.** *In the setting of Corollary 8.551, use the same parameters and define the families*

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

as in Corollary 8.380. Assume all hypotheses of Corollary 8.551, except that the seventeenth confined bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

satisfy the same logarithmic exponent bound. The adjacent seventeenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

and all retained spread-profile hypotheses from Corollary 8.551 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.380 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new seventeenth bridge-return family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent seventeenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained sixteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent sixteenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.553 (Seventeenth record-absorbed radial)**.** *In the setting of Corollary 8.552, use the same parameters and define*

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j},$$
$$D^{\mathrm{rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i},$$
$$A^{\mathrm{long,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i}$$

as in Corollary 8.381. Assume all hypotheses of Corollary 8.552, except that the seventeenth inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. The seventeenth bridge-return spread profiles

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j,h}$$

over the same finite h-range are retained. Also retain the adjacent seventeenth return-spread family and all other spread-profile hypotheses from Corollary 8.552. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.381 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained seventeenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the adjacent seventeenth return-spread family

$$R^{\mathrm{spr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.554 (Eighteenth persistent radial)**.** *In the setting of Corollary 8.553, use the same parameters and define*

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j},$$
$$R^{\mathrm{spr,recbr,ch}}_{i,j}$$

as in Corollary 8.382. Assume all hypotheses of Corollary 8.553, except that the seventeenth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ and $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R^{\mathrm{spr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. Retain the seventeenth bridge-return spread profiles over the same finite h-range, the adjacent seventeenth return-spread family, and all other spread-profile hypotheses of the referenced corollary. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.382 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the new adjacent eighteenth return-spread family

$$R^{\mathrm{spr,recbr,ch}},$$

the retained seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced prefix-record term, the long-run coefficient, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.555 (Eighteenth bounded-gap radial)**.** *In the setting of Corollary 8.554, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}$$

as in Corollary 8.383. Assume all hypotheses of Corollary 8.554, except that the persistent-prefix profile P^{per17} bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent eighteenth return-spread family and all retained spread-profile hypotheses from Corollary 8.554 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.383 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,ch}},$$

the retained adjacent eighteenth return-spread family, the retained seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.556 (Eighteenth fixed-gap bridge radial)**.** *In the setting of Corollary 8.555, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}$$

as in Corollary 8.384. Assume all hypotheses of the bounded-gap radial criterion, except that the eighteenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent eighteenth return-spread family and all retained spread-profile hypotheses from the referenced bounded-gap radial criterion are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.384 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,ch}},$$

the retained adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.557 (Eighteenth confined bridge radial)**.** *In the setting of Corollary 8.556, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

as in Corollary 8.385. Assume all hypotheses of the fixed-gap radial criterion, except that the eighteenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent eighteenth return-spread family and all retained spread-profile hypotheses from the referenced fixed-gap radial criterion are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.385 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

the retained adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.558 (Eighteenth inward-free radial)**.** *In the setting of Corollary 8.557, use the same parameters and define the families*

$$P^{\mathrm{pre,br,free,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,ch}}_{i,j,h}$$

as in Corollary 8.386. Assume all hypotheses of the confined bridge radial criterion, except that the eighteenth confined bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,recbr,ch}},$$

both profiles

$$P^{\mathrm{pre,br,free,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,ch}}_{i,j,h}$$

satisfy the same logarithmic exponent bound. The adjacent eighteenth return-spread family and all retained spread-profile hypotheses from the referenced confined bridge radial criterion are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.386 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,ch}},$$

the new eighteenth bridge-return family

$$R^{\mathrm{spr,br,recbr,ch}},$$

the retained adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.559 (Eighteenth record-absorbed radial)**.** *In the setting of Corollary 8.558, use the same parameters and define*

$$P^{\mathrm{pre,rec,br,recbr,ch}}_{i,j},$$
$$D^{\mathrm{rec,br,recbr,ch}}_{i},$$
$$A^{\mathrm{long,br,recbr,ch}}_{i}$$

as in Corollary 8.387. Assume all hypotheses of the inward-free radial criterion, except that the eighteenth inward-free bridge profile bound is replaced by the requirement that, for every $0 \le i < N$ and $0 \le j < J_{\sigma_i,\beta_i}$,

$$P^{\mathrm{pre,rec,br,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. The eighteenth bridge-return spread profiles

$$R^{\mathrm{spr,br,recbr,ch}}_{i,j,h}$$

over the same finite h-range are retained. Also retain the adjacent eighteenth return-spread family and all other spread-profile hypotheses from the referenced inward-free radial criterion. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.387 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new prefix-record profiles

$$P^{\mathrm{pre,rec,br,recbr,ch}},$$

the retained eighteenth bridge-return spread family

$$R^{\mathrm{spr,br,recbr,ch}},$$

the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined

staircase and bridge profiles, long-run profiles, and exact-level return profiles. The newly defined prefix-record and long-run coefficients, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.560 (Nineteenth persistent radial)**.** *In the setting of Corollary 8.559, use the same parameters and define*

$$P^{\mathrm{pre,per,br,recbr,ch}}_{i,j},$$
$$R^{\mathrm{spr,recbr,ch}}_{i,j}$$

as in Corollary 8.388. Assume all hypotheses of the record-absorbed radial criterion, except that the eighteenth prefix-record profile bound is replaced by the following two requirements. For every $0 \le i < N$ *and* $0 \le j < J_{\sigma_i,\beta_i}$,

$$P^{\mathrm{pre,per,br,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound, and, for every $0 \le i < N$ *and* $0 \le j \le J_{\sigma_i,\beta_i}$,

$$R^{\mathrm{spr,recbr,ch}}_{i,j}$$

satisfies the same logarithmic exponent bound. Retain the eighteenth bridge-return spread profiles over the same finite h-range, the adjacent eighteenth return-spread family, and all other spread-profile hypotheses of the referenced corollary. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.388 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new persistent-prefix profiles

$$P^{\mathrm{pre,per,br,recbr,ch}},$$

the new adjacent nineteenth return-spread family

$$R^{\mathrm{spr,recbr,ch}},$$

the retained eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced prefix-record term, the long-run coefficient, all retained finite-ladder coefficients, and all B_i- and K_i-cutoffs are fixed. The sums over the new adjacent return-spread indices and the retained bridge-return h-ranges are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.561 (Nineteenth bounded-gap radial)**.** *In the setting of Corollary 8.560, use the same parameters and define*

$$B_i^{\mathrm{br,recbr,ch}},$$

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}$$

as in Corollary 8.389. Assume all hypotheses of Corollary 8.560, except that the persistent-prefix profile P^{per18} bound is replaced by the requirement that, for every $0\le i<N$ and $0\le j<J_{\sigma_i,\beta_i}$,

$$P_{i,j}^{\mathrm{pre,pair,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent nineteenth return-spread family and all retained spread-profile hypotheses from Corollary 8.560 are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.389 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new bounded-gap prefix-pair profiles

$$P^{\mathrm{pre,pair,br,recbr,ch}},$$

the retained adjacent nineteenth return-spread family, the retained eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the retained seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced persistent-prefix term, the long-run coefficient, the new bounded-gap cutoff, all retained finite-ladder coefficients, and all K_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.562 (Nineteenth fixed-gap bridge radial)**.** *In the setting of Corollary 8.561, use the same parameters and define*

$$K_i^{\mathrm{gap,br,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}$$

as in Corollary 8.390. Assume all hypotheses of the bounded-gap radial criterion, except that the nineteenth bounded-gap prefix-pair profile bound is replaced by the requirement that, for every $0\le i<N$, $0\le j<J_{\sigma_i,\beta_i}$, and

$$1\le h\le B_i^{\mathrm{br,recbr,ch}},$$

$P_{i,j,h}^{\mathrm{pre,gap,br,recbr,ch}}$

satisfies the same logarithmic exponent bound. The adjacent nineteenth return-spread family and all retained spread-profile hypotheses from the referenced bounded-gap radial criterion are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.390 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new fixed-gap prefix-pair profiles

$$P^{\mathrm{pre,gap,br,recbr,ch}},$$

the retained adjacent nineteenth return-spread family, the eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the retained seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced bounded-gap term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H}\to 0 \qquad (p\uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda=\lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.563 (Nineteenth confined bridge radial)**.** *In the setting of Corollary 8.562, use the same parameters and define*

$$P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}$$

as in Corollary 8.391. Assume all hypotheses of the fixed-gap radial criterion, except that the nineteenth fixed-gap prefix-pair profile bound is replaced by the requirement that, for every $0\le i<N$, $0\le j<J_{\sigma_i,\beta_i}$, *and*

$$1\le h\le B_i^{\mathrm{br,recbr,ch}},$$

$$P_{i,j,h}^{\mathrm{pre,br,recbr,ch}}$$

satisfies the same logarithmic exponent bound. The adjacent nineteenth return-spread family and all retained spread-profile hypotheses from the referenced fixed-gap radial criterion are retained. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ *holds.*

Proof. Corollary 8.391 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail, finite sums of the new confined bridge profiles

$$P^{\mathrm{pre,br,recbr,ch}},$$

the retained adjacent nineteenth return-spread family, the eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the retained seventeenth bridge-return spread family,

the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced fixed-gap term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.564 (Nineteenth inward-free bridge radial)**.** *In the setting of Corollary 8.563, use the same parameters and define the families*

$$P^{\mathrm{pre,br,free,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,ch}}_{i,j,h}$$

as in Corollary 8.392. Assume all hypotheses of the confined bridge radial criterion, except that the nineteenth confined bridge profile bound is replaced by the following requirement. For every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, *and*

$$1 \le h \le B_i^{\mathrm{br,recbr,ch}},$$

both profiles

$$P^{\mathrm{pre,br,free,recbr,ch}}_{i,j,h},$$
$$R^{\mathrm{spr,br,recbr,ch}}_{i,j,h}$$

satisfy the same logarithmic exponent bound. The adjacent nineteenth return-spread family and all retained spread-profile hypotheses from the referenced confined bridge radial criterion are retained. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ *holds.*

Proof. Corollary 8.392 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail, finite sums of the new inward-free bridge profiles

$$P^{\mathrm{pre,br,free,recbr,ch}},$$

the new nineteenth bridge-return family

$$R^{\mathrm{spr,br,recbr,ch}},$$

the retained adjacent nineteenth return-spread family, the eighteenth bridge-return spread family, the adjacent eighteenth return-spread family, the retained seventeenth bridge-return spread family, the adjacent seventeenth return-spread family, the retained sixteenth bridge-return spread family, the adjacent sixteenth return-spread family, the retained fifteenth bridge-return spread family, the adjacent fifteenth return-spread family, all older return-spread and bridge-return families, confined

staircase and bridge profiles, long-run profiles, and exact-level return profiles. The coefficient of the replaced confined bridge term, the long-run coefficient, the fixed-gap cutoff, all retained finite-ladder coefficients, and all older K_i- and B_i-cutoffs are fixed. All new sums are finite, and the only nonconstant multiplicative losses are the same factors $W_{\beta_i,p}$ as before. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.565 (Seventh persistent subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.488 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.488 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.566 (Seventh bounded-gap subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.489 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.489 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.567 (Seventh fixed-gap subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.490 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.490 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.568 (Seventh confined bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.491 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.491 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.569 (Seventh inward-free bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.492 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.492 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.570 (Seventh record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.493 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.493 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.571 (Eighth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.494 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.494 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.572 (Eighth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.495 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.495 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.573 (Eighth fixed-gap subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.496 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.496 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.574 (Eighth confined bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.497 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.497 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.575 (Eighth inward-free subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.498 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.498 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.576 (Eighth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.499 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.499 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.577 (Ninth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.500 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.500 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.578 (Ninth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.501 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.501 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.579 (Ninth fixed-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.502 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.502 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.580 (Ninth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.503 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.503 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.581 (Ninth inward-free subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.504 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.504 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.582 (Ninth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.505 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.505 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.583 (Tenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.506 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.506 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.584 (Tenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.507 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.507 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.585 (Tenth fixed-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.508 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.508 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.586 (Tenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.509 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.509 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.587 (Tenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.510 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.510 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.588 (Tenth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.511 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.511 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.589 (Eleventh persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.512 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.512 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.590 (Eleventh bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.513 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.513 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.591 (Eleventh fixed-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.514 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.514 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.592 (Eleventh confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.515 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.515 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.593 (Eleventh inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.516 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.516 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.594 (Eleventh record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.517 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.517 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.595 (Twelfth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.518 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.518 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.596 (Twelfth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.519 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.519 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.597 (Twelfth fixed-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.520 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.520 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.598 (Twelfth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.521 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.521 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.599 (Twelfth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.522 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.522 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.600 (Twelfth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.523 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.523 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.601 (Thirteenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.524 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.524 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.602 (Thirteenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.525 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.525 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.603 (Thirteenth fixed-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.526 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.526 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.604 (Thirteenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.527 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.527 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.605 (Thirteenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.528 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.528 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.606 (Thirteenth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.529 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.529 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.607 (Fourteenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.530 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.530 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.608 (Fourteenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.531 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.531 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.609 (Fourteenth fixed-gap bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.532 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.532 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.610 (Fourteenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.533 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.533 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.611 (Fourteenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.534 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.534 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.612 (Fourteenth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.535 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.535 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.613 (Fifteenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.536 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.536 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.614 (Fifteenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.537 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.537 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.615 (Fifteenth fixed-gap bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.538 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.538 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.616 (Fifteenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.539 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.539 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.617 (Fifteenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.540 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.540 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.618 (Fifteenth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.541 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.541 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.619 (Sixteenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.542 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.542 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.620 (Sixteenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.543 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.543 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.621 (Sixteenth fixed-gap bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.544 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.544 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.622 (Sixteenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.545 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.545 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.623 (Sixteenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.546 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.546 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.624 (Sixteenth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.547 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.547 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.625 (Seventeenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.548 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.548 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.626 (Seventeenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.549 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.549 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.627 (Seventeenth fixed-gap bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.550 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.550 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.628 (Seventeenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.551 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.551 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.629 (Seventeenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.552 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.552 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.630 (Seventeenth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.553 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.553 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.631 (Eighteenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.554 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.554 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.632 (Eighteenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.555 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.555 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.633 (Eighteenth fixed-gap bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.556 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.556 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.634 (Eighteenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.557 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.557 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.635 (Eighteenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.558 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.558 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.636 (Eighteenth record-absorbed subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.559 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.559 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.637 (Nineteenth persistent subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.560 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.560 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.638 (Nineteenth bounded-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.561 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.561 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.639 (Nineteenth fixed-gap bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.562 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.562 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.640 (Nineteenth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.563 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.563 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.641 (Nineteenth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.564 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.564 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.642 (Sixth record-absorbed inward-free bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.487 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.487 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.643 (Sixth inward-free bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.486 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.486 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.644 (Sixth confined bridge subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.485 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.485 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.645 (Sixth fixed-gap subdivision)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.484 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.484 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.646 (Sixth gap subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.483 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.483 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.647 (Sixth persistent bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.482 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.482 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta=\delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.648 (Fifth record-absorbed inward-free bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.481 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}}>0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}=\left\lfloor\frac{\rho}{2\delta_{\mathrm{fin}}}\right\rfloor+1,$$

and assume, with $\log_{D_H}0=-\infty$, that for every $0\le a<m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)}\frac{\log_{D_H}\mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H}\overline{\chi}_p^H}+a\delta_{\mathrm{fin}}\log_{D_H}Q<-c_s\log_{D_H}p_{\mathrm{c}}(H)-(2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G)<p_{2\to2}(G).$$

Proof. Corollary 8.481 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta=\delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.649 (Inward-free bridge-record-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.480 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}}>0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}=\left\lfloor\frac{\rho}{2\delta_{\mathrm{fin}}}\right\rfloor+1,$$

and assume, with $\log_{D_H}0=-\infty$, that for every $0\le a<m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)}\frac{\log_{D_H}\mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H}\overline{\chi}_p^H}+a\delta_{\mathrm{fin}}\log_{D_H}Q<-c_s\log_{D_H}p_{\mathrm{c}}(H)-(2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G)<p_{2\to2}(G).$$

Proof. Corollary 8.480 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta=\delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.650 (Confined bridge-record-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.479 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}}>0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}=\left\lfloor\frac{\rho}{2\delta_{\mathrm{fin}}}\right\rfloor+1,$$

and assume, with $\log_{D_H}0=-\infty$, that for every $0\le a<m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)}\frac{\log_{D_H}\mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H}\overline{\chi}_p^H}+a\delta_{\mathrm{fin}}\log_{D_H}Q<-c_s\log_{D_H}p_{\mathrm{c}}(H)-(2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G)<p_{2\to2}(G).$$

Proof. Corollary 8.479 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.651 (Fixed-gap bridge-record-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.478 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.478 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.652 (Bounded-gap bridge-record-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.477 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.477 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.653 (Persistent bridge-record-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.476 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.476 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.654 (Record-absorbed inward-free bridge-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.475 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.475 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.655 (Inward-free bridge-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.474 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.474 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.656 (Confined bridge-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.473 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.473 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.657 (Fixed-gap bridge-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.472 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.472 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.658 (Bounded-gap bridge-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.471 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.471 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.659 (Persistent bridge-record-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.470 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.470 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.660 (Record-absorbed inward-free bridge-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.469 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.469 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.661 (Inward-free bridge-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.468 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.468 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.662 (Confined bridge-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.467 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.467 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.663 (Fixed-gap bridge-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.466 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.466 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.664 (Bounded-gap bridge-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.465 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.465 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.665 (Persistent bridge-record-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.464 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.464 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.666 (Record-absorbed inward-free bridge-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.463 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.463 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.667 (Inward-free bridge-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.462 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.462 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.668 (Confined bridge-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.461 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.461 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.669 (Fixed-gap bridge-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.460 hold for some choice of tail parameters. Let* $\delta_{\mathrm{fin}} > 0$*, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$*, that for every* $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$*,*

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.460 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.670 (Bounded-gap bridge-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.459 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.459 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.671 (Persistent bridge-record-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.458 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.458 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.672 (Record-absorbed inward-free bridge-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.457 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.457 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.673 (Inward-free bridge-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.456 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.456 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.674 (Confined bridge-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.455 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.455 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.675 (Fixed-gap bridge-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.454 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.454 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.676 (Bounded-gap bridge-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.453 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.453 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.677 (Persistent bridge-record chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.452 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.452 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.678 (Record-absorbed inward-free bridge persistent chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.451 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.451 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.679 (Inward-free bridge persistent chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.450 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.450 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.680 (Confined-bridge persistent chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.449 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.449 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.681 (Fixed-gap persistent chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.448 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.448 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.682 (Bounded-gap persistent chain-absorbed subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.447 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.447 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.683 (Persistent chain-absorbed staircase-chain/bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.446 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.446 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.684 (Chain-absorbed staircase-chain/bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.445 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.445 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.685 (Staircase-chain/bridge subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.444 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^H_p} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.444 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.686 (Staircase-chain-pair radial tail criterion)**.** *In the setting of Corollary 8.444, use the same parameters and define B_i, C_i, $P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h}$, $P^{\mathrm{pre,br,sh}}_{i,j,h}$, $R^{\mathrm{spr},++}_{i,j}$, $T^{\mathrm{long}}_{i,j}$, $S^{\mathrm{ret}}_{i,j}$, and $\widehat{X}^{\mathrm{ret}}_{i,j}$ as there. Define D_i, $K^{\sharp}_i$, $P^{\mathrm{pre,stair,loc,longchain,sh}}_{i,j,q,h}$, and $P^{\mathrm{pre,stair,loc,chainbr,sh}}_{i,j,q,h}$ as in Corollary 8.393. Assume all hypotheses of Corollary 8.444, except that the locally short staircase-chain profile bound for $q = 3,4$ is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $q \in \{3,4\}$, the profiles*

$$P^{\mathrm{pre,stair,loc,longchain,sh}}_{i,j,q,h} \qquad (1 \le h \le q)$$

and

$$P^{\mathrm{pre,stair,loc,chainbr,sh}}_{i,j,q,h} \qquad (q < h \le D_i)$$

satisfy the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.393 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the longer staircase-chain, confined staircase-chain-bridge, confined locally short staircase-bridge, confined fixed-gap bridge $h \ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the coefficient 3 on the long-run family is fixed. The additional sums over $q \in \{3,4\}$ and $1 \le h \le D_i$ are finite and independent of p. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.687 (Staircase-chain-pair subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.686 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.686 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.688 (Second-stage staircase-chain radial tail criterion)**.** *In the setting of Corollary 8.686, use the same parameters, and define the inherited finite-ladder profiles*

$$B_i,\ C_i,\ D_i,\ K_i^{\sharp},\quad P_{i,j,q,h}^{\mathrm{pre,stair,loc,chainbr,sh}},\quad P_{i,j,h}^{\mathrm{pre,stair,loc,brdg,sh}},\quad P_{i,j,h}^{\mathrm{pre,br,sh}},$$

$$R_{i,j}^{\mathrm{spr},++},\quad T_{i,j}^{\mathrm{long}},\quad S_{i,j}^{\mathrm{ret}},\quad \widehat{X}_{i,j}^{\mathrm{ret}}$$

as there. Define E_i, $K_i^{\flat}$, $P_{i,j,q,h,a}^{\mathrm{pre,stair,loc,nextchain,sh}}$, and $P_{i,j,q,h,a}^{\mathrm{pre,stair,loc,nextchainbr,sh}}$ as in Corollary 8.394. Assume all hypotheses of Corollary 8.686, except that the longer staircase-chain profile bound is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, $q \in \{3,4\}$, and $1 \le h \le q$, the profiles

$$P_{i,j,q,h,a}^{\mathrm{pre,stair,loc,nextchain,sh}} \qquad (1 \le a \le q+h)$$

and

$$P_{i,j,q,h,a}^{\mathrm{pre,stair,loc,nextchainbr,sh}} \qquad (q+h < a \le E_i)$$

satisfy the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$ holds.

Proof. Corollary 8.394 bounds $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}$ by the terminal far tail and finite sums of the next-stage staircase-chain, next-stage confined staircase-chain-bridge, previous confined staircase-chain-bridge, confined locally short staircase-bridge, confined fixed-gap bridge $h \ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the coefficient 3 on the long-run family is fixed. The additional sums over q, h, a are finite and independent of p. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}_{Q,\lambda_0}^{G\to H,\mathrm{rad}}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.689 (Second-stage staircase-chain subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.688 hold for some choice of tail parameters. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m^{G\to H}_{\rho,\delta_{\mathrm{fin}}} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m^{G\to H}_{\rho,\delta_{\mathrm{fin}}}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}^{G\to H,\mathrm{rad}}_{a,\delta_{\mathrm{fin}}}(s,p)}{\log_{D_H} \overline{\chi}^{H}_{p}} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.688 gives the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.690 (Finite-stage chain tail criterion)**.** *In the setting of Corollary 8.444, use the same parameters and define the inherited finite-ladder profiles*

$$B_i,\ C_i,\quad P^{\mathrm{pre,stair,loc,brdg,sh}}_{i,j,h},\quad P^{\mathrm{pre,br,sh}}_{i,j,h},\quad R^{\mathrm{spr},++}_{i,j},\quad T^{\mathrm{long}}_{i,j},\quad S^{\mathrm{ret}}_{i,j},\quad \widehat{X}^{\mathrm{ret}}_{i,j}$$

as there. Fix $n \ge 0$, and define

$$K^{\mathrm{ch}}_{i,r},\quad B^{\mathrm{ch}}_{i,r},\quad P^{\mathrm{pre,stair,loc,nchain,sh}}_{i,j,q,\mathbf{a}},\quad P^{\mathrm{pre,stair,loc,nchainbr,sh}}_{i,j,q,r,\mathbf{a},h}$$

as in Corollary 8.395. Assume all hypotheses of Corollary 8.444, except that the locally short staircase-chain profile bound for $q = 3,4$ is replaced by the requirement that, for every $0 \le i < N$, $0 \le j < J_{\sigma_i,\beta_i}$, and $q \in \{3,4\}$, the profiles

$$P^{\mathrm{pre,stair,loc,nchain,sh}}_{i,j,q,\mathbf{a}} \qquad (\mathbf{a} \in \mathcal{A}_n(q))$$

and

$$P^{\mathrm{pre,stair,loc,nchainbr,sh}}_{i,j,q,r,\mathbf{a},h} \qquad \big(1 \le r \le n,\ \mathbf{a} \in \mathcal{A}_{r-1}(q),\ a_{r-1} < h \le B^{\mathrm{ch}}_{i,r}\big)$$

satisfy the same logarithmic exponent bound. Then the exponent hypothesis on $\mathfrak{L}^{G\to H,\mathrm{rad}}_{Q,\rho}(s,p)$ holds.

Proof. Corollary 8.395 bounds $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}$ by the terminal far tail and finite sums of the terminal finite-stage staircase-chain profiles, the accumulated finite-stage confined staircase-chain bridge profiles, confined locally short staircase-bridge profiles, confined fixed-gap bridge $h \ge 2$, long-run, spread-return, and ordinary exact-level return profile families. The only nonconstant multiplicative losses are the factors $W_{\beta_i,p}$, and the coefficient 3 on the long-run family is fixed. All additional sums are finite for fixed n and independent of p. Since

$$\frac{\log_{D_H} W_{\beta_i,p}}{L_H} \to 0 \qquad (p \uparrow p_{\mathrm{c}}(H)),$$

the displayed assumptions imply the far-tail exponent hypothesis on $\mathfrak{Y}^{G\to H,\mathrm{rad}}_{Q,\lambda_0}(s,p)$. Together with the near-tail bin estimates, the hypotheses of Corollary 8.125 hold with $\lambda = \lambda_0$, giving the claimed tail exponent bound. □

Corollary 8.691 (Finite-stage chain subdivision criterion)**.** *In the setting of Corollary 8.123, suppose that the hypotheses of Corollary 8.690 hold for some choice of tail parameters and finite chain-amplification depth n. Let $\delta_{\mathrm{fin}} > 0$, set*

$$m_{\rho,\delta_{\mathrm{fin}}}^{G\to H} = \left\lfloor \frac{\rho}{2\delta_{\mathrm{fin}}} \right\rfloor + 1,$$

and assume, with $\log_{D_H} 0 = -\infty$, that for every $0 \le a < m_{\rho,\delta_{\mathrm{fin}}}^{G\to H}$,

$$\limsup_{p\uparrow p_{\mathrm{c}}(H)} \frac{\log_{D_H} \mathfrak{D}_{a,\delta_{\mathrm{fin}}}^{G\to H,\mathrm{rad}}(s,p)}{\log_{D_H} \overline{\chi}_p^H} + a\delta_{\mathrm{fin}} \log_{D_H} Q < -c_s \log_{D_H} p_{\mathrm{c}}(H) - (2+7\alpha_\omega)s.$$

Then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. Corollary 8.690 gives the exponent hypothesis on $\mathfrak{L}_{Q,\rho}^{G\to H,\mathrm{rad}}(s,p)$. The displayed finite-bin assumptions are the finite-profile assumptions in Corollary 8.124, with $\delta = \delta_{\mathrm{fin}}$. Applying that corollary gives the conclusion. □

Corollary 8.692 (Near-half-logarithmic pointwise root-annulus obstruction)**.** *Under the graph and volume assumptions of Corollary 8.69, suppose that $\alpha_\omega = \log_\Delta \omega > 0$. There is $0 < \epsilon \le 1$ such that for every $\lambda > 0$, every $\eta > 0$, every $B > 0$, every $\theta > 0$ satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_{c}, there are data satisfying all conclusions of Corollary 8.68 such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad U_\lambda = \frac{1}{2\alpha_\omega} + \lambda, \qquad r_p = \lfloor \theta L \rfloor, \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

at least one of the following alternatives holds:

$$2^j > U_\lambda L,$$

or there is a vertex $a \in V$ such that

$$d(x,a) \in I_{j,p}^{\mathrm{gate}}$$

and

$$\tau_p(x,a) \ge D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ is the constant from Corollary 8.75. Consequently, if G is connected and $\tau_p^{\max}$ is defined as in Corollary 7.34, then in the second alternative there is an integer $n \in I_{j,p}^{\mathrm{gate}}$ such that

$$\tau_p^{\max}(n) \ge D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega}.$$

Proof. Apply Corollary 8.79. If the super-half-logarithmic scale alternative fails, then the average of $\tau_p(x,\cdot)$ over the non-empty annulus $\{a : d(x,a) \in I_{j,p}^{\mathrm{gate}}\}$ is at least the displayed threshold. Some vertex in the annulus therefore attains at least this value. The final assertion follows by taking $n = d(x,a)$. □

Proposition 8.693 (Near-half-logarithmic pointwise root-annulus criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 < \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\lambda > 0$, $\eta > 0$, $B > 0$, $\theta > 0$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

such that the following two exclusions hold. First, no $p \in [p_1, p_{\mathrm{c}})$ admits expanded gate-annulus data satisfying all conclusions of Corollary 8.68 and

$$2^j > \left(\frac{1}{2\alpha_\omega} + \lambda \right) \log_\Delta \overline{\chi}_p.$$

Second, for every $p \in [p_1, p_{\mathrm{c}})$, every pair of vertices x, a, and every integer $j > J + H$ with

$$B \log_\Delta \overline{\chi}_p < 2^j \leq \left(\frac{1}{2\alpha_\omega} + \lambda \right) \log_\Delta \overline{\chi}_p,$$

if

$$L = \log_\Delta \overline{\chi}_p, \qquad I_{j,p}^{\mathrm{gate}} = \left(2^j - \lfloor \theta L \rfloor, 2^{j+1} + \lfloor \theta L \rfloor\right],$$

and $d(x,a) \in I_{j,p}^{\mathrm{gate}}$, then

$$\tau_p(x,a) < D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}\, L^{-3\eta}\, \overline{\chi}_p^{-2\lambda\alpha_\omega - 2\theta - 5\theta\alpha_\omega},$$

where $D_{\epsilon,\eta,J,\lambda,C_{\mathrm{vol}},\omega}$ is the constant from Corollary 8.75. Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. We argue by contraposition. Suppose that

$$\limsup_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.692, and choose the parameters from the assumed criterion for this ϵ. Take $p_0 \geq p_1$ sufficiently close to p_{c} that Corollary 8.692 applies. The corollary gives expanded gate-annulus data satisfying either the super-half-logarithmic scale condition, contradicting the first exclusion, or a pointwise root-annulus lower bound. In the latter case the inherited weighted separated-fan conclusion gives $2^j > B \log_\Delta \overline{\chi}_p$, so the second exclusion applies and gives the contradiction. Hence $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Remark 8.694. The pointwise formulation above is only a diagnostic. As Proposition 8.789 shows, global strict pointwise-supremum decay criteria can become vacuous because every geodesic gives the lower bound $\tau_p^{\max}(n) \geq p^n$. The annular average criterion in Proposition 8.80 retains more information and is the stronger residual target.

Lemma 8.695 (Gate-annulus mass-to-operator lower bound)**.** *Let G be connected and locally finite, let $I \subseteq [0,\infty)$, and let $A, Y \subset V$ be finite non-empty sets. Then*

$$\left\|T_p^I\right\|_{2\to 2} \geq \frac{\sum_{a\in A}\sum_{\substack{y\in Y\\ d(y,a)\in I}} \tau_p(y,a)}{\sqrt{|A|\,|Y|}}.$$

Proof. Let $\mathbf{1}_A$ and $\mathbf{1}_Y$ be the indicator functions of A and Y. Since T_p^I is a positive symmetric kernel,

$$\left\|T_p^I\right\|_{2\to 2} \geq \frac{\left\langle \mathbf{1}_Y, T_p^I \mathbf{1}_A\right\rangle}{\|\mathbf{1}_Y\|_2\,\|\mathbf{1}_A\|_2}.$$

The numerator is exactly the displayed double sum, and the denominator is $\sqrt{|A|\,|Y|}$. □

Corollary 8.696 (Volume-sensitive gate-annular operator dichotomy)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B\alpha_\omega < 1,$$

every $U > B$, every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and an interval

$$I_{j,p}^{\mathrm{gate}} = \left(2^j - r_p, 2^{j+1} + r_p\right], \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

such that, with $L = \log_\Delta \overline{\chi}_p$, either

$$2^j > UL,$$

or

$$\left\| T_p^{I_{j,p}^{\mathrm{gate}}} \right\|_{2\to 2} \geq C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}\, L^{-6\eta}\, \overline{\chi}_p^{\beta(\theta,U,\omega)}$$

where

$$C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{256 C_{\mathrm{vol}}^7}\left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^6 2^{6\eta J} U^{-6\eta}$$

and

$$\beta(\theta,U,\omega) = 2 - 4\theta - 8\theta\alpha_\omega - 3U\alpha_\omega - \frac{3}{2}\theta\alpha_\omega.$$

Proof. Use the ϵ from Corollary 8.68. Since $p_c > 1/\Delta$, choose p_0 close enough to p_c that every selected $p \in [p_0, p_c)$ satisfies $p \geq 1/\Delta$ and $L \geq 1$. Choose the expanded gate-annulus data from Corollary 8.68. If $2^j > UL$, the first alternative holds, so assume that $2^j \leq UL$.

Let

$$A = \{a_z : z \in Z^\sharp\}$$

and let

$$Y = \{y \in K : \text{ there exists } z \in Z^\sharp \text{ with } d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}.$$

The weighted separated-fan separation gives disjointness of the balls $B(z, r_p)$, and since $d(a_z, z) = r_p$, the map $z \mapsto a_z$ is injective on $Z^\sharp$. Thus the root mass lower bound and $\tau_p \leq 1$ imply

$$|A| \geq \sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j^{\mathrm{vol}}(p),$$

where

$$\kappa_j^{\mathrm{vol}}(p) = \frac{p^{2r_p}\sigma_j^3}{16C_{\mathrm{vol}}^3}\overline{\chi}_p^{1-4\theta\alpha_\omega}.$$

For every $z \in Z^\sharp$, the K-mass lower bound gives

$$\sum_{\substack{y \in Y \\ d(y,a_z) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a_z) \geq \kappa_j^{\mathrm{vol}}(p).$$

Consequently,

$$\sum_{a \in A} \sum_{\substack{y \in Y \\ d(y,a) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a) \geq |A|\kappa_j^{\mathrm{vol}}(p) \geq \left(\kappa_j^{\mathrm{vol}}(p)\right)^2.$$

The metric conclusions give $A \subseteq B(x, 2^{j+1} + r_p)$. If $y \in Y$, then for some $z \in Z^\sharp$,

$$d(x, y) \leq d(x, a_z) + d(a_z, y) \leq 2^{j+2} + 2r_p,$$

so $Y \subseteq B(x, 2^{j+2} + 2r_p)$. Hence

$$|A| \leq C_{\mathrm{vol}}\omega^{2^{j+1}+r_p}, \qquad |Y| \leq C_{\mathrm{vol}}\omega^{2^{j+2}+2r_p}.$$

Lemma 8.695 therefore yields

$$\left\| T_p^{I_{j,p}^{\mathrm{gate}}} \right\|_{2\to 2} \geq \frac{\left(\kappa_j^{\mathrm{vol}}(p)\right)^2}{C_{\mathrm{vol}}\omega^{3\cdot 2^j + (3/2)r_p}}.$$

Since $p \geq 1/\Delta$ and $r_p \leq \theta L$,

$$\kappa_j^{\mathrm{vol}}(p) \geq \frac{\sigma_j^3}{16C_{\mathrm{vol}}^3}\overline{\chi}_p^{1-2\theta-4\theta\alpha_\omega}.$$

Since $2^j \leq UL$, we also have $j \leq \log_2(UL)$, and hence

$$\sigma_j^6 \geq \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^6 2^{6\eta J} U^{-6\eta} L^{-6\eta}.$$

Finally,

$$\omega^{-3\cdot 2^j - (3/2)r_p} \geq \overline{\chi}_p^{-3U\alpha_\omega - (3/2)\theta\alpha_\omega}.$$

Combining the last three displays gives the stated lower bound. □

Corollary 8.697 (Gate support-profile product)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B\alpha_\omega < 1,$$

every $U > B$, every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are data satisfying all conclusions of Corollary 8.68 such that, with $L = \log_\Delta \overline{\chi}_p$ and

$$Y = \{y \in K :\ \text{there exists } z \in Z^\sharp \text{ with } d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\},$$

either

$$2^j > UL,$$

or

$$|Y|\,\Theta_p(I_{j,p}^{\mathrm{gate}}) \geq E_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}\, L^{-9\eta}\, \overline{\chi}_p^{\zeta(\theta,\omega)}$$

where

$$E_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{4096 C_{\mathrm{vol}}^9}\left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^9 2^{9\eta J} U^{-9\eta}$$

and

$$\zeta(\theta,\omega) = 1 - 6\theta - 12\theta\alpha_\omega.$$

Proof. Use the ϵ from Corollary 8.68. Since $p_{\mathrm{c}} > 1/\Delta$, choose p_0 close enough to p_{c} that every selected $p \in [p_0, p_{\mathrm{c}})$ satisfies $p \geq 1/\Delta$ and $L \geq 1$. Choose the expanded gate-annulus data. If $2^j > UL$, the first alternative holds, so assume $2^j \leq UL$.

Let $A = \{a_z : z \in Z^\sharp\}$. The weighted separated-fan separation gives disjointness of the balls $B(z, r_p)$, and since $d(a_z, z) = r_p$, the map $z \mapsto a_z$ is injective on $Z^\sharp$. Hence

$$|A| \geq \sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j^{\mathrm{vol}}(p).$$

Moreover, the K-mass lower bound gives

$$M := \sum_{a \in A} \sum_{\substack{y \in Y \\ d(y,a) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a) \geq |A| \kappa_j^{\mathrm{vol}}(p).$$

For $y \in Y$, put

$$s_y = \sum_{\substack{a \in A \\ d(y,a) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a).$$

Cauchy–Schwarz gives

$$\sum_{y \in Y} s_y^2 \geq \frac{M^2}{|Y|} \geq \frac{|A|^2 \left(\kappa_j^{\mathrm{vol}}(p)\right)^2}{|Y|}.$$

Using $K = A$ in Definition 8.21, we get

$$\Theta_p(I^{\mathrm{gate}}_{j,p}) \geq \frac{1}{\overline{\chi}_p^2|A|}\sum_{y\in Y} s_y^2 \geq \frac{\left(\kappa_j^{\mathrm{vol}}(p)\right)^3}{\overline{\chi}_p^2|Y|}.$$

Since $p \geq 1/\Delta$ and $r_p \leq \theta L$,

$$\kappa_j^{\mathrm{vol}}(p) \geq \frac{\sigma_j^3}{16C_{\mathrm{vol}}^3}\overline{\chi}_p^{1-2\theta-4\theta\alpha_\omega}.$$

Since $2^j \leq UL$, we also have $j \leq \log_2(UL)$, and hence

$$\sigma_j^9 \geq \left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^9 2^{9\eta J}U^{-9\eta}L^{-9\eta}.$$

Combining the last three displays gives the support-profile product lower bound. □

Remark 8.698. Corollary 8.697 separates the two losses in the square-profile route. The expanded gate data force a large product

$$|Y|\,\Theta_p(I^{\mathrm{gate}}_{j,p}).$$

Corollary 8.703 then divides by the crude ball-volume bound on Y. Thus any improvement of the square-profile threshold must either prove annular profile decay at the gate scale or replace the ball-volume estimate for Y by a sharper support-spread bound.

Corollary 8.699 (Support-profile escape)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$, every $U > B$, every $\theta > 0$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

and

$$\delta(\theta, U, \omega) := 2 - 6\theta - 15\theta\alpha_\omega - 6U\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are data satisfying all conclusions of Corollary 8.68 and

$$2^j > U\log_\Delta \overline{\chi}_p.$$

Proof. Use the ϵ from Corollary 8.68. Put

$$F_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{4096C_{\mathrm{vol}}^{10}}\left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^9 2^{9\eta J}U^{-9\eta}.$$

Since $\delta(\theta,U,\omega) > 0$, choose $p_* < p_{\mathrm{c}}$ sufficiently close to p_{c} that the expanded gate-annulus obstruction applies to every $p_0 \in [p_*, p_{\mathrm{c}})$, every selected $p \in [p_*, p_{\mathrm{c}})$ satisfies $p \geq 1/\Delta$ and $L = \log_\Delta \overline{\chi}_p \geq 1$, and

$$F_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}C_{\mathrm{vol}}^{-1}L^{-9\eta}\overline{\chi}_p^{\delta(\theta,U,\omega)} > 1.$$

Fix $p_0 \in [p_*, p_{\mathrm{c}})$, and choose the expanded gate-annulus data. Suppose, for contradiction, that $2^j \leq UL$. Let

$$A = \{a_z : z \in Z^\sharp\}$$

and

$$Y = \{y \in K : \text{ there exists } z \in Z^\sharp \text{ with } d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}.$$

As in the proof of Corollary 8.697, the specific finite set A satisfies

$$\frac{\mathcal{R}_p^{I_{j,p}^{\mathrm{gate}}}(A)}{\overline{\chi}_p} \geq \frac{\left(\kappa_j^{\mathrm{vol}}(p)\right)^3}{\overline{\chi}_p^2 |Y|}.$$

Lemma 8.22, applied to this same set A, gives

$$\frac{\mathcal{R}_p^{I_{j,p}^{\mathrm{gate}}}(A)}{\overline{\chi}_p} \leq \frac{|A|}{\overline{\chi}_p}.$$

Therefore

$$|Y| \geq \frac{\left(\kappa_j^{\mathrm{vol}}(p)\right)^3}{\overline{\chi}_p |A|}.$$

The metric conclusions give

$$A \subseteq B(x, 2^{j+1} + r_p), \qquad Y \subseteq B(x, 2^{j+2} + 2r_p).$$

Under the cap $2^j \leq UL$, this yields

$$|A| \leq C_{\mathrm{vol}} \overline{\chi}_p^{(2U+\theta)\alpha_\omega}, \qquad |Y| \leq C_{\mathrm{vol}} \overline{\chi}_p^{(4U+2\theta)\alpha_\omega}.$$

Since $p \geq 1/\Delta$, $r_p \leq \theta L$, and $j \leq \log_2(UL)$, we also have

$$\left(\kappa_j^{\mathrm{vol}}(p)\right)^3 \geq \frac{1}{4096 C_{\mathrm{vol}}^9} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^9 2^{9\eta J} U^{-9\eta} L^{-9\eta} \overline{\chi}_p^{3-6\theta-12\theta\alpha_\omega}.$$

Combining the last four displays gives

$$|Y| \geq F_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} L^{-9\eta} \overline{\chi}_p^{2-6\theta-13\theta\alpha_\omega-2U\alpha_\omega}.$$

Together with the upper bound on $|Y|$, this contradicts the choice of p_*, because the exponent difference is

$$2 - 6\theta - 13\theta\alpha_\omega - 2U\alpha_\omega - (4U + 2\theta)\alpha_\omega = \delta(\theta, U, \omega).$$

Hence $2^j > UL$, as required. □

Remark 8.700. The condition in Corollary 8.699 is non-empty precisely in the same optimized range as the Schur-calibrated argument: letting $\theta \downarrow 0$, it becomes $6U \log_\Delta \omega < 2$, i.e.

$$U < \frac{1}{3 \log_\Delta \omega}.$$

This is weaker as a pure scale statement than Corollary 8.70, which reaches $U < 1/(2 \log_\Delta \omega)$ using only the anchor count. The support-profile form is still useful because it records a two-replica support-volume contradiction for the specific gate set A, rather than only a cardinality contradiction or a consequence of the annular Schur bound.

Corollary 8.701 (Capped support-profile feasibility)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^{R} \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$ and

$$\delta(\theta, U, \omega) = 2 - 6\theta - 15\theta\alpha_\omega - 6U\alpha_\omega.$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$, every $U > B$, every $\theta > 0$ satisfying

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are data satisfying all conclusions of Corollary 8.68 such that, with $L = \log_\Delta \overline{\chi}_p$, either

$$2^j > UL,$$

or

$$G_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}\, L^{-9\eta}\overline{\chi}_p^{\delta(\theta,U,\omega)} \leq 1,$$

where

$$G_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{4096 C_{\mathrm{vol}}^{11}} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^9 2^{9\eta J} U^{-9\eta}.$$

Consequently, for fixed admissible parameters, a capped branch occurring arbitrarily close to criticality forces $\delta(\theta, U, \omega) \leq 0$.

Proof. Use the ϵ from Corollary 8.68. Take p_0 close enough to p_{c} that the expanded gate-annulus obstruction applies, every selected $p \in [p_0, p_{\mathrm{c}})$ satisfies $p \geq 1/\Delta$, and $L = \log_\Delta \overline{\chi}_p \geq 1$. Choose the expanded gate-annulus data. If $2^j > UL$, the first alternative holds, so assume $2^j \leq UL$.

The proof of Corollary 8.699 gives

$$|Y| \geq F_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}\, L^{-9\eta}\overline{\chi}_p^{2-6\theta-13\theta\alpha_\omega-2U\alpha_\omega},$$

where

$$F_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{4096 C_{\mathrm{vol}}^{10}} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^9 2^{9\eta J} U^{-9\eta}.$$

The metric containment $Y \subseteq B(x, 2^{j+2} + 2r_p)$ and the cap $2^j \leq UL$ give

$$|Y| \leq C_{\mathrm{vol}}\overline{\chi}_p^{(4U+2\theta)\alpha_\omega}.$$

Combining the last two displays gives the stated feasibility inequality.

If the capped branch occurs for a sequence $p \uparrow p_{\mathrm{c}}$, then

$$G_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}\, L^{-9\eta}\overline{\chi}_p^{\delta(\theta,U,\omega)} \leq 1$$

along that sequence. Since $\overline{\chi}_p \to \infty$ and $L = \log_\Delta \overline{\chi}_p$, this is impossible when $\delta(\theta, U, \omega) > 0$. Hence $\delta(\theta, U, \omega) \leq 0$. □

Corollary 8.702 (Optimized support-profile gate escape)**.** *Under the graph and volume assumptions of Corollary 8.699, put*

$$\alpha_\omega = \log_\Delta \omega.$$

There is $0 < \epsilon \le 1$ *such that the following holds. If* $\alpha_\omega > 0$*, then for every*

$$0 < U < \frac{1}{3\alpha_\omega},$$

every $\eta > 0$*, every pair of integers* $J, H \ge 0$*, and all* p_0 *sufficiently close to* p_c*, there are data satisfying all conclusions of Corollary 8.68 and*

$$2^j > U \log_\Delta \overline{\chi}_p.$$

If $\alpha_\omega = 0$*, the same conclusion holds for every finite* $U > 0$*.*

Proof. Use the ϵ from Corollary 8.699. First suppose that $\alpha_\omega > 0$ and $0 < U < 1/(3\alpha_\omega)$. Choose $B = U/2$, and then choose $\theta > 0$ so small that

$$\theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

and

$$2 - 6\theta - 15\theta\alpha_\omega - 6U\alpha_\omega > 0.$$

This is possible because $2 - 6U\alpha_\omega > 0$. Also

$$2B\alpha_\omega = U\alpha_\omega < \frac{1}{3} < 1.$$

Corollary 8.699 applies with these parameters and gives the desired data.

If $\alpha_\omega = 0$, fix any finite $U > 0$, choose $B = U/2$, and choose

$$0 < \theta < \min\{B, 1/3\}.$$

Then

$$2 - 6\theta - 15\theta\alpha_\omega - 6U\alpha_\omega = 2 - 6\theta > 0,$$

while the other parameter inequalities are automatic. Applying Corollary 8.699 again proves the claim. □

Corollary 8.703 (Volume-sensitive gate-annular square-profile dichotomy)**.** *Under the assumptions of Proposition 8.37, suppose in addition that* G *has maximum degree at most* $\Delta \ge 2$*, and that there are constants* $C_{\mathrm{vol}} < \infty$ *and* $1 \le \omega < \Delta$ *such that*

$$|B(u, R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

Put $\alpha_\omega = \log_\Delta \omega$*. There is* $0 < \epsilon \le 1$ *such that for every* $\eta > 0$*, every* $B > 0$ *satisfying*

$$2B\alpha_\omega < 1,$$

every $U > B$*, every* $\theta > 0$ *satisfying*

$$\theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and an interval

$$I_{j,p}^{\mathrm{gate}} = \left(2^j - r_p, 2^{j+1} + r_p\right], \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

such that, with $L = \log_\Delta \overline{\chi}_p$, either

$$2^j > UL,$$

or

$$\Theta_p(I_{j,p}^{\mathrm{gate}}) \geq D_{\epsilon,\eta,J,U,C_{\mathrm{vol}}}\, L^{-9\eta}\, \overline{\chi}_p^{\gamma(\theta,U,\omega)}$$

where

$$D_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{4096 C_{\mathrm{vol}}^{10}} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^9 2^{9\eta J} U^{-9\eta}$$

and

$$\gamma(\theta, U, \omega) = 1 - 6\theta - 14\theta\alpha_\omega - 4U\alpha_\omega.$$

Proof. Use the ϵ from Corollary 8.68. Since $p_c > 1/\Delta$, choose p_0 close enough to p_c that every selected $p \in [p_0, p_c)$ satisfies $p \geq 1/\Delta$ and $L \geq 1$. Choose the expanded gate-annulus data from Corollary 8.68. If $2^j > UL$, the first alternative holds, so assume that $2^j \leq UL$.

Let

$$A = \{a_z : z \in Z^\sharp\}$$

and let

$$Y = \{y \in K : \text{ there exists } z \in Z^\sharp \text{ with } d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}.$$

As in the proof of Corollary 8.696, the map $z \mapsto a_z$ is injective on $Z^\sharp$, and hence

$$|A| \geq \kappa_j^{\mathrm{vol}}(p).$$

Moreover,

$$M := \sum_{a \in A} \sum_{\substack{y \in Y \\ d(y,a) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a) \geq |A| \kappa_j^{\mathrm{vol}}(p).$$

For $y \in Y$, put

$$s_y = \sum_{\substack{a \in A \\ d(y,a) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a).$$

Then Cauchy–Schwarz gives

$$\sum_{y \in Y} s_y^2 \geq \frac{M^2}{|Y|} \geq \frac{|A|^2 \left(\kappa_j^{\mathrm{vol}}(p)\right)^2}{|Y|}.$$

Using $K = A$ in Definition 8.21, we obtain

$$\Theta_p(I_{j,p}^{\mathrm{gate}}) \geq \frac{1}{\overline{\chi}_p^2 |A|} \sum_{y \in Y} s_y^2 \geq \frac{|A| \left(\kappa_j^{\mathrm{vol}}(p)\right)^2}{\overline{\chi}_p^2 |Y|} \geq \frac{\left(\kappa_j^{\mathrm{vol}}(p)\right)^3}{\overline{\chi}_p^2 |Y|}.$$

The metric conclusions give $Y \subseteq B(x, 2^{j+2} + 2r_p)$, and therefore

$$|Y| \leq C_{\mathrm{vol}} \omega^{2^{j+2} + 2r_p}.$$

Since $p \ge 1/\Delta$ and $r_p \le \theta L$,

$$\kappa_j^{\mathrm{vol}}(p) \ge \frac{\sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-2\theta-4\theta\alpha_\omega}.$$

Since $2^j \le UL$, we also have $j \le \log_2(UL)$, and hence

$$\sigma_j^9 \ge \left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^9 2^{9\eta J} U^{-9\eta} L^{-9\eta}.$$

Finally,

$$\omega^{-2^{j+2}-2r_p} \ge \overline{\chi}_p^{-4U\alpha_\omega - 2\theta\alpha_\omega}.$$

Combining the last four displays gives the stated lower bound. □

Remark 8.704. Lemma 8.22 gives $\Theta_p(I) \le 1$ for every annulus I. Thus, whenever $\gamma(\theta, U, \omega) > 0$, the second branch in Corollary 8.703 is impossible sufficiently close to criticality. Optimizing with $\theta \downarrow 0$ gives the square-profile escape range $U < 1/(4 \log_\Delta \omega)$ when $\omega > 1$, weaker than the Schur-calibrated operator range in Corollary 8.711 but normalized in the two-replica profile.

Remark 8.705. Lemma 8.24 explains why a purely weighted re-run of the square-profile argument does not improve the exponent by itself. Applying it with $w = \mathbf{1}_A$ gives the lower bound used above. Applying it instead with the root weights $w(a_z) = \tau_p(x, a_z)$ gives the same worst-case estimate, because the available root-weight input is only $\sum_z \tau_p(x, a_z) \ge \kappa_j^{\mathrm{vol}}(p)$. Any improvement of the square-profile threshold must therefore prove either a larger selected gate count or a larger total selected root weight; changing weights alone is lossless but not stronger.

Proposition 8.706 (Capped gate-annular operator criterion)**.** *Let G have maximum degree at most $\Delta \ge 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \le \omega < \Delta$ such that*

$$|B(u, R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. Suppose that for every $0 < \epsilon \le 1$ there exist $\eta > 0$, $B > 0$, $U > B$, $\theta > 0$, integers $J, H \ge 0$, and $p_1 < p_{\mathrm{c}}$ such that

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

and the following two exclusions hold.

First, no $p \in [p_1, p_{\mathrm{c}})$ admits data satisfying all conclusions of Corollary 8.68 and

$$2^j > U \log_\Delta \overline{\chi}_p.$$

Second, for every $p \in [p_1, p_{\mathrm{c}})$ and every integer $j > J + H$ such that

$$2^j \le U \log_\Delta \overline{\chi}_p,$$

if

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor, \qquad I_{j,p}^{\mathrm{gate}} = \left(2^j - r_p, 2^{j+1} + r_p\right],$$

then

$$\left\| T_p^{I_{j,p}^{\mathrm{gate}}} \right\|_{2\to 2} < C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} L_p^{-6\eta} \overline{\chi}_p^{\beta(\theta,U,\omega)}, \qquad L_p = \log_\Delta \overline{\chi}_p,$$

where

$$C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{256 C_{\mathrm{vol}}^7} \left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^6 2^{6\eta J} U^{-6\eta}$$

and

$$\beta(\theta, U, \omega) = 2 - 4\theta - 8\theta\alpha_\omega - 3U\alpha_\omega - \frac{3}{2}\theta\alpha_\omega.$$

Then

$$\lim_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. We prove the first claim by contraposition. Suppose that

$$\limsup_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Let ϵ be the constant supplied by Corollary 8.68, and choose the parameters from the assumed criterion for this ϵ. Take $p_0 \geq p_1$ sufficiently close to p_c that the expanded gate-annulus obstruction applies, and choose its data.

If $2^j > U \log_\Delta \overline{\chi}_p$, this data violates the first exclusion. Thus $2^j \leq U \log_\Delta \overline{\chi}_p$. Repeating the proof of Corollary 8.696 with this same data gives

$$\left\| T_p^{I_{j,p}^{\mathrm{gate}}} \right\|_{2\to 2} \geq C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} L_p^{-6\eta} \overline{\chi}_p^{\beta(\theta,U,\omega)},$$

contradicting the second exclusion. Hence $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

The L^2-gap conclusion follows from Corollary 8.794. □

Remark 8.707. The first exclusion in Proposition 8.706 is an exclusion of large scales carrying the full expanded gate-annulus obstruction data. It is not the meaningless assertion that no large integer j exists. This is the point of keeping the gate-incidence data visible after the pointwise audit.

Lemma 8.708 (Annular Schur bound)**.** *Let G be connected and locally finite. For every $p < p_c$ and every $I \subseteq [0, \infty)$,*

$$\left\| T_p^I \right\|_{2\to 2} \leq \overline{\chi}_p.$$

Proof. The row sums of T_p^I are bounded above by the row sums of T_p, hence

$$\sup_x \sum_y T_p^I(x, y) \leq \overline{\chi}_p.$$

The kernel T_p^I is symmetric, so the same bound holds for column sums. Schur's test gives the claim. □

Corollary 8.709 (Capped branch escape)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Put $\alpha_\omega = \log_\Delta \omega$. *There is* $0 < \epsilon \le 1$ *such that for every* $\eta > 0$, *every* $B > 0$, *every* $U > B$, *every* $\theta > 0$ *satisfying*

$$2B\alpha_\omega < 1, \qquad \theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

and

$$\beta(\theta, U, \omega) = 2 - 4\theta - 8\theta\alpha_\omega - 3U\alpha_\omega - \frac{3}{2}\theta\alpha_\omega > 1,$$

every pair of integers $J, H \ge 0$, *and all* p_0 *sufficiently close to* p_c, *there are data satisfying all conclusions of Corollary 8.68 and*

$$2^j > U \log_\Delta \overline{\chi}_p.$$

Proof. Use the ϵ from Corollary 8.68. Let

$$C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} = \frac{1}{256 C_{\mathrm{vol}}^7} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^6 2^{6\eta J} U^{-6\eta}.$$

Since $\overline{\chi}_p \to \infty$ and $\beta(\theta, U, \omega) > 1$, choose $p_* < p_c$ sufficiently close to p_c that the expanded gate-annulus obstruction applies to every $p_0 \in [p_*, p_c)$, and whenever $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$,

$$C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} L^{-6\eta} \overline{\chi}_p^{\beta(\theta,U,\omega)} > \overline{\chi}_p.$$

Fix $p_0 \in [p_*, p_c)$, and choose the expanded gate-annulus data. Suppose, for contradiction, that $2^j \le U \log_\Delta \overline{\chi}_p$. The proof of Corollary 8.696, applied to this same data, gives

$$\left\| T_p^{I_{j,p}^{\mathrm{gate}}} \right\|_{2\to 2} \ge C_{\epsilon,\eta,J,U,C_{\mathrm{vol}}} L^{-6\eta} \overline{\chi}_p^{\beta(\theta,U,\omega)} > \overline{\chi}_p.$$

This contradicts Lemma 8.708. Hence $2^j > U \log_\Delta \overline{\chi}_p$. □

Remark 8.710. The exponent condition in Corollary 8.709 is non-empty. For example, if

$$3U \log_\Delta \omega < 1,$$

then choosing $0 < \theta < B < U$ sufficiently small gives $\beta(\theta, U, \omega) > 1$.

Corollary 8.711 (Optimized logarithmic gate-scale escape)**.** *Under the assumptions of Corollary 8.709, put*

$$\alpha_\omega = \log_\Delta \omega.$$

There is $0 < \epsilon \le 1$ *such that the following holds. If* $\alpha_\omega > 0$, *then for every*

$$0 < U < \frac{1}{3\alpha_\omega},$$

every $\eta > 0$, *every pair of integers* $J, H \ge 0$, *and all* p_0 *sufficiently close to* p_c, *there are data satisfying all conclusions of Corollary 8.68 and*

$$2^j > U \log_\Delta \overline{\chi}_p.$$

If $\alpha_\omega = 0$, *the same conclusion holds for every finite* $U > 0$.

Proof. Use the ϵ from Corollary 8.709. First suppose that $\alpha_\omega > 0$ and $0 < U < 1/(3\alpha_\omega)$. Choose $B = U/2$, and then choose $\theta > 0$ so small that

$$\theta < B, \qquad 1 - 4\theta\alpha_\omega > 0,$$

and

$$2 - 4\theta - 8\theta\alpha_\omega - 3U\alpha_\omega - \frac{3}{2}\theta\alpha_\omega > 1.$$

This is possible because $1 - 3U\alpha_\omega > 0$. Also

$$2B\alpha_\omega = U\alpha_\omega < \frac{1}{3} < 1.$$

Corollary 8.709 applies with these parameters and gives the desired data.

If $\alpha_\omega = 0$, fix any finite $U > 0$, choose $B = U/2$, and choose $0 < \theta < \min\{B, 1/4\}$. Then

$$\beta(\theta, U, \omega) = 2 - 4\theta > 1,$$

while the other parameter inequalities are automatic. Applying Corollary 8.709 again proves the claim. □

Corollary 8.712 (Volume-sensitive root-weighted gate-incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 4\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are $p \in [p_0, p_{\mathrm{c}})$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

$$I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

and

$$\kappa_j^{\mathrm{vol}}(p) = \frac{p^{2r_p}\sigma_j^3}{16C_{\mathrm{vol}}^3}\overline{\chi}_p^{1-4\theta\log_\Delta \omega},$$

the tuple (K, x, j, Z_θ) satisfies the conclusions of Corollary 8.57,

$$d(a_z, z) = r_p, \qquad d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z^\sharp),$$

$$\sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j^{\mathrm{vol}}(p),$$

and

$$|\{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \geq \kappa_j^{\mathrm{vol}}(p) \qquad (z \in Z^\sharp).$$

Proof. Choose the data by Corollary 8.68. For each $z \in Z^\sharp$,

$$\kappa_j^{\mathrm{vol}}(p) \le \sum_{\substack{y \in K \\ d(y,a_z) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a_z) \le |\{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}|,$$

since $\tau_p(y, a_z) \le 1$. The other properties are inherited. □

Corollary 8.713 (Volume-sensitive deterministic anchored gate-incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \ge 1$ such that*

$$|B(u, R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 4\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

$$I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

and

$$\kappa_j^{\mathrm{vol}}(p) = \frac{p^{2r_p} \sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1 - 4\theta \log_\Delta \omega},$$

the tuple (K, x, j, Z_θ) satisfies the conclusions of Corollary 8.57,

$$|Z^\sharp| \ge \kappa_j^{\mathrm{vol}}(p),$$

$$d(a_z, z) = r_p, \qquad d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z^\sharp),$$

and

$$|\{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \ge \kappa_j^{\mathrm{vol}}(p) \qquad (z \in Z^\sharp).$$

Proof. Choose the data by Corollary 8.712. Since $\tau_p(x, a_z) \le 1$,

$$|Z^\sharp| \ge \sum_{z \in Z^\sharp} \tau_p(x, a_z) \ge \kappa_j^{\mathrm{vol}}(p).$$

The remaining properties are inherited. □

Corollary 8.714 (Volume-sensitive uniform deterministic gate-incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 2\theta - 4\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are $p \in [p_0, p_{\mathrm{c}})$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

$$I^{\mathrm{gate}}_{j,p} = (2^j - r_p, 2^{j+1} + r_p],$$

and

$$\widehat{\kappa}^{\mathrm{vol}}_j(p) = \frac{\sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-2\theta-4\theta\log_\Delta \omega},$$

the tuple (K, x, j, Z_θ) satisfies the conclusions of Corollary 8.57,

$$|Z^\sharp| \geq \widehat{\kappa}^{\mathrm{vol}}_j(p),$$

$$d(a_z, z) = r_p, \qquad d(x, a_z) \in I^{\mathrm{gate}}_{j,p} \qquad (z \in Z^\sharp),$$

and

$$|\{y \in K : d(y, a_z) \in I^{\mathrm{gate}}_{j,p}\}| \geq \widehat{\kappa}^{\mathrm{vol}}_j(p) \qquad (z \in Z^\sharp).$$

Proof. Choose p_* close enough to p_{c} that the conclusion of Corollary 8.713 applies to every $p_0 \in [p_*, p_{\mathrm{c}})$, and also $p_* \geq 1/\Delta$. This is possible by Lemma 8.732. Fix $p_0 \in [p_*, p_{\mathrm{c}})$, and choose the deterministic gate-incidence data. Since $p \geq p_0 \geq 1/\Delta$,

$$p^{2r_p} \geq \Delta^{-2r_p} \geq \overline{\chi}_p^{-2\theta}.$$

Hence the threshold $\kappa^{\mathrm{vol}}_j(p)$ from Corollary 8.713 satisfies

$$\kappa^{\mathrm{vol}}_j(p) = \frac{p^{2r_p}\sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-4\theta\log_\Delta \omega} \geq \widehat{\kappa}^{\mathrm{vol}}_j(p).$$

The displayed lower bounds with $\widehat{\kappa}^{\mathrm{vol}}_j(p)$ follow. □

Corollary 8.715 (Volume-sensitive localized anchored incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_\Delta \omega < 1,$$

every $\theta > 0$ satisfying

$$\theta < B, \qquad 1 - 2\theta - 4\theta \log_\Delta \omega > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, a vertex $x \in V$, finite non-empty sets $A, Y, Z \subset V$, and a bijection $a : Z \to A$ such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}],$$
$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$
$$I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

and

$$\widehat{\kappa}_j^{\mathrm{vol}}(p) = \frac{\sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-2\theta-4\theta\log_\Delta \omega},$$

we have

$$2^j > BL, \qquad |A| \geq \widehat{\kappa}_j^{\mathrm{vol}}(p), \qquad |Y| \geq \widehat{\kappa}_j^{\mathrm{vol}}(p),$$
$$d(x,z) \in I_j, \qquad d(a_z, z) = r_p, \qquad d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z),$$
$$d(z, z') > 2r_p \qquad (z, z' \in Z,\ z \neq z'),$$
$$Y \subseteq B(x, 2^{j+2} + 2r_p),$$

and

$$|\{y \in Y : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \geq \widehat{\kappa}_j^{\mathrm{vol}}(p) \qquad (z \in Z).$$

Proof. Choose $K, x, j, Z_\theta, Z^\sharp$, and $a : Z^\sharp \to V$, by Corollary 8.714. Put

$$Z = Z^\sharp, \qquad A = \{a_z : z \in Z^\sharp\},$$

and

$$Y = \bigcup_{z \in Z^\sharp} \{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}.$$

If $z, z' \in Z^\sharp$ and $z \neq z'$, then $a_z \neq a_{z'}$, since $a_z = a_{z'}$ would imply $d(z, z') \leq 2r_p$, contradicting the separated fan condition. Thus $a : Z \to A$ is a bijection and

$$|A| = |Z^\sharp| \geq \widehat{\kappa}_j^{\mathrm{vol}}(p).$$

The same lower bound for $|Y|$ follows from the pointwise incidence lower bound for any fixed $z \in Z^\sharp$.

For $y \in Y$, choose $z \in Z^\sharp$ with $d(y, a_z) \in I_{j,p}^{\mathrm{gate}}$. Since also $d(x, a_z) \in I_{j,p}^{\mathrm{gate}}$,

$$d(x, y) \leq d(x, a_z) + d(a_z, y) \leq 2^{j+2} + 2r_p.$$

The remaining displayed properties are inherited from the volume-sensitive uniform deterministic gate-incidence obstruction and the separated fan conditions. □

Corollary 8.716 (Volume-sensitive improved logarithmic gate scale)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^{R} \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_{\Delta} \omega < 1,$$

every $\theta > 0$ satisfying $\theta < B$, every $a > 0$ satisfying

$$(2a+\theta)\log_{\Delta}\omega < 1 - 2\theta - 4\theta \log_{\Delta}\omega,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, the volume-sensitive localized anchored incidence obstruction of Corollary 8.715 can be chosen so that, with $L = \log_{\Delta} \overline{\chi}_p$,

$$2^{j} \geq aL.$$

Proof. Let ϵ be the constant from Corollary 8.715. Fix η, B, θ, a, J, H as in the statement, and put

$$c_a = \frac{1}{16 C_{\mathrm{vol}}^{3}} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^{2}\right]^{3} 2^{3\eta J} a^{-3\eta}.$$

Choose $p_* < p_{\mathrm{c}}$ sufficiently close to p_{c} that the volume-sensitive localized anchored incidence obstruction applies to every $p_0 \in [p_*, p_{\mathrm{c}})$, and whenever $p \in [p_*, p_{\mathrm{c}})$, with $L = \log_{\Delta} \overline{\chi}_p$,

$$c_a L^{-3\eta} \overline{\chi}_p^{\,1-2\theta-4\theta\log_{\Delta}\omega} > C_{\mathrm{vol}} \overline{\chi}_p^{\,(2a+\theta)\log_{\Delta}\omega}.$$

This is possible by the strict exponent inequality in the statement.

Fix $p_0 \in [p_*, p_{\mathrm{c}})$, and choose the localized anchored incidence data. Suppose for contradiction that $2^{j} < aL$. Then $j \leq \log_2(aL)$, and the definition of σ_j gives

$$\widehat{\kappa}_j^{\mathrm{vol}}(p) = \frac{\sigma_j^{3}}{16 C_{\mathrm{vol}}^{3}} \overline{\chi}_p^{\,1-2\theta-4\theta\log_{\Delta}\omega} \geq c_a L^{-3\eta} \overline{\chi}_p^{\,1-2\theta-4\theta\log_{\Delta}\omega}.$$

On the other hand, $A \subseteq B(x, 2^{j+1} + r_p)$, so the volume-growth bound gives

$$|A| \leq C_{\mathrm{vol}}\omega^{2^{j+1}+r_p} < C_{\mathrm{vol}} \overline{\chi}_p^{\,(2a+\theta)\log_{\Delta}\omega}.$$

This contradicts $|A| \geq \widehat{\kappa}_j^{\mathrm{vol}}(p)$. Hence $2^{j} \geq aL$. □

Corollary 8.717 (Volume-sensitive matched anchored incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^{R} \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $B > 0$ satisfying

$$2B \log_{\Delta} \omega < 1,$$

every $\theta > 0$ satisfying $\theta < B$, every $a > 0$ satisfying

$$(2a + \theta)\log_\Delta \omega < 1 - 2\theta - 4\theta \log_\Delta \omega,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, a vertex $x \in V$, and vertices

$$z_1, \dots, z_m, \qquad a_1, \dots, a_m, \qquad y_1, \dots, y_m$$

such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

$$\widehat{\kappa}_j^{\mathrm{vol}}(p) = \frac{\sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-2\theta-4\theta\log_\Delta \omega}, \qquad m = \left\lceil \widehat{\kappa}_j^{\mathrm{vol}}(p) \right\rceil,$$

we have

$$2^j \geq aL,$$

the vertices $z_1, \dots, z_m$ are pairwise distinct and satisfy

$$d(z_i, z_k) > 2r_p \qquad (i \neq k),$$

the vertices $a_1, \dots, a_m$ are pairwise distinct, the vertices $y_1, \dots, y_m$ are pairwise distinct, and, for every $1 \leq i \leq m$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p).$$

Proof. Choose the localized anchored incidence data from Corollary 8.716. Form the finite bipartite graph $\mathfrak{B}$ with left part A, right part Y, and an edge between $a_z \in A$ and $y \in Y$ when

$$d(y, a_z) \in I_{j,p}^{\mathrm{gate}}.$$

The corollary gives $|A| \geq \widehat{\kappa}_j^{\mathrm{vol}}(p)$, and the bijection $a : Z \to A$ gives, for every $a_z \in A$,

$$\deg_{\mathfrak{B}}(a_z) = |\{y \in Y : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \geq \widehat{\kappa}_j^{\mathrm{vol}}(p).$$

Since cardinalities and degrees are integers,

$$|A| \geq m, \qquad \deg_{\mathfrak{B}}(a) \geq m \quad (a \in A), \qquad m = \left\lceil \widehat{\kappa}_j^{\mathrm{vol}}(p) \right\rceil.$$

Choose $A_0 \subseteq A$ with $|A_0| = m$. For every non-empty $S \subseteq A_0$, the neighbourhood $N(S)$ contains the neighbourhood of any vertex of S, and hence $|N(S)| \geq m \geq |S|$. Hall's theorem gives a matching saturating A_0. Write the matched left vertices as $a_1, \dots, a_m$, the matched right vertices as $y_1, \dots, y_m$, and set $z_i = a^{-1}(a_i)$. The matching gives the distinctness of the a_i's and y_i's, the bijection $a : Z \to A$ gives the distinctness of the z_i's, and all displayed metric properties are inherited from the localized anchored incidence configuration. □

Proposition 8.718 (Volume-sensitive matched anchored incidence criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $B > 0$, $\theta > 0$, $a > 0$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that

$$2B \log_\Delta \omega < 1, \qquad \theta < B,$$

$$(2a+\theta)\log_\Delta \omega < 1 - 2\theta - 4\theta \log_\Delta \omega,$$

and no $p \in [p_1, p_{\mathrm{c}})$ admits an integer $j > J + H$, a vertex $x \in V$, and vertices

$$z_1, \dots, z_m, \qquad a_1, \dots, a_m, \qquad y_1, \dots, y_m$$

with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \lfloor \theta L \rfloor,$$

$$\widehat{\kappa}_j^{\mathrm{vol}}(p) = \frac{\sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1 - 2\theta - 4\theta \log_\Delta \omega}, \qquad m = \left\lceil \widehat{\kappa}_j^{\mathrm{vol}}(p) \right\rceil,$$

such that all of the following hold:

$$2^j \geq aL,$$

the vertices $z_1, \dots, z_m$ are pairwise distinct and satisfy

$$d(z_i, z_k) > 2r_p \qquad (i \neq k),$$

the vertices $a_1, \dots, a_m$ are pairwise distinct, the vertices $y_1, \dots, y_m$ are pairwise distinct, and, for every $1 \leq i \leq m$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p).$$

Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.717, with the L^2-gap conclusion supplied by Corollary 8.794. □

Proposition 8.719 (Gate-energy exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/4$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p satisfies*

$$\mathcal{G}_{p,r_p}^{(2^j, 2^{j+1}]}(x, K, Z) \geq \frac{\sigma_j^3}{8\Delta} \overline{\chi}_p^{2-2\theta}$$

for any $p \in [p_1, p_c)$, where σ_j and r_p are the parameters in Definition 8.58. Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.63, with the L^2-gap conclusion again supplied by Corollary 8.794. □

Corollary 8.720 (Point-gated fan obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/4$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, and maps $a, b : Z_\theta \to V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

(K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p,

$$d(a_z, z) = d(b_z, z) = r_p \qquad (z \in Z_\theta),$$

and

$$\mathcal{H}_{p,r_p}^{I_j}(x, K, Z_\theta, a, b) \geq \frac{\sigma_j^3}{8\Delta^3} \overline{\chi}_p^{2-4\theta}.$$

Proof. Choose p, K, j, x, Z_θ by Corollary 8.63. Since $d(x, z) \in I_j$ for every $z \in Z_\theta$ and $r_p < 2^j$, each sphere $\{v : d(v, z) = r_p\}$ is non-empty. Let

$$M_{r_p} = \sup_{z \in V} |\{v : d(v, z) = r_p\}|.$$

Lemma 8.53 gives maps $a, b : Z_\theta \to V$ with

$$\mathcal{H}_{p,r_p}^{I_j}(x, K, Z_\theta, a, b) \geq M_{r_p}^{-2} \mathcal{G}_{p,r_p}^{I_j}(x, K, Z_\theta).$$

The maximum-degree bound gives

$$M_{r_p} \leq \Delta^{r_p+1} \leq \Delta \overline{\chi}_p^{\theta}.$$

Combining this with Corollary 8.63 proves the displayed point-gated lower bound. □

Proposition 8.721 (Point-gated exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/4$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p admits maps $a, b : Z \to V$, with $d(a_z, z) = d(b_z, z) = r_p$, satisfying*

$$\mathcal{H}_{p,r_p}^{(2^j,2^{j+1}]}(x, K, Z, a, b) \geq \frac{\sigma_j^3}{8\Delta^3}\overline{\chi}_p^{2-4\theta}$$

for any $p \in [p_1, p_{\mathrm{c}})$, where σ_j and r_p are the parameters in Definition 8.58. Then

$$\lim_{p\uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p\uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.720, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.722 (One-gate fan obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/4$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are $p \in [p_0, p_{\mathrm{c}})$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, and a map $a : Z_\theta \to V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

(K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p,

$$d(a_z, z) = r_p \qquad (z \in Z_\theta),$$

and

$$\mathcal{J}_{p,r_p}^{I_j}(x, K, Z_\theta, a) \geq p^{2r_p}\frac{\sigma_j^3}{8\Delta^3}\overline{\chi}_p^{2-4\theta}.$$

Proof. Choose $p, K, j, x, Z_\theta, a, b$ by Corollary 8.720. Lemma 8.54 gives

$$\mathcal{J}_{p,r_p}^{I_j}(x, K, Z_\theta, a) \geq p^{2r_p}\mathcal{H}_{p,r_p}^{I_j}(x, K, Z_\theta, a, b),$$

and the point-gated lower bound gives the claim. □

Proposition 8.723 (One-gate exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/4$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p admits a map $a : Z \to V$, with $d(a_z, z) = r_p$, satisfying*

$$\mathcal{J}_{p,r_p}^{(2^j,2^{j+1}]}(x, K, Z, a) \geq p^{2r_p}\frac{\sigma_j^3}{8\Delta^3}\overline{\chi}_p^{2-4\theta}$$

for any $p \in [p_1, p_c)$, where σ_j and r_p are the parameters in Definition 8.58. Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.722, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.724 (High one-gate obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/4$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad \sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

(K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p,

$$d(a_z, z) = r_p \qquad (z \in Z^\sharp),$$

and, putting

$$\kappa_j(p) = \frac{p^{2r_p}\sigma_j^3}{16\Delta^3}\overline{\chi}_p^{1-4\theta},$$

we have

$$\sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j(p)$$

and

$$\sum_{\substack{y \in K \\ d(y,z) \in I_j}} \tau_p(y, a_z) \geq \kappa_j(p) \qquad (z \in Z^\sharp).$$

Proof. Choose p, K, j, x, Z_θ and a map $a : Z_\theta \to V$ by Corollary 8.722, and set

$$L = \frac{p^{2r_p}\sigma_j^3}{8\Delta^3}\overline{\chi}_p^{2-4\theta}.$$

Since (K, x, j, Z_θ) is separated, the balls $B(z, r_p)$, $z \in Z_\theta$, are pairwise disjoint. Hence the vertices a_z are distinct, and

$$\sum_{z \in Z_\theta} \tau_p(x, a_z) \leq \overline{\chi}_p.$$

Apply Lemma 8.55 to this map a. The resulting set $Z^\sharp$ satisfies the two displayed lower bounds, since $L/(2\overline{\chi}_p) = \kappa_j(p)$. The root lower bound also makes $Z^\sharp$ non-empty. □

Proposition 8.725 (High one-gate exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/4$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p admits a non-empty subset $Z' \subseteq Z$ and a map $a : Z' \to V$, with $d(a_z, z) = r_p$, satisfying*

$$\sum_{z \in Z'} \tau_p(x, a_z) \geq \kappa_j(p)$$

and

$$\sum_{\substack{y \in K \\ d(y,z) \in (2^j, 2^{j+1}]}} \tau_p(y, a_z) \geq \kappa_j(p) \qquad (z \in Z')$$

for any $p \in [p_1, p_{\mathrm{c}})$, where σ_j and r_p are the parameters in Definition 8.58 and

$$\kappa_j(p) = \frac{p^{2r_p} \sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-4\theta}.$$

Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2 \to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.724, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.726 (Expanded gate obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/4$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are $p \in [p_0, p_{\mathrm{c}})$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with*

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

and with $\sigma_j, r_p, \kappa_j(p)$ as in Corollary 8.724, the tuple (K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p,

$$d(a_z, z) = r_p \qquad (z \in Z^\sharp),$$
$$d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z^\sharp),$$
$$\sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j(p),$$

and

$$\sum_{\substack{y \in K \\ d(y,a_z) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a_z) \geq \kappa_j(p) \qquad (z \in Z^\sharp).$$

Proof. Choose the data by Corollary 8.724. If $z \in Z^\sharp$, then $d(a_z, z) = r_p$, while the separated fan conditions give $d(x, z) \in I_j$. The triangle inequality gives

$$d(x, a_z) \in (2^j - r_p, 2^{j+1} + r_p] = I_{j,p}^{\mathrm{gate}}.$$

Similarly, every $y \in K$ with $d(y, z) \in I_j$ satisfies

$$d(y, a_z) \in I_{j,p}^{\mathrm{gate}}.$$

Thus the pointwise K-mass lower bound from Corollary 8.724 remains valid when the summation is enlarged to $d(y, a_z) \in I_{j,p}^{\mathrm{gate}}$. The remaining displayed properties are inherited. □

Proposition 8.727 (Expanded gate-annulus exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/4$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p admits a non-empty subset $Z' \subseteq Z$ and a map $a : Z' \to V$, with $d(a_z, z) = r_p$, satisfying*

$$d(x, a_z) \in (2^j - r_p, 2^{j+1} + r_p] \qquad (z \in Z'),$$

$$\sum_{z \in Z'} \tau_p(x, a_z) \geq \kappa_j(p),$$

and

$$\sum_{\substack{y \in K \\ d(y,a_z) \in (2^j - r_p, 2^{j+1} + r_p]}} \tau_p(y, a_z) \geq \kappa_j(p) \qquad (z \in Z')$$

for any $p \in [p_1, p_c)$, where σ_j, r_p are the parameters in Definition 8.58 and

$$\kappa_j(p) = \frac{p^{2r_p} \sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-4\theta}.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.726, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.728 (Root-weighted gate-incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/4$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with*

$$I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

and with $\sigma_j, r_p, \kappa_j(p)$ as in Corollary 8.724, the tuple (K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p,

$$d(a_z, z) = r_p, \qquad d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z^\sharp),$$

$$\sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j(p),$$

and

$$|\{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \geq \kappa_j(p) \qquad (z \in Z^\sharp).$$

Proof. Choose the data by Corollary 8.726. For each $z \in Z^\sharp$,

$$\kappa_j(p) \leq \sum_{\substack{y \in K \\ d(y,a_z) \in I_{j,p}^{\mathrm{gate}}}} \tau_p(y, a_z) \leq |\{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}|,$$

since $\tau_p(y, a_z) \leq 1$. The other properties are inherited. □

Proposition 8.729 (Gate-incidence exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/4$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p admits a non-empty subset $Z' \subseteq Z$ and a map $a : Z' \to V$, with $d(a_z, z) = r_p$, satisfying*

$$d(x, a_z) \in (2^j - r_p, 2^{j+1} + r_p] \qquad (z \in Z'),$$

$$\sum_{z \in Z'} \tau_p(x, a_z) \geq \kappa_j(p),$$

and

$$|\{y \in K : d(y, a_z) \in (2^j - r_p, 2^{j+1} + r_p]\}| \geq \kappa_j(p) \qquad (z \in Z')$$

for any $p \in [p_1, p_{\mathrm{c}})$, where σ_j, r_p are the parameters in Definition 8.58 and

$$\kappa_j(p) = \frac{p^{2r_p} \sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-4\theta}.$$

Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.728, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.730 (Deterministic anchored gate-incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/4$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with*

$$I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

and with $\sigma_j, r_p, \kappa_j(p)$ as in Corollary 8.724, the tuple (K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p,

$$|Z^\sharp| \geq \kappa_j(p),$$

$$d(a_z, z) = r_p, \qquad d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z^\sharp),$$

and

$$|\{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \geq \kappa_j(p) \qquad (z \in Z^\sharp).$$

Proof. Choose the data by Corollary 8.728. Since $\tau_p(x, a_z) \leq 1$,

$$|Z^\sharp| \geq \sum_{z \in Z^\sharp} \tau_p(x, a_z) \geq \kappa_j(p).$$

The remaining properties are inherited. □

Proposition 8.731 (Deterministic gate-incidence exclusion criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/4$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p admits a non-empty subset $Z' \subseteq Z$ and a map $a : Z' \to V$, with*

$$|Z'| \geq \kappa_j(p),$$

$$d(a_z, z) = r_p, \qquad d(x, a_z) \in (2^j - r_p, 2^{j+1} + r_p] \qquad (z \in Z'),$$

and

$$|\{y \in K : d(y, a_z) \in (2^j - r_p, 2^{j+1} + r_p]\}| \geq \kappa_j(p) \qquad (z \in Z')$$

for any $p \in [p_1, p_c)$, where σ_j, r_p are the parameters in Definition 8.58 and

$$\kappa_j(p) = \frac{p^{2r_p} \sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-4\theta}.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.730, with the L^2-gap conclusion supplied by Corollary 8.794. □

Lemma 8.732 (Bounded-degree lower bound on criticality)**.** *If G has maximum degree at most $\Delta \geq 2$, then*

$$p_c(G) \geq \frac{1}{\Delta - 1} \geq \frac{1}{\Delta}.$$

Proof. For $p < 1/(\Delta - 1)$, the expected size of the open cluster of any vertex x is at most the union bound over self-avoiding paths:

$$\chi_p(x) \leq 1 + \sum_{n \geq 1} \Delta(\Delta - 1)^{n-1} p^n < \infty.$$

Thus there is no infinite cluster at x almost surely. Since x was arbitrary, $p \leq p_c(G)$. Letting $p \uparrow 1/(\Delta - 1)$ gives the first bound, and the second is immediate from $\Delta \geq 2$. □

Corollary 8.733 (Uniform deterministic gate-incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/6$, every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, an integer $j > J + H$, a vertex $x \in K$, a finite non-empty set $Z_\theta \subset V$, a non-empty subset $Z^\sharp \subseteq Z_\theta$, and a map $a : Z^\sharp \to V$ such that, with*

$$I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

with σ_j, r_p as in Definition 8.58, and with

$$\widehat{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-6\theta},$$

the tuple (K, x, j, Z_θ) is an $(\epsilon, \eta, \theta, J, H)$-separated fan at p,

$$|Z^\sharp| \geq \widehat{\kappa}_j(p),$$

$$d(a_z, z) = r_p, \qquad d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z^\sharp),$$

and

$$|\{y \in K : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \geq \widehat{\kappa}_j(p) \qquad (z \in Z^\sharp).$$

Proof. Choose p_* close enough to p_c that the conclusion of Corollary 8.730 applies to every $p_0 \in [p_*, p_c)$, and also $p_* \geq 1/\Delta$. This is possible by Lemma 8.732. Fix $p_0 \in [p_*, p_c)$, and choose the deterministic gate-incidence data. Since $p \geq p_0 \geq 1/\Delta$,

$$p^{2r_p} \geq \Delta^{-2r_p} \geq \overline{\chi}_p^{-2\theta}.$$

Hence the threshold $\kappa_j(p)$ from Corollary 8.730 satisfies

$$\kappa_j(p) = \frac{p^{2r_p} \sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-4\theta} \geq \widehat{\kappa}_j(p).$$

The displayed lower bounds with $\widehat{\kappa}_j(p)$ follow. □

Proposition 8.734 (Uniform deterministic gate-incidence criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/6$, integers $J, H \geq 0$, and $p_1 < p_c$ such that no $(\epsilon, \eta, \theta, J, H)$-separated fan (K, x, j, Z) at p admits a non-empty subset $Z' \subseteq Z$ and a map $a : Z' \to V$, with*

$$|Z'| \geq \widehat{\kappa}_j(p),$$

$$d(a_z,z)=r_p,\qquad d(x,a_z)\in(2^j-r_p,2^{j+1}+r_p]\qquad (z\in Z'),$$

and

$$|\{y\in K: d(y,a_z)\in(2^j-r_p,2^{j+1}+r_p]\}|\geq \widehat{\kappa}_j(p)\qquad (z\in Z')$$

for any $p\in[p_1,p_c)$*, where* σ_j, r_p *are the parameters in Definition 8.58 and*

$$\widehat{\kappa}_j(p)=\frac{\sigma_j^3}{16\Delta^3}\overline{\chi}_p^{1-6\theta}.$$

Then

$$\lim_{p\uparrow p_c}\frac{\mathcal{R}_p^*}{\overline{\chi}_p}=0.$$

If, in addition, G *is connected, locally finite, quasi-transitive, and*

$$\limsup_{p\uparrow p_c}(p_c-p)\overline{\chi}_p<\infty,$$

then

$$p_c(G)<p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.733, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.735 (Localized anchored incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that* G *has maximum degree at most* $\Delta\geq 2$*. There is* $0<\epsilon\leq 1$ *such that for every* $\eta>0$*, every* $0<\theta<1/6$*, every pair of integers* $J,H\geq 0$*, and all* p_0 *sufficiently close to* p_c*, there are* $p\in[p_0,p_c)$*, an integer* $j>J+H$*, a vertex* $x\in V$*, finite non-empty sets* $A,Y,Z\subset V$*, and a bijection* $a:Z\to A$ *such that, with*

$$I_j=(2^j,2^{j+1}],\qquad I_{j,p}^{\mathrm{gate}}=(2^j-r_p,2^{j+1}+r_p],$$

with σ_j, r_p *as in Definition 8.58, and with*

$$\widehat{\kappa}_j(p)=\frac{\sigma_j^3}{16\Delta^3}\overline{\chi}_p^{1-6\theta},$$

we have

$$2^j\geq\frac{1}{4}\log_\Delta\overline{\chi}_p,\qquad |A|\geq\widehat{\kappa}_j(p),\qquad |Y|\geq\widehat{\kappa}_j(p),$$

$$d(x,z)\in I_j,\qquad d(a_z,z)=r_p,\qquad d(x,a_z)\in I_{j,p}^{\mathrm{gate}}\qquad (z\in Z),$$

$$d(z,z')>2r_p\qquad (z,z'\in Z,\ z\neq z'),$$

$$Y\subseteq B(x,2^{j+2}+2r_p),$$

and

$$|\{y\in Y: d(y,a_z)\in I_{j,p}^{\mathrm{gate}}\}|\geq\widehat{\kappa}_j(p)\qquad (z\in Z).$$

Proof. Choose $K,x,j,Z_\theta,Z^\sharp$, and $a:Z^\sharp\to V$, by Corollary 8.733. Put

$$Z=Z^\sharp,\qquad A=\{a_z: z\in Z^\sharp\},$$

and

$$Y=\bigcup_{z\in Z^\sharp}\{y\in K: d(y,a_z)\in I_{j,p}^{\mathrm{gate}}\}.$$

If $z, z' \in Z^\sharp$ and $z \neq z'$, then $a_z \neq a_{z'}$, since $a_z = a_{z'}$ would imply $d(z, z') \leq 2r_p$, contradicting the separated fan condition. Thus $a : Z \to A$ is a bijection and

$$|A| = |Z^\sharp| \geq \widehat{\kappa}_j(p).$$

The same lower bound for $|Y|$ follows from the pointwise incidence lower bound for any fixed $z \in Z^\sharp$.

For $y \in Y$, choose $z \in Z^\sharp$ with $d(y, a_z) \in I_{j,p}^{\mathrm{gate}}$. Since also $d(x, a_z) \in I_{j,p}^{\mathrm{gate}}$,

$$d(x, y) \leq d(x, a_z) + d(a_z, y) \leq 2^{j+2} + 2r_p.$$

The remaining displayed properties are inherited from the uniform deterministic gate-incidence obstruction and the separated fan conditions. □

Proposition 8.736 (Localized anchored incidence criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/6$, integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that for no $p \in [p_1, p_{\mathrm{c}})$ do there exist an integer $j > J + H$, a vertex $x \in V$, finite non-empty sets $A, Y, Z \subset V$, and a bijection $a : Z \to A$ such that, writing*

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$
$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

and

$$\widehat{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-6\theta},$$

all of the following hold:

$$2^j \geq \frac{1}{4} \log_\Delta \overline{\chi}_p, \qquad |A| \geq \widehat{\kappa}_j(p), \qquad |Y| \geq \widehat{\kappa}_j(p),$$

$$d(x, z) \in I_j, \qquad d(a_z, z) = r_p, \qquad d(x, a_z) \in I_{j,p}^{\mathrm{gate}} \qquad (z \in Z),$$
$$d(z, z') > 2r_p \qquad (z, z' \in Z,\ z \neq z'),$$
$$Y \subseteq B(x, 2^{j+2} + 2r_p),$$

and

$$|\{y \in Y : d(y, a_z) \in I_{j,p}^{\mathrm{gate}}\}| \geq \widehat{\kappa}_j(p) \qquad (z \in Z).$$

Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.735, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.737 (Improved logarithmic gate scale)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/7$, every*

$$0 < a < \frac{1 - 7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, the localized anchored incidence obstruction of Corollary 8.735 can be chosen so that

$$2^j \geq a \log_\Delta \overline{\chi}_p.$$

Proof. Let ϵ be the constant from Corollary 8.735. Fix η, θ, a, J, H as in the statement, and put

$$c_a = \frac{1}{16\Delta^3} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J} a^{-3\eta}.$$

Choose $p_* < p_c$ sufficiently close to p_c that the localized anchored incidence obstruction applies to every $p_0 \in [p_*, p_c)$, and whenever $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$,

$$c_a L^{-3\eta} \overline{\chi}_p^{1-6\theta} > \Delta \overline{\chi}_p^{2a+\theta}.$$

This is possible because $2a + \theta < 1 - 6\theta$.

Fix $p_0 \in [p_*, p_c)$, and choose the localized anchored incidence data. Suppose for contradiction that $2^j < aL$. Then $j \leq \log_2(aL)$, and the definition of σ_j gives

$$\widehat{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-6\theta} \geq c_a L^{-3\eta} \overline{\chi}_p^{1-6\theta}.$$

On the other hand, $A \subseteq B(x, 2^{j+1} + r_p)$, so bounded degree gives

$$|A| \leq \Delta^{2^{j+1}+r_p+1} < \Delta \overline{\chi}_p^{2a+\theta}.$$

This contradicts $|A| \geq \widehat{\kappa}_j(p)$. Hence $2^j \geq a \log_\Delta \overline{\chi}_p$. □

Proposition 8.738 (Improved gate-scale incidence criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/7$,*

$$0 < a < \frac{1 - 7\theta}{2},$$

integers $J, H \geq 0$, and $p_1 < p_c$ such that no $p \in [p_1, p_c)$ admits a localized anchored incidence configuration satisfying the conditions in Proposition 8.736 and the additional scale condition

$$2^j \geq a \log_\Delta \overline{\chi}_p.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.737, with the L^2-gap conclusion supplied by Corollary 8.794. □

Lemma 8.739 (Left-degree matching extraction)**.** *Let $\mathfrak{B}$ be a finite bipartite graph with left part A and right part Y. Let $m \geq 1$ be an integer. If $|A| \geq m$ and every $a \in A$ has at least m neighbours in Y, then $\mathfrak{B}$ contains a matching of size m.*

Proof. Choose $A_0 \subseteq A$ with $|A_0| = m$. For every non-empty $S \subseteq A_0$, the neighbourhood $N(S)$ contains the neighbourhood of any one vertex of S, and hence $|N(S)| \geq m \geq |S|$. Hall's theorem therefore gives a matching saturating A_0. □

Corollary 8.740 (Matched anchored incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/7$, every*

$$0 < a < \frac{1 - 7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, a vertex $x \in V$, and vertices

$$z_1, \dots, z_m, \qquad a_1, \dots, a_m, \qquad y_1, \dots, y_m$$

such that, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

with σ_j, r_p as in Definition 8.58, and with

$$\widehat{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-6\theta}, \qquad m = \lceil \widehat{\kappa}_j(p) \rceil,$$

we have

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

the vertices $z_1, \dots, z_m$ are pairwise distinct and satisfy

$$d(z_i, z_k) > 2r_p \qquad (i \neq k),$$

the vertices $a_1, \dots, a_m$ are pairwise distinct, the vertices $y_1, \dots, y_m$ are pairwise distinct, and, for every $1 \leq i \leq m$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$
$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p).$$

Proof. Choose the localized anchored incidence data from Corollary 8.737. Form the finite bipartite graph $\mathfrak{B}$ with left part A, right part Y, and an edge between $a_z \in A$ and $y \in Y$ when

$$d(y, a_z) \in I_{j,p}^{\mathrm{gate}}.$$

Since $|A| \geq \widehat{\kappa}_j(p)$ and each left degree is at least $\widehat{\kappa}_j(p)$, while cardinalities and degrees are integers, we have

$$|A| \geq m, \qquad \deg_{\mathfrak{B}}(a) \geq m \quad (a \in A), \qquad m = \lceil \widehat{\kappa}_j(p) \rceil.$$

Lemma 8.739 gives a matching of size m. Write the matched left vertices as $a_1, \dots, a_m$, the matched right vertices as $y_1, \dots, y_m$, and set $z_i = a^{-1}(a_i)$. The matching gives the distinctness of the a_i's and y_i's, the bijection $a : Z \to A$ gives the distinctness of the z_i's, and all displayed metric properties are inherited from the localized anchored incidence configuration. □

Proposition 8.741 (Matched anchored incidence criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/7$,*

$$0 < a < \frac{1 - 7\theta}{2},$$

integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that no $p \in [p_1, p_{\mathrm{c}})$ admits an integer $j > J + H$, a vertex $x \in V$, and vertices

$$z_1, \dots, z_m, \qquad a_1, \dots, a_m, \qquad y_1, \dots, y_m$$

with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\widehat{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^3} \overline{\chi}_p^{1-6\theta}, \qquad m = \left\lceil \widehat{\kappa}_j(p) \right\rceil,$$

such that all of the following hold:

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

the vertices $z_1, \dots, z_m$ are pairwise distinct and satisfy

$$d(z_i, z_k) > 2r_p \qquad (i \neq k),$$

the vertices $a_1, \dots, a_m$ are pairwise distinct, the vertices $y_1, \dots, y_m$ are pairwise distinct, and, for every $1 \leq i \leq m$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p).$$

Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.740, with the L^2-gap conclusion supplied by Corollary 8.794. □

Lemma 8.742 (Bounded-degree packing extraction)**.** *Let G have maximum degree at most $\Delta \geq 2$. If $S \subseteq V$ is finite and $R \geq 0$ is an integer, then there is $S' \subseteq S$ such that*

$$d(u, v) > R \qquad (u, v \in S',\ u \neq v)$$

and

$$|S'| \geq |S| \Delta^{-R-1}.$$

Proof. Choose vertices greedily from S, and after choosing u discard all remaining vertices of $S \cap B(u,R)$. Each step discards at most Δ^{R+1} vertices, so the number of chosen vertices is at least $|S|\Delta^{-R-1}$. The construction gives the displayed separation. □

Corollary 8.743 (Separated matched incidence obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/12$, every*

$$0 < a < \frac{1-7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, a vertex $x \in V$, and vertices

$$z_1, \dots, z_n, \qquad a_1, \dots, a_n, \qquad y_1, \dots, y_n$$

such that, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

with σ_j, r_p as in Definition 8.58, and with

$$\widetilde{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^5}\overline{\chi}_p^{1-12\theta}, \qquad n = \lceil \widetilde{\kappa}_j(p) \rceil,$$

we have

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \neq k),$$

and, for every $1 \leq i \leq n$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p).$$

Proof. Choose the matched anchored incidence obstruction from Corollary 8.740, with matched triples indexed by a set M of size

$$m = \lceil \widehat{\kappa}_j(p) \rceil, \qquad \widehat{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^3}\overline{\chi}_p^{1-6\theta}.$$

Apply Lemma 8.742 to the anchor set $\{z_i : i \in M\}$ with $R = 4r_p$. This gives $M_0 \subseteq M$ such that the anchors $\{z_i : i \in M_0\}$ are $4r_p$-separated and

$$|M_0| \geq m\Delta^{-4r_p-1}.$$

Apply the same lemma to the set $\{y_i : i \in M_0\}$ with $R = 2r_p$. This gives $M_1 \subseteq M_0$ such that the vertices $\{y_i : i \in M_1\}$ are $2r_p$-separated and

$$|M_1| \geq m\Delta^{-6r_p-2}.$$

Since $m \geq \widehat{\kappa}_j(p)$ and $\Delta^{6r_p} \leq \overline{\chi}_p^{6\theta}$,

$$|M_1| \geq \frac{\sigma_j^3}{16\Delta^5}\overline{\chi}_p^{1-12\theta} = \widetilde{\kappa}_j(p).$$

Thus $|M_1| \geq n$, after replacing M_1 by an arbitrary subset of size n. Relabel this subset as $\{1, \dots, n\}$.

The z_i's are $4r_p$-separated by construction, the y_i's are $2r_p$-separated by construction, and

$$d(a_i, a_k) \geq d(z_i, z_k) - d(a_i, z_i) - d(a_k, z_k) > 2r_p \qquad (i \neq k).$$

All remaining metric properties are inherited from Corollary 8.740. □

Proposition 8.744 (Separated matched incidence criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/12$,*

$$0 < a < \frac{1 - 7\theta}{2},$$

integers $J, H \geq 0$, and $p_1 < p_c$ such that no $p \in [p_1, p_c)$ admits an integer $j > J + H$, a vertex $x \in V$, and vertices

$$z_1, \dots, z_n, \qquad a_1, \dots, a_n, \qquad y_1, \dots, y_n$$

with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\widetilde{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^5} \overline{\chi}_p^{1-12\theta}, \qquad n = \left\lceil \widetilde{\kappa}_j(p) \right\rceil,$$

such that all of the following hold:

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \neq k),$$

and, for every $1 \leq i \leq n$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p).$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.743, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.745 (Disjoint gate-tube obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/12$, every*

$$0 < a < \frac{1 - 7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, the separated matched incidence obstruction of Corollary 8.743 can be chosen together with simple paths $\gamma_1, \dots, \gamma_n$ such that each γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

and the paths $\gamma_1, \dots, \gamma_n$ are pairwise vertex-disjoint.

Proof. Choose the separated matched incidence obstruction. For each i, choose a geodesic γ_i from a_i to z_i, which has length r_p since $d(a_i, z_i) = r_p$. Every vertex of γ_i lies in $B(z_i, r_p)$. If γ_i and γ_k intersect for $i \neq k$, then

$$d(z_i, z_k) \leq 2r_p,$$

contradicting the separated matched obstruction, where $d(z_i, z_k) > 4r_p$. □

Proposition 8.746 (Disjoint gate-tube criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/12$,*

$$0 < a < \frac{1 - 7\theta}{2},$$

integers $J, H \geq 0$, and $p_1 < p_c$ such that no $p \in [p_1, p_c)$ admits vertices satisfying the metric and size conditions displayed in Proposition 8.744, together with simple paths $\gamma_1, \dots, \gamma_n$ such that each γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

and the paths $\gamma_1, \dots, \gamma_n$ are pairwise vertex-disjoint. Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.745, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.747 (Cross-separated tube obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/14$, every*

$$3\theta < a < \frac{1 - 7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, a vertex $x \in V$, vertices

$$z_1, \dots, z_q, \qquad a_1, \dots, a_q, \qquad y_1, \dots, y_q,$$

and simple paths $\gamma_1, \dots, \gamma_q$ such that, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

with σ_j, r_p as in Definition 8.58, and with

$$\kappa_j^{\mathrm{xsep}}(p) = \frac{\sigma_j^3}{48\Delta^6} \overline{\chi}_p^{1-14\theta}, \qquad q = \left\lceil \kappa_j^{\mathrm{xsep}}(p) \right\rceil,$$

we have

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$
$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \neq k),$$
$$d(a_i, y_k) > 2r_p \qquad (1 \leq i, k \leq q),$$

and, for every $1 \leq i \leq q$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$
$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p),$$

while γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

and the paths $\gamma_1, \dots, \gamma_q$ are pairwise vertex-disjoint.

Proof. Choose the disjoint gate-tube obstruction from Corollary 8.745, with triples indexed by a set M of size

$$n = \left\lceil \widetilde{\kappa}_j(p) \right\rceil, \qquad \widetilde{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^5} \overline{\chi}_p^{1-12\theta}.$$

Form a graph $\mathfrak{C}$ on the index set M, joining distinct $i, k \in M$ if

$$d(a_i, y_k) \leq 2r_p \qquad \text{or} \qquad d(a_k, y_i) \leq 2r_p.$$

For a fixed i, there are at most Δ^{2r_p+1} indices k with $d(a_i, y_k) \leq 2r_p$, and at most Δ^{2r_p+1} indices k with $d(y_i, a_k) \leq 2r_p$. Hence $\mathfrak{C}$ has maximum degree at most $2\Delta^{2r_p+1}$. A greedy independent-set extraction gives $M' \subseteq M$ such that no two distinct indices of M' are adjacent in $\mathfrak{C}$ and

$$|M'| \geq \frac{n}{2\Delta^{2r_p+1} + 1} \geq \frac{n}{3\Delta^{2r_p+1}}.$$

Since $n \geq \widetilde{\kappa}_j(p)$ and $\Delta^{2r_p} \leq \overline{\chi}_p^{2\theta}$,

$$|M'| \geq \frac{\sigma_j^3}{48\Delta^6} \overline{\chi}_p^{1-14\theta} = \kappa_j^{\mathrm{xsep}}(p).$$

Replace M' by a subset of size $q = \lceil \kappa_j^{\mathrm{xsep}}(p) \rceil$ and relabel it as $\{1, \dots, q\}$.

For distinct indices i, k, the independent-set property gives $d(a_i, y_k) > 2r_p$. For $i = k$, put $L = \log_\Delta \overline{\chi}_p$. Since $d(a_i, y_i) > 2^j - r_p$, $2^j \geq aL$, $r_p \leq \theta L$, and $a > 3\theta$,

$$d(a_i, y_i) > (a - \theta)L > 2\theta L \geq 2r_p.$$

All other displayed properties are inherited from the disjoint gate-tube obstruction. □

Proposition 8.748 (Cross-separated tube criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/14$,*

$$3\theta < a < \frac{1-7\theta}{2},$$

integers $J, H \geq 0$, and $p_1 < p_{\mathrm{c}}$ such that no $p \in [p_1, p_{\mathrm{c}})$ admits an integer $j > J + H$, a vertex $x \in V$, vertices

$$z_1, \dots, z_q, \qquad a_1, \dots, a_q, \qquad y_1, \dots, y_q,$$

and simple paths $\gamma_1, \dots, \gamma_q$, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\kappa_j^{\mathrm{xsep}}(p) = \frac{\sigma_j^3}{48\Delta^6} \overline{\chi}_p^{1-14\theta}, \qquad q = \left\lceil \kappa_j^{\mathrm{xsep}}(p) \right\rceil,$$

such that all of the following hold:

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \neq k),$$

$$d(a_i, y_k) > 2r_p \qquad (1 \leq i, k \leq q),$$

and, for every $1 \leq i \leq q$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p),$$

while γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

and the paths $\gamma_1, \dots, \gamma_q$ are pairwise vertex-disjoint. Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p) \overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.747, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.749 (Endpoint tube obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/16$, every*

$$4\theta < a < \frac{1-7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, a vertex $x \in V$, vertices

$$z_1, \dots, z_s, \qquad a_1, \dots, a_s, \qquad y_1, \dots, y_s,$$

and simple paths $\gamma_1, \dots, \gamma_s$ such that, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

with σ_j, r_p as in Definition 8.58, and with

$$\kappa_j^{\mathrm{end}}(p) = \frac{\sigma_j^3}{144\Delta^7}\overline{\chi}_p^{1-16\theta}, \qquad s = \left\lceil \kappa_j^{\mathrm{end}}(p) \right\rceil,$$

we have

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \neq k),$$

$$d(a_i, y_k) > 2r_p, \qquad d(z_i, y_k) > 2r_p \qquad (1 \leq i, k \leq s),$$

and, for every $1 \leq i \leq s$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p),$$

while γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

and the paths $\gamma_1, \dots, \gamma_s$ are pairwise vertex-disjoint.

Proof. Choose the cross-separated tube obstruction from Corollary 8.747, with triples indexed by a set M of size

$$q = \left\lceil \kappa_j^{\mathrm{xsep}}(p) \right\rceil, \qquad \kappa_j^{\mathrm{xsep}}(p) = \frac{\sigma_j^3}{48\Delta^6}\overline{\chi}_p^{1-14\theta}.$$

Form a graph $\mathfrak{D}$ on the index set M, joining distinct $i, k \in M$ if

$$d(z_i, y_k) \leq 2r_p \qquad \text{or} \qquad d(z_k, y_i) \leq 2r_p.$$

For a fixed i, there are at most Δ^{2r_p+1} indices k with $d(z_i, y_k) \leq 2r_p$, and at most Δ^{2r_p+1} indices k with $d(y_i, z_k) \leq 2r_p$. Hence $\mathfrak{D}$ has maximum degree at most $2\Delta^{2r_p+1}$, and a greedy independent-set extraction gives $M' \subseteq M$ with

$$|M'| \geq \frac{q}{2\Delta^{2r_p+1} + 1} \geq \frac{q}{3\Delta^{2r_p+1}}.$$

Since $q \ge \kappa_j^{\mathrm{xsep}}(p)$ and $\Delta^{2r_p} \le \overline{\chi}_p^{2\theta}$,

$$|M'| \ge \frac{\sigma_j^3}{144\Delta^7}\overline{\chi}_p^{1-16\theta} = \kappa_j^{\mathrm{end}}(p).$$

Replace M' by a subset of size $s = \lceil \kappa_j^{\mathrm{end}}(p) \rceil$, and relabel it as $\{1, \dots, s\}$.

For distinct indices i, k, the independent-set property gives $d(z_i, y_k) > 2r_p$. For $i = k$, put $L = \log_\Delta \overline{\chi}_p$. Since $d(z_i, y_i) \ge d(a_i, y_i) - d(a_i, z_i)$, $d(a_i, y_i) > 2^j - r_p$, $2^j \ge aL$, $r_p \le \theta L$, and $a > 4\theta$,

$$d(z_i, y_i) > 2^j - 2r_p \ge (a - 2\theta)L > 2\theta L \ge 2r_p.$$

All other displayed properties are inherited from the cross-separated tube obstruction. □

Proposition 8.750 (Endpoint tube criterion)**.** *Let G have maximum degree at most $\Delta \ge 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \le 1$ there exist $\eta > 0$, $0 < \theta < 1/16$,*

$$4\theta < a < \frac{1 - 7\theta}{2},$$

integers $J, H \ge 0$, and $p_1 < p_{\mathrm{c}}$ such that no $p \in [p_1, p_{\mathrm{c}})$ admits an integer $j > J + H$, a vertex $x \in V$, vertices

$$z_1, \dots, z_s, \qquad a_1, \dots, a_s, \qquad y_1, \dots, y_s,$$

and simple paths $\gamma_1, \dots, \gamma_s$, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\kappa_j^{\mathrm{end}}(p) = \frac{\sigma_j^3}{144\Delta^7}\overline{\chi}_p^{1-16\theta}, \qquad s = \left\lceil \kappa_j^{\mathrm{end}}(p) \right\rceil,$$

such that all of the following hold:

$$2^j \ge a \log_\Delta \overline{\chi}_p,$$

$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \ne k),$$

$$d(a_i, y_k) > 2r_p, \qquad d(z_i, y_k) > 2r_p \qquad (1 \le i, k \le s),$$

and, for every $1 \le i \le s$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p),$$

while γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

and the paths $\gamma_1, \dots, \gamma_s$ are pairwise vertex-disjoint. Then

$$\lim_{p \uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.749, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.751 (Buffered tube obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/19$, every*

$$5\theta < a < \frac{1-7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, a vertex $x \in V$, vertices

$$z_1, \dots, z_t, \qquad a_1, \dots, a_t, \qquad y_1, \dots, y_t,$$

and simple paths $\gamma_1, \dots, \gamma_t$ such that, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

with σ_j, r_p as in Definition 8.58, and with

$$\kappa_j^{\mathrm{buf}}(p) = \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-19\theta}, \qquad t = \left\lceil \kappa_j^{\mathrm{buf}}(p) \right\rceil,$$

we have

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \neq k),$$

$$d(a_i, y_k) > 2r_p, \qquad d(z_i, y_k) > 2r_p \qquad (1 \leq i, k \leq t),$$

and, for every $1 \leq i \leq t$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p),$$

while γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

the paths $\gamma_1, \dots, \gamma_t$ are pairwise vertex-disjoint, and

$$d(v, y_k) > 2r_p \qquad (1 \leq i, k \leq t,\ v \in \gamma_i).$$

Proof. Choose the endpoint tube obstruction from Corollary 8.749, with triples indexed by a set M of size

$$s = \left\lceil \kappa_j^{\mathrm{end}}(p) \right\rceil, \qquad \kappa_j^{\mathrm{end}}(p) = \frac{\sigma_j^3}{144\Delta^7}\overline{\chi}_p^{1-16\theta}.$$

Form a graph $\mathfrak{E}$ on the index set M, joining distinct $i, k \in M$ if

$$d(z_i, y_k) \leq 3r_p \qquad \text{or} \qquad d(z_k, y_i) \leq 3r_p.$$

For a fixed i, there are at most Δ^{3r_p+1} indices k with $d(z_i, y_k) \leq 3r_p$, and at most Δ^{3r_p+1} indices k with $d(y_i, z_k) \leq 3r_p$. Hence $\mathfrak{E}$ has maximum degree at most $2\Delta^{3r_p+1}$, and a greedy independent-set extraction gives $M' \subseteq M$ with

$$|M'| \geq \frac{s}{2\Delta^{3r_p+1} + 1} \geq \frac{s}{3\Delta^{3r_p+1}}.$$

Since $s \ge \kappa_j^{\mathrm{end}}(p)$ and $\Delta^{3r_p} \le \overline{\chi}_p^{3\theta}$,

$$|M'| \ge \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-19\theta} = \kappa_j^{\mathrm{buf}}(p).$$

Replace M' by a subset of size $t = \lceil \kappa_j^{\mathrm{buf}}(p) \rceil$, and relabel it as $\{1, \dots, t\}$.

For distinct i, k, the independent-set property gives $d(z_i, y_k) > 3r_p$. For $i = k$, put $L = \log_\Delta \overline{\chi}_p$. Since

$$d(z_i, y_i) > 2^j - 2r_p, \qquad 2^j \ge aL, \qquad r_p \le \theta L,$$

and $a > 5\theta$, we have

$$d(z_i, y_i) > (a - 2\theta)L > 3\theta L \ge 3r_p.$$

Thus $d(z_i, y_k) > 3r_p$ for all i, k. If $v \in \gamma_i$, then $d(v, z_i) \le r_p$, and hence

$$d(v, y_k) \ge d(z_i, y_k) - r_p > 2r_p.$$

All other displayed properties are inherited from the endpoint tube obstruction. □

Proposition 8.752 (Buffered tube criterion)**.** *Let G have maximum degree at most $\Delta \ge 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. Suppose that for every $0 < \epsilon \le 1$ there exist $\eta > 0$, $0 < \theta < 1/19$,*

$$5\theta < a < \frac{1 - 7\theta}{2},$$

integers $J, H \ge 0$, and $p_1 < p_{\mathrm{c}}$ such that no $p \in [p_1, p_{\mathrm{c}})$ admits an integer $j > J + H$, a vertex $x \in V$, vertices

$$z_1, \dots, z_t, \qquad a_1, \dots, a_t, \qquad y_1, \dots, y_t,$$

and simple paths $\gamma_1, \dots, \gamma_t$, with

$$I_j = (2^j, 2^{j+1}], \qquad I_{j,p}^{\mathrm{gate}} = (2^j - r_p, 2^{j+1} + r_p],$$

$$\sigma_j = \frac{\epsilon}{2}(1 - 2^{-\eta/2})^2 2^{-\eta(j-J)}, \qquad r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\kappa_j^{\mathrm{buf}}(p) = \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-19\theta}, \qquad t = \left\lceil \kappa_j^{\mathrm{buf}}(p) \right\rceil,$$

such that all of the following hold:

$$2^j \ge a \log_\Delta \overline{\chi}_p,$$

$$d(z_i, z_k) > 4r_p, \qquad d(a_i, a_k) > 2r_p, \qquad d(y_i, y_k) > 2r_p \qquad (i \ne k),$$

$$d(a_i, y_k) > 2r_p, \qquad d(z_i, y_k) > 2r_p \qquad (1 \le i, k \le t),$$

and, for every $1 \le i \le t$,

$$d(x, z_i) \in I_j, \qquad d(a_i, z_i) = r_p, \qquad d(x, a_i) \in I_{j,p}^{\mathrm{gate}},$$

$$d(y_i, a_i) \in I_{j,p}^{\mathrm{gate}}, \qquad y_i \in B(x, 2^{j+2} + 2r_p),$$

while γ_i has length r_p, runs from a_i to z_i, satisfies

$$\gamma_i \subseteq B(z_i, r_p),$$

the paths $\gamma_1, \dots, \gamma_t$ are pairwise vertex-disjoint, and

$$d(v, y_k) > 2r_p \qquad (1 \le i, k \le t,\ v \in \gamma_i).$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.751, with the L^2-gap conclusion supplied by Corollary 8.794. □

Lemma 8.753 (Independent path-opening count)**.** *Let $\gamma_1, \dots, \gamma_t$ be pairwise edge-disjoint simple paths, each of length r. In Bernoulli-p percolation, let*

$$N = \#\{1 \le i \le t : \text{ every edge of } \gamma_i \text{ is open}\}.$$

Then N is binomial with mean $\mu = tp^r$, and

$$\mathbf{P}_p\left(N < \frac{\mu}{2}\right) \le \exp\left[-\frac{\mu}{8}\right].$$

Proof. The path-opening events depend on disjoint edge sets and are therefore independent. Each has probability p^r, so $N \sim \mathrm{Bin}(t, p^r)$. The displayed inequality is the standard Chernoff lower-tail bound for a binomial random variable. □

Corollary 8.754 (Open buffered tube obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$. There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every $0 < \theta < 1/20$, every*

$$5\theta < a < \frac{1 - 7\theta}{2},$$

every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, with

$$N_{\mathrm{open}} = \#\{1 \le i \le t : \text{ every edge of } \gamma_i \text{ is open}\}$$

and

$$\lambda_j^{\mathrm{open}}(p) = \frac{\sigma_j^3}{864\Delta^8}\overline{\chi}_p^{1-20\theta},$$

we have

$$\mathbf{P}_p\left(N_{\mathrm{open}} \ge \lambda_j^{\mathrm{open}}(p)\right) \ge 1 - \exp\left[-\frac{\lambda_j^{\mathrm{open}}(p)}{4}\right].$$

Proof. Use the ϵ from Corollary 8.751. By Lemma 8.732, we have $p_c > 1/\Delta$; hence we may take p_0 close enough to p_c that every selected $p \in [p_0, p_c)$ satisfies $p \geq 1/\Delta$. Choose the buffered tube configuration from Corollary 8.751. The paths $\gamma_1, \dots, \gamma_t$ are pairwise vertex-disjoint, hence edge-disjoint, and each has length r_p. Thus Lemma 8.753 applies with

$$\mu = tp^{r_p}.$$

Since $t \geq \kappa_j^{\mathrm{buf}}(p)$ and

$$p^{r_p} \geq \Delta^{-r_p} \geq \overline{\chi}_p^{-\theta},$$

we have

$$\mu \geq \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-20\theta} = 2\lambda_j^{\mathrm{open}}(p).$$

Consequently

$$\mathbf{P}_p\left(N_{\mathrm{open}} < \lambda_j^{\mathrm{open}}(p)\right) \leq \mathbf{P}_p\left(N_{\mathrm{open}} < \frac{\mu}{2}\right) \leq \exp\left[-\frac{\mu}{8}\right] \leq \exp\left[-\frac{\lambda_j^{\mathrm{open}}(p)}{4}\right],$$

as claimed. □

Corollary 8.755 (Buffered tube mass obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/20$, every*

$$5\theta < a < \frac{1-7\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that

$$\sum_{i=1}^{t} \tau_p(a_i, z_i) \geq \kappa_j^{\mathrm{tube}}(p),$$

where

$$\kappa_j^{\mathrm{tube}}(p) = \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-20\theta}.$$

Proof. Use the ϵ from Corollary 8.751. As in the proof of Corollary 8.754, take p_0 close enough to p_c that the selected p satisfies $p \geq 1/\Delta$, and choose the buffered tube configuration. For each i, the path γ_i has length r_p and runs from a_i to z_i. Therefore

$$\tau_p(a_i, z_i) \geq p^{r_p} \geq \Delta^{-r_p} \geq \overline{\chi}_p^{-\theta}.$$

Since

$$t \geq \kappa_j^{\mathrm{buf}}(p) = \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-19\theta},$$

we obtain

$$\sum_{i=1}^{t} \tau_p(a_i, z_i) \geq t\overline{\chi}_p^{-\theta} \geq \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-20\theta}.$$

□

Proposition 8.756 (Buffered tube mass criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/20$,*

$$5\theta < a < \frac{1-7\theta}{2},$$

integers $J, H \geq 0$, and $p_1 < p_c$ such that no $p \in [p_1, p_c)$ admits a buffered tube configuration satisfying the geometric conditions in Proposition 8.752 and the additional mass bound

$$\sum_{i=1}^{t} \tau_p(a_i, z_i) \geq \frac{\sigma_j^3}{432\Delta^8} \overline{\chi}_p^{1-20\theta}.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.755, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.757 (Open-tube scale dichotomy)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/20$, every*

$$5\theta < a < \frac{1-7\theta}{2},$$

every $b > 0$ satisfying

$$\rho := 1 - 20\theta - 3\eta b > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that

$$2^j \geq a \log_\Delta \overline{\chi}_p,$$

and at least one of the following alternatives holds:

$$2^j > \overline{\chi}_p^b,$$

or, with

$$c_{\epsilon,\eta,J,\Delta} = \frac{1}{864\Delta^8} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J},$$

$$\mathbf{P}_p\left(N_{\mathrm{open}} \geq c_{\epsilon,\eta,J,\Delta}\overline{\chi}_p^\rho\right) \geq 1 - \exp\left[-\frac{c_{\epsilon,\eta,J,\Delta}}{4}\overline{\chi}_p^\rho\right].$$

Proof. Choose the configuration supplied by Corollary 8.754. If $2^j > \overline{\chi}_p^b$, then the first alternative holds. Otherwise $j \le b \log_2 \overline{\chi}_p$, and therefore

$$\sigma_j^3 \ge \left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J}\overline{\chi}_p^{-3\eta b}.$$

Thus

$$\lambda_j^{\mathrm{open}}(p) = \frac{\sigma_j^3}{864\Delta^8}\overline{\chi}_p^{1-20\theta} \ge c_{\epsilon,\eta,J,\Delta}\overline{\chi}_p^{\rho}.$$

The second alternative follows from Corollary 8.754. □

Corollary 8.758 (Endpoint-anchor scale dichotomy)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$. There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every $0 < \theta < 1/31$, every*

$$5\theta < a < B < \frac{1-21\theta}{2},$$

every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, writing

$$L = \log_\Delta \overline{\chi}_p,$$

we have

$$2^j \ge aL,$$

and at least one of the following alternatives holds:

$$2^j > BL,$$

or

$$\sum_{i=1}^{t} \tau_p(y_i, z_i) \ge c_{\epsilon,\eta,J,\Delta,B} L^{-3\eta}\overline{\chi}_p^{1-21\theta-2B},$$

where

$$c_{\epsilon,\eta,J,\Delta,B} = \frac{1}{432\Delta^8}\left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J} B^{-3\eta}.$$

Proof. Use the ϵ from Corollary 8.751. As in the proof of Corollary 8.754, take p_0 close enough to p_c that the selected p satisfies $p \ge 1/\Delta$, and choose the buffered tube configuration. Corollary 8.751 already gives $2^j \ge aL$. If $2^j > BL$, then the first alternative holds. Assume therefore that $2^j \le BL$.

For every i, choose a geodesic from y_i to a_i and concatenate it with γ_i. Since

$$d(y_i, a_i) \le 2^{j+1} + r_p, \qquad d(a_i, z_i) = r_p,$$

there is a path from y_i to z_i of length at most $2^{j+1} + 2r_p$. Hence

$$\tau_p(y_i, z_i) \ge p^{2^{j+1}+2r_p} \ge \Delta^{-2^{j+1}-2r_p} \ge \overline{\chi}_p^{-2B-2\theta}.$$

Moreover, $2^j \le BL$ implies

$$\sigma_j^3 \ge \left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J}(BL)^{-3\eta}.$$

Since

$$t \geq \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-19\theta},$$

we get

$$\sum_{i=1}^{t} \tau_p(y_i, z_i) \geq c_{\epsilon,\eta,J,\Delta,B} L^{-3\eta}\overline{\chi}_p^{1-21\theta-2B}.$$

□

Corollary 8.759 (Buffered tube logarithmic scale amplification)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/29$, every*

$$5\theta < a < B < \frac{1-19\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, writing

$$L = \log_\Delta \overline{\chi}_p,$$

we have

$$2^j > BL.$$

Proof. Use the ϵ from Corollary 8.751. Put

$$c^{\mathrm{buf}}_{\epsilon,\eta,J,\Delta,B} = \frac{1}{432\Delta^8}\left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J} B^{-3\eta}.$$

Since $2B < 1 - 19\theta$, choose $p_* < p_c$ sufficiently close to p_c that whenever $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$,

$$c^{\mathrm{buf}}_{\epsilon,\eta,J,\Delta,B} L^{-3\eta}\overline{\chi}_p^{1-19\theta} > \Delta\overline{\chi}_p^{2B}.$$

This is possible because $\overline{\chi}_p \to \infty$.

Fix $p_0 \in [p_*, p_c)$, and choose the buffered tube configuration from Corollary 8.751. Suppose for contradiction that $2^j \leq BL$. Then $j \leq \log_2(BL)$, and hence

$$\sigma_j^3 \geq \left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J}(BL)^{-3\eta}.$$

Since $t = \lceil \kappa_j^{\mathrm{buf}}(p) \rceil$, this gives

$$t \geq c^{\mathrm{buf}}_{\epsilon,\eta,J,\Delta,B} L^{-3\eta}\overline{\chi}_p^{1-19\theta}.$$

On the other hand, the vertices $z_1, \ldots, z_t$ are distinct and satisfy $d(x, z_i) \leq 2^{j+1}$. Bounded degree therefore gives

$$t \leq |B(x, 2^{j+1})| \leq \Delta^{2^{j+1}+1} \leq \Delta\overline{\chi}_p^{2B},$$

contradicting the choice of p_*. Thus $2^j > BL$. □

Corollary 8.760 (Matched-count buffered scale amplification)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/19$, every*

$$5\theta < B < \frac{1-6\theta}{2},$$

every

$$5\theta < a < \min\left\{B, \frac{1-7\theta}{2}\right\},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, writing

$$L = \log_\Delta \overline{\chi}_p,$$

we have

$$2^j > BL.$$

Proof. Use the ϵ from Corollary 8.740. Put

$$c^{\mathrm{match}}_{\epsilon,\eta,J,\Delta,B} = \frac{1}{16\Delta^3}\left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J} B^{-3\eta}.$$

Since $2B < 1 - 6\theta$, choose $p_* < p_c$ sufficiently close to p_c that whenever $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$,

$$c^{\mathrm{match}}_{\epsilon,\eta,J,\Delta,B} L^{-3\eta} \overline{\chi}_p^{1-6\theta} > \Delta \overline{\chi}_p^{2B}.$$

Fix $p_0 \in [p_*, p_c)$, and choose the matched anchored incidence obstruction with the auxiliary lower-scale parameter a. Suppose first that $2^j \leq BL$. Then $j \leq \log_2(BL)$, and hence

$$\widehat{\kappa}_j(p) = \frac{\sigma_j^3}{16\Delta^3}\overline{\chi}_p^{1-6\theta} \geq c^{\mathrm{match}}_{\epsilon,\eta,J,\Delta,B} L^{-3\eta}\overline{\chi}_p^{1-6\theta}.$$

The matched anchors $z_1, \ldots, z_m$ are distinct and satisfy $d(x, z_i) \leq 2^{j+1}$. Therefore

$$m \leq |B(x, 2^{j+1})| \leq \Delta^{2^{j+1}+1} \leq \Delta \overline{\chi}_p^{2B},$$

contradicting $m = \lceil \widehat{\kappa}_j(p) \rceil \geq \widehat{\kappa}_j(p)$ and the choice of p_*. Thus the matched obstruction has $2^j > BL$.

Starting from this fixed matched obstruction, run the deterministic thinning and path-selection steps in the proofs of Corollaries 8.743, 8.745, 8.747, 8.749, and 8.751. These steps keep the same p, x, j and only pass to subsets of the matched triples, adding the geodesic tubes. The cardinality estimates are exactly those in the cited proofs, and the same-index separation checks use the stronger bound $2^j > BL$ together with $B > 5\theta$. The resulting buffered tube configuration therefore satisfies all conditions of Corollary 8.751 and still has $2^j > BL$. □

Corollary 8.761 (Volume-sensitive matched-count scale amplification)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $\omega \geq 1$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Assume also that $\omega < \Delta$. There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/19$, every $B > 5\theta$ satisfying

$$2B \log_\Delta \omega < 1 - 2\theta - 4\theta \log_\Delta \omega,$$

every

$$5\theta < a < B \quad \textit{satisfying} \quad (2a + \theta)\log_\Delta \omega < 1 - 2\theta - 4\theta \log_\Delta \omega,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, writing

$$L = \log_\Delta \overline{\chi}_p,$$

we have

$$2^j > BL.$$

In particular, if $\Delta \geq 3$, one may take $\omega = \Delta - 1$, and the condition on B becomes

$$B < \frac{1 - 2\theta - 4\theta \log_\Delta(\Delta - 1)}{2 \log_\Delta(\Delta - 1)}.$$

Proof. Use the ϵ from Corollary 8.717. Write

$$\alpha = \log_\Delta \omega,$$

and put

$$c^{\mathrm{vmatch}}_{\epsilon,\eta,J,C_{\mathrm{vol}},B} = \frac{1}{16 C_{\mathrm{vol}}^3} \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J} B^{-3\eta}.$$

Since $2B\alpha < 1 - 2\theta - 4\theta\alpha$ and $\alpha < 1$, choose $p_* < p_c$ sufficiently close to p_c that whenever $p \in [p_*, p_c)$, with $L = \log_\Delta \overline{\chi}_p$,

$$c^{\mathrm{vmatch}}_{\epsilon,\eta,J,C_{\mathrm{vol}},B} L^{-3\eta} \overline{\chi}_p^{1-2\theta-4\theta\alpha} > C_{\mathrm{vol}} \overline{\chi}_p^{2B\alpha}$$

and

$$\frac{1}{432 C_{\mathrm{vol}}^3 \Delta^5} \overline{\chi}_p^{1-15\theta-4\theta\alpha} > \frac{1}{432 \Delta^8} \overline{\chi}_p^{1-19\theta}.$$

Fix $p_0 \in [p_*, p_c)$, and choose the volume-sensitive matched anchored incidence obstruction with the lower-scale parameter a. If $2^j \leq BL$, then $j \leq \log_2(BL)$, and hence

$$\widehat{\kappa}^{\mathrm{vol}}_j(p) \geq c^{\mathrm{vmatch}}_{\epsilon,\eta,J,C_{\mathrm{vol}},B} L^{-3\eta} \overline{\chi}_p^{1-2\theta-4\theta\alpha}.$$

The matched anchors are distinct and lie in $B(x, 2^{j+1})$, so the volume-growth assumption gives

$$m \leq |B(x, 2^{j+1})| \leq C_{\mathrm{vol}} \omega^{2^{j+1}} \leq C_{\mathrm{vol}} \overline{\chi}_p^{2B\alpha}.$$

This contradicts $m \geq \widehat{\kappa}^{\mathrm{vol}}_j(p)$ and the choice of p_*. Thus the matched obstruction has $2^j > BL$.

Now run the same deterministic thinning and path-selection steps as in the proof of Corollary 8.760. These steps preserve the same p, x, j, and $B > 5\theta$ supplies the same-index separation checks needed in the later buffered-tube extraction. The total loss from the packing and independent-set extractions after the matched stage is at most a factor $27\Delta^{13r_p+5}$. Since

$$m \geq \frac{\sigma_j^3}{16 C_{\mathrm{vol}}^3} \overline{\chi}_p^{1-2\theta-4\theta\alpha}$$

and $\Delta^{13r_p} \leq \overline{\chi}_p^{13\theta}$, the final thinned index set has size at least

$$\frac{\sigma_j^3}{432 C_{\mathrm{vol}}^3 \Delta^5} \overline{\chi}_p^{1-15\theta-4\theta\alpha}.$$

By the second near-critical choice of p_*, this is at least $\kappa_j^{\mathrm{buf}}(p)$. Passing to an arbitrary subset of size $\lceil \kappa_j^{\mathrm{buf}}(p) \rceil$ gives a buffered tube configuration satisfying the conditions of Corollary 8.751, and it retains $2^j > BL$.

For the final assertion, when $\Delta \geq 3$ and $R \geq 0$,

$$|B(u,R)| \leq 1 + \Delta \sum_{k=1}^{R} (\Delta-1)^{k-1} \leq \left(1 + \frac{\Delta}{\Delta-2}\right)(\Delta-1)^R.$$

Thus the preceding part applies with $C_{\mathrm{vol}} = 1 + \Delta/(\Delta-2)$ and $\omega = \Delta - 1$. □

Corollary 8.762 (Endpoint-anchor logarithmic alternative collapse)**.** *Under the assumptions and parameter range of Corollary 8.758, the buffered tube configuration can be chosen so that*

$$2^j > B \log_\Delta \overline{\chi}_p.$$

In particular, within that parameter range the second alternative in Corollary 8.758 is not needed.

Proof. The parameter assumptions in Corollary 8.758 give

$$0 < \theta < 1/31 < 1/19, \qquad 5\theta < a < B < \frac{1-21\theta}{2} < \frac{1-6\theta}{2},$$

and also $a < \min\{B, (1-7\theta)/2\}$. Hence Corollary 8.760 applies with the same a and B, giving a buffered tube configuration satisfying $2^j > B \log_\Delta \overline{\chi}_p$. □

Corollary 8.763 (Volume-sensitive endpoint-anchor collapse)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/31$, every

$$5\theta < a < B < \frac{1-21\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, with

$$L = \log_\Delta \overline{\chi}_p,$$

we have

$$2^j > BL.$$

In particular, if $\Delta \geq 3$, one may take $\omega = \Delta - 1$.

Proof. Put $\alpha = \log_\Delta \omega$. Since $\omega < \Delta$, we have $\alpha < 1$. The endpoint-anchor parameter range gives

$$0 < \theta < 1/31 < 1/19, \qquad B > 5\theta, \qquad 5\theta < a < B.$$

Moreover,

$$2B\alpha < 2B < 1 - 21\theta < 1 - 2\theta - 4\theta\alpha,$$

where the last strict inequality follows from $\alpha < 1$. Similarly,

$$(2a+\theta)\alpha < 2a+\theta < 2B+\theta < 1-20\theta < 1-2\theta-4\theta\alpha.$$

Thus all hypotheses of Corollary 8.761 are satisfied, and that corollary gives the desired buffered tube configuration with $2^j > BL$.

For the final assertion, when $\Delta \geq 3$, the tree-volume estimate in Corollary 8.761 applies with $\omega = \Delta - 1$. □

Proposition 8.764 (Volume-sensitive large-scale buffered-tube criterion)**.** *Let G have maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

Suppose that for every $0 < \epsilon \leq 1$ there exist $\eta > 0$, $0 < \theta < 1/31$,

$$5\theta < a < B < \frac{1-21\theta}{2},$$

integers $J, H \geq 0$, and $p_1 < p_c$ such that no $p \in [p_1, p_c)$ admits an integer $j > J+H$ and a buffered tube configuration satisfying the conditions of Corollary 8.751 and

$$2^j > B \log_\Delta \overline{\chi}_p.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. This is the contrapositive of Corollary 8.763, with the L^2-gap conclusion supplied by Corollary 8.794. □

Corollary 8.765 (Volume-sensitive large-scale open buffered tube obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/31$, every

$$5\theta < a < B < \frac{1-21\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J+H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, with

$$L = \log_\Delta \overline{\chi}_p,$$

$$2^j > BL,$$

and, with

$$N_{\mathrm{open}} = \#\{1 \le i \le t : \text{ every edge of } \gamma_i \text{ is open}\}$$

and

$$\lambda_j^{\mathrm{open}}(p) = \frac{\sigma_j^3}{864\Delta^8}\overline{\chi}_p^{1-20\theta},$$

we have

$$\mathbf{P}_p\left(N_{\mathrm{open}} \ge \lambda_j^{\mathrm{open}}(p)\right) \ge 1 - \exp\left[-\frac{\lambda_j^{\mathrm{open}}(p)}{4}\right].$$

Proof. Use the ϵ from Corollary 8.763. By Lemma 8.732, we have $p_{\mathrm{c}} > 1/\Delta$; hence we may take p_0 close enough to p_{c} that every selected $p \in [p_0, p_{\mathrm{c}})$ satisfies $p \ge 1/\Delta$. Choose the large-scale buffered tube configuration from Corollary 8.763. The paths $\gamma_1, \dots, \gamma_t$ are pairwise vertex-disjoint, hence edge-disjoint, and each has length r_p. Lemma 8.753 therefore applies with

$$\mu = tp^{r_p}.$$

Since $t \ge \kappa_j^{\mathrm{buf}}(p)$ and

$$p^{r_p} \ge \Delta^{-r_p} \ge \overline{\chi}_p^{-\theta},$$

we have

$$\mu \ge \frac{\sigma_j^3}{432\Delta^8}\overline{\chi}_p^{1-20\theta} = 2\lambda_j^{\mathrm{open}}(p).$$

Consequently

$$\mathbf{P}_p\left(N_{\mathrm{open}} < \lambda_j^{\mathrm{open}}(p)\right) \le \mathbf{P}_p\left(N_{\mathrm{open}} < \frac{\mu}{2}\right) \le \exp\left[-\frac{\mu}{8}\right] \le \exp\left[-\frac{\lambda_j^{\mathrm{open}}(p)}{4}\right].$$

□

Corollary 8.766 (Volume-sensitive large-scale open-tube dichotomy)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \ge 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \le \omega < \Delta$ such that*

$$|B(u, R)| \le C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \ge 0).$$

There is $0 < \epsilon \le 1$ such that for every $\eta > 0$, every $0 < \theta < 1/31$, every

$$5\theta < a < B < \frac{1 - 21\theta}{2},$$

every $b > 0$ satisfying

$$\rho := 1 - 20\theta - 3\eta b > 0,$$

every pair of integers $J, H \ge 0$, and all p_0 sufficiently close to p_{c}, there are $p \in [p_0, p_{\mathrm{c}})$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, with

$$L = \log_\Delta \overline{\chi}_p,$$

$$2^j > BL,$$

and at least one of the following alternatives holds:

$$2^j > \overline{\chi}_p^b,$$

or, with

$$c_{\epsilon,\eta,J,\Delta} = \frac{1}{864\Delta^8}\left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J},$$

$$\mathbf{P}_p\left(N_{\mathrm{open}} \geq c_{\epsilon,\eta,J,\Delta}\overline{\chi}_p^\rho\right) \geq 1 - \exp\left[-\frac{c_{\epsilon,\eta,J,\Delta}}{4}\overline{\chi}_p^\rho\right].$$

Proof. Choose the configuration supplied by Corollary 8.765. It already satisfies $2^j > BL$. If $2^j > \overline{\chi}_p^b$, then the first alternative holds. Otherwise $j \leq b\log_2 \overline{\chi}_p$, and hence

$$\sigma_j^3 \geq \left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J}\overline{\chi}_p^{-3\eta b}.$$

Therefore

$$\lambda_j^{\mathrm{open}}(p) = \frac{\sigma_j^3}{864\Delta^8}\overline{\chi}_p^{1-20\theta} \geq c_{\epsilon,\eta,J,\Delta}\overline{\chi}_p^\rho.$$

The second alternative now follows from Corollary 8.765. □

Corollary 8.767 (Volume-sensitive large-scale tube-mass dichotomy)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/31$, every

$$5\theta < a < B < \frac{1-21\theta}{2},$$

every $b > 0$ satisfying

$$\rho := 1 - 20\theta - 3\eta b > 0,$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, with

$$L = \log_\Delta \overline{\chi}_p,$$

$$2^j > BL,$$

and at least one of the following alternatives holds:

$$2^j > \overline{\chi}_p^b,$$

or, with

$$d_{\epsilon,\eta,J,\Delta} = \frac{1}{432\Delta^8}\left[\frac{\epsilon}{2}(1-2^{-\eta/2})^2\right]^3 2^{3\eta J},$$

$$\sum_{i=1}^{t}\tau_p(a_i,z_i) \geq d_{\epsilon,\eta,J,\Delta}\overline{\chi}_p^\rho.$$

Proof. Use the ϵ from Corollary 8.763. By Lemma 8.732, take p_0 close enough to p_c that every selected $p \in [p_0, p_c)$ satisfies $p \geq 1/\Delta$. Choose the large-scale buffered tube configuration from Corollary 8.763. It already satisfies $2^j > BL$. If $2^j > \overline{\chi}_p^b$, then the first alternative holds, so assume $2^j \leq \overline{\chi}_p^b$. Then

$$j \leq b \log_2 \overline{\chi}_p$$

and hence

$$\sigma_j^3 \geq \left[\frac{\epsilon}{2}(1 - 2^{-\eta/2})^2\right]^3 2^{3\eta J} \overline{\chi}_p^{-3\eta b}.$$

For every i, the path γ_i has length r_p and joins a_i to z_i, so

$$\tau_p(a_i, z_i) \geq p^{r_p} \geq \Delta^{-r_p} \geq \overline{\chi}_p^{-\theta}.$$

Since the buffered tube configuration has

$$t \geq \frac{\sigma_j^3}{432\Delta^8} \overline{\chi}_p^{1-19\theta},$$

we get

$$\sum_{i=1}^{t} \tau_p(a_i, z_i) \geq \frac{\sigma_j^3}{432\Delta^8} \overline{\chi}_p^{1-20\theta} \geq d_{\epsilon,\eta,J,\Delta} \overline{\chi}_p^{\rho}.$$

□

Corollary 8.768 (Endpoint-anchor annular mass dichotomy)**.** *Under the assumptions and notation of Corollary 8.758, the same conclusion remains valid if the second alternative is replaced by the stronger distance-localized alternative*

$$\sum_{\substack{1 \leq i \leq t \\ d(y_i, z_i) \in I_{j,p}^{\mathrm{ea}}}} \tau_p(y_i, z_i) = \sum_{i=1}^{t} \tau_p(y_i, z_i) \geq c_{\epsilon,\eta,J,\Delta,B} L^{-3\eta} \overline{\chi}_p^{1-21\theta-2B},$$

where

$$I_{j,p}^{\mathrm{ea}} = (2^j - 2r_p, 2^{j+1} + 2r_p].$$

Proof. Choose the data supplied by Corollary 8.758. If $2^j > BL$, then the first alternative holds, so assume the second alternative in that corollary holds. For every i, the buffered tube configuration gives

$$d(y_i, a_i) \in (2^j - r_p, 2^{j+1} + r_p], \qquad d(a_i, z_i) = r_p.$$

Therefore

$$d(y_i, z_i) \geq d(y_i, a_i) - d(a_i, z_i) > 2^j - 2r_p$$

and

$$d(y_i, z_i) \leq d(y_i, a_i) + d(a_i, z_i) \leq 2^{j+1} + 2r_p.$$

Thus $d(y_i, z_i) \in I_{j,p}^{\mathrm{ea}}$ for every i, and the distance-localized lower bound is exactly the second lower bound from Corollary 8.758. □

Definition 8.769 (Annular crossing mass)**.** For finite non-empty $K \subset V$, a set $Z \subseteq V$, and $I \subseteq [0, \infty)$, define

$$\mathcal{B}_p^I(K, Z) = \frac{1}{|K|} \sum_{z \in Z} \sum_{\substack{x \in K \\ d(x,z) \in I}} Q_p(x, z).$$

Proposition 8.770 (Weighted annular crossing obstruction)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $\epsilon > 0$ such that for every $0 < \alpha < \beta$, every integer $J \geq 0$, and every $p_0 < p_c$, there are $p \in [p_0, p_c)$, a finite non-empty $K \subset V$, and sets $Z_j \subset V$, $j \geq J$, such that $|K| > 2\epsilon\overline{\chi}_p$, and

$$\sum_{j \geq J} 2^{(\alpha+\beta)(j-J)} \mathcal{B}_p^{(2^j, 2^{j+1}]}(K, Z_j) \geq \frac{\epsilon^2}{4}(1 - 2^{-\alpha})^2(1 - 2^{\alpha-\beta}).$$

Proof. Use the same ϵ, p, K, and thresholds λ_j as in Proposition 8.30, and set

$$Z_j = Z_p^{(2^j, 2^{j+1}]}(K, \lambda_j).$$

For every $j \geq J$, the definition of Z_j gives

$$\mathcal{B}_p^{(2^j, 2^{j+1}]}(K, Z_j) \geq \lambda_j \frac{|Z_j|}{|K|}.$$

Writing $c = (1 - 2^{-\alpha})\epsilon$, we have

$$\lambda_j = \frac{c}{2}(1 - 2^{\alpha-\beta})2^{-\beta(j-J)}$$

and Proposition 8.30 gives

$$\sum_{j \geq J} 2^{\alpha(j-J)} \frac{|Z_j|}{|K|} \geq \frac{c}{2}.$$

Therefore

$$\begin{aligned} \sum_{j \geq J} 2^{(\alpha+\beta)(j-J)} \mathcal{B}_p^{(2^j, 2^{j+1}]}(K, Z_j) &\geq \frac{c}{2}(1 - 2^{\alpha-\beta}) \sum_{j \geq J} 2^{\alpha(j-J)} \frac{|Z_j|}{|K|} \\ &\geq \frac{c^2}{4}(1 - 2^{\alpha-\beta}), \end{aligned}$$

which is the desired bound. □

Definition 8.771 (Annular two-point operators)**.** For $I \subseteq [0, \infty)$, define

$$T_p^I(x, y) = \tau_p(x, y)\mathbf{1}\{d(x, y) \in I\}.$$

Lemma 8.772 (Separated pairs lower bound annular operators)**.** *Let $I \subseteq [0, \infty)$, let $p < p_c$, and let*

$$u_1, \ldots, u_m, \qquad v_1, \ldots, v_m$$

be vertices such that the u_i's are distinct, the v_i's are distinct, and $d(u_i, v_i) \in I$ for every $1 \leq i \leq m$. Then

$$\left\|T_p^I\right\|_{2 \to 2} \geq \frac{1}{m} \sum_{i=1}^m \tau_p(u_i, v_i)$$

and

$$\Theta_p(I) \geq \frac{1}{\overline{\chi}_p^2 m} \sum_{i=1}^m \tau_p(u_i, v_i)^2.$$

Proof. Put $U = \{u_1, \dots, u_m\}$ and $V = \{v_1, \dots, v_m\}$. Since both sets have size m, the vectors $\mathbf{1}_U/\sqrt{m}$ and $\mathbf{1}_V/\sqrt{m}$ have norm one. Positivity gives

$$\begin{aligned}\left\|T_p^I\right\|_{2\to 2} &\geq \frac{1}{m}\left\langle \mathbf{1}_V, T_p^I \mathbf{1}_U \right\rangle \\ &= \frac{1}{m}\sum_{i=1}^{m}\sum_{k=1}^{m} \tau_p(u_k, v_i)\mathbf{1}\{d(u_k, v_i) \in I\} \\ &\geq \frac{1}{m}\sum_{i=1}^{m} \tau_p(u_i, v_i).\end{aligned}$$

For the square-profile bound, use $K = U$ in Definition 8.21. At the distinct vertices v_i, the summand contains the diagonal contribution $\tau_p(u_i, v_i)$. Hence

$$\mathcal{R}_p^I(U) \geq \frac{1}{\overline{\chi}_p m}\sum_{i=1}^{m} \tau_p(u_i, v_i)^2.$$

Dividing by $\overline{\chi}_p$ and taking the supremum over finite K gives the displayed lower bound for $\Theta_p(I)$. □

Corollary 8.773 (Volume-sensitive gate-shell operator obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/31$, every

$$5\theta < a < B < \frac{1 - 21\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_{c}, there are $p \in [p_0, p_{\mathrm{c}})$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad I_p^{\mathrm{tube}} = \{r_p\},$$

we have

$$2^j > BL,$$

$$\left\|T_p^{I_p^{\mathrm{tube}}}\right\|_{2\to 2} \geq \overline{\chi}_p^{-\theta},$$

and

$$\Theta_p(I_p^{\mathrm{tube}}) \geq \overline{\chi}_p^{-2-2\theta}.$$

Proof. Use the ϵ from Corollary 8.763. By Lemma 8.732, take p_0 close enough to p_{c} that every selected $p \in [p_0, p_{\mathrm{c}})$ satisfies $p \geq 1/\Delta$. Choose the large-scale buffered tube configuration from Corollary 8.763. It satisfies $2^j > BL$. The vertices $a_1, \dots, a_t$ are distinct and $z_1, \dots, z_t$ are distinct by the separation conditions, and

$$d(a_i, z_i) = r_p \qquad (1 \leq i \leq t).$$

Moreover each γ_i is a path of length r_p from a_i to z_i, so

$$\tau_p(a_i, z_i) \geq p^{r_p} \geq \Delta^{-r_p} \geq \overline{\chi}_p^{-\theta}.$$

Apply Lemma 8.772 with $u_i = a_i$, $v_i = z_i$, and $I = I_p^{\mathrm{tube}}$. The operator lower bound follows from the average lower bound on the $\tau_p(a_i, z_i)$'s, and the square-profile lower bound follows from

$$\frac{1}{\overline{\chi}_p^2 t} \sum_{i=1}^{t} \tau_p(a_i, z_i)^2 \geq \overline{\chi}_p^{-2-2\theta}.$$

□

Lemma 8.774 (Annular operator control)**.** *For every $I \subseteq [0, \infty)$ and every $p < p_{\mathrm{c}}$,*

$$\Theta_p(I) \leq \left(\frac{\left\| T_p^I \right\|_{2\to 2}}{\overline{\chi}_p} \right)^2.$$

Proof. For every finite non-empty K,

$$\frac{\mathcal{R}_p^I(K)}{\overline{\chi}_p} = \frac{\left\| T_p^I \mathbf{1}_K \right\|_2^2}{\overline{\chi}_p^2 |K|} \leq \left(\frac{\left\| T_p^I \right\|_{2\to 2}}{\overline{\chi}_p} \right)^2.$$

Taking the supremum over K gives the claim. □

Corollary 8.775 (Dyadic annular operator criterion)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$. For $j \geq J$, let $I_j = (2^j, 2^{j+1}]$. Suppose that for some $\alpha > 0$,*

$$\lim_{J\to\infty} \limsup_{p\uparrow p_{\mathrm{c}}} \sum_{j\geq J} 2^{\alpha(j-J)} \left(\frac{\left\| T_p^{I_j} \right\|_{2\to 2}}{\overline{\chi}_p} \right)^2 = 0.$$

Then

$$\lim_{p\uparrow p_{\mathrm{c}}} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is connected, locally finite, quasi-transitive, and

$$\limsup_{p\uparrow p_{\mathrm{c}}} (p_{\mathrm{c}} - p)\overline{\chi}_p < \infty,$$

then

$$p_{\mathrm{c}}(G) < p_{2\to 2}(G).$$

Proof. The displayed assumption and Lemma 8.774 imply the hypothesis of Corollary 8.28. Hence $\mathcal{R}_p^*/\overline{\chi}_p \to 0$. Under the additional hypotheses, the L^2-gap conclusion follows from Corollary 8.794. □

Lemma 8.776 (Pointwise shell control of annular operators)**.** *Let G be connected and locally finite. For every $I \subseteq [0, \infty)$ and every $p < p_{\mathrm{c}}$,*

$$\left\| T_p^I \right\|_{2\to 2} \leq \sum_{n\in I\cap\mathbb{N}} \tau_p^{\max}(n) \left\| S_n^G \right\|_{2\to 2},$$

with the convention that the right-hand side may be infinite.

Proof. For $n \geq 0$, let

$$H_{p,n}(x,y) = \tau_p(x,y)\mathbf{1}\{d(x,y) = n\}.$$

Then $0 \leq H_{p,n} \leq \tau_p^{\max}(n) S_n^G$ entrywise. Positivity gives

$$\|H_{p,n}\|_{2\to 2} \leq \tau_p^{\max}(n) \left\|S_n^G\right\|_{2\to 2}.$$

Since $T_p^I = \sum_{n\in I\cap\mathbb{N}} H_{p,n}$ pointwise, the triangle inequality in operator norm gives the claim, first for finite sums and then by monotone convergence of positive kernels. □

Corollary 8.777 (Volume-sensitive logarithmic shell-growth obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G is connected and locally finite, has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

There is $0 < \epsilon \leq 1$ such that for every $\eta > 0$, every $0 < \theta < 1/31$, every

$$5\theta < a < B < \frac{1-21\theta}{2},$$

every pair of integers $J, H \geq 0$, and all p_0 sufficiently close to p_c, there are $p \in [p_0, p_c)$, an integer $j > J + H$, and a buffered tube configuration satisfying the conditions of Corollary 8.751 such that, with

$$L = \log_\Delta \overline{\chi}_p, \qquad r_p = \lfloor \theta L \rfloor,$$

we have

$$2^j > BL$$

and

$$\tau_p^{\max}(r_p) \left\|S_{r_p}^G\right\|_{2\to 2} \geq \overline{\chi}_p^{-\theta}.$$

Proof. Choose the configuration supplied by Corollary 8.773. It satisfies $2^j > BL$ and

$$\left\|T_p^{\{r_p\}}\right\|_{2\to 2} \geq \overline{\chi}_p^{-\theta}.$$

Lemma 8.776, applied to $I = \{r_p\}$, gives

$$\left\|T_p^{\{r_p\}}\right\|_{2\to 2} \leq \tau_p^{\max}(r_p) \left\|S_{r_p}^G\right\|_{2\to 2}.$$

Combining the two inequalities proves the shell-growth lower bound. □

Proposition 8.778 (Volume-sensitive logarithmic shell-growth criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u,R)| \leq C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \geq 0).$$

If there exists $0 < \theta < 1/31$ such that, with

$$r_p = \lfloor \theta \log_\Delta \overline{\chi}_p \rfloor,$$

$$\limsup_{p\uparrow p_c} \overline{\chi}_p^{\theta} \tau_p^{\max}(r_p) \left\|S_{r_p}^G\right\|_{2\to 2} < 1,$$

then

$$\lim_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. We prove the first claim by contraposition. Suppose that

$$\limsup_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0.$$

Choose a, B with

$$5\theta < a < B < \frac{1-21\theta}{2},$$

which is possible since $\theta < 1/31$, and fix arbitrary $\eta > 0$ and integers $J, H \geq 0$. Corollary 8.777 then gives, for every p_0 sufficiently close to p_c, some $p \in [p_0, p_c)$ such that

$$\overline{\chi}_p^{\theta} \tau_p^{\max}(r_p) \left\| S_{r_p}^G \right\|_{2\to 2} \geq 1.$$

Hence the displayed limsup is at least 1, contradicting the hypothesis. Thus $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

Under the additional quasi-transitivity and mean-field susceptibility assumptions, the L^2-gap conclusion follows from Corollary 8.794. □

Corollary 8.779 (Volume-sensitive logarithmic shell exponent obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G is connected and locally finite, has maximum degree at most $\Delta \geq 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

For every $0 < \theta < 1/31$, if

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

then

$$\limsup_{p\uparrow p_c} \left(\tau_p^{\max}(r_p) \left\| S_{r_p}^G \right\|_{2\to 2} \right)^{1/r_p} \geq \Delta^{-1}.$$

Proof. Fix $0 < \theta < 1/31$, and choose a, B with

$$5\theta < a < B < \frac{1-21\theta}{2}.$$

Apply Corollary 8.777 with arbitrary $\eta > 0$ and $J = H = 0$. For all p_0 sufficiently close to p_c, it gives some $p \in [p_0, p_c)$ such that, with

$$A_p = \tau_p^{\max}(r_p) \left\| S_{r_p}^G \right\|_{2\to 2},$$

we have

$$A_p \geq \overline{\chi}_p^{-\theta}.$$

Letting $p_0 \uparrow p_c$ and choosing one such p for each p_0 gives a sequence $p_n \uparrow p_c$. Since $\overline{\chi}_{p_n} \to \infty$,

$$\frac{r_{p_n}}{\theta \log_\Delta \overline{\chi}_{p_n}} \longrightarrow 1.$$

Along this sequence,

$$A_{p_n}^{1/r_{p_n}} \geq \overline{\chi}_{p_n}^{-\theta/r_{p_n}} \longrightarrow \Delta^{-1}.$$

This proves the displayed limsup lower bound. □

Proposition 8.780 (Volume-sensitive logarithmic shell exponent criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

If there exists $0 < \theta < 1/31$ such that, with

$$r_p = \lfloor \theta \log_\Delta \overline{\chi}_p \rfloor,$$

$$\limsup_{p \uparrow p_c} \left(\tau_p^{\max}(r_p) \left\| S_{r_p}^G \right\|_{2\to 2} \right)^{1/r_p} < \Delta^{-1},$$

then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. We prove the first claim by contraposition. If

$$\limsup_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then Corollary 8.779 gives

$$\limsup_{p \uparrow p_c} \left(\tau_p^{\max}(r_p) \left\| S_{r_p}^G \right\|_{2\to 2} \right)^{1/r_p} \geq \Delta^{-1},$$

contradicting the hypothesis. Therefore $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

Under the additional quasi-transitivity and mean-field susceptibility assumptions, the L^2-gap conclusion follows from Corollary 8.794. □

Proposition 8.781 (Factored logarithmic shell exponent criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Suppose that there exist $0 < \theta < 1/31$ *and positive constants* α, ρ *such that, with*

$$r_p = \lfloor \theta \log_\Delta \overline{\chi}_p \rfloor ,$$

$$\limsup_{p \uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \alpha, \qquad \limsup_{p \uparrow p_c} \left\| S_{r_p}^G \right\|_{2\to 2}^{1/r_p} < \rho,$$

and

$$\alpha\rho < \Delta^{-1}.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For p sufficiently close to p_c, the two displayed limsup assumptions give

$$\tau_p^{\max}(r_p)^{1/r_p} \le \alpha, \qquad \left\| S_{r_p}^G \right\|_{2\to 2}^{1/r_p} \le \rho.$$

Consequently

$$\left(\tau_p^{\max}(r_p) \left\| S_{r_p}^G \right\|_{2\to 2} \right)^{1/r_p} \le \alpha\rho < \Delta^{-1}.$$

The first conclusion follows from Proposition 8.780. The L^2-gap conclusion again follows from Corollary 8.794. □

Corollary 8.782 (Volume-growth pointwise logarithmic decay criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \ge 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \le \omega < \Delta$ such that*

$$|B(u, R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

If there exists $0 < \theta < 1/31$ *such that, with*

$$r_p = \lfloor \theta \log_\Delta \overline{\chi}_p \rfloor ,$$

$$\limsup_{p \uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \frac{1}{\Delta\omega},$$

then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. The sphere kernel S_n^G is symmetric and every row sum is

$$|\{v \in V : d(u,v) = n\}| \le |B(u,n)| \le C_{\mathrm{vol}}\omega^n.$$

Schur's test therefore gives

$$\left\|S_n^G\right\|_{2\to 2} \le C_{\mathrm{vol}}\omega^n \qquad (n \ge 0),$$

and hence

$$\limsup_{p\uparrow p_c} \left\|S_{r_p}^G\right\|_{2\to 2}^{1/r_p} \le \omega.$$

Choose α and ρ such that

$$\limsup_{p\uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \alpha, \qquad \omega < \rho, \qquad \alpha\rho < \Delta^{-1}.$$

This is possible by the strict pointwise-decay hypothesis. The result follows from Proposition 8.781. □

Corollary 8.783 (Critical logarithmic shell exponent obstruction)**.** *Under the assumptions of Proposition 8.37, suppose in addition that G is connected and locally finite, has maximum degree at most $\Delta \ge 2$, and that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \le \omega < \Delta$ such that*

$$|B(u,R)| \le C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \ge 0).$$

For every $0 < \theta < 1/31$, if

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

then

$$\limsup_{p\uparrow p_c} \left(\tau_p^{\max}(r_p)\left\|S_{r_p}^G\right\|_{2\to 2}\right)^{1/r_p} \ge p_c.$$

Proof. Fix $0 < \theta < 1/31$ and $q < p_c$. Choose a, B with

$$5\theta < a < B < \frac{1-21\theta}{2}.$$

Let $p_0 < p_c$ be close enough to p_c that $p_0 \ge q$ and that Corollary 8.763 applies with, say, $\eta = 1$ and $J = H = 0$. Choose the resulting large-scale buffered tube configuration. The vertices $a_1, \dots, a_t$ are distinct, the vertices $z_1, \dots, z_t$ are distinct, and $d(a_i, z_i) = r_p$. Since $p \ge p_0 \ge q$, each tube path gives

$$\tau_p(a_i, z_i) \ge p^{r_p} \ge q^{r_p}.$$

Lemma 8.772, applied with $I = \{r_p\}$, gives

$$\left\|T_p^{\{r_p\}}\right\|_{2\to 2} \ge q^{r_p}.$$

The pointwise shell-control lemma gives

$$\left\|T_p^{\{r_p\}}\right\|_{2\to 2} \le \tau_p^{\max}(r_p)\left\|S_{r_p}^G\right\|_{2\to 2},$$

and hence

$$\left(\tau_p^{\max}(r_p)\left\|S_{r_p}^G\right\|_{2\to 2}\right)^{1/r_p} \ge q.$$

Letting $p_0 \uparrow p_c$ produces a sequence $p \uparrow p_c$, and then letting $q \uparrow p_c$ proves the claim. □

Proposition 8.784 (Critical logarithmic shell exponent criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

If there exists $0 < \theta < 1/31$ such that, with

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\limsup_{p \uparrow p_c} \left(\tau_p^{\max}(r_p) \left\| S_{r_p}^G \right\|_{2\to 2} \right)^{1/r_p} < p_c,$$

then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. The first claim is the contrapositive of Corollary 8.783. The L^2-gap conclusion follows from Corollary 8.794. □

Proposition 8.785 (Critical factored logarithmic shell criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \geq 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \leq \omega < \Delta$ such that*

$$|B(u, R)| \leq C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \geq 0).$$

Suppose that there exist $0 < \theta < 1/31$ and positive constants α, ρ such that, with

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\limsup_{p \uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \alpha, \qquad \limsup_{p \uparrow p_c} \left\| S_{r_p}^G \right\|_{2\to 2}^{1/r_p} < \rho,$$

and

$$\alpha \rho < p_c.$$

Then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p \uparrow p_c} (p_c - p) \overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For p sufficiently close to p_c,

$$\left(\tau_p^{\max}(r_p)\left\|S_{r_p}^G\right\|_{2\to 2}\right)^{1/r_p} \le \alpha\rho < p_c.$$

Apply Proposition 8.784. □

Corollary 8.786 (Spherical obstruction)**.** *Assume the hypotheses of Corollary 8.783. Define*

$$\rho_{\mathrm{sph},2}(G) = \limsup_{n\to\infty} \left\|S_n^G\right\|_{2\to 2}^{1/n}.$$

For every $0 < \theta < 1/31$*, if*

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

then

$$\limsup_{p\uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} \ge \frac{p_c}{\rho_{\mathrm{sph},2}(G)}.$$

Proof. Let L denote the limsup on the left-hand side, and fix $\rho > \rho_{\mathrm{sph},2}(G)$. Since $\overline{\chi}_p \to \infty$, we have $r_p \to \infty$ as $p \uparrow p_c$. Hence, for p sufficiently close to p_c,

$$\left\|S_{r_p}^G\right\|_{2\to 2} \le \rho^{r_p}.$$

Therefore

$$\limsup_{p\uparrow p_c} \left(\tau_p^{\max}(r_p)\left\|S_{r_p}^G\right\|_{2\to 2}\right)^{1/r_p} \le \rho L.$$

Corollary 8.783 gives the opposite lower bound p_c for this product limsup, so $L \ge p_c/\rho$. Letting $\rho \downarrow \rho_{\mathrm{sph},2}(G)$ proves the claim. □

Corollary 8.787 (Spherical logarithmic criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \ge 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \le \omega < \Delta$ such that*

$$|B(u,R)| \le C_{\mathrm{vol}}\omega^R \qquad (u \in V,\ R \ge 0).$$

Define

$$\rho_{\mathrm{sph},2}(G) = \limsup_{n\to\infty} \left\|S_n^G\right\|_{2\to 2}^{1/n}.$$

If there exists $0 < \theta < 1/31$ such that, with

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\limsup_{p\uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \frac{p_c}{\rho_{\mathrm{sph},2}(G)},$$

then

$$\lim_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Choose $\alpha, \rho > 0$ such that

$$\limsup_{p \uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \alpha, \qquad \rho_{\mathrm{sph},2}(G) < \rho, \qquad \alpha\rho < p_c.$$

Since $r_p \to \infty$,

$$\limsup_{p \uparrow p_c} \left\| S_{r_p}^G \right\|_{2\to 2}^{1/r_p} \le \rho_{\mathrm{sph},2}(G) < \rho.$$

The result follows from Proposition 8.785. □

Corollary 8.788 (Critical volume-growth pointwise logarithmic decay criterion)**.** *Let G be connected and locally finite with maximum degree at most $\Delta \ge 2$, suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$, and suppose that there are constants $C_{\mathrm{vol}} < \infty$ and $1 \le \omega < \Delta$ such that*

$$|B(u, R)| \le C_{\mathrm{vol}} \omega^R \qquad (u \in V,\ R \ge 0).$$

If there exists $0 < \theta < 1/31$ such that, with

$$r_p = \left\lfloor \theta \log_\Delta \overline{\chi}_p \right\rfloor,$$

$$\limsup_{p \uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \frac{p_c}{\omega},$$

then

$$\lim_{p \uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p \uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. As in the proof of Corollary 8.782,

$$\limsup_{p \uparrow p_c} \left\| S_{r_p}^G \right\|_{2\to 2}^{1/r_p} \le \omega.$$

Choose $\alpha, \rho > 0$ such that

$$\limsup_{p \uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} < \alpha, \qquad \omega < \rho, \qquad \alpha\rho < p_c.$$

Apply Proposition 8.785. □

Proposition 8.789 (Geodesic lower-bound audit)**.** *Let G be an infinite connected locally finite graph. For every $n \ge 0$ and every $p \in [0, 1]$,*

$$\tau_p^{\max}(n) \ge p^n, \qquad \left\| S_n^G \right\|_{2\to 2} \ge 1.$$

Consequently, if $r_p \to \infty$ as $p \uparrow p_c$, then

$$\limsup_{p \uparrow p_c} \tau_p^{\max}(r_p)^{1/r_p} \ge p_c$$

and

$$\limsup_{p\uparrow p_c}\left(\tau_p^{\max}(r_p)\left\|S_{r_p}^G\right\|_{2\to 2}\right)^{1/r_p}\geq p_c.$$

In particular, the strict product hypothesis in Proposition 8.784 is never satisfied on an infinite graph. Since $\rho_{\mathrm{sph},2}(G)\geq 1$ and every admissible volume-growth constant ω in the criteria above satisfies $\omega\geq 1$, the strict pointwise hypotheses in Corollaries 8.787 and 8.788 are also never satisfied.

Proof. Fix n. Since G is infinite, connected, and locally finite, every vertex has some vertex at distance n. If $d(x,y)=n$, then any fixed geodesic from x to y is open with probability p^n, and this event implies $x\leftrightarrow y$. Hence $\tau_p^{\max}(n)\geq p^n$.

For the sphere kernel, choose x with $S_n(x)\neq\varnothing$. Then

$$\left\|S_n^G\mathbf{1}_{\{x\}}\right\|_2=|S_n(x)|^{1/2}\geq 1,$$

so $\left\|S_n^G\right\|_{2\to 2}\geq 1$. Applying these two bounds with $n=r_p$ gives

$$\tau_p^{\max}(r_p)^{1/r_p}\geq p,\qquad\left(\tau_p^{\max}(r_p)\left\|S_{r_p}^G\right\|_{2\to 2}\right)^{1/r_p}\geq p.$$

Taking $p\uparrow p_c$ proves the two limsup lower bounds.

The final assertions follow immediately. The product criterion asks for a strict upper bound below p_c, contradicting the product lower bound. The spherical and volume-growth pointwise criteria ask for strict upper bounds below $p_c/\rho_{\mathrm{sph},2}(G)$ and p_c/ω, respectively; these thresholds are at most p_c. □

Remark 8.790. Proposition 8.789 is a warning about loss of information in the pointwise supremum. The preceding obstruction statements remain true, but after replacing the structured endpoint configuration by $\tau_p^{\max}$, the resulting strict criteria are vacuous. A viable proof must keep some averaged, localized, or matched-pair information from the finite-set obstruction rather than trying to beat the open-geodesic lower bound for the worst pair at distance r_p.

Corollary 8.791 (Pointwise dyadic annular criterion)**.** *Let G be connected, locally finite, quasi-transitive, and of bounded degree, and suppose that*

$$\limsup_{p\uparrow p_c}(p_c-p)\overline{\chi}_p<\infty.$$

For $j\geq J$, put $I_j=(2^j,2^{j+1}]$. If for some $\alpha>0$,

$$\lim_{J\to\infty}\limsup_{p\uparrow p_c}\sum_{j\geq J}2^{\alpha(j-J)}\left(\frac{\sum_{n\in I_j\cap\mathbb{N}}\tau_p^{\max}(n)\left\|S_n^G\right\|_{2\to 2}}{\overline{\chi}_p}\right)^2=0,$$

then

$$p_c(G)<p_{2\to 2}(G).$$

Proof. Lemma 8.776 implies the hypothesis of Corollary 8.775. Since quasi-transitivity gives $\overline{\chi}_p\to\infty$ as $p\uparrow p_c$, that corollary gives the claim. □

Corollary 8.792 (Dyadic operator obstruction)**.** *Let G have bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\limsup_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} > 0,$$

then there is $\epsilon > 0$ such that for every $\alpha > 0$,

$$\liminf_{J\to\infty} \limsup_{p\uparrow p_c} \sum_{j\geq J} 2^{\alpha(j-J)} \left(\frac{\left\| T_p^{(2^j,2^{j+1}]} \right\|_{2\to 2}}{\overline{\chi}_p} \right)^2 \geq (1-2^{-\alpha})\epsilon.$$

If, in addition, G is connected and locally finite, then

$$\liminf_{J\to\infty} \limsup_{p\uparrow p_c} \sum_{j\geq J} 2^{\alpha(j-J)} \left(\frac{\sum_{2^j<n\leq 2^{j+1}} \tau_p^{\max}(n) \left\| S_n^G \right\|_{2\to 2}}{\overline{\chi}_p} \right)^2 \geq (1-2^{-\alpha})\epsilon$$

for every $\alpha > 0$. In particular, the two displayed obstructions hold if G is connected, locally finite, quasi-transitive, and of bounded degree,

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

and $p_c(G) = p_{2\to 2}(G)$.

Proof. The first lower bound follows by combining Proposition 8.29 with Lemma 8.774. The pointwise shell lower bound then follows from Lemma 8.776.

For the final assertion, quasi-transitivity gives $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If $\mathcal{R}_p^*/\overline{\chi}_p \to 0$, then Corollary 8.794 gives $p_c(G) < p_{2\to 2}(G)$, contradicting $p_c(G) = p_{2\to 2}(G)$. Hence the hypothesis of the first part holds. □

Proposition 8.793 (Critical shell summability gives far-leg tightness)**.** *Let G be connected, locally finite, and of bounded degree, and suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If*

$$\sum_{n\geq 0} \tau_{p_c}^{\max}(n) \left\| S_n^G \right\|_{2\to 2} < \infty,$$

then

$$\lim_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

If, in addition, G is quasi-transitive and

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For $p < p_c$, monotonicity gives $\tau_p^{\max}(n) \leq \tau_{p_c}^{\max}(n)$. Hence Lemma 8.776, applied to (R, ∞), gives

$$\left\| T_p^{(R,\infty)} \right\|_{2\to 2} \leq \sum_{n>R} \tau_{p_c}^{\max}(n) \left\| S_n^G \right\|_{2\to 2}.$$

The right-hand side tends to zero as $R \to \infty$, uniformly in $p < p_c$. For every finite non-empty K,

$$\frac{\mathcal{R}_p^{>R}(K)}{\overline{\chi}_p} = \frac{\left\| T_p^{(R,\infty)} \mathbf{1}_K \right\|_2^2}{\overline{\chi}_p^2 |K|} \le \left(\frac{\left\| T_p^{(R,\infty)} \right\|_{2\to 2}}{\overline{\chi}_p} \right)^2 .$$

Since $\overline{\chi}_p \ge 1$, it follows that $\lim_{R\to\infty} \limsup_{p\uparrow p_c} \Theta_p(R) = 0$. The tail-limit criterion, Corollary 8.20, gives $\mathcal{R}_p^*/\overline{\chi}_p \to 0$.

Under the additional quasi-transitivity and mean-field assumptions, the L^2-gap conclusion follows from Corollary 8.794. □

Corollary 8.794 (Sublinear two-replica finite-set criterion)**.** *Let G be connected, locally finite, and quasi-transitive. Suppose that*

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty$$

and

$$\lim_{p\uparrow p_c} \frac{\mathcal{R}_p^*}{\overline{\chi}_p} = 0.$$

Then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Proposition 8.13 gives finite-set escape, and the first assumption is the mean-field susceptibility input (MF). The conclusion follows from Theorem 4.1. □

Corollary 8.795 (Far-leg tightness criterion for an L^2 gap)**.** *Let G be connected, locally finite, quasi-transitive, and of bounded degree. Suppose that*

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty$$

and that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. For $R < \infty$, let $\Theta_p(R)$ be as in Proposition 8.19. If for every $\delta > 0$ there are $R < \infty$ and $p_0 < p_c$ such that

$$\Theta_p(R) \le \delta \qquad (p \in [p_0, p_c)),$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Proposition 8.19 gives $\mathcal{R}_p^*/\overline{\chi}_p \to 0$. The conclusion then follows from Corollary 8.794. □

Corollary 8.796 (Operator tail criterion)**.** *Let G be connected, locally finite, quasi-transitive, and of bounded degree. Suppose that*

$$\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty$$

and that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. For $R < \infty$, define the truncated two-point operator

$$T_p^{>R}(x, y) = \tau_p(x, y)\mathbf{1}\{d(x, y) > R\}.$$

If

$$\lim_{R\to\infty} \limsup_{p\uparrow p_c} \frac{\left\| T_p^{>R} \right\|_{2\to 2}}{\overline{\chi}_p} = 0,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For every finite non-empty K,

$$\frac{\mathcal{R}_p^{>R}(K)}{\overline{\chi}_p} = \frac{\left\|T_p^{>R}\mathbf{1}_K\right\|_2^2}{\overline{\chi}_p^2|K|} \le \left(\frac{\left\|T_p^{>R}\right\|_{2\to 2}}{\overline{\chi}_p}\right)^2.$$

Taking the supremum over K gives the same bound on $\Theta_p(R)$. The operator-tail assumption therefore implies the far-leg tightness hypothesis in Corollary 8.795. □

Corollary 8.797 (Shell operator tail criterion)**.** *Let G satisfy the assumptions of Corollary 8.796. For $n \ge 0$, define the exact-distance two-point shell*

$$H_{p,n}(x,y) = \tau_p(x,y)\mathbf{1}\{d(x,y) = n\}.$$

If

$$\lim_{R\to\infty} \limsup_{p\uparrow p_c} \frac{1}{\overline{\chi}_p} \sum_{n>R} \|H_{p,n}\|_{2\to 2} = 0,$$

then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For every R,

$$T_p^{>R} = \sum_{n>R} H_{p,n}$$

pointwise. Hence, by the triangle inequality in operator norm,

$$\left\|T_p^{>R}\right\|_{2\to 2} \le \sum_{n>R} \|H_{p,n}\|_{2\to 2},$$

with the inequality understood as trivial if the right-hand side is infinite. The assumed shell-tail estimate implies the hypothesis of Corollary 8.796. □

Proposition 8.798 (Self-avoiding shell tails)**.** *Let G be connected, locally finite, and of bounded degree, and suppose that*

$$\gamma_{\mathrm{SAW},2}(G) < 1/p_c(G).$$

Then the exact-distance two-point shells from Corollary 8.797 satisfy

$$\lim_{R\to\infty} \sup_{0\le p<p_c} \sum_{n>R} \|H_{p,n}\|_{2\to 2} = 0.$$

In particular, if G also satisfies the hypotheses of Corollary 8.797, then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. Choose θ such that $\gamma_{\mathrm{SAW},2} < \theta < 1/p_c$. By the definition of $\gamma_{\mathrm{SAW},2}$, there is $A < \infty$ such that

$$\|C_m\|_{2\to 2} \le A\theta^m \qquad (m \ge 0),$$

where C_m is the length-m self-avoiding endpoint kernel.

If $d(x,y) = n$ and $x \leftrightarrow y$, then the open cluster contains an open self-avoiding path from x to y of some length $m \ge n$. The union bound gives the entrywise estimate

$$H_{p,n}(x,y) \le \sum_{m\ge n} p^m C_m(x,y).$$

Since all kernels are nonnegative, entrywise domination implies domination of ℓ^2 operator norms. Thus for $p < p_c$,

$$\|H_{p,n}\|_{2\to 2} \le A \sum_{m\ge n} (p\theta)^m.$$

Summing over $n > R$ and using $p\theta \le p_c\theta < 1$ gives

$$\begin{aligned}\sum_{n>R} \|H_{p,n}\|_{2\to 2} &\le A \sum_{n>R} \sum_{m\ge n} (p\theta)^m \\ &\le A \sum_{m>R} (m-R)(p_c\theta)^m.\end{aligned}$$

The final tail tends to zero as $R \to \infty$, uniformly over $p < p_c$. Since $\overline{\chi}_p \ge 1$, the shell-tail hypothesis of Corollary 8.797 follows whenever its remaining assumptions hold. □

Lemma 8.799 (Two-replica cluster interpretation)**.** *Let ω_1, ω_2 be two independent Bernoulli-p percolation configurations, and let $C_i(z)$ be the open cluster of z in ω_i. Then, for every finite non-empty $K \subset V$,*

$$\overline{\chi}_p |K|\, \mathcal{R}_p(K) = \sum_{z\in V} \mathbb{E}_p^{\otimes 2}\big[|K \cap C_1(z)|\, |K \cap C_2(z)|\big],$$

with equality in $[0,\infty]$.

Proof. Expanding the square and using independence gives

$$\begin{aligned}\sum_{z\in V} \mathbb{E}_p^{\otimes 2}\big[|K \cap C_1(z)|\, |K \cap C_2(z)|\big] &= \sum_{z\in V} \sum_{x,y\in K} \mathbb{P}_p(x \leftrightarrow z)\mathbb{P}_p(y \leftrightarrow z) \\ &= \sum_{z\in V} \left(\sum_{x\in K} \tau_p(x,z) \right)^2.\end{aligned}$$

This is the defining numerator of $\mathcal{R}_p(K)$. □

Lemma 8.800 (Far-leg cluster interpretation)**.** *Let ω_1, ω_2 and $C_i(z)$ be as in Lemma 8.799. Then, for every $R < \infty$ and every finite non-empty $K \subset V$,*

$$\overline{\chi}_p |K|\, \mathcal{R}_p^{>R}(K) = \sum_{z\in V} \mathbb{E}_p^{\otimes 2}\big[|(K \setminus B(z,R)) \cap C_1(z)|\, |(K \setminus B(z,R)) \cap C_2(z)|\big],$$

with equality in $[0,\infty]$.

Proof. For fixed z, independence gives

$$\begin{aligned}&\mathbb{E}_p^{\otimes 2}\big[|(K \setminus B(z,R)) \cap C_1(z)|\, |(K \setminus B(z,R)) \cap C_2(z)|\big] \\ &\quad = \left(\sum_{\substack{x\in K \\ d(x,z)>R}} \tau_p(x,z) \right)^2.\end{aligned}$$

Summing over z gives the defining numerator of $\mathcal{R}_p^{>R}(K)$. □

Proposition 8.801 (Two-replica profile controls finite-set growth)**.** *Assume G has maximum degree at most $\Delta < \infty$. Let*

$$S_p^{\mathrm{amb}}(K) = \sum_{x,y \in K} \tau_p(x,y).$$

For every finite non-empty $K \subset V$ and $0 \le p \le q < p_c$,

$$\frac{S_q^{\mathrm{amb}}(K)}{|K|} \le \frac{S_p^{\mathrm{amb}}(K)}{|K|} + \Delta \int_p^q \overline{\chi}_s \, \mathcal{R}_s(K)\, ds.$$

Proof. First work in a finite induced subgraph $W \supseteq K$, and let T_s^W be the two-point matrix for percolation restricted to W. As in the proof of Proposition 7.36, Russo's formula and BK give

$$\frac{d}{ds} T_s^W \le T_s^W A_W T_s^W$$

entrywise, where A_W is the adjacency matrix of W. Therefore

$$\begin{aligned} \frac{d}{ds} \left\langle \mathbf{1}_K, T_s^W \mathbf{1}_K \right\rangle &\le \left\langle T_s^W \mathbf{1}_K, A_W T_s^W \mathbf{1}_K \right\rangle \\ &\le \Delta \left\| T_s^W \mathbf{1}_K \right\|_2^2 \le \Delta \left\| T_s \mathbf{1}_K \right\|_2^2 \\ &= \Delta \overline{\chi}_s |K| \, \mathcal{R}_s(K). \end{aligned}$$

Integrating from p to q, dividing by $|K|$, and then letting $W \uparrow V$ gives the claimed inequality by monotone convergence. □

Remark 8.802. Lemma 8.799 explains the name: large $\mathcal{R}_p(K)$ means that, after summing over possible roots z, two independent clusters rooted at the same z both collect a large amount of mass from K. Proposition 8.801 shows that this is also the quantity that appears naturally in the Russo–BK differential growth of the finite-set two-point mass. This makes $\mathcal{R}_p^*$ a percolation-specific target, not just an operator-norm proxy.

Proposition 8.803 (Two-replica boundedness implies finite-set escape)**.** *Let G be connected and locally finite, and suppose that*

$$\overline{\chi}_p \to \infty \qquad \text{as } p \uparrow p_c.$$

If

$$\limsup_{p \uparrow p_c} \mathcal{R}_p^* < \infty,$$

then finite-set escape holds.

Proof. Let $C < \infty$ and $p_0 < p_c$ be such that $\mathcal{R}_p^* \le C$ for $p \in [p_0, p_c)$. For every finite non-empty $K \subset V$, Cauchy–Schwarz gives

$$\begin{aligned} \frac{1}{\overline{\chi}_p |K|} \sum_{x,y \in K} \tau_p(x,y) &= \frac{\langle \mathbf{1}_K, T_p \mathbf{1}_K \rangle}{\overline{\chi}_p |K|} \\ &\le \frac{|K|^{1/2} \left\| T_p \mathbf{1}_K \right\|_2}{\overline{\chi}_p |K|} \\ &= \left(\frac{\mathcal{R}_p(K)}{\overline{\chi}_p} \right)^{1/2} \le \left(\frac{C}{\overline{\chi}_p} \right)^{1/2}. \end{aligned}$$

The right-hand side tends to zero as $p \uparrow p_c$, uniformly in K. This is finite-set escape. □

Theorem 8.804 (Two-replica intermediate theorem)**.** *Let G be connected, locally finite, quasi-transitive, nonamenable, and of bounded degree. Assume:*

(i) $\limsup_{p\uparrow p_c} \mathcal{R}_p^* < \infty$*;*

(ii) $\limsup_{p\uparrow p_c} (p_c - p)\overline{\chi}_p < \infty$.

Then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. For quasi-transitive graphs, $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. Proposition 8.803 gives finite-set escape. The second assumption is the mean-field susceptibility input (MF), so Theorem 4.1 applies. □

Proposition 8.805 (A strict L^2 gap forces the two-replica bound)**.** *Suppose that $\overline{\chi}_p \to \infty$ as $p \uparrow p_c$. If there exists $q > p_c$ such that T_q is bounded on $\ell^2(V)$, then*

$$\lim_{p\uparrow p_c} \mathcal{R}_p^* = 0.$$

In particular, $p_c(G) < p_{2\to 2}(G)$ implies the two-replica hypothesis in Theorem 8.804.

Proof. Fix $q > p_c$ with $M = \|T_q\|_{2\to 2} < \infty$. By monotonicity of connection probabilities, $0 \le T_p \le T_q$ pointwise for every $p < p_c$. Hence, for every finite non-empty K,

$$\mathcal{R}_p(K) = \frac{\|T_p \mathbf{1}_K\|_2^2}{\overline{\chi}_p |K|} \le \frac{M^2}{\overline{\chi}_p}.$$

Taking the supremum over finite K and letting $p \uparrow p_c$ proves the claim. □

Remark 8.806. Proposition 8.13 shows that the finite-set escape scale is exactly $\mathcal{R}_p^* = o(\overline{\chi}_p)$. The bounded profile assumed in Theorem 8.804 is a stronger, more usable target: it tests T_p only on indicators of finite sets and gives the layer-square estimate needed in the separator-grid setup. At the other end, Proposition 8.805 shows that any proof of $p_c < p_{2\to 2}$ would force the still stronger conclusion $\mathcal{R}_p^* \to 0$. Thus the two-replica profile gives a calibrated ladder: sublinear growth is equivalent to finite-set escape, bounded growth is a plausible backscattering target, and vanishing growth is a necessary consequence of the desired strict L^2 gap.

9 A publishable intermediate target

The global conjecture may be too hard to close in one jump. A serious intermediate result would still be valuable if it covers a natural class not already handled by the known methods. The following abstract theorem indicates what such a result could look like.

Definition 9.1 (Uniform critical scattering property)**.** A quasi-transitive graph G has the uniform critical scattering property if for every sequence $p_n \uparrow p_c$ and every sequence of finite sets $K_n \subset V$, there are partitions of K_n that are $(o(1), o(1))$-scattering for Q_{p_n}.

Definition 9.2 (Uniform critical separator-convolution property)**.** A quasi-transitive graph G has the uniform critical separator-convolution property if for every sequence $p_n \uparrow p_c$ and every sequence of finite sets $K_n \subset V$, there are buffered partitions

$$K_n = K_{n,0} \sqcup K_{n,1} \sqcup \cdots \sqcup K_{n,m_n}$$

and proper path separators $W_{ij}^{(n)}$ between $K_{n,i}$ and $K_{n,j}$, $i,j \geq 1$, $i \neq j$, such that

$$\mathcal{B}_{p_n}(K_{n,0}; K_n) \to 0,$$

$$\frac{1}{\overline{\chi}_{p_n}|K_n|} \sum_{i=1}^{m_n} \sum_{x,y \in K_{n,i}} \tau_{p_n}(x,y) \to 0$$

and

$$\mathcal{C}_{p_n}(K_{n,1}, \dots, K_{n,m_n}; W^{(n)}) \to 0.$$

Theorem 9.3 (Abstract intermediate theorem)**.** *Let G be connected, locally finite, quasi-transitive, and nonamenable. Assume:*

(i) G has the uniform critical scattering property;

(ii) $\limsup_{p \uparrow p_c}(p_c - p)\overline{\chi}_p < \infty$.

Then $p_c(G) < p_{2\to 2}(G)$, and hence $p_c(G) < p_u(G)$.

Proof. The scattering property and Proposition 8.4 give finite-set escape. The susceptibility assumption is exactly (MF). The conclusion follows from Theorem 4.1. □

Theorem 9.4 (Separator-convolution intermediate theorem)**.** *Let G be connected, locally finite, quasi-transitive, and nonamenable. Assume:*

(i) G has the uniform critical separator-convolution property;

(ii) $\limsup_{p \uparrow p_c}(p_c - p)\overline{\chi}_p < \infty$.

Then

$$p_c(G) < p_{2\to 2}(G).$$

Proof. The separator-convolution property and Proposition 8.6 give finite-set escape. The susceptibility assumption is (MF), so Theorem 4.1 applies. □

Remark 9.5. These intermediate theorems are not the final goal, because their assumptions may hide most of the difficulty. They are nevertheless useful targets: a future paper could be publishable if it proves one of the uniform critical separator properties for a previously untreated geometric or group-theoretic class and verifies the susceptibility input.

10 From trapped mass to almost invariant sets

This section formulates the desired contradiction mechanism. It is not yet proved in the required generality.

Conjecture 10.1 (Trapped two-point mass creates graph almost-invariance)**.** *Let G be connected, locally finite, quasi-transitive, and bounded degree. If finite-set escape fails for $Q_p = T_p/\overline{\chi}_p$, then G has a Folner sequence. Equivalently, G is amenable.*

Proposition 10.2. *Assume Conjecture 10.1. Let G be connected, locally finite, quasi-transitive, and nonamenable. If (MF) holds, then $p_c(G) < p_{2\to 2}(G)$.*

Proof. Since G is nonamenable, it has no Folner sequence. By the contrapositive of Conjecture 10.1, finite-set escape holds. Together with the assumed mean-field susceptibility bound (MF), Theorem 4.1 gives $p_c(G) < p_{2\to 2}(G)$. □

Remark 10.3. Proposition 10.2 does not solve the open problem; it isolates the missing bridge. The bridge is plausible because failure of escape gives finite sets retaining a positive fraction of a normalized, symmetric, sub-Markov kernel. A general principle of nonamenability says that finite sets cannot retain random-walk mass uniformly. The challenge is that Q_p is not the random-walk kernel, and its relation to graph boundary is subtle.

11 A concrete spectral route

Let P be the simple random walk operator on G. For bounded-degree nonamenable quasi-transitive graphs, $\|P\|_{2\to 2} < 1$. A direct proof of finite-set escape would follow from an inequality of the following type.

Conjecture 11.1 (Kernel-to-walk comparison)**.** *For every bounded-degree quasi-transitive graph G, there are functions $\Phi_p : [0,1] \to [0,1]$ with $\Phi_p(t) \to 0$ as $t \downarrow 0$, uniformly as $p \uparrow p_c$, such that for every finite non-empty $K \subset V$,*

$$\mathcal{M}_{Q_p}(K) \le \Phi_p\left(\frac{|\partial_E K|}{|K|}\right) + o_{p\uparrow p_c}(1).$$

This exact formulation is probably too optimistic: in a nonamenable graph, the boundary ratio is bounded below, not small. The useful direction would be the contrapositive: if $\mathcal{M}_{Q_p}(K)$ is large, then $|\partial_E K|/|K|$ must be small. A more realistic form is therefore:

Conjecture 11.2 (Mass-trapping isoperimetric inverse)**.** *There exists a function Ψ with $\Psi(\epsilon) \downarrow 0$ as $\epsilon \downarrow 0$ such that the following holds. If*

$$\mathcal{M}_{Q_p}(K) \ge \epsilon$$

for a sequence $p \uparrow p_c$, then K contains a subset K' with

$$\frac{|\partial_E K'|}{|K'|} \le \Psi(\epsilon) + o_{p\uparrow p_c}(1).$$

Proposition 11.3. *If Conjecture 11.2 holds with $\Psi(\epsilon) < h(G)$ for every fixed $\epsilon > 0$ after passing sufficiently close to p_c, then finite-set escape holds on every nonamenable quasi-transitive graph G.*

Proof. Suppose finite-set escape fails. By Lemma 5.2, there are $\epsilon > 0$, $p_n \uparrow p_c$, and finite sets K_n with $\mathcal{M}_{Q_{p_n}}(K_n) \ge \epsilon$. Conjecture 11.2 gives subsets K_n' with boundary ratios eventually below $h(G)$, contradicting the definition of the Cheeger constant. □

The following elementary example shows why this inverse cannot be a theorem about arbitrary symmetric adapted kernels on nonamenable graphs.

Proposition 11.4 (Kernel-only nonamenability is insufficient)**.** *Let $\Gamma = F_2 \times \mathbb{Z}$. For every $\delta > 0$, there is a finitely supported symmetric probability measure μ whose support generates Γ, and a finite set $K \subset \Gamma$, such that*

$$\frac{1}{|K|}\sum_{x,y\in K} \mu(x^{-1}y) \ge 1 - \delta.$$

Proof. Let a, b generate F_2 and let t generate $\mathbb{Z}$. For $L \geq 1$, let u_L be the uniform probability measure on $\{t^j : |j| \leq L\}$, and let ν be the uniform probability measure on $\{a, a^{-1}, b, b^{-1}, t, t^{-1}\}$. Put

$$\mu = (1-\eta)u_L + \eta\nu.$$

Then μ is symmetric, finitely supported, and generating whenever $\eta > 0$.

Let $K_M = \{t^j : |j| \leq M\}$, with $M > L$. If $|i| \leq M - L$, then $t^i\{t^j : |j| \leq L\} \subseteq K_M$. Hence

$$\frac{1}{|K_M|}\sum_{x,y\in K_M}\mu(x^{-1}y) = \frac{1}{|K_M|}\sum_{x\in K_M}\mu(x^{-1}K_M) \geq (1-\eta)\frac{2(M-L)+1}{2M+1}.$$

Choosing η small and then M/L large makes the right-hand side at least $1-\delta$. □

Remark 11.5. Thus Kesten's theorem, property T, or a spectral gap for a fixed simple random walk cannot by themselves prove finite-set escape for Q_p. The kernel Q_p must be used as a percolation two-point kernel, with its BK, differential, or near-critical structure; otherwise mass can hide for a long time inside a large amenable direction even when the ambient group is nonamenable.

12 Known cases and stress tests

Any proposed proof must recover the known positive cases without accidentally proving a false amenable statement.

12.1 Regular trees and killed growth

The killed-growth condition is not vacuous. On a regular tree, finite subtrees have many boundary vertices, and the average number of long killed self-avoiding paths grows like the square root of the ambient connective constant.

Proposition 12.1 (Killed growth on regular trees)**.** *Let T_d be the d-regular tree with $d \geq 3$. Then*

$$\kappa_{\mathrm{kill}}(T_d) = \sqrt{d-1}.$$

Consequently,

$$\kappa_{\mathrm{kill}}(T_d) < 1/p_{\mathrm{c}}(T_d).$$

Proof. Put $q = d - 1$. Let R_n be the spherical operator on T_d,

$$R_n f(x) = \sum_{y:\mathrm{dist}(x,y)=n} f(y).$$

The standard spherical-polynomial estimate on the d-regular tree gives

$$\|R_n\|_{2\to 2} \leq C_d(n+1)q^{n/2}. \tag{1}$$

Indeed, $R_n = P_n(A)$, where A is the adjacency operator, the spectrum of A is contained in $[-2\sqrt{q}, 2\sqrt{q}]$, and the polynomials P_n are the tree-spherical Chebyshev polynomials. The Chebyshev bound on this interval yields (1).

For any finite $K \subset T_d$, every killed self-avoiding path of length n in K has endpoints at ambient distance n. Therefore

$$|K|A_n(K) \leq \sum_{x,y\in K}\mathbf{1}\{\mathrm{dist}(x,y) = n\} = \langle \mathbf{1}_K, R_n\mathbf{1}_K\rangle \leq C_d(n+1)q^{n/2}|K|.$$

Taking the supremum over finite K and then the nth root gives $\kappa_{\mathrm{kill}}(T_d) \le \sqrt{q}$.

For the reverse inequality, let B_m be the ball of radius m around a root o. If x and y lie at distance m from o in two different principal branches, then the geodesic from x to y is a killed self-avoiding path of length $2m$ inside B_m. There are $d(d-1)q^{2m-2}$ ordered pairs of this type, while $|B_m| \le C'_d q^m$. Thus $A_{2m}(B_m) \ge c_d q^m$, and hence $\kappa_{\mathrm{kill}}(T_d) \ge \sqrt{q}$.

Finally, $p_{\mathrm{c}}(T_d) = q^{-1}$, so $\sqrt{q} < q = 1/p_{\mathrm{c}}(T_d)$. □

Proposition 12.2 (Finite-set escape on regular trees)**.** *Let T_d be the d-regular tree with $d \ge 3$. Then finite-set escape holds on T_d:*

$$\lim_{p \uparrow p_{\mathrm{c}}(T_d)} \sup_{K \subset V \text{ finite}} \frac{1}{|K|} \sum_{x,y \in K} Q_p(x,y) = 0.$$

Proof. Again put $q = d-1$. On T_d, the two-point function is exactly

$$\tau_p(x,y) = p^{\mathrm{dist}(x,y)}$$

and

$$\overline{\chi}_p = 1 + d \sum_{n \ge 1} q^{n-1} p^n = \frac{1+p}{1-qp}.$$

Using the spherical operators R_n from Proposition 12.1, we have, for every finite K,

$$\frac{1}{|K|} \sum_{x,y \in K} Q_p(x,y) = \frac{1}{\overline{\chi}_p |K|} \sum_{n \ge 0} p^n \left\langle \mathbf{1}_K, R_n \mathbf{1}_K \right\rangle.$$

By (1),

$$\frac{1}{|K|} \sum_{x,y \in K} Q_p(x,y) \le \frac{C_d}{\overline{\chi}_p} \sum_{n \ge 0} (n+1)(p\sqrt{q})^n.$$

As $p \uparrow p_{\mathrm{c}} = q^{-1}$, the series remains bounded because $p\sqrt{q} \le q^{-1/2} < 1$, while $\overline{\chi}_p \to \infty$. The bound is uniform in K, proving finite-set escape. □

Proposition 12.3 (Two-replica profile on regular trees)**.** *Let T_d be the d-regular tree with $d \ge 3$. Then*

$$\lim_{p \uparrow p_{\mathrm{c}}(T_d)} \mathcal{R}^*_p = 0.$$

In particular, $\sup_{p < p_{\mathrm{c}}(T_d)} \mathcal{R}^*_p < \infty$.

Proof. Put $q = d-1$, so $p_{\mathrm{c}} = q^{-1}$. As in Proposition 12.4, the tree spherical bound (1) gives

$$M_d := \sup_{0 \le p \le p_{\mathrm{c}}} \|T_p\|_{2 \to 2} < \infty.$$

For every finite non-empty $K \subset V(T_d)$,

$$\mathcal{R}_p(K) = \frac{\|T_p \mathbf{1}_K\|_2^2}{\overline{\chi}_p |K|} \le \frac{M_d^2}{\overline{\chi}_p}.$$

On the tree,

$$\overline{\chi}_p = \frac{1+p}{1-qp} \to \infty \qquad (p \uparrow q^{-1}),$$

so the displayed bound tends to zero uniformly over finite K. □

Proposition 12.4 (Local separator-convolution on regular trees)**.** *Let T_d be the d-regular tree with $d \geq 3$. Then T_d has the local separator-convolution property near $p_{\mathrm{c}}(T_d)$.*

Proof. Put $q = d - 1$, so $p_{\mathrm{c}} = 1/q$. Since $\tau_p(x, y) = p^{\mathrm{dist}(x,y)}$, we have

$$T_p = \sum_{n \geq 0} p^n R_n,$$

where R_n is the tree sphere kernel. By (1),

$$M_d := \sup_{0 \leq p \leq p_{\mathrm{c}}} \|T_p\|_{2 \to 2} < \infty.$$

Fix $\delta > 0$. Choose an integer L such that $2/L \leq \delta$, and put $R = 2L$. Since $\overline{\chi}_p \to \infty$ as $p \uparrow p_{\mathrm{c}}$, choose $p_0 < p_{\mathrm{c}}$ such that

$$\frac{M_d^2}{\overline{\chi}_p} \leq \delta \qquad (p \in [p_0, p_{\mathrm{c}})).$$

Root the tree at o, and for $r \in \{0, \ldots, L-1\}$ let

$$S_r = \{x : \mathrm{dist}(o, x) \equiv r \pmod{L}\}.$$

Given a finite non-empty $K \subset V$, choose r with $|K \cap S_r| \leq |K|/L$, and set $K_0 = K \cap S_r$. The non-buffer pieces are the non-empty intersections of $K \setminus K_0$ with the connected components of $T_d \setminus S_r$. Each component is contained either in the initial root slab before the first level of S_r, or between two consecutive levels of S_r, so every such component has diameter at most $2L$. Hence the pieces have diameter at most R.

For any two distinct non-buffer pieces, take $W_{ij} = S_r$. This is a proper path separator: it is disjoint from both pieces, and every path between different components of $T_d \setminus S_r$ hits S_r. The buffer mass is bounded by the row sum:

$$\mathcal{B}_p(K_0; K) = \frac{2}{\overline{\chi}_p |K|} \sum_{x \in K_0, y \in K} \tau_p(x, y) \leq \frac{2|K_0|}{|K|} \leq \frac{2}{L} \leq \delta.$$

For the separator-convolution term, write $f = \mathbf{1}_{K \setminus K_0}$. Since the kernels are nonnegative,

$$\begin{aligned} \overline{\chi}_p |K| \, \mathcal{C}_p(K_1, \ldots, K_m; W) &= \sum_{i \neq j} \sum_{z \in S_r} U_i^p(z) U_j^p(z) \\ &\leq \sum_{z \in S_r} \left(\sum_i U_i^p(z) \right)^2 \\ &\leq \|T_p f\|_2^2 \leq M_d^2 \, \|f\|_2^2 \leq M_d^2 |K|. \end{aligned}$$

Thus

$$\mathcal{C}_p(K_1, \ldots, K_m; W) \leq \frac{M_d^2}{\overline{\chi}_p} \leq \delta.$$

This proves the local separator-convolution property. □

Proposition 12.5 (Finite-fiber tree-grid separators)**.** *Let G be a connected bounded-degree graph, and let $s \geq 0$. Suppose that there are a bounded-degree tree $\mathbb{T}$, a root $o \in V(\mathbb{T})$, and a map*

$$\pi : V(G) \to V(\mathbb{T} \square \mathbb{Z}^s)$$

such that

$$b := \sup_{u \in V(\mathbb{T}\square\mathbb{Z}^s)} |\pi^{-1}(u)| < \infty$$

and, whenever $x \sim y$ in G,

$$\operatorname{dist}_{\mathbb{T}\square\mathbb{Z}^s}(\pi(x), \pi(y)) \le 1.$$

If

$$\limsup_{p \uparrow p_c(G)} \frac{\|T_p\|_{2\to 2}^2}{\overline{\chi}_p} < \infty,$$

then G has the local separator-convolution property near $p_c(G)$.

Proof. Let $\Delta_{\mathbb{T}}$ be a degree bound for $\mathbb{T}$, and let

$$A = 1 + \limsup_{p \uparrow p_c(G)} \frac{\|T_p\|_{2\to 2}^2}{\overline{\chi}_p}.$$

Fix $\delta > 0$. Choose an integer L such that

$$\frac{(s+1)(2+A)}{L} \le \delta,$$

and choose $p_0 < p_c(G)$ such that

$$\frac{\|T_p\|_{2\to 2}^2}{\overline{\chi}_p} \le A \qquad (p \in [p_0, p_c(G))).$$

Write a vertex of $\mathbb{T}\square\mathbb{Z}^s$ as $(t, n_1, \dots, n_s)$, and define

$$\rho_0(t,n) = \operatorname{dist}_{\mathbb{T}}(o,t), \qquad \rho_j(t,n) = n_j \quad (1 \le j \le s).$$

For $\mathbf{a} = (a_0, \dots, a_s) \in \{0, \dots, L-1\}^{s+1}$, put

$$\Sigma_{\mathbf{a}} = \{u \in V(\mathbb{T}\square\mathbb{Z}^s) : \rho_j(u) \equiv a_j \pmod{L} \text{ for some } 0 \le j \le s\}, \qquad S_{\mathbf{a}} = \pi^{-1}(\Sigma_{\mathbf{a}}).$$

The connected components of $(\mathbb{T}\square\mathbb{Z}^s) \setminus \Sigma_{\mathbf{a}}$ have at most

$$N_L = 1 + \Delta_{\mathbb{T}} \sum_{\ell=0}^{L-1} (\Delta_{\mathbb{T}} - 1)^{\ell}$$

choices for the tree coordinate and at most L^s choices for the Euclidean coordinates, with the evident interpretation when $\Delta_{\mathbb{T}} = 1$. Thus each base component has at most $N_L L^s$ vertices. Since the fibers of π have size at most b, every component of $G \setminus S_{\mathbf{a}}$ has at most $bN_L L^s$ vertices. Indeed, the projection of a connected set in $G \setminus S_{\mathbf{a}}$ is connected in $(\mathbb{T}\square\mathbb{Z}^s) \setminus \Sigma_{\mathbf{a}}$, because adjacent vertices in G project to equal or adjacent vertices of the base graph.

Given a finite non-empty $K \subset V(G)$, write $K_{\mathbf{a}} = K \cap S_{\mathbf{a}}$. For each fixed vertex of G, the proportion of residue vectors $\mathbf{a}$ for which it lies in $S_{\mathbf{a}}$ is at most $(s+1)/L$. Hence

$$\frac{1}{L^{s+1}} \sum_{\mathbf{a}} \frac{2|K_{\mathbf{a}}|}{|K|} \le \frac{2(s+1)}{L}.$$

Also, by positivity of T_p,

$$\frac{1}{L^{s+1}}\sum_{\mathbf{a}}\frac{1}{\overline{\chi}_p|K|}\sum_{z\in S_{\mathbf{a}}}\big(T_p\mathbf{1}_K(z)\big)^2\le\frac{s+1}{L}\frac{\|T_p\mathbf{1}_K\|_2^2}{\overline{\chi}_p|K|}\le\frac{(s+1)A}{L}.$$

Hence for some $\mathbf{a}$,

$$\frac{2|K_{\mathbf{a}}|}{|K|}+\frac{1}{\overline{\chi}_p|K|}\sum_{z\in S_{\mathbf{a}}}\big(T_p\mathbf{1}_K(z)\big)^2\le\frac{(s+1)(2+A)}{L}\le\delta.$$

Set $K_0=K_{\mathbf{a}}$, and let the non-buffer pieces be the non-empty intersections of $K\setminus K_0$ with the connected components of $G\setminus S_{\mathbf{a}}$. Each such piece has graph diameter at most

$$R=bN_LL^s-1.$$

For two distinct non-buffer pieces, take $W_{ij}=S_{\mathbf{a}}$. This is a proper path separator because the pieces lie in distinct components of $G\setminus S_{\mathbf{a}}$.

The buffer mass is bounded by

$$\mathcal{B}_p(K_0;K)\le\frac{2|K_0|}{|K|}.$$

For the separator-convolution term, put $f=\mathbf{1}_{K\setminus K_0}$. Since $T_pf\le T_p\mathbf{1}_K$ pointwise,

$$\begin{aligned}\overline{\chi}_p|K|\,\mathcal{C}_p(K_1,\dots,K_m;W)&=\sum_{i\neq j}\sum_{z\in S_{\mathbf{a}}}U_i^p(z)U_j^p(z)\\&\le\sum_{z\in S_{\mathbf{a}}}\left(\sum_i U_i^p(z)\right)^2\\&=\sum_{z\in S_{\mathbf{a}}}\big(T_pf(z)\big)^2\\&\le\sum_{z\in S_{\mathbf{a}}}\big(T_p\mathbf{1}_K(z)\big)^2.\end{aligned}$$

The choice of $\mathbf{a}$ therefore gives both

$$\mathcal{B}_p(K_0;K)\le\delta,\qquad\mathcal{C}_p(K_1,\dots,K_m;W)\le\delta.$$

Together with the diameter bound, this is the local separator-convolution property. □

Proposition 12.6 (Tree-grid separators from two-replica boundedness)**.** *In the setup of Proposition 12.5, replace the analytic hypothesis there by*

$$\limsup_{p\uparrow p_c(G)}\mathcal{R}_p^*<\infty.$$

Then G has the local separator-convolution property near $p_c(G)$.

Proof. Let $C<\infty$ and $p_1<p_c(G)$ be such that $\mathcal{R}_p^*\le C$ for $p\in[p_1,p_c(G))$. Repeat the construction of the periodic tree-Euclidean grids $\Sigma_{\mathbf{a}}$, their lifts $S_{\mathbf{a}}$, and the constant N_L from the proof of Proposition 12.5. Choose L so large that

$$\frac{(s+1)(2+C)}{L}\le\delta.$$

For a given finite non-empty K, the buffer average is still at most $2(s+1)/L$. Moreover, since each vertex of G belongs to at most an $(s+1)/L$ fraction of the lifted grids,

$$\frac{1}{L^{s+1}}\sum_{\mathbf{a}}\frac{1}{\overline{\chi}_p|K|}\sum_{z\in S_{\mathbf{a}}}\big(T_p\mathbf{1}_K(z)\big)^2 \le \frac{s+1}{L}\frac{\|T_p\mathbf{1}_K\|_2^2}{\overline{\chi}_p|K|} \le \frac{(s+1)C}{L}.$$

Thus some $\mathbf{a}$ has both buffer contribution and layer-square contribution with total at most δ. The components of $G\setminus S_{\mathbf{a}}$ have at most bN_LL^s vertices as before, so the resulting pieces have uniformly bounded diameter. The same separator convolution estimate used in Proposition 12.5 proves the local separator-convolution property. □

Remark 12.7. The two preceding finite-fiber tree-grid propositions are calibrations rather than solutions of the general problem. The global L^2-type hypothesis in Proposition 12.5 is strong; when $\overline{\chi}_p\to\infty$, it already implies finite-set escape by the elementary spectral bound

$$\frac{\langle\mathbf{1}_K,T_p\mathbf{1}_K\rangle}{\overline{\chi}_p|K|}\le\frac{\|T_p\|_{2\to2}}{\overline{\chi}_p}.$$

The two-replica hypothesis in Proposition 12.6 also implies finite-set escape directly by Proposition 8.803; its value here is that it matches the exact square mass appearing in the separator-convolution term. The point is geometric: finite fibers over periodic tree-Euclidean grids automatically provide small buffers and uniformly local pieces. In the separator-convolution route, the genuinely missing estimate is control of square two-point mass on the chosen separating grid without assuming the conclusion-like global ℓ^2 bound. In this setting it would be enough to prove the layer-square estimate

$$\lim_{L\to\infty}\limsup_{p\uparrow p_c}\sup_{K\subset V\text{ finite}}\min_{\mathbf{a}\in\{0,\dots,L-1\}^{s+1}}\frac{1}{\overline{\chi}_p|K|}\sum_{z\in S_{\mathbf{a}}}\big(T_p\mathbf{1}_K(z)\big)^2=0.$$

The buffer term is already small on average over the shifts $\mathbf{a}$. Thus, for product-like geometries, the separator program asks for a two-replica backscattering estimate: two independent connections from the same separating grid should not both carry a macroscopic amount of near-critical mass into the same finite test set. Proposition 7.29 rewrites this target, for graphs admitting a fibre decomposition over a base graph, as a restricted base-sphere estimate.

Proposition 12.8 (Boundary energy on regular trees)**.** *Let T_d be the d-regular tree with $d\ge3$. Then*

$$\lim_{p\uparrow p_c(T_d)}\sup_{K\subset V\ \text{finite}}\mathcal{E}_p(K)=0.$$

Proof. Put $q=d-1$. The spherical estimate (1) gives

$$\sup_{0\le p\le q^{-1}}\|T_p\|_{2\to2}\le C_d\sum_{n\ge0}(n+1)q^{-n/2}<\infty.$$

For a finite set K, put

$$w(a)=\mathbf{1}_{\partial_{\mathrm{in}}K}(a)\chi_p^K(a).$$

Since $\tau_p^K(a,x)\le\tau_p(a,x)$, we have

$$0\le w\le T_p\mathbf{1}_K$$

pointwise, and hence $\|w\|_2 \le C_d \|\mathbf{1}_K\|_2$. The exterior boundary weight $\mu_{p,K}$ is obtained from w by summing over at most d neighbouring boundary vertices at each exterior vertex, so

$$\|\mu_{p,K}\|_2^2 \le d^2 \|w\|_2^2 \le C_d'|K|.$$

Moreover

$$\|Q_p\|_{2\to 2} = \frac{\|T_p\|_{2\to 2}}{\overline{\chi}_p} \le \frac{C_d}{\overline{\chi}_p}.$$

Therefore

$$\mathcal{E}_p(K) = \frac{p^2}{|K|}\langle \mu_{p,K}, Q_p \mu_{p,K}\rangle \le \frac{C_d''}{\overline{\chi}_p}.$$

Since $\overline{\chi}_p \to \infty$ as $p \uparrow p_c(T_d)$, the bound tends to zero uniformly in K. □

Remark 12.9. The proof identifies the right obstruction for extending (KCG). The relevant comparison is not with the full number $d(d-1)^{n-1}$ of ambient self-avoiding paths from one vertex, but with the spherical L^2-norm $q^{n/2}$. This suggests that a useful general version of (KCG) should be tied to Haagerup-type radial operator bounds or to an L^2 growth exponent, rather than to pointwise connective constants. Proposition 12.8 also shows that the boundary-energy condition is calibrated correctly on trees, although its proof uses the already-known L^2-boundedness of T_p up to p_c and is therefore not a general substitute for the missing argument.

12.2 Hyperbolic graphs

Hutchcroft proved $p_c < p_u$ and the stronger triangle-condition conclusions for nonamenable Gromov hyperbolic quasi-transitive graphs [Hut19]. In the language of this draft, hyperbolicity provides enough large-scale separation to prove finite-set escape.

This case is the model for the separator program, but it also shows why the general problem is hard. Hyperbolicity supplies a canonical boundary and a strong form of thin-triangle geometry. General nonamenable Cayley graphs may have no such boundary, may contain large flat or amenable-looking regions, and may not admit any obvious half-space decomposition. A general proof must replace geometric boundary separation with an intrinsic probabilistic or operator-theoretic mechanism.

12.3 Nonunimodular transitive graphs

Graphs admitting a nonunimodular quasitransitive automorphism group are also not part of the remaining obstruction. For such an action, let Δ be the modular function and define the tilted susceptibility threshold p_t by finiteness of

$$\chi_{p,1/2} = \sum_x \tau_p(o,x)\Delta(o,x)^{1/2}.$$

Hutchcroft proved $p_c < p_t$ in this setting [Hut20]. The general tilted-to-L^2 comparison gives $p_t \le p_{2\to 2}$ [Hut19]. Hence

$$p_c(G) < p_{2\to 2}(G)$$

for every graph in this nonunimodular class.

This matters for the project because nonunimodularity supplies an external height function and a tilted mass-transport principle. The unresolved cases without a nonunimodular quasitransitive action do not have this structure available globally. The backscattering and Hammersley–Welsh mechanism of Hutchcroft–Pan therefore needs a replacement for the tilt before it can apply there.

12.4 Products and uniqueness threshold phenomena

For products such as $T \times \mathbb{Z}^d$, the non-uniqueness phase and the behavior at p_{u} are far better understood than in arbitrary graphs. Recent work proves refined dimension-jump and $p_{2\to 2}$-threshold identifications in these product and lamplighter settings [HP24a]. These examples are essential tests for any proposed general proof: they contain large amenable directions but remain nonamenable because of the tree factor.

They also test overly naive boundary arguments. A box in the $\mathbb{Z}^d$-direction has small boundary relative to its volume inside that factor, while the product graph remains nonamenable because of expansion in the tree direction. Therefore a proof based only on local Euclidean-looking boundary ratios will fail. The normalized kernel Q_p must see the nonamenable direction at the correct scale.

12.5 Amenable graphs

On amenable quasi-transitive graphs, $p_{\mathrm{c}} = p_{\mathrm{u}}$. Therefore no argument can show finite-set escape from quasi-transitivity alone. Any proof must use nonamenability in a genuinely quantitative way.

This is a useful guard against false proofs. If an argument uses only bounded degree, transitivity, positivity, symmetry, BK-type inequalities, and subcriticality, it is almost certainly proving too much. Somewhere the proof must use a spectral gap, a Cheeger lower bound, a cost/nonamenability input, or an equivalent obstruction to Folner sequences.

Proposition 12.10 (Two-replica profile diverges on the line)**.** *For $G = \mathbb{Z}$,*

$$\lim_{p\uparrow 1} \mathcal{R}_p^* = \infty.$$

Proof. On $\mathbb{Z}$, $p_{\mathrm{c}} = 1$,

$$\tau_p(x,y) = p^{|x-y|}, \qquad \overline{\chi}_p = \sum_{m\in\mathbb{Z}} p^{|m|} = \frac{1+p}{1-p}.$$

Let $K_N = \{1,\dots,N\}$, and take

$$N = \left\lceil 4(1-p)^{-1} \right\rceil.$$

If $z \in \{N/4,\dots,3N/4\} \cap \mathbb{Z}$, then z is at distance at least $N/4 - 1$ from both endpoints up to harmless rounding. Thus, with $M = \lfloor N/4 \rfloor - 1$,

$$T_p \mathbf{1}_{K_N}(z) = \sum_{x=1}^{N} p^{|x-z|} \geq \sum_{|m|\leq M} p^{|m|} = \overline{\chi}_p - \frac{2p^{M+1}}{1-p}.$$

For p sufficiently close to 1, the choice of N gives

$$\frac{2p^{M+1}}{1-p} \leq \frac{1}{2}\overline{\chi}_p,$$

and hence $T_p \mathbf{1}_{K_N}(z) \geq \overline{\chi}_p/2$ for all such z. There are at least $N/2$ such z, so

$$\mathcal{R}_p(K_N) = \frac{\|T_p \mathbf{1}_{K_N}\|_2^2}{\overline{\chi}_p N} \geq \frac{1}{8}\,\overline{\chi}_p$$

for all p sufficiently close to 1. Since $\overline{\chi}_p \to \infty$, the supremum over finite K diverges. □

12.6 Planar site conjectures

Recent planar site-percolation work has resolved and refuted variants of older Benjamini-Schramm planar questions in settings that are not the same as Conjecture 2.1 [Li26]. These papers are relevant as evidence that the area is active and that end-structure can be decisive. They do not remove the need to solve the quasi-transitive bond-percolation nonamenability conjecture.

13 Failed-proof guardrails

The following tempting reductions should not be accepted without new input.

F1. *Row-sum normalization alone is insufficient.* Lemma 5.1 gives $\|Q_p\|_{2\to 2} \le 1$, but finite-set escape asks for a much stronger delocalization statement for normalized indicators.

F2. *Nonamenability of G is not automatically nonamenability of Q_p.* The kernel Q_p is long range and depends on p. Its internal mass on finite sets need not be controlled directly by the edge-boundary of those sets without an additional comparison theorem.

F3. *Mean-field susceptibility is not free.* The condition $(p_c - p)\overline{\chi}_p = O(1)$ is part of Hutchcroft's criterion. A complete proof must establish it, bypass it, or replace the criterion.

F4. *Known special cases should not be relabeled as the general theorem.* Hyperbolic, nonunimodular, planar, product, and subgroup relativization results are major, but none is the full theorem targeted here.

14 Immediate technical tasks

The next stages of the project are concrete.

T1. Prove or disprove Conjecture 11.2 in the stated form. The first check is whether products $T \times \mathbb{Z}^d$ force a different inverse profile.

T2. Re-read the proof of Hutchcroft's hyperbolic case and rewrite the geometric half-space argument as an abstract separator lemma.

T3. Search for a class of nonamenable quasi-transitive graphs not covered by current results but satisfying the abstract separator lemma.

T4. Try to prove the mean-field susceptibility bound (MF) from nonamenability alone, or find a replacement for Theorem 4.1 that avoids it.

T5. Keep an explicit failed-proof log so that recurring false reductions are not retried.

15 Conclusion

The selected target is a real open problem in probability theory. A complete solution would be publishable and significant. The working route is to prove that nonamenability forces finite-set escape for the normalized subcritical two-point kernel. This draft proves only elementary reductions and identifies the core missing bridge. Subsequent work should focus on turning trapped two-point mass into graph almost-invariance, or finding a different route to $p_c < p_{2\to 2}$.

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